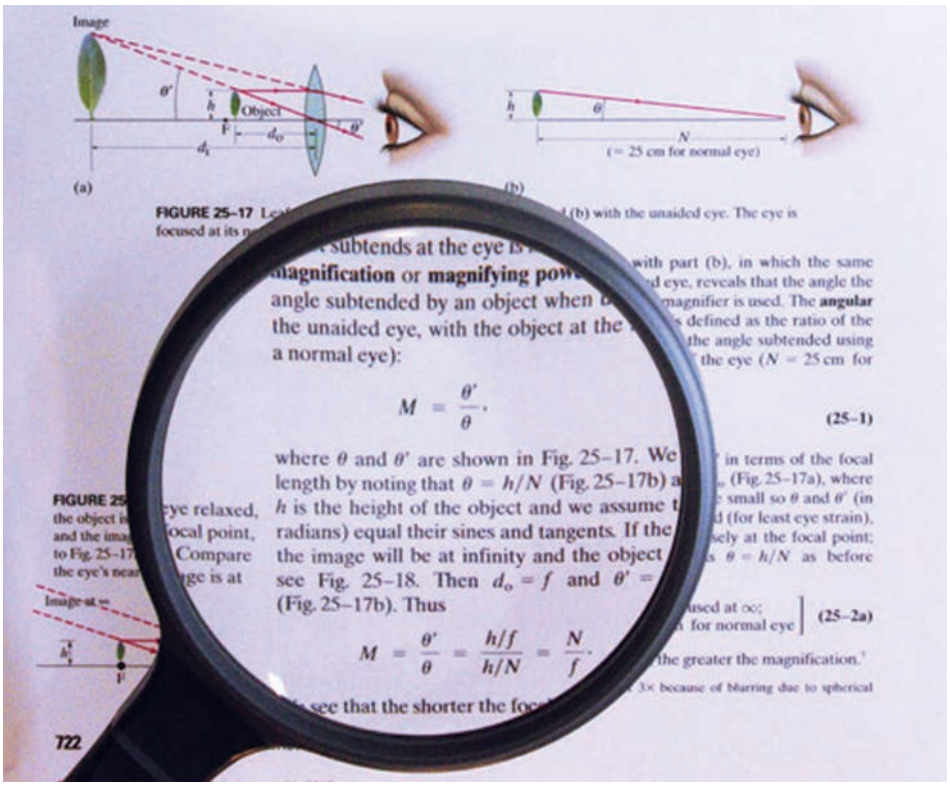


Of the many optical devices we discuss in this Chapter, the magnifying glass is the simplest. Here it is magnifying part of page 722 of this Chapter, which describes how the magnifying glass works according to the ray model. In this Chapter we examine film and digital cameras, the human eye, eyeglasses, telescopes, and microscopes as well as image resolution, X-rays, and CT scans.

CHAPTER 25

Optical Instruments

CHAPTER-OPENING QUESTION—Guess now!

Because of diffraction, a light microscope has a useful magnification of about

(a) 50×; **(b)** 100×; **(c)** 500×; **(d)** 2000×; **(e)** 5000×;

and the smallest objects it can resolve have a size of about

(a) 10 nm; **(b)** 100 nm; **(c)** 500 nm; **(d)** 2500 nm; **(e)** 5500 nm.

In our discussion of the behavior of light in the two previous Chapters, we also described a few instruments such as the spectrometer and the Michelson interferometer. In this Chapter, we will discuss some more common instruments, most of which use lenses, including the camera, telescope, microscope, and the human eye. To describe their operation, we will use ray diagrams as we did in Chapter 23. However, we will see that understanding some aspects of their operation will require the wave nature of light.

CONTENTS

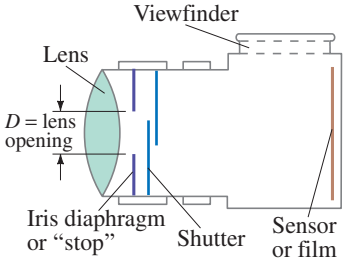
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25–2 The Human Eye; Corrective Lenses
25–3 Magnifying Glass
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25–1 Cameras: Film and Digital

The basic elements of a **camera** are a lens, a light-tight box, a shutter to let light pass through the lens only briefly, and in a digital camera an electronic sensor or in a traditional camera a piece of film (Fig. 25–1). When the shutter is opened for a brief "exposure," light from external objects in the field of view is focused by the lens as an image on the sensor or film.

You can see the image yourself if you remove the back of a conventional camera, keeping the shutter open, and view through a piece of tissue paper (on which an image can form) placed where the film should be.

FIGURE 25–1 A simple camera.



PHYSICS APPLIED

Cameras, film and digital

Film contains a thin **emulsion** (= a coating) with light-sensitive chemicals that change when light strikes them. The film is then developed by chemicals dissolved in water, which causes the most changed areas (brightest light) to turn opaque, so the image is recorded on the film.[†] We might call film "chemical photography" as compared to digital, which is electronic.

Digital Cameras, Electronic Sensors (CCD, CMOS)

In a **digital camera**, the film is replaced by a semiconductor sensor. Two types are common: **CCD** (*charge-coupled device*) and **CMOS** (*complementary metal oxide semiconductor*). A CCD sensor is made up of millions of tiny semiconductor **pixels** ("picture elements")—see Fig. 24–49. A 12-MP (12-megapixel) sensor might contain about 4000 pixels horizontally by 3000 pixels vertically over an area of perhaps 16×12 mm, and preferably larger such as 36×24 mm like 35-mm film. Light reaching any pixel liberates electrons within the semiconductor[‡] which are stored as charge in that pixel's capacitance. The more intense the light, the more charge accumulates during the brief exposure time. After exposure, the charge on each pixel has to be "read" (measured) and stored. A reader circuit first reads the charge on the pixel capacitance right next to it. Immediately after, the charge on each pixel is electronically transferred to its adjacent pixel, towards the reader which reads each pixel charge in sequence, one-by-one. Hence the name "charge-coupled device." All this information (the brightness of each pixel) goes to a central processor that stores it and allows re-formation of the image later onto the camera's screen, a computer screen, or a printer. After all the pixel charge information is transferred to memory (Section 21–8), a new picture can be taken.

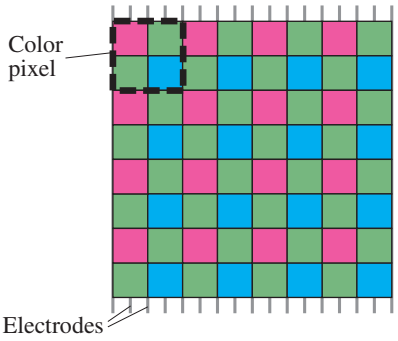


FIGURE 25–2 Portion of a typical Bayer array sensor. A square group of four pixels $\begin{matrix}R&G\\G&B\end{matrix}$ is sometimes called a "color pixel."

A CMOS sensor also uses a silicon semiconductor, and incorporates transistor electronics within each pixel, allowing parallel readout, somewhat like the similar MOSFET array that was shown in Fig. 21–29.

Sensor sizes are typically in the ratio 4:3 or 3:2. A larger sensor is better because it can hold more pixels, and/or each pixel can be larger and hold more charge (free electrons) to provide a wider range of brightness, better color accuracy, and better sensitivity in low-light conditions.

In the most common array of pixels, referred to as a **Bayer mosaic**, color is achieved by red, green, and blue filters over alternating pixels as shown in Fig. 25–2, similar to what a color LCD or CRT screen does (Sections 17–11 and 24–11). The sensor type shown in Fig. 25–2 contains twice as many green pixels as red or blue (because green seems to have a stronger influence on the human eye's sensation of sharpness). The computer-analyzed color at many pixels is often an average with nearest-neighbor colors to reduce memory size (= compression, see page 489).

Each different color of pixel in a Bayer array is counted as a *separate* pixel. In contrast, in an LCD screen (Sections 17–11 and 24–11), a group of three subpixels is counted as one pixel, a more conservative count.

An alternative technology, called "Foveon," uses a semiconductor layer system. Different wavelengths of light penetrate silicon to different depths, as shown in Fig. 25–3: blue wavelengths are absorbed in the top layer, allowing green and red light to pass through. Longer wavelengths (green) are absorbed in the second layer, and the bottom layer detects the longest wavelengths (red). All three colors are detected by each "tri-pixel" site, resulting in better color resolution and fewer artifacts.

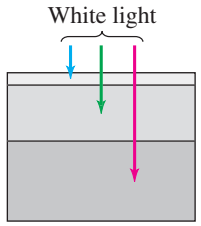


FIGURE 25–3 A layered or "Foveon" tri-pixel that includes all three colors, arranged vertically so light can pass through all three subpixels.

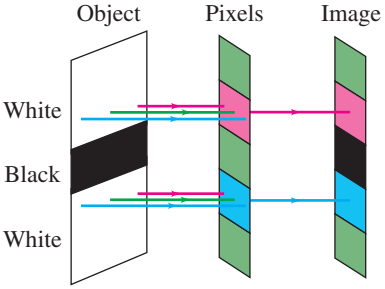


FIGURE 25–4 Suppose we take a picture that includes a thin black line (our object) on a white background. The *image* of this black line has a colored "halo" (red above, blue below) due to the mosaic arrangement of color filter pixels, as shown by the colors transmitted to the image. Computer averaging minimizes such color problems (the green at top and bottom of image may average with nearby pixels to give white or nearly so) but the image is consequently "softened" or blurred.

Digital Artifacts

Digital cameras can produce image artifacts (artificial effects in the image not present in the original) resulting from the electronic sensing of the image. One example using the Bayer mosaic of pixels (Fig. 25–2) is described in Fig. 25–4.

[†]This is called a *negative*, because the black areas correspond to bright objects and vice versa. The same process occurs during printing to produce a black-and-white "positive" picture from the negative. Color film has three emulsion layers (or dyes) corresponding to the three primary colors.

[‡]Specifically, photons of light knock electrons in the valence band up to the conduction band. This material on semiconductors is covered in Chapter 29.

Camera Adjustments

There are three main adjustments on good-quality cameras: shutter speed, *f*-stop, and focusing. Although most cameras today make these adjustments automatically, it is valuable to understand these adjustments to use a camera effectively. For special or top-quality work, a manual camera is indispensable (Fig. 25–5).



FIGURE 25–5 On this camera, the *f*-stops and the focusing ring are on the camera lens. Shutter speeds are selected on the small wheel on top of the camera body.

Exposure time or shutter speed This refers to how quickly the digital sensor can make an accurate reading of the images, or how long the shutter of a camera is open and the film or sensor is exposed. It could vary from a second or more ("time exposures") to $\frac{1}{1000}$ s or faster. To avoid blurring from camera movement, exposure times shorter than $\frac{1}{100}$ s are normally needed. If the object is moving, even shorter exposure times are needed to "stop" the action ($\frac{1}{1000}$ s or less). If the exposure time is not fast enough, the image will be blurred by camera shake. Blurring in low light conditions is more of a problem with cell-phone cameras whose inexpensive sensors need to have the shutter open longer to collect enough light. Digital still cameras or cell phones that take short videos must have a fast enough "sampling" time and fast "clearing" (of the charge) so as to take pictures at least 15 frames per second; preferable is 24 fps (like film) or 30, 60, or 120 fps like TV refresh rates (25, 50, or 100 in areas like Europe where 50 Hz is the normal line voltage frequency).

***f*-stop** The amount of light reaching the sensor or film depends on the area of the **lens opening** as well as shutter speed, and must be carefully controlled to avoid **underexposure** (too little light so the picture is dark and only the brightest objects show up) or **overexposure** (too much light, so that bright objects have a lack of contrast and "washed-out" appearance). A high quality camera controls the exposure with a "stop" or iris diaphragm, whose opening is of variable diameter, placed behind the lens (Fig. 25–1). The lens opening is controlled (automatically or manually) to compensate for bright or dark lighting conditions, the sensitivity of the sensor or film,[†] and for different shutter speeds. The size of the opening is specified by the ***f*-stop** or ***f*-number**, defined as

$$f\text{-stop} = \frac{f}{D},$$

where f is the focal length of the lens and D is the diameter of the lens opening (see Fig. 25–1). For example, when a 50-mm-focal-length lens has an opening $D = 25\,\text{mm}$, then $f/D = 50\,\text{mm}/25\,\text{mm} = 2$, so we say it is set at $f/2$. When this lens is set at $f/5.6$, the opening is only 9 mm $(50/9 = 5.6)$. For faster shutter speeds, or low light conditions, a wider lens opening must be used to get a proper exposure, which corresponds to a smaller *f*-stop number. The smaller the *f*-stop number, the larger the opening and the more light passes through the lens to the sensor or film. The smallest *f*-number of a lens (largest opening) is referred to as the *speed* of the lens. The best lenses may have a speed of $f/2.0$, or even faster. The advantage of a fast lens is that it allows pictures to be taken under poor lighting conditions. Good quality lenses consist of several elements to reduce the defects present in simple thin lenses (Section 25–6). Standard *f*-stops are

$$1.0,\ 1.4,\ 2.0,\ 2.8,\ 4.0,\ 5.6,\ 8,\ 11,\ 16,\ 22,\ \text{and}\ 32$$

(Fig. 25–5). Each of these stops corresponds to a diameter reduction by a factor of $\sqrt{2} \approx 1.4$. Because the amount of light reaching the film is proportional to the *area* of the opening, and therefore proportional to the diameter squared, each standard *f*-stop corresponds to a factor of 2 change in light intensity reaching the film.

[†]Different films have different sensitivities to light, referred to as the "film speed" and specified as an "ISO (or ASA) number." A "faster" film is more sensitive and needs less light to produce a good image, but is grainier which you see when the image is enlarged. Digital cameras may have a "gain" or "ISO" adjustment for sensitivity. A typical everyday ISO might be 200 or so. Adjusting a CCD to be "faster" (high ISO like 3200) for low light conditions results in "noise," resulting in graininess just as in film cameras.



(a)



(b)

FIGURE 25–6 Photos taken with a camera lens (a) focused on a nearby object with distant object blurry, and (b) focused on a more distant object with nearby object blurry.

Focusing Focusing is the operation of placing the lens at the correct position relative to the sensor or film for the sharpest image. The image distance is smallest for objects at infinity (the symbol ∞ is used for infinity) and is equal to the focal length, as we saw in Section 23–7. For closer objects, the image distance is greater than the focal length, as can be seen from the lens equation, $1/f = 1/d_o + 1/d_i$ (Eq. 23–8). To focus on nearby objects, the lens must therefore be moved away from the sensor or film, and this is usually done on a manual camera by turning a ring on the lens.

If the lens is focused on a nearby object, a sharp image of it will be formed, but the image of distant objects may be blurry (Fig. 25–6). The rays from a point on the distant object will be out of focus—instead of a point, they will form a circle on the sensor or film as shown (exaggerated) in Fig. 25–7. The distant object will thus produce an image consisting of overlapping circles and will be blurred. These circles are called **circles of confusion**. To have near and distant objects sharp in the same photo, you (or the camera) can try setting the lens focus at an intermediate position. For a given distance setting, there is a range of distances over which the circles of confusion will be small enough that the images will be reasonably sharp. This is called the **depth of field**. For a sensor or film width of 36 mm (including 35-mm film cameras), the depth of field is usually based on a maximum circle of confusion diameter of 0.030 mm, even 0.02 mm or 0.01 mm for critical work or very large photographs. The depth of field varies with the lens opening. If the lens opening is smaller, only rays through the central part of the lens are accepted, and these form smaller circles of confusion for a given object distance. Hence, at smaller lens openings, a greater range of object distances will fit within the circle of confusion criterion, so the depth of field is greater. Smaller lens openings, however, result in reduced resolution due to diffraction (discussed later in this Chapter). Best resolution is typically found around $f/5.6$ or $f/8$.

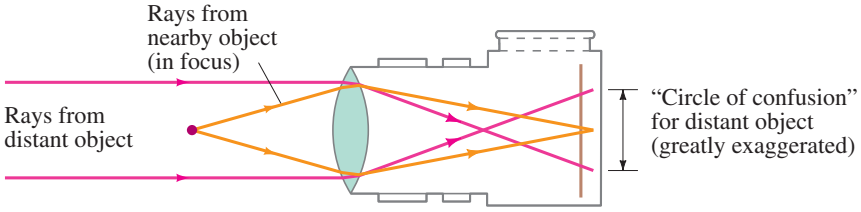


FIGURE 25–7 When the lens is positioned to focus on a nearby object, points on a distant object produce circles and are therefore blurred. (The effect is shown greatly exaggerated.)

EXAMPLE 25–1 **Camera focus.** How far must a 50.0-mm-focal-length camera lens be moved from its infinity setting to sharply focus an object 3.00 m away?

APPROACH For an object at infinity, the image is at the focal point, by definition (Section 23–7). For an object distance of 3.00 m, we use the thin lens equation, Eq. 23–8, to find the image distance (distance of lens to film or sensor).

SOLUTION When focused at infinity, the lens is 50.0 mm from the film. When focused at $d_o = 3.00\,\text{m}$, the image distance is given by the lens equation,

$$\frac{1}{d_i} = \frac{1}{f} - \frac{1}{d_o} = \frac{1}{50.0\,\text{mm}} - \frac{1}{3000\,\text{mm}} = \frac{3000 - 50}{(3000)(50.0)\,\text{mm}} = \frac{2950}{150{,}000\,\text{mm}}.$$

We solve for d_i and find $d_i = 50.8\,\text{mm}$, so the lens needs to move 0.8 mm away from the film or digital sensor.

EXERCISE A If the lens of Example 25–1 is 50.4 mm from the film or sensor, what is the object distance for sharp focus?

CONCEPTUAL EXAMPLE 25–2 **Shutter speed.** To improve the depth of field, you "stop down" your camera lens by two *f*-stops from $f/4$ to $f/8$. What should you do to the shutter speed to maintain the same exposure?

RESPONSE The amount of light admitted by the lens is proportional to the area of the lens opening. Reducing the lens opening by two *f*-stops reduces the diameter by a factor of 2, and the area by a factor of 4. To maintain the same exposure, the shutter must be open four times as long. If the shutter speed had been $\frac{1}{500}$ s, you would have to increase the exposure time to $\frac{1}{125}$ s.

* Picture Sharpness

The sharpness of a picture depends not only on accurate focusing and short exposure times, but also on the graininess of the film, or number of pixels on a digital sensor. Fine-grained films and tiny pixels are "slower," meaning they require longer exposures for a given light level. All pixels are rarely used because digital cameras have averaging (or "compression") programs, such as JPEG, which reduce memory size by averaging over pixels where little contrast is detected. But some detail is inevitably lost. For example, a small blue lake may seem uniform, and coding 600 pixels as identical takes less memory than specifying all 600. Any slight variation of the water surface is lost. Full "RAW" data uses more memory. Film records everything (down to its grain size), as does RAW if your camera offers it. The processor also averages over pixels in low light conditions, resulting in a less sharp photo.

The quality of the lens strongly affects the image quality, and we discuss lens resolution and diffraction effects in Sections 25–6 and 25–7. The sharpness, or **resolution**, of a lens is often given as so many lines per millimeter, measured by photographing a standard set of parallel black lines on a white background (sometimes said as "line pairs/mm") on fine-grain film or high quality sensor, or as so many dots per inch (dpi). The minimum spacing of distinguishable lines or dots gives the resolution. A lens that can give 50 lines/mm is reasonable, 100 lines/mm is very good (= 100 dots/mm $\approx$ 2500 dpi). Electronic sensors also have a resolution and it is sometimes given as line pairs across the full sensor width.

A "full" Bayer pixel (upper left in Fig. 25–2) is 4 regular pixels: for example, to make a white dot as part of a white line (between two black lines when determining lens resolution), all 4 Bayer pixels (RGGB) would have to be bright. For a Foveon, all three colors of one pixel need to be bright to produce a white dot.[†]

EXAMPLE 25–3 **Pixels and resolution.** A digital camera offers a maximum resolution of 4000 $\times$ 3000 pixels on a 32 mm $\times$ 24 mm sensor. How sharp should the lens be to make use of this sensor resolution in RAW?

APPROACH We find the number of pixels per millimeter and require the lens to be at least that good.

SOLUTION We can either take the image height (3000 pixels in 24 mm) or the width (4000 pixels in 32 mm):

$$\frac{3000 \text{ pixels}}{24 \text{ mm}} = 125 \text{ pixels/mm}.$$

We would like the lens to match this resolution of 125 lines or dots per mm, which would be a quite good lens. If the lens is not this good, fewer pixels and less memory could be used.

NOTE Increasing lens resolution is a tougher problem today than is squeezing more pixels on a CCD or CMOS sensor. The sensor for high quality cameras must also be physically larger for better image accuracy and greater light sensitivity in low light conditions.

[†]Consider a 4000 $\times$ 3000 pixel array. For a Foveon, each "full pixel" (Fig. 25–3) has all 3 colors, each of which can be counted as a pixel, so it may be considered as 4000 $\times$ 3000 $\times$ 3 = 36 MP. For a Bayer sensor, Fig. 25–2, 4000 $\times$ 3000 is 12 MP (6 MP of green, 3 MP each of red and blue). There are more green pixels because they are most important in our eyes' ability to note resolution. So the distance between green pixels is a rough guide to the sharpness of a Bayer. To match a 4000 $\times$ 3000 Foveon (36 MP, or 12 MP of tri-pixel sites), a Bayer would need to have about 24 MP (because it would then have 12 MP of green). This "equivalence" is only a rough approximation.

PHYSICS APPLIED
When is a photo sharp?

EXAMPLE 25–4 **Blown-up photograph.** A photograph looks sharp at normal viewing distances if the dots or lines are resolved to perhaps 10 dots/mm. Would an 8 × 10-inch enlargement of a photo taken by the camera in Example 25–3 seem sharp?

APPROACH We assume the image is 4000 × 3000 pixels on a 32 × 24-mm sensor as in Example 25–3, or 125 pixels/mm. We make an enlarged photo 8 × 10 in. = 20 cm × 25 cm.

SOLUTION The short side of the sensor is 24 mm = 2.4 cm long, and that side of the photograph is 8 inches or 20 cm. Thus the size is increased by a factor of 20 cm/2.4 cm ≈ 8× (or 25 cm/3.2 cm ≈ 8×). To fill the 8 × 10-in. paper, we assume the enlargement is 8×. The pixels are thus enlarged 8×. So the pixel count of 125/mm on the sensor becomes 125/8 = 15 per mm on the print. Hence an 8 × 10-inch print would be a sharp photograph. We could go 50% larger—11 × 14 or maybe even 12 × 18 inches.

In order to make very large photographic prints, large-format cameras are used such as 6 cm × 6 cm ($2\frac{1}{4}$ inch square)—either film or sensor—and even 4 × 5 inch and 8 × 10 inch (using sheet film or glass plates).

EXERCISE B The criterion of 0.030 mm as the diameter of a circle of confusion as acceptable sharpness is how many dots per mm on the sensor?

Telephotos and Wide-angles

Camera lenses are categorized into normal, telephoto, and wide angle, according to focal length and film size. A **normal lens** covers the sensor or film with a field of view that corresponds approximately to that of normal vision. A "normal" lens for 35-mm film has a focal length of 50 mm. The best digital cameras aim for a sensor of the same size[†] (24 mm × 36 mm). (If the sensor is smaller, digital cameras sometimes specify focal lengths to correspond with classic 35-mm cameras.) **Telephoto lenses** act like telescopes to magnify images. They have longer focal lengths than a normal lens: as we saw in Section 23–8 (Eq. 23–9), the height of the image for a given object distance is proportional to the image distance, and the image distance will be greater for a lens with longer focal length. For distant objects, the image height is very nearly proportional to the focal length. Thus a 200-mm telephoto lens for use with a 35-mm camera gives a 4× magnification over the normal 50-mm lens. A **wide-angle lens** has a shorter focal length than normal: a wider field of view is included, and objects appear smaller. A **zoom lens** is one whose focal length can be changed (by changing the distance between the thin lenses that make up the compound lens) so that you seem to zoom up to, or away from, the subject as you change the focal length.

Digital cameras may have an **optical zoom** meaning the lens can change focal length and maintain resolution. But an "electronic" or **digital zoom** just enlarges the dots (pixels) with loss of sharpness.

Different types of viewing systems are used in cameras. In some cameras, you view through a small window just above the lens as in Fig. 25–1. In a **single-lens reflex** camera (SLR), you actually view through the lens with the use of prisms and mirrors (Fig. 25–8). A mirror hangs at a 45° angle behind the lens and flips up out of the way just before the shutter opens. SLRs have the advantage that you can see almost exactly what you will get. Digital cameras use an LCD display, and it too can show what you will get on the photo if it is carefully designed.

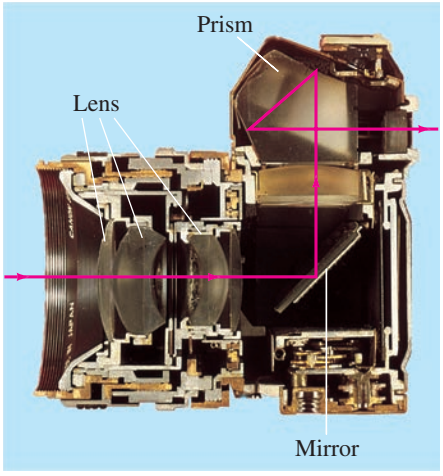


FIGURE 25–8 Single-lens reflex (SLR) camera, showing how the image is viewed through the lens with the help of a movable mirror and prism.

[†]A "35-mm camera" uses film that is physically 35 mm wide; that 35 mm is not to be confused with a focal length. 35-mm film has sprocket holes, so only 24 mm of its height is used for the photo; the width is usually 36 mm for stills. Thus one frame is 36 mm × 24 mm. Movie frames on 35-mm film are 24 mm × 18 mm.

25–2 The Human Eye; Corrective Lenses

The human eye resembles a camera in its basic structure (Fig. 25–9), but is far more sophisticated. The interior of the eye is filled with a transparent gel-like substance called the *vitreous humor* with index of refraction $n = 1.337$. Light enters this enclosed volume through the cornea and lens. Between the cornea and lens is a watery fluid, the aqueous humor (*aqua* is "water" in Latin) with $n = 1.336$. A diaphragm, called the **iris** (the colored part of your eye), adjusts automatically to control the amount of light entering the eye, similar to a camera. The hole in the iris through which light passes (the **pupil**) is black because no light is reflected from it (it's a hole), and very little light is reflected back out from the interior of the eye. The **retina**, which plays the role of the film or sensor in a camera, is on the curved rear surface of the eye. The retina consists of a complex array of nerves and receptors known as *rods* and *cones* which act to change light energy into electrical signals that travel along the nerves. The reconstruction of the image from all these tiny receptors is done mainly in the brain, although some analysis may also be done in the complex interconnected nerve network at the retina itself. At the center of the retina is a small area called the **fovea**, about 0.25 mm in diameter, where the cones are very closely packed and the sharpest image and best color discrimination are found.

PHYSICS APPLIED
The eye

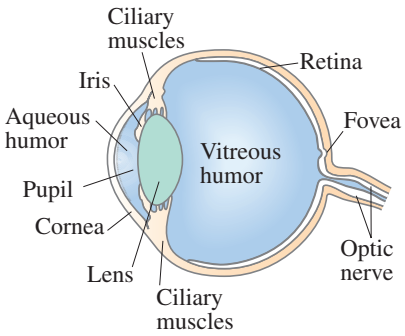
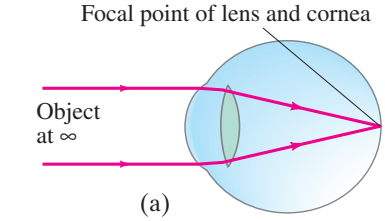


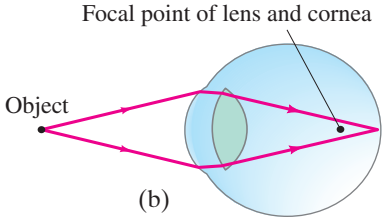
FIGURE 25–9 Diagram of a human eye.

Unlike a camera, the eye contains no shutter. The equivalent operation is carried out by the nervous system, which analyzes the signals to form images at the rate of about 30 per second. This can be compared to motion picture or television cameras, which operate by taking a series of still pictures at a rate of 24 (movies) and 60 or 30 (U.S. television) per second. Their rapid projection on the screen gives the appearance of motion.

The lens of the eye ($n = 1.386$ to 1.406) does little of the bending of the light rays. Most of the refraction is done at the front surface of the **cornea** ($n = 1.376$) at its interface with air ($n = 1.0$). The lens acts as a fine adjustment for focusing at different distances. This is accomplished by the ciliary muscles (Fig. 25–9), which change the curvature of the lens so that its focal length is changed. To focus on a distant object, the ciliary muscles of the eye are relaxed and the lens is thin, as shown in Fig. 25–10a, and parallel rays focus at the focal point (on the retina). To focus on a nearby object, the muscles contract, causing the center of the lens to thicken, Fig. 25–10b, thus shortening the focal length so that images of nearby objects can be focused on the retina, behind the new focal point. This focusing adjustment is called **accommodation**.

FIGURE 25–10 Accommodation by a normal eye: (a) lens relaxed, focused at infinity; (b) lens thickened, focused on a nearby object.





The closest distance at which the eye can focus clearly is called the **near point** of the eye. For young adults it is typically 25 cm, although younger children can often focus on objects as close as 10 cm. As people grow older, the ability to accommodate is reduced and the near point increases. A given person's **far point** is the farthest distance at which an object can be seen clearly. For some purposes it is useful to speak of a **normal eye** (a sort of average over the population), defined as an eye having a near point of 25 cm and a far point of infinity. To check your own near point, place this book close to your eye and slowly move it away until the type is sharp.

The "normal" eye is sort of an ideal. Many people have eyes that do not accommodate within the "normal" range of 25 cm to infinity, or have some other defect. Two common defects are nearsightedness and farsightedness. Both can be corrected to a large extent with lenses—either eyeglasses or contact lenses.

In **nearsightedness**, or **myopia**, the human eye can focus only on nearby objects. The far point is not infinity but some shorter distance, so distant objects are not seen clearly. Nearsightedness is usually caused by an eyeball that is too long, although sometimes it is the curvature of the cornea that is too great. In either case, images of distant objects are focused in front of the retina.

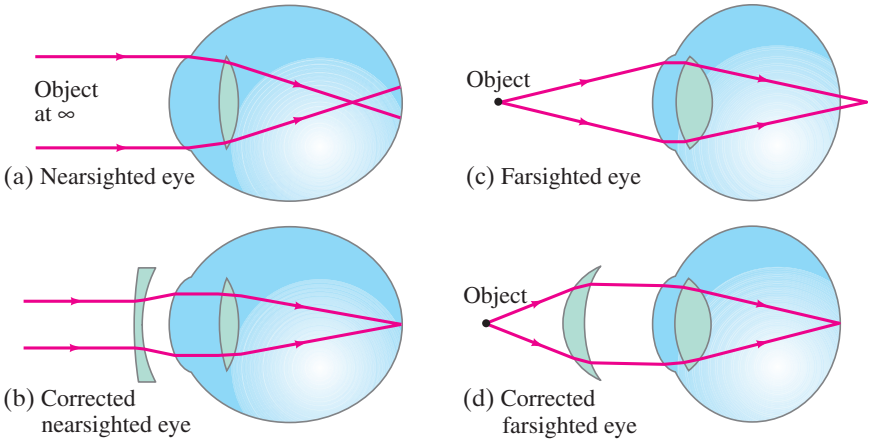


FIGURE 25–11 Correcting eye defects with lenses. (a) A nearsighted eye, which cannot focus clearly on distant objects (focal point is in front of retina), can be corrected (b) by use of a diverging lens. (c) A farsighted eye, which cannot focus clearly on nearby objects (focus point behind retina), can be corrected (d) by use of a converging lens.

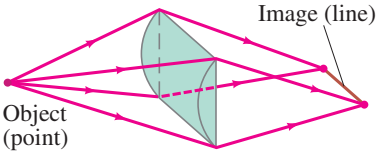
PHYSICS APPLIED
Corrective lenses

A diverging lens, because it causes parallel rays to diverge, allows the rays to be focused at the retina (Figs. 25–11a and b) and thus can correct nearsightedness.

In **farsightedness**, or *hyperopia*, the eye cannot focus on nearby objects. Although distant objects are usually seen clearly, the near point is somewhat greater than the "normal" 25 cm, which makes reading difficult. This defect is caused by an eyeball that is too short or (less often) by a cornea that is not sufficiently curved. It is corrected by a converging lens, Figs. 25–11c and d. Similar to hyperopia is *presbyopia*, which is the lessening ability of the eye to accommodate as a person ages, and the near point moves out. Converging lenses also compensate for this.

Astigmatism is usually caused by an out-of-round cornea or lens so that point objects are focused as short lines, which blurs the image. It is as if the cornea were spherical with a cylindrical section superimposed. As shown in Fig. 25–12, a cylindrical lens focuses a point into a line parallel to its axis. An astigmatic eye may focus rays in one plane, such as the vertical plane, at a shorter distance than it does for rays in a horizontal plane. Astigmatism is corrected with the use of a compensating cylindrical lens. Lenses for eyes that are nearsighted or farsighted as well as astigmatic are ground with superimposed spherical and cylindrical surfaces, so that the radius of curvature of the correcting lens is different in different planes.

FIGURE 25–12 A cylindrical lens forms a line image of a point object because it is converging in one plane only.



EXAMPLE 25–5 **Farsighted eye.** A farsighted eye has a near point of 100 cm. Reading glasses must have what lens power so that a newspaper can be read at a distance of 25 cm? Assume the lens is very close to the eye.

APPROACH When the object is placed 25 cm from the lens ($= d_o$), we want the image to be 100 cm away on the *same* side of the lens (so the eye can focus it), and so the image is virtual, as shown in Fig. 25–13, and $d_i = -100\,\text{cm}$ will be negative. We use the thin lens equation (Eq. 23–8) to determine the needed focal length. Optometrists' prescriptions specify the power ($P = 1/f$, Eq. 23–7) given in diopters ($1\,\text{D} = 1\,\text{m}^{-1}$).

SOLUTION Given that $d_o = 25\,\text{cm}$ and $d_i = -100\,\text{cm}$, the thin lens equation gives

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{25\,\text{cm}} + \frac{1}{-100\,\text{cm}} = \frac{4-1}{100\,\text{cm}} = \frac{1}{33\,\text{cm}}.$$

So $f = 33\,\text{cm} = 0.33\,\text{m}$. The power P of the lens is $P = 1/f = +3.0\,\text{D}$. The plus sign indicates that it is a converging lens.

NOTE We chose the image position to be where the eye can actually focus. The lens needs to put the image there, given the desired placement of the object (newspaper).

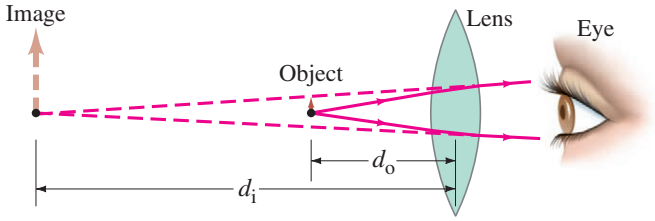


FIGURE 25–13 Lens of reading glasses (Example 25–5).

EXAMPLE 25–6 **Nearsighted eye.** A nearsighted eye has near and far points of 12 cm and 17 cm, respectively. (*a*) What lens power is needed for this person to see distant objects clearly, and (*b*) what then will be the near point? Assume that the lens is 2.0 cm from the eye (typical for eyeglasses).

APPROACH For a distant object $(d_o = \infty)$, the lens must put the image at the far point of the eye as shown in Fig. 25–14a, 17 cm in front of the eye. We can use the thin lens equation to find the focal length of the lens, and from this its lens power. The new near point (as shown in Fig. 25–14b) can be calculated for the lens by again using the thin lens equation.

SOLUTION (*a*) For an object at infinity $(d_o = \infty)$, the image must be in front of the lens 17 cm from the eye or $(17\text{ cm} - 2\text{ cm}) = 15\text{ cm}$ from the lens; hence $d_i = -15\text{ cm}$. We use the thin lens equation to solve for the focal length of the needed lens:

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{\infty} + \frac{1}{-15\text{ cm}} = -\frac{1}{15\text{ cm}}.$$

So $f = -15\text{ cm} = -0.15\text{ m}$ or $P = 1/f = -6.7\text{ D}$. The minus sign indicates that it must be a diverging lens for the myopic eye.

(*b*) The near point when glasses are worn is where an object is placed (d_o) so that the lens forms an image at the "near point of the naked eye," namely 12 cm from the eye. That image point is $(12\text{ cm} - 2\text{ cm}) = 10\text{ cm}$ in front of the lens, so $d_i = -0.10\text{ m}$ and the thin lens equation gives

$$\frac{1}{d_o} = \frac{1}{f} - \frac{1}{d_i} = -\frac{1}{0.15\text{ m}} + \frac{1}{0.10\text{ m}} = \frac{-2 + 3}{0.30\text{ m}} = \frac{1}{0.30\text{ m}}.$$

So $d_o = 30\text{ cm}$, which means the near point when the person is wearing glasses is 30 cm in front of the lens, or 32 cm from the eye.

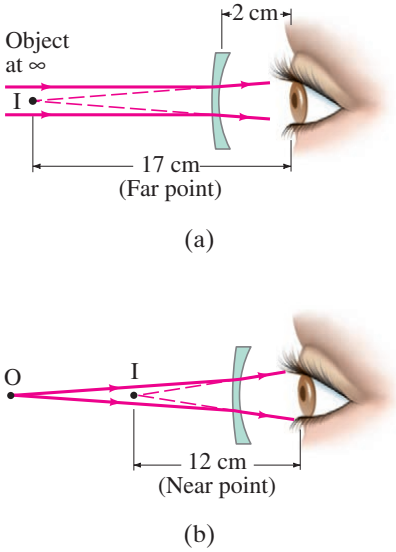


FIGURE 25–14 Example 25–6.

Contact Lenses

PHYSICS APPLIED
Contact lenses—different f and P

Suppose contact lenses are used to correct the eye in Example 25–6. Since contacts are placed directly on the cornea, we would not subtract out the 2.0 cm for the image distances. That is, for distant objects $d_i = f = -17\text{ cm}$, so $P = 1/f = -5.9\text{ D}$. The new near point would be 41 cm. Thus we see that a contact lens and an eyeglass lens will require slightly different powers, or focal lengths, for the same eye because of their different placements relative to the eye. We also see that glasses in this case give a better near point than contacts.

EXERCISE C What power of contact lens is needed for an eye to see distant objects if its far point is 25 cm?

Underwater Vision

PHYSICS APPLIED
Underwater vision

When your eyes are under water, distant underwater objects look blurry because at the water–cornea interface, the difference in indices of refraction is very small: $n = 1.33$ for water, 1.376 for the cornea. Hence light rays are bent very little and are focused far behind the retina, Fig. 25–15a. If you wear goggles or a face mask, you restore an air–cornea interface ($n = 1.0$ and 1.376, respectively) and the rays can be focused, Fig. 25–15b.

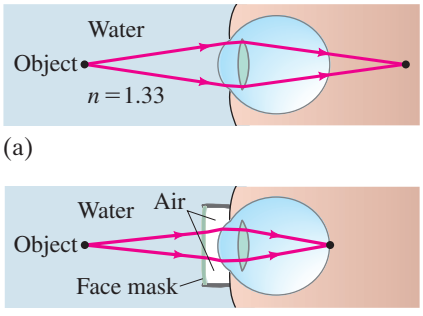


FIGURE 25–15 (a) Under water, we see a blurry image because light rays are bent much less than in air. (b) If we wear goggles, we again have an air–cornea interface and can see clearly.

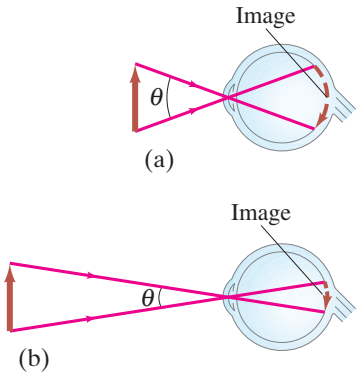


FIGURE 25–16 When the same object is viewed at a shorter distance, the image on the retina is greater, so the object appears larger and more detail can be seen. The angle θ that the object subtends in (a) is greater than in (b). *Note*: This is not a normal ray diagram because we are showing only one ray from each point.

25–3 Magnifying Glass

Much of the remainder of this Chapter will deal with optical devices that are used to produce magnified images of objects. We first discuss the **simple magnifier**, or **magnifying glass**, which is simply a converging lens (see Chapter-Opening Photo, page 713).

How large an object appears, and how much detail we can see on it, depends on the size of the image it makes on the retina. This, in turn, depends on the angle subtended by the object at the eye. For example, a penny held 30 cm from the eye looks twice as tall as one held 60 cm away because the angle it subtends is twice as great (Fig. 25–16). When we want to examine detail on an object, we bring it up close to our eyes so that it subtends a greater angle. However, our eyes can accommodate only up to a point (the near point), and we will assume a standard distance of $N = 25$ cm as the near point in what follows.

A magnifying glass allows us to place the object closer to our eye so that it subtends a greater angle. As shown in Fig. 25–17a, the object is placed at the focal point or just within it. Then the converging lens produces a virtual image, which must be at least 25 cm from the eye if the eye is to focus on it. If the eye is relaxed, the image will be at infinity, and in this case the object is exactly at the focal point. (You make this slight adjustment yourself when you "focus" on the object by moving the magnifying glass.)

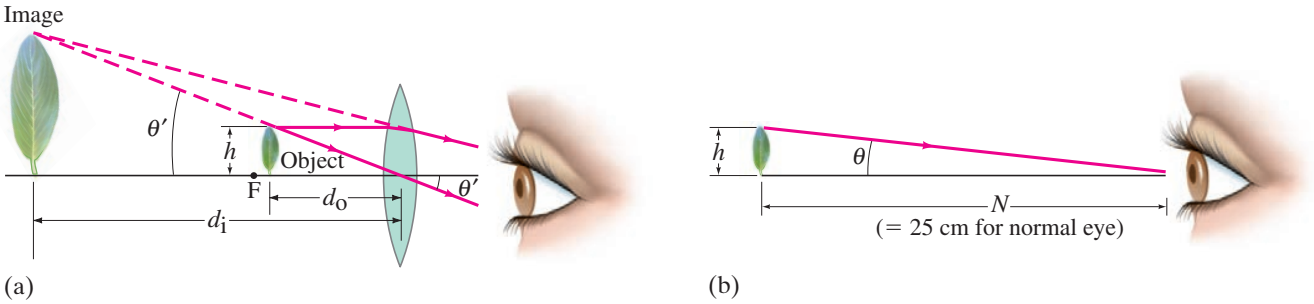


FIGURE 25–17 Leaf viewed (a) through a magnifying glass, and (b) with the unaided eye. The eye is focused at its near point in both cases.

A comparison of part (a) of Fig. 25–17 with part (b), in which the same object is viewed at the near point with the unaided eye, reveals that the angle the object subtends at the eye is much larger when the magnifier is used. The **angular magnification** or **magnifying power**, M, of the lens is defined as the ratio of the angle subtended by an object when using the lens, to the angle subtended using the unaided eye, with the object at the near point N of the eye ($N = 25$ cm for a normal eye):

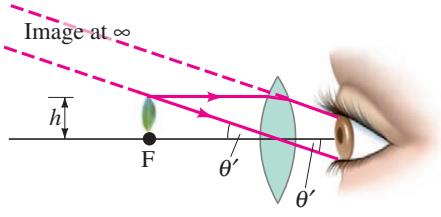
$$M = \frac{\theta'}{\theta}, \qquad \textbf{(25–1)}$$

where θ and θ' are shown in Fig. 25–17. We can write M in terms of the focal length by noting that $\theta = h/N$ (Fig. 25–17b) and $\theta' = h/d_o$ (Fig. 25–17a), where h is the height of the object and we assume the angles are small so θ and θ' (in radians) equal their sines and tangents. If the eye is relaxed (for least eye strain), the image will be at infinity and the object will be precisely at the focal point; see Fig. 25–18. Then $d_o = f$ and $\theta' = h/f$, whereas $\theta = h/N$ as before (Fig. 25–17b). Thus

$$M = \frac{\theta'}{\theta} = \frac{h/f}{h/N} = \frac{N}{f}. \qquad \left[\begin{array}{c}\text{eye focused at } \infty; \\ N = 25\text{ cm for normal eye}\end{array}\right] \qquad \textbf{(25–2a)}$$

We see that the shorter the focal length of the lens, the greater the magnification.†

FIGURE 25–18 With the eye relaxed, the object is placed at the focal point, and the image is at infinity. Compare to Fig. 25–17a where the image is at the eye's near point.



†Simple single-lens magnifiers are limited to about 2 or 3× because of blurring due to spherical aberration (Section 25–6).

The magnification of a given lens can be increased a bit by moving the lens and adjusting your eye so it focuses on the image at the eye's near point. In this case, $d_i = -N$ (see Fig. 25–17a) if your eye is very near the magnifier. Then the object distance d_o is given by

$$\frac{1}{d_o} = \frac{1}{f} - \frac{1}{d_i} = \frac{1}{f} + \frac{1}{N}.$$

We see from this equation that $d_o = fN/(f + N) < f$, as shown in Fig. 25–17a, since $N/(f + N)$ must be less than 1. With $\theta' = h/d_o$ the magnification is

$$M = \frac{\theta'}{\theta} = \frac{h/d_o}{h/N} = N\left(\frac{1}{d_o}\right) = N\left(\frac{1}{f} + \frac{1}{N}\right)$$

or

$$M = \frac{N}{f} + 1. \qquad \left[\begin{array}{c}\text{eye focused at near point, } N;\\ N = 25\text{ cm for normal eye}\end{array}\right] \quad \textbf{(25–2b)}$$

We see that the magnification is slightly greater when the eye is focused at its near point, as compared to when it is relaxed.

EXAMPLE 25–7 ESTIMATE **A jeweler's "loupe."** An 8-cm-focal-length converging lens is used as a "jeweler's loupe," which is a magnifying glass. Estimate (*a*) the magnification when the eye is relaxed, and (*b*) the magnification if the eye is focused at its near point $N = 25$ cm.

APPROACH The magnification when the eye is relaxed is given by Eq. 25–2a. When the eye is focused at its near point, we use Eq. 25–2b and we assume the lens is near the eye.

SOLUTION (*a*) With the relaxed eye focused at infinity,

$$M = \frac{N}{f} = \frac{25\text{ cm}}{8\text{ cm}} \approx 3\times.$$

(*b*) The magnification when the eye is focused at its near point ($N = 25$ cm), and the lens is near the eye, is

$$M = 1 + \frac{N}{f} = 1 + \frac{25}{8} \approx 4\times.$$

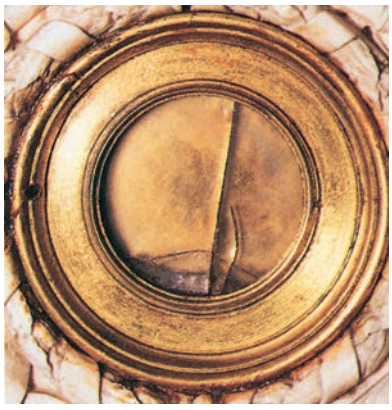
25–4 Telescopes

A telescope is used to magnify objects that are very far away. In most cases, the object can be considered to be at infinity.

Galileo, although he did not invent it,[†] developed the telescope into a usable and important instrument. He was the first to examine the heavens with the telescope (Fig. 25–19), and he made world-shaking discoveries, including the moons of Jupiter, the phases of Venus, sunspots, the structure of the Moon's surface, and that the Milky Way is made up of a huge number of individual stars.

[†]Galileo built his first telescope in 1609 after having heard of such an instrument existing in Holland. The first telescopes magnified only three to four times, but Galileo soon made a 30-power instrument. The first Dutch telescopes date from about 1604 and probably were copies of an Italian telescope built around 1590. Kepler (see Chapter 5) gave a ray description (in 1611) of the Keplerian telescope, which is named for him because he first described it, although he did not build it.

FIGURE 25–19 (a) Objective lens (mounted now in an ivory frame) from the telescope with which Galileo made his world-shaking discoveries, including the moons of Jupiter. (b) Telescopes made by Galileo (1609).



(a)



(b)

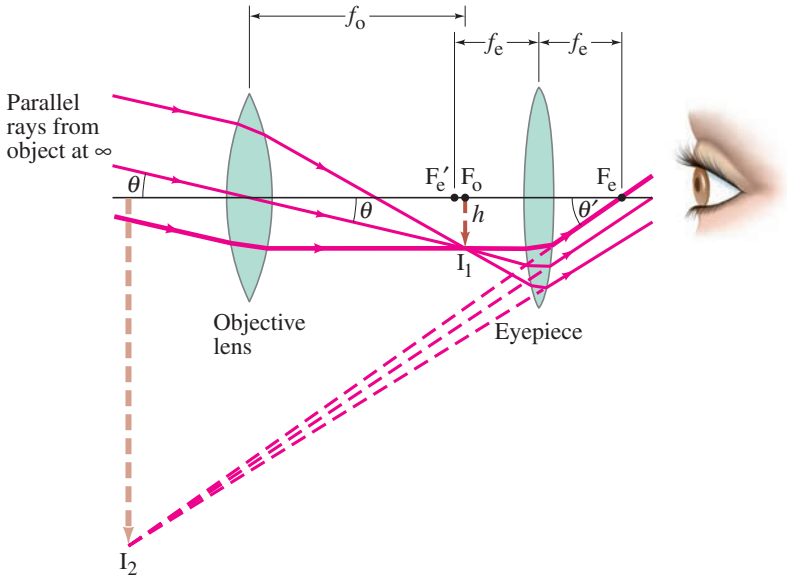


FIGURE 25–20 Astronomical telescope (refracting). Parallel light from one point on a distant object $(d_o = \infty)$ is brought to a focus by the objective lens in its focal plane. This image (I_1) is magnified by the eyepiece to form the final image I_2. Only two of the rays shown entering the objective are standard rays (2 and 3) as described in Fig. 23–37.

Several types of **astronomical telescope** exist. The common **refracting** type, sometimes called **Keplerian**, contains two converging lenses located at opposite ends of a long tube, as illustrated in Fig. 25–20. The lens closest to the object is called the **objective lens** (focal length f_o) and forms a real image I_1 of the distant object in the plane of its focal point F_o (or near it if the object is not at infinity). The second lens, called the **eyepiece** (focal length f_e), acts as a magnifier. That is, the eyepiece magnifies the image I_1 formed by the objective lens to produce a second, greatly magnified image, I_2, which is virtual and inverted. If the viewing eye is relaxed, the eyepiece is adjusted so the image I_2 is at infinity. Then the real image I_1 is at the focal point F_e' of the eyepiece, and the distance between the lenses is $f_o + f_e$ for an object at infinity.

PROBLEM SOLVING
Distance between lenses = $f_o + f_e$ for relaxed eye

To find the total angular magnification of this telescope, we note that the angle an object subtends as viewed by the unaided eye is just the angle θ subtended at the telescope objective. From Fig. 25–20 we can see that $\theta \approx h/f_o$, where h is the height of the image I_1 and we assume θ is small so that $\tan\theta \approx \theta$. Note, too, that the thickest of the three rays drawn in Fig. 25–20 is parallel to the axis before it strikes the eyepiece and therefore is refracted through the eyepiece focal point F_e on the far side. Thus, $\theta' \approx h/f_e$ and the **total magnifying power** (that is, angular magnification, which is what is always quoted) of this telescope is

$$M = \frac{\theta'}{\theta} = \frac{(h/f_e)}{(h/f_o)} = -\frac{f_o}{f_e}, \qquad \left[\begin{array}{c}\text{telescope}\\ \text{magnification}\end{array}\right] \quad \textbf{(25–3)}$$

where we used Eq. 25–1 and we inserted a minus sign to indicate that the image is inverted. To achieve a large magnification, the objective lens should have a long focal length and the eyepiece a short focal length.

FIGURE 25–21 This large refracting telescope was built in 1897 and is housed at Yerkes Observatory in Wisconsin. The objective lens is 102 cm (40 inches) in diameter, and the telescope tube is about 19 m long. Example 25–8.



EXAMPLE 25–8 **Telescope magnification.** The largest optical refracting telescope in the world is located at the Yerkes Observatory in Wisconsin, Fig. 25–21. It is referred to as a "40-inch" telescope, meaning that the diameter of the objective is 40 in., or 102 cm. The objective lens has a focal length of 19 m, and the eyepiece has a focal length of 10 cm. (*a*) Calculate the total magnifying power of this telescope. (*b*) Estimate the length of the telescope.

APPROACH Equation 25–3 gives the magnification. The length of the telescope is the distance between the two lenses.

SOLUTION (*a*) From Eq. 25–3 we find

$$M = -\frac{f_o}{f_e} = -\frac{19\text{ m}}{0.10\text{ m}} = -190\times.$$

(*b*) For a relaxed eye, the image I_1 is at the focal point of both the eyepiece and the objective lenses. The distance between the two lenses is thus $f_o + f_e \approx 19$ m, which is essentially the length of the telescope.

EXERCISE D A 40× telescope has a 1.2-cm focal length eyepiece. What is the focal length of the objective lens?

For an astronomical telescope to produce bright images of faint stars, the objective lens must be large to allow in as much light as possible. Indeed, the diameter of the objective lens (and hence its "light-gathering power") is an important parameter for an astronomical telescope, which is why the largest ones are specified by giving the objective diameter (such as the 10-meter Keck telescope in Hawaii). The construction and grinding of large lenses is very difficult. Therefore, the largest telescopes are **reflecting telescopes** which use a curved mirror as the objective, Fig. 25–22. A mirror has only one surface to be ground and can be supported along its entire surface[†] (a large lens, supported at its edges, would sag under its own weight). Often, the eyepiece lens or mirror (see Fig. 25–22) is removed so that the real image formed by the objective mirror can be recorded directly on film or on an electronic sensor (CCD or CMOS, Section 25–1).

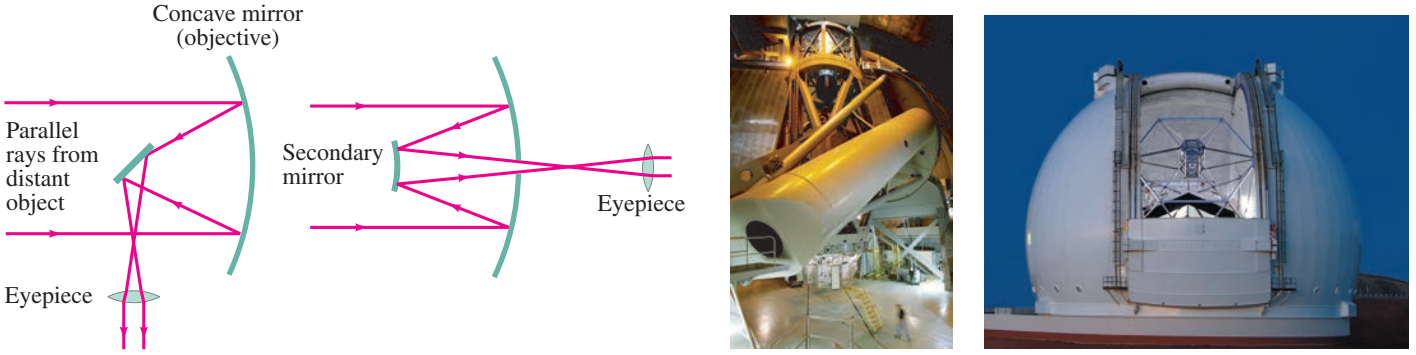


FIGURE 25–22 A concave mirror can be used as the objective of an astronomical telescope. Arrangement (a) is called the Newtonian focus, and (b) the Cassegrainian focus. Other arrangements are also possible. (c) The 200-inch (mirror diameter) Hale telescope on Palomar Mountain in California. (d) The 10-meter Keck telescope on Mauna Kea, Hawaii. The Keck combines thirty-six 1.8-meter six-sided mirrors into the equivalent of a very large single reflector, 10 m in diameter.

A **terrestrial telescope**, for viewing objects on Earth, must provide an upright image—seeing normal objects upside down would be difficult (much less important for viewing stars). Two designs are shown in Fig. 25–23. The **Galilean** type, which Galileo used for his great astronomical discoveries, has a diverging lens as eyepiece which intercepts the converging rays from the objective lens before they reach a focus, and acts to form a virtual upright image, Fig. 25–23a. This design is still used in opera glasses. The tube is reasonably short, but the field of view is small. The second type, shown in Fig. 25–23b, is often called a **spyglass** and makes use of a third convex lens that acts to make the image upright as shown. A spyglass must be quite long. The most practical design today is the **prism binocular** which was shown in Fig. 23–28. The objective and eyepiece are converging lenses. The prisms reflect the rays by total internal reflection and shorten the physical size of the device, and they also act to produce an upright image. One prism reinverts the image in the vertical plane, the other in the horizontal plane.

[†]Another advantage of mirrors is that they exhibit no chromatic aberration because the light doesn't pass through them; and they can be ground into a parabolic shape to correct for spherical aberration (Section 25–6). The reflecting telescope was first proposed by Newton.

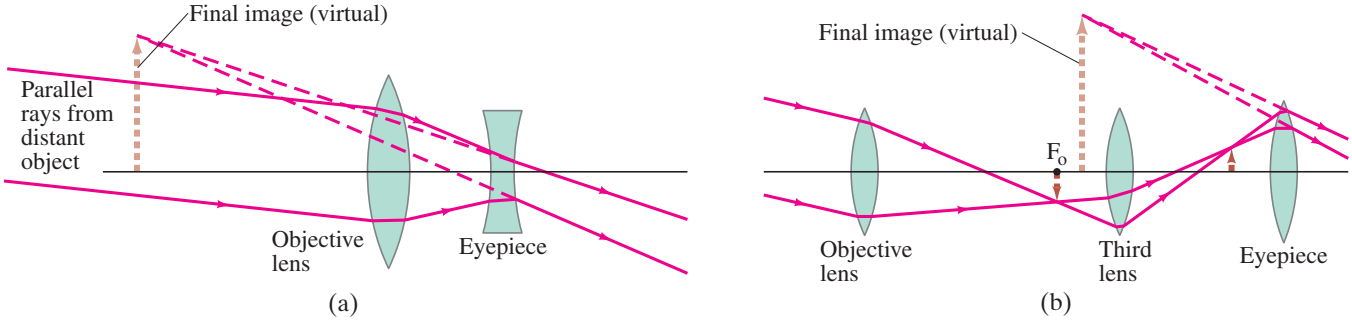


FIGURE 25–23 Terrestrial telescopes that produce an upright image: (a) Galilean; (b) spyglass, or erector type.

25–5 Compound Microscope

PHYSICS APPLIED
Microscopes

The compound **microscope**, like the telescope, has both objective and eyepiece (or ocular) lenses, Fig. 25–24. The design is different from that for a telescope because a microscope is used to view objects that are very close, so the object distance is very small. The object is placed just beyond the objective's focal point as shown in Fig. 25–24a. The image I_1 formed by the objective lens is real, quite far from the objective lens, and much enlarged. The eyepiece is positioned so that this image is near the eyepiece focal point F_e. The image I_1 is magnified by the eyepiece into a very large virtual image, I_2, which is seen by the eye and is inverted. Modern microscopes use a third "tube" lens behind the objective, but we will analyze the simpler arrangement shown in Fig. 25–24a.

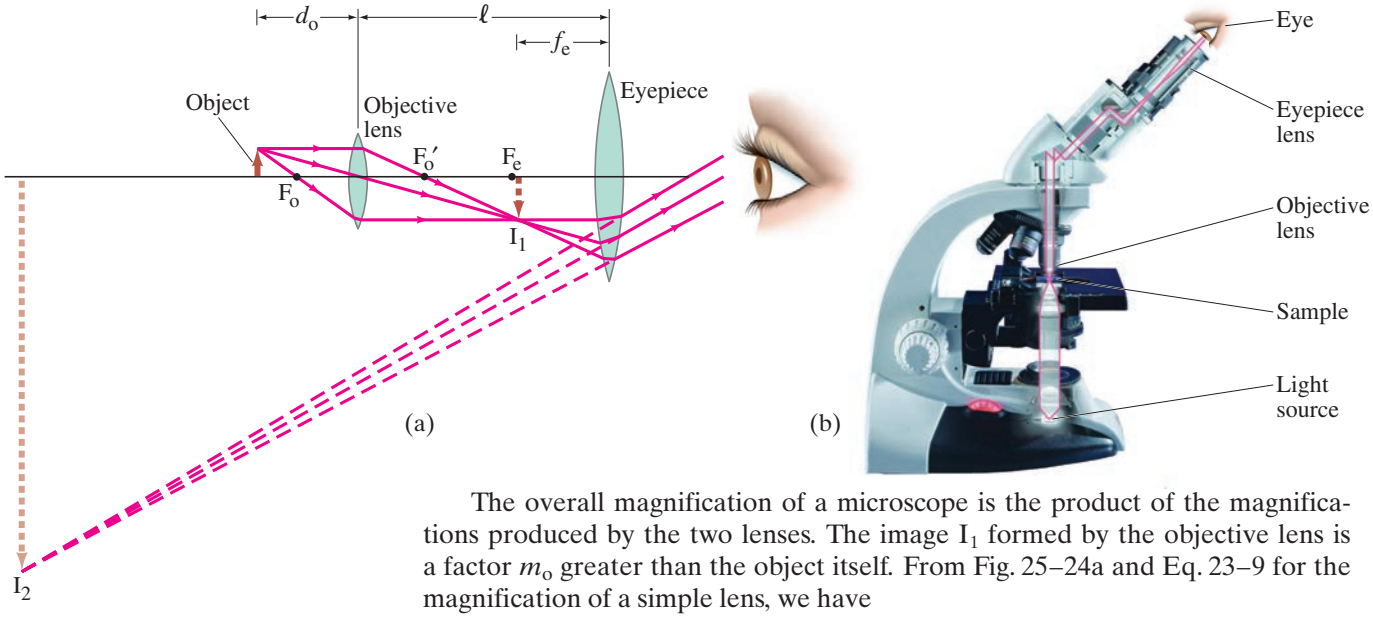


FIGURE 25–24 Compound microscope: (a) ray diagram, (b) photograph (illumination comes from below, outlined in red, then up through the slide holding the sample or object).

The overall magnification of a microscope is the product of the magnifications produced by the two lenses. The image I_1 formed by the objective lens is a factor m_o greater than the object itself. From Fig. 25–24a and Eq. 23–9 for the magnification of a simple lens, we have

$$m_o = \frac{h_i}{h_o} = \frac{d_i}{d_o} = \frac{\ell - f_e}{d_o}, \qquad \textbf{(25–4)}$$

where d_o and d_i are the object and image distances for the objective lens, ℓ is the distance between the lenses (equal to the length of the barrel), and we ignored the minus sign in Eq. 23–9 which only tells us that the image is inverted. We set $d_i = \ell - f_e$, which is exact only if the eye is relaxed, so that the image I_1 is at the eyepiece focal point F_e. The eyepiece acts like a simple magnifier. If we assume that the eye is relaxed, the eyepiece angular magnification M_e is (from Eq. 25–2a)

$$M_e = \frac{N}{f_e}, \qquad \textbf{(25–5)}$$

where the near point $N = 25\,\text{cm}$ for the normal eye. Since the eyepiece enlarges the image formed by the objective, the overall angular magnification M is the product of the magnification of the objective lens, m_o, times the angular magnification, M_e, of the eyepiece lens (Eqs. 25–4 and 25–5):

$$M = M_e m_o = \left(\frac{N}{f_e}\right)\left(\frac{\ell - f_e}{d_o}\right) \qquad \left[\begin{matrix}\text{microscope}\\ \text{magnification}\end{matrix}\right] \quad \textbf{(25–6a)}$$

$$\approx \frac{N\ell}{f_e f_o}. \qquad [f_o \text{ and } f_e \ll \ell] \quad \textbf{(25–6b)}$$

The approximation, Eq. 25–6b, is accurate when f_e and f_o are small compared to ℓ, so $\ell - f_e \approx \ell$, and the object is near F_o so $d_o \approx f_o$ (Fig. 25–24a). This is a good approximation for large magnifications, which are obtained when f_o and f_e are very small (they are in the denominator of Eq. 25–6b). To make lenses of very short focal length, compound lenses involving several elements must be used to avoid serious aberrations, as discussed in the next Section.

EXAMPLE 25–9 **Microscope.** A compound microscope consists of a 10× eyepiece and a 50× objective 17.0 cm apart. Determine (*a*) the overall magnification, (*b*) the focal length of each lens, and (*c*) the position of the object when the final image is in focus with the eye relaxed. Assume a normal eye, so $N = 25$ cm.

APPROACH The overall magnification is the product of the eyepiece magnification and the objective magnification. The focal length of the eyepiece is found from Eq. 25–2a or 25–5 for the magnification of a simple magnifier. For the objective lens, it is easier to next find d_o (part *c*) using Eq. 25–4 before we find f_o.

SOLUTION (*a*) The overall magnification is $(10\times)(50\times) = 500\times$.

(*b*) The eyepiece focal length is (Eq. 25–5) $f_e = N/M_e = 25\text{ cm}/10 = 2.5\text{ cm}$. Next we solve Eq. 25–4 for d_o, and find

$$d_o = \frac{\ell - f_e}{m_o} = \frac{(17.0\text{ cm} - 2.5\text{ cm})}{50} = 0.29\text{ cm}.$$

Then, from the thin lens equation for the objective with $d_i = \ell - f_e = 14.5$ cm (see Fig. 25–24a),

$$\frac{1}{f_o} = \frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{0.29\text{ cm}} + \frac{1}{14.5\text{ cm}} = 3.52\text{ cm}^{-1};$$

so $f_o = 1/(3.52\text{ cm}^{-1}) = 0.28$ cm.

(*c*) We just calculated $d_o = 0.29$ cm, which is very close to f_o.

25–6 Aberrations of Lenses and Mirrors

In Chapter 23 we developed a theory of image formation by a thin lens. We found, for example, that all rays from each point on an object are brought to a single point as the image point. This result, and others, were based on approximations for a thin lens, mainly that all rays make small angles with the axis and that we can use $\sin\theta \approx \theta$. Because of these approximations, we expect deviations from the simple theory, which are referred to as **lens aberrations**. There are several types of aberration; we will briefly discuss each of them separately, but all may be present at one time.

PHYSICS APPLIED
Lens aberrations

Consider an object at any point (even at infinity) on the axis of a lens with spherical surfaces. Rays from this point that pass through the outer regions of the lens are brought to a focus at a different point from those that pass through the center of the lens. This is called **spherical aberration**, and is shown exaggerated in Fig. 25–25. Consequently, the image seen on a screen or film will not be a point but a tiny circular patch of light. If the sensor or film is placed at the point C, as indicated, the circle will have its smallest diameter, which is referred to as the **circle of least confusion**. Spherical aberration is present whenever spherical surfaces are used. It can be reduced by using nonspherical (= aspherical) lens surfaces, but grinding such lenses is difficult and expensive. Spherical aberration can be reduced by the use of several lenses in combination, and by using primarily the central part of lenses.

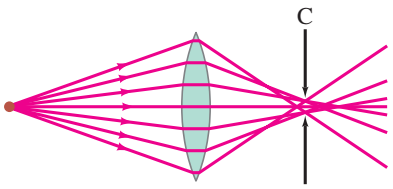


FIGURE 25–25 Spherical aberration (exaggerated). Circle of least confusion is at C.

For object points off the lens axis, additional aberrations occur. Rays passing through the different parts of the lens cause spreading of the image that is noncircular. There are two effects: **coma** (because the image of a point is comet-shaped rather than a tiny circle) and **off-axis astigmatism**.† Furthermore, the image points for objects off the axis but at the same distance from the lens do not fall on a flat plane but on a curved surface—that is, the focal plane is not flat. (We expect this because the points on a flat plane, such as the film in a camera, are not equidistant from the lens.) This aberration is known as **curvature of field** and is a problem in cameras and other devices where the sensor or film is a flat plane. In the eye, however, the retina is curved, which compensates for this effect.

†Although the effect is the same as for astigmatism in the eye (Section 25–2), the cause is different. Off-axis astigmatism is no problem in the eye because objects are clearly seen only at the fovea, on the lens axis.

Another aberration, **distortion**, is a result of variation of magnification at different distances from the lens axis. Thus a straight-line object some distance from the axis may form a curved image. A square grid of lines may be distorted to produce "barrel distortion," or "pincushion distortion," Fig. 25–26. The former is common in extreme wide-angle lenses.

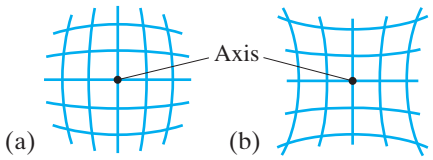


FIGURE 25–26 Distortion: lenses may image a square grid of perpendicular lines to produce (a) barrel distortion or (b) pincushion distortion. These distortions can be seen in the photograph of Fig. 23–31d for a simple lens.

FIGURE 25–27 Chromatic aberration. Different colors are focused at different points.

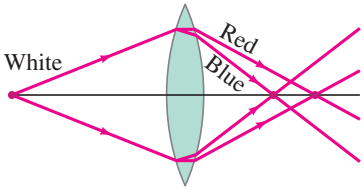
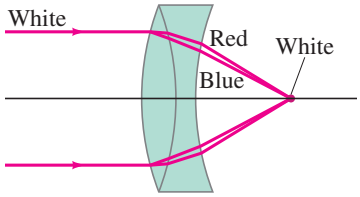


FIGURE 25–28 Achromatic doublet.



PHYSICS APPLIED
Aberration in the human eye

All the above aberrations occur for monochromatic light and hence are referred to as *monochromatic aberrations*. Normal light is not monochromatic, and there will also be **chromatic aberration**. This aberration arises because of dispersion—the variation of index of refraction of transparent materials with wavelength (Section 24–4). For example, blue light is bent more than red light by glass. So if white light is incident on a lens, the different colors are focused at different points, Fig. 25–27, and have slightly different magnifications resulting in colored fringes in the image. Chromatic aberration can be eliminated for any two colors (and reduced greatly for all others) by the use of two lenses made of different materials with different indices of refraction and dispersion. Normally one lens is converging and the other diverging, and they are often cemented together (Fig. 25–28). Such a lens combination is called an **achromatic doublet** (or "color-corrected" lens).

To reduce aberrations, high-quality lenses are **compound lenses** consisting of many simple lenses, referred to as **elements**. A typical high-quality camera lens may contain six to eight (or more) elements. For simplicity we will usually indicate lenses in diagrams as if they were simple lenses.

The human eye is also subject to aberrations, but they are minimal. Spherical aberration, for example, is minimized because (1) the cornea is less curved at the edges than at the center, and (2) the lens is less dense at the edges than at the center. Both effects cause rays at the outer edges to be bent less strongly, and thus help to reduce spherical aberration. Chromatic aberration is partially compensated for because the lens absorbs the shorter wavelengths appreciably and the retina is less sensitive to the blue and violet wavelengths. This is just the region of the spectrum where dispersion—and thus chromatic aberration—is greatest (Fig. 24–14).

Spherical mirrors (Section 23–3) also suffer aberrations including spherical aberration (see Fig. 23–13). Mirrors can be ground in a parabolic shape to correct for aberrations, but they are much harder to make and therefore very expensive. Spherical mirrors do not, however, exhibit chromatic aberration because the light does not pass through them (no refraction, no dispersion).

25–7 Limits of Resolution; Circular Apertures

The ability of a lens to produce distinct images of two point objects very close together is called the **resolution** of the lens. The closer the two images can be and still be seen as distinct (rather than overlapping blobs), the higher the resolution. The resolution of a camera lens, for example, is often specified as so many dots or lines per millimeter, as mentioned in Section 25–1.

Two principal factors limit the resolution of a lens. The first is lens aberrations. As we just saw, because of spherical and other aberrations, a point object is not a point on the image but a tiny blob. Careful design of compound lenses can reduce aberrations significantly, but they cannot be eliminated entirely. The second factor that limits resolution is *diffraction*, which cannot be corrected for because it is a natural result of the wave nature of light. We discuss it now.

In Section 24–5, we saw that because light travels as a wave, light from a point source passing through a slit is spread out into a diffraction pattern (Figs. 24–19 and 24–21). A lens, because it has edges, acts like a round slit. When a lens forms the image of a point object, the image is actually a tiny diffraction pattern. Thus *an image would be blurred even if aberrations were absent*.

In the analysis that follows, we assume that the lens is free of aberrations, so we can concentrate on diffraction effects and how much they limit the resolution of a lens. In Fig. 24–21 we saw that the diffraction pattern produced by light passing through a rectangular slit has a central maximum in which most of the light falls. This central peak falls to a minimum on either side of its center at an angle θ given by

$$\sin\theta = \lambda/D$$

(this is Eq. 24–3a), where D is the slit width and λ the wavelength of light used. θ is the angular half-width of the central maximum, and for small angles (in radians) can be written

$$\theta \approx \sin\theta = \frac{\lambda}{D}.$$

There are also low-intensity fringes beyond.

For a lens, or any circular hole, the image of a point object will consist of a *circular* central peak (called the *diffraction spot* or *Airy disk*) surrounded by faint circular fringes, as shown in Fig. 25–29a. The central maximum has an angular half-width given by

$$\theta = \frac{1.22\lambda}{D},$$

where D is the diameter of the circular opening. [This is a theoretical result for a perfect circle or lens. For real lenses or circles, the factor is on the order of 1 to 2.] This formula differs from that for a slit (Eq. 24–3) by the factor 1.22. This factor appears because the width of a circular hole is not uniform (like a rectangular slit) but varies from its diameter D to zero. A mathematical analysis shows that the "average" width is $D/1.22$. Hence we get the equation above rather than Eq. 24–3. The intensity of light in the diffraction pattern from a point source of light passing through a circular opening is shown in Fig. 25–30. The image for a non-point source is a superposition of such patterns. For most purposes we need consider only the central spot, since the concentric rings are so much dimmer.

If two point objects are very close, the diffraction patterns of their images will overlap as shown in Fig. 25–29b. As the objects are moved closer, a separation is reached where you can't tell if there are two overlapping images or a single image. The separation at which this happens may be judged differently by different observers. However, a generally accepted criterion is that proposed by Lord Rayleigh (1842–1919). This **Rayleigh criterion** states that *two images are just resolvable when the center of the diffraction disk of one image is directly over the first minimum in the diffraction pattern of the other*. This is shown in Fig. 25–31. Since the first minimum is at an angle $\theta = 1.22\lambda/D$ from the central maximum, Fig. 25–31 shows that two objects can be considered *just resolvable* if they are separated by at least an angle θ given by

$$\theta = \frac{1.22\lambda}{D}. \qquad \begin{bmatrix}\text{2 points just resolvable;}\\ \theta \text{ in radians}\end{bmatrix} \qquad \textbf{(25–7)}$$

In this equation, D is the diameter of the lens, and applies also to a mirror diameter. This is the limit on resolution set by the wave nature of light due to diffraction. A smaller angle means better resolution: you can make out closer objects. We see from Eq. 25–7 that using a shorter wavelength λ can reduce θ and thus increase resolution.

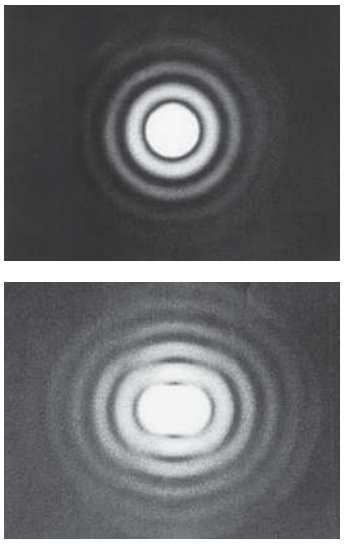


FIGURE 25–29 Photographs of images (greatly magnified) formed by a lens, showing the diffraction pattern of an image for: (a) a single point object; (b) two point objects whose images are barely resolved.

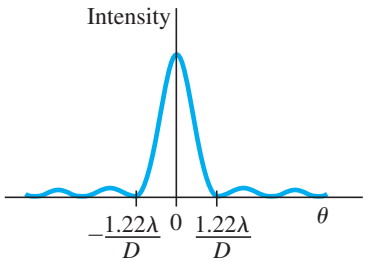


FIGURE 25–30 Intensity of light across the diffraction pattern of a circular hole.

FIGURE 25–31 The *Rayleigh criterion*. Two images are just resolvable when the center of the diffraction peak of one is directly over the first minimum in the diffraction pattern of the other. The two point objects O and O′ subtend an angle θ at the lens; only one ray (it passes through the center of the lens) is drawn for each object, to indicate the center of the diffraction pattern of its image.

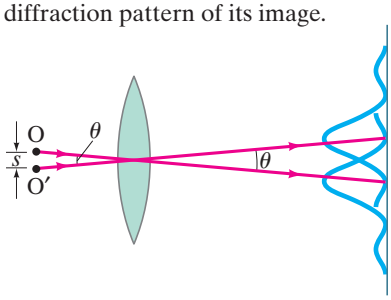




FIGURE 25–32 Hubble Space Telescope, with Earth in the background. The flat orange panels are solar cells that collect energy from the Sun to power the equipment.

EXAMPLE 25–10 Hubble Space Telescope. The Hubble Space Telescope (HST) is a reflecting telescope that was placed in orbit above the Earth's atmosphere, so its resolution would not be limited by turbulence in the atmosphere (Fig. 25–32). Its objective diameter is 2.4 m. For visible light, say $\lambda = 550\,\text{nm}$, estimate the improvement in resolution the Hubble offers over Earth-bound telescopes, which are limited in resolution by movement of the Earth's atmosphere to about half an arc second. (Each degree is divided into 60 minutes each containing 60 seconds, so $1^\circ = 3600$ arc seconds.)

APPROACH Angular resolution for the Hubble is given (in radians) by Eq. 25–7. The resolution for Earth telescopes is given, and we first convert it to radians so we can compare.

SOLUTION Earth-bound telescopes are limited to an angular resolution of

$$\theta = \frac{1}{2}\left(\frac{1}{3600}\right)^\circ\left(\frac{2\pi\,\text{rad}}{360^\circ}\right) = 2.4 \times 10^{-6}\,\text{rad}.$$

The Hubble, on the other hand, is limited by diffraction (Eq. 25–7) which for $\lambda = 550\,\text{nm}$ is

$$\theta = \frac{1.22\lambda}{D} = \frac{1.22(550 \times 10^{-9}\,\text{m})}{2.4\,\text{m}} = 2.8 \times 10^{-7}\,\text{rad},$$

thus giving almost ten times better resolution $(2.4 \times 10^{-6}\,\text{rad}/2.8 \times 10^{-7}\,\text{rad} \approx 9\times)$.

EXAMPLE 25–11 ESTIMATE Eye resolution. You are in an airplane at an altitude of 10,000 m. If you look down at the ground, estimate the minimum separation s between objects that you could distinguish. Could you count cars in a parking lot? Consider only diffraction, and assume your pupil is about 3.0 mm in diameter and $\lambda = 550\,\text{nm}$.

PHYSICS APPLIED
How well the eye can see

APPROACH We use the Rayleigh criterion, Eq. 25–7, to estimate θ. The separation s of objects is $s = \ell\theta$, where $\ell = 10^4\,\text{m}$ and θ is in radians.

SOLUTION In Eq. 25–7, we set $D = 3.0\,\text{mm}$ for the opening of the eye:

$$s = \ell\theta = \ell\frac{1.22\lambda}{D}$$

$$= (10^4\,\text{m})\frac{(1.22)(550 \times 10^{-9}\,\text{m})}{3.0 \times 10^{-3}\,\text{m}} = 2.2\,\text{m}.$$

Yes, you could just resolve a car (roughly 2 m wide by 3 or 4 m long) and so could count the number of cars in the lot.

EXERCISE E Someone claims a spy satellite camera can see 3-cm-high newspaper headlines from an altitude of 100 km. If diffraction were the only limitation ($\lambda = 550\,\text{nm}$), use Eq. 25–7 to determine what diameter lens the camera would have.

25–8 Resolution of Telescopes and Microscopes; the λ Limit

You might think that a microscope or telescope could be designed to produce any desired magnification, depending on the choice of focal lengths and quality of the lenses. But this is not possible, because of diffraction. An increase in magnification above a certain point merely results in magnification of the diffraction patterns. This can be highly misleading since we might think we are seeing details of an object when we are really seeing details of the diffraction pattern.

To examine this problem, we apply the Rayleigh criterion: two objects (or two nearby points on one object) are just resolvable if they are separated by an angle θ (Fig. 25–31) given by Eq. 25–7:

$$\theta = \frac{1.22\lambda}{D}.$$

This formula is valid for either a microscope or a telescope, where D is the diameter of the objective lens or mirror. For a telescope, the resolution is specified by stating θ as given by this equation.[†]

EXAMPLE 25–12 **Telescope resolution (radio wave vs. visible light).** What is the theoretical minimum angular separation of two stars that can just be resolved by (*a*) the 200-inch telescope on Palomar Mountain (Fig. 25–22c); and (*b*) the Arecibo radiotelescope (Fig. 25–33), whose diameter is 300 m and whose radius of curvature is also 300 m. Assume $\lambda = 550\,\text{nm}$ for the visible-light telescope in part (*a*), and $\lambda = 4\,\text{cm}$ (the shortest wavelength at which the radiotelescope has operated) in part (*b*).

APPROACH We apply the Rayleigh criterion (Eq. 25–7) for each telescope.

SOLUTION (*a*) With $D = 200\,\text{in.} = 5.1\,\text{m}$, we have from Eq. 25–7 that

$$\theta = \frac{1.22\lambda}{D} = \frac{(1.22)(5.50 \times 10^{-7}\,\text{m})}{(5.1\,\text{m})} = 1.3 \times 10^{-7}\,\text{rad},$$

or 0.75×10^{-5} deg. (Note that this is equivalent to resolving two points less than 1 cm apart from a distance of 100 km!)

(*b*) For radio waves with $\lambda = 0.04\,\text{m}$ emitted by stars, the resolution is

$$\theta = \frac{(1.22)(0.04\,\text{m})}{(300\,\text{m})} = 1.6 \times 10^{-4}\,\text{rad}.$$

The resolution is less because the wavelength is so much larger, but the larger objective collects more radiation and thus detects fainter objects.

NOTE In both cases, we determined the limit set by diffraction. The resolution for a visible-light Earth-bound telescope is not this good because of aberrations and, more importantly, turbulence in the atmosphere. In fact, large-diameter objectives are not justified by increased resolution, but by their greater light-gathering ability—they allow more light in, so fainter objects can be seen. Radiotelescopes are not hindered by atmospheric turbulence, and the resolution found in (*b*) is a good estimate.



FIGURE 25–33 The 300-meter radiotelescope in Arecibo, Puerto Rico, uses radio waves (Fig. 22–8) instead of visible light.

PHYSICS APPLIED
Why large-diameter objectives

For a microscope, it is more convenient to specify the actual distance, s, between two points that are just barely resolvable: see Fig. 25–31. Since objects are normally placed near the focal point of the microscope objective, the angle subtended by two objects is $\theta = s/f$, so $s = f\theta$. If we combine this with Eq. 25–7, we obtain the **resolving power** (**RP**) of a microscope

$$\text{RP} = s = f\theta = \frac{1.22\lambda f}{D}, \qquad \text{[microscope]} \quad \textbf{(25–8)}$$

where f is the objective lens' focal length (not frequency) and D its diameter. The distance s is called the resolving power of the lens because it is the minimum separation of two object points that can just be resolved—assuming the highest quality lens since this limit is imposed by the wave nature of light. A smaller RP means better resolution, better detail.

[†]Earth-bound telescopes with large-diameter objectives are usually limited not by diffraction but by other effects such as turbulence in the atmosphere. The resolution of a high-quality microscope, on the other hand, normally *is* limited by diffraction; microscope objectives are complex compound lenses containing many elements of small diameter (since f is small), thus reducing aberrations.

Diffraction sets an ultimate limit on the detail that can be seen on any object. In Eq. 25–8 for the resolving power of a microscope, the focal length of the lens cannot practically be made less than (approximately) the radius of the lens ($= D/2$), and even that is very difficult (see the lensmaker's equation, Eq. 23–10). In this best case, Eq. 25–8 gives, with $f \approx D/2$,

$$\text{RP} \approx \frac{\lambda}{2}. \qquad \textbf{(25–9)}$$

Thus we can say, to within a factor of 2 or so, that

Wavelength limits resolution

it is not possible to resolve detail of objects smaller than the wavelength of the radiation being used.

This is an important and useful rule of thumb.

Compound lenses in microscopes are now designed so well that the actual limit on resolution is often set by diffraction—that is, by the wavelength of the light used. To obtain greater detail, one must use radiation of shorter wavelength. The use of UV radiation can increase the resolution by a factor of perhaps 2. Far more important, however, was the discovery in the early twentieth century that electrons have wave properties (Chapter 27) and that their wavelengths can be very small. The wave nature of electrons is utilized in the electron microscope (Section 27–9), which can magnify 100 to 1000 times more than a visible-light microscope because of the much shorter wavelengths. X-rays, too, have very short wavelengths and are often used to study objects in great detail (Section 25–11).

25–9 Resolution of the Human Eye and Useful Magnification

The resolution of the human eye is limited by several factors, all of roughly the same order of magnitude. The resolution is best at the fovea, where the cone spacing is smallest, about 3 μm ($= 3000$ nm). The diameter of the pupil varies from about 0.1 cm to about 0.8 cm. So for $\lambda = 550$ nm (where the eye's sensitivity is greatest), the diffraction limit is about $\theta \approx 1.22\lambda/D \approx 8 \times 10^{-5}$ rad to 6×10^{-4} rad. The eye is about 2 cm long, giving a resolving power (Eq. 25–8) of $s \approx (2 \times 10^{-2}\,\text{m})(8 \times 10^{-5}\,\text{rad}) \approx 2\,\mu\text{m}$ at best, to about 10 μm at worst (pupil small). Spherical and chromatic aberration also limit the resolution to about 10 μm. The net result is that the eye can just resolve objects whose angular separation is around

$$5 \times 10^{-4}\,\text{rad}. \qquad \left[\begin{array}{c}\text{best eye}\\ \text{resolution}\end{array}\right]$$

This corresponds to objects separated by 1 cm at a distance of about 20 m.

The typical near point of a human eye is about 25 cm. At this distance, the eye can just resolve objects that are $(25\,\text{cm})(5 \times 10^{-4}\,\text{rad}) \approx 10^{-4}\,\text{m} = \frac{1}{10}\,\text{mm}$ apart.[†] Since the best light microscopes can resolve objects no smaller than about 200 nm at best (Eq. 25–9 for violet light, $\lambda = 400$ nm), the useful magnification [$=$ (resolution by naked eye)/(resolution by microscope)] is limited to about

$$\frac{10^{-4}\,\text{m}}{200 \times 10^{-9}\,\text{m}} \approx 500\times. \qquad \left[\begin{array}{c}\text{maximum useful}\\ \text{microscope magnification}\end{array}\right]$$

In practice, magnifications of about 1000× are often used to minimize eyestrain. Any greater magnification would simply make visible the diffraction pattern produced by the microscope objective lens.

EXERCISE F Return to the Chapter-Opening Question, page 713, and answer it again now. Try to explain why you may have answered differently the first time.

[†]A nearsighted eye that needs -8 or -10 D lenses can have a near point of 8 or 10 cm, and a higher resolution up close (without glasses) of a factor of $2\frac{1}{2}$ or 3, or $\approx \frac{1}{25}\,\text{mm} \approx 40\,\mu\text{m}$.

*25–10 Specialty Microscopes and Contrast

All the resolving power a microscope can attain will be useless if the object to be seen cannot be distinguished from the background. The difference in brightness between the image of an object and the image of the surroundings is called **contrast**. Achieving high contrast is an important problem in microscopy and other forms of imaging. The problem arises in biology, for example, because cells consist largely of water and are almost uniformly transparent to light. We now briefly discuss two special types of microscope that can increase contrast: the interference and phase-contrast microscopes.

An **interference microscope** makes use of the wave properties of light in a direct way to increase contrast in a transparent object. Consider a transparent object—say, a bacterium in water (Fig. 25–34). Coherent light enters uniformly from the left and is in phase at all points such as a and b. If the object is as transparent as the water, the beam leaving at d will be as bright as that at c. There will be no contrast and the object will not be seen. However, if the object's refractive index is slightly different from that of the surrounding medium, the wavelength within the object will be altered as shown. Hence light waves at points c and d will differ in phase, if not in amplitude. The interference microscope changes this difference in phase into a difference of amplitude which our eyes can detect. Light that passes through the sample is superimposed onto a reference beam that does not pass through the object, so that they interfere. One way of doing this is shown in Fig. 25–35. Light from a source is split into two equal beams by a half-silvered mirror, MS_1. One beam passes through the object, and the second (comparison beam) passes through an identical system without the object. The two meet again and are superposed by the half-silvered mirror MS_2 before entering the eyepiece and eye. The path length (and amplitude) of the comparison beam is adjustable so that the background can be dark; that is, full destructive interference occurs. Light passing through the object (beam bd in Fig. 25–34) will also interfere with the comparison beam. But because of its different phase, the interference will not be completely destructive. Thus it will appear brighter than the background. Where the object varies in thickness, the phase difference between beams ac and bd in Fig. 25–34 will be different, thus affecting the amount of interference. Hence *variation in the thickness of the object will appear as variations in brightness in the image*.

A **phase-contrast microscope** also makes use of interference and differences in phase to produce a high-contrast image. Contrast is achieved by a circular glass *phase plate* that has a groove (or a raised portion) in the shape of a ring, positioned so undeviated source rays pass through it, but rays deviated by the object do not pass through this ring. Because the rays deviated by the object travel through a different thickness of glass than the undeviated source rays, the two can be out of phase and can interfere destructively at the object image plane. Thus the image of the object can contrast sharply with the background. Phase-contrast microscope images tend to have "halos" around them (as a result of diffraction from the phase-plate opening), so care must be taken in the interpretation of images.

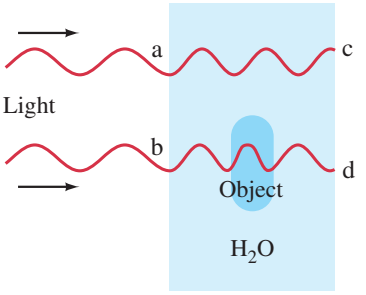
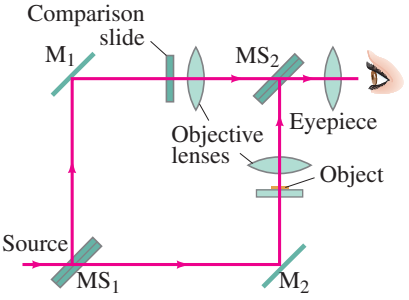


FIGURE 25–34 Object—say, a bacterium—in a water solution.

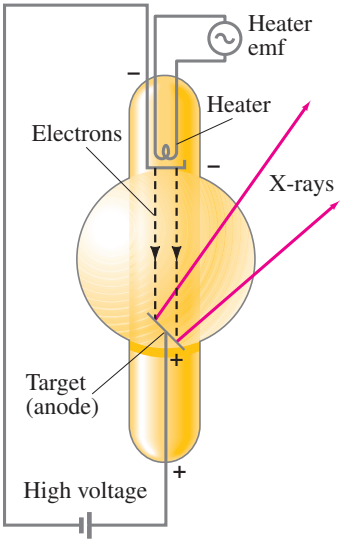
PHYSICS APPLIED
Interference microscope

FIGURE 25–35 Diagram of an interference microscope.



PHYSICS APPLIED
Phase-contrast microscope

FIGURE 25–36 X-ray tube. Electrons emitted by a heated filament in a vacuum tube are accelerated by a high voltage. When they strike the surface of the anode, the "target," X-rays are emitted.



25–11 X-Rays and X-Ray Diffraction

In 1895, W. C. Roentgen (1845–1923) discovered that when electrons were accelerated by a high voltage in a vacuum tube and allowed to strike a glass or metal surface inside the tube, fluorescent minerals some distance away would glow, and photographic film would become exposed. Roentgen attributed these effects to a new type of radiation (different from cathode rays). They were given the name **X-rays** after the algebraic symbol *x*, meaning an unknown quantity. He soon found that X-rays penetrated through some materials better than through others, and within a few weeks he presented the first X-ray photograph (of his wife's hand). The production of X-rays today is usually done in a tube (Fig. 25–36) similar to Roentgen's, using voltages of typically 30 kV to 150 kV.

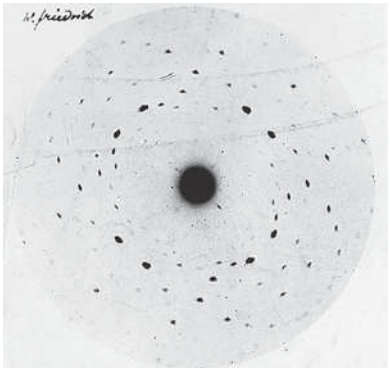


FIGURE 25–37 This X-ray diffraction pattern is one of the first observed by Max von Laue in 1912 when he aimed a beam of X-rays at a zinc sulfide crystal. The diffraction pattern was detected directly on a photographic plate.

Investigations into the nature of X-rays indicated they were not charged particles (such as electrons) since they could not be deflected by electric or magnetic fields. It was suggested that they might be a form of invisible light. However, they showed no diffraction or interference effects using ordinary gratings. Indeed, if their wavelengths were much smaller than the typical grating spacing of 10^{-6} m $(= 10^{3}$ nm), no effects would be expected. Around 1912, Max von Laue (1879–1960) suggested that if the atoms in a crystal were arranged in a regular array (see Fig. 13–2a), such a crystal might serve as a diffraction grating for very short wavelengths on the order of the spacing between atoms, estimated to be about 10^{-10} m $(= 10^{-1}$ nm). Experiments soon showed that X-rays scattered from a crystal did indeed show the peaks and valleys of a diffraction pattern (Fig. 25–37). Thus it was shown, in a single blow, that X-rays have a wave nature and that atoms are arranged in a regular way in crystals. Today, X-rays are recognized as electromagnetic radiation with wavelengths in the range of about 10^{-2} nm to 10 nm, the range readily produced in an X-ray tube.

*X-Ray Diffraction

We saw in Sections 25–7 and 25–8 that light of shorter wavelength provides greater resolution when we are examining an object microscopically. Since X-rays have much shorter wavelengths than visible light, they should in principle offer much greater resolution. However, there seems to be no effective material to use as lenses for the very short wavelengths of X-rays. Instead, the clever but complicated technique of **X-ray diffraction** (or **crystallography**) has proved very effective for examining the microscopic world of atoms and molecules. In a simple crystal such as NaCl, the atoms are arranged in an orderly cubical fashion, Fig. 25–38, with atoms spaced a distance d apart. Suppose that a beam of X-rays is incident on the crystal at an angle ϕ to the surface, and that the two rays shown are reflected from two subsequent planes of atoms as shown. The two rays will constructively interfere if the extra distance ray I travels is a whole number of wavelengths farther than the distance ray II travels. This extra distance is $2d \sin\phi$. Therefore, constructive interference will occur when

$$m\lambda = 2d \sin\phi, \qquad m = 1, 2, 3, \cdots, \tag{25–10}$$

where m can be any integer. (Notice that ϕ is *not* the angle with respect to the normal to the surface.) This is called the **Bragg equation** after W. L. Bragg (1890–1971), who derived it and who, together with his father, W. H. Bragg (1862–1942), developed the theory and technique of X-ray diffraction by crystals in 1912–1913. If the X-ray wavelength is known and the angle ϕ is measured, the distance d between atoms can be obtained. This is the basis for X-ray crystallography.

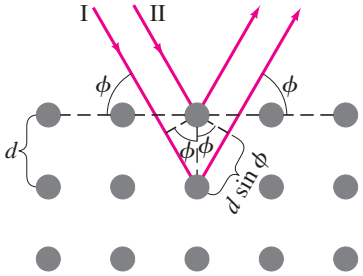
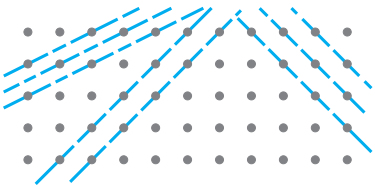


FIGURE 25–38 X-ray diffraction by a crystal.

FIGURE 25–39 X-rays can be diffracted from many possible planes within a crystal.



EXERCISE G When X-rays of wavelength 0.10×10^{-9} m are scattered from a sodium chloride crystal, a second-order diffraction peak is observed at 21°. What is the spacing between the planes of atoms for this scattering?

Actual X-ray diffraction patterns are quite complicated. First of all, a crystal is a three-dimensional object, and X-rays can be diffracted from different planes at different angles within the crystal, as shown in Fig. 25–39. Although the analysis is complex, a great deal can be learned from X-ray diffraction about any substance that can be put in crystalline form.

X-ray diffraction has been very useful in determining the structure of biologically important molecules, such as the double helix structure of DNA, worked out by James Watson and Francis Crick in 1953. See Fig. 25–40, and for models of the double helix, Figs. 16–39a and 16–40. Around 1960, the first detailed structure of a protein molecule, myoglobin, was elucidated with the aid of X-ray diffraction. Soon the structure of an important constituent of blood, hemoglobin, was worked out, and since then the structures of a great many molecules have been determined with the help of X-rays.

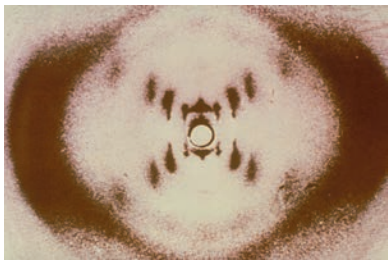


FIGURE 25–40 X-ray diffraction photo of DNA molecules taken by Rosalind Franklin in the early 1950s. The cross of spots suggested that DNA is a helix.

*25–12 X-Ray Imaging and Computed Tomography (CT Scan)

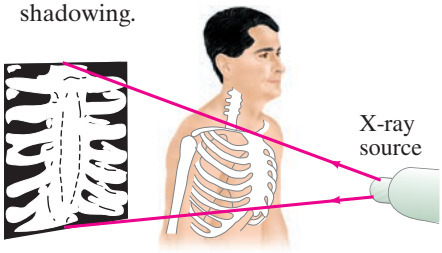
*Normal X-Ray Image

For a conventional medical or dental X-ray photograph, the X-rays emerging from the tube (Fig. 25–36) pass through the body and are detected on photographic film, a digital sensor, or a fluorescent screen, Fig. 25–41. The rays travel in very nearly straight lines through the body with minimal deviation since at X-ray wavelengths there is little diffraction or refraction. There is absorption (and scattering), however; and the difference in absorption by different structures in the body is what gives rise to the image produced by the transmitted rays. The less the absorption, the greater the transmission and the darker the film. The image is, in a sense, a "shadow" of what the rays have passed through. The X-ray image is *not* produced by focusing rays with lenses as for the instruments discussed earlier in this Chapter.

PHYSICS APPLIED
Normal X-ray image

CAUTION
Normal X-ray image is a sort of shadow (no lenses are involved)

FIGURE 25–41 Conventional X-ray imaging, which is essentially shadowing.



*Tomography Images (CT)

In conventional X-ray images, a body's thickness is projected onto film or a sensor; structures overlap and in many cases are difficult to distinguish. In the 1970s, a revolutionary X-ray technique was developed called **computed tomography** (CT), which produces an image of a *slice* through the body. (The word **tomography** comes from the Greek: *tomos* = slice, *graph* = picture.) Structures and lesions previously impossible to visualize can now be seen with remarkable clarity. The principle behind CT is shown in Fig. 25–42: a thin collimated (parallel) beam of X-rays passes through the body to a detector that measures the transmitted intensity. Measurements are made at a large number of points as the source and detector are moved past the body together. The apparatus is then rotated slightly about the body axis and again scanned; this is repeated at (perhaps) 1° intervals for 180°. The intensity of the transmitted beam for the many points of each scan, and for each angle, is sent to a computer that reconstructs the image of the slice. Note that the imaged slice is perpendicular to the long axis of the body. For this reason, CT is sometimes called **computerized axial tomography** (CAT), although the abbreviation CAT, as in CAT scan, can also be read as **computer-assisted tomography**.

PHYSICS APPLIED
Computed tomography images (CT or CAT scans)

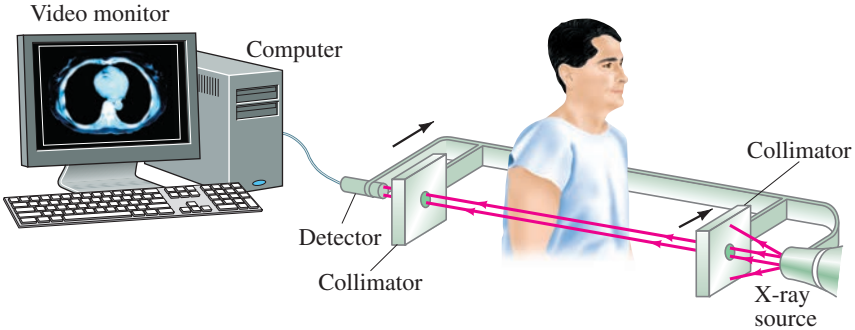


FIGURE 25–42 Tomographic imaging: the X-ray source and detector move together across the body, the transmitted intensity being measured at a large number of points. Then the "source–detector" assembly is rotated slightly (say, 1°) around a vertical axis, and another scan is made. This process is repeated for perhaps 180°. The computer reconstructs the image of the slice and it is presented on a TV or computer monitor.

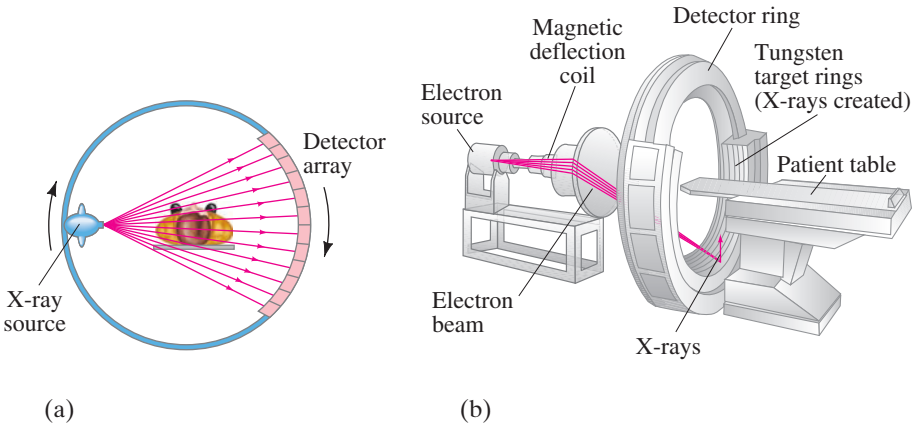


FIGURE 25–43 (a) Fan-beam scanner. Rays transmitted through the entire body are measured simultaneously at each angle. The source and detector rotate to take measurements at different angles. In another type of fan-beam scanner, there are detectors around the entire 360° of the circle which remain fixed as the source moves. (b) In still another type, a beam of electrons from a source is directed by magnetic fields at tungsten targets surrounding the patient.

The use of a single detector as in Fig. 25–42 would require a few minutes for the many scans needed to form a complete image. Much faster scanners use a fan beam, Fig. 25–43a, in which beams passing through the entire cross section of the body are detected simultaneously by many detectors. The source and detectors are then rotated about the patient, and an image requires only a few seconds. Even faster, and therefore useful for heart scans, are fixed source machines wherein an electron beam is directed (by magnetic fields) to tungsten targets surrounding the patient, creating the X-rays. See Fig. 25–43b.

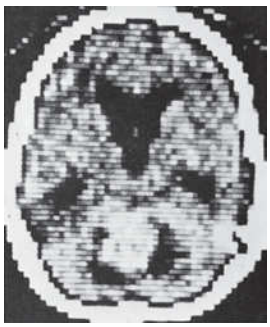
*Image Formation

But how is the image formed? We can think of the slice to be imaged as being divided into many tiny picture elements (or **pixels**), which could be squares (as in Fig. 24–49). For CT, the width of each pixel is chosen according to the width of the detectors and/or the width of the X-ray beams, and this determines the resolution of the image, which might be 1 mm. An X-ray detector measures the intensity of the transmitted beam. Subtracting this value from the intensity of the beam at the source yields the total absorption (called a "projection") along that beam line. Complicated mathematical techniques are used to analyze all the absorption projections for the huge number of beam scans measured (see the next Subsection), obtaining the absorption at each pixel and assigning each a "grayness value" according to how much radiation was absorbed. The image is made up of tiny spots (pixels) of varying shades of gray. Often the amount of absorption is color-coded. The colors in the resulting **false-color** image have nothing to do, however, with the actual color of the object. The actual images are monochromatic (various shades of gray, depending on the absorption). Only *visible* light has color; X-rays do not.

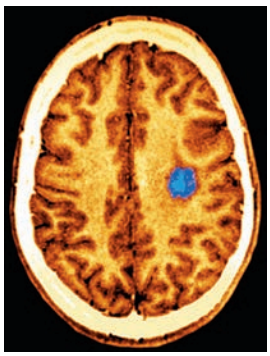
Figure 25–44 illustrates what actual CT images look like. It is generally agreed that CT scanning has revolutionized some areas of medicine by providing much less invasive, and/or more accurate, diagnosis.

Computed tomography can also be applied to ultrasound imaging (Section 12–9) and to emissions from radioisotopes and nuclear magnetic resonance (Sections 31–8 and 31–9).

FIGURE 25–44 Two CT images, with different resolutions, each showing a cross section of a brain. Photo (a) is of low resolution; photo (b), of higher resolution, shows a brain tumor, and uses false color to highlight it.



(a)



(b)

*Tomographic Image Reconstruction

How can the "grayness" of each pixel be determined even though all we can measure is the total absorption along each beam line in the slice? It can be done only by using the many beam scans made at a great many different angles. Suppose the image is to be an array of 100 × 100 elements for a total of 10^4 pixels. If we have 100 detectors and measure the absorption projections at 100 different angles, then we get 10^4 pieces of information. From this information, an image can be reconstructed, but not precisely. If more angles are measured, the reconstruction of the image can be done more accurately.

To suggest how mathematical reconstruction is done, we consider a very simple case using the **iterative** technique ("to iterate" is from the Latin "to repeat"). Suppose our sample slice is divided into the simple 2×2 pixels as shown in Fig. 25–45. The number inside each pixel represents the amount of absorption by the material in that area (say, in tenths of a percent): that is, 4 represents twice as much absorption as 2. But we cannot directly measure these values—they are the unknowns we want to solve for. All we can measure are the projections—the total absorption along each beam line—and these are shown in Fig. 25–45 outside the yellow squares as the sum of the absorptions for the pixels along each line at four different angles. These projections (given at the tip of each arrow) are what we can measure, and we now want to work back from them to see how close we can get to the true absorption value for each pixel. We start our analysis with each pixel being assigned a zero value, Fig. 25–46a. In the iterative technique, we use the projections to estimate the absorption value in each square, and repeat for each angle. The angle 1 projections are 7 and 13 (Fig. 25–45). We divide each of these equally between their two squares: each square in the left column of Fig. 25–46a gets $3\frac{1}{2}$ (half of 7), and each square in the right column gets $6\frac{1}{2}$ (half of 13).

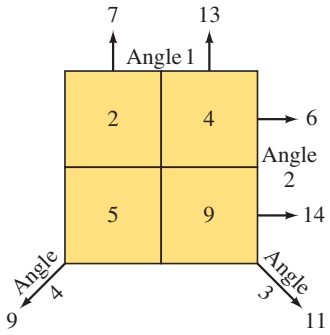
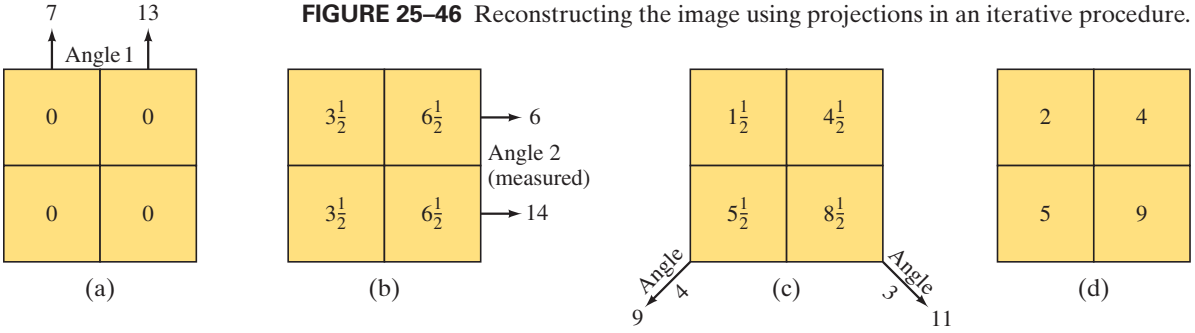


FIGURE 25–45 A simple 2×2 image showing true absorption values (inside the squares) and measured projections.

FIGURE 25–46 Reconstructing the image using projections in an iterative procedure.



Next we use the projections at angle 2. We calculate the difference between the measured projections at angle 2 (6 and 14) and the projections based on the previous estimate (top row: $3\frac{1}{2} + 6\frac{1}{2} = 10$; same for bottom row). Then we distribute this difference equally to the squares in that row. For the top row, we have

$$3\tfrac{1}{2} + \frac{6 - 10}{2} = 1\tfrac{1}{2} \quad \text{and} \quad 6\tfrac{1}{2} + \frac{6 - 10}{2} = 4\tfrac{1}{2};$$

and for the bottom row,

$$3\tfrac{1}{2} + \frac{14 - 10}{2} = 5\tfrac{1}{2} \quad \text{and} \quad 6\tfrac{1}{2} + \frac{14 - 10}{2} = 8\tfrac{1}{2}.$$

These values are inserted as shown in Fig. 25–46c. Next, the projection at angle 3 ($= 11$), combined with the difference as above, gives

$$\text{(upper left)} \quad 1\tfrac{1}{2} + \frac{11 - 10}{2} = 2 \quad \text{and} \quad \text{(lower right)} \quad 8\tfrac{1}{2} + \frac{11 - 10}{2} = 9;$$

and then for angle 4 we have

$$\text{(lower left)} \quad 5\tfrac{1}{2} + \frac{9 - 10}{2} = 5 \quad \text{and} \quad \text{(upper right)} \quad 4\tfrac{1}{2} + \frac{9 - 10}{2} = 4.$$

The result, shown in Fig. 25–46d, corresponds exactly to the true values. (In real situations, the true values are not known, which is why these computer techniques are required.) To obtain these numbers exactly, we used six pieces of information (two each at angles 1 and 2, one each at angles 3 and 4). For the much larger number of pixels used for actual images, exact values are generally not attained. Many iterations may be needed, and the calculation is considered sufficiently precise when the difference between calculated and measured projections is sufficiently small. The above example illustrates the "convergence" of the process: the first iteration (b to c in Fig. 25–46) changed the values by 2, the last iteration (c to d) by only $\frac{1}{2}$.

Summary

A **camera** lens forms an image on film, or on an electronic sensor (CCD or CMOS) in a digital camera. Light is allowed in briefly through a shutter. The image is focused by moving the lens relative to the film or sensor, and the ***f*-stop** (or lens opening) must be adjusted for the brightness of the scene and the chosen shutter speed. The *f*-stop is defined as the ratio of the focal length to the diameter of the lens opening.

The human **eye** also adjusts for the available light—by opening and closing the iris. It focuses not by moving the lens, but by adjusting the shape of the lens to vary its focal length. The image is formed on the retina, which contains an array of receptors known as rods and cones.

Diverging eyeglass or contact lenses are used to correct the defect of a nearsighted eye, which cannot focus well on distant objects. Converging lenses are used to correct for defects in which the eye cannot focus on close objects.

A **simple magnifier** is a converging lens that forms a virtual image of an object placed at (or within) the focal point. The **angular magnification**, when viewed by a relaxed normal eye, is

$$M = \frac{N}{f}, \qquad \textbf{(25–2a)}$$

where f is the focal length of the lens and N is the near point of the eye (25 cm for a "normal" eye).

An **astronomical telescope** consists of an **objective lens** or mirror, and an **eyepiece** that magnifies the real image formed by the objective. The **magnification** is equal to the ratio of the objective and eyepiece focal lengths, and the image is inverted:

$$M = -\frac{f_o}{f_e}. \qquad \textbf{(25–3)}$$

A compound **microscope** also uses objective and eyepiece lenses, and the final image is inverted. The total magnification is the product of the magnifications of the two lenses and is approximately

$$M \approx \frac{N\ell}{f_e f_o}, \qquad \textbf{(25–6b)}$$

where ℓ is the distance between the lenses, N is the near point of the eye, and f_o and f_e are the focal lengths of objective and eyepiece, respectively.

Microscopes, telescopes, and other optical instruments are limited in the formation of sharp images by **lens aberrations**. These include **spherical aberration**, in which rays passing through the edge of a lens are not focused at the same point as those that pass near the center; and **chromatic aberration**, in which different colors are focused at different points. Compound lenses, consisting of several elements, can largely correct for aberrations.

The wave nature of light also limits the sharpness, or **resolution**, of images. Because of diffraction, it is *not possible to discern details smaller than the wavelength* of the radiation being used. The useful magnification of a light microscope is limited by diffraction to about 500×.

[***X-rays** are a form of electromagnetic radiation of very short wavelength. They are produced when high-speed electrons, accelerated by high voltage in an evacuated tube, strike a glass or metal target.]

[***Computed tomography** (CT or CAT scan) uses many narrow X-ray beams through a section of the body to construct an image of that section.]

Questions

1. Why must a camera lens be moved farther from the sensor or film to focus on a closer object?
2. Why is the depth of field greater, and the image sharper, when a camera lens is "stopped down" to a larger *f*-number? Ignore diffraction.
3. Describe how diffraction affects the statement of Question 2. [*Hint*: See Eq. 24–3 or 25–7.]
4. Why are bifocals needed mainly by older persons and not generally by younger people?
5. Will a nearsighted person who wears corrective lenses in her glasses be able to see clearly underwater when wearing those glasses? Use a diagram to show why or why not.
6. You can tell whether people are nearsighted or farsighted by looking at the width of their face through their glasses. If a person's face appears narrower through the glasses (Fig. 25–47), is the person farsighted or nearsighted? Try to explain, but also check experimentally with friends who wear glasses.



FIGURE 25–47
Question 6.

7. In attempting to discern distant details, people will sometimes squint. Why does this help?
8. Is the image formed on the retina of the human eye upright or inverted? Discuss the implications of this for our perception of objects.
9. The human eye is much like a camera—yet, when a camera shutter is left open and the camera is moved, the image will be blurred. But when you move your head with your eyes open, you still see clearly. Explain.
10. Reading glasses use converging lenses. A simple magnifier is also a converging lens. Are reading glasses therefore magnifiers? Discuss the similarities and differences between converging lenses as used for these two different purposes.
11. Nearsighted people often look over (or under) their glasses when they want to see something small up close, like a cell phone screen. Why?
12. Spherical aberration in a thin lens is minimized if rays are bent equally by the two surfaces. If a planoconvex lens is used to form a real image of an object at infinity, which surface should face the object? Use ray diagrams to show why.
13. Explain why chromatic aberration occurs for thin lenses but not for mirrors.
14. Inexpensive microscopes for children's use usually produce images that are colored at the edges. Why?

15. Which aberrations present in a simple lens are not present (or are greatly reduced) in the human eye?

16. By what factor can you improve resolution, other things being equal, if you use blue light ($\lambda = 450\,\text{nm}$) rather than red (700 nm)?

17. Atoms have diameters of about 10^{-8} cm. Can visible light be used to "see" an atom? Explain.

18. Which color of visible light would give the best resolution in a microscope? Explain.

19. For both converging and diverging lenses, discuss how the focal length for red light differs from that for violet light.

20. The 300-meter radiotelescope in Arecibo, Puerto Rico (Fig. 25–33), is the world's largest radiotelescope, but many other radiotelescopes are also very large. Why are radiotelescopes so big? Why not make optical telescopes that are equally large? (The largest optical telescopes have diameters of about 10 meters.)

MisConceptual Questions

1. The image of a nearby object formed by a camera lens is
(*a*) at the lens' focal point.
(*b*) always blurred.
(*c*) at the same location as the image of an object at infinity.
(*d*) farther from the lens than the lens' focal point.

2. What is a megapixel in a digital camera?
(*a*) A large spot on the detector where the image is focused.
(*b*) A special kind of lens that gives a sharper image.
(*c*) A number related to how many photographs the camera can store.
(*d*) A million light-sensitive spots on the detector.
(*e*) A number related to how fast the camera can take pictures.

3. When a nearsighted person looks at a distant object through her glasses, the image produced by the glasses should be
(*a*) about 25 cm from her eye.
(*b*) at her eye's far point.
(*c*) at her eye's near point.
(*d*) at the far point for a normal eye.

4. If the distance from your eye's lens to the retina is shorter than for a normal eye, you will struggle to see objects that are
(*a*) nearby. (*c*) colorful.
(*b*) far away. (*d*) moving fast.

5. The image produced on the retina of the eye is _____ compared to the object being viewed.
(*a*) inverted. (*c*) sideways.
(*b*) upright. (*d*) enlarged.

6. How do eyeglasses help a nearsighted person see more clearly?
(*a*) Diverging lenses bend light entering the eye, so the image focuses farther from the front of the eye.
(*b*) Diverging lenses bend light entering the eye, so the image focuses closer to the front of the eye.
(*c*) Converging lenses bend light entering the eye, so the image focuses farther from the front of the eye.
(*d*) Converging lenses bend light entering the eye, so the image focuses closer to the front of the eye.
(*e*) Lenses adjust the distance from the cornea to the back of the eye.

7. When you closely examine an object through a magnifying glass, the magnifying glass
(*a*) makes the object bigger.
(*b*) makes the object appear closer than it actually is.
(*c*) makes the object appear farther than it actually is.
(*d*) causes additional light rays to be emitted by the object.

8. It would be impossible to build a microscope that could use visible light to see the molecular structure of a crystal because.
(*a*) lenses with enough magnification cannot be made.
(*b*) lenses cannot be ground with fine enough precision.
(*c*) lenses cannot be placed in the correct place with enough precision.
(*d*) diffraction limits the resolving power to about the size of the wavelength of the light used.
(*e*) More than one of the above is correct.

9. Why aren't white-light microscopes made with a magnification of 3000×?
(*a*) Lenses can't be made large enough.
(*b*) Lenses can't be made small enough.
(*c*) Lenses can't be made with short enough focal lengths.
(*d*) Lenses can't be made with long enough focal lengths.
(*e*) Diffraction limits useful magnification to several times less than this.

10. The resolving power of a microscope is greatest when the object being observed is illuminated by
(*a*) ultraviolet light. (*c*) visible light.
(*b*) infrared light. (*d*) radio waves.

11. Which of the following statements is true?
(*a*) A larger-diameter lens can better resolve two distant points.
(*b*) Red light can better resolve two distant points than blue light can.
(*c*) It is easier to resolve distant objects than nearer objects.
(*d*) Objects that are closer together are easier to resolve than objects that are farther apart.

12. While you are photographing a dog, it begins to move away. What must you do to keep it in focus?
(*a*) Increase the *f*-stop value.
(*b*) Decrease the *f*-stop value.
(*c*) Move the lens away from the sensor or film.
(*d*) Move the lens closer to the sensor or film.
(*e*) None of the above.

13. A converging lens, like the type used in a magnifying glass,
(*a*) always produces a magnified image (image taller than the object).
(*b*) can also produce an image smaller than the object.
(*c*) always produces an upright image.
(*d*) can also produce an inverted image (upside down).
(*e*) None of these statements are true.

Problems

25–1 Camera

1. (I) A properly exposed photograph is taken at $f/16$ and $\frac{1}{100}$ s. What lens opening is required if the shutter speed is $\frac{1}{400}$ s?
2. (I) A television camera lens has a 17-cm focal length and a lens diameter of 6.0 cm. What is its f-number?
3. (I) A 65-mm-focal-length lens has f-stops ranging from $f/1.4$ to $f/22$. What is the corresponding range of lens diaphragm diameters?
4. (I) A light meter reports that a camera setting of $\frac{1}{500}$ s at $f/5.6$ will give a correct exposure. But the photographer wishes to use $f/11$ to increase the depth of field. What should the shutter speed be?
5. (II) For a camera equipped with a 55-mm-focal-length lens, what is the object distance if the image height equals the object height? How far is the object from the image on the film?
6. (II) A nature photographer wishes to shoot a 34-m-tall tree from a distance of 65 m. What focal-length lens should be used if the image is to fill the 24-mm height of the sensor?
7. (II) A 200-mm-focal-length lens can be adjusted so that it is 200.0 mm to 208.2 mm from the film. For what range of object distances can it be adjusted?
8. (II) How large is the image of the Sun on film used in a camera with (*a*) a 28-mm-focal-length lens, (*b*) a 50-mm-focal-length lens, and (*c*) a 135-mm-focal-length lens? (*d*) If the 50-mm lens is considered normal for this camera, what relative magnification does each of the other two lenses provide? The Sun has diameter 1.4×10^6 km, and it is 1.5×10^8 km away.
9. (II) If a 135-mm telephoto lens is designed to cover object distances from 1.30 m to ∞, over what distance must the lens move relative to the plane of the sensor or film?
10. (III) Show that for objects very far away (assume infinity), the magnification of any camera lens is proportional to its focal length.

25–2 Eye and Corrective Lenses

11. (I) A human eyeball is about 2.0 cm long and the pupil has a maximum diameter of about 8.0 mm. What is the "speed" of this lens?
12. (II) A person struggles to read by holding a book at arm's length, a distance of 52 cm away. What power of reading glasses should be prescribed for her, assuming they will be placed 2.0 cm from the eye and she wants to read at the "normal" near point of 25 cm?
13. (II) Reading glasses of what power are needed for a person whose near point is 125 cm, so that he can read a computer screen at 55 cm? Assume a lens–eye distance of 1.8 cm.
14. (II) An eye is corrected by a -5.50-D lens, 2.0 cm from the eye. (*a*) Is this eye near- or farsighted? (*b*) What is this eye's far point without glasses?
15. (II) A person's right eye can see objects clearly only if they are between 25 cm and 85 cm away. (*a*) What power of contact lens is required so that objects far away are sharp? (*b*) What will be the near point with the lens in place?
16. (II) About how much longer is the nearsighted eye in Example 25–6 than the 2.0 cm of a normal eye?
17. (II) A person has a far point of 14 cm. What power glasses would correct this vision if the glasses were placed 2.0 cm from the eye? What power contact lenses, placed on the eye, would the person need?
18. (II) One lens of a nearsighted person's eyeglasses has a focal length of -26.0 cm and the lens is 1.8 cm from the eye. If the person switches to contact lenses placed directly on the eye, what should be the focal length of the corresponding contact lens?
19. (II) What is the focal length of the eye–lens system when viewing an object (*a*) at infinity, and (*b*) 34 cm from the eye? Assume that the lens–retina distance is 2.0 cm.
20. (III) The closely packed cones in the fovea of the eye have a diameter of about 2 μm. For the eye to discern two images on the fovea as distinct, assume that the images must be separated by at least one cone that is not excited. If these images are of two point-like objects at the eye's 25-cm near point, how far apart are these barely resolvable objects? Assume the eye's diameter (cornea-to-fovea distance) is 2.0 cm.
21. (III) A nearsighted person has near and far points of 10.6 and 20.0 cm, respectively. If she puts on contact lenses with power $P = -4.00$ D, what are her new near and far points?

25–3 Magnifying Glass

22. (I) What is the focal length of a magnifying glass of 3.2× magnification for a relaxed normal eye?
23. (I) What is the magnification of a lens used with a relaxed eye if its focal length is 16 cm?
24. (I) A magnifier is rated at 3.5× for a normal eye focusing on an image at the near point. (*a*) What is its focal length? (*b*) What is its focal length if the 3.5× refers to a relaxed eye?
25. (II) Sherlock Holmes is using an 8.20-cm-focal-length lens as his magnifying glass. To obtain maximum magnification, where must the object be placed (assume a normal eye), and what will be the magnification?
26. (II) A small insect is placed 4.85 cm from a +5.00-cm-focal-length lens. Calculate (*a*) the position of the image, and (*b*) the angular magnification.
27. (II) A 3.80-mm-wide bolt is viewed with a 9.60-cm-focal-length lens. A normal eye views the image at its near point. Calculate (*a*) the angular magnification, (*b*) the width of the image, and (*c*) the object distance from the lens.
28. (II) A magnifying glass with a focal length of 9.2 cm is used to read print placed at a distance of 8.0 cm. Calculate (*a*) the position of the image; (*b*) the angular magnification.
29. (III) A writer uses a converging lens of focal length $f = 12$ cm as a magnifying glass to read fine print on his book contract. Initially, the writer holds the lens above the fine print so that its image is at infinity. To get a better look, he then moves the lens so that the image is at his 25-cm near point. How far, and in what direction (toward or away from the fine print) did the writer move the lens? Assume his eye is adjusted to remain always very near the magnifying glass.
30. (III) A magnifying glass is rated at 3.0× for a normal eye that is relaxed. What would be the magnification for a relaxed eye whose near point is (*a*) 75 cm, and (*b*) 15 cm? Explain the differences.

25–4 Telescopes

31. (I) What is the magnification of an astronomical telescope whose objective lens has a focal length of 82 cm, and whose eyepiece has a focal length of 2.8 cm? What is the overall length of the telescope when adjusted for a relaxed eye?

32. (I) The overall magnification of an astronomical telescope is desired to be 25×. If an objective of 88-cm focal length is used, what must be the focal length of the eyepiece? What is the overall length of the telescope when adjusted for use by the relaxed eye?

33. (II) A 7.0× binocular has 3.5-cm-focal-length eyepieces. What is the focal length of the objective lenses?

34. (II) An astronomical telescope has an objective with focal length 75 cm and a +25-D eyepiece. What is the total magnification?

35. (II) An astronomical telescope has its two lenses spaced 82.0 cm apart. If the objective lens has a focal length of 78.5 cm, what is the magnification of this telescope? Assume a relaxed eye.

36. (II) A Galilean telescope adjusted for a relaxed eye is 36.8 cm long. If the objective lens has a focal length of 39.0 cm, what is the magnification?

37. (II) What is the magnifying power of an astronomical telescope using a reflecting mirror whose radius of curvature is 6.1 m and an eyepiece whose focal length is 2.8 cm?

38. (II) The Moon's image appears to be magnified 150× by a reflecting astronomical telescope with an eyepiece having a focal length of 3.1 cm. What are the focal length and radius of curvature of the main (objective) mirror?

39. (II) A 120× astronomical telescope is adjusted for a relaxed eye when the two lenses are 1.10 m apart. What is the focal length of each lens?

40. (II) An astronomical telescope longer than about 50 cm is not easy to hold by hand. Estimate the maximum angular magnification achievable for a telescope designed to be handheld. Assume its eyepiece lens, if used as a magnifying glass, provides a magnification of 5× for a relaxed eye with near point $N = 25$ cm.

41. (III) A 6.5× pair of binoculars has an objective focal length of 26 cm. If the binoculars are focused on an object 4.0 m away (from the objective), what is the magnification? (The 6.5× refers to objects at infinity; Eq. 25–3 holds only for objects at infinity and not for nearby ones.)

25–5 Microscopes

42. (I) A microscope uses an eyepiece with a focal length of 1.70 cm. Using a normal eye with a final image at infinity, the barrel length is 17.5 cm and the focal length of the objective lens is 0.65 cm. What is the magnification of the microscope?

43. (I) A 720× microscope uses a 0.40-cm-focal-length objective lens. If the barrel length is 17.5 cm, what is the focal length of the eyepiece? Assume a normal eye and that the final image is at infinity.

44. (I) A 17-cm-long microscope has an eyepiece with a focal length of 2.5 cm and an objective with a focal length of 0.33 cm. What is the approximate magnification?

45. (II) A microscope has a 14.0× eyepiece and a 60.0× objective lens 20.0 cm apart. Calculate (*a*) the total magnification, (*b*) the focal length of each lens, and (*c*) where the object must be for a normal relaxed eye to see it in focus.

46. (II) Repeat Problem 45 assuming that the final image is located 25 cm from the eyepiece (near point of a normal eye).

47. (II) A microscope has a 1.8-cm-focal-length eyepiece and a 0.80-cm objective. Assuming a relaxed normal eye, calculate (*a*) the position of the object if the distance between the lenses is 14.8 cm, and (*b*) the total magnification.

48. (III) An inexpensive instructional lab microscope allows the user to select its objective lens to have a focal length of 32 mm, 15 mm, or 3.9 mm. It also has two possible eyepieces with magnifications 5× and 15×. Each objective forms a real image 160 mm beyond its focal point. What are the largest and smallest overall magnifications obtainable with this instrument?

25–6 Lens Aberrations

49. (II) An achromatic lens is made of two very thin lenses, placed in contact, that have focal lengths $f_1 = -27.8$ cm and $f_2 = +25.3$ cm. (*a*) Is the combination converging or diverging? (*b*) What is the net focal length?

***50.** (III) A planoconvex lens (Fig. 23–31a) has one flat surface and the other has $R = 14.5$ cm. This lens is used to view a red and yellow object which is 66.0 cm away from the lens. The index of refraction of the glass is 1.5106 for red light and 1.5226 for yellow light. What are the locations of the red and yellow images formed by the lens? [*Hint*: See Section 23–10.]

25–7 to 25–9 Resolution Limits

51. (I) What is the angular resolution limit (degrees) set by diffraction for the 100-inch (254-cm mirror diameter) Mt. Wilson telescope ($\lambda = 560$ nm)?

52. (I) What is the resolving power of a microscope ($\lambda = 550$ nm) with a 5-mm-diameter objective which has $f = 9$ mm?

53. (II) Two stars 18 light-years away are barely resolved by a 66-cm (mirror diameter) telescope. How far apart are the stars? Assume $\lambda = 550$ nm and that the resolution is limited by diffraction.

54. (II) The nearest neighboring star to the Sun is about 4 light-years away. If a planet happened to be orbiting this star at an orbital radius equal to that of the Earth–Sun distance, what minimum diameter would an Earth-based telescope's aperture have to be in order to obtain an image that resolved this star–planet system? Assume the light emitted by the star and planet has a wavelength of 550 nm.

55. (II) If you could shine a very powerful flashlight beam toward the Moon, estimate the diameter of the beam when it reaches the Moon. Assume that the beam leaves the flashlight through a 5.0-cm aperture, that its white light has an average wavelength of 550 nm, and that the beam spreads due to diffraction only.

56. (II) The normal lens on a 35-mm camera has a focal length of 50.0 mm. Its aperture diameter varies from a maximum of 25 mm ($f/2$) to a minimum of 3.0 mm ($f/16$). Determine the resolution limit set by diffraction for ($f/2$) and ($f/16$). Specify as the number of lines per millimeter resolved on the detector or film. Take $\lambda = 560$ nm.

57. (III) Suppose that you wish to construct a telescope that can resolve features 6.5 km across on the Moon, 384,000 km away. You have a 2.0-m-focal-length objective lens whose diameter is 11.0 cm. What focal-length eyepiece is needed if your eye can resolve objects 0.10 mm apart at a distance of 25 cm? What is the resolution limit set by the size of the objective lens (that is, by diffraction)? Use $\lambda = 560$ nm.

*25–11 X-Ray Diffraction

***58.** (II) X-rays of wavelength 0.138 nm fall on a crystal whose atoms, lying in planes, are spaced 0.285 nm apart. At what angle ϕ (relative to the surface, Fig. 25–38) must the X-rays be directed if the first diffraction maximum is to be observed?

***59.** (II) First-order Bragg diffraction is observed at 23.8° relative to the crystal surface, with spacing between atoms of 0.24 nm. (*a*) At what angle will second order be observed? (*b*) What is the wavelength of the X-rays?

***60.** (II) If X-ray diffraction peaks corresponding to the first three orders ($m = 1,\ 2$, and 3) are measured, can both the X-ray wavelength λ and lattice spacing d be determined? Prove your answer.

*25–12 Imaging by Tomography

***61.** (II) (*a*) Suppose for a conventional X-ray image that the X-ray beam consists of parallel rays. What would be the magnification of the image? (*b*) Suppose, instead, that the X-rays come from a point source (as in Fig. 25–41) that is 15 cm in front of a human body which is 25 cm thick, and the film is pressed against the person's back. Determine and discuss the range of magnifications that result.

General Problems

62. A **pinhole** camera uses a tiny pinhole instead of a lens. Show, using ray diagrams, how reasonably sharp images can be formed using such a pinhole camera. In particular, consider two point objects 2.0 cm apart that are 1.0 m from a 1.0-mm-diameter pinhole. Show that on a piece of film 7.0 cm behind the pinhole the two objects produce two separate circles that do not overlap.

63. An astronomical telescope has a magnification of 7.5×. If the two lenses are 28 cm apart, determine the focal length of each lens.

64. (*a*) How far away can a human eye distinguish two car headlights 2.0 m apart? Consider only diffraction effects and assume an eye pupil diameter of 6.0 mm and a wavelength of 560 nm. (*b*) What is the minimum angular separation an eye could resolve when viewing two stars, considering only diffraction effects? In reality, it is about 1′ of arc. Why is it not equal to your answer in (*b*)?

65. Figure 25–48 was taken from the NIST Laboratory (National Institute of Standards and Technology) in Boulder, CO, 2.0 km from the hiker in the photo. The Sun's image was 15 mm across on the film. Estimate the focal length of the camera lens (actually a telescope). The Sun has diameter 1.4×10^6 km, and it is 1.5×10^8 km away.



FIGURE 25–48
Problem 65.

66. A 1.0-cm-diameter lens with a focal length of 35 cm uses blue light to image two objects 15 m away that are very close together. What is the closest those objects can be to each other and still be imaged as separate objects?

67. A movie star catches a reporter shooting pictures of her at home. She claims the reporter was trespassing. To prove her point, she gives as evidence the film she seized. Her 1.65-m height is 8.25 mm high on the film, and the focal length of the camera lens was 220 mm. How far away from the subject was the reporter standing?

68. A child has a near point of 15 cm. What is the maximum magnification the child can obtain using a 9.5-cm-focal-length magnifier? What magnification can a normal eye obtain with the same lens? Which person sees more detail?

69. A woman can see clearly with her right eye only when objects are between 45 cm and 135 cm away. Prescription bifocals should have what powers so that she can see distant objects clearly (upper part) and be able to read a book 25 cm away (lower part) with her right eye? Assume that the glasses will be 2.0 cm from the eye.

70. What is the magnifying power of a +4.0-D lens used as a magnifier? Assume a relaxed normal eye.

71. A physicist lost in the mountains tries to make a telescope using the lenses from his reading glasses. They have powers of +2.0 D and +5.5 D, respectively. (*a*) What maximum magnification telescope is possible? (*b*) Which lens should be used as the eyepiece?

72. A person with normal vision adjusts a microscope for a good image when her eye is relaxed. She then places a camera where her eye was. For what object distance should the camera be set? Explain.

73. A 50-year-old man uses +2.5-D lenses to read a newspaper 25 cm away. Ten years later, he must hold the paper 38 cm away to see clearly with the same lenses. What power lenses does he need now in order to hold the paper 25 cm away? (Distances are measured from the lens.)

74. Two converging lenses, one with $f = 4.0\,\text{cm}$ and the other with $f = 48\,\text{cm}$, are made into a telescope. (*a*) What are the length and magnification? Which lens should be the eyepiece? (*b*) Assume these lenses are now combined to make a microscope; if the magnification needs to be 25×, how long would the microscope be?

75. An X-ray tube operates at 95 kV with a current of 25 mA and nearly all the electron energy goes into heat. If the specific heat of the 0.065-kg anode plate is $0.11\,\text{kcal/kg}\cdot\text{C}^\circ$, what will be the temperature rise per minute if no cooling water is used? (See Fig. 25–36.)

76. Human vision normally covers an angle of roughly 40° horizontally. A "normal" camera lens then is defined as follows: When focused on a distant horizontal object which subtends an angle of 40°, the lens produces an image that extends across the full horizontal extent of the camera's light-recording medium (film or electronic sensor). Determine the focal length *f* of the "normal" lens for the following types of cameras: (*a*) a 35-mm camera that records images on film 36 mm wide; (*b*) a digital camera that records images on a charge-coupled device (CCD) 1.60 cm wide.

77. The objective lens and the eyepiece of a telescope are spaced 85 cm apart. If the eyepiece is +19 D, what is the total magnification of the telescope?

78. Spy planes fly at extremely high altitudes (25 km) to avoid interception. If their cameras are to discern features as small as 5 cm, what is the minimum aperture of the camera lens to afford this resolution? (Use $\lambda = 580\,\text{nm}$.)

79. X-rays of wavelength 0.0973 nm are directed at an unknown crystal. The second diffraction maximum is recorded when the X-rays are directed at an angle of 21.2° relative to the crystal surface. What is the spacing between crystal planes?

80. The Hubble Space Telescope, with an objective diameter of 2.4 m, is viewing the Moon. Estimate the minimum distance between two objects on the Moon that the Hubble can distinguish. Consider diffraction of light with wavelength 550 nm. Assume the Hubble is near the Earth.

81. You want to design a spy satellite to photograph license plate numbers. Assuming it is necessary to resolve points separated by 5 cm with 550-nm light, and that the satellite orbits at a height of 130 km, what minimum lens aperture (diameter) is required?

82. Given two 12-cm-focal-length lenses, you attempt to make a crude microscope using them. While holding these lenses a distance 55 cm apart, you position your microscope so that its objective lens is distance d_o from a small object. Assume your eye's near point $N = 25\,\text{cm}$. (*a*) For your microscope to function properly, what should d_o be? (*b*) Assuming your eye is relaxed when using it, what magnification *M* does your microscope achieve? (*c*) Since the length of your microscope is not much greater than the focal lengths of its lenses, the approximation $M \approx N\ell/f_e f_o$ is not valid. If you apply this approximation to your microscope, what % error do you make in your microscope's true magnification?

***83.** The power of one lens in a pair of eyeglasses is −3.5 D. The radius of curvature of the outside surface is 16.0 cm. What is the radius of curvature of the inside surface? The lens is made of plastic with $n = 1.62$.

Search and Learn

1. Digital cameras may offer an optical zoom or a digital zoom. An optical zoom uses a variable focal-length lens, so only the central part of the field of view fills the entire sensor; a digital zoom electronically includes only the central pixels of the sensor, so objects are larger in the final picture. Discuss which is better, and why.

2. Redo Examples 25–3 and 25–4 assuming the sensor has only 6 MP. Explain the different results and their impact on finished photographs.

3. An astronomical telescope, Fig. 25–20, produces an inverted image. One way to make a telescope that produces an upright image is to insert a third lens between the objective and the eyepiece, Fig. 25–23b. To have the same magnification, the non-inverting telescope will be longer. Suppose lenses of focal length 150 cm, 1.5 cm, and 10 cm are available. Where should these three lenses be placed to make a non-inverting telescope with magnification 100×?

ANSWERS TO EXERCISES

A: 6.3 m.
B: 33 dots/mm.
C: $P = -4.0\,\text{D}$.
D: 48 cm.
E: 2 m.
F: (*c*) as stated on page 732; (*c*) by the λ rule.
G: 0.28 nm.

A science fantasy book called *Mr Tompkins in Wonderland* (1940), by physicist George Gamow, imagined a world in which the speed of light was only 10 m/s (20 mi/h). Mr Tompkins had studied relativity and when he began "speeding" on a bicycle, he "expected that he would be immediately shortened, and was very happy about it as his increasing figure had lately caused him some anxiety. To his great surprise, however, nothing happened to him or to his cycle. On the other hand, the picture around him completely changed. The streets grew shorter, the windows of the shops began to look like narrow slits, and the policeman on the corner became the thinnest man he had ever seen. 'By Jove!' exclaimed Mr Tompkins excitedly, 'I see the trick now. This is where the word *relativity* comes in.'"

Relativity does indeed predict that objects moving relative to us at high speed, close to the speed of light c, are shortened in length. We don't notice it as Mr Tompkins did, because $c = 3 \times 10^8$ m/s is incredibly fast. We will study length contraction, time dilation, simultaneity non-agreement, and how energy and mass are equivalent $(E = mc^2)$.



CHAPTER 26

The Special Theory of Relativity

CONTENTS

26–1 Galilean–Newtonian Relativity
26–2 Postulates of the Special Theory of Relativity
26–3 Simultaneity
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26–5 Length Contraction
26–6 Four-Dimensional Space–Time
26–7 Relativistic Momentum
26–8 The Ultimate Speed
26–9 $E = mc^2$; Mass and Energy
26–10 Relativistic Addition of Velocities
26–11 The Impact of Special Relativity

CHAPTER-OPENING QUESTION—Guess now!

A rocket is headed away from Earth at a speed of 0.80c. The rocket fires a small payload at a speed of 0.70c (relative to the rocket) aimed away from Earth. How fast is the payload moving relative to Earth?

(a) 1.50c;
(b) a little less than 1.50c;
(c) a little over c;
(d) a little under c;
(e) 0.75c.

Physics at the end of the nineteenth century looked back on a period of great progress. The theories developed over the preceding three centuries had been very successful in explaining a wide range of natural phenomena. Newtonian mechanics beautifully explained the motion of objects on Earth and in the heavens. Furthermore, it formed the basis for successful treatments of fluids, wave motion, and sound. Kinetic theory explained the behavior of gases and other materials. Maxwell's theory of electromagnetism embodied all of electric and magnetic phenomena, and it predicted the existence of electromagnetic waves that would behave just like light—so light came to be thought of as an electromagnetic wave. Indeed, it seemed that the natural world, as seen through the eyes of physicists, was very well explained. A few puzzles remained, but it was felt that these would soon be explained using already known principles.

It did not turn out so simply. Instead, these puzzles were to be solved only by the introduction, in the early part of the twentieth century, of two revolutionary new theories that changed our whole conception of nature: the *theory of relativity* and *quantum theory.*

Physics as it was known at the end of the nineteenth century (what we've covered up to now in this book) is referred to as **classical physics**. The new physics that grew out of the great revolution at the turn of the twentieth century is now called **modern physics**. In this Chapter, we present the special theory of relativity, which was first proposed by Albert Einstein (1879–1955; Fig. 26–1) in 1905. In Chapter 27, we introduce the equally momentous quantum theory.

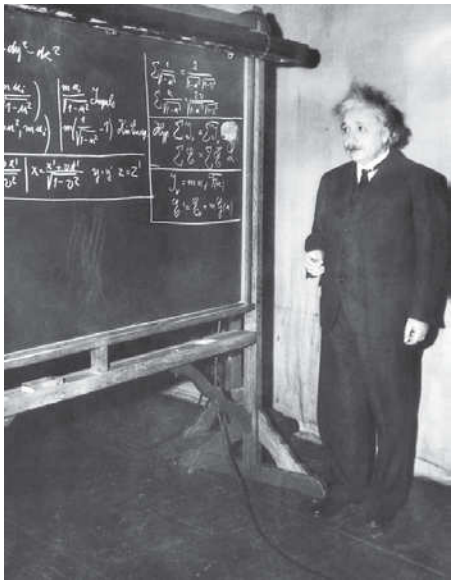


FIGURE 26–1 Albert Einstein (1879–1955), one of the great minds of the twentieth century, was the creator of the special and general theories of relativity.

26–1 Galilean–Newtonian Relativity

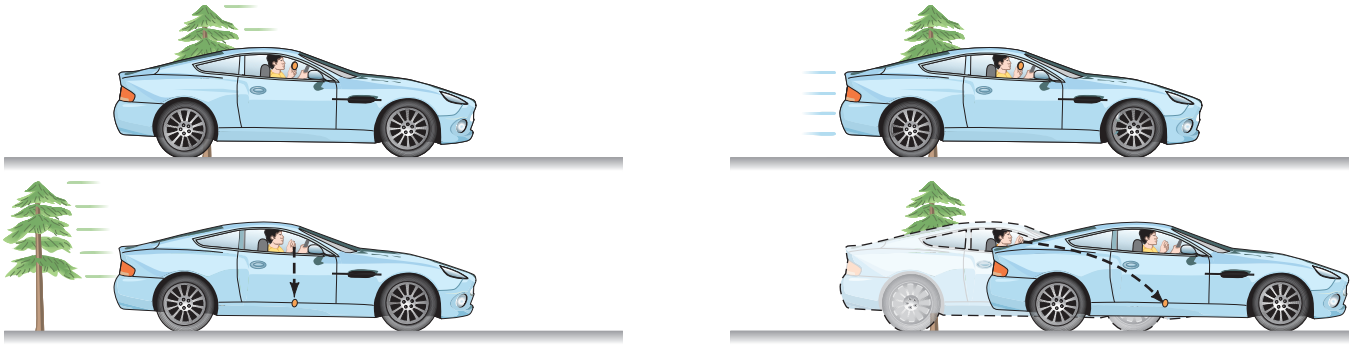
Einstein's special theory of relativity deals with how we observe events, particularly how objects and events are observed from different frames of reference.[†] This subject had already been explored by Galileo and Newton.

The special theory of relativity deals with events that are observed and measured from so-called **inertial reference frames** (Section 4–2 and Appendix C), which are reference frames in which Newton's first law is valid: if an object experiences no net force, the object either remains at rest or continues in motion with constant speed in a straight line. It is usually easiest to analyze events when they are observed and measured by observers at rest in an inertial frame. The Earth, though not quite an inertial frame (it rotates), is close enough that for most purposes we can approximate it as an inertial frame. Rotating or otherwise accelerating frames of reference are noninertial frames,[‡] and won't concern us in this Chapter (they are dealt with in Einstein's general theory of relativity, as we will see in Chapter 33).

A reference frame that moves with constant velocity with respect to an inertial frame is itself also an inertial frame, since Newton's laws hold in it as well. When we say that we observe or make measurements from a certain reference frame, it means that we are at rest in that reference frame.

[†]A reference frame is a set of coordinate axes fixed to some object such as the Earth, a train, or the Moon. See Section 2–1.

[‡]On a rotating platform (say a merry-go-round), for example, a ball at rest starts moving outward even though no object exerts a force on it. This is therefore not an inertial frame. See Appendix C, Fig. C–1.



(a)
Reference frame = car

(b)
Reference frame = Earth

FIGURE 26–2 A coin is dropped by a person in a moving car. The upper views show the moment of the coin's release, the lower views are a short time later. (a) In the reference frame of the car, the coin falls straight down (and the tree moves to the left). (b) In a reference frame fixed on the Earth, the coin has an initial velocity (= to car's) and follows a curved (parabolic) path.

Both Galileo and Newton were aware of what we now call the **relativity principle** applied to mechanics: that *the basic laws of physics are the same in all inertial reference frames*. You may have recognized its validity in everyday life. For example, objects move in the same way in a smoothly moving (constant-velocity) train or airplane as they do on Earth. (This assumes no vibrations or rocking which would make the reference frame noninertial.) When you walk, drink a cup of soup, play pool, or drop a pencil on the floor while traveling in a train, airplane, or ship moving at constant velocity, the objects move just as they do when you are at rest on Earth. Suppose you are in a car traveling rapidly at constant velocity. If you drop a coin from above your head inside the car, how will it fall? It falls straight downward with respect to the car, and hits the floor directly below the point of release, Fig. 26–2a. This is just how objects fall on the Earth—straight down—and thus our experiment in the moving car is in accord with the relativity principle. (If you drop the coin out the car's window, this won't happen because the moving air drags the coin backward relative to the car.)

CAUTION
Laws are the same, but paths may be different in different reference frames

Note in this example, however, that to an observer on the Earth, the coin follows a curved path, Fig. 26–2b. The actual path followed by the coin is different as viewed from different frames of reference. This does not violate the relativity principle because this principle states that the *laws* of physics are the same in all inertial frames. The same law of gravity, and the same laws of motion, apply in both reference frames. The acceleration of the coin is the same in both reference frames. The difference in Figs. 26–2a and b is that in the Earth's frame of reference, the coin has an initial velocity (equal to that of the car). The laws of physics therefore predict it will follow a parabolic path like any projectile (Chapter 3). In the car's reference frame, there is no initial velocity, and the laws of physics predict that the coin will fall straight down. The laws are the same in both reference frames, although the specific paths are different.

CAUTION
Length and time intervals are absolute (pre-relativity)

Galilean–Newtonian relativity involves certain unprovable assumptions that make sense from everyday experience. It is assumed that the lengths of objects are the same in one reference frame as in another, and that time passes at the same rate in different reference frames. In classical mechanics, then, space and time intervals are considered to be **absolute**: their measurement does not change from one reference frame to another. The mass of an object, as well as all forces, are assumed to be unchanged by a change in inertial reference frame.

CAUTION
Position and velocity are different in different reference frames, but length is the same (classical)

The position of an object, however, is different when specified in different reference frames, and so is velocity. For example, a person may walk inside a bus toward the front with a speed of 2 m/s. But if the bus moves 10 m/s with respect to the Earth, the person is then moving with a speed of 12 m/s with respect to the Earth. The acceleration of an object, however, is the same in any inertial reference frame according to classical mechanics. This is because the change in velocity, and the time interval, will be the same. For example, the person in the bus may accelerate from 0 to 2 m/s in 1.0 seconds, so $a = 2\ \text{m/s}^2$ in the reference frame of the bus. With respect to the Earth, the acceleration is

$$(12\ \text{m/s} - 10\ \text{m/s})/(1.0\ \text{s}) = 2\ \text{m/s}^2,$$

which is the same.

Since neither F, m, nor a changes from one inertial frame to another, Newton's second law, $F = ma$, does not change. Thus Newtons' second law satisfies the relativity principle. The other laws of mechanics also satisfy the relativity principle.

That the laws of mechanics are the same in all inertial reference frames implies that no one inertial frame is special in any sense. We express this important conclusion by saying that **all inertial reference frames are equivalent** for the description of mechanical phenomena. No one inertial reference frame is any better than another. A reference frame fixed to a car or an aircraft traveling at constant velocity is as good as one fixed on the Earth. When you travel smoothly at constant velocity in a car or airplane, it is just as valid to say you are at rest and the Earth is moving as it is to say the reverse.[†] There is no experiment you can do to tell which frame is "really" at rest and which is moving. Thus, there is no way to single out one particular reference frame as being at absolute rest.

A complication arose, however, in the last half of the nineteenth century. Maxwell's comprehensive and successful theory of electromagnetism (Chapter 22) predicted that light is an electromagnetic wave. Maxwell's equations gave the velocity of light c as $3.00 \times 10^8\,\text{m/s}$; and this is just what is measured. The question then arose: in what reference frame does light have precisely the value predicted by Maxwell's theory? It was assumed that light would have a different speed in different frames of reference. For example, if observers could travel on a rocket ship at a speed of $1.0 \times 10^8\,\text{m/s}$ away from a source of light, we might expect them to measure the speed of the light reaching them to be $(3.0 \times 10^8\,\text{m/s}) - (1.0 \times 10^8\,\text{m/s}) = 2.0 \times 10^8\,\text{m/s}$. But Maxwell's equations have no provision for relative velocity. They predicted the speed of light to be $c = 3.0 \times 10^8\,\text{m/s}$, which seemed to imply that there must be some preferred reference frame where c would have this value.

We discussed in Chapters 11 and 12 that waves can travel on water and along ropes or strings, and sound waves travel in air and other materials. Nineteenth-century physicists viewed the material world in terms of the laws of mechanics, so it was natural for them to assume that light too must travel in some *medium*. They called this transparent medium the **ether** and assumed it permeated all space.[‡] It was therefore assumed that the velocity of light given by Maxwell's equations must be with respect to the ether.[§]

Scientists soon set out to determine the speed of the Earth relative to this absolute frame, whatever it might be. A number of clever experiments were designed. The most direct were performed by A. A. Michelson and E. W. Morley in the 1880s. They measured the difference in the speed of light in different directions using Michelson's interferometer (Section 24–9). They expected to find a difference depending on the orientation of their apparatus with respect to the ether. For just as a boat has different speeds relative to the land when it moves upstream, downstream, or across the stream, so too light would be expected to have different speeds depending on the velocity of the ether past the Earth.

Strange as it may seem, they detected no difference at all. This was a great puzzle. A number of explanations were put forth over a period of years, but they led to contradictions or were otherwise not generally accepted. This **null result** was one of the great puzzles at the end of the nineteenth century.

Then in 1905, Albert Einstein proposed a radical new theory that reconciled these many problems in a simple way. But at the same time, as we shall see, it completely changed our ideas of space and time.

[†]We use the reasonable approximation that Earth is an inertial reference frame.

[‡]The medium for light waves could not be air, since light travels from the Sun to Earth through nearly empty space. Therefore, another medium was postulated, the ether. The ether was not only transparent but, because of difficulty in detecting it, was assumed to have zero density.

[§]Also, it appeared that Maxwell's equations did *not* satisfy the relativity principle: They were simplest in the frame where $c = 3.00 \times 10^8\,\text{m/s}$, in a reference frame at rest in the ether. In any other reference frame, extra terms were needed to account for relative velocity. Although other laws of physics obeyed the relativity principle, the laws of electricity and magnetism apparently did not. Einstein's second postulate (next Section) resolved this problem: Maxwell's equations do satisfy relativity.

26–2 Postulates of the Special Theory of Relativity

The problems that existed at the start of the twentieth century with regard to electromagnetic theory and Newtonian mechanics were beautifully resolved by Einstein's introduction of the special theory of relativity in 1905. Unaware of the Michelson–Morley null result, Einstein was motivated by certain questions regarding electromagnetic theory and light waves. For example, he asked himself: "What would I see if I rode a light beam?" The answer was that instead of a traveling electromagnetic wave, he would see alternating electric and magnetic fields at rest whose magnitude changed in space, but did not change in time. Such fields, he realized, had never been detected and indeed were not consistent with Maxwell's electromagnetic theory. He argued, therefore, that it was unreasonable to think that the speed of light relative to any observer could be reduced to zero, or in fact reduced at all. This idea became the second postulate of his theory of relativity.

In his famous 1905 paper, Einstein proposed doing away with the idea of the ether and the accompanying assumption of a preferred or absolute reference frame at rest. This proposal was embodied in two postulates. The first was an extension of the Galilean–Newtonian relativity principle to include not only the laws of mechanics but also those of the rest of physics, including electricity and magnetism:

RELATIVITY PRINCIPLE

> ***First postulate (the relativity principle)*: The laws of physics have the same form in all inertial reference frames.**

The first postulate can also be stated as: *there is no experiment you can do in an inertial reference frame to determine if you are at rest or moving uniformly at constant velocity.*

The second postulate is consistent with the first:

SPEED OF LIGHT PRINCIPLE

> ***Second postulate (constancy of the speed of light)*: Light propagates through empty space with a definite speed *c* independent of the speed of the source or observer.**

These two postulates form the foundation of Einstein's **special theory of relativity**. It is called "special" to distinguish it from his later "general theory of relativity," which deals with noninertial (accelerating) reference frames (Chapter 33). The special theory, which is what we discuss here, deals only with inertial frames.

The second postulate may seem hard to accept, for it seems to violate common sense. First of all, we have to think of light traveling through empty space. Giving up the ether is not too hard, however, since it had never been detected. But the second postulate also tells us that the speed of light in vacuum is always the same, 3.00×10^8 m/s, no matter what the speed of the observer or the source. Thus, a person traveling toward or away from a source of light will measure the same speed for that light as someone at rest with respect to the source. This conflicts with our everyday experience: we would expect to have to add in the velocity of the observer. On the other hand, perhaps we can't expect our everyday experience to be helpful when dealing with the high velocity of light. Furthermore, the null result of the Michelson–Morley experiment is fully consistent with the second postulate.[†]

Einstein's proposal has a certain beauty. By doing away with the idea of an absolute reference frame, it was possible to reconcile classical mechanics with Maxwell's electromagnetic theory. The speed of light predicted by Maxwell's equations *is* the speed of light in vacuum in *any* reference frame.

Einstein's theory required us to give up common sense notions of space and time, and in the following Sections we will examine some strange but interesting consequences of special relativity. Our arguments for the most part will be simple ones.

[†]The Michelson–Morley experiment can also be considered as evidence for the first postulate, since it was intended to measure the motion of the Earth relative to an absolute reference frame. Its failure to do so implies the absence of any such preferred frame.

We will use a technique that Einstein himself did: we will imagine very simple experimental situations in which little mathematics is needed. In this way, we can see many of the consequences of relativity theory without getting involved in detailed calculations. Einstein called these **thought experiments**.

26–3 Simultaneity

An important consequence of the theory of relativity is that we can no longer regard time as an absolute quantity. No one doubts that time flows onward and never turns back. But according to relativity, the time interval between two events, and even whether or not two events are simultaneous, depends on the observer's reference frame. By an **event**, which we use a lot here, we mean something that happens at a particular place and at a particular time.

Two events are said to occur simultaneously if they occur at exactly the same time. But how do we know if two events occur precisely at the same time? If they occur at the same point in space—such as two apples falling on your head at the same time—it is easy. But if the two events occur at widely separated places, it is more difficult to know whether the events are simultaneous since we have to take into account the time it takes for the light from them to reach us. Because light travels at finite speed, a person who sees two events must calculate back to find out when they actually occurred. For example, if two events are *observed* to occur at the same time, but one actually took place farther from the observer than the other, then the more distant one must have occurred earlier, and the two events were not simultaneous.

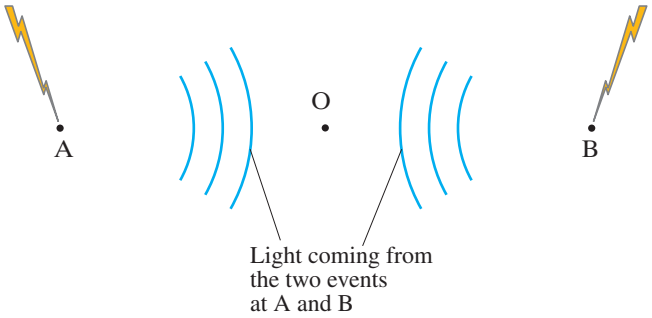
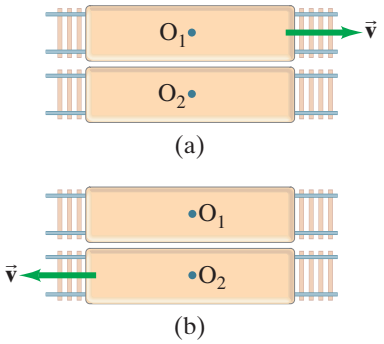


FIGURE 26–3 A moment after lightning strikes at points A and B, the pulses of light (shown as blue waves) are traveling toward the observer O, but O "sees" the lightning only when the light reaches O.

We now imagine a simple thought experiment. Assume an observer, called O, is located exactly halfway between points A and B where two events occur, Fig. 26–3. Suppose the two events are lightning that strikes the points A and B, as shown. For brief events like lightning, only short pulses of light (blue in Fig. 26–3) will travel outward from A and B and reach O. Observer O "sees" the events when the pulses of light reach point O. If the two pulses reach O at the same time, then the two events had to be simultaneous. This is because (i) the two light pulses travel at the same speed (postulate 2), and (ii) the distance OA equals OB, so the time for the light to travel from A to O and from B to O must be the same. Observer O can then definitely state that the two events occurred simultaneously. On the other hand, if O sees the light from one event before that from the other, then the former event occurred first.

The question we really want to examine is this: if two events are simultaneous to an observer in one reference frame, are they also simultaneous to another observer moving with respect to the first? Let us call the observers O_1 and O_2 and assume they are fixed in reference frames 1 and 2 that move with speed v relative to one another. These two reference frames can be thought of as two rockets or two trains (Fig. 26–4). O_2 says that O_1 is moving to the right with speed v, as in Fig. 26–4a; and O_1 says O_2 is moving to the left with speed v, as in Fig. 26–4b. Both viewpoints are legitimate according to the relativity principle. [There is no third point of view that will tell us which one is "really" moving.]

FIGURE 26–4 Observers O_1 and O_2, on two different trains (two different reference frames), are moving with relative speed v. (a) O_2 says that O_1 is moving to the right. (b) O_1 says that O_2 is moving to the left. Both viewpoints are legitimate: it all depends on your reference frame.



Now suppose that observers O_1 and O_2 observe and measure two lightning strikes. The lightning bolts mark both trains where they strike: at A_1 and B_1 on O_1's train, and at A_2 and B_2 on O_2's train, Fig. 26–5a. For simplicity, we assume that O_1 is exactly halfway between A_1 and B_1, and O_2 is halfway between A_2 and B_2. Let us first put ourselves in O_2's reference frame, so we observe O_1 moving to the right with speed v. Let us also assume that the two events occur *simultaneously* in O_2's frame, and just at the instant when O_1 and O_2 are opposite each other, Fig. 26–5a. A short time later, Fig. 26–5b, light from A_2 and from B_2 reach O_2 at the same time (we assumed this). Since O_2 knows (or measures) the distances O_2A_2 and O_2B_2 as equal, O_2 knows the two events are simultaneous in the O_2 reference frame.

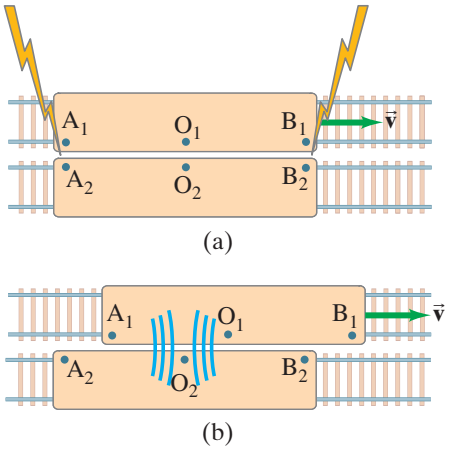


FIGURE 26–5 Thought experiment on simultaneity. In both (a) and (b) we are in the reference frame of observer O_2, who sees the reference frame of O_1 moving to the right. In (a), one lightning bolt strikes the two reference frames at A_1 and A_2, and a second lightning bolt strikes at B_1 and B_2. (b) A moment later, the light (shown in blue) from the two events reaches O_2 at the same time. So according to observer O_2, the two bolts of lightning struck simultaneously. But in O_1's reference frame, the light from B_1 has already reached O_1, whereas the light from A_1 has not yet reached O_1. So in O_1's reference frame, the event at B_1 must have preceded the event at A_1. Simultaneity in time is not absolute.

But what does observer O_1 observe and measure? From our (O_2) reference frame, we can predict what O_1 will observe. We see that O_1 moves to the right during the time the light is traveling to O_1 from A_1 and B_1. As shown in Fig. 26–5b, we can see from our O_2 reference frame that the light from B_1 has already passed O_1, whereas the light from A_1 has not yet reached O_1. That is, O_1 observes the light coming from B_1 before observing the light coming from A_1. Given (i) that light travels at the same speed c in any direction and in any reference frame, and (ii) that the distance O_1A_1 equals O_1B_1, then observer O_1 can only conclude that the event at B_1 occurred before the event at A_1. The two events are *not* simultaneous for O_1, even though they are for O_2.

We thus find that two events which take place at different locations and are simultaneous to one observer, are actually not simultaneous to a second observer who moves relative to the first.

It may be tempting to ask: "Which observer is right, O_1 or O_2?" The answer, according to relativity, is that they are *both* right. There is no "best" reference frame we can choose to determine which observer is right. Both frames are equally good. We can only conclude that *simultaneity is not an absolute concept*, but is relative. We are not aware of this lack of agreement on simultaneity in everyday life because the effect is noticeable only when the relative speed of the two reference frames is very large (near c), or the distances involved are very large.

26–4 Time Dilation and the Twin Paradox

The fact that two events simultaneous to one observer may not be simultaneous to a second observer suggests that time itself is not absolute. Could it be that time passes differently in one reference frame than in another? This is, indeed, just what Einstein's theory of relativity predicts, as the following thought experiment shows.

Figure 26–6 shows a spaceship traveling past Earth at high speed. The point of view of an observer on the spaceship is shown in part (a), and that of an observer on Earth in part (b). Both observers have accurate clocks. The person on the spaceship (Fig. 26–6a) flashes a light and measures the time it takes the light to travel directly across the spaceship and return after reflecting from a mirror (the rays are drawn at a slight angle for clarity). In the reference frame of the spaceship, the

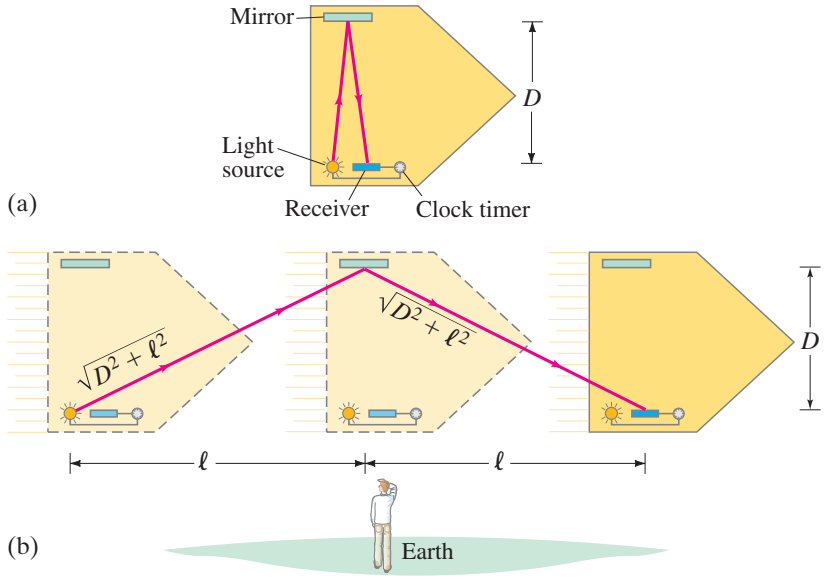


FIGURE 26–6 Time dilation can be shown by a thought experiment: the time it takes for light to travel across a spaceship and back is longer for the observer on Earth (b) than for the observer on the spaceship (a).

light travels a distance $2D$ at speed c, Fig. 26–6a; so the time required to go across and back, Δt_0, is

$$\Delta t_0 = \frac{2D}{c}.$$

The observer on Earth, Fig. 26–6b, observes the same process. But to this observer, the spaceship is moving. So the light travels the diagonal path shown going across the spaceship, reflecting off the mirror, and returning to the sender. Although the light travels at the same speed to this observer (the second postulate), it travels a greater distance. Hence the time required, as measured by the observer on Earth, will be *greater* than that measured by the observer on the spaceship.

Let us determine the time interval Δt measured by the observer on Earth between sending and receiving the light. In time Δt, the spaceship travels a distance $2\ell = v\,\Delta t$ where v is the speed of the spaceship (Fig. 26–6b). The light travels a total distance on its diagonal path (Pythagorean theorem) of $2\sqrt{D^2 + \ell^2} = c\,\Delta t$, where $\ell = v\,\Delta t/2$. Therefore

$$c\,\Delta t = 2\sqrt{D^2 + \ell^2} = 2\sqrt{D^2 + v^2(\Delta t)^2/4}.$$

We square both sides to find $c^2(\Delta t)^2 = 4D^2 + v^2(\Delta t)^2$, and solve for $(\Delta t)^2$:

$$(\Delta t)^2 = 4D^2/(c^2 - v^2)$$

so

$$\Delta t = \frac{2D}{c\sqrt{1 - v^2/c^2}}.$$

We combine this equation for Δt with the formula for Δt_0 above, $\Delta t_0 = 2D/c$:

$$\Delta t = \frac{\Delta t_0}{\sqrt{1 - v^2/c^2}}. \qquad \textbf{(26–1a)}$$

TIME DILATION

Since $\sqrt{1 - v^2/c^2}$ is always less than 1, we see that $\Delta t > \Delta t_0$. That is, the time interval between the two events (the sending of the light, and its reception on the spaceship) is *greater* for the observer on Earth than for the observer on the spaceship. This is a general result of the theory of relativity, and is known as **time dilation**. The time dilation effect can be stated as

> **clocks moving relative to an observer are measured to run more slowly, as compared to clocks at rest.**

However, we should not think that the clocks are somehow at fault.

> **Time is actually measured to pass more slowly in any moving reference frame as compared to your own.**

This remarkable result is an inevitable outcome of the two postulates of the special theory of relativity.

The factor $1/\sqrt{1 - v^2/c^2}$ occurs so often in relativity that we often give it the shorthand symbol γ (the Greek letter "gamma"), and write Eq. 26–1a as

$$\Delta t = \gamma \, \Delta t_0 \qquad \textbf{(26–1b)}$$

where

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}. \qquad \textbf{(26–2)}$$

Note that γ is never less than one, and has no units. At normal speeds, $\gamma = 1$ to many decimal places. In general, $\gamma \geq 1$.

TABLE 26–1 Values of γ

v	γ
0	1.00000 …
$0.01c$	1.00005
$0.10c$	1.005
$0.50c$	1.15
$0.90c$	2.3
$0.99c$	7.1

Values for $\gamma = 1/\sqrt{1 - v^2/c^2}$ at a few speeds v are given in Table 26–1. γ is never less than 1.00 and exceeds 1.00 significantly only at very high speeds, much above let's say 10^6 m/s (for which $\gamma = 1.000006$).

The concept of time dilation may be hard to accept, for it contradicts our experience. We can see from Eq. 26–1 that the time dilation effect is indeed negligible unless v is reasonably close to c. If v is much less than c, then the term v^2/c^2 is much smaller than the 1 in the denominator of Eq. 26–1, and then $\Delta t \approx \Delta t_0$ (see Example 26–2). The speeds we experience in everyday life are much smaller than c, so it is little wonder we don't ordinarily notice time dilation. But experiments that have tested the time dilation effect have confirmed Einstein's predictions. In 1971, for example, extremely precise atomic clocks were flown around the Earth in jet planes. The speed of the planes (10^3 km/h) was much less than c, so the clocks had to be accurate to nanoseconds (10^{-9} s) in order to detect any time dilation. They were this accurate, and they confirmed Eqs. 26–1 to within experimental error. Time dilation had been confirmed decades earlier, however, by observations on "elementary particles" which have very small masses (typically 10^{-30} to 10^{-27} kg) and so require little energy to be accelerated to speeds close to the speed of light, c. Many of these elementary particles are not stable and decay after a time into lighter particles. One example is the muon, whose mean lifetime is 2.2 μs when at rest. Careful experiments showed that when a muon is traveling at high speeds, its lifetime is measured to be longer than when it is at rest, just as predicted by the time dilation formula.

EXAMPLE 26–1 Lifetime of a moving muon. (a) What will be the mean lifetime of a muon as measured in the laboratory if it is traveling at $v = 0.60c = 1.80 \times 10^8$ m/s with respect to the laboratory? A muon's mean lifetime at rest is $2.20\,\mu\text{s} = 2.20 \times 10^{-6}$ s. (b) How far does a muon travel in the laboratory, on average, before decaying?

APPROACH If an observer were to move along with the muon (the muon would be at rest to this observer), the muon would have a mean life of 2.20×10^{-6} s. To an observer in the lab, the muon lives longer because of time dilation. We find the mean lifetime using Eq. 26–1 and the average distance using $d = v \, \Delta t$.

SOLUTION (a) From Eq. 26–1 with $v = 0.60c$, we have

$$\Delta t = \frac{\Delta t_0}{\sqrt{1 - v^2/c^2}}$$

$$= \frac{2.20 \times 10^{-6} \text{ s}}{\sqrt{1 - 0.36c^2/c^2}} = \frac{2.20 \times 10^{-6} \text{ s}}{\sqrt{0.64}} = 2.8 \times 10^{-6} \text{ s}.$$

(b) Relativity predicts that a muon with speed 1.80×10^8 m/s would travel an average distance $d = v \, \Delta t = (1.80 \times 10^8 \text{ m/s})(2.8 \times 10^{-6} \text{ s}) = 500$ m, and this is the distance that is measured experimentally in the laboratory.

NOTE At a speed of 1.8×10^8 m/s, classical physics would tell us that with a mean life of 2.2 μs, an average muon would travel $d = vt = (1.8 \times 10^8 \text{ m/s})(2.2 \times 10^{-6} \text{ s}) = 400$ m. This is shorter than the distance measured.

EXERCISE A What is the muon's mean lifetime (Example 26–1) if it is traveling at $v = 0.90c$? (a) 0.42 μs; (b) 2.3 μs; (c) 5.0 μs; (d) 5.3 μs; (e) 12.0 μs.

We need to clarify how to use Eq. 26–1, $\Delta t = \gamma \Delta t_0$, and the meaning of Δt and Δt_0. The equation is true only when Δt_0 represents the time interval between the two events *in a reference frame where an observer at rest sees the two events occur at the same point in space* (as in Fig. 26–6a where the two events are the light flash being sent and being received). This time interval, Δt_0, is called the **proper time**. Then Δt in Eqs. 26–1 represents the time interval between the two events as measured in a reference frame *moving* with speed v with respect to the first. In Example 26–1 above, Δt_0 (and not Δt) was set equal to 2.2×10^{-6} s because it is only in the rest frame of the muon that the two events ("birth" and "decay") occur at the same point in space. The proper time Δt_0 is the shortest time between the events any observer can measure. In any other moving reference frame, the time Δt is greater.

CAUTION
Proper time Δt_0 is for 2 events at the same point *in space*

CAUTION
Proper time is shortest: $\Delta t > \Delta t_0$

EXAMPLE 26–2 **Time dilation at 100 km/h.** Let us check time dilation for everyday speeds. A car traveling 100 km/h covers a certain distance in 10.00 s according to the driver's watch. What does an observer at rest on Earth measure for the time interval?

APPROACH The car's speed relative to Earth, written in meters per second, is $100 \text{ km/h} = (1.00 \times 10^5 \text{ m})/(3600 \text{ s}) = 27.8 \text{ m/s}$. The driver is at rest in the reference frame of the car, so we set $\Delta t_0 = 10.00$ s in the time dilation formula.

SOLUTION We use Eq. 26–1a:

$$\Delta t = \frac{\Delta t_0}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{10.00 \text{ s}}{\sqrt{1 - \left(\frac{27.8 \text{ m/s}}{3.00 \times 10^8 \text{ m/s}}\right)^2}}$$

$$= \frac{10.00 \text{ s}}{\sqrt{1 - (8.59 \times 10^{-15})}}.$$

If you put these numbers into a calculator, you will obtain $\Delta t = 10.00$ s, because the denominator differs from 1 by such a tiny amount. The time measured by an observer fixed on Earth would show no difference from that measured by the driver, even with the best instruments. A computer that could calculate to a large number of decimal places would reveal a slight difference between Δt and Δt_0.

NOTE We can estimate the difference using the binomial expansion (Appendix A–5),

$$(1 \pm x)^n \approx 1 \pm nx. \qquad [\text{for } x \ll 1]$$

PROBLEM SOLVING
Use of the binomial expansion

In our time dilation formula, we have the factor $\gamma = (1 - v^2/c^2)^{-\frac{1}{2}}$. Thus[†]

$$\Delta t = \gamma \Delta t_0 = \Delta t_0 \left(1 - \frac{v^2}{c^2}\right)^{-\frac{1}{2}} \approx \Delta t_0 \left(1 + \frac{1}{2}\frac{v^2}{c^2}\right)$$

$$\approx 10.00 \text{ s} \left[1 + \frac{1}{2}\left(\frac{27.8 \text{ m/s}}{3.00 \times 10^8 \text{ m/s}}\right)^2\right]$$

$$\approx 10.00 \text{ s} + 4 \times 10^{-14} \text{ s}.$$

So the difference between Δt and Δt_0 is predicted to be 4×10^{-14} s, an extremely small amount.

EXERCISE B A certain atomic clock keeps precise time on Earth. If the clock is taken on a spaceship traveling at a speed $v = 0.60c$, does this clock now run slow according to the people (*a*) on the spaceship, (*b*) on Earth?

[†]Recall that $1/x^n$ is written as x^{-n}, such as $1/x^2 = x^{-2}$, Appendix A–2.

EXAMPLE 26–3 **Reading a magazine on a spaceship.** A passenger on a fictional high-speed spaceship traveling between Earth and Jupiter at a steady speed of $0.75c$ reads a magazine which takes 10.0 min according to her watch. (*a*) How long does this take as measured by Earth-based clocks? (*b*) How much farther is the spaceship from Earth at the end of reading the article than it was at the beginning?

APPROACH (*a*) The time interval in one reference frame is related to the time interval in the other by Eq. 26–1a or b. (*b*) At constant speed, distance is speed × time. Because there are two time intervals (Δt and Δt_0) we will get two distances, one for each reference frame. [This surprising result is explored in the next Section (26–5).]

SOLUTION (*a*) The given 10.0-min time interval is the proper time Δt_0—starting and finishing the magazine happen at the same place on the spaceship. Earth clocks measure

$$\Delta t = \frac{\Delta t_0}{\sqrt{1 - (v^2/c^2)}} = \frac{10.00 \text{ min}}{\sqrt{1 - (0.75)^2}} = 15.1 \text{ min}.$$

(*b*) In the Earth frame, the rocket travels a distance $D = v\,\Delta t = (0.75c)(15.1 \text{ min}) = (0.75)(3.0 \times 10^8 \text{ m/s})(15.1 \text{ min} \times 60 \text{ s/min}) = 2.04 \times 10^{11} \text{ m}$. In the spaceship's frame, the Earth is moving away from the spaceship at $0.75c$, but the time is only 10.0 min, so the distance is measured to be $D_0 = v\,\Delta t_0 = (2.25 \times 10^8 \text{ m/s})(600 \text{ s}) = 1.35 \times 10^{11} \text{ m}$.

Space Travel?

Time dilation has aroused interesting speculation about space travel. According to classical (Newtonian) physics, to reach a star 100 light-years away would not be possible for ordinary mortals (1 light-year is the distance light can travel in 1 year $= 3.0 \times 10^8 \text{ m/s} \times 3.16 \times 10^7 \text{ s} = 9.5 \times 10^{15} \text{ m}$). Even if a spaceship could travel at close to the speed of light, it would take over 100 years to reach such a star. But time dilation tells us that the time involved could be less. In a spaceship traveling at $v = 0.999c$, the time for such a trip would be only about $\Delta t_0 = \Delta t \sqrt{1 - v^2/c^2} = (100 \text{ yr})\sqrt{1 - (0.999)^2} = 4.5 \text{ yr}$. Thus time dilation allows such a trip, but the enormous practical problems of achieving such speeds may not be possible to overcome, certainly not in the near future.

When we talk in this Chapter and in the Problems about spaceships moving at speeds close to c, it is for understanding and for fun, but not realistic, although for tiny elementary particles such high speeds *are* realistic.

In this example, 100 years would pass on Earth, whereas only 4.5 years would pass for the astronaut on the trip. Is it just the clocks that would slow down for the astronaut? No.

> *All processes, including aging and other life processes, run more slowly for the astronaut as measured by an Earth observer. But to the astronaut, time would pass in a normal way.*

The astronaut would experience 4.5 years of normal sleeping, eating, reading, and so on. And people on Earth would experience 100 years of ordinary activity.

Twin Paradox

Not long after Einstein proposed the special theory of relativity, an apparent paradox was pointed out. According to this **twin paradox**, suppose one of a pair of 20-year-old twins takes off in a spaceship traveling at very high speed to a distant star and back again, while the other twin remains on Earth. According to the Earth twin, the astronaut twin will age less. Whereas 20 years might pass for the Earth twin, perhaps only 1 year (depending on the spacecraft's speed) would pass for the traveler. Thus, when the traveler returns, the earthbound twin could expect to be 40 years old whereas the traveling twin would be only 21.

This is the viewpoint of the twin on the Earth. But what about the traveling twin? If all inertial reference frames are equally good, won't the traveling twin make all the claims the Earth twin does, only in reverse? Can't the astronaut twin claim that since the Earth is moving away at high speed, time passes more slowly on Earth and the twin on Earth will age less? This is the opposite of what the Earth twin predicts. They cannot both be right, for after all the spacecraft returns to Earth and a direct comparison of ages and clocks can be made.

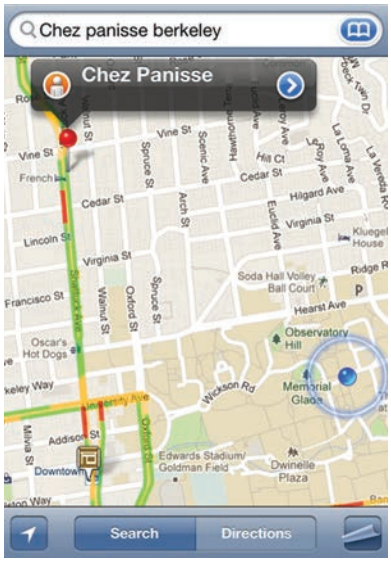
There is, however, no contradiction here. One of the viewpoints is indeed incorrect. The consequences of the special theory of relativity—in this case, time dilation—can be applied only by observers in an inertial reference frame. The Earth is such a frame (or nearly so), whereas the spacecraft is not. The spacecraft accelerates at the start and end of its trip and when it turns around at the far point of its journey. Part of the time, the astronaut twin may be in an inertial frame (and is justified in saying the Earth twin's clocks run slow). But during the accelerations, the twin on the spacecraft is not in an inertial frame. So she cannot use special relativity to predict their relative ages when she returns to Earth. The Earth twin stays in the same inertial frame, and we can thus trust her predictions based on special relativity. Thus, there is no paradox. The prediction of the Earth twin that the traveling twin ages less is the correct one.

*Global Positioning System (GPS)

PHYSICS APPLIED
Global positioning system (GPS)

Airplanes, cars, boats, and hikers use **global positioning system** (**GPS**) receivers to tell them quite accurately where they are at a given moment (Fig. 26–7). There are more than 30 global positioning system satellites that send out precise time signals using atomic clocks. Your receiver compares the times received from at least four satellites, all of whose times are carefully synchronized to within 1 part in 10^{13}. By comparing the time differences with the known satellite positions and the fixed speed of light, the receiver can determine how far it is from each satellite and thus where it is on the Earth. It can do this to an accuracy of a few meters, if it has been constructed to make corrections such as the one below due to relativity.

FIGURE 26–7 A visiting professor of physics uses the GPS on her smart phone to find a restaurant (red dot). Her location in the physics department is the blue dot. Traffic on some streets is also shown (green = good, orange = slow, red = heavy traffic) which comes in part by tracking cell phone movements.



CONCEPTUAL EXAMPLE 26–4 **A relativity correction to GPS.** GPS satellites move at about 4 km/s = 4000 m/s. Show that a good GPS receiver needs to correct for time dilation if it is to produce results consistent with atomic clocks accurate to 1 part in 10^{13}.

RESPONSE Let us calculate the magnitude of the time dilation effect by inserting $v = 4000\ \text{m/s}$ into Eq. 26–1a:

$$\Delta t = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}\Delta t_0 = \frac{1}{\sqrt{1 - \left(\frac{4 \times 10^3\ \text{m/s}}{3 \times 10^8\ \text{m/s}}\right)^2}}\Delta t_0$$
$$= \frac{1}{\sqrt{1 - 1.8 \times 10^{-10}}}\Delta t_0 .$$

We use the binomial expansion: $(1 \pm x)^n \approx 1 \pm nx$ for $x \ll 1$ (see Appendix A–5) which here is $(1 - x)^{-\frac{1}{2}} \approx 1 + \frac{1}{2}x$. That is

$$\Delta t = \left(1 + \tfrac{1}{2}(1.8 \times 10^{-10})\right)\Delta t_0 = \left(1 + 9 \times 10^{-11}\right)\Delta t_0 .$$

The time "error" divided by the time interval is

$$\frac{(\Delta t - \Delta t_0)}{\Delta t_0} = 1 + 9 \times 10^{-11} - 1 = 9 \times 10^{-11} \approx 1 \times 10^{-10}.$$

Time dilation, if not accounted for, would introduce an error of about 1 part in 10^{10}, which is 1000 times greater than the precision of the atomic clocks. Not correcting for time dilation means a receiver could give much poorer position accuracy.

NOTE GPS devices must make other corrections as well, including effects associated with general relativity.

26–5 Length Contraction

Time intervals are not the only things different in different reference frames. Space intervals—lengths and distances—are different as well, according to the special theory of relativity, and we illustrate this with a thought experiment.

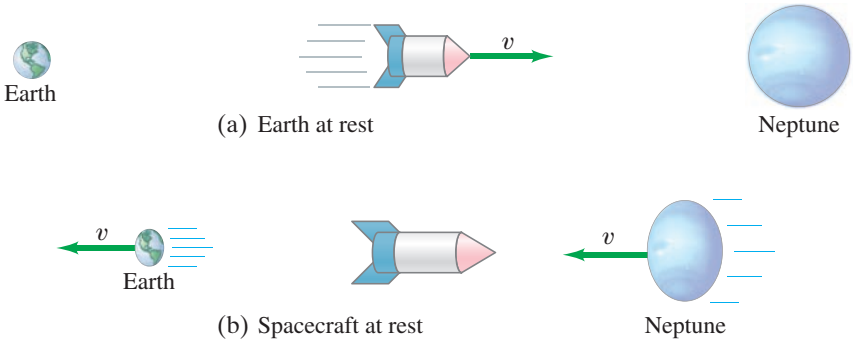


FIGURE 26–8 (a) A spaceship traveling at very high speed v from Earth to the planet Neptune, as seen from Earth's frame of reference. (b) According to an observer on the spaceship, Earth and Neptune are moving at the very high speed v: Earth leaves the spaceship, and a time Δt_0 later Neptune arrives at the spaceship.

Observers on Earth watch a spacecraft traveling at speed v from Earth to, say, Neptune, Fig. 26–8a. The distance between the planets, as measured by the Earth observers, is ℓ_0. The time required for the trip, measured from Earth, is

$$\Delta t = \frac{\ell_0}{v}. \qquad \text{[Earth observer]}$$

In Fig. 26–8b we see the point of view of observers on the spacecraft. In this frame of reference, the spaceship is at rest; Earth and Neptune move[†] with speed v. The time between departure of Earth and arrival of Neptune, as observed from the spacecraft, is the "proper time" Δt_0 (page 753), because these two events occur at the same point in space (i.e., at the spacecraft). Therefore the time interval is less for the spacecraft observers than for the Earth observers. That is, because of time dilation (Eq. 26–1a), the time for the trip as viewed by the spacecraft is

$$\begin{aligned} \Delta t_0 &= \Delta t\sqrt{1 - v^2/c^2} \\ &= \Delta t/\gamma. \end{aligned} \qquad \text{[spacecraft observer]}$$

Because the spacecraft observers measure the same speed but less time between these two events, they also measure the distance as less. If we let ℓ be the distance between the planets as viewed by the spacecraft observers, then $\ell = v\,\Delta t_0$, which we can rewrite as $\ell = v\,\Delta t_0 = v\,\Delta t\sqrt{1 - v^2/c^2} = \ell_0\sqrt{1 - v^2/c^2}$. Thus we have the important result that

LENGTH CONTRACTION

$$\ell = \ell_0\sqrt{1 - v^2/c^2} \qquad \textbf{(26–3a)}$$

or, using γ (Eq. 26–2),

$$\ell = \frac{\ell_0}{\gamma}. \qquad \textbf{(26–3b)}$$

This is a general result of the special theory of relativity and applies to lengths of objects as well as to distance between objects. The result can be stated most simply in words as:

the length of an object moving relative to an observer is measured to be shorter along its direction of motion than when it is at rest.

CAUTION

Proper length is measured in reference frame where the two positions are at rest

This is called **length contraction**. The length ℓ_0 in Eqs. 26–3 is called the **proper length**. It is the length of the object (or distance between two points whose positions are measured at the same time) as determined by *observers at rest* with respect to the object. Equations 26–3 give the length ℓ that will be measured by observers when the object travels past them at speed v.

[†]We assume v is much greater than the relative speed of Neptune and Earth (which we thus ignore).

It is important to note that length contraction occurs *only along the direction of motion*. For example, the moving spaceship in Fig. 26–8a is shortened in length, but its height is the same as when it is at rest.

Length contraction, like time dilation, is not noticeable in everyday life because the factor $\sqrt{1 - v^2/c^2}$ in Eq. 26–3a differs significantly from 1.00 only when v is very large.

EXAMPLE 26–5 **Painting's contraction.** A rectangular painting measures 1.00 m tall and 1.50 m wide, Fig. 26–9a. It is hung on the side wall of a spaceship which is moving past the Earth at a speed of $0.90c$. (*a*) What are the dimensions of the picture according to the captain of the spaceship? (*b*) What are the dimensions as seen by an observer on the Earth?

APPROACH We apply the length contraction formula, Eq. 26–3a, to the dimension parallel to the motion; v is the speed of the painting relative to the Earth observer.

SOLUTION (*a*) The painting is at rest $(v = 0)$ on the spaceship so it (as well as everything else in the spaceship) looks perfectly normal to everyone on the spaceship. The captain sees a 1.00-m by 1.50-m painting.

(*b*) Only the dimension in the direction of motion is shortened, so the height is unchanged at 1.00 m, Fig. 26–9b. The length, however, is contracted to

$$\ell = \ell_0\sqrt{1 - \frac{v^2}{c^2}}$$
$$= (1.50\text{ m})\sqrt{1 - (0.90)^2} = 0.65\text{ m}.$$

So the picture has dimensions $1.00\text{ m} \times 0.65\text{ m}$ to an observer on Earth.

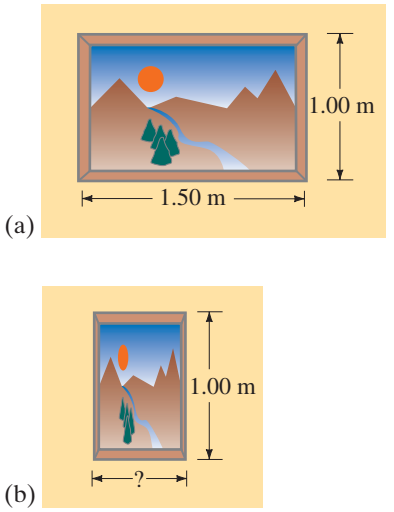


FIGURE 26–9 Example 26–5.

EXAMPLE 26–6 **A fantasy supertrain.** A very fast train with a "proper length" of $\ell_0 = 500\text{ m}$ (measured by people at rest on the train) is passing through a tunnel that is 200 m long according to observers on the ground. Let us imagine the train's speed to be so great that the train fits completely within the tunnel as seen by observers on the ground. That is, the engine is just about to emerge from one end of the tunnel at the time the last car disappears into the other end. What is the train's speed?

APPROACH Since the train just fits inside the tunnel, its length measured by the person on the ground is $\ell = 200\text{ m}$. The length contraction formula, Eq. 26–3a or b, can thus be used to solve for v.

SOLUTION Substituting $\ell = 200\text{ m}$ and $\ell_0 = 500\text{ m}$ into Eq. 26–3a gives

$$200\text{ m} = 500\text{ m}\sqrt{1 - \frac{v^2}{c^2}};$$

dividing both sides by 500 m and squaring, we get

$$(0.40)^2 = 1 - \frac{v^2}{c^2}$$

or

$$\frac{v}{c} = \sqrt{1 - (0.40)^2}$$

and

$$v = 0.92c.$$

NOTE No real train could go this fast. But it is fun to think about.

NOTE An observer on the *train* would *not* see the two ends of the train inside the tunnel at the same time. Recall that observers moving relative to each other do not agree about simultaneity. (See Example 26–7, next.)

EXERCISE C What is the length of the tunnel as measured by observers on the train in Example 26–6?

CONCEPTUAL EXAMPLE 26–7 **Resolving the train and tunnel length.** Observers at rest on the Earth see a very fast 200-m-long train pass through a 200-m-long tunnel (as in Example 26–6) so that the train momentarily disappears from view inside the tunnel. Observers on the train measure the train's length to be 500 m and the tunnel's length to be only 80 m (Exercise C, using Eq. 26–3a). Clearly a 500-m-long train cannot fit inside an 80-m-long tunnel. How is this apparent inconsistency explained?

RESPONSE Events simultaneous in one reference frame may not be simultaneous in another. Let the engine emerging from one end of the tunnel be "event A," and the last car disappearing into the other end of the tunnel "event B." To observers in the Earth frame, events A and B are simultaneous. To observers on the train, however, the events are not simultaneous. In the train's frame, event A occurs before event B. As the engine emerges from the tunnel, observers on the train observe the last car as still $500\text{ m} - 80\text{ m} = 420\text{ m}$ from the entrance to the tunnel.

26–6 Four-Dimensional Space–Time

Let us imagine a person is on a train moving at a very high speed, say $0.65c$, Fig. 26–10. This person begins a meal at 7:00 and finishes at 7:15, according to a clock on the train. The two events, beginning and ending the meal, take place at the same point on the train, so the "proper time" between these two events is 15 min. To observers on Earth, the plate is moving and the meal will take longer—20 min according to Eqs. 26–1. Let us assume that the meal was served on a 20-cm-diameter plate (its "proper length"). To observers on the Earth, the plate is moving and is only 15 cm wide (length contraction). Thus, to observers on the Earth, the meal looks smaller but lasts longer.

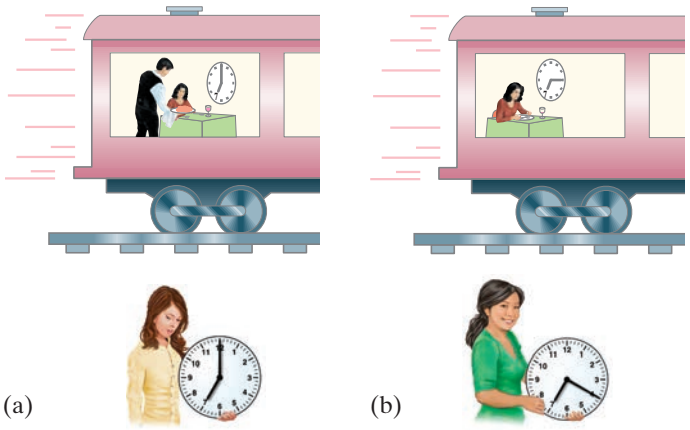


FIGURE 26–10 According to an accurate clock on a fast-moving train, a person (a) begins dinner at 7:00 and (b) finishes at 7:15. At the beginning of the meal, two observers on Earth set their watches to correspond with the clock on the train. These observers measure the eating time as 20 minutes.

In a sense the two effects, time dilation and length contraction, balance each other. When viewed from the Earth, what an object seems to lose in size it gains in length of time it lasts. Space, or length, is exchanged for time.

Considerations like this led to the idea of **four-dimensional space–time**: space takes up three dimensions and time is a fourth dimension. Space and time are intimately connected. Just as when we squeeze a balloon we make one dimension larger and another smaller, so when we examine objects and events from different reference frames, a certain amount of space is exchanged for time, or vice versa.

Although the idea of four dimensions may seem strange, it refers to the idea that any object or event is specified by four quantities—three to describe where in space, and one to describe when in time. The really unusual aspect of four-dimensional space–time is that space and time can intermix: a little of one can be exchanged for a little of the other when the reference frame is changed.

[In Galilean–Newtonian relativity, the time interval between two events, Δt, and the distance between two events or points, Δx, are invariant quantities no matter what inertial reference frame they are viewed from. Neither of these quantities is invariant according to Einstein's relativity. But there is an invariant quantity in four-dimensional space–time, called the **space–time interval**, which is $(\Delta s)^2 = (c\,\Delta t)^2 - (\Delta x)^2$.]

26–7 Relativistic Momentum

So far in this Chapter, we have seen that two basic mechanical quantities, length and time intervals, need modification because they are relative—their value depends on the reference frame from which they are measured. We might expect that other physical quantities might need some modification according to the theory of relativity, such as momentum and energy.

The analysis of collisions between two particles shows that if we want to preserve the law of conservation of momentum in relativity, we must redefine momentum as

$$p = \frac{mv}{\sqrt{1 - v^2/c^2}} = \gamma mv. \qquad \textbf{(26–4)}$$

Here γ is shorthand for $1/\sqrt{1 - v^2/c^2}$ as before (Eq. 26–2). For speeds much less than the speed of light, Eq. 26–4 gives the classical momentum, $p = mv$.

Relativistic momentum has been tested many times on tiny elementary particles (such as muons), and it has been found to behave in accord with Eq. 26–4.

EXAMPLE 26–8 **Momentum of moving electron.** Compare the momentum of an electron to its classical value when it has a speed of (*a*) 4.00×10^7 m/s in the CRT of an old TV set, and (*b*) $0.98c$ in an accelerator used for cancer therapy.

APPROACH We use Eq. 26–4 for the momentum of a moving electron.

SOLUTION (*a*) At $v = 4.00 \times 10^7$ m/s, the electron's momentum is

$$p = \frac{mv}{\sqrt{1 - \dfrac{v^2}{c^2}}} = \frac{mv}{\sqrt{1 - \dfrac{(4.00 \times 10^7\ \text{m/s})^2}{(3.00 \times 10^8\ \text{m/s})^2}}} = 1.01mv.$$

The factor $\gamma = 1/\sqrt{1 - v^2/c^2} \approx 1.01$, so the momentum is only about 1% greater than the classical value. (If we put in the mass of an electron, $m = 9.11 \times 10^{-31}$ kg, the momentum is $p = 1.01mv = 3.68 \times 10^{-23}$ kg·m/s, compared to 3.64×10^{-23} kg·m/s classically.)

(*b*) With $v = 0.98c$, the momentum is

$$p = \frac{mv}{\sqrt{1 - \dfrac{v^2}{c^2}}} = \frac{mv}{\sqrt{1 - \dfrac{(0.98c)^2}{c^2}}} = \frac{mv}{\sqrt{1 - (0.98)^2}} = 5.0mv.$$

An electron traveling at 98% the speed of light has $\gamma = 5.0$ and a momentum 5.0 times its classical value.

*Rest Mass and Relativistic Mass

CAUTION
Most physicists prefer to consider the mass of a particle as fixed

The relativistic definition of momentum, Eq. 26–4, has sometimes been interpreted as an increase in the mass of an object. In this interpretation, a particle can have a **relativistic mass**, m_{rel}, which increases with speed according to

$$m_{\text{rel}} = \frac{m}{\sqrt{1 - v^2/c^2}}.$$

In this "mass-increase" formula, m is referred to as the **rest mass** of the object. With this interpretation, *the mass of an object appears to increase as its speed increases*. But there are problems with relativistic mass. If we plug it into formulas like $F = ma$ or $\text{KE} = \frac{1}{2}mv^2$, we obtain formulas that do not agree with experiment. (If we write Newton's second law in its more general form, $\vec{\mathbf{F}} = \Delta\vec{\mathbf{p}}/\Delta t$, that would get a correct result.) Also, be careful *not* to think a mass acquires more particles or more molecules as its speed becomes very large. It doesn't. Today, most physicists prefer not to use relativistic mass, so an object has only one mass (its rest mass), and it is only the momentum that increases with speed.

Whenever we talk about the mass of an object, we will always mean its rest mass (a fixed value). [But see Problem 39.]

26–8 The Ultimate Speed

A basic result of the special theory of relativity is that the speed of an object cannot equal or exceed the speed of light. That the speed of light is a natural speed limit in the universe can be seen from any of Eqs. 26–1, 26–3, or 26–4. It is perhaps easiest to see from Eq. 26–4. As an object is accelerated to greater and greater speeds, its momentum becomes larger and larger. Indeed, if v were to equal c, the denominator in this equation would be zero, and the momentum would be infinite. To accelerate an object up to $v = c$ would thus require infinite energy, and so is not possible.

26–9 $E = mc^2$; Mass and Energy

If momentum needs to be modified to fit with relativity as we just saw in Eq. 26–4, then we might expect that energy would also need to be rethought. Indeed, Einstein not only developed a new formula for kinetic energy, but also found a new relation between mass and energy, and the startling idea that mass is a form of energy.

We start with the work-energy principle (Chapter 6), hoping it is still valid in relativity and will give verifiable results. That is, we assume the net work done on a particle is equal to its change in kinetic energy (KE). Using this principle, Einstein showed that at high speeds the formula $\text{KE} = \frac{1}{2}mv^2$ is not correct. Instead, Einstein showed that the kinetic energy of a particle of mass m traveling at speed v is given by

$$\text{KE} = \frac{mc^2}{\sqrt{1 - v^2/c^2}} - mc^2. \qquad \textbf{(26–5a)}$$

In terms of $\gamma = 1/\sqrt{1 - v^2/c^2}$ we can rewrite Eq. 26–5a as

$$\text{KE} = \gamma mc^2 - mc^2 = (\gamma - 1)mc^2. \qquad \textbf{(26–5b)}$$

Equation 26–5a requires some interpretation. The first term increases with the speed v of the particle. The second term, mc^2, is constant; it is called the **rest energy** of the particle, and represents a form of energy that a particle has even when at rest. Note that if a particle is at rest ($v = 0$) the first term in Eq. 26–5a becomes mc^2, so $\text{KE} = 0$ as it should.

We can rearrange Eq. 26–5b to get

$$\gamma mc^2 = mc^2 + \text{KE}.$$

We call γmc^2 the *total energy* E of the particle (assuming no potential energy), because it equals the rest energy plus the kinetic energy:

$$E = \text{KE} + mc^2. \tag{26–6a}$$

The total energy[†] can also be written, using Eqs. 26–5, as

$$E = \gamma mc^2 = \frac{mc^2}{\sqrt{1 - v^2/c^2}}. \tag{26–6b}$$

For a particle at rest in a given reference frame, KE is zero in Eq. 26–6a, so the total energy is its rest energy:

$$E = mc^2. \tag{26–7}$$

MASS RELATED TO ENERGY

Here we have Einstein's famous formula, $E = mc^2$. This formula mathematically relates the concepts of energy and mass. But if this idea is to have any physical meaning, then mass ought to be convertible to other forms of energy and vice versa. Einstein suggested that this might be possible, and indeed changes of mass to other forms of energy, and vice versa, have been experimentally confirmed countless times in nuclear and elementary particle physics. For example, an electron and a positron (= a positive electron, see Section 32–3) have often been observed to collide and disappear, producing pure electromagnetic radiation. The amount of electromagnetic energy produced is found to be exactly equal to that predicted by Einstein's formula, $E = mc^2$. The reverse process is also commonly observed in the laboratory: electromagnetic radiation under certain conditions can be converted into material particles such as electrons (see Section 27–6 on pair production). On a larger scale, the energy produced in nuclear power plants is a result of the loss in mass of the uranium fuel as it undergoes the process called fission (Chapter 31). Even the radiant energy we receive from the Sun is an example of $E = mc^2$; the Sun's mass is continually decreasing as it radiates electromagnetic energy outward.

The relation $E = mc^2$ is now believed to apply to all processes, although the changes are often too small to measure. That is, when the energy of a system changes by an amount ΔE, the mass of the system changes by an amount Δm given by

$$\Delta E = (\Delta m)(c^2). \tag{26–8}$$

In a nuclear reaction where an energy E is required or released, the masses of the reactants and the products will be different by $\Delta m = \Delta E/c^2$.

EXAMPLE 26–9 **Pion's kinetic energy.** A π^0 meson $(m = 2.4 \times 10^{-28}\,\text{kg})$ travels at a speed $v = 0.80c = 2.4 \times 10^8\,\text{m/s}$. What is its kinetic energy? Compare to a classical calculation.

APPROACH We use Eq. 26–5 and compare to $\frac{1}{2}mv^2$.

SOLUTION We substitute values into Eq. 26–5a

$$\begin{aligned}\text{KE} &= mc^2\left(\frac{1}{\sqrt{1 - v^2/c^2}} - 1\right) \\ &= (2.4 \times 10^{-28}\,\text{kg})(3.0 \times 10^8\,\text{m/s})^2\left(\frac{1}{(1 - 0.64)^{\frac{1}{2}}} - 1\right) \\ &= 1.4 \times 10^{-11}\,\text{J}.\end{aligned}$$

PROBLEM SOLVING
Relativistic kinetic energy

Notice that the units of mc^2 are $\text{kg}\cdot\text{m}^2/\text{s}^2$, which is the joule.

NOTE Classically $\text{KE} = \frac{1}{2}mv^2 = \frac{1}{2}(2.4 \times 10^{-28}\,\text{kg})(2.4 \times 10^8\,\text{m/s})^2 = 6.9 \times 10^{-12}\,\text{J}$, about half as much, but this is not a correct result. Note that $\frac{1}{2}\gamma mv^2$ also does not work.

[†]This is for a "free particle," without forces and potential energy. Potential energy terms can be added.

EXAMPLE 26–10 **Energy from nuclear decay.** The energy required or released in nuclear reactions and decays comes from a change in mass between the initial and final particles. In one type of radioactive decay (Chapter 30), an atom of uranium $(m = 232.03716\,\text{u})$ decays to an atom of thorium $(m = 228.02874\,\text{u})$ plus an atom of helium $(m = 4.00260\,\text{u})$ where the masses given are in atomic mass units $(1\,\text{u} = 1.6605 \times 10^{-27}\,\text{kg})$. Calculate the energy released in this decay.

APPROACH The initial mass minus the total final mass gives the mass loss in atomic mass units (u); we convert that to kg, and multiply by c^2 to find the energy released, $\Delta E = \Delta m\, c^2$.

SOLUTION The initial mass is 232.03716 u, and after the decay the mass is 228.02874 u + 4.00260 u = 232.03134 u, so there is a loss of mass of 0.00582 u. This mass, which equals $(0.00582\,\text{u})(1.66 \times 10^{-27}\,\text{kg}) = 9.66 \times 10^{-30}\,\text{kg}$, is changed into energy. By $\Delta E = \Delta m\, c^2$, we have

$$\begin{aligned}\Delta E &= (9.66 \times 10^{-30}\,\text{kg})(3.0 \times 10^{8}\,\text{m/s})^2 \\ &= 8.70 \times 10^{-13}\,\text{J}.\end{aligned}$$

Since $1\,\text{MeV} = 1.60 \times 10^{-13}\,\text{J}$ (Section 17–4), the energy released is 5.4 MeV.

In the tiny world of atoms and nuclei, it is common to quote energies in eV (electron volts) or multiples such as MeV $(10^6\,\text{eV})$. Momentum (see Eq. 26–4) can be quoted in units of eV/c (or MeV/c). And mass can be quoted (from $E = mc^2$) in units of eV/c^2 (or MeV/c^2). Note the use of c to keep the units correct. The masses of the electron and the proton can be shown to be $0.511\,\text{MeV}/c^2$ and $938\,\text{MeV}/c^2$, respectively. For example, for the electron, $mc^2 = (9.11 \times 10^{-31}\,\text{kg})(2.998 \times 10^{8}\,\text{m/s})^2/(1.602 \times 10^{-13}\,\text{J/MeV}) = 0.511\,\text{MeV}$. See also the Table inside the front cover.

EXAMPLE 26–11 **A 1-TeV proton.** The Tevatron accelerator at Fermilab in Illinois can accelerate protons to a kinetic energy of 1.0 TeV $(10^{12}\,\text{eV})$. What is the speed of such a proton?

APPROACH We solve the kinetic energy formula, Eq. 26–5a, for v.

SOLUTION The rest energy of a proton is $mc^2 = 938\,\text{MeV}$ or $9.38 \times 10^{8}\,\text{eV}$. Compared to the kinetic energy of $10^{12}\,\text{eV}$, the rest energy can be neglected, so we simplify Eq. 26–5a to

$$\text{KE} \approx \frac{mc^2}{\sqrt{1 - v^2/c^2}}.$$

We solve this for v in the following steps:

$$\begin{aligned}\sqrt{1 - \frac{v^2}{c^2}} &= \frac{mc^2}{\text{KE}}; \\ 1 - \frac{v^2}{c^2} &= \left(\frac{mc^2}{\text{KE}}\right)^2; \\ \frac{v^2}{c^2} &= 1 - \left(\frac{mc^2}{\text{KE}}\right)^2 = 1 - \left(\frac{9.38 \times 10^{8}\,\text{eV}}{1.0 \times 10^{12}\,\text{eV}}\right)^2; \\ v &= \sqrt{1 - (9.38 \times 10^{-4})^2}\,c \\ &= 0.99999956\,c.\end{aligned}$$

So the proton is traveling at a speed very nearly equal to c.

At low speeds, $v \ll c$, the relativistic formula for kinetic energy reduces to the classical one, as we now show by using the binomial expansion (Appendix A): $(1 \pm x)^n = 1 \pm nx + \cdots$, keeping only two terms because $x = v/c$ is very much less than 1. With $n = -\frac{1}{2}$ we expand the square root in Eq. 26–5a

$$\text{KE} = mc^2\left(\frac{1}{\sqrt{1 - v^2/c^2}} - 1\right)$$

so that

$$\text{KE} \approx mc^2\left(1 + \frac{1}{2}\frac{v^2}{c^2} + \cdots - 1\right) \approx \tfrac{1}{2}mv^2.$$

The dots in the first expression represent very small terms in the expansion which we neglect since we assumed that $v \ll c$. Thus at low speeds, the relativistic form for kinetic energy reduces to the classical form, $\text{KE} = \frac{1}{2}mv^2$. This makes relativity a viable theory in that it can predict accurate results at low speed as well as at high. Indeed, the other equations of special relativity also reduce to their classical equivalents at ordinary speeds: length contraction, time dilation, and modifications to momentum as well as kinetic energy, all disappear for $v \ll c$ since $\sqrt{1 - v^2/c^2} \approx 1$.

A useful relation between the total energy E of a particle and its momentum p can also be derived. The momentum of a particle of mass m and speed v is given by Eq. 26–4

$$p = \gamma mv = \frac{mv}{\sqrt{1 - v^2/c^2}}.$$

The total energy is

$$E = \text{KE} + mc^2$$

or

$$E = \gamma mc^2 = \frac{mc^2}{\sqrt{1 - v^2/c^2}}.$$

We square this equation (and we insert "$v^2 - v^2$" which is zero, but will help us):

$$E^2 = \frac{m^2c^2c^2}{1 - v^2/c^2} = \frac{m^2c^2(c^2 - v^2 + v^2)}{1 - v^2/c^2} = \frac{m^2c^2v^2}{1 - v^2/c^2} + \frac{m^2c^2(c^2 - v^2)}{1 - v^2/c^2}$$
$$= p^2c^2 + \frac{m^2c^4(1 - v^2/c^2)}{1 - v^2/c^2}$$

or

$$E^2 = p^2c^2 + m^2c^4. \qquad \textbf{(26–9)}$$

Thus, the total energy can be written in terms of the momentum p, or in terms of the kinetic energy (Eq. 26–6a), where we have assumed there is no potential energy.

*Invariant Energy–Momentum

We can rewrite Eq. 26–9 as $E^2 - p^2c^2 = m^2c^4$. Since the mass m of a given particle is the same in any reference frame, we see that the quantity $E^2 - p^2c^2$ must also be the same in any reference frame. Thus, at any given moment the total energy E and momentum p of a particle will be different in different reference frames, but the quantity $E^2 - p^2c^2$ will have the same value in all inertial reference frames. We say that the quantity $E^2 - p^2c^2$ is **invariant**.

When Do We Use Relativistic Formulas?

From a practical point of view, we do not have much opportunity in our daily lives to use the mathematics of relativity. For example, the γ factor, $\gamma = 1/\sqrt{1 - v^2/c^2}$, has a value of 1.005 when $v = 0.10c$. Thus, for speeds even as high as $0.10c = 3.0 \times 10^7$ m/s, the factor $\sqrt{1 - v^2/c^2}$ in relativistic formulas gives a numerical correction of less than 1%. For speeds less than $0.10c$, or unless mass and energy are interchanged, we don't usually need the more complicated relativistic formulas, and can use the simpler classical formulas.

If you are given a particle's mass m and its kinetic energy KE, you can do a quick calculation to determine if you need to use relativistic formulas or if classical ones are good enough. You simply compute the ratio KE/mc^2 because (Eq. 26–5b)

$$\frac{\text{KE}}{mc^2} = \gamma - 1 = \frac{1}{\sqrt{1 - v^2/c^2}} - 1.$$

If this ratio comes out to be less than, say, 0.01, then $\gamma \leq 1.01$ and relativistic equations will correct the classical ones by about 1%. If your expected precision is no better than 1%, classical formulas are good enough. But if your precision is 1 part in 1000 (0.1%) then you would want to use relativistic formulas. If your expected precision is only 10%, you need relativity if $(\text{KE}/mc^2) \gtrsim 0.1$.

EXERCISE D For 1% accuracy, does an electron with KE = 100 eV need to be treated relativistically? [*Hint*: The mass of an electron is 0.511 MeV.]

26–10 Relativistic Addition of Velocities

Consider a rocket ship that travels away from the Earth with speed v, and assume that this rocket has fired off a second rocket that travels at speed u' with respect to the first (Fig. 26–11). We might expect that the speed u of rocket 2 with respect to Earth is $u = v + u'$, which in the case shown in Fig. 26–11 is $u = 0.60c + 0.60c = 1.20c$. But, as discussed in Section 26–8, no object can travel faster than the speed of light in any reference frame. Indeed, Einstein showed that since length and time are different in different reference frames, the classical addition-of-velocities formula is no longer valid. Instead, the correct formula is

$$u = \frac{v + u'}{1 + vu'/c^2} \qquad \begin{bmatrix}\vec{\mathbf{u}} \text{ and } \vec{\mathbf{v}} \text{ along} \\ \text{the same direction}\end{bmatrix} \qquad \textbf{(26–10)}$$

for motion along a straight line. We derive this formula in Appendix E. If u' is in the opposite direction from v, then u' must have a minus sign in the above equation so $u = (v - u')/(1 - vu'/c^2)$.

EXAMPLE 26–12 **Relative velocity, relativistically.** Calculate the speed of rocket 2 in Fig. 26–11 with respect to Earth.

APPROACH We combine the speed of rocket 2 relative to rocket 1 with the speed of rocket 1 relative to Earth, using the relativistic Eq. 26–10 because the speeds are high and they are along the same line.

SOLUTION Rocket 2 moves with speed $u' = 0.60c$ with respect to rocket 1. Rocket 1 has speed $v = 0.60c$ with respect to Earth. The speed of rocket 2 with respect to Earth is (Eq. 26–10)

$$u = \frac{0.60c + 0.60c}{1 + \dfrac{(0.60c)(0.60c)}{c^2}} = \frac{1.20c}{1.36} = 0.88c.$$

NOTE The speed of rocket 2 relative to Earth is less than c, as it must be.

We can see that Eq. 26–10 reduces to the classical form for velocities small compared to the speed of light since $1 + vu'/c^2 \approx 1$ for v and $u' \ll c$. Thus, $u \approx v + u'$, as in classical physics (Chapter 3).

Let us test our formula at the other extreme, that of the speed of light. Suppose that rocket 1 in Fig. 26–11 sends out a beam of light so that $u' = c$. Equation 26–10 tells us that the speed of this light relative to Earth is

$$u = \frac{0.60c + c}{1 + \dfrac{(0.60c)(c)}{c^2}} = \frac{1.60c}{1.60} = c,$$

which is fully consistent with the second postulate of relativity.

EXERCISE E Use Eq. 26–10 to calculate the speed of rocket 2 in Fig. 26–11 relative to Earth if it was shot from rocket 1 at a speed $u' = 3000 \text{ km/s} = 0.010c$. Assume rocket 1 had a speed $v = 6000 \text{ km/s} = 0.020c$.

EXERCISE F Return to the Chapter-Opening Question, page 744, and answer it again now. Try to explain why you may have answered differently the first time.

CAUTION

Relative velocities do not add simply, as in classical mechanics ($v \ll c$)

Relativistic addition of velocities formula ($\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$ along same line)

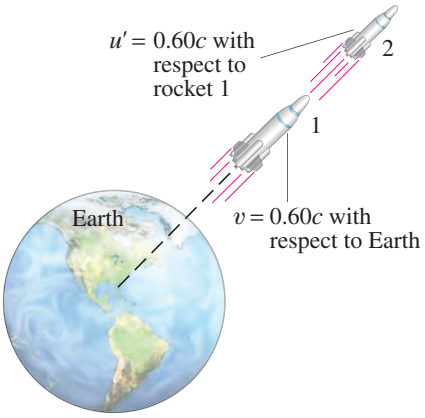


FIGURE 26–11 Rocket 1 leaves Earth at $v = 0.60c$. Rocket 2 is fired from rocket 1 with speed $u' = 0.60c$. What is the speed of rocket 2 with respect to the Earth? Example 26–12.

26–11 The Impact of Special Relativity

A great many experiments have been performed to test the predictions of the special theory of relativity. Within experimental error, no contradictions have been found. Scientists have therefore accepted relativity as an accurate description of nature.

At speeds much less than the speed of light, the relativistic formulas reduce to the old classical ones, as we have discussed. We would, of course, hope—or rather, insist—that this be true since Newtonian mechanics works so well for objects moving with speeds $v \ll c$. This insistence that a more general theory (such as relativity) give the same results as a more restricted theory (such as classical mechanics which works for $v \ll c$) is called the **correspondence principle**. The two theories must correspond where their realms of validity overlap. Relativity thus does not contradict classical mechanics. Rather, it is a more general theory, of which classical mechanics is now considered to be a limiting case.

The importance of relativity is not simply that it gives more accurate results, especially at very high speeds. Much more than that, it has changed the way we view the world. The concepts of space and time are now seen to be relative, and intertwined with one another, whereas before they were considered absolute and separate. Even our concepts of matter and energy have changed: either can be converted to the other. The impact of relativity extends far beyond physics. It has influenced the other sciences, and even the world of art and literature; it has, indeed, entered the general culture.

The special theory of relativity we have studied in this Chapter deals with inertial (nonaccelerating) reference frames. In Chapter 33 we will discuss briefly the more complicated "general theory of relativity" which can deal with non-inertial reference frames.

Summary

An **inertial reference frame** is one in which Newton's law of inertia holds. Inertial reference frames move at constant velocity relative to one another. Accelerating reference frames are **noninertial**.

The **special theory of relativity** is based on two principles: the **relativity principle**, which states that the laws of physics are the same in all inertial reference frames, and the principle of the **constancy of the speed of light**, which states that the speed of light in empty space has the same value in all inertial reference frames.

One consequence of relativity theory is that two events that are simultaneous in one reference frame may not be simultaneous in another. Other effects are **time dilation**: moving clocks are measured to run slow; and **length contraction**: the length of a moving object is measured to be shorter (in its direction of motion) than when it is at rest. Quantitatively,

$$\Delta t = \frac{\Delta t_0}{\sqrt{1 - v^2/c^2}} = \gamma\, \Delta t_0 \qquad \textbf{(26–1)}$$

$$\ell = \ell_0 \sqrt{1 - v^2/c^2} = \frac{\ell_0}{\gamma} \qquad \textbf{(26–3)}$$

where ℓ and Δt are the length and time interval of objects (or events) observed as they move by at the speed v; ℓ_0 and Δt_0 are the **proper length** and **proper time**—that is, the same quantities as measured in the rest frame of the objects or events. The quantity γ is shorthand for

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}. \qquad \textbf{(26–2)}$$

The theory of relativity has changed our notions of space and time, and of momentum, energy, and mass. Space and time are seen to be intimately connected, with time being the fourth dimension in addition to the three dimensions of space.

The **momentum** of an object is given by

$$p = \gamma m v = \frac{m v}{\sqrt{1 - v^2/c^2}}. \qquad \textbf{(26–4)}$$

Mass and energy are interconvertible. The equation

$$E = mc^2 \qquad \textbf{(26–7)}$$

tells how much energy E is needed to create a mass m, or vice versa. Said another way, $E = mc^2$ is the amount of energy an object has because of its mass m. The law of conservation of energy must include mass as a form of energy.

The kinetic energy KE of an object moving at speed v is given by

$$\text{KE} = \frac{mc^2}{\sqrt{1 - v^2/c^2}} - mc^2 = (\gamma - 1)mc^2 \qquad \textbf{(26–5)}$$

where m is the mass of the object. The total energy E, if there is no potential energy, is

$$E = \text{KE} + mc^2 = \gamma mc^2. \qquad \textbf{(26–6)}$$

The momentum p of an object is related to its total energy E (assuming no potential energy) by

$$E^2 = p^2c^2 + m^2c^4. \qquad \textbf{(26–9)}$$

Velocity addition also must be done in a special way. All these relativistic effects are significant only at high speeds, close to the speed of light, which itself is the ultimate speed in the universe.

Questions

1. You are in a windowless car in an exceptionally smooth train moving at constant velocity. Is there any physical experiment you can do in the train car to determine whether you are moving? Explain.
2. You might have had the experience of being at a red light when, out of the corner of your eye, you see the car beside you creep forward. Instinctively you stomp on the brake pedal, thinking that you are rolling backward. What does this say about absolute and relative motion?
3. A worker stands on top of a railroad car moving at constant velocity and throws a heavy ball straight up (from his point of view). Ignoring air resistance, explain whether the ball will land back in his hand or behind him.
4. Does the Earth really go around the Sun? Or is it also valid to say that the Sun goes around the Earth? Discuss in view of the relativity principle (that there is no best reference frame). Explain. See Section 5–8.
5. If you were on a spaceship traveling at $0.6c$ away from a star, at what speed would the starlight pass you?
6. The time dilation effect is sometimes expressed as "moving clocks run slowly." Actually, this effect has nothing to do with motion affecting the functioning of clocks. What then does it deal with?
7. Does time dilation mean that time actually passes more slowly in moving reference frames or that it only *seems* to pass more slowly?
8. A young-looking woman astronaut has just arrived home from a long trip. She rushes up to an old gray-haired man and in the ensuing conversation refers to him as her son. How might this be possible?
9. If you were traveling away from Earth at speed $0.6c$, would you notice a change in your heartbeat? Would your mass, height, or waistline change? What would observers on Earth using telescopes say about you?
10. Do time dilation and length contraction occur at ordinary speeds, say 90 km/h?
11. Suppose the speed of light were infinite. What would happen to the relativistic predictions of length contraction and time dilation?
12. Explain how the length contraction and time dilation formulas might be used to indicate that c is the limiting speed in the universe.
13. Discuss how our everyday lives would be different if the speed of light were only 25 m/s.
14. The drawing at the start of this Chapter shows the street as seen by Mr Tompkins, for whom the speed of light is $c = 20$ mi/h. What does Mr Tompkins look like to the people standing on the street (Fig. 26–12)? Explain.

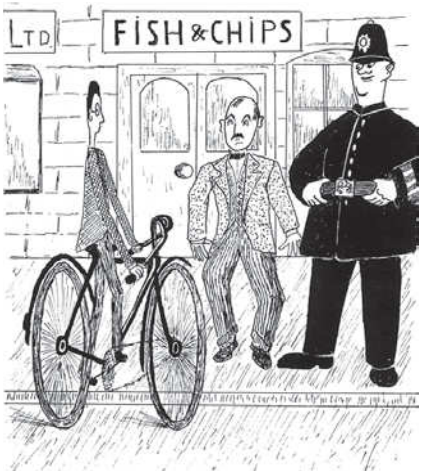


FIGURE 26–12
Question 14.
Mr Tompkins as seen by people on the sidewalk. See also Chapter-Opening Figure on page 744.

15. An electron is limited to travel at speeds less than c. Does this put an upper limit on the momentum of an electron? If so, what is this upper limit? If not, explain.
16. Can a particle of nonzero mass attain the speed of light? Explain.
17. Does the equation $E = mc^2$ conflict with the conservation of energy principle? Explain.
18. If mass is a form of energy, does this mean that a spring has more mass when compressed than when relaxed? Explain.
19. It is not correct to say that "matter can neither be created nor destroyed." What must we say instead?
20. Is our intuitive notion that velocities simply add, as in Section 3–8, completely wrong?

MisConceptual Questions

1. The fictional rocket ship *Adventure* is measured to be 50 m long by the ship's captain inside the rocket. When the rocket moves past a space dock at $0.5c$, space-dock personnel measure the rocket ship to be 43.3 m long. What is its proper length?
   (*a*) 50 m. (*b*) 43.3 m. (*c*) 93.3 m. (*d*) 13.3 m.

2. As rocket ship *Adventure* (MisConceptual Question 1) passes by the space dock, the ship's captain flashes a flashlight at 1.00-s intervals as measured by space-dock personnel. How often does the flashlight flash relative to the captain?
   (*a*) Every 1.15 s. (*b*) Every 1.00 s. (*c*) Every 0.87 s.
   (*d*) We need to know the distance between the ship and the space dock.

3. For the flashing of the flashlight in MisConceptual Question 2, what time interval is the proper time interval?
   (*a*) 1.15 s. (*b*) 1.00 s. (*c*) 0.87 s. (*d*) 0.13 s.

4. The rocket ship of MisConceptual Question 1 travels to a star many light-years away, then turns around and returns at the same speed. When it returns to the space dock, who would have aged less: the space-dock personnel or ship's captain?
   (*a*) The space-dock personnel.
   (*b*) The ship's captain.
   (*c*) Both the same amount, because both sets of people were moving relative to each other.
   (*d*) We need to know how far away the star is.

5. An Earth observer notes that clocks on a passing spacecraft run slowly. The person on the spacecraft
   (*a*) agrees her clocks move slower than those on Earth.
   (*b*) feels normal, and her heartbeat and eating habits are normal.
   (*c*) observes that Earth clocks are moving slowly.
   (*d*) The real time is in between the times measured by the two observers.
   (*e*) Both (*a*) and (*b*).
   (*f*) Both (*b*) and (*c*).

6. Spaceships A and B are traveling directly toward each other at a speed $0.5c$ relative to the Earth, and each has a headlight aimed toward the other ship. What value do technicians on ship B get by measuring the speed of the light emitted by ship A's headlight?
(*a*) $0.5c$. (*b*) $0.75c$. (*c*) $1.0c$. (*d*) $1.5c$.
7. Relativistic formulas for time dilation, length contraction, and mass are valid
(*a*) only for speeds less than $0.10c$.
(*b*) only for speeds greater than $0.10c$.
(*c*) only for speeds very close to c.
(*d*) for all speeds.
8. Which of the following will two observers in inertial reference frames always agree on? (Choose all that apply.)
(*a*) The time an event occurred.
(*b*) The distance between two events.
(*c*) The time interval between the occurence of two events.
(*d*) The speed of light.
(*e*) The validity of the laws of physics.
(*f*) The simultaneity of two events.
9. Two observers in different inertial reference frames moving relative to each other at nearly the speed of light see the same two events but, using precise equipment, record different time intervals between the two events. Which of the following is true of their measurements?
(*a*) One observer is incorrect, but it is impossible to tell which one.
(*b*) One observer is incorrect, and it is possible to tell which one.
(*c*) Both observers are incorrect.
(*d*) Both observers are correct.
10. You are in a rocket ship going faster and faster. As your speed increases and your velocity gets closer to the speed of light, which of the following do you observe in your frame of reference?
(*a*) Your mass increases.
(*b*) Your length shortens in the direction of motion.
(*c*) Your wristwatch slows down.
(*d*) All of the above.
(*e*) None of the above.
11. You are in a spaceship with no windows, radios, or other means to check outside. How could you determine whether your spaceship is at rest or moving at constant velocity?
(*a*) By determining the apparent velocity of light in the spaceship.
(*b*) By checking your precision watch. If it's running slow, then the ship is moving.
(*c*) By measuring the lengths of objects in the spaceship. If they are shortened, then the ship is moving.
(*d*) Give up, because you can't tell.
12. The period of a pendulum attached in a spaceship is 2 s while the spaceship is parked on Earth. What is the period to an observer on Earth when the spaceship moves at $0.6c$ with respect to the Earth?
(*a*) Less than 2 s.
(*b*) More than 2 s.
(*c*) 2 s.
13. Two spaceships, each moving at a speed $0.75c$ relative to the Earth, are headed directly toward each other. What do occupants of one ship measure the speed of other ship to be?
(*a*) $0.96c$. (*b*) $1.0c$. (*c*) $1.5c$. (*d*) $1.75c$. (*e*) $0.75c$.

For assigned homework and other learning materials, go to the MasteringPhysics website.

Problems

26–4 and 26–5 Time Dilation, Length Contraction

1. (I) A spaceship passes you at a speed of $0.850c$. You measure its length to be 44.2 m. How long would it be when at rest?
2. (I) A certain type of elementary particle travels at a speed of 2.70×10^8 m/s. At this speed, the average lifetime is measured to be 4.76×10^{-6} s. What is the particle's lifetime at rest?
3. (II) You travel to a star 135 light-years from Earth at a speed of 2.90×10^8 m/s. What do you measure this distance to be?
4. (II) What is the speed of a pion if its average lifetime is measured to be 4.40×10^{-8} s? At rest, its average lifetime is 2.60×10^{-8} s.
5. (II) In an Earth reference frame, a star is 49 light-years away. How fast would you have to travel so that to you the distance would be only 35 light-years?
6. (II) At what speed v will the length of a 1.00-m stick look 10.0% shorter (90.0 cm)?
7. (II) At what speed do the relativistic formulas for (*a*) length and (*b*) time intervals differ from classical values by 1.00%? (This is a reasonable way to estimate when to use relativistic calculations rather than classical.)
8. (II) You decide to travel to a star 62 light-years from Earth at a speed that tells you the distance is only 25 light-years. How many years would it take you to make the trip?
9. (II) A friend speeds by you in her spacecraft at a speed of $0.720c$. It is measured in your frame to be 4.80 m long and 1.35 m high. (*a*) What will be its length and height at rest? (*b*) How many seconds elapsed on your friend's watch when 20.0 s passed on yours? (*c*) How fast did you appear to be traveling according to your friend? (*d*) How many seconds elapsed on your watch when she saw 20.0 s pass on hers?
10. (II) A star is 21.6 light-years from Earth. How long would it take a spacecraft traveling $0.950c$ to reach that star as measured by observers: (*a*) on Earth, (*b*) on the spacecraft? (*c*) What is the distance traveled according to observers on the spacecraft? (*d*) What will the spacecraft occupants compute their speed to be from the results of (*b*) and (*c*)?
11. (II) A fictional news report stated that starship *Enterprise* had just returned from a 5-year voyage while traveling at $0.70c$. (*a*) If the report meant 5.0 years of *Earth time*, how much time elapsed on the ship? (*b*) If the report meant 5.0 years of *ship time*, how much time passed on Earth?
12. (II) A box at rest has the shape of a cube 2.6 m on a side. This box is loaded onto the flat floor of a spaceship and the spaceship then flies past us with a horizontal speed of $0.80c$. What is the volume of the box as we observe it?
13. (III) Escape velocity from the Earth is 11.2 km/s. What would be the percent decrease in length of a 68.2-m-long spacecraft traveling at that speed as seen from Earth?

14. (III) An unstable particle produced in an accelerator experiment travels at constant velocity, covering 1.00 m in 3.40 ns in the lab frame before changing ("decaying") into other particles. In the rest frame of the particle, determine (*a*) how long it lived before decaying, (*b*) how far it moved before decaying.

26–7 Relativistic Momentum

15. (I) What is the momentum of a proton traveling at $v = 0.68c$?

16. (II) (*a*) A particle travels at $v = 0.15c$. By what percentage will a calculation of its momentum be wrong if you use the classical formula? (*b*) Repeat for $v = 0.75c$.

17. (II) A particle of mass m travels at a speed $v = 0.22c$. At what speed will its momentum be doubled?

18. (II) An unstable particle is at rest and suddenly decays into two fragments. No external forces act on the particle or its fragments. One of the fragments has a speed of $0.60c$ and a mass of 6.68×10^{-27} kg, while the other has a mass of 1.67×10^{-27} kg. What is the speed of the less massive fragment?

19. (II) What is the percent change in momentum of a proton that accelerates from (*a*) $0.45c$ to $0.85c$, (*b*) $0.85c$ to $0.98c$?

26–9 $E = mc^2$; Mass and Energy

20. (I) Calculate the rest energy of an electron in joules and in MeV $(1\,\text{MeV} = 1.60 \times 10^{-13}\,\text{J})$.

21. (I) When a uranium nucleus at rest breaks apart in the process known as *fission* in a nuclear reactor, the resulting fragments have a total kinetic energy of about 200 MeV. How much mass was lost in the process?

22. (I) The total annual energy consumption in the United States is about 1×10^{20} J. How much mass would have to be converted to energy to fuel this need?

23. (I) Calculate the mass of a proton $(1.67 \times 10^{-27}\,\text{kg})$ in MeV/c^2.

24. (II) Calculate the kinetic energy and momentum of a proton traveling 2.90×10^8 m/s.

25. (II) What is the momentum of a 950-MeV proton (that is, its kinetic energy is 950 MeV)?

26. (II) What is the speed of an electron whose kinetic energy is 1.12 MeV?

27. (II) (*a*) How much work is required to accelerate a proton from rest up to a speed of $0.985c$? (*b*) What would be the momentum of this proton?

28. (II) At what speed will an object's kinetic energy be 33% of its rest energy?

29. (II) Determine the speed and the momentum of an electron $(m = 9.11 \times 10^{-31}\,\text{kg})$ whose KE equals its rest energy.

30. (II) A proton is traveling in an accelerator with a speed of 1.0×10^8 m/s. By what factor does the proton's kinetic energy increase if its speed is doubled?

31. (II) How much energy can be obtained from conversion of 1.0 gram of mass? How much mass could this energy raise to a height of 1.0 km above the Earth's surface?

32. (II) To accelerate a particle of mass m from rest to speed $0.90c$ requires work W_1. To accelerate the particle from speed $0.90c$ to $0.99c$ requires work W_2. Determine the ratio W_2/W_1.

33. (II) Suppose there was a process by which two photons, each with momentum 0.65 MeV/c, could collide and make a single particle. What is the maximum mass that the particle could possess?

34. (II) What is the speed of a proton accelerated by a potential difference of 165 MV?

35. (II) The kinetic energy of a particle is 45 MeV. If the momentum is 121 MeV/c, what is the particle's mass?

36. (II) Calculate the speed of a proton $(m = 1.67 \times 10^{-27}\,\text{kg})$ whose kinetic energy is exactly half (*a*) its total energy, (*b*) its rest energy.

37. (II) Calculate the kinetic energy and momentum of a proton $(m = 1.67 \times 10^{-27}\,\text{kg})$ traveling 8.65×10^7 m/s. By what percentages would your calculations have been in error if you had used classical formulas?

38. (II) Suppose a spacecraft of mass 17,000 kg is accelerated to $0.15c$. (*a*) How much kinetic energy would it have? (*b*) If you used the classical formula for kinetic energy, by what percentage would you be in error?

39. (II) A negative muon traveling at 53% the speed of light collides head on with a positive muon traveling at 65% the speed of light. The two muons (each of mass 105.7 MeV/c^2) annihilate, and produce how much electromagnetic energy?

40. (III) The americium nucleus, $^{241}_{95}\text{Am}$, decays to a neptunium nucleus, $^{237}_{93}\text{Np}$, by emitting an alpha particle of mass 4.00260 u and kinetic energy 5.5 MeV. Estimate the mass of the neptunium nucleus, ignoring its recoil, given that the americium mass is 241.05682 u.

41. (III) Show that the kinetic energy KE of a particle of mass m is related to its momentum p by the equation

$$p = \sqrt{\text{KE}^2 + 2\text{KE}\,mc^2}/c.$$

***42.** (III) What magnetic field B is needed to keep 998-GeV protons revolving in a circle of radius 1.0 km? Use the relativistic mass. The proton's "rest mass" is 0.938 GeV/c^2. $(1\,\text{GeV} = 10^9\,\text{eV}.)$ [*Hint*: In relativity, $m_{\text{rel}}v^2/r = qvB$ is still valid in a magnetic field, where $m_{\text{rel}} = \gamma m$.]

26–10 Relativistic Addition of Velocities

43. (I) A person on a rocket traveling at $0.40c$ (with respect to the Earth) observes a meteor come from behind and pass her at a speed she measures as $0.40c$. How fast is the meteor moving with respect to the Earth?

44. (II) Two spaceships leave Earth in opposite directions, each with a speed of $0.60c$ with respect to Earth. (*a*) What is the velocity of spaceship 1 relative to spaceship 2? (*b*) What is the velocity of spaceship 2 relative to spaceship 1?

45. (II) A spaceship leaves Earth traveling at $0.65c$. A second spaceship leaves the first at a speed of $0.82c$ with respect to the first. Calculate the speed of the second ship with respect to Earth if it is fired (*a*) in the same direction the first spaceship is already moving, (*b*) directly backward toward Earth.

46. (II) An observer on Earth sees an alien vessel approach at a speed of $0.60c$. The fictional starship *Enterprise* comes to the rescue (Fig. 26–13), overtaking the aliens while moving directly toward Earth at a speed of $0.90c$ relative to Earth. What is the relative speed of one vessel as seen by the other?

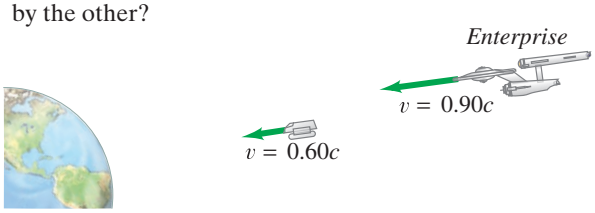


FIGURE 26–13 Problem 46.

47. (II) A spaceship in distress sends out two escape pods in opposite directions. One travels at a speed $v_1 = +0.70c$ in one direction, and the other travels at a speed $v_2 = -0.80c$ in the other direction, as observed from the spaceship. What speed does the first escape pod measure for the second escape pod?

48. (II) Rocket A passes Earth at a speed of $0.65c$. At the same time, rocket B passes Earth moving $0.95c$ relative to Earth in the same direction as A. How fast is B moving relative to A when it passes A?

49. (II) Your spaceship, traveling at $0.90c$, needs to launch a probe out the forward hatch so that its speed relative to the planet that you are approaching is $0.95c$. With what speed must it leave your ship?

General Problems

50. What is the speed of a particle when its kinetic energy equals its rest energy? Does the mass of the particle affect the result?

51. The nearest star to Earth is Proxima Centauri, 4.3 light-years away. (*a*) At what constant velocity must a spacecraft travel from Earth if it is to reach the star in 4.9 years, as measured by travelers on the spacecraft? (*b*) How long does the trip take according to Earth observers?

52. According to the special theory of relativity, the factor γ that determines the length contraction and the time dilation is given by $\gamma = 1/\sqrt{1 - v^2/c^2}$. Determine the numerical values of γ for an object moving at speed $v = 0.01c$, $0.05c$, $0.10c$, $0.20c$, $0.30c$, $0.40c$, $0.50c$, $0.60c$, $0.70c$, $0.80c$, $0.90c$, $0.95c$, and $0.99c$. Make a graph of γ versus v.

53. A healthy astronaut's heart rate is 60 beats/min. Flight doctors on Earth can monitor an astronaut's vital signs remotely while in flight. How fast would an astronaut be flying away from Earth if the doctor measured her having a heart rate of 25 beats/min?

54. What minimum amount of electromagnetic energy is needed to produce an electron and a positron together? A positron is a particle with the same mass as an electron, but has the opposite charge. (Note that electric charge is conserved in this process. See Section 27–6.)

55. How many grams of matter would have to be totally destroyed to run a 75-W lightbulb for 1.0 year?

56. A free neutron can decay into a proton, an electron, and a neutrino. Assume the neutrino's mass is zero; the other masses can be found in the Table inside the front cover. Determine the total kinetic energy shared among the three particles when a neutron decays at rest.

57. An electron $(m = 9.11 \times 10^{-31}\,\text{kg})$ is accelerated from rest to speed v by a conservative force. In this process, its potential energy decreases by $6.20 \times 10^{-14}\,\text{J}$. Determine the electron's speed, v.

58. The Sun radiates energy at a rate of about $4 \times 10^{26}\,\text{W}$. (*a*) At what rate is the Sun's mass decreasing? (*b*) How long does it take for the Sun to lose a mass equal to that of Earth? (*c*) Estimate how long the Sun could last if it radiated constantly at this rate.

59. How much energy would be required to break a helium nucleus into its constituents, two protons and two neutrons? The masses of a proton (including an electron), a neutron, and neutral helium are, respectively, 1.00783 u, 1.00867 u, and 4.00260 u. (This energy difference is called the *total binding energy* of the ^4_2He nucleus.)

60. Show analytically that a particle with momentum p and energy E has a speed given by

$$v = \frac{pc^2}{E} = \frac{pc}{\sqrt{m^2c^2 + p^2}}.$$

61. Two protons, each having a speed of $0.990c$ in the laboratory, are moving toward each other. Determine (*a*) the momentum of each proton in the laboratory, (*b*) the total momentum of the two protons in the laboratory, and (*c*) the momentum of one proton as seen by the other proton.

62. When two moles of hydrogen molecules (H_2) and one mole of oxygen molecules (O_2) react to form two moles of water (H_2O), the energy released is 484 kJ. How much does the mass decrease in this reaction? What % of the total original mass is this?

63. Make a graph of the kinetic energy versus momentum for (*a*) a particle of nonzero mass, and (*b*) a particle with zero mass.

64. In a nuclear reaction two identical particles are created, traveling in opposite directions. If the speed of each particle is $0.82c$, relative to the laboratory frame of reference, what is one particle's speed relative to the other particle?

65. A 36,000-kg spaceship is to travel to the vicinity of a star 6.6 light-years from Earth. Passengers on the ship want the (one-way) trip to take no more than 1.0 year. How much work must be done on the spaceship to bring it to the speed necessary for this trip?

66. Suppose a 14,500-kg spaceship left Earth at a speed of $0.90c$. What is the spaceship's kinetic energy? Compare with the total U.S. annual energy consumption (about 10^{20} J).

67. A pi meson of mass m_π decays at rest into a muon (mass m_μ) and a neutrino of negligible or zero mass. Show that the kinetic energy of the muon is $\text{KE}_\mu = (m_\pi - m_\mu)^2c^2/(2m_\pi)$.

68. An astronaut on a spaceship traveling at $0.75c$ relative to Earth measures his ship to be 23 m long. On the ship, he eats his lunch in 28 min. (*a*) What length is the spaceship according to observers on Earth? (*b*) How long does the astronaut's lunch take to eat according to observers on Earth?

69. An electron is accelerated so that its kinetic energy is greater than its rest energy mc^2 by a factor of (*a*) 5.00, (*b*) 999. What is the speed of the electron in each case?

70. You are traveling in a spaceship at a speed of $0.70c$ away from Earth. You send a laser beam toward the Earth traveling at velocity c relative to you. What do observers on the Earth measure for the speed of the laser beam?

71. A farm boy studying physics believes that he can fit a 13.0-m-long pole into a 10.0-m-long barn if he runs fast enough, carrying the pole. Can he do it? Explain in detail. How does this fit with the idea that when he is running the barn looks even shorter than 10.0 m?

72. An atomic clock is taken to the North Pole, while another stays at the Equator. How far will they be out of synchronization after 2.0 years has elapsed? [*Hint*: Use the binomial expansion, Appendix A.]

73. An airplane travels 1300 km/h around the Earth in a circle of radius essentially equal to that of the Earth, returning to the same place. Using special relativity, estimate the difference in time to make the trip as seen by Earth and by airplane observers. [*Hint*: Use the binomial expansion, Appendix A.]

Search and Learn

1. Determine about how fast Mr Tompkins is traveling in the Chapter-Opening Photograph. Do you agree with the picture in terms of the way Mr Tompkins would see the world? Explain. [*Hint*: Assume the bank clock and Stop sign facing us are round according to the people on the sidewalk.]

2. Examine the experiment of Fig. 26–5 from O_1's reference frame. In this case, O_1 will be at rest and will see the lightning bolt at B_1 and B_2, before the lightning bolt at A_1 and A_2. Will O_1 recognize that O_2, who is moving with speed v to the left, will see the two events as simultaneous? Explain in detail, drawing diagrams equivalent to Fig. 26–5. [*Hint*: Include length contraction.]

3. Using Example 26–2 as a guide, show that for objects that move slowly in comparison to c, the length contraction formula is roughly $\ell \approx \ell_0(1 - \frac{1}{2}v^2/c^2)$. Use this approximation to find the "length shortening" $\Delta\ell = \ell_0 - \ell$ of the train in Example 26–6 if the train travels at 100 km/h (rather than $0.92c$).

4. In Example 26–5, the spaceship is moving at $0.90c$ in the horizontal direction relative to an observer on the Earth. If instead the spaceship moved at $0.90c$ directed at 30° above the horizontal, what would be the painting's dimensions as seen by the observer on Earth?

5. Protons from outer space crash into the Earth's atmosphere at a high rate. These protons create particles that eventually decay into other particles called *muons*. This cosmic debris travels through the atmosphere. Every second, dozens of muons pass through your body. If a muon is created 30 km above the Earth's surface, what minimum speed and kinetic energy must the muon have in order to hit Earth's surface? A muon's mean lifetime (at rest) is 2.20 μs and its mass is 105.7 MeV/c^2.

6. As a rough rule, anything traveling faster than about $0.1c$ is called *relativistic*—that is, special relativity is a significant effect. Determine the speed of an electron in a hydrogen atom (radius 0.53×10^{-10} m) and state whether or not it is relativistic. (Treat the electron as though it were in a circular orbit around the proton. See hint for Problem 42.)

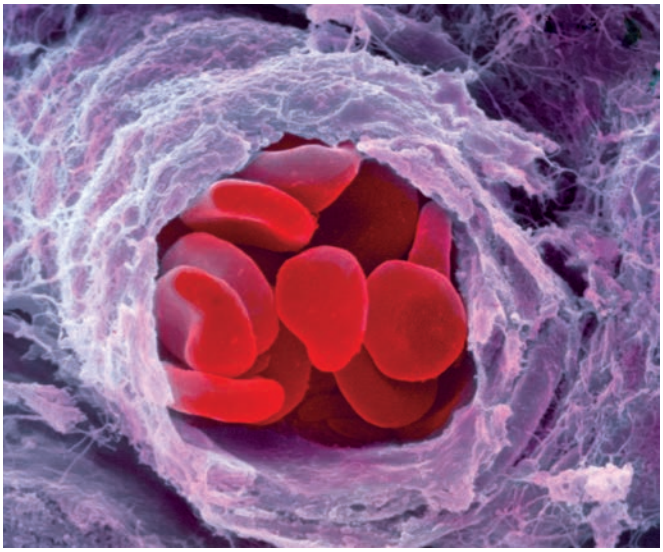
ANSWERS TO EXERCISES

A: (*c*).
B: (*a*) No; (*b*) yes.
C: 80 m.
D: No: $\text{KE}/mc^2 \approx 2 \times 10^{-4}$.
E: $0.030c$, same as classical, to an accuracy of better than 0.1%.
F: (*d*).

Electron microscopes (EM) produce images using electrons which have wave properties just as light does. Because the wavelength of electrons can be much smaller than that of visible light, much greater resolution and magnification can be obtained. A scanning electron microscope (SEM) can produce images with a three-dimensional quality.

All EM images are monochromatic (black and white). Artistic coloring has been added here, as is common. On the left is an SEM image of a blood clot forming (yellow-color web) due to a wound. White blood cells are colored green here for visibility. On the right, red blood cells in a small artery. A red blood cell travels about 15 km a day inside our bodies and lives roughly 4 months before damage or rupture. Humans contain 4 to 6 liters of blood, and 2 to 3×10^{13} red blood cells.





CHAPTER 27

Early Quantum Theory and Models of the Atom

CHAPTER-OPENING QUESTION—Guess now!

It has been found experimentally that

(a) light behaves as a wave.
(b) light behaves as a particle.
(c) electrons behave as particles.
(d) electrons behave as waves.
(e) all of the above are true.
(f) only (a) and (b) are true.
(g) only (a) and (c) are true.
(h) none of the above are true.

The second aspect of the revolution that shook the world of physics in the early part of the twentieth century was the quantum theory (the other was Einstein's theory of relativity). Unlike the special theory of relativity, the revolution of quantum theory required almost three decades to unfold, and many scientists contributed to its development. It began in 1900 with Planck's quantum hypothesis, and culminated in the mid-1920s with the theory of quantum mechanics of Schrödinger and Heisenberg which has been so effective in explaining the structure of matter. The discovery of the electron in the 1890s, with which we begin this Chapter, might be said to mark the beginning of modern physics, and is a sort of precursor to the quantum theory.

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27–1 Discovery and Properties of the Electron

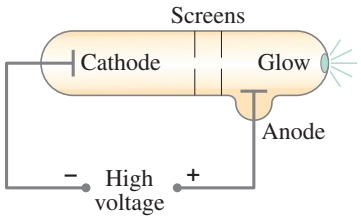


FIGURE 27–1 Discharge tube. In some models, one of the screens is the anode (positive plate).

Toward the end of the nineteenth century, studies were being done on the discharge of electricity through rarefied gases. One apparatus, diagrammed in Fig. 27–1, was a glass tube fitted with electrodes and evacuated so only a small amount of gas remained inside. When a very high voltage was applied to the electrodes, a dark space seemed to extend outward from the cathode (negative electrode) toward the opposite end of the tube; and that far end of the tube would glow. If one or more screens containing a small hole were inserted as shown, the glow was restricted to a tiny spot on the end of the tube. It seemed as though something being emitted by the cathode traveled across to the opposite end of the tube. These "somethings" were named **cathode rays**.

There was much discussion at the time about what these rays might be. Some scientists thought they might resemble light. But the observation that the bright spot at the end of the tube could be deflected to one side by an electric or magnetic field suggested that cathode rays were charged particles; and the direction of the deflection was consistent with a negative charge. Furthermore, if the tube contained certain types of rarefied gas, the path of the cathode rays was made visible by a slight glow.

Estimates of the charge e of the cathode-ray particles, as well as of their charge-to-mass ratio e/m, had been made by 1897. But in that year, J. J. Thomson (1856–1940) was able to measure e/m directly, using the apparatus shown in Fig. 27–2. Cathode rays are accelerated by a high voltage and then pass between a pair of parallel plates built into the tube. Another voltage applied to the parallel plates produces an electric field $\vec{\mathbf{E}}$, and a pair of coils produces a magnetic field $\vec{\mathbf{B}}$. If $E = B = 0$, the cathode rays follow path b in Fig. 27–2.

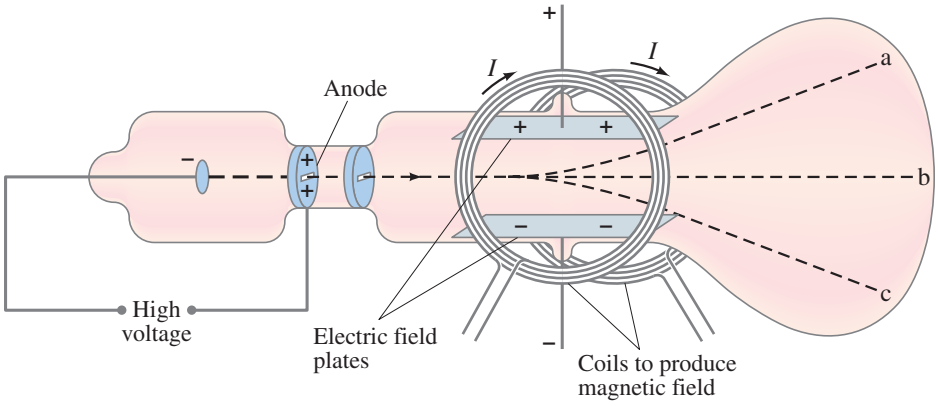


FIGURE 27–2 Cathode rays deflected by electric and magnetic fields. (See also Section 17–11 on the CRT.)

When only the electric field is present, say with the upper plate positive, the cathode rays are deflected upward as in path a in Fig. 27–2. If only a magnetic field exists, say inward, the rays are deflected downward along path c. These observations are just what is expected for a negatively charged particle. The force on the rays due to the magnetic field is $F = evB$, where e is the charge and v is the velocity of the cathode rays (Eq. 20–4). In the absence of an electric field, the rays are bent into a curved path, and applying Newton's second law $F = ma$ with a = centripetal acceleration gives

$$evB = m\frac{v^2}{r},$$

and thus

$$\frac{e}{m} = \frac{v}{Br}.$$

The radius of curvature r can be measured and so can B. The velocity v can be found by applying an electric field in addition to the magnetic field. The electric

field E is adjusted so that the cathode rays are undeflected and follow path b in Fig. 27–2. In this situation the upward force due to the electric field, $F = eE$, is balanced by the downward force due to the magnetic field, $F = evB$. We equate the two forces, $eE = evB$, and find

$$v = \frac{E}{B}.$$

Combining this with the above equation we have

$$\frac{e}{m} = \frac{E}{B^2 r}. \qquad \textbf{(27–1)}$$

The quantities on the right side can all be measured, and although e and m could not be determined separately, the ratio e/m could be determined. The accepted value today is $e/m = 1.76 \times 10^{11}\,\text{C/kg}$. Cathode rays soon came to be called **electrons**.

Discovery in Science

The "discovery" of the electron, like many others in science, is not quite so obvious as discovering gold or oil. Should the discovery of the electron be credited to the person who first saw a glow in the tube? Or to the person who first called them cathode rays? Perhaps neither one, for they had no conception of the electron as we know it today. In fact, the credit for the discovery is generally given to Thomson, but not because he was the first to see the glow in the tube. Rather it is because he believed that this phenomenon was due to tiny negatively charged particles and made careful measurements on them. Furthermore he argued that these particles were constituents of atoms, and not ions or atoms themselves as many thought, and he developed an electron theory of matter. His view is close to what we accept today, and this is why Thomson is credited with the "discovery." Note, however, that neither he nor anyone else ever actually saw an electron itself. We discuss this briefly, for it illustrates the fact that discovery in science is not always a clear-cut matter. In fact some philosophers of science think the word "discovery" is often not appropriate, such as in this case.

Electron Charge Measurement

Thomson believed that an electron was not an atom, but rather a constituent, or part, of an atom. Convincing evidence for this came soon with the determination of the charge and the mass of the cathode rays. Thomson's student J. S. Townsend made the first direct (but rough) measurements of e in 1897. But it was the more refined **oil-drop experiment** of Robert A. Millikan (1868–1953) that yielded a precise value for the charge on the electron and showed that charge comes in discrete amounts. In this experiment, tiny droplets of mineral oil carrying an electric charge were allowed to fall under gravity between two parallel plates, Fig. 27–3. The electric field E between the plates was adjusted until the drop was suspended in midair. The downward pull of gravity, mg, was then just balanced by the upward force due to the electric field. Thus $qE = mg$ so the charge $q = mg/E$. The mass of the droplet was determined by measuring its terminal velocity in the absence of the electric field. Often the droplet was charged negatively, but sometimes it was positive, suggesting that the droplet had acquired or lost electrons (by friction, leaving the atomizer). Millikan's painstaking observations and analysis presented convincing evidence that any charge was an integral multiple of a smallest charge, e, that was ascribed to the electron, and that the value of e was $1.6 \times 10^{-19}\,\text{C}$. This value of e, combined with the measurement of e/m, gives the mass of the electron to be $(1.6 \times 10^{-19}\,\text{C})/(1.76 \times 10^{11}\,\text{C/kg}) = 9.1 \times 10^{-31}\,\text{kg}$. This mass is less than a thousandth the mass of the smallest atom, and thus confirmed the idea that the electron is only a part of an atom. The accepted value today for the mass of the electron is

$$m_e = 9.11 \times 10^{-31}\,\text{kg}.$$

The experimental result that any charge is an integral multiple of e means that electric charge is *quantized* (exists only in discrete amounts).

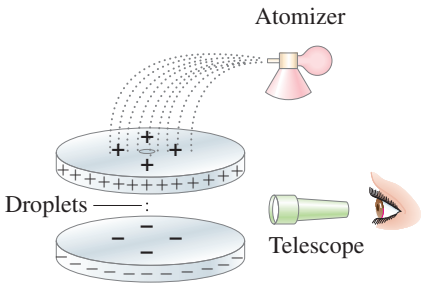


FIGURE 27–3 Millikan's oil-drop experiment.

27–2 Blackbody Radiation; Planck's Quantum Hypothesis

Blackbody Radiation

One of the observations that was unexplained at the end of the nineteenth century was the spectrum of light emitted by hot objects. We saw in Section 14–8 that all objects emit radiation whose total intensity is proportional to the fourth power of the Kelvin (absolute) temperature (T^4). At normal temperatures ($\approx$ 300 K), we are not aware of this electromagnetic radiation because of its low intensity. At higher temperatures, there is sufficient infrared radiation that we can feel heat if we are close to the object. At still higher temperatures (on the order of 1000 K), objects actually glow, such as a red-hot electric stove burner or the heating element in a toaster. At temperatures above 2000 K, objects glow with a yellow or whitish color, such as white-hot iron and the filament of a lightbulb. The light emitted contains a continuous range of wavelengths or frequencies, and the spectrum is a plot of intensity vs. wavelength or frequency. As the temperature increases, the electromagnetic radiation emitted by objects not only increases in total intensity but has its peak intensity at higher and higher frequencies.

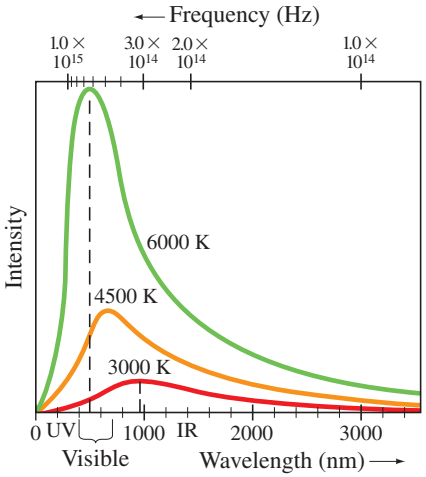


FIGURE 27–4 Measured spectra of wavelengths and frequencies emitted by a blackbody at three different temperatures.

The spectrum of light emitted by a hot dense object is shown in Fig. 27–4 for an idealized **blackbody**. A blackbody is a body that, when cool, would absorb all the radiation falling on it (and so would appear black under reflection when illuminated by other sources). The radiation such an idealized blackbody would emit when hot and luminous, called **blackbody radiation** (though not necessarily black in color), approximates that from many real objects. The 6000-K curve in Fig. 27–4, corresponding to the temperature of the surface of the Sun, peaks in the visible part of the spectrum. For lower temperatures, the total intensity drops considerably and the peak occurs at longer wavelengths (or lower frequencies). This is why objects glow with a red color at around 1000 K. It is found experimentally that the wavelength at the peak of the spectrum, λ_P, is related to the Kelvin temperature T by

$$\lambda_P T = 2.90 \times 10^{-3}\,\text{m}\cdot\text{K}. \qquad \textbf{(27–2)}$$

This is known as **Wien's law**.

EXAMPLE 27–1 **The Sun's surface temperature.** Estimate the temperature of the surface of our Sun, given that the Sun emits light whose peak intensity occurs in the visible spectrum at around 500 nm.

APPROACH We assume the Sun acts as a blackbody, and use $\lambda_P = 500\,\text{nm}$ in Wien's law (Eq. 27–2).

SOLUTION Wien's law gives

$$T = \frac{2.90 \times 10^{-3}\,\text{m}\cdot\text{K}}{\lambda_P} = \frac{2.90 \times 10^{-3}\,\text{m}\cdot\text{K}}{500 \times 10^{-9}\,\text{m}} \approx 6000\,\text{K}.$$

EXAMPLE 27–2 **Star color.** Suppose a star has a surface temperature of 32,500 K. What color would this star appear?

APPROACH We assume the star emits radiation as a blackbody, and solve for λ_P in Wien's law, Eq. 27–2.

SOLUTION From Wien's law we have

$$\lambda_P = \frac{2.90 \times 10^{-3}\,\text{m}\cdot\text{K}}{T} = \frac{2.90 \times 10^{-3}\,\text{m}\cdot\text{K}}{3.25 \times 10^{4}\,\text{K}} = 89.2\,\text{nm}.$$

The peak is in the UV range of the spectrum, and will be way to the left in Fig. 27–4. In the visible region, the curve will be descending, so the shortest visible wavelengths will be strongest. Hence the star will appear bluish (or blue-white).

NOTE This example helps us to understand why stars have different colors (reddish for the coolest stars; orangish, yellow, white, bluish for "hotter" stars.)

EXERCISE A What is the color of an object at 4000 K?

Planck's Quantum Hypothesis

In the year 1900, Max Planck (1858–1947) proposed a theory that was able to reproduce the graphs of Fig. 27–4. His theory, still accepted today, made a new and radical assumption: that the energy of the oscillations of atoms within molecules cannot have just any value; instead each has energy which is a multiple of a minimum value related to the frequency of oscillation by

$$E = hf.$$

Here h is a new constant, now called **Planck's constant**, whose value was estimated by Planck by fitting his formula for the blackbody radiation curve to experiment. The value accepted today is

$$h = 6.626 \times 10^{-34}\,\text{J}\cdot\text{s}.$$

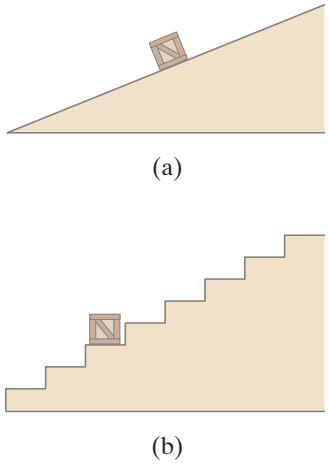
Planck's assumption suggests that the energy of any molecular vibration could be only a whole number multiple of hf:

$$E = nhf, \qquad n = 1, 2, 3, \cdots, \tag{27–3}$$

where n is called a **quantum number** ("quantum" means "discrete amount" as opposed to "continuous"). This idea is often called **Planck's quantum hypothesis**, although little attention was brought to this point at the time. In fact, it appears that Planck considered it more as a mathematical device to get the "right answer" rather than as an important discovery. Planck himself continued to seek a classical explanation for the introduction of h. The recognition that this was an important and radical innovation did not come until later, after about 1905 when others, particularly Einstein, entered the field.

The quantum hypothesis, Eq. 27–3, states that the energy of an oscillator can be $E = hf$, or $2hf$, or $3hf$, and so on, but there cannot be vibrations with energies between these values. That is, energy would not be a continuous quantity as had been believed for centuries; rather it is **quantized**—it exists only in discrete amounts. The smallest amount of energy possible (hf) is called the **quantum of energy**. Recall from Chapter 11 that the energy of an oscillation is proportional to the amplitude squared. Another way of expressing the quantum hypothesis is that not just any amplitude of vibration is possible. The possible values for the amplitude are related to the frequency f.

A simple analogy may help. Compare a ramp, on which a box can be placed at any height, to a flight of stairs on which the box can have only certain discrete amounts of potential energy, as shown in Fig. 27–5.



(a) (b)

FIGURE 27–5 Ramp versus stair analogy. (a) On a ramp, a box can have continuous values of potential energy. (b) But on stairs, the box can have only discrete (quantized) values of energy.

27–3 Photon Theory of Light and the Photoelectric Effect

In 1905, the same year that he introduced the special theory of relativity, Einstein made a bold extension of the quantum idea by proposing a new theory of light. Planck's work had suggested that the vibrational energy of molecules in a radiating object is quantized with energy $E = nhf$, where n is an integer and f is the frequency of molecular vibration. Einstein argued that when light is emitted by a molecular oscillator, the molecule's vibrational energy of nhf must decrease by an amount hf (or by $2hf$, etc.) to another integer times hf, such as $(n - 1)hf$. Then to conserve energy, the light ought to be emitted in packets, or *quanta*, each with an energy

$$E = hf, \tag{27–4}$$

Photon energy

where f is here the frequency of the emitted light. Again h is Planck's constant. Because all light ultimately comes from a radiating source, this idea suggests that *light is transmitted as tiny particles*, or **photons** as they are now called, as well as via the waves predicted by Maxwell's electromagnetic theory. The photon theory of light was also a radical departure from classical ideas. Einstein proposed a test of the quantum theory of light: quantitative measurements on the photoelectric effect.

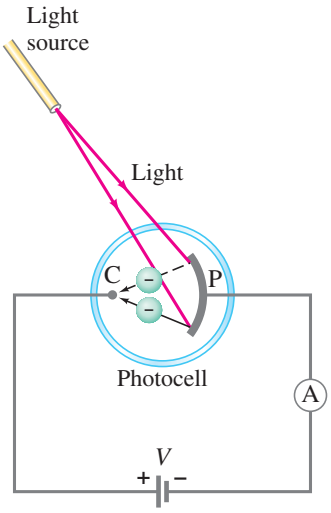


FIGURE 27–6 The photoelectric effect.

When light shines on a metal surface, electrons are found to be emitted from the surface. This effect is called the **photoelectric effect** and it occurs in many materials, but is most easily observed with metals. It can be observed using the apparatus shown in Fig. 27–6. A metal plate P and a smaller electrode C are placed inside an evacuated glass tube, called a **photocell**. The two electrodes are connected to an ammeter and a source of emf, as shown. When the photocell is in the dark, the ammeter reads zero. But when light of sufficiently high frequency illuminates the plate, the ammeter indicates a current flowing in the circuit. We explain completion of the circuit by imagining that electrons, ejected from the plate by the impinging light, flow across the tube from the plate to the "collector" C as indicated in Fig. 27–6.

That electrons should be emitted when light shines on a metal is consistent with the electromagnetic (EM) wave theory of light: the electric field of an EM wave could exert a force on electrons in the metal and eject some of them. Einstein pointed out, however, that the wave theory and the photon theory of light give very different predictions on the details of the photoelectric effect. For example, one thing that can be measured with the apparatus of Fig. 27–6 is the maximum kinetic energy (KE_{max}) of the emitted electrons. This can be done by using a variable voltage source and reversing the terminals so that electrode C is negative and P is positive. The electrons emitted from P will be repelled by the negative electrode, but if this reverse voltage is small enough, the fastest electrons will still reach C and there will be a current in the circuit. If the reversed voltage is increased, a point is reached where the current reaches zero—no electrons have sufficient kinetic energy to reach C. This is called the *stopping potential*, or *stopping voltage*, V_0, and from its measurement, KE_{max} can be determined using conservation of energy (loss of kinetic energy = gain in potential energy):

$$\text{KE}_{\text{max}} = eV_0.$$

Now let us examine the details of the photoelectric effect from the point of view of the wave theory versus Einstein's particle theory.

First the wave theory, assuming monochromatic light. The two important properties of a light wave are its intensity and its frequency (or wavelength). When these two quantities are varied, the wave theory makes the following predictions:

Wave theory predictions

1. If the light intensity is increased, the number of electrons ejected and their maximum kinetic energy should be increased because the higher intensity means a greater electric field amplitude, and the greater electric field should eject electrons with higher speed.
2. The frequency of the light should not affect the kinetic energy of the ejected electrons. Only the intensity should affect KE_{max}.

The photon theory makes completely different predictions. First we note that in a monochromatic beam, all photons have the same energy $(= hf)$. Increasing the intensity of the light beam means increasing the number of photons in the beam, but does not affect the energy of each photon as long as the frequency is not changed. According to Einstein's theory, an electron is ejected from the metal by a collision with a single photon. In the process, all the photon energy is transferred to the electron and the photon ceases to exist. Since electrons are held in the metal by attractive forces, some minimum energy W_0 is required just to get an electron out through the surface. W_0 is called the **work function**, and is a few electron volts $(1\,\text{eV} = 1.6 \times 10^{-19}\,\text{J})$ for most metals. If the frequency f of the incoming light is so low that hf is less than W_0, then the photons will not have enough energy to eject any electrons at all. If $hf > W_0$, then electrons will be ejected and energy will be conserved in the process. That is, the input energy (of the photon), hf, will equal the outgoing kinetic energy KE of the electron plus the energy required to get it out of the metal, W:

$$hf = \text{KE} + W. \qquad \textbf{(27–5a)}$$

The least tightly held electrons will be emitted with the most kinetic energy (KE_{max}),

in which case W in this equation becomes the work function W_0, and KE becomes KE_{max}:

$$hf = KE_{max} + W_0. \qquad \text{[least bound electrons]} \quad \textbf{(27–5b)}$$

Many electrons will require more energy than the bare minimum (W_0) to get out of the metal, and thus the kinetic energy of such electrons will be less than the maximum.

From these considerations, the photon theory makes the following predictions:

Photon theory predictions

1. An increase in intensity of the light beam means more photons are incident, so more electrons will be ejected; but since the energy of each photon is not changed, the maximum kinetic energy of electrons is not changed by an increase in intensity.
2. If the frequency of the light is increased, the maximum kinetic energy of the electrons increases linearly, according to Eq. 27–5b. That is,

$$KE_{max} = hf - W_0.$$

This relationship is plotted in Fig. 27–7.

3. If the frequency f is less than the "cutoff" frequency f_0, where $hf_0 = W_0$, no electrons will be ejected, no matter how great the intensity of the light.

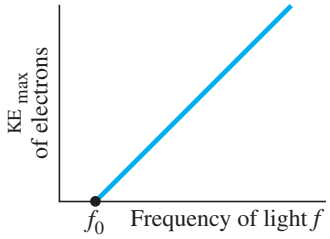


FIGURE 27–7 Photoelectric effect: the maximum kinetic energy of ejected electrons increases linearly with the frequency of incident light. No electrons are emitted if $f < f_0$.

These predictions of the photon theory are very different from the predictions of the wave theory. In 1913–1914, careful experiments were carried out by R. A. Millikan. The results were fully in agreement with Einstein's photon theory.

One other aspect of the photoelectric effect also confirmed the photon theory. If extremely low light intensity is used, the wave theory predicts a time delay before electron emission so that an electron can absorb enough energy to exceed the work function. The photon theory predicts no such delay—it only takes one photon (if its frequency is high enough) to eject an electron—and experiments showed no delay. This too confirmed Einstein's photon theory.

EXAMPLE 27–3 **Photon energy.** Calculate the energy of a photon of blue light, $\lambda = 450\,\text{nm}$ in air (or vacuum).

APPROACH The photon has energy $E = hf$ (Eq. 27–4) where $f = c/\lambda$ (Eq. 22–4).

SOLUTION Since $f = c/\lambda$, we have

$$E = hf = \frac{hc}{\lambda} = \frac{(6.63 \times 10^{-34}\,\text{J}\cdot\text{s})(3.00 \times 10^{8}\,\text{m/s})}{(4.5 \times 10^{-7}\,\text{m})} = 4.4 \times 10^{-19}\,\text{J},$$

or $(4.4 \times 10^{-19}\,\text{J})/(1.60 \times 10^{-19}\,\text{J/eV}) = 2.8\,\text{eV}$. (See definition of eV in Section 17–4, $1\,\text{eV} = 1.60 \times 10^{-19}\,\text{J}$.)

EXAMPLE 27–4 **ESTIMATE** **Photons from a lightbulb.** Estimate how many visible light photons a 100-W lightbulb emits per second. Assume the bulb has a typical efficiency of about 3% (that is, 97% of the energy goes to heat).

APPROACH Let's assume an average wavelength in the middle of the visible spectrum, $\lambda \approx 500\,\text{nm}$. The energy of each photon is $E = hf = hc/\lambda$. Only 3% of the 100-W power is emitted as visible light, or $3\,\text{W} = 3\,\text{J/s}$. The number of photons emitted per second equals the light output of 3 J/s divided by the energy of each photon.

SOLUTION The energy emitted in one second $(= 3\,\text{J})$ is $E = Nhf$ where N is the number of photons emitted per second and $f = c/\lambda$. Hence

$$N = \frac{E}{hf} = \frac{E\lambda}{hc} = \frac{(3\,\text{J})(500 \times 10^{-9}\,\text{m})}{(6.63 \times 10^{-34}\,\text{J}\cdot\text{s})(3.00 \times 10^{8}\,\text{m/s})} \approx 8 \times 10^{18}$$

per second, or almost 10^{19} photons emitted per second, an enormous number.

EXERCISE B A beam contains infrared light of a single wavelength, 1000 nm, and monochromatic UV at 100 nm, both of the same intensity. Are there more 100-nm photons or more 1000-nm photons?

EXAMPLE 27–5 Photoelectron speed and energy. What is the kinetic energy and the speed of an electron ejected from a sodium surface whose work function is $W_0 = 2.28\,\text{eV}$ when illuminated by light of wavelength (*a*) 410 nm, (*b*) 550 nm?

APPROACH We first find the energy of the photons ($E = hf = hc/\lambda$). If the energy is greater than W_0, then electrons will be ejected with varying amounts of KE, with a maximum of $\text{KE}_{\text{max}} = hf - W_0$.

SOLUTION (*a*) For $\lambda = 410\,\text{nm}$,

$$hf = \frac{hc}{\lambda} = 4.85 \times 10^{-19}\,\text{J} \qquad \text{or} \qquad 3.03\,\text{eV}.$$

The maximum kinetic energy an electron can have is given by Eq. 27–5b, $\text{KE}_{\text{max}} = 3.03\,\text{eV} - 2.28\,\text{eV} = 0.75\,\text{eV}$, or $(0.75\,\text{eV})(1.60 \times 10^{-19}\,\text{J/eV}) = 1.2 \times 10^{-19}\,\text{J}$. Since $\text{KE} = \frac{1}{2}mv^2$ where $m = 9.1 \times 10^{-31}\,\text{kg}$,

$$v_{\text{max}} = \sqrt{\frac{2\text{KE}}{m}} = 5.1 \times 10^5\,\text{m/s}.$$

Most ejected electrons will have less KE and less speed than these maximum values.

(*b*) For $\lambda = 550\,\text{nm}$, $hf = hc/\lambda = 3.61 \times 10^{-19}\,\text{J} = 2.26\,\text{eV}$. Since this photon energy is less than the work function, no electrons are ejected.

NOTE In (*a*) we used the nonrelativistic equation for kinetic energy. If v had turned out to be more than about $0.1c$, our calculation would have been inaccurate by more than a percent or so, and we would probably prefer to redo it using the relativistic form (Eq. 26–5).

EXERCISE C Determine the lowest frequency and the longest wavelength needed to emit electrons from sodium.

By converting units, we can show that the energy of a photon in electron volts, when given the wavelength λ in nm, is

$$E\,(\text{eV}) = \frac{1.240 \times 10^3\,\text{eV}\cdot\text{nm}}{\lambda\,(\text{nm})}. \qquad \text{[photon energy in eV]}$$

Applications of the Photoelectric Effect

The photoelectric effect, besides playing an important historical role in confirming the photon theory of light, also has many practical applications. Burglar alarms and automatic doors often make use of the photocell circuit of Fig. 27–6. When a person interrupts the beam of light, the sudden drop in current in the circuit activates a switch—often a solenoid—which operates a bell or opens the door. UV or IR light is sometimes used in burglar alarms because of its invisibility. Many smoke detectors use the photoelectric effect to detect tiny amounts of smoke that interrupt the flow of light and so alter the electric current. Photographic light meters use this circuit as well. Photocells are used in many other devices, such as absorption spectrophotometers, to measure light intensity. One type of film sound track is a variably shaded narrow section at the side of the film, Fig. 27–8. Light passing through the film is thus "modulated," and the output electrical signal of the photocell detector follows the frequencies on the sound track. For many applications today, the vacuum-tube photocell of Fig. 27–6 has been replaced by a semiconductor device known as a **photodiode** (Section 29–9). In these semiconductors, the absorption of a photon liberates a bound electron so it can move freely, which changes the conductivity of the material and the current through a photodiode is altered.

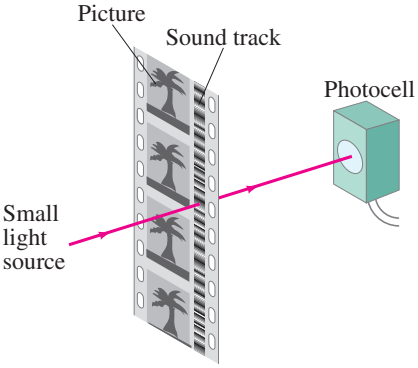


FIGURE 27–8 Optical sound track on movie film. In the projector, light from a small source (different from that for the picture) passes through the sound track on the moving film.

27–4 Energy, Mass, and Momentum of a Photon

We have just seen (Eq. 27–4) that the total energy of a single photon is given by $E = hf$. Because a photon always travels at the speed of light, it is truly a relativistic particle. Thus we must use relativistic formulas for dealing with its mass, energy, and momentum. The momentum of any particle of mass m is given by $p = mv/\sqrt{1 - v^2/c^2}$. Since $v = c$ for a photon, the denominator is zero. To avoid having an infinite momentum, we conclude that the photon's mass must be zero: $m = 0$. This makes sense too because a photon can never be at rest (it always moves at the speed of light). A photon's kinetic energy is its total energy:

$$\text{KE} = E = hf. \qquad \text{[photon]}$$

The momentum of a photon can be obtained from the relativistic formula (Eq. 26–9) $E^2 = p^2c^2 + m^2c^4$ where we set $m = 0$, so $E^2 = p^2c^2$ or

$$p = \frac{E}{c}. \qquad \text{[photon]}$$

CAUTION
Momentum of photon is not *mv*

Since $E = hf$ for a photon, its momentum is related to its wavelength by

$$p = \frac{E}{c} = \frac{hf}{c} = \frac{h}{\lambda}. \qquad \textbf{(27–6)}$$

EXAMPLE 27–6 ESTIMATE **Photon momentum and force.** Suppose the 10^{19} photons emitted per second from the 100-W lightbulb in Example 27–4 were all focused onto a piece of black paper and absorbed. (*a*) Calculate the momentum of one photon and (*b*) estimate the force all these photons could exert on the paper.

APPROACH Each photon's momentum is obtained from Eq. 27–6, $p = h/\lambda$. Next, each absorbed photon's momentum changes from $p = h/\lambda$ to zero. We use Newton's second law, $F = \Delta p/\Delta t$, to get the force. Let $\lambda = 500\,\text{nm}$.

SOLUTION (*a*) Each photon has a momentum

$$p = \frac{h}{\lambda} = \frac{6.63 \times 10^{-34}\,\text{J}\cdot\text{s}}{500 \times 10^{-9}\,\text{m}} = 1.3 \times 10^{-27}\,\text{kg}\cdot\text{m/s}.$$

(*b*) Using Newton's second law for $N = 10^{19}$ photons (Example 27–4) whose momentum changes from h/λ to 0, we obtain

$$F = \frac{\Delta p}{\Delta t} = \frac{Nh/\lambda - 0}{1\,\text{s}} = N\frac{h}{\lambda} \approx (10^{19}\,\text{s}^{-1})(10^{-27}\,\text{kg}\cdot\text{m/s}) \approx 10^{-8}\,\text{N}.$$

NOTE This is a tiny force, but we can see that a very strong light source could exert a measurable force, and near the Sun or a star the force due to photons in electromagnetic radiation could be considerable. See Section 22–6.

EXAMPLE 27–7 **Photosynthesis.** In *photosynthesis*, pigments such as chlorophyll in plants capture the energy of sunlight to change CO_2 to useful carbohydrate. About nine photons are needed to transform one molecule of CO_2 to carbohydrate and O_2. Assuming light of wavelength $\lambda = 670\,\text{nm}$ (chlorophyll absorbs most strongly in the range 650 nm to 700 nm), how efficient is the photosynthetic process? The reverse chemical reaction releases an energy of 4.9 eV/molecule of CO_2, so 4.9 eV is needed to transform CO_2 to carbohydrate.

PHYSICS APPLIED
Photosynthesis

APPROACH The efficiency is the minimum energy required (4.9 eV) divided by the actual energy absorbed, nine times the energy (hf) of one photon.

SOLUTION The energy of nine photons, each of energy $hf = hc/\lambda$, is $(9)(6.63 \times 10^{-34}\,\text{J}\cdot\text{s})(3.00 \times 10^{8}\,\text{m/s})/(6.7 \times 10^{-7}\,\text{m}) = 2.7 \times 10^{-18}\,\text{J}$ or 17 eV. Thus the process is about $(4.9\,\text{eV}/17\,\text{eV}) = 29\%$ efficient.

*27–5 Compton Effect

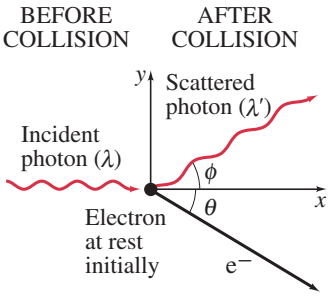


FIGURE 27–9 The Compton effect. A single photon of wavelength λ strikes an electron in some material, knocking it out of its atom. The scattered photon has less energy (some energy is given to the electron) and hence has a longer wavelength λ' (shown exaggerated). Experiments found scattered X-rays of just the wavelengths predicted by conservation of energy and momentum using the photon model.

Besides the photoelectric effect, a number of other experiments were carried out in the early twentieth century which also supported the photon theory. One of these was the **Compton effect** (1923) named after its discoverer, A. H. Compton (1892–1962). Compton aimed short-wavelength light (actually X-rays) at various materials, and detected light scattered at various angles. He found that the scattered light had a slightly longer wavelength than did the incident light, and therefore a slightly lower frequency indicating a loss of energy. He explained this result on the basis of the photon theory as incident photons colliding with electrons of the material, Fig. 27–9. Using Eq. 27–6 for momentum of a photon, Compton applied the laws of conservation of momentum and energy to the collision of Fig. 27–9 and derived the following equation for the wavelength of the scattered photons:

$$\lambda' = \lambda + \frac{h}{m_e c}(1 - \cos\phi), \qquad \textbf{(27–7)}$$

where m_e is the mass of the electron. (The quantity $h/m_e c$, which has the dimensions of length, is called the **Compton wavelength** of the electron.) We see that the predicted wavelength of scattered photons depends on the angle ϕ at which they are detected. Compton's measurements of 1923 were consistent with this formula. The wave theory of light predicts no such shift: an incoming electromagnetic wave of frequency f should set electrons into oscillation at frequency f; and such oscillating electrons would reemit EM waves of this same frequency f (Section 22–2), which would not change with angle (ϕ). Hence the Compton effect adds to the firm experimental foundation for the photon theory of light.

EXERCISE D When a photon scatters off an electron by the Compton effect, which of the following increases: its energy, frequency, wavelength?

EXAMPLE 27–8 **X-ray scattering.** X-rays of wavelength 0.140 nm are scattered from a very thin slice of carbon. What will be the wavelengths of X-rays scattered at (*a*) 0°, (*b*) 90°, (*c*) 180°?

APPROACH This is an example of the Compton effect, and we use Eq. 27–7 to find the wavelengths.

SOLUTION (*a*) For $\phi = 0°$, $\cos\phi = 1$ and $1 - \cos\phi = 0$. Then Eq. 27–7 gives $\lambda' = \lambda = 0.140$ nm. This makes sense since for $\phi = 0°$, there really isn't any collision as the photon goes straight through without interacting.

(*b*) For $\phi = 90°$, $\cos\phi = 0$, and $1 - \cos\phi = 1$. So

$$\lambda' = \lambda + \frac{h}{m_e c} = 0.140\text{ nm} + \frac{6.63 \times 10^{-34}\text{ J}\cdot\text{s}}{(9.11 \times 10^{-31}\text{ kg})(3.00 \times 10^{8}\text{ m/s})}$$
$$= 0.140\text{ nm} + 2.4 \times 10^{-12}\text{ m} = 0.142\text{ nm};$$

that is, the wavelength is longer by one Compton wavelength ($= h/m_e c^2 = 0.0024$ nm for an electron).

(*c*) For $\phi = 180°$, which means the photon is scattered backward, returning in the direction from which it came (a direct "head-on" collision), $\cos\phi = -1$, and $1 - \cos\phi = 2$. So

$$\lambda' = \lambda + 2\frac{h}{m_e c} = 0.140\text{ nm} + 2(0.0024\text{ nm}) = 0.145\text{ nm}.$$

NOTE The maximum shift in wavelength occurs for backward scattering, and it is twice the Compton wavelength.

PHYSICS APPLIED
Measuring bone density

The Compton effect has been used to diagnose bone disease such as osteoporosis. Gamma rays, which are photons of even shorter wavelength than X-rays, coming from a radioactive source are scattered off bone material. The total intensity of the scattered radiation is proportional to the density of electrons, which is in turn proportional to the bone density. A low bone density may indicate osteoporosis.

27–6 Photon Interactions; Pair Production

When a photon passes through matter, it interacts with the atoms and electrons. There are four important types of interactions that a photon can undergo:

1. The *photoelectric effect*: A photon may knock an electron out of an atom and in the process the photon disappears.
2. The photon may knock an atomic electron to a higher energy state in the atom if its energy is not sufficient to knock the electron out altogether. In this process the photon also disappears, and all its energy is given to the atom. Such an atom is then said to be in an *excited state*, and we shall discuss it more later.
3. The photon can be scattered from an electron (or a nucleus) and in the process lose some energy; this is the *Compton effect* (Fig. 27–9). But notice that the photon is not slowed down. It still travels with speed c, but its frequency will be lower because it has lost some energy.
4. *Pair production*: A photon can actually create matter, such as the production of an electron and a positron, Fig. 27–10. (A positron has the same mass as an electron, but the opposite charge, $+e$.)

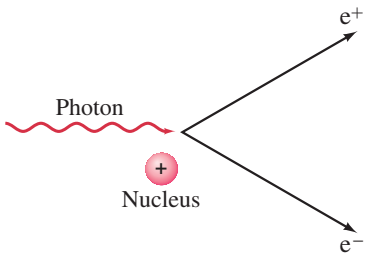


FIGURE 27–10 Pair production: a photon disappears and produces an electron and a positron.

In process 4, **pair production**, the photon disappears in the process of creating the electron–positron pair. This is an example of mass being created from pure energy, and it occurs in accord with Einstein's equation $E = mc^2$. Notice that a photon cannot create an electron alone since electric charge would not then be conserved. The inverse of pair production also occurs: if a positron comes close to an electron, the two quickly **annihilate** each other and their energy, including their mass, appears as electromagnetic energy of photons. Because positrons are not as plentiful in nature as electrons, they usually do not last long.

Electron–positron annihilation is the basis for the type of medical imaging known as PET, as discussed in Section 31–8.

EXAMPLE 27–9 **Pair production.** (a) What is the minimum energy of a photon that can produce an electron–positron pair? (b) What is this photon's wavelength?

APPROACH The minimum photon energy E equals the rest energy (mc^2) of the two particles created, via Einstein's famous equation $E = mc^2$ (Eq. 26–7). There is no energy left over, so the particles produced will have zero kinetic energy. The wavelength is $\lambda = c/f$ where $E = hf$ for the original photon.

SOLUTION (a) Because $E = mc^2$, and the mass created is equal to two electron masses, the photon must have energy

$$E = 2(9.11 \times 10^{-31}\,\text{kg})(3.00 \times 10^{8}\,\text{m/s})^2 = 1.64 \times 10^{-13}\,\text{J} = 1.02\,\text{MeV}$$

($1\,\text{MeV} = 10^6\,\text{eV} = 1.60 \times 10^{-13}\,\text{J}$). A photon with less energy cannot undergo pair production.

(b) Since $E = hf = hc/\lambda$, the wavelength of a 1.02-MeV photon is

$$\lambda = \frac{hc}{E} = \frac{(6.63 \times 10^{-34}\,\text{J}\cdot\text{s})(3.00 \times 10^{8}\,\text{m/s})}{(1.64 \times 10^{-13}\,\text{J})} = 1.2 \times 10^{-12}\,\text{m},$$

which is 0.0012 nm. Such photons are in the gamma-ray (or very short X-ray) region of the electromagnetic spectrum (Fig. 22–8).

NOTE Photons of higher energy (shorter wavelength) can also create an electron–positron pair, with the excess energy becoming kinetic energy of the particles.

Pair production cannot occur in empty space, for momentum could not be conserved. In Example 27–9, for instance, energy is conserved, but only enough energy was provided to create the electron–positron pair at rest and thus with zero momentum, which could not equal the initial momentum of the photon. Indeed, it can be shown that at any energy, an additional massive object, such as an atomic nucleus (Fig. 27–10), must take part in the interaction to carry off some of the momentum.

27–7 Wave–Particle Duality; the Principle of Complementarity



FIGURE 27–11 Niels Bohr (right), walking with Enrico Fermi along the Appian Way outside Rome. This photo shows one important way physics is done.

The photoelectric effect, the Compton effect, and other experiments have placed the particle theory of light on a firm experimental basis. But what about the classic experiments of Young and others (Chapter 24) on interference and diffraction which showed that the wave theory of light also rests on a firm experimental basis?

We seem to be in a dilemma. Some experiments indicate that light behaves like a wave; others indicate that it behaves like a stream of particles. These two theories seem to be incompatible, but both have been shown to have validity. Physicists finally came to the conclusion that this duality of light must be accepted as a fact of life. It is referred to as the **wave–particle duality**. Apparently, light is a more complex phenomenon than just a simple wave or a simple beam of particles.

To clarify the situation, the great Danish physicist Niels Bohr (1885–1962, Fig. 27–11) proposed his famous **principle of complementarity**. It states that to understand an experiment, sometimes we find an explanation using wave theory and sometimes using particle theory. Yet we must be aware of both the wave and particle aspects of light if we are to have a full understanding of light. Therefore these two aspects of light complement one another.

It is not easy to "visualize" this duality. We cannot readily picture a combination of wave and particle. Instead, we must recognize that the two aspects of light are different "faces" that light shows to experimenters.

Part of the difficulty stems from how we think. Visual pictures (or models) in our minds are based on what we see in the everyday world. We apply the concepts of waves and particles to light because in the macroscopic world we see that energy is transferred from place to place by these two methods. We cannot see directly whether light is a wave or particle, so we do indirect experiments. To explain the experiments, we apply the models of waves or of particles to the nature of light. But these are abstractions of the human mind. When we try to conceive of what light really "is," we insist on a visual picture. Yet there is no reason why light should conform to these models (or visual images) taken from the macroscopic world. The "true" nature of light—if that means anything—is not possible to visualize. The best we can do is recognize that our knowledge is limited to the indirect experiments, and that in terms of everyday language and images, light reveals both wave and particle properties.

CAUTION

Not correct to say light is a wave and/or a particle. Light can **act** *like a wave or like a particle*

It is worth noting that Einstein's equation $E = hf$ itself links the particle and wave properties of a light beam. In this equation, E refers to the energy of a particle; and on the other side of the equation, we have the frequency f of the corresponding wave.

27–8 Wave Nature of Matter

In 1923, Louis de Broglie (1892–1987) extended the idea of the wave–particle duality. He appreciated the *symmetry* in nature, and argued that if light sometimes behaves like a wave and sometimes like a particle, then perhaps those things in nature thought to be particles—such as electrons and other material objects—might also have wave properties. De Broglie proposed that the wavelength of a material particle would be related to its momentum in the same way as for a photon, Eq. 27–6, $p = h/\lambda$. That is, for a particle having linear momentum $p = mv$, the wavelength λ is given by

de Broglie wavelength

$$\lambda = \frac{h}{p}, \qquad \textbf{(27–8)}$$

and is valid classically ($p = mv$ for $v \ll c$) and relativistically $\left(p = \gamma mv = mv/\sqrt{1 - v^2/c^2}\right)$. This is sometimes called the **de Broglie wavelength** of a particle.

EXAMPLE 27–10 **Wavelength of a ball.** Calculate the de Broglie wavelength of a 0.20-kg ball moving with a speed of 15 m/s.

APPROACH We use Eq. 27–8.

SOLUTION $\lambda = \dfrac{h}{p} = \dfrac{h}{mv} = \dfrac{(6.6 \times 10^{-34}\,\text{J}\cdot\text{s})}{(0.20\,\text{kg})(15\,\text{m/s})} = 2.2 \times 10^{-34}\,\text{m}.$

Ordinary objects, such as the ball of Example 27–10, have unimaginably small wavelengths. Even if the speed is extremely small, say 10^{-4} m/s, the wavelength would be about 10^{-29} m. Indeed, the wavelength of any ordinary object is much too small to be measured and detected. The problem is that the properties of waves, such as interference and diffraction, are significant only when the size of objects or slits is not much larger than the wavelength. And there are no known objects or slits to diffract waves only 10^{-30} m long, so the wave properties of ordinary objects go undetected.

But tiny elementary particles, such as electrons, are another matter. Since the mass m appears in the denominator of Eq. 27–8, a very small mass should have a much larger wavelength.

EXAMPLE 27–11 **Wavelength of an electron.** Determine the wavelength of an electron that has been accelerated through a potential difference of 100 V.

APPROACH If the kinetic energy is much less than the rest energy, we can use the classical formula, $\text{KE} = \frac{1}{2}mv^2$ (see end of Section 26–9). For an electron, $mc^2 = 0.511$ MeV. We then apply conservation of energy: the kinetic energy acquired by the electron equals its loss in potential energy. After solving for v, we use Eq. 27–8 to find the de Broglie wavelength.

SOLUTION The gain in kinetic energy equals the loss in potential energy: $\Delta\text{PE} = eV - 0$. Thus $\text{KE} = eV$, so $\text{KE} = 100\,\text{eV}$. The ratio $\text{KE}/mc^2 = 100\,\text{eV}/(0.511 \times 10^6\,\text{eV}) \approx 10^{-4}$, so relativity is not needed. Thus

$$\frac{1}{2}mv^2 = eV$$

and

$$v = \sqrt{\frac{2\,eV}{m}} = \sqrt{\frac{(2)(1.6 \times 10^{-19}\,\text{C})(100\,\text{V})}{(9.1 \times 10^{-31}\,\text{kg})}} = 5.9 \times 10^6\,\text{m/s}.$$

Then

$$\lambda = \frac{h}{mv} = \frac{(6.63 \times 10^{-34}\,\text{J}\cdot\text{s})}{(9.1 \times 10^{-31}\,\text{kg})(5.9 \times 10^6\,\text{m/s})} = 1.2 \times 10^{-10}\,\text{m},$$

or 0.12 nm.

EXERCISE E As a particle travels faster, does its de Broglie wavelength decrease, increase, or remain the same?

EXERCISE F Return to the Chapter-Opening Question, page 771, and answer it again now. Try to explain why you may have answered differently the first time.

Electron Diffraction

From Example 27–11, we see that electrons can have wavelengths on the order of 10^{-10} m, and even smaller. Although small, this wavelength can be detected: the spacing of atoms in a crystal is on the order of 10^{-10} m and the orderly array of atoms in a crystal could be used as a type of diffraction grating, as was done earlier for X-rays (see Section 25–11). C. J. Davisson and L. H. Germer performed the crucial experiment: they scattered electrons from the surface of a metal crystal and, in early 1927, observed that the electrons were scattered into a pattern of regular peaks. When they interpreted these peaks as a diffraction pattern, the wavelength of the diffracted electron wave was found to be just that predicted by de Broglie, Eq. 27–8. In the same year, G. P. Thomson (son of J. J. Thomson) used a different experimental arrangement and also detected diffraction of electrons. (See Fig. 27–12. Compare it to X-ray diffraction, Section 25–11.) Later experiments showed that protons, neutrons, and other particles also have wave properties.

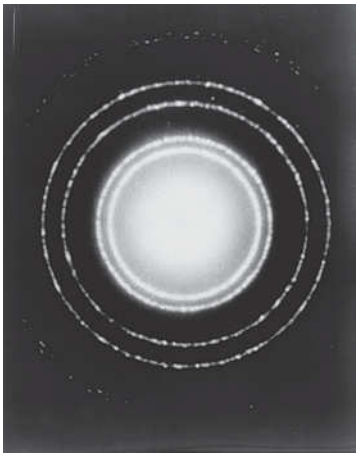


FIGURE 27–12 Diffraction pattern of electrons scattered from aluminum foil, as recorded on film.

Thus the wave–particle duality applies to material objects as well as to light. The principle of complementarity applies to matter as well. That is, we must be aware of both the particle and wave aspects in order to have an understanding of matter, including electrons. But again we must recognize that a visual picture of a "wave–particle" is not possible.

PHYSICS APPLIED
Electron diffraction

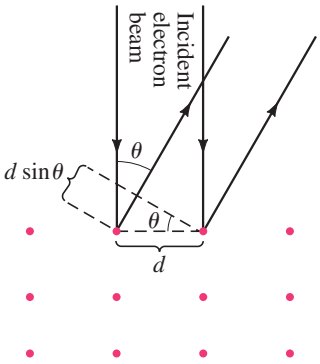


FIGURE 27–13 Example 27–12. The red dots represent atoms in an orderly array in a solid.

EXAMPLE 27–12 **Electron diffraction.** The wave nature of electrons is manifested in experiments where an electron beam interacts with the atoms on the surface of a solid, especially crystals. By studying the angular distribution of the diffracted electrons, one can indirectly measure the geometrical arrangement of atoms. Assume that the electrons strike perpendicular to the surface of a solid (see Fig. 27–13), and that their energy is low, KE = 100 eV, so that they interact only with the surface layer of atoms. If the smallest angle at which a diffraction maximum occurs is at 24°, what is the separation d between the atoms on the surface?

SOLUTION Treating the electrons as waves, we need to determine the condition where the difference in path traveled by the wave diffracted from adjacent atoms is an integer multiple of the de Broglie wavelength, so that constructive interference occurs. The path length difference is $d \sin\theta$ (Fig. 27–13); so for the smallest value of θ we must have

$$d \sin\theta = \lambda.$$

However, λ is related to the (non-relativistic) kinetic energy KE by

$$\text{KE} = \frac{p^2}{2m_e} = \frac{h^2}{2m_e\lambda^2}.$$

Thus

$$\lambda = \frac{h}{\sqrt{2m_e\,\text{KE}}}$$

$$= \frac{(6.63 \times 10^{-34}\,\text{J}\cdot\text{s})}{\sqrt{2(9.11 \times 10^{-31}\,\text{kg})(100\,\text{eV})(1.6 \times 10^{-19}\,\text{J/eV})}} = 0.123\,\text{nm}.$$

The surface inter-atomic spacing is

$$d = \frac{\lambda}{\sin\theta} = \frac{0.123\,\text{nm}}{\sin 24^\circ} = 0.30\,\text{nm}.$$

NOTE Experiments of this type verify both the wave nature of electrons and the orderly array of atoms in crystalline solids.

What Is an Electron?

We might ask ourselves: "What is an electron?" The early experiments of J. J. Thomson (Section 27–1) indicated a glow in a tube, and that glow moved when a magnetic field was applied. The results of these and other experiments were best interpreted as being caused by tiny negatively charged particles which we now call electrons. No one, however, has actually seen an electron directly. The drawings we sometimes make of electrons as tiny spheres with a negative charge on them are merely convenient pictures (now recognized to be inaccurate). Again we must rely on experimental results, some of which are best interpreted using the particle model and others using the wave model. These models are mere pictures that we use to extrapolate from the macroscopic world to the tiny microscopic world of the atom. And there is no reason to expect that these models somehow reflect the reality of an electron. We thus use a wave or a particle model (whichever works best in a situation) so that we can talk about what is happening. But we should not be led to believe that an electron *is* a wave or a particle. Instead we could say that an electron is the set of its properties that we can measure. Bertrand Russell said it well when he wrote that an electron is "a logical construction."

27–9 Electron Microscopes

The idea that electrons have wave properties led to the development of the **electron microscope** (EM), which can produce images of much greater magnification than a light microscope. Figures 27–14 and 27–15 are diagrams of two types, developed around the middle of the twentieth century: the **transmission electron microscope** (TEM), which produces a two-dimensional image, and the **scanning electron microscope** (SEM), which produces images with a three-dimensional quality.

PHYSICS APPLIED
Electron microscope

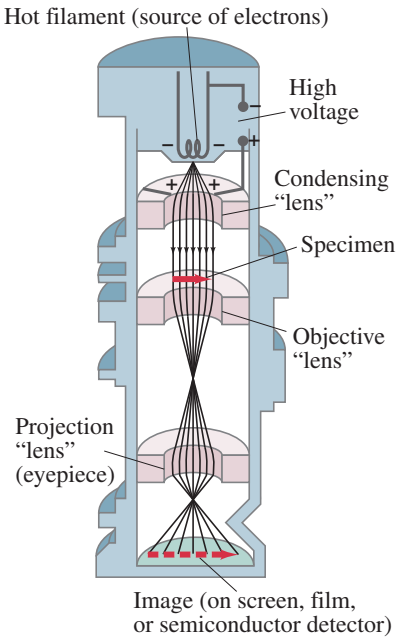


FIGURE 27–14 Transmission electron microscope. The magnetic field coils are designed to be "magnetic lenses," which bend the electron paths and bring them to a focus, as shown. The sensors of the image measure electron intensity only, no color.

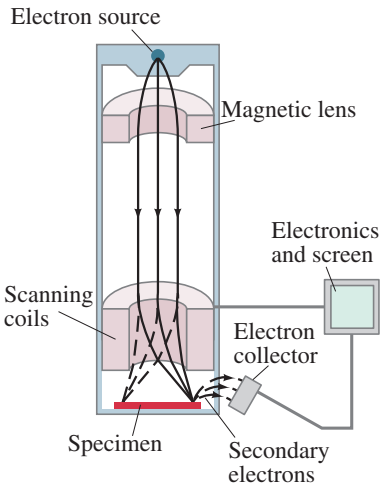


FIGURE 27–15 Scanning electron microscope. Scanning coils move an electron beam back and forth across the specimen. Secondary electrons produced when the beam strikes the specimen are collected and their intensity affects the brightness of pixels in a monitor to produce a picture.

In both types, the objective and eyepiece lenses are actually magnetic fields that exert forces on the electrons to bring them to a focus. The fields are produced by carefully designed current-carrying coils of wire. Photographs using each type are shown in Fig. 27–16. EMs measure the intensity of electrons, producing monochromatic photos. Color is often added artificially to highlight.

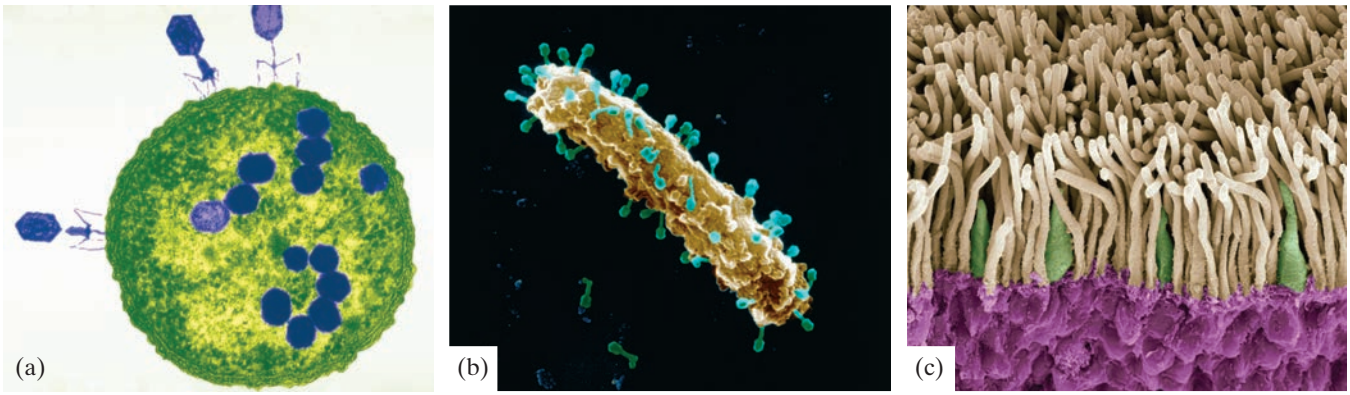


FIGURE 27–16 Electron micrographs, in false color, of (a) viruses attacking a cell of the bacterium *Escherichia coli* (TEM, ≈ 50,000×). (b) Same subject by an SEM (≈ 35,000×). (c) SEM image of an eye's retina (Section 25–2); the rods and cones have been colored beige and green, respectively. Part (c) is also on the cover of this book.

As discussed in Sections 25–7 and 25–8, the maximum resolution of details on an object is about the size of the wavelength of the radiation used to view it. Electrons accelerated by voltages on the order of 10^5 V have wavelengths of about 0.004 nm. The maximum resolution obtainable would be on this order, but in practice, aberrations in the magnetic lenses limit the resolution in transmission electron microscopes to about 0.1 to 0.5 nm. This is still 1000 times better than a visible-light microscope, and corresponds to a useful magnification of about a million. Such magnifications are difficult to achieve, and more common magnifications are 10^4 to 10^5. The maximum resolution of a scanning electron microscope is less, typically 5 to 10 nm although new high-resolution SEMs approach 1 nm.

PHYSICS APPLIED
STM and AFM

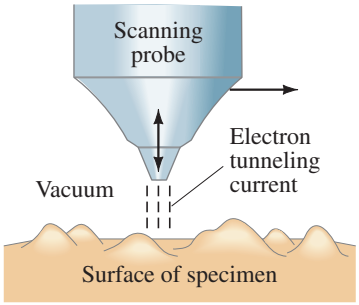


FIGURE 27–17 The probe tip of a scanning tunneling electron microscope, as it is moved horizontally, automatically moves up and down to maintain a constant tunneling current, and this motion is translated into an image of the surface.

The **scanning tunneling electron microscope** (STM), developed in the 1980s, contains a tiny probe, whose tip may be only one (or a few) atoms wide, that is moved across the specimen to be examined in a series of linear passes. The tip, as it scans, remains very close to the surface of the specimen, about 1 nm above it, Fig. 27–17. A small voltage applied between the probe and the surface causes electrons to leave the surface and pass through the vacuum to the probe, by a process known as *tunneling* (discussed in Section 30–12). This "tunneling" current is very sensitive to the gap width, so a feedback mechanism can be used to raise and lower the probe to maintain a constant electron current. The probe's vertical motion, following the surface of the specimen, is then plotted as a function of position, scan after scan, producing a three-dimensional image of the surface. Surface features as fine as the size of an atom can be resolved: a resolution better than 50 pm (0.05 nm) laterally and 0.01 to 0.001 nm vertically. This kind of resolution has given a great impetus to the study of the surface structure of materials. The "topographic" image of a surface actually represents the distribution of electron charge.

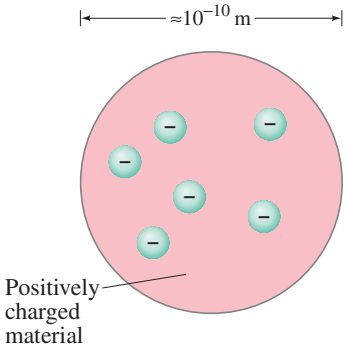
The **atomic force microscope** (AFM), developed in the 1980s, is in many ways similar to an STM, but can be used on a wider range of sample materials. Instead of detecting an electric current, the AFM measures the force between a cantilevered tip and the sample, a force which depends strongly on the tip–sample separation at each point. The tip is moved as for the STM.

27–10 Early Models of the Atom

The idea that matter is made up of atoms was accepted by most scientists by 1900. With the discovery of the electron in the 1890s, scientists began to think of the atom itself as having a structure with electrons as part of that structure. We now discuss how our modern view of the atom developed, and the quantum theory with which it is intertwined.†

A typical model of the atom in the 1890s visualized the atom as a homogeneous sphere of positive charge inside of which there were tiny negatively charged electrons, a little like plums in a pudding, Fig. 27–18.

FIGURE 27–18 Plum-pudding model of the atom.



Around 1911, Ernest Rutherford (1871–1937) and his colleagues performed experiments whose results contradicted the plum-pudding model of the atom. In these experiments a beam of positively charged alpha (α) particles was directed at a thin sheet of metal foil such as gold, Fig. 27–19. (These newly discovered α particles were emitted by certain radioactive materials and were soon shown to be doubly ionized helium atoms—that is, having a charge of $+2e$.) It was expected from the plum-pudding model that the alpha particles would not be deflected significantly because electrons are so much lighter than alpha particles, and the alpha particles should not have encountered any massive concentration of positive charge to strongly repel them. The experimental results completely contradicted these predictions. It was found that most of the alpha particles passed through the foil unaffected, as if the foil were mostly empty space.

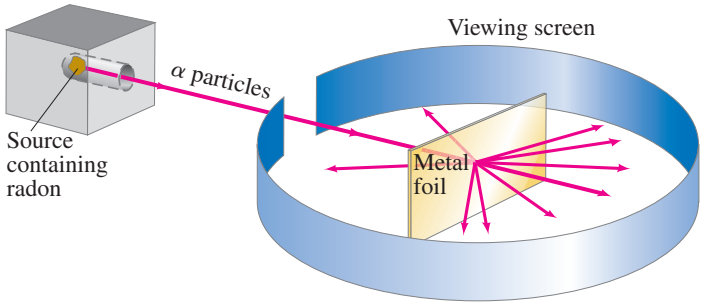


FIGURE 27–19 Experimental setup for Rutherford's experiment: α particles emitted by radon are deflected by the atoms of a thin metal foil and a few rebound backward.

†Some readers may say: "Tell us the facts as we know them today, and don't bother us with the historical background and its outmoded theories." Such an approach would ignore the creative aspect of science and thus give a false impression of how science develops. Moreover, it is not really possible to understand today's view of the atom without insight into the concepts that led to it.

And of those deflected, a few were deflected at very large angles—some even backward, nearly in the direction from which they had come. This could happen, Rutherford reasoned, only if the positively charged alpha particles were being repelled by a massive positive charge concentrated in a very small region of space (see Fig. 27–20). He hypothesized that the atom must consist of a tiny but massive positively charged nucleus, containing over 99.9% of the mass of the atom, surrounded by much lighter electrons some distance away. The electrons would be moving in orbits about the nucleus—much as the planets move around the Sun—because if they were at rest, they would fall into the nucleus due to electrical attraction. See Fig. 27–21. Rutherford's experiments suggested that the nucleus must have a radius of about 10^{-15} to 10^{-14} m. From kinetic theory, and especially Einstein's analysis of Brownian motion (see Section 13–1), the radius of atoms was estimated to be about 10^{-10} m. Thus the electrons would seem to be at a distance from the nucleus of about 10,000 to 100,000 times the radius of the nucleus itself. (If the nucleus were the size of a baseball, the atom would have the diameter of a big city several kilometers across.) So an atom would be mostly empty space.

Rutherford's **planetary model** of the atom (also called the **nuclear model** of the atom) was a major step toward how we view the atom today. It was not, however, a complete model and presented some major problems, as we shall see.

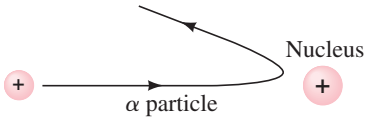
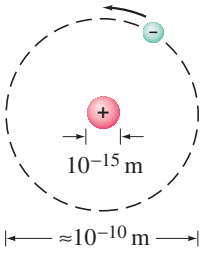


FIGURE 27–20 Backward rebound of α particles in Fig. 27–19 explained as the repulsion from a heavy positively charged nucleus.

FIGURE 27–21 Rutherford's model of the atom: electrons orbit a tiny positive nucleus (not to scale). The atom is visualized as mostly empty space.



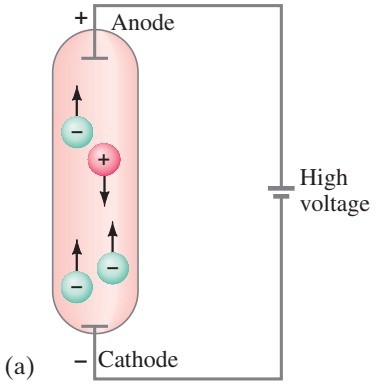
27–11 Atomic Spectra: Key to the Structure of the Atom

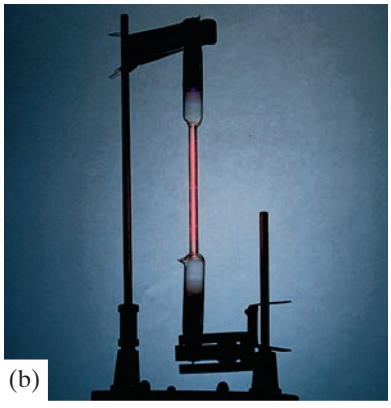
Earlier in this Chapter we saw that heated solids (as well as liquids and dense gases) emit light with a continuous spectrum of wavelengths. This radiation is assumed to be due to oscillations of atoms and molecules, which are largely governed by the interaction of each atom or molecule with its neighbors.

Rarefied gases can also be excited to emit light. This is done by intense heating, or more commonly by applying a high voltage to a "discharge tube" containing the gas at low pressure, Fig. 27–22. The radiation from excited gases had been observed early in the nineteenth century, and it was found that the spectrum was not continuous. Rather, excited gases emit light of only certain wavelengths, and when this light is analyzed through the slit of a spectroscope or spectrometer, a **line spectrum** is seen rather than a continuous spectrum. The line spectra emitted by a number of elements in the visible region are shown below in Fig. 27–23, and in Chapter 24, Fig. 24–28. The **emission spectrum** is characteristic of the material and can serve as a type of "fingerprint" for identification of the gas.

We also saw (Chapter 24) that if a continuous spectrum passes through a rarefied gas, dark lines are observed in the emerging spectrum, at wavelengths corresponding to lines normally emitted by the gas. This is called an **absorption spectrum** (Fig. 27–23c), and it became clear that gases can absorb light at the same frequencies at which they emit. Using film sensitive to ultraviolet and to infrared light, it was found that gases emit and absorb discrete frequencies in these regions as well as in the visible.

FIGURE 27–22 Gas-discharge tube: (a) diagram; (b) photo of an actual discharge tube for hydrogen.





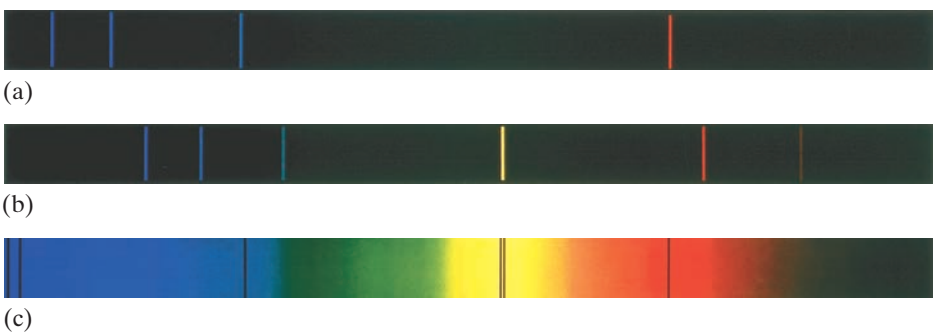


FIGURE 27–23 Emission spectra of the gases (a) atomic hydrogen, (b) helium, and (c) the *solar absorption* spectrum.

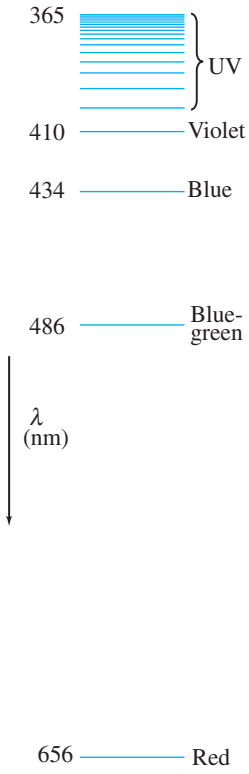


FIGURE 27–24 Balmer series of lines for hydrogen.

In low-density gases, the atoms are far apart on average and hence the light emitted or absorbed is assumed to be by *individual atoms* rather than through interactions between atoms, as in a solid, liquid, or dense gas. Thus the line spectra serve as a key to the structure of the atom: any theory of atomic structure must be able to explain why atoms emit light only of discrete wavelengths, and it should be able to predict what these wavelengths are.

Hydrogen is the simplest atom—it has only one electron orbiting its nucleus. It also has the simplest spectrum. The spectrum of most atoms shows little apparent regularity. But the spacing between lines in the hydrogen spectrum decreases in a regular way, Fig. 27–24. Indeed, in 1885, J. J. Balmer (1825–1898) showed that the four lines in the visible portion of the hydrogen spectrum (with measured wavelengths 656 nm, 486 nm, 434 nm, and 410 nm) have wavelengths that fit the formula

$$\frac{1}{\lambda} = R\left(\frac{1}{2^2} - \frac{1}{n^2}\right), \qquad n = 3, 4, \cdots. \tag{27–9}$$

Here n takes on the values 3, 4, 5, 6 for the four visible lines, and R, called the **Rydberg constant**, has the value $R = 1.0974 \times 10^7\ \text{m}^{-1}$. Later it was found that this **Balmer series** of lines extended into the UV region, ending at $\lambda = 365$ nm, as shown in Fig. 27–24. Balmer's formula, Eq. 27–9, also worked for these lines with higher integer values of n. The lines near 365 nm become too close together to distinguish, but the limit of the series at 365 nm corresponds to $n = \infty$ (so $1/n^2 = 0$ in Eq. 27–9).

Later experiments on hydrogen showed that there were similar series of lines in the UV and IR regions, and each series had a pattern just like the Balmer series, but at different wavelengths, Fig. 27–25. Each of these series was found to

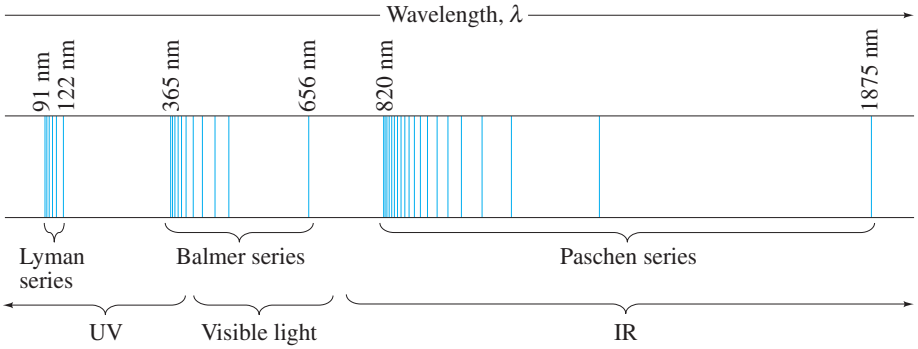


FIGURE 27–25 Line spectrum of atomic hydrogen. Each series fits the formula $\frac{1}{\lambda} = R\left(\frac{1}{n'^2} - \frac{1}{n^2}\right)$ where
$n' = 1$ for the Lyman series,
$n' = 2$ for the Balmer series,
$n' = 3$ for the Paschen series,
and so on; n can take on all integer values from $n = n' + 1$ up to infinity. The only lines in the visible region of the electromagnetic spectrum are part of the Balmer series.

fit a formula with the same form as Eq. 27–9 but with the $1/2^2$ replaced by $1/1^2$, $1/3^2$, $1/4^2$, and so on. For example, the **Lyman series** contains lines with wavelengths from 91 nm to 122 nm (in the UV region) and fits the formula

$$\frac{1}{\lambda} = R\left(\frac{1}{1^2} - \frac{1}{n^2}\right), \qquad n = 2, 3, \cdots.$$

The wavelengths of the **Paschen series** (in the IR region) fit

$$\frac{1}{\lambda} = R\left(\frac{1}{3^2} - \frac{1}{n^2}\right), \qquad n = 4, 5, \cdots.$$

The Rutherford model was unable to explain why atoms emit line spectra. It had other difficulties as well. According to the Rutherford model, electrons orbit the nucleus, and since their paths are curved the electrons are accelerating. Hence they should give off light like any other accelerating electric charge (Chapter 22).

Since light carries off energy and energy is conserved, the electron's own energy must decrease to compensate. Hence electrons would be expected to spiral into the nucleus. As they spiraled inward, their frequency would increase in a short time and so too would the frequency of the light emitted. Thus the two main difficulties of the Rutherford model are these: (1) it predicts that light of a continuous range of frequencies will be emitted, whereas experiment shows line spectra; (2) it predicts that atoms are unstable—electrons would quickly spiral into the nucleus—but we know that atoms in general are stable, because there is stable matter all around us.

Clearly Rutherford's model was not sufficient. Some sort of modification was needed, and Niels Bohr provided it in a model that included the quantum hypothesis. Although the Bohr model has been superseded, it did provide a crucial stepping stone to our present understanding. And some aspects of the Bohr model are still useful today, so we examine it in detail in the next Section.

27–12 The Bohr Model

Bohr had studied in Rutherford's laboratory for several months in 1912 and was convinced that Rutherford's planetary model of the atom had validity. But in order to make it work, he felt that the newly developing quantum theory would somehow have to be incorporated in it. The work of Planck and Einstein had shown that in heated solids, the energy of oscillating electric charges must change discontinuously—from one discrete energy state to another, with the emission of a quantum of light. Perhaps, Bohr argued, the electrons in an atom also cannot lose energy continuously, but must do so in quantum "jumps." In working out his model during the next year, Bohr postulated that electrons move about the nucleus in circular orbits, but that only certain orbits are allowed. He further postulated that an electron in each orbit would have a definite energy and would move in the orbit *without radiating energy* (even though this violated classical ideas since accelerating electric charges are supposed to emit EM waves; see Chapter 22). He thus called the possible orbits **stationary states**. In this **Bohr model**, light is emitted only when an electron jumps from a higher (upper) stationary state to another of lower energy, Fig. 27–26. When such a transition occurs, a single photon of light is emitted whose energy, by energy conservation, is given by

$$hf = E_u - E_\ell, \qquad \textbf{(27–10)}$$

where E_u refers to the energy of the upper state and E_ℓ the energy of the lower state.

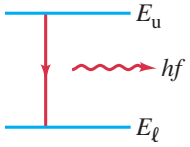


FIGURE 27–26 An atom emits a photon (energy = hf) when its energy changes from E_u to a lower energy E_ℓ.

In 1912–13, Bohr set out to determine what energies these orbits would have in the simplest atom, hydrogen; the spectrum of light emitted could then be predicted from Eq. 27–10. In the Balmer formula he had the key he was looking for. Bohr quickly found that his theory would agree with the Balmer formula if he assumed that the electron's angular momentum L is quantized and equal to an integer n times $h/2\pi$. As we saw in Chapter 8 angular momentum is given by $L = I\omega$, where I is the moment of inertia and ω is the angular velocity. For a single particle of mass m moving in a circle of radius r with speed v, $I = mr^2$ and $\omega = v/r$; hence, $L = I\omega = (mr^2)(v/r) = mvr$. Bohr's **quantum condition** is

$$L = mvr_n = n\frac{h}{2\pi}, \qquad n = 1, 2, 3, \cdots, \qquad \textbf{(27–11)}$$

where n is an integer and r_n is the radius of the nth possible orbit. The allowed orbits are numbered 1, 2, 3, $\cdots$, according to the value of n, which is called the **principal quantum number** of the orbit.

Equation 27–11 did not have a firm theoretical foundation. Bohr had searched for some "quantum condition," and such tries as $E = hf$ (where E represents the energy of the electron in an orbit) did not give results in accord with experiment. Bohr's reason for using Eq. 27–11 was simply that it worked; and we now look at how. In particular, let us determine what the Bohr theory predicts for the measurable wavelengths of emitted light.

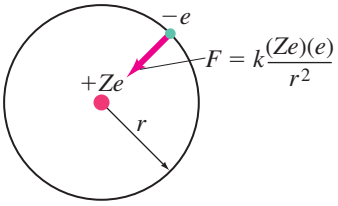


FIGURE 27–27 Electric force (Coulomb's law) keeps the negative electron in orbit around the positively charged nucleus.

An electron in a circular orbit of radius r (Fig. 27–27) would have a centripetal acceleration v^2/r produced by the electrical force of attraction between the negative electron and the positive nucleus. This force is given by Coulomb's law,

$$F = k\frac{(Ze)(e)}{r^2},$$

where $k = 1/4\pi\epsilon_0 = 8.99 \times 10^9\,\text{N}\cdot\text{m}^2/\text{C}^2$. The charge on the electron is $q_1 = -e$, and that on the nucleus is $q_2 = +Ze$, where Ze is the charge on the nucleus: $+e$ is the charge on a proton, Z is the number of protons in the nucleus (called "atomic number," Section 28–7).[†] For the hydrogen atom, $Z = +1$.

In Newton's second law, $F = ma$, we substitute Coulomb's law for F and $a = v^2/r_n$ for a particular allowed orbit of radius r_n, and obtain

$$F = ma$$
$$k\frac{Ze^2}{r_n^2} = \frac{mv^2}{r_n}.$$

We solve this for r_n,

$$r_n = \frac{kZe^2}{mv^2},$$

and then substitute for v from Eq. 27–11 (which says $v = nh/2\pi mr_n$):

$$r_n = \frac{kZe^2 4\pi^2 m r_n^2}{n^2h^2}.$$

We solve for r_n (it appears on both sides, so we cancel one of them) and find

$$r_n = \frac{n^2h^2}{4\pi^2 mkZe^2} = \frac{n^2}{Z}r_1 \qquad n = 1, 2, 3\cdots, \qquad \textbf{(27–12)}$$

where n is an integer (Eq. 27–11), and

$$r_1 = \frac{h^2}{4\pi^2 mke^2}.$$

Equation 27–12 gives the radii of all possible orbits. The smallest orbit is for $n = 1$, and for hydrogen $(Z = 1)$ has the value

$$r_1 = \frac{(1)^2(6.626 \times 10^{-34}\,\text{J}\cdot\text{s})^2}{4\pi^2(9.11 \times 10^{-31}\,\text{kg})(8.99 \times 10^9\,\text{N}\cdot\text{m}^2/\text{C}^2)(1.602 \times 10^{-19}\,\text{C})^2}$$
$$r_1 = 0.529 \times 10^{-10}\,\text{m}. \qquad \textbf{(27–13)}$$

The radius of the smallest orbit in hydrogen, r_1, is sometimes called the **Bohr radius**. From Eq. 27–12, we see that the radii of the larger orbits[‡] increase as n^2, so

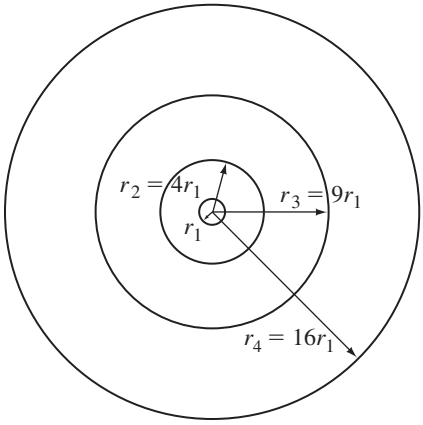
$$r_2 = 4r_1 = 2.12 \times 10^{-10}\,\text{m},$$
$$r_3 = 9r_1 = 4.76 \times 10^{-10}\,\text{m},$$
$$\vdots$$
$$r_n = n^2 r_1, \qquad n = 1, 2, 3, \cdots.$$

The first four orbits are shown in Fig. 27–28. Notice that, according to Bohr's model, an electron can exist only in the orbits given by Eq. 27–12. There are no allowable orbits in between.

For an atom with $Z \neq 1$, we can write the orbital radii, r_n, using Eq. 27–12:

$$r_n = \frac{n^2}{Z}(0.529 \times 10^{-10}\,\text{m}), \qquad n = 1, 2, 3, \cdots. \qquad \textbf{(27–14)}$$

FIGURE 27–28 The four smallest orbits in the Bohr model of hydrogen; $r_1 = 0.529 \times 10^{-10}\,\text{m}$.



[†]We include Z in our derivation so that we can treat other single-electron ("hydrogenlike") atoms such as the ions He^+ $(Z = 2)$ and Li^{2+} $(Z = 3)$. Helium in the neutral state has two electrons; if one electron is missing, the remaining He^+ ion consists of one electron revolving around a nucleus of charge $+2e$. Similarly, doubly ionized lithium, Li^{2+}, also has a single electron, and in this case $Z = 3$.

[‡]Be careful not to believe that these well-defined orbits actually exist. Today electrons are better thought of as forming "clouds," as discussed in Chapter 28.

In each of its possible orbits, the electron in a Bohr model atom would have a definite energy, as the following calculation shows. The total energy equals the sum of the kinetic and potential energies. The potential energy of the electron is given by $\text{PE} = qV = -eV$, where V is the potential due to a point charge $+Ze$ as given by Eq. 17–5: $V = kQ/r = kZe/r$. So

$$\text{PE} = -eV = -k\frac{Ze^2}{r}.$$

The total energy E_n for an electron in the nth orbit of radius r_n is the sum of the kinetic and potential energies:

$$E_n = \tfrac{1}{2}mv^2 - \frac{kZe^2}{r_n}.$$

When we substitute v from Eq. 27–11 and r_n from Eq. 27–12 into this equation, we obtain

$$E_n = -\frac{2\pi^2 Z^2 e^4 m k^2}{h^2}\frac{1}{n^2} \qquad n = 1, 2, 3, \cdots. \tag{27–15a}$$

If we evaluate the constant term in Eq. 27–15a and convert it to electron volts, as is customary in atomic physics, we obtain

$$E_n = -(13.6\text{ eV})\frac{Z^2}{n^2}, \qquad n = 1, 2, 3, \cdots. \tag{27–15b}$$

The lowest energy level $(n = 1)$ for hydrogen $(Z = 1)$ is

$$E_1 = -13.6\text{ eV}.$$

Since n^2 appears in the denominator of Eq. 27–15b, the energies of the larger orbits in hydrogen $(Z = 1)$ are given by

$$E_n = \frac{-13.6\text{ eV}}{n^2}.$$

For example,

$$E_2 = \frac{-13.6\text{ eV}}{4} = -3.40\text{ eV},$$

$$E_3 = \frac{-13.6\text{ eV}}{9} = -1.51\text{ eV}.$$

We see that not only are the orbit radii quantized, but from Eqs. 27–15, so is the energy. The quantum number n that labels the orbit radii also labels the energy levels. The lowest **energy level** or **energy state** has energy E_1, and is called the **ground state**. The higher states, E_2, E_3, and so on, are called **excited states**. The fixed energy levels are also called **stationary states**.

Notice that although the energy for the larger orbits has a smaller numerical value, all the energies are less than zero. Thus, -3.4 eV is a higher energy than -13.6 eV. Hence the orbit closest to the nucleus (r_1) has the lowest energy (the most negative). The reason the energies have negative values has to do with the way we defined the zero for potential energy. For two point charges, $\text{PE} = kq_1q_2/r$ corresponds to zero potential energy when the two charges are infinitely far apart (Section 17–5). Thus, an electron that can just barely be free from the atom by reaching $r = \infty$ (or, at least, far from the nucleus) with zero kinetic energy will have $E = \text{KE} + \text{PE} = 0 + 0 = 0$, corresponding to $n = \infty$ in Eqs. 27–15. If an electron is free and has kinetic energy, then $E > 0$. To remove an electron that is part of an atom requires an energy input (otherwise atoms would not be stable). Since $E \geq 0$ for a free electron, then an electron bound to an atom needs to have $E < 0$. That is, energy must be added to bring its energy up, from a negative value to at least zero in order to free it.

The minimum energy required to remove an electron from an atom initially in the ground state is called the **binding energy** or **ionization energy**. The ionization energy for hydrogen has been measured to be 13.6 eV, and this corresponds precisely to removing an electron from the lowest state, $E_1 = -13.6$ eV, up to $E = 0$ where it can be free.

Spectra Lines Explained

It is useful to show the various possible energy values as horizontal lines on an energy-level diagram. This is shown for hydrogen in Fig. 27–29. The electron in a hydrogen atom can be in any one of these levels according to Bohr theory. But it could never be in between, say at $-9.0\,\text{eV}$. At room temperature, nearly all H atoms will be in the ground state $(n = 1)$. At higher temperatures, or during an electric discharge when there are many collisions between free electrons and atoms, many atoms can be in excited states $(n > 1)$. Once in an excited state, an atom's electron can jump down to a lower state, and give off a photon in the process. This is, according to the Bohr model, the origin of the emission spectra of excited gases.

Note that above $E = 0$, an electron is free and can have any energy (E is not quantized). Thus there is a continuum of energy states above $E = 0$, as indicated in the energy-level diagram of Fig. 27–29.

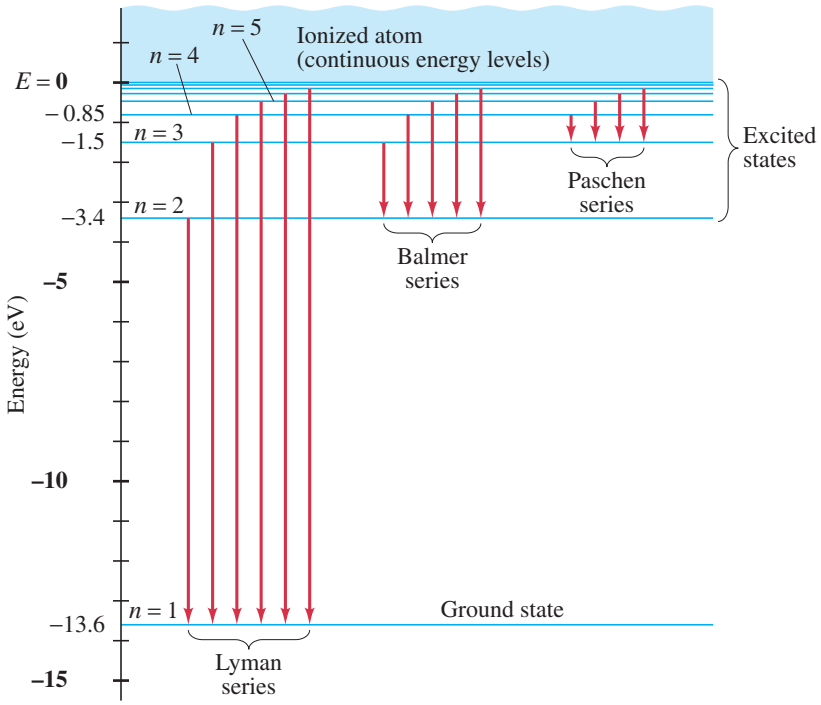


FIGURE 27–29 Energy-level diagram for the hydrogen atom, showing the transitions for the spectral lines of the Lyman, Balmer, and Paschen series (Fig. 27–25). Each vertical arrow represents an atomic transition that gives rise to the photons of one spectral line (a single wavelength or frequency).

The vertical arrows in Fig. 27–29 represent the transitions or jumps that correspond to the various observed spectral lines. For example, an electron jumping from the level $n = 3$ to $n = 2$ would give rise to the 656-nm line in the Balmer series, and the jump from $n = 4$ to $n = 2$ would give rise to the 486-nm line (see Fig. 27–24). We can predict wavelengths of the spectral lines emitted according to Bohr theory by combining Eq. 27–10 with Eq. 27–15. Since $hf = hc/\lambda$, we have from Eq. 27–10

$$\frac{1}{\lambda} = \frac{hf}{hc} = \frac{1}{hc}(E_n - E_{n'}),$$

where n refers to the upper state and n' to the lower state. Then using Eq. 27–15,

$$\frac{1}{\lambda} = \frac{2\pi^2 Z^2 e^4 m k^2}{h^3 c}\left(\frac{1}{n'^2} - \frac{1}{n^2}\right). \qquad \textbf{(27–16)}$$

This theoretical formula has the same form as the experimental Balmer formula, Eq. 27–9, with $n' = 2$. Thus we see that the Balmer series of lines corresponds to transitions or "jumps" that bring the electron down to the second energy level. Similarly, $n' = 1$ corresponds to the Lyman series and $n' = 3$ to the Paschen series (see Fig. 27–29).

When the constant in Eq. 27–16 is evaluated with $Z = 1$, it is found to have the measured value of the Rydberg constant, $R = 1.0974 \times 10^7\,\text{m}^{-1}$ in Eq. 27–9, in accord with experiment (see Problem 47).

The great success of Bohr's model is that it gives an explanation for why atoms emit line spectra, and accurately predicts the wavelengths of emitted light for hydrogen. The Bohr model also explains absorption spectra: photons of just the right wavelength can knock an electron from one energy level to a higher one. To conserve energy, only photons that have just the right energy will be absorbed. This explains why a continuous spectrum of light entering a gas will emerge with dark (absorption) lines at frequencies that correspond to emission lines (Fig. 27–23c).

The Bohr theory also ensures the stability of atoms. It establishes stability by decree: the ground state is the lowest state for an electron and there is no lower energy level to which it can go and emit more energy. Finally, as we saw above, the Bohr theory accurately predicts the ionization energy of 13.6 eV for hydrogen. However, the Bohr model was not so successful for other atoms, and has been superseded as we shall discuss in the next Chapter. We discuss the Bohr model because it *was* an important start and because we still use the concept of stationary states, the ground state, and transitions between states. Also, the terminology used in the Bohr model is still used by chemists and spectroscopists.

EXAMPLE 27–13 **Wavelength of a Lyman line.** Use Fig. 27–29 to determine the wavelength of the first Lyman line, the transition from $n = 2$ to $n = 1$. In what region of the electromagnetic spectrum does this lie?

APPROACH We use Eq. 27–10, $hf = E_u - E_\ell$, with the energies obtained from Fig. 27–29 to find the energy and the wavelength of the transition. The region of the electromagnetic spectrum is found using the EM spectrum in Fig. 22–8.

SOLUTION In this case, $hf = E_2 - E_1 = \{-3.4\,\text{eV} - (-13.6\,\text{eV})\} = 10.2\,\text{eV} = (10.2\,\text{eV})(1.60 \times 10^{-19}\,\text{J/eV}) = 1.63 \times 10^{-18}\,\text{J}$. Since $\lambda = c/f$, we have

$$\lambda = \frac{c}{f} = \frac{hc}{E_2 - E_1} = \frac{(6.63 \times 10^{-34}\,\text{J}\cdot\text{s})(3.00 \times 10^8\,\text{m/s})}{1.63 \times 10^{-18}\,\text{J}} = 1.22 \times 10^{-7}\,\text{m},$$

or 122 nm, which is in the UV region of the EM spectrum, Fig. 22–8. See also Fig. 27–25, where this value is confirmed experimentally.

NOTE An alternate approach: use Eq. 27–16 to find λ, and get the same result.

EXAMPLE 27–14 **Wavelength of a Balmer line.** Use the Bohr model to determine the wavelength of light emitted when a hydrogen atom makes a transition from the $n = 6$ to the $n = 2$ energy level.

APPROACH We can use Eq. 27–16 or its equivalent, Eq. 27–9, with $R = 1.097 \times 10^7\,\text{m}^{-1}$.

SOLUTION We find

$$\frac{1}{\lambda} = (1.097 \times 10^7\,\text{m}^{-1})\left(\frac{1}{4} - \frac{1}{36}\right) = 2.44 \times 10^6\,\text{m}^{-1}.$$

So $\lambda = 1/(2.44 \times 10^6\,\text{m}^{-1}) = 4.10 \times 10^{-7}\,\text{m}$ or 410 nm. This is the fourth line in the Balmer series, Fig. 27–24, and is violet in color.

EXAMPLE 27–15 **Absorption wavelength.** Use Fig. 27–29 to determine the maximum wavelength that hydrogen in its ground state can absorb. What would be the next smaller wavelength that would work?

APPROACH Maximum wavelength corresponds to minimum energy, and this would be the jump from the ground state up to the first excited state (Fig. 27–29). The next smaller wavelength occurs for the jump from the ground state to the second excited state.

SOLUTION The energy needed to jump from the ground state to the first excited state is $13.6\,\text{eV} - 3.4\,\text{eV} = 10.2\,\text{eV}$; the required wavelength, as we saw in Example 27–13, is 122 nm. The energy to jump from the ground state to the second excited state is $13.6\,\text{eV} - 1.5\,\text{eV} = 12.1\,\text{eV}$, which corresponds to a wavelength

$$\lambda = \frac{c}{f} = \frac{hc}{hf} = \frac{hc}{E_3 - E_1} = \frac{(6.63 \times 10^{-34}\,\text{J}\cdot\text{s})(3.00 \times 10^8\,\text{m/s})}{(12.1\,\text{eV})(1.60 \times 10^{-19}\,\text{J/eV})} = 103\,\text{nm}.$$

EXAMPLE 27–16 **He^+ ionization energy.** (*a*) Use the Bohr model to determine the ionization energy of the He^+ ion, which has a single electron. (*b*) Also calculate the maximum wavelength a photon can have to cause ionization. The helium atom is the second atom, after hydrogen, in the Periodic Table (next Chapter); its nucleus contains 2 protons and normally has 2 electrons circulating around it, so $Z = 2$.

APPROACH We want to determine the minimum energy required to lift the electron from its ground state and to barely reach the free state at $E = 0$. The ground state energy of He^+ is given by Eq. 27–15b with $n = 1$ and $Z = 2$.

SOLUTION (*a*) Since all the symbols in Eq. 27–15b are the same as for the calculation for hydrogen, except that Z is 2 instead of 1, we see that E_1 will be $Z^2 = 2^2 = 4$ times the E_1 for hydrogen:

$$E_1 = 4(-13.6\,\text{eV}) = -54.4\,\text{eV}.$$

Thus, to ionize the He^+ ion should require 54.4 eV, and this value agrees with experiment.

(*b*) The maximum wavelength photon that can cause ionization will have energy $hf = 54.4\,\text{eV}$ and wavelength

$$\lambda = \frac{c}{f} = \frac{hc}{hf} = \frac{(6.63 \times 10^{-34}\,\text{J}\cdot\text{s})(3.00 \times 10^8\,\text{m/s})}{(54.4\,\text{eV})(1.60 \times 10^{-19}\,\text{J/eV})} = 22.8\,\text{nm}.$$

If $\lambda > 22.8\,\text{nm}$, ionization can not occur.

NOTE If the atom absorbed a photon of greater energy (wavelength shorter than 22.8 nm), the atom could still be ionized and the freed electron would have kinetic energy of its own.

In this Example 27–16, we saw that E_1 for the He^+ ion is four times more negative than that for hydrogen. Indeed, the energy-level diagram for He^+ looks just like that for hydrogen, Fig. 27–29, except that the numerical values for each energy level are four times larger. Note, however, that we are talking here about the He^+ *ion*. Normal (neutral) helium has two electrons and its energy level diagram is entirely different.

CONCEPTUAL EXAMPLE 27–17 **Hydrogen at 20°C.** (*a*) Estimate the average kinetic energy of whole hydrogen atoms (not just the electrons) at room temperature. (*b*) Use the result to explain why, at room temperature, very few H atoms are in excited states and nearly all are in the ground state, and hence emit no light.

RESPONSE According to kinetic theory (Chapter 13), the average kinetic energy of atoms or molecules in a gas is given by Eq. 13–8:

$$\overline{\text{KE}} = \tfrac{3}{2}kT,$$

where $k = 1.38 \times 10^{-23}\,\text{J/K}$ is Boltzmann's constant, and T is the kelvin (absolute) temperature. Room temperature is about $T = 300\,\text{K}$, so

$$\overline{\text{KE}} = \tfrac{3}{2}(1.38 \times 10^{-23}\,\text{J/K})(300\,\text{K}) = 6.2 \times 10^{-21}\,\text{J},$$

or, in electron volts:

$$\overline{\text{KE}} = \frac{6.2 \times 10^{-21}\,\text{J}}{1.6 \times 10^{-19}\,\text{J/eV}} = 0.04\,\text{eV}.$$

The average KE of an atom as a whole is thus very small compared to the energy between the ground state and the next higher energy state (13.6 eV − 3.4 eV = 10.2 eV). Any atoms in excited states quickly fall to the ground state and emit light. Once in the ground state, collisions with other atoms can transfer energy of only 0.04 eV on the average. A small fraction of atoms can have much more energy (see Section 13–10 on the distribution of molecular speeds), but even a kinetic energy that is 10 times the average is not nearly enough to excite atoms into states above the ground state. Thus, at room temperature, practically all atoms are in the ground state. Atoms can be excited to upper states by very high temperatures, or by applying a high voltage so a current of high energy electrons passes through the gas as in a discharge tube (Fig. 27–22).

Correspondence Principle

We should note that Bohr made some radical assumptions that were at variance with classical ideas. He assumed that electrons in fixed orbits do not radiate light even though they are accelerating (moving in a circle), and he assumed that angular momentum is quantized. Furthermore, he was not able to say how an electron moved when it made a transition from one energy level to another. On the other hand, there is no real reason to expect that in the tiny world of the atom electrons would behave as ordinary-sized objects do. Nonetheless, he felt that where quantum theory overlaps with the macroscopic world, it should predict classical results. This is the **correspondence principle**, already mentioned in regard to relativity (Section 26–11). This principle does work for Bohr's theory of the hydrogen atom. The orbit sizes and energies are quite different for $n = 1$ and $n = 2$, say. But orbits with $n = 100{,}000{,}000$ and $100{,}000{,}001$ would be very close in radius and energy (see Fig. 27–29). Indeed, transitions between such large orbits, which would approach macroscopic sizes, would be imperceptible. Such orbits would thus appear to be continuously spaced, which is what we expect in the everyday world.

Finally, it must be emphasized that the well-defined orbits of the Bohr model do not actually exist. The Bohr model is only a model, not reality. The idea of electron orbits was rejected a few years later, and today electrons are thought of (Chapter 28) as forming "probability clouds."

27–13 de Broglie's Hypothesis Applied to Atoms

Bohr's theory was largely of an *ad hoc* nature. Assumptions were made so that theory would agree with experiment. But Bohr could give no reason why the orbits were quantized, nor why there should be a stable ground state. Finally, ten years later, a reason was proposed by Louis de Broglie. We saw in Section 27–8 that in 1923, de Broglie proposed that material particles, such as electrons, have a wave nature; and that this hypothesis was confirmed by experiment several years later.

One of de Broglie's original arguments in favor of the wave nature of electrons was that it provided an explanation for Bohr's theory of the hydrogen atom. According to de Broglie, a particle of mass m moving with a nonrelativistic speed v would have a wavelength (Eq. 27–8) of

$$\lambda = \frac{h}{mv}.$$

Each electron orbit in an atom, he proposed, is actually a standing wave. As we saw in Chapter 11, when a violin or guitar string is plucked, a vast number of wavelengths are excited. But only certain ones—those that have nodes at the ends—are sustained. These are the *resonant* modes of the string. Waves with other wavelengths interfere with themselves upon reflection and their amplitudes quickly drop to zero. With electrons moving in circles, according to Bohr's theory, de Broglie argued that the electron wave was a *circular* standing wave that closes on itself, Fig. 27–30a. If the wavelength of a wave does not close on itself, as in Fig. 27–30b, destructive interference takes place as the wave travels around the loop, and the wave quickly dies out. Thus, the only waves that persist are those for which the circumference of the circular orbit contains a whole number of wavelengths, Fig. 27–31. The circumference of a Bohr orbit of radius r_n is $2\pi r_n$, so to have constructive interference, we need

$$2\pi r_n = n\lambda, \qquad n = 1, 2, 3, \cdots.$$

When we substitute $\lambda = h/mv$, we get $2\pi r_n = nh/mv$, or

$$mvr_n = \frac{nh}{2\pi}.$$

This is just the *quantum condition* proposed by Bohr on an *ad hoc* basis, Eq. 27–11. It is from this equation that the discrete orbits and energy levels were derived.

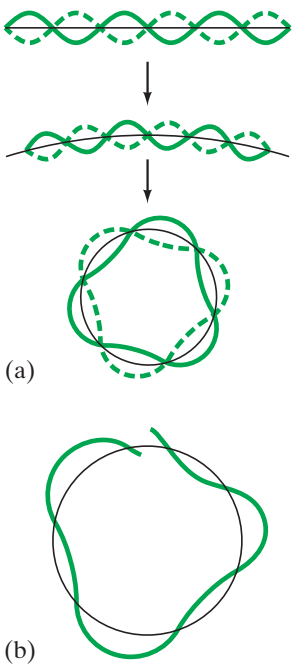
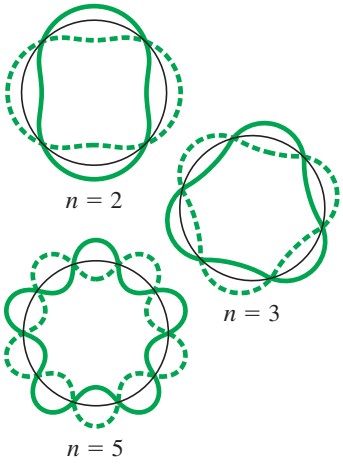


FIGURE 27–30 (a) An ordinary standing wave compared to a circular standing wave. (b) When a wave does not close (and hence interferes destructively with itself), it rapidly dies out.

FIGURE 27–31 Standing circular waves for two, three, and five wavelengths on the circumference; n, the number of wavelengths, is also the quantum number.



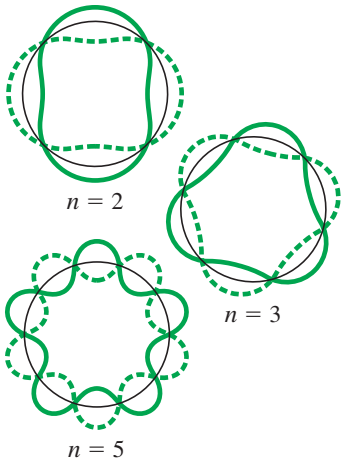


FIGURE 27–31 (Repeated.) Standing circular waves for two, three, and five wavelengths on the circumference; n, the number of wavelengths, is also the quantum number.

Now we have a first explanation for the quantized orbits and energy states in the Bohr model: they are due to the wave nature of the electron, and only resonant "standing" waves can persist.† This implies that the *wave–particle duality* is at the root of atomic structure.

In viewing the circular electron waves of Fig. 27–31, the electron is not to be thought of as following the oscillating wave pattern. In the Bohr model of hydrogen, the electron moves in a circle. The circular wave, on the other hand, represents the *amplitude* of the electron "matter wave," and in Fig. 27–31 the wave amplitude is shown superimposed on the circular path of the particle orbit for convenience.

Bohr's theory worked well for hydrogen and for one-electron ions. But it did not prove successful for multi-electron atoms. Bohr theory could not predict line spectra even for the next simplest atom, helium. It could not explain why some emission lines are brighter than others, nor why some lines are split into two or more closely spaced lines ("fine structure"). A new theory was needed and was indeed developed in the 1920s. This new and radical theory is called *quantum mechanics*. It finally solved the problem of atomic structure, but it gives us a very different view of the atom: the idea of electrons in well-defined orbits was replaced with the idea of electron "clouds." This new theory of quantum mechanics has given us a wholly different view of the basic mechanisms underlying physical processes.

†We note, however, that Eq. 27–11 is no longer considered valid, as discussed in the next Chapter.

Summary

The electron was discovered using an evacuated cathode ray tube. The measurement of the charge-to-mass ratio (e/m) of the electron was done using magnetic and electric fields. The charge e on the electron was first measured in the Millikan oil-drop experiment and then its mass was obtained from the measured value of the e/m ratio.

Quantum theory has its origins in **Planck's quantum hypothesis** that molecular oscillations are **quantized**: their energy E can only be integer (n) multiples of hf, where h is Planck's constant and f is the natural frequency of oscillation:

$$E = nhf. \qquad \textbf{(27–3)}$$

This hypothesis explained the spectrum of radiation emitted by a **blackbody** at high temperature.

Einstein proposed that for some experiments, light could be pictured as being emitted and absorbed as **quanta** (particles), which we now call **photons**, each with energy

$$E = hf \qquad \textbf{(27–4)}$$

and momentum

$$p = \frac{E}{c} = \frac{hf}{c} = \frac{h}{\lambda}. \qquad \textbf{(27–6)}$$

He proposed the photoelectric effect as a test for the photon theory of light. In the **photoelectric effect**, the photon theory says that each incident photon can strike an electron in a material and eject it if the photon has sufficient energy. The maximum energy of ejected electrons is then linearly related to the frequency of the incident light.

The photon theory is also supported by the **Compton effect** and the observation of electron–positron **pair production**.

The **wave–particle duality** refers to the idea that light and matter (such as electrons) have both wave and particle properties. The wavelength of an object is given by

$$\lambda = \frac{h}{p}, \qquad \textbf{(27–8)}$$

where p is the momentum of the object ($p = mv$ for a particle of mass m and speed v).

The **principle of complementarity** states that we must be aware of both the particle and wave properties of light and of matter for a complete understanding of them.

Electron microscopes (EM) make use of the wave properties of electrons to form an image: their "lenses" are magnetic. Various types of EM exist: some can magnify 100,000× (1000× better than a light microscope); others can give a 3-D image.

Early models of the atom include Rutherford's planetary (or nuclear) model of an atom which consists of a tiny but massive positively charged nucleus surrounded (at a relatively great distance) by electrons.

To explain the **line spectra** emitted by atoms, as well as the stability of atoms, the **Bohr model** postulated that: (1) electrons bound in an atom can only occupy orbits for which the angular momentum is quantized, which results in discrete values for the radius and energy; (2) an electron in such a **stationary state** emits no radiation; (3) if an electron jumps to a lower state, it emits a photon whose energy equals the difference in energy between the two states; (4) the angular momentum L of atomic electrons is quantized by the rule $L = nh/2\pi$, where n is an integer called the **quantum number**. The $n = 1$ state is the **ground state**, which in hydrogen has an energy $E_1 = -13.6\,\text{eV}$. Higher values of n correspond to **excited states**, and their energies are

$$E_n = -(13.6\,\text{eV})\frac{Z^2}{n^2}, \qquad \textbf{(27–15b)}$$

where Ze is the charge on the nucleus. Atoms are excited to these higher states by collisions with other atoms or electrons, or by absorption of a photon of just the right frequency.

De Broglie's hypothesis that electrons (and other matter) have a wavelength $\lambda = h/mv$ gave an explanation for Bohr's quantized orbits by bringing in the wave–particle duality: the orbits correspond to circular standing waves in which the circumference of the orbit equals a whole number of wavelengths.

Questions

1. Does a lightbulb at a temperature of 2500 K produce as white a light as the Sun at 6000 K? Explain.

2. If energy is radiated by all objects, why can we not see them in the dark? (See also Section 14–8.)

3. What can be said about the relative temperatures of whitish-yellow, reddish, and bluish stars? Explain.

4. Darkrooms for developing black-and-white film were sometimes lit by a red bulb. Why red? Explain if such a bulb would work in a darkroom for developing color film.

5. If the threshold wavelength in the photoelectric effect increases when the emitting metal is changed to a different metal, what can you say about the work functions of the two metals?

6. Explain why the existence of a cutoff frequency in the photoelectric effect more strongly favors a particle theory rather than a wave theory of light.

7. UV light causes sunburn, whereas visible light does not. Suggest a reason.

8. The work functions for sodium and cesium are 2.28 eV and 2.14 eV, respectively. For incident photons of a given frequency, which metal will give a higher maximum kinetic energy for the electrons? Explain.

9. Explain how the photoelectric circuit of Fig. 27–6 could be used in (*a*) a burglar alarm, (*b*) a smoke detector, (*c*) a photographic light meter.

10. (*a*) Does a beam of infrared photons always have less energy than a beam of ultraviolet photons? Explain. (*b*) Does a single photon of infrared light always have less energy than a single photon of ultraviolet light? Why?

11. Light of 450-nm wavelength strikes a metal surface, and a stream of electrons emerges from the metal. If light of the same intensity but of wavelength 400 nm strikes the surface, are more electrons emitted? Does the energy of the emitted electrons change? Explain.

***12.** If an X-ray photon is scattered by an electron, does the photon's wavelength change? If so, does it increase or decrease? Explain.

***13.** In both the photoelectric effect and in the Compton effect, a photon collides with an electron causing the electron to fly off. What is the difference between the two processes?

14. Why do we say that light has wave properties? Why do we say that light has particle properties?

15. Why do we say that electrons have wave properties? Why do we say that electrons have particle properties?

16. What are the differences between a photon and an electron? Be specific: make a list.

17. If an electron and a proton travel at the same speed, which has the shorter wavelength? Explain.

18. An electron and a proton are accelerated through the same voltage. Which has the longer wavelength? Explain why.

19. In Rutherford's planetary model of the atom, what keeps the electrons from flying off into space?

20. When a wide spectrum of light passes through hydrogen gas at room temperature, absorption lines are observed that correspond only to the Lyman series. Why don't we observe the other series?

21. How can you tell if there is oxygen near the surface of the Sun?

22. (*a*) List at least three successes of the Bohr model of the atom, according to Section 27–12. (*b*) List at least two observations that the Bohr model could not explain, according to Section 27–13.

23. According to Section 27–11, what were the two main difficulties of the Rutherford model of the atom?

24. Is it possible for the de Broglie wavelength of a "particle" to be greater than the dimensions of the particle? To be smaller? Is there any direct connection? Explain.

25. How can the spectrum of hydrogen contain so many lines when hydrogen contains only one electron?

26. Explain how the closely spaced energy levels for hydrogen near the top of Fig. 27–29 correspond to the closely spaced spectral lines at the top of Fig. 27–24.

27. In a helium atom, which contains two electrons, do you think that on average the electrons are closer to the nucleus or farther away than in a hydrogen atom? Why?

28. The Lyman series is brighter than the Balmer series, because this series of transitions ends up in the most common state for hydrogen, the ground state. Why then was the Balmer series discovered first?

29. Use conservation of momentum to explain why photons emitted by hydrogen atoms have slightly less energy than that predicted by Eq. 27–10.

30. State if a continuous or a line spectrum is produced by each of the following: (*a*) a hot solid object; (*b*) an excited, rarefied gas; (*c*) a hot liquid; (*d*) light from a hot solid that passes through a cooler rarefied gas; (*e*) a hot dense gas. For each, if a line spectrum is produced, is it an emission or an absorption spectrum?

31. Suppose we obtain an emission spectrum for hydrogen at very high temperature (when some of the atoms are in excited states), and an absorption spectrum at room temperature, when all atoms are in the ground state. Will the two spectra contain identical lines?

MisConceptual Questions

1. Which of the following statements is true regarding how blackbody radiation changes as the temperature of the radiating object increases?
(*a*) Both the maximum intensity and the peak wavelength increase.
(*b*) The maximum intensity increases, and the peak wavelength decreases.
(*c*) Both the maximum intensity and the peak wavelength decrease.
(*d*) The maximum intensity decreases, and the peak wavelength increases.

2. As red light shines on a piece of metal, no electrons are released. When the red light is slowly changed to shorter-wavelength light (basically progressing through the rainbow), nothing happens until yellow light shines on the metal, at which point electrons are released from the metal. If this metal is replaced with a metal having a higher work function, which light would have the best chance of releasing electrons from the metal?
(*a*) Blue.
(*b*) Red.
(*c*) Yellow would still work fine.
(*d*) We need to know more about the metals involved.

3. A beam of red light and a beam of blue light have equal intensities. Which statement is true?
(*a*) There are more photons in the blue beam.
(*b*) There are more photons in the red beam.
(*c*) Both beams contain the same number of photons.
(*d*) The number of photons is not related to intensity.

4. Which of the following is necessarily true?
(*a*) Red light has more energy than violet light.
(*b*) Violet light has more energy than red light.
(*c*) A single photon of red light has more energy than a single photon of violet light.
(*d*) A single photon of violet light has more energy than a single photon of red light.
(*e*) None of the above.
(*f*) A combination of the above (specify).

5. If a photon of energy E ejects electrons from a metal with kinetic energy KE, then a photon with energy $E/2$
(*a*) will eject electrons with kinetic energy KE/2.
(*b*) will eject electrons with an energy greater than KE/2.
(*c*) will eject electrons with an energy less than KE/2.
(*d*) might not eject any electrons.

6. If the momentum of an electron were doubled, how would its wavelength change?
(*a*) No change.
(*b*) It would be halved.
(*c*) It would double.
(*d*) It would be quadrupled.
(*e*) It would be reduced to one-fourth.

7. Which of the following can be thought of as either a wave or a particle?
(*a*) Light.
(*b*) An electron.
(*c*) A proton.
(*d*) All of the above.

8. When you throw a baseball, its de Broglie wavelength is
(*a*) the same size as the ball.
(*b*) about the same size as an atom.
(*c*) about the same size as an atom's nucleus.
(*d*) much smaller than the size of an atom's nucleus.

9. Electrons and photons of light are similar in that
(*a*) both have momentum given by h/λ.
(*b*) both exhibit wave–particle duality.
(*c*) both are used in diffraction experiments to explore structure.
(*d*) All of the above.
(*e*) None of the above.

10. In Rutherford's famous set of experiments described in Section 27–10, the fact that some alpha particles were deflected at large angles indicated that (choose all that apply)
(*a*) the nucleus was positive.
(*b*) charge was quantized.
(*c*) the nucleus was concentrated in a small region of space.
(*d*) most of the atom is empty space.
(*e*) None of the above.

11. Which of the following electron transitions between two energy states (n) in the hydrogen atom corresponds to the emission of a photon with the longest wavelength?
(*a*) $2 \rightarrow 5$.
(*b*) $5 \rightarrow 2$.
(*c*) $5 \rightarrow 8$.
(*d*) $8 \rightarrow 5$.

12. If we set the potential energy of an electron and a proton to be zero when they are an infinite distance apart, then the lowest energy a bound electron in a hydrogen atom can have is
(*a*) 0.
(*b*) -13.6 eV.
(*c*) any possible value.
(*d*) any value between -13.6 eV and 0.

13. Which of the following is the currently accepted model of the atom?
(*a*) The plum-pudding model.
(*b*) The Rutherford atom.
(*c*) The Bohr atom.
(*d*) None of the above.

14. Light has all of the following except:
(*a*) mass.
(*b*) momentum.
(*c*) kinetic energy.
(*d*) frequency.
(*e*) wavelength.

Problems

27–1 Discovery of the Electron

1. (I) What is the value of e/m for a particle that moves in a circle of radius 14 mm in a 0.86-T magnetic field if a perpendicular 640-V/m electric field will make the path straight?
2. (II) (*a*) What is the velocity of a beam of electrons that go undeflected when passing through crossed (perpendicular) electric and magnetic fields of magnitude 1.88×10^4 V/m and 2.60×10^{-3} T, respectively? (*b*) What is the radius of the electron orbit if the electric field is turned off?
3. (II) An oil drop whose mass is 2.8×10^{-15} kg is held at rest between two large plates separated by 1.0 cm (Fig. 27–3), when the potential difference between the plates is 340 V. How many excess electrons does this drop have?

27–2 Blackbodies; Planck's Quantum Hypothesis

4. (I) How hot is a metal being welded if it radiates most strongly at 520 nm?
5. (I) Estimate the peak wavelength for radiation emitted from (*a*) ice at 0°C, (*b*) a floodlamp at 3100 K, (*c*) helium at 4 K, assuming blackbody emission. In what region of the EM spectrum is each?
6. (I) (*a*) What is the temperature if the peak of a blackbody spectrum is at 18.0 nm? (*b*) What is the wavelength at the peak of a blackbody spectrum if the body is at a temperature of 2200 K?
7. (I) An HCl molecule vibrates with a natural frequency of 8.1×10^{13} Hz. What is the difference in energy (in joules and electron volts) between successive values of the oscillation energy?
8. (II) The steps of a flight of stairs are 20.0 cm high (vertically). If a 62.0-kg person stands with both feet on the same step, what is the gravitational potential energy of this person, relative to the ground, on (*a*) the first step, (*b*) the second step, (*c*) the third step, (*d*) the nth step? (*e*) What is the change in energy as the person descends from step 6 to step 2?
9. (II) Estimate the peak wavelength of light emitted from the pupil of the human eye (which approximates a blackbody) assuming normal body temperature.

27–3 and 27–4 Photons and the Photoelectric Effect

10. (I) What is the energy of photons (joules) emitted by a 91.7-MHz FM radio station?
11. (I) What is the energy range (in joules and eV) of photons in the visible spectrum, of wavelength 400 nm to 750 nm?
12. (I) A typical gamma ray emitted from a nucleus during radioactive decay may have an energy of 320 keV. What is its wavelength? Would we expect significant diffraction of this type of light when it passes through an everyday opening, such as a door?
13. (I) Calculate the momentum of a photon of yellow light of wavelength 5.80×10^{-7} m.
14. (I) What is the momentum of a $\lambda = 0.014$ nm X-ray photon?
15. (II) About 0.1 eV is required to break a "hydrogen bond" in a protein molecule. Calculate the minimum frequency and maximum wavelength of a photon that can accomplish this.
16. (II) What minimum frequency of light is needed to eject electrons from a metal whose work function is 4.8×10^{-19} J?
17. (II) The human eye can respond to as little as 10^{-18} J of light energy. For a wavelength at the peak of visual sensitivity, 550 nm, how many photons lead to an observable flash?
18. (II) What is the longest wavelength of light that will emit electrons from a metal whose work function is 2.90 eV?
19. (II) The work functions for sodium, cesium, copper, and iron are 2.3, 2.1, 4.7, and 4.5 eV, respectively. Which of these metals will not emit electrons when visible light shines on it?
20. (II) In a photoelectric-effect experiment it is observed that no current flows unless the wavelength is less than 550 nm. (*a*) What is the work function of this material? (*b*) What stopping voltage is required if light of wavelength 400 nm is used?
21. (II) What is the maximum kinetic energy of electrons ejected from barium $(W_0 = 2.48\text{ eV})$ when illuminated by white light, $\lambda = 400$ to 750 nm?
22. (II) Barium has a work function of 2.48 eV. What is the maximum kinetic energy of electrons if the metal is illuminated by UV light of wavelength 365 nm? What is their speed?
23. (II) When UV light of wavelength 255 nm falls on a metal surface, the maximum kinetic energy of emitted electrons is 1.40 eV. What is the work function of the metal?
24. (II) The threshold wavelength for emission of electrons from a given surface is 340 nm. What will be the maximum kinetic energy of ejected electrons when the wavelength is changed to (*a*) 280 nm, (*b*) 360 nm?
25. (II) When 250-nm light falls on a metal, the current through a photoelectric circuit (Fig. 27–6) is brought to zero at a stopping voltage of 1.64 V. What is the work function of the metal?
26. (II) In a photoelectric experiment using a clean sodium surface, the maximum energy of the emitted electrons was measured for a number of different incident frequencies, with the following results.

Frequency ($\times 10^{14}$ Hz)	Energy (eV)
11.8	2.60
10.6	2.11
9.9	1.81
9.1	1.47
8.2	1.10
6.9	0.57

Plot the graph of these results and find: (*a*) Planck's constant; (*b*) the cutoff frequency of sodium; (*c*) the work function.

27. (II) Show that the energy E (in electron volts) of a photon whose wavelength is λ (nm) is given by

$$E = \frac{1.240 \times 10^3\ \text{eV}\cdot\text{nm}}{\lambda\ (\text{nm})}.$$

Use at least 4 significant figures for values of h, c, e (see inside front cover).

*27–5 Compton Effect

*28. (I) A high-frequency photon is scattered off of an electron and experiences a change of wavelength of 1.7×10^{-4} nm. At what angle must a detector be placed to detect the scattered photon (relative to the direction of the incoming photon)?

*29. (II) The quantity h/mc, which has the dimensions of length, is called the *Compton wavelength*. Determine the Compton wavelength for (*a*) an electron, (*b*) a proton. (*c*) Show that if a photon has wavelength equal to the Compton wavelength of a particle, the photon's energy is equal to the rest energy of the particle, mc^2.

*30. (II) X-rays of wavelength $\lambda = 0.140\,\text{nm}$ are scattered from carbon. What is the expected Compton wavelength shift for photons detected at angles (relative to the incident beam) of exactly (*a*) 45°, (*b*) 90°, (*c*) 180°?

27–6 Pair Production

31. (I) How much total kinetic energy will an electron–positron pair have if produced by a 3.64-MeV photon?

32. (II) What is the longest wavelength photon that could produce a proton–antiproton pair? (Each has a mass of $1.67 \times 10^{-27}\,\text{kg}$.)

33. (II) What is the minimum photon energy needed to produce a $\mu^+\mu^-$ pair? The mass of each μ (muon) is 207 times the mass of an electron. What is the wavelength of such a photon?

34. (II) An electron and a positron, each moving at $3.0 \times 10^5\,\text{m/s}$, collide head on, disappear, and produce two photons, each with the same energy and momentum moving in opposite directions. Determine the energy and momentum of each photon.

35. (II) A gamma-ray photon produces an electron and a positron, each with a kinetic energy of 285 keV. Determine the energy and wavelength of the photon.

27–8 Wave Nature of Matter

36. (I) Calculate the wavelength of a 0.21-kg ball traveling at 0.10 m/s.

37. (I) What is the wavelength of a neutron $(m = 1.67 \times 10^{-27}\,\text{kg})$ traveling at $8.5 \times 10^4\,\text{m/s}$?

38. (II) Through how many volts of potential difference must an electron, initially at rest, be accelerated to achieve a wavelength of 0.27 nm?

39. (II) Calculate the ratio of the kinetic energy of an electron to that of a proton if their wavelengths are equal. Assume that the speeds are nonrelativistic.

40. (II) An electron has a de Broglie wavelength $\lambda = 4.5 \times 10^{-10}\,\text{m}$. (*a*) What is its momentum? (*b*) What is its speed? (*c*) What voltage was needed to accelerate it from rest to this speed?

41. (II) What is the wavelength of an electron of energy (*a*) 10 eV, (*b*) 100 eV, (*c*) 1.0 keV?

42. (II) Show that if an electron and a proton have the same nonrelativistic kinetic energy, the proton has the shorter wavelength.

43. (II) Calculate the de Broglie wavelength of an electron if it is accelerated from rest by 35,000 V as in Fig. 27–2. Is it relativistic? How does its wavelength compare to the size of the "neck" of the tube, typically 5 cm? Do we have to worry about diffraction problems blurring the picture on the CRT screen?

27–9 Electron Microscope

44. (II) What voltage is needed to produce electron wavelengths of 0.26 nm? (Assume that the electrons are nonrelativistic.)

45. (II) Electrons are accelerated by 2850 V in an electron microscope. Estimate the maximum possible resolution of the microscope.

27–11 and 27–12 Spectra and the Bohr Model

46. (I) For the three hydrogen transitions indicated below, with n being the initial state and n' being the final state, is the transition an absorption or an emission? Which is higher, the initial state energy or the final state energy of the atom? Finally, which of these transitions involves the largest energy photon? (*a*) $n = 1$, $n' = 3$; (*b*) $n = 6$, $n' = 2$; (*c*) $n = 4$, $n' = 5$.

47. (I) How much energy is needed to ionize a hydrogen atom in the $n = 3$ state?

48. (I) The second longest wavelength in the Paschen series in hydrogen (Fig. 27–29) corresponds to what transition?

49. (I) Calculate the ionization energy of doubly ionized lithium, Li^{2+}, which has $Z = 3$ (and is in the ground state).

50. (I) Evaluate the Rydberg constant R using the Bohr model (compare Eqs. 27–9 and 27–16) and show that its value is $R = 1.0974 \times 10^7\,\text{m}^{-1}$. (Use values inside front cover to 5 or 6 significant figures.)

51. (II) What is the longest wavelength light capable of ionizing a hydrogen atom in the ground state?

52. (II) What wavelength photon would be required to ionize a hydrogen atom in the ground state and give the ejected electron a kinetic energy of 11.5 eV?

53. (II) In the Sun, an ionized helium (He^+) atom makes a transition from the $n = 6$ state to the $n = 2$ state, emitting a photon. Can that photon be absorbed by hydrogen atoms present in the Sun? If so, between what energy states will the hydrogen atom transition occur?

54. (II) Construct the energy-level diagram for the He^+ ion (like Fig. 27–29).

55. (II) Construct the energy-level diagram for doubly ionized lithium, Li^{2+}.

56. (II) Determine the electrostatic potential energy and the kinetic energy of an electron in the ground state of the hydrogen atom.

57. (II) An excited hydrogen atom could, in principle, have a radius of 1.00 cm. What would be the value of n for a Bohr orbit of this size? What would its energy be?

58. (II) Is the use of nonrelativistic formulas justified in the Bohr atom? To check, calculate the electron's velocity, v, in terms of c, for the ground state of hydrogen, and then calculate $\sqrt{1 - v^2/c^2}$.

59. (III) Show that the magnitude of the electrostatic potential energy of an electron in any Bohr orbit of a hydrogen atom is twice the magnitude of its kinetic energy in that orbit.

60. (III) Suppose an electron was bound to a proton, as in the hydrogen atom, but by the gravitational force rather than by the electric force. What would be the radius, and energy, of the first Bohr orbit?

General Problems

61. The Big Bang theory (Chapter 33) states that the beginning of the universe was accompanied by a huge burst of photons. Those photons are still present today and make up the so-called cosmic microwave background radiation. The universe radiates like a blackbody with a temperature today of about 2.7 K. Calculate the peak wavelength of this radiation.

62. At low temperatures, nearly all the atoms in hydrogen gas will be in the ground state. What minimum frequency photon is needed if the photoelectric effect is to be observed?

63. A beam of 72-eV electrons is scattered from a crystal, as in X-ray diffraction, and a first-order peak is observed at $\theta = 38°$. What is the spacing between planes in the diffracting crystal? (See Section 25–11.)

64. A microwave oven produces electromagnetic radiation at $\lambda = 12.2$ cm and produces a power of 720 W. Calculate the number of microwave photons produced by the microwave oven each second.

65. Sunlight reaching the Earth's atmosphere has an intensity of about $1300\ \text{W/m}^2$. Estimate how many photons per square meter per second this represents. Take the average wavelength to be 550 nm.

66. A beam of red laser light ($\lambda = 633$ nm) hits a black wall and is fully absorbed. If this light exerts a total force $F = 5.8$ nN on the wall, how many photons per second are hitting the wall?

67. A flashlight emits 2.5 W of light. As the light leaves the flashlight in one direction, a reaction force is exerted on the flashlight in the opposite direction. Estimate the size of this reaction force.

68. A **photomultiplier tube** (a very sensitive light sensor), is based on the photoelectric effect: incident photons strike a metal surface and the resulting ejected electrons are collected. By counting the number of collected electrons, the number of incident photons (i.e., the incident light intensity) can be determined. (*a*) If a photomultiplier tube is to respond properly for incident wavelengths throughout the visible range (410 nm to 750 nm), what is the maximum value for the work function W_0 (eV) of its metal surface? (*b*) If W_0 for its metal surface is above a certain threshold value, the photomultiplier will only function for incident ultraviolet wavelengths and be unresponsive to visible light. Determine this threshold value (eV).

69. An electron and a positron collide head on, annihilate, and create two 0.85-MeV photons traveling in opposite directions. What were the initial kinetic energies of electron and positron?

70. By what potential difference must (*a*) a proton $(m = 1.67 \times 10^{-27}\ \text{kg})$, and (*b*) an electron $(m = 9.11 \times 10^{-31}\ \text{kg})$, be accelerated from rest to have a wavelength $\lambda = 4.0 \times 10^{-12}$ m?

71. In some of Rutherford's experiments (Fig. 27–19) the α particles $(\text{mass} = 6.64 \times 10^{-27}\ \text{kg})$ had a kinetic energy of 4.8 MeV. How close could they get to the surface of a gold nucleus (radius $\approx 7.0 \times 10^{-15}$ m, charge $= +79e$)? Ignore the recoil motion of the nucleus.

72. By what fraction does the mass of an H atom decrease when it makes an $n = 3$ to $n = 1$ transition?

73. Calculate the ratio of the gravitational force to the electric force for the electron in the ground state of a hydrogen atom. Can the gravitational force be reasonably ignored?

74. Electrons accelerated from rest by a potential difference of 12.3 V pass through a gas of hydrogen atoms at room temperature. What wavelengths of light will be emitted?

75. In a particular photoelectric experiment, a stopping potential of 2.10 V is measured when ultraviolet light of wavelength 270 nm is incident on the metal. Using the same setup, what will the new stopping potential be if blue light of wavelength 440 nm is used, instead?

76. Neutrons can be used in diffraction experiments to probe the lattice structure of crystalline solids. Since the neutron's wavelength needs to be on the order of the spacing between atoms in the lattice, about 0.3 nm, what should the speed of the neutrons be?

77. In Chapter 22, the intensity of light striking a surface was related to the electric field of the associated electromagnetic wave. For photons, the intensity is the number of photons striking a 1-m^2 area per second. Suppose 1.0×10^{12} photons of 497-nm light are incident on a 1-m^2 surface every second. What is the intensity of the light? Using the wave model of light, what is the maximum electric field of the electromagnetic wave?

78. The intensity of the Sun's light in the vicinity of the Earth is about $1350\ \text{W/m}^2$. Imagine a spacecraft with a mirrored square sail of dimension 1.0 km. Estimate how much thrust (in newtons) this craft will experience due to collisions with the Sun's photons. [*Hint*: Assume the photons bounce off the sail with no change in the magnitude of their momentum.]

79. Light of wavelength 280 nm strikes a metal whose work function is 2.2 eV. What is the shortest de Broglie wavelength for the electrons that are produced as photoelectrons?

80. What would be the theoretical limit of resolution for an electron microscope whose electrons are accelerated through 110 kV? (Relativistic formulas should be used.)

81. Assume hydrogen atoms in a gas are initially in their ground state. If free electrons with kinetic energy 12.75 eV collide with these atoms, what photon wavelengths will be emitted by the gas?

82. Visible light incident on a diffraction grating with slit spacing of 0.010 mm has the first maximum at an angle of 3.6° from the central peak. If electrons could be diffracted by the same grating, what electron velocity would produce the same diffraction pattern as the visible light?

83. (*a*) Suppose an unknown element has an absorption spectrum with lines corresponding to 2.5, 4.7, and 5.1 eV above its ground state and an ionization energy of 11.5 eV. Draw an energy level diagram for this element. (*b*) If a 5.1-eV photon is absorbed by an atom of this substance, in which state was the atom before absorbing the photon? What will be the energies of the photons that can subsequently be emitted by this atom?

84. A photon of momentum $3.53 \times 10^{-28}\,\text{kg}\cdot\text{m/s}$ is emitted from a hydrogen atom. To what spectrum series does this photon belong, and from what energy level was it ejected?

85. Light of wavelength 464 nm falls on a metal which has a work function of 2.28 eV. (*a*) How much voltage should be applied to bring the current to zero? (*b*) What is the maximum speed of the emitted electrons? (*c*) What is the de Broglie wavelength of these electrons?

86. An electron accelerated from rest by a 96-V potential difference is injected into a $3.67 \times 10^{-4}\,\text{T}$ magnetic field where it travels in an 18-cm-diameter circle. Calculate e/m from this information.

87. Estimate the number of photons emitted by the Sun in a year. (Take the average wavelength to be 550 nm and the intensity of sunlight reaching the Earth (outer atmosphere) as $1350\,\text{W/m}^2$.)

88. Apply Bohr's assumptions to the Earth–Moon system to calculate the allowed energies and radii of motion. Given the known distance between the Earth and Moon, is the quantization of the energy and radius apparent?

89. At what temperature would the average kinetic energy (Chapter 13) of a molecule of hydrogen gas (H_2) be sufficient to excite a hydrogen atom out of the ground state?

Search and Learn

1. Imagine the following Young's double-slit experiment using matter rather than light: electrons are accelerated through a potential difference of 12 V, pass through two closely spaced slits separated by a distance d, and create an interference pattern. (*a*) Using Example 27–11 and Section 24–3 as guides, find the required value for d if the first-order interference fringe is to be produced at an angle of 10°. (*b*) Given the approximate size of atoms, would it be possible to construct the required two-slit set-up for this experiment?

2. Does each of the following support the wave nature or the particle nature of light? (*a*) The existence of the cutoff frequency in the photoelectric effect; (*b*) Young's double-slit experiment; (*c*) the shift in the photon frequency in Compton scattering; (*d*) the diffraction of light.

3. (*a*) From Sections 22–3, 24–4, and 27–3, estimate the minimum energy (eV) that initiates the chemical process on the retina responsible for vision. (*b*) Estimate the threshold photon energy above which the eye registers no sensation of sight.

4. (*a*) A rubidium atom ($m = 85\,\text{u}$) is at rest with one electron in an excited energy level. When the electron jumps to the ground state, the atom emits a photon of wavelength $\lambda = 780\,\text{nm}$. Determine the resulting (nonrelativistic) recoil speed v of the atom. (*b*) The recoil speed sets the lower limit on the temperature to which an ideal gas of rubidium atoms can be cooled in a laser-based **atom trap**. Using the kinetic theory of gases (Chapter 13), estimate this "lowest achievable" temperature.

5. Suppose a particle of mass m is confined to a one-dimensional box of width L. According to quantum theory, the particle's wave (with $\lambda = h/mv$) is a standing wave with nodes at the edges of the box. (*a*) Show the possible modes of vibration on a diagram. (*b*) Show that the kinetic energy of the particle has quantized energies given by $\text{KE} = n^2h^2/8mL^2$, where n is an integer. (*c*) Calculate the ground-state energy ($n = 1$) for an electron confined to a box of width $0.50 \times 10^{-10}\,\text{m}$. (*d*) What is the ground-state energy, and speed, of a baseball ($m = 140\,\text{g}$) in a box 0.65 m wide? (*e*) An electron confined to a box has a ground-state energy of 22 eV. What is the width of the box? [*Hint*: See Sections 27–8, 27–13, and 11–12.]

ANSWERS TO EXERCISES

A: $\lambda_p = 725\,\text{nm}$, so red.
B: More 1000-nm photons (each has lower energy).
C: $5.50 \times 10^{14}\,\text{Hz}$, 545 nm.
D: Only λ.
E: Decrease.
F: (*e*).



A neon tube is a thin glass tube, moldable into various shapes, filled with neon (or other) gas that glows with a particular color when a current at high voltage passes through it. Gas atoms, excited to upper energy levels, jump down to lower energy levels and emit light (photons) whose wavelengths (color) are characteristic of the type of gas.

In this Chapter we study what quantum mechanics tells us about atoms, their energy levels, and the effect of the exclusion principle for atoms with more than one electron. We also discuss interesting applications such as lasers and holography.

CHAPTER 28

Quantum Mechanics of Atoms

CHAPTER-OPENING QUESTION—Guess now!

The uncertainty principle states that

(a) no measurement can be perfect because it is technologically impossible to make perfect measuring instruments.

(b) it is impossible to measure exactly where a particle is, unless it is at rest.

(c) it is impossible to simultaneously know both the position and the momentum of a particle with complete certainty.

(d) a particle cannot actually have a completely certain value of momentum.

Bohr's model of the atom gave us a first (though rough) picture of what an atom is like. It proposed explanations for why there is emission and absorption of light by atoms at only certain wavelengths. The wavelengths of the line spectra and the ionization energy for hydrogen (and one-electron ions) are in excellent agreement with experiment. But the Bohr model had important limitations. It was not able to predict line spectra for more complex atoms—atoms with more than one electron—not even for the neutral helium atom, which has only two electrons. Nor could it explain why emission lines, when viewed with great precision, consist of two or more very closely spaced lines (referred to as *fine structure*). The Bohr model also did not explain why some spectral lines were brighter than others. And it could not explain the bonding of atoms in molecules or in solids and liquids.

From a theoretical point of view, too, the Bohr model was not satisfactory: it was a strange mixture of classical and quantum ideas. Moreover, the wave–particle duality was not really resolved.

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FIGURE 28–1 Erwin Schrödinger with Lise Meitner. (She was a codiscoverer of nuclear fission, Chapter 31.)



FIGURE 28–2 Werner Heisenberg (center) on Lake Como (Italy) with Enrico Fermi (left) and Wolfgang Pauli (right).

We mention these limitations of the Bohr model not to disparage it—for it was a landmark in the history of science. Rather, we mention them to show why, in the early 1920s, it became increasingly evident that a new, more comprehensive theory was needed. It was not long in coming. Less than two years after de Broglie gave us his matter–wave hypothesis, Erwin Schrödinger (1887–1961; Fig. 28–1) and Werner Heisenberg (1901–1976; Fig. 28–2) independently developed a new comprehensive theory.

28–1 Quantum Mechanics—A New Theory

The new theory, called **quantum mechanics**, has been extremely successful. It unifies the wave–particle duality into a single consistent theory and has successfully dealt with the spectra emitted by complex atoms, even the fine details. It explains the relative brightness of spectral lines and how atoms form molecules. It is also a much more general theory that covers all quantum phenomena from blackbody radiation to atoms and molecules. It has explained a wide range of natural phenomena and from its predictions many new practical devices have become possible. Indeed, it has been so successful that it is accepted today by nearly all physicists as the fundamental theory underlying physical processes.

Quantum mechanics deals mainly with the microscopic world of atoms and light. But this new theory, when it is applied to macroscopic phenomena, must be able to produce the old classical laws. This, the **correspondence principle** (already mentioned in Section 27–12), is satisfied fully by quantum mechanics.

This doesn't mean we should throw away classical theories such as Newton's laws. In the everyday world, classical laws are far easier to apply and they give sufficiently accurate descriptions. But when we deal with high speeds, close to the speed of light, we must use the theory of relativity; and when we deal with the tiny world of the atom, we use quantum mechanics.

Although we won't go into the detailed mathematics of quantum mechanics, we will discuss the main ideas and how they involve the wave and particle properties of matter to explain atomic structure and other applications.

28–2 The Wave Function and Its Interpretation; the Double-Slit Experiment

The important properties of any wave are its wavelength, frequency, and amplitude. For an electromagnetic wave, the frequency (or wavelength) determines whether the light is in the visible spectrum or not, and if so, what color it is. We also have seen that the frequency is a measure of the energy of the corresponding photon, $E = hf$ (Eq. 27–4). The amplitude or displacement of an electromagnetic wave at any point is the strength of the electric (or magnetic) field at that point, and is related to the intensity of the wave (the brightness of the light).

For material particles such as electrons, quantum mechanics relates the wavelength to momentum according to de Broglie's formula, $\lambda = h/p$, Eq. 27–8. But what corresponds to the *amplitude* or *displacement* of a matter wave? The amplitude of an electromagnetic wave is represented by the electric and magnetic fields, E and B. In quantum mechanics, this role is played by the **wave function**, which is given the symbol Ψ (the Greek capital letter psi, pronounced "sigh"). Thus Ψ represents the wave displacement, as a function of time and position, of a new kind of field which we might call a "matter" field or a matter wave.

To understand how to interpret the wave function Ψ, we make an analogy with light using the wave–particle duality.

We saw in Chapter 11 that the intensity I of any wave is proportional to the square of the amplitude. This holds true for light waves as well, as we saw in Chapter 22. That is,

$$I \propto E^2,$$

where E is the electric field strength. From the *particle* point of view, the intensity of a light beam (of given frequency) is proportional to the number of photons, N, that pass through a given area per unit time. The more photons there are, the greater the intensity. Thus

$$I \propto E^2 \propto N.$$

This proportion can be turned around so that we have

$$N \propto E^2.$$

That is, the number of photons (striking a page of this book, say) is proportional to the square of the electric field strength.

If the light beam is very weak, only a few photons will be involved. Indeed, it is possible to "build up" a photograph in a camera using very weak light so the effect of photons arriving can be seen. If we are dealing with only one photon, the relationship above $(N \propto E^2)$ can be interpreted in a slightly different way. At any point, the square of the electric field strength E^2 is a measure of the *probability* that a photon will be at that location. At points where E^2 is large, there is a high probability the photon will be there; where E^2 is small, the probability is low.

We can interpret matter waves in the same way, as was first suggested by Max Born (1882–1970) in 1927. The wave function Ψ may vary in magnitude from point to point in space and time. If Ψ describes a collection of many electrons, then Ψ^2 at any point will be proportional to the number of electrons expected to be found at that point. When dealing with small numbers of electrons we can't make very exact predictions, so Ψ^2 takes on the character of a probability. If Ψ, which depends on time and position, represents a single electron (say, in an atom), then Ψ^2 is interpreted like this: Ψ^2 *at a certain point in space and time represents the probability of finding the electron at the given position and time.* Thus Ψ^2 is often referred to as the **probability density** or **probability distribution**.

Double-Slit Interference Experiment for Electrons

To understand this better, we take as a thought experiment the familiar double-slit experiment, and consider it both for light and for electrons.

Consider two slits whose size and separation are on the order of the wavelength of whatever we direct at them, either light or electrons, Fig. 28–3. We know very well what would happen in this case for light, since this is just Young's double-slit experiment (Section 24–3): an interference pattern would be seen on the screen behind. If light were replaced by electrons with wavelength comparable to the slit size, they too would produce an interference pattern (recall Fig. 27–12). In the case of light, the pattern would be visible to the eye or could be recorded on film, semiconductor sensor, or screen. For electrons, a fluorescent screen could be used (it glows where an electron strikes).

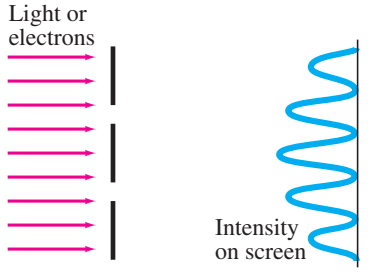


FIGURE 28–3 Parallel beam, of light or electrons, falls on two slits whose sizes are comparable to the wavelength. An interference pattern is observed.

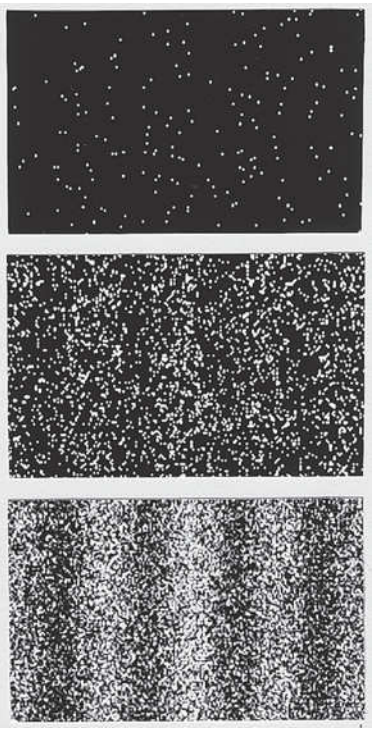


FIGURE 28–4 Young's double-slit experiment done with electrons—note that the pattern is not evident with only a few electrons (top photo), but with more and more electrons (second and third photos), the familiar double-slit interference pattern (Chapter 24) is seen.

If we reduced the flow of electrons (or photons) so they passed through the slits one at a time, we would see a flash each time one struck the screen. At first, the flashes would seem random. Indeed, there is no way to predict just where any one electron would hit the screen. If we let the experiment run for a long time, and kept track of where each electron hit the screen, we would soon see a pattern emerging—the interference pattern predicted by the wave theory; see Fig. 28–4. Thus, although we could not predict where a given electron would strike the screen, we could predict probabilities. (The same can be said for photons.) The probability, as we saw, is proportional to Ψ^2. Where Ψ^2 is zero, we would get a minimum in the interference pattern. And where Ψ^2 is a maximum, we would get a peak in the interference pattern.

The interference pattern would thus occur even when electrons (or photons) passed through the slits one at a time. So the interference pattern could not arise from the interaction of one electron with another. It is as if an electron passed through both slits at the same time, interfering with itself. This is possible because an electron is not precisely a particle. It is as much a wave as it is a particle, and a wave could travel through both slits at once. But what would happen if we covered one of the slits so we knew that the electron passed through the other slit, and a little later we covered the second slit so the electron had to have passed through the first slit? The result would be that no interference pattern would be seen. We would see, instead, two bright areas (or diffraction patterns) on the screen behind the slits.

If both slits are open, the screen shows an interference pattern as if each electron passed through both slits, like a wave. Yet each electron would make a tiny spot on the screen as if it were a particle.

The main point of this discussion is this: if we treat electrons (and other particles) as if they were waves, then Ψ represents the wave amplitude. If we treat them as particles, then we must treat them on a *probabilistic* basis. The square of the wave function, Ψ^2, gives the probability of finding a given electron at a given point. We cannot predict—or even follow—the path of a single electron precisely through space and time.

28–3 The Heisenberg Uncertainty Principle

Whenever a measurement is made, some uncertainty is always involved. For example, you cannot make an absolutely exact measurement of the length of a table. Even with a measuring stick that has markings 1 mm apart, there will be an inaccuracy of perhaps $\frac{1}{2}$ mm or so. More precise instruments will produce more precise measurements. But there is always some uncertainty involved in a measurement, no matter how good the measuring device. We expect that by using more precise instruments, the uncertainty in a measurement can be made indefinitely small.

But according to quantum mechanics, there is actually a limit to the precision of certain measurements. This limit is not a restriction on how well instruments can be made; rather, it is inherent in nature. It is the result of two factors: the wave–particle duality, and the unavoidable interaction between the thing observed and the observing instrument. Let us look at this in more detail.

To make a measurement on an object without disturbing it, at least a little, is not possible. Consider trying to locate a lost Ping-pong ball in a dark room: you could probe about with your hand or a stick, or you could shine a light and detect the photons reflecting off the ball. When you search with your hand or a stick, you find the ball's position when you touch it, but at the same time you unavoidably bump it, and give it some momentum. Thus you won't know its *future* position. If you search for the Ping-pong ball using light, in order to "see" the ball at least one photon (really, quite a few) must scatter from it, and the reflected photon must enter your eye or some other detector. When a photon strikes an ordinary-sized object, it only slightly alters the motion or position of the object.

But a photon striking a tiny object like an electron transfers enough momentum to greatly change the electron's motion and position in an unpredictable way. The mere act of measuring the position of an object at one time makes our knowledge of its future position imprecise.

Now let us see where the wave–particle duality comes in. Imagine a thought experiment in which we are trying to measure the position of an object, say an electron, with photons, Fig. 28–5. (The arguments would be similar if we were using, instead, an electron microscope.) As we saw in Chapter 25, objects can be seen to a precision at best of about the wavelength of the radiation used due to diffraction. If we want a precise position measurement, we must use a short wavelength. But a short wavelength corresponds to high frequency and large momentum ($p = h/\lambda$); and the more momentum the photons have, the more momentum they can give the object when they strike it. If we use photons of longer wavelength, and correspondingly smaller momentum, the object's motion when struck by the photons will not be affected as much. But the longer wavelength means lower resolution, so the object's position will be less accurately known. Thus the act of observing produces an uncertainty in both the *position* and the *momentum* of the electron. This is the essence of the *uncertainty principle* first enunciated by Heisenberg in 1927.

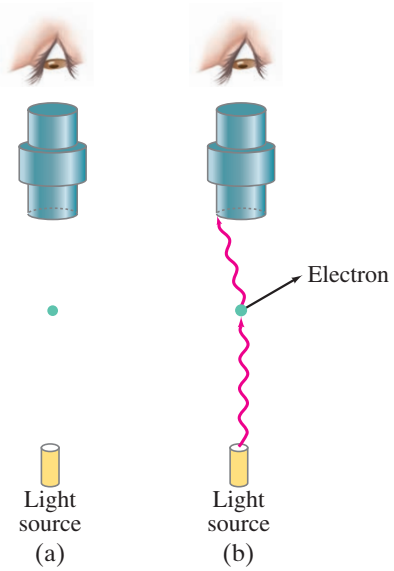


FIGURE 28–5 Thought experiment for observing an electron with a powerful light microscope. At least one photon must scatter from the electron (transferring some momentum to it) and enter the microscope.

Quantitatively, we can make an approximate calculation of the magnitude of the uncertainties. If we use light of wavelength λ, the position can be measured at best to a precision of about λ. That is, the uncertainty in the position measurement, Δx, is approximately

$$\Delta x \approx \lambda.$$

Suppose that the object can be detected by a single photon. The photon has a momentum $p_x = h/\lambda$ (Eq. 27–6). When the photon strikes our object, it will give some or all of this momentum to the object, Fig. 28–5. Therefore, the final x momentum of our object will be uncertain in the amount

$$\Delta p_x \approx \frac{h}{\lambda}$$

since we can't tell how much momentum will be transferred. The product of these uncertainties is

$$(\Delta x)(\Delta p_x) \approx (\lambda)\left(\frac{h}{\lambda}\right) \approx h.$$

The uncertainties could be larger than this, depending on the apparatus and the number of photons needed for detection. A more careful mathematical calculation shows the product of the uncertainties as, at best, about

$$(\Delta x)(\Delta p_x) \gtrapprox \frac{h}{2\pi}. \qquad \textbf{(28–1)}$$

UNCERTAINTY PRINCIPLE (position and momentum)

This is a mathematical statement of the **Heisenberg uncertainty principle**, or, as it is sometimes called, the **indeterminancy principle**. It tells us that we cannot measure both the position *and* momentum of an object precisely at the same time. The more accurately we try to measure the position, so that Δx is small, the greater will be the uncertainty in momentum, Δp_x. If we try to measure the momentum very accurately, then the uncertainty in the position becomes large. The uncertainty principle does not forbid individual precise measurements, however. For example, in principle we could measure the position of an object exactly. But then its momentum would be completely unknown. Thus, although we might know the position of the object exactly at one instant, we could have no idea at all where it would be a moment later. The uncertainties expressed here are inherent in nature, and reflect the best precision theoretically attainable even with the best instruments.

CAUTION

Uncertainties not due to instrument deficiency, but inherent in nature (wave–particle)

EXERCISE A Return to the Chapter-Opening Question, page 803, and answer it again now. Try to explain why you may have answered differently the first time.

Another useful form of the uncertainty principle relates energy and time, and we examine this as follows. The object to be detected has an uncertainty in position $\Delta x \approx \lambda$. The photon that detects it travels with speed c, and it takes a time $\Delta t \approx \Delta x/c \approx \lambda/c$ to pass through the distance of uncertainty. Hence, the measured time when our object is at a given position is uncertain by about

$$\Delta t \approx \frac{\lambda}{c}.$$

Since the photon can transfer some or all of its energy $(= hf = hc/\lambda)$ to our object, the uncertainty in energy of our object as a result is

$$\Delta E \approx \frac{hc}{\lambda}.$$

The product of these two uncertainties is

$$(\Delta E)(\Delta t) \approx \left(\frac{hc}{\lambda}\right)\left(\frac{\lambda}{c}\right) \approx h.$$

A more careful calculation gives

UNCERTAINTY PRINCIPLE (energy and time)

$$(\Delta E)(\Delta t) \gtrsim \frac{h}{2\pi}. \qquad \textbf{(28–2)}$$

This form of the uncertainty principle tells us that the energy of an object can be uncertain (or can be interpreted as briefly nonconserved) by an amount ΔE for a time $\Delta t \approx h/(2\pi\,\Delta E)$.

The quantity $(h/2\pi)$ appears so often in quantum mechanics that for convenience it is given the symbol $\hbar$ ("h-bar"). That is,

$$\hbar = \frac{h}{2\pi} = \frac{6.626 \times 10^{-34}\,\text{J}\cdot\text{s}}{2\pi} = 1.055 \times 10^{-34}\,\text{J}\cdot\text{s}.$$

By using this notation, Eqs. 28–1 and 28–2 for the uncertainty principle can be written

$$(\Delta x)(\Delta p_x) \gtrsim \hbar$$

and

$$(\Delta E)(\Delta t) \gtrsim \hbar.$$

We have been discussing the position and velocity of an electron as if it were a particle. But it isn't simply a particle. Indeed, we have the uncertainty principle because an electron—and matter in general—has wave as well as particle properties. What the uncertainty principle really tells us is that if we insist on thinking of the electron as a particle, then there are certain limitations on this simplified view—namely, that the position and velocity cannot both be known precisely at the same time; and even that the electron does not *have* a precise position and momentum at the same time (because it is not simply a particle). Similarly, the energy can be uncertain (or nonconserved) by an amount ΔE for a time $\Delta t \approx \hbar/\Delta E$.

Because Planck's constant, h, is so small, the uncertainties expressed in the uncertainty principle are usually negligible on the macroscopic level. But at the level of atomic sizes, the uncertainties are significant. Because we consider ordinary objects to be made up of atoms containing nuclei and electrons, the uncertainty principle is relevant to our understanding of all of nature. The uncertainty principle expresses, perhaps most clearly, the probabilistic nature of quantum mechanics. It thus is often used as a basis for philosophic discussion.

EXAMPLE 28–1 **Position uncertainty of electron.** An electron moves in a straight line with a constant speed $v = 1.10 \times 10^6\,\text{m/s}$ which has been measured to a precision of 0.10%. What is the maximum precision with which its position could be simultaneously measured?

APPROACH The momentum is $p = mv$, and the uncertainty in p is $\Delta p = 0.0010p$. The uncertainty principle (Eq. 28–1) gives us the smallest uncertainty in position Δx using the equals sign.

SOLUTION The momentum of the electron is

$$p = mv = (9.11 \times 10^{-31}\,\text{kg})(1.10 \times 10^6\,\text{m/s}) = 1.00 \times 10^{-24}\,\text{kg}\cdot\text{m/s}.$$

The uncertainty in the momentum is 0.10% of this, or $\Delta p = 1.0 \times 10^{-27}\,\text{kg}\cdot\text{m/s}$. From the uncertainty principle, the best simultaneous position measurement will have an uncertainty of

$$\Delta x \approx \frac{\hbar}{\Delta p} = \frac{1.055 \times 10^{-34}\,\text{J}\cdot\text{s}}{1.0 \times 10^{-27}\,\text{kg}\cdot\text{m/s}} = 1.1 \times 10^{-7}\,\text{m},$$

or 110 nm.

NOTE This is about 1000 times the diameter of an atom.

EXERCISE B An electron's position is measured with a precision of $0.50 \times 10^{-10}\,\text{m}$. Find the minimum uncertainty in its momentum and velocity.

EXAMPLE 28–2 **Position uncertainty of a baseball.** What is the uncertainty in position, imposed by the uncertainty principle, on a 150-g baseball thrown at $(93 \pm 2)\,\text{mi/h} = (42 \pm 1)\,\text{m/s}$?

APPROACH The uncertainty in the speed is $\Delta v = 1\,\text{m/s}$. We multiply Δv by m to get Δp and then use the uncertainty principle, solving for Δx.

SOLUTION The uncertainty in the momentum is

$$\Delta p = m\,\Delta v = (0.150\,\text{kg})(1\,\text{m/s}) = 0.15\,\text{kg}\cdot\text{m/s}.$$

Hence the uncertainty in a position measurement could be as small as

$$\Delta x = \frac{\hbar}{\Delta p} = \frac{1.055 \times 10^{-34}\,\text{J}\cdot\text{s}}{0.15\,\text{kg}\cdot\text{m/s}} = 7 \times 10^{-34}\,\text{m}.$$

NOTE This distance is far smaller than any we could imagine observing or measuring. It is trillions of trillions of times smaller than an atom. Indeed, the uncertainty principle sets no relevant limit on measurement for macroscopic objects.

EXAMPLE 28–3 **ESTIMATE** **J/ψ lifetime calculated.** The J/ψ meson, discovered in 1974, was measured to have an average mass of $3100\,\text{MeV}/c^2$ (note the use of energy units since $E = mc^2$) and a mass "width" of $63\,\text{keV}/c^2$. By this we mean that the masses of different J/ψ mesons were actually measured to be slightly different from one another. This mass "width" is related to the very short lifetime of the J/ψ before it decays into other particles. From the uncertainty principle, if the particle exists for only a time Δt, its mass (or rest energy) will be uncertain by $\Delta E \approx \hbar/\Delta t$. Estimate the J/ψ lifetime.

APPROACH We use the energy–time version of the uncertainty principle, Eq. 28–2.

SOLUTION The uncertainty of $63\,\text{keV}/c^2$ in the J/ψ's mass is an uncertainty in its rest energy, which in joules is

$$\Delta E = (63 \times 10^3\,\text{eV})(1.60 \times 10^{-19}\,\text{J/eV}) = 1.01 \times 10^{-14}\,\text{J}.$$

Then we expect its lifetime τ $(= \Delta t$ using Eq. 28–2) to be

$$\tau \approx \frac{\hbar}{\Delta E} = \frac{1.055 \times 10^{-34}\,\text{J}\cdot\text{s}}{1.01 \times 10^{-14}\,\text{J}} \approx 1 \times 10^{-20}\,\text{s}.$$

Lifetimes this short are difficult to measure directly, and the assignment of very short lifetimes depends on this use of the uncertainty principle.

28–4 Philosophic Implications; Probability versus Determinism

The classical Newtonian view of the world is a deterministic one (see Section 5–8). One of its basic ideas is that once the position and velocity of an object are known at a particular time, its future position can be predicted if the forces on it are known. For example, if a stone is thrown a number of times with the same initial velocity and angle, and the forces on it remain the same, the path of the projectile will always be the same. If the forces are known (gravity and air resistance, if any), the stone's path can be precisely predicted. This mechanistic view implies that the future unfolding of the universe, assumed to be made up of particulate objects, is completely determined.

This classical deterministic view of the physical world has been radically altered by quantum mechanics. As we saw in the analysis of the double-slit experiment (Section 28–2), electrons all treated in the same way will not all end up in the same place. According to quantum mechanics, certain probabilities exist that an electron will arrive at different points. This is very different from the classical view, in which the path of a particle is precisely predictable from the initial position and velocity and the forces exerted on it. According to quantum mechanics, the position and velocity of an object cannot even be known accurately at the same time. This is expressed in the uncertainty principle, and arises because basic entities, such as electrons, are not considered simply as particles: they have wave properties as well. Quantum mechanics allows us to calculate only the probability[†] that, say, an electron (when thought of as a particle) will be observed at various places. Quantum mechanics says there is some inherent unpredictability in nature. This is very different from the deterministic view of classical mechanics.

Because matter is considered to be made up of atoms, even ordinary-sized objects are expected to be governed by probability, rather than by strict determinism. For example, quantum mechanics predicts a finite (but negligibly small) probability that when you throw a stone, its path might suddenly curve upward instead of following the downward-curved parabola of normal projectile motion. Quantum mechanics predicts with extremely high probability that ordinary objects will behave just as the classical laws of physics predict. But these predictions are considered probabilities, not absolute certainties. The reason that macroscopic objects behave in accordance with classical laws with such high probability is due to the large number of molecules involved: when large numbers of objects are present in a statistical situation, deviations from the average (or most probable) approach zero. It is the average configuration of vast numbers of molecules that follows the so-called fixed laws of classical physics with such high probability, and gives rise to an apparent "determinism." Deviations from classical laws are observed when small numbers of molecules are dealt with. We can say, then, that although there are no precise deterministic laws in quantum mechanics, there are statistical laws based on probability.

It is important to note that there is a difference between the probability imposed by quantum mechanics and that used in the nineteenth century to understand thermodynamics and the behavior of gases in terms of molecules (Chapters 13 and 15). In thermodynamics, probability is used because there are far too many particles to keep track of. But the molecules are still assumed to move and interact in a deterministic way following Newton's laws. Probability in quantum mechanics is quite different; it is seen as *inherent* in nature, and not as a limitation on our abilities to calculate or to measure.

[†]Note that these probabilities can be calculated precisely, just like predictions of probabilities at rolling dice or dealing cards; but they are unlike predictions of probabilities at sporting events, which are only estimates.

The view presented here is the generally accepted one and is called the **Copenhagen interpretation** of quantum mechanics in honor of Niels Bohr's home, since it was largely developed there through discussions between Bohr and other prominent physicists.

Because electrons are not simply particles, they cannot be thought of as following particular paths in space and time. This suggests that a description of matter in space and time may not be completely correct. This deep and far-reaching conclusion has been a lively topic of discussion among philosophers. Perhaps the most important and influential philosopher of quantum mechanics was Bohr. He argued that a space–time description of actual atoms and electrons is not possible. Yet a description of experiments on atoms or electrons must be given in terms of space and time and other concepts familiar to ordinary experience, such as waves and particles. We must not let our *descriptions* of experiments lead us into believing that atoms or electrons themselves actually move in space and time as classical particles.

28–5 Quantum-Mechanical View of Atoms

At the beginning of this Chapter, we discussed the limitations of the Bohr model of atomic structure. Now we examine the quantum-mechanical theory of atoms, which is a far more complete theory than the old Bohr model. Although the Bohr model has been discarded as an accurate description of nature, nonetheless, quantum mechanics reaffirms certain aspects of the older theory, such as that electrons in an atom exist only in discrete states of definite energy, and that a photon of light is emitted (or absorbed) when an electron makes a transition from one state to another. But quantum mechanics is a much deeper theory, and has provided us with a very different view of the atom. According to quantum mechanics, electrons do not exist in well-defined circular orbits as in the Bohr model. Rather, the electron (because of its wave nature) can be thought of as spread out in space as a "**cloud**." The size and shape of the electron cloud can be calculated for a given state of an atom. For the ground state in the hydrogen atom, the electron cloud is spherically symmetric, as shown in Fig. 28–6. The electron cloud at its higher densities roughly indicates the "size" of an atom. But just as a cloud may not have a distinct border, atoms do not have a precise boundary or a well-defined size. Not all electron clouds have a spherical shape, as we shall see later in this Chapter.

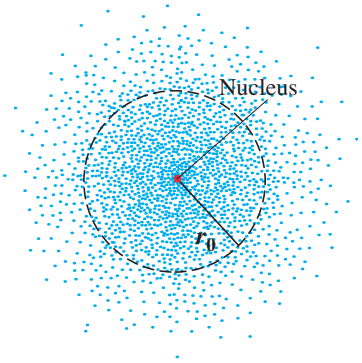


FIGURE 28–6 Electron cloud or "probability distribution" for the ground state of the hydrogen atom, as seen from afar. The dots represent a hypothetical detection of an electron at each point: dots closer together represent more probable presence of an electron (denser cloud). The dashed circle represents the Bohr radius r_0.

The electron cloud can be interpreted from either the particle or the wave viewpoint. Remember that by a particle we mean something that is localized in space—it has a definite position at any given instant. By contrast, a wave is spread out in space. The electron cloud, spread out in space as in Fig. 28–6, is a result of the wave nature of electrons. Electron clouds can also be interpreted as **probability distributions** (or **probability density**) for a particle. As we saw in Section 28–3, we cannot predict the path an electron will follow (thinking of it as a particle). After one measurement of its position we cannot predict exactly where it will be at a later time. We can only calculate the probability that it will be found at different points. If you were to make 500 different measurements of the position of an electron in a hydrogen atom, the majority of the results would show the electron at points where the probability is high (dark area in Fig. 28–6). Only occasionally would the electron be found where the probability is low. The electron cloud or probability distribution becomes small (or thin) at places, especially far away, but never becomes zero. So quantum mechanics suggests that an atom is *not* mostly empty space, and that there is no truly empty space in the universe.

28–6 Quantum Mechanics of the Hydrogen Atom; Quantum Numbers

We now look more closely at what quantum mechanics tells us about the hydrogen atom. Much of what we say here also applies to more complex atoms, which are discussed in the next Section.

Quantum mechanics is a much more sophisticated and successful theory than Bohr's. Yet in a few details they agree. Quantum mechanics predicts the same basic energy levels (Fig. 27–29) for the hydrogen atom as does the Bohr model. That is,

$$E_n = -\frac{13.6\,\text{eV}}{n^2}, \qquad n = 1, 2, 3, \cdots,$$

where n is an integer. In the simple Bohr model, there was only one quantum number, n. In quantum mechanics, four different quantum numbers are needed to specify each state in the atom:

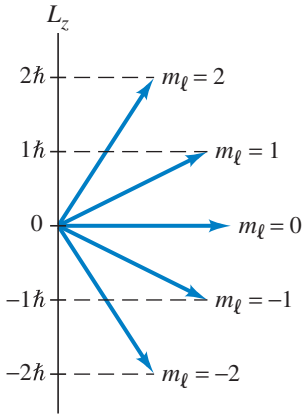
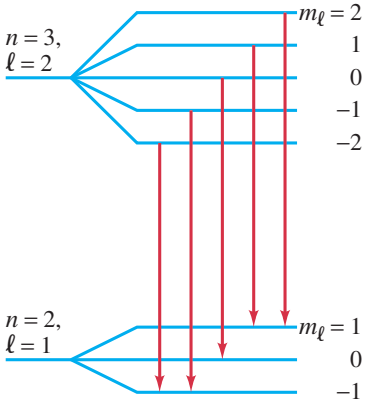


FIGURE 28–7 Quantization of angular momentum direction for $\ell = 2$. (Magnitude of $\vec{\mathbf{L}}$ is $L = \sqrt{6}\,\hbar$.)

FIGURE 28–8 Energy levels (not to scale). When a magnetic field is applied, the $n = 3$, $\ell = 2$ energy level is split into five separate levels, corresponding to the five values of m_ℓ (2, 1, 0, −1, −2). An $n = 2$, $\ell = 1$ level is split into three levels ($m_\ell = 1, 0, -1$). Transitions can occur between levels (not all transitions are shown), with photons of several slightly different frequencies being given off (the Zeeman effect).



(1) The quantum number, n, from the Bohr model is found also in quantum mechanics and is called the **principal quantum number**. It can have any integer value from 1 to ∞. The total energy of a state in the hydrogen atom depends on n, as we saw above.

(2) The **orbital quantum number**, ℓ, is related to the magnitude of the angular momentum of the electron; ℓ can take on integer values from 0 to $(n - 1)$. For the ground state, $n = 1$, ℓ can only be zero.† For $n = 3$, ℓ can be 0, 1, or 2. The actual magnitude of the angular momentum L is related to the quantum number ℓ by

$$L = \sqrt{\ell(\ell + 1)}\,\hbar \qquad \textbf{(28–3)}$$

(where again $\hbar = h/2\pi$). The value of ℓ has almost no effect on the total energy in the hydrogen atom; only n does to any appreciable extent (but see *fine structure* below). In atoms with two or more electrons, the energy does depend on ℓ as well as n, as we shall see.

(3) The **magnetic quantum number**, m_ℓ, is related to the *direction* of the electron's angular momentum, and it can take on integer values ranging from $-\ell$ to $+\ell$. For example, if $\ell = 2$, then m_ℓ can be -2, -1, 0, $+1$, or $+2$. Since angular momentum is a vector, it is not surprising that both its magnitude and its direction would be quantized. For $\ell = 2$, the five different directions allowed can be represented by the diagram of Fig. 28–7. This limitation on the direction of $\vec{\mathbf{L}}$ is often called **space quantization**. In quantum mechanics, the direction of the angular momentum is usually specified by giving its component along the z axis (this choice is arbitrary). Then L_z is related to m_ℓ by the equation

$$L_z = m_\ell \hbar.$$

The values of L_x and L_y are not definite, however. The name for m_ℓ derives not from theory (which relates it to L_z), but from experiment. It was found that when a gas-discharge tube was placed in a magnetic field, the spectral lines were split into several very closely spaced lines. This splitting, known as the **Zeeman effect**, implies that the energy levels must be split (Fig. 28–8), and thus that the energy of a state depends not only on n but also on m_ℓ when a magnetic field is applied—hence the name "magnetic quantum number."

†This replaces Bohr theory, which assigned $\ell = 1$ to the ground state (Eq. 27–11).

(4) Finally, there is the **spin quantum number**, m_s, which for an electron can have only two values, $m_s = +\frac{1}{2}$ and $m_s = -\frac{1}{2}$. The existence of this quantum number did not come out of Schrödinger's original wave theory, as did n, ℓ, and m_ℓ. Instead, a subsequent modification by P. A. M. Dirac (1902–1984) explained its presence as a relativistic effect. The first hint that m_s was needed, however, came from experiment. A careful study of the spectral lines of hydrogen showed that each actually consisted of two (or more) very closely spaced lines even in the absence of an external magnetic field. It was at first hypothesized that this tiny splitting of energy levels, called **fine structure**, was due to angular momentum associated with a spinning of the electron. That is, the electron might spin on its axis as well as orbit the nucleus, just as the Earth spins on its axis as it orbits the Sun. The interaction between the tiny current of the spinning electron could then interact with the magnetic field due to the orbiting charge and cause the small observed splitting of energy levels. (The energy thus depends slightly on m_ℓ and m_s.)[†] Today we consider the picture of a spinning electron as not legitimate. We cannot even view an electron as a localized object, much less a spinning one. What is important is that the electron can have two different states due to some intrinsic property that behaves like an angular momentum, and we still call this property "spin." The two possible values of m_s ($+\frac{1}{2}$ and $-\frac{1}{2}$) are often said to be "spin up" and "spin down," referring to the two possible directions of the spin angular momentum.

The possible values of the four quantum numbers for an electron in the hydrogen atom are summarized in Table 28–1.

TABLE 28–1 Quantum Numbers for an Electron

Name	Symbol	Possible Values
Principal	n	$1, 2, 3, \cdots, \infty$.
Orbital	ℓ	For a given n: ℓ can be $0, 1, 2, \cdots, n-1$.
Magnetic	m_ℓ	For given n and ℓ: m_ℓ can be $\ell, \ell-1, \cdots, 0, \cdots, -\ell$.
Spin	m_s	For each set of n, ℓ, and m_ℓ: m_s can be $+\frac{1}{2}$ or $-\frac{1}{2}$.

CONCEPTUAL EXAMPLE 28–4 **Possible states for $n = 3$.** How many different states are possible for an electron with principal quantum number $n = 3$?

RESPONSE For $n = 3$, ℓ can have the values $\ell = 2, 1, 0$. For $\ell = 2$, m_ℓ can be $2, 1, 0, -1, -2$, which is five different possibilities. For each of these, m_s can be either up or down ($+\frac{1}{2}$ or $-\frac{1}{2}$); so for $\ell = 2$, there are $2 \times 5 = 10$ states. For $\ell = 1$, m_ℓ can be $1, 0, -1$, and since m_s can be $+\frac{1}{2}$ or $-\frac{1}{2}$ for each of these, we have 6 more possible states. Finally, for $\ell = 0$, m_ℓ can only be 0, and there are only 2 states corresponding to $m_s = +\frac{1}{2}$ and $-\frac{1}{2}$. The total number of states is $10 + 6 + 2 = 18$, as detailed in the following Table:

n	ℓ	m_ℓ	m_s	n	ℓ	m_ℓ	m_s
3	2	2	$\frac{1}{2}$	3	1	1	$\frac{1}{2}$
3	2	2	$-\frac{1}{2}$	3	1	1	$-\frac{1}{2}$
3	2	1	$\frac{1}{2}$	3	1	0	$\frac{1}{2}$
3	2	1	$-\frac{1}{2}$	3	1	0	$-\frac{1}{2}$
3	2	0	$\frac{1}{2}$	3	1	−1	$\frac{1}{2}$
3	2	0	$-\frac{1}{2}$	3	1	−1	$-\frac{1}{2}$
3	2	−1	$\frac{1}{2}$	3	0	0	$\frac{1}{2}$
3	2	−1	$-\frac{1}{2}$	3	0	0	$-\frac{1}{2}$
3	2	−2	$\frac{1}{2}$				
3	2	−2	$-\frac{1}{2}$				

EXERCISE C An electron has $n = 4$, $\ell = 2$. Which of the following values of m_ℓ are possible: 4, 3, 2, 1, 0, −1, −2, −3, −4?

[†]Fine structure is said to be due to a **spin–orbit interaction**.

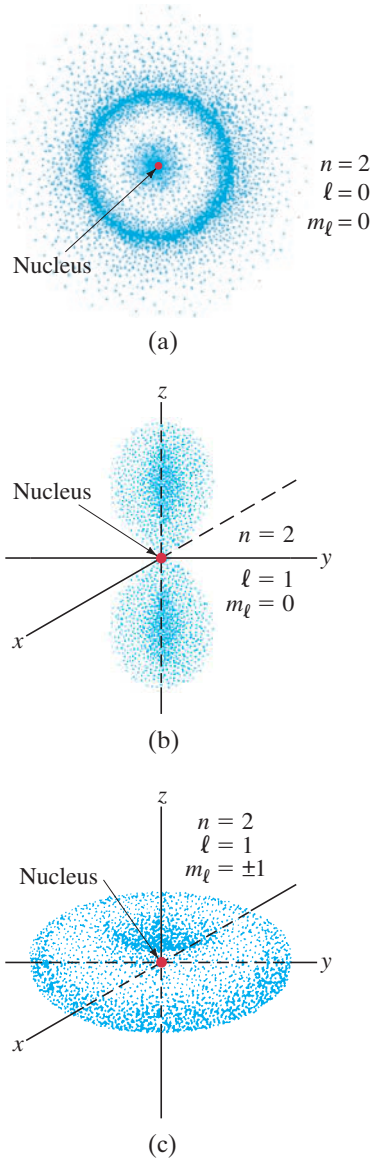


FIGURE 28–9 Electron cloud, or probability distribution, for $n = 2$ states in hydrogen. [The donut-shaped orbit in (c) is the sum of two dumbbell-shaped orbits, as in (b), along the x and y axes added together.]

EXAMPLE 28–5 ***E* and *L* for *n* = 3.** Determine (*a*) the energy and (*b*) the orbital angular momentum for an electron in each of the hydrogen atom states with $n = 3$, as in Example 28–4.

APPROACH The energy of a state depends only on n, except for the very small corrections mentioned above, which we will ignore. Energy is calculated as in the Bohr model, $E_n = -13.6\,\text{eV}/n^2$. For angular momentum we use Eq. 28–3.

SOLUTION (*a*) Since $n = 3$ for all these states, they all have the same energy,

$$E_3 = -\frac{13.6\,\text{eV}}{(3)^2} = -1.51\,\text{eV}.$$

(*b*) For $\ell = 0$, Eq. 28–3 gives

$$L = \sqrt{\ell(\ell+1)}\,\hbar = 0.$$

For $\ell = 1$,

$$L = \sqrt{1(1+1)}\,\hbar = \sqrt{2}\,\hbar = 1.49 \times 10^{-34}\,\text{J}\cdot\text{s}.$$

For $\ell = 2$, $L = \sqrt{2(2+1)}\,\hbar = \sqrt{6}\,\hbar$.

NOTE Atomic angular momenta are generally given as a multiple of $\hbar$ ($\sqrt{2}\,\hbar$ or $\sqrt{6}\,\hbar$ in this case), rather than in SI units.

EXERCISE D What are the energy and angular momentum of the electron in a hydrogen atom with $n = 6$, $\ell = 4$?

Although ℓ and m_ℓ do not significantly affect the energy levels in hydrogen, they do affect the electron probability distribution in space. For $n = 1$, ℓ and m_ℓ can only be zero and the electron distribution is as shown in Fig. 28–6. For $n = 2$, ℓ can be 0 or 1. The distribution for $n = 2$, $\ell = 0$ is shown in Fig. 28–9a, and it is seen to differ from that for the ground state (Fig. 28–6), although it is still spherically symmetric. For $n = 2$, $\ell = 1$, the distributions are not spherically symmetric as shown in Figs. 28–9b (for $m_\ell = 0$) and 28–9c (for $m_\ell = +1$ or -1).

Although the spatial distributions of the electron can be calculated for the various states, it is difficult to measure them experimentally. Most of the experimental information about atoms has come from a careful examination of the emission spectra under various conditions as in Figs. 27–23 and 24–28.

[Chemists refer to atomic states, and especially the shape in space of their probability distributions, as **orbitals**. Each atomic orbital is characterized by its quantum numbers n, ℓ, and m_ℓ, and can hold one or two electrons ($m_s = +\frac{1}{2}$ or $m_s = -\frac{1}{2}$); *s*-orbitals ($\ell = 0$) are spherically symmetric, Figs. 28–6 and 28–9a; *p*-orbitals ($\ell = 1$) can be dumbbell shaped with lobes, Fig. 28–9b, or donut shaped if combining $m_\ell = +1$ and $m_\ell = -1$, Fig. 28–9c.]

Selection Rules: Allowed and Forbidden Transitions

Another prediction of quantum mechanics is that when a photon is emitted or absorbed, transitions can occur only between states with values of ℓ that differ by exactly one unit:

$$\Delta\ell = \pm 1.$$

According to this **selection rule**, an electron in an $\ell = 2$ state can jump only to a state with $\ell = 1$ or $\ell = 3$. It cannot jump to a state with $\ell = 2$ or $\ell = 0$. A transition such as $\ell = 2$ to $\ell = 0$ is called a **forbidden transition**. Actually, such a transition is not absolutely forbidden and can occur, but only with very low probability compared to **allowed transitions**—those that satisfy the selection rule $\Delta\ell = \pm 1$. Since the orbital angular momentum of an H atom must change by one unit when it emits a photon, conservation of angular momentum tells us that the photon must carry off angular momentum. Indeed, experimental evidence of many sorts shows that the photon can be assigned a spin angular momentum of $1\hbar$.

28–7 Multielectron Atoms; the Exclusion Principle

We have discussed the hydrogen atom in detail because it is the simplest to deal with. Now we briefly discuss more complex atoms, those that contain more than one electron. Their energy levels can be determined experimentally from an analysis of their emission spectra. The energy levels are *not* the same as in the H atom, because the electrons interact with each other as well as with the nucleus. Each electron in a complex atom still occupies a particular state characterized by the quantum numbers n, ℓ, m_ℓ, and m_s. For atoms with more than one electron, the energy levels depend on both n and ℓ.

The number of electrons in a neutral atom is called its **atomic number**, Z; Z is also the number of positive charges (protons) in the nucleus, and determines what kind of atom it is. That is, Z determines the fundamental properties that distinguish one type of atom from another.

To understand the possible arrangements of electrons in an atom, a new principle was needed. It was introduced by Wolfgang Pauli (1900–1958; Fig. 28–2) and is called the **Pauli exclusion principle**. It states:

No two electrons in an atom can occupy the same quantum state.

Thus, no two electrons in an atom can have exactly the same set of the quantum numbers n, ℓ, m_ℓ, and m_s. The Pauli exclusion principle forms the basis not only for understanding atoms, but also for understanding molecules and bonding, and other phenomena as well. (See also note at end of this Section.)

Let us now look at the structure of some of the simpler atoms when they are in the ground state. After hydrogen, the next simplest atom is *helium* with two electrons. Both electrons can have $n = 1$, because one can have spin up $\left(m_s = +\frac{1}{2}\right)$ and the other spin down $\left(m_s = -\frac{1}{2}\right)$, thus satisfying the exclusion principle. Since $n = 1$, then ℓ and m_ℓ must be zero (Table 28–1, page 813). Thus the two electrons have the quantum numbers indicated at the top of Table 28–2.

Lithium has three electrons, two of which can have $n = 1$. But the third cannot have $n = 1$ without violating the exclusion principle. Hence the third electron must have $n = 2$. It happens that the $n = 2$, $\ell = 0$ level has a lower energy than $n = 2$, $\ell = 1$. So the electrons in the ground state have the quantum numbers indicated in Table 28–2. The quantum numbers of the third electron could also be, say, $(n, \ell, m_\ell, m_s) = (3, 1, -1, \frac{1}{2})$. But the atom in this case would be in an excited state, because it would have greater energy. It would not be long before it jumped to the ground state with the emission of a photon. At room temperature, unless extra energy is supplied (as in a discharge tube), the vast majority of atoms are in the ground state.

We can continue in this way to describe the quantum numbers of each electron in the ground state of larger and larger atoms. The quantum numbers for sodium, with its eleven electrons, are shown in Table 28–2.

EXERCISE E Construct a Table of the ground-state quantum numbers for beryllium, $Z = 4$ (like those in Table 28–2).

Figure 28–10 shows a simple energy level diagram where occupied states are shown as up or down arrows $\left(m_s = +\frac{1}{2} \text{ or } -\frac{1}{2}\right)$, and possible empty states are shown as a small circle.

TABLE 28–2 Ground-State Quantum Numbers

Helium, $Z = 2$

n	ℓ	m_ℓ	m_s
1	0	0	$\frac{1}{2}$
1	0	0	$-\frac{1}{2}$

Lithium, $Z = 3$

n	ℓ	m_ℓ	m_s
1	0	0	$\frac{1}{2}$
1	0	0	$-\frac{1}{2}$
2	0	0	$\frac{1}{2}$

Sodium, $Z = 11$

n	ℓ	m_ℓ	m_s
1	0	0	$\frac{1}{2}$
1	0	0	$-\frac{1}{2}$
2	0	0	$\frac{1}{2}$
2	0	0	$-\frac{1}{2}$
2	1	1	$\frac{1}{2}$
2	1	1	$-\frac{1}{2}$
2	1	0	$\frac{1}{2}$
2	1	0	$-\frac{1}{2}$
2	1	−1	$\frac{1}{2}$
2	1	−1	$-\frac{1}{2}$
3	0	0	$\frac{1}{2}$

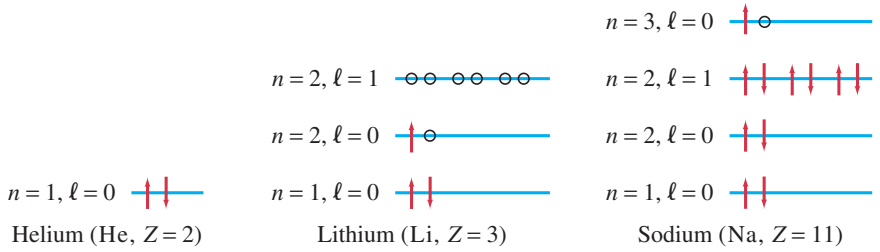


FIGURE 28–10 Energy level diagrams (not to scale) showing occupied states (arrows) and unoccupied states (○) for the ground states of He, Li, and Na. Note that we have shown the $n = 2$, $\ell = 1$ level of Li even though it is empty.

TABLE 28–3 Value of ℓ

Value of ℓ	Letter Symbol	Maximum Number of Electrons in Subshell
0	s	2
1	p	6
2	d	10
3	f	14
4	g	18
5	h	22
⋮	⋮	⋮

TABLE 28–4 Electron Configuration of Some Elements

Z (Number of Electrons)	Element†	Ground State Configuration (outer electrons)
1	H	$1s^1$
2	He	$1s^2$
3	Li	$2s^1$
4	Be	$2s^2$
5	B	$2s^2 2p^1$
6	C	$2s^2 2p^2$
7	N	$2s^2 2p^3$
8	O	$2s^2 2p^4$
9	F	$2s^2 2p^5$
10	Ne	$2s^2 2p^6$
11	Na	$3s^1$
12	Mg	$3s^2$
13	Al	$3s^2 3p^1$
14	Si	$3s^2 3p^2$
15	P	$3s^2 3p^3$
16	S	$3s^2 3p^4$
17	Cl	$3s^2 3p^5$
18	Ar	$3s^2 3p^6$
19	K	$3d^0 4s^1$
20	Ca	$3d^0 4s^2$
21	Sc	$3d^1 4s^2$
22	Ti	$3d^2 4s^2$
23	V	$3d^3 4s^2$
24	Cr	$3d^5 4s^1$
25	Mn	$3d^5 4s^2$
26	Fe	$3d^6 4s^2$

†Names of elements can be found in Appendix B.

The ground-state configuration for all atoms is given in the **Periodic Table**, which is displayed inside the back cover of this book, and discussed in the next Section.

[The *exclusion principle* applies to identical particles whose spin quantum number is a half-integer ($\frac{1}{2}$, $\frac{3}{2}$, and so on), including electrons, protons, and neutrons; such particles are called **fermions**, after Enrico Fermi who derived a statistical theory describing them. A basic assumption is that all electrons are **identical**, indistinguishable one from another. Similarly, all protons are identical, all neutrons are identical, and so on. The exclusion principle does not apply to particles with integer spin (0, 1, 2, and so on), such as the photon and π meson, all of which are referred to as **bosons** (after Satyendranath Bose, who derived a statistical theory for them).]

28–8 The Periodic Table of Elements

More than a century ago, Dmitri Mendeleev (1834–1907) arranged the (then) known elements into what we now call the **Periodic Table** of the elements. The atoms were arranged according to increasing mass, but also so that elements with similar chemical properties would fall in the same column. Today's version is shown inside the back cover of this book. Each square contains the atomic number Z, the symbol for the element, and the atomic mass (in atomic mass units). Finally, the lower left corner shows the configuration of the ground state of the atom. This requires some explanation. Electrons with the same value of n are referred to as being in the same **shell**. Electrons with $n = 1$ are in one shell (the K shell), those with $n = 2$ are in a second shell (the L shell), those with $n = 3$ are in the third (M) shell, and so on. Electrons with the same values of n and ℓ are referred to as being in the same **subshell**. Letters are often used to specify the value of ℓ as shown in Table 28–3. That is, $\ell = 0$ is the s subshell; $\ell = 1$ is the p subshell; $\ell = 2$ is the d subshell; beginning with $\ell = 3$, the letters follow the alphabet, f, g, h, i, and so on. (The first letters s, p, d, and f were originally abbreviations of "sharp," "principal," "diffuse," and "fundamental," terms referring to the experimental spectra.)

The Pauli exclusion principle limits the number of electrons possible in each shell and subshell. For any value of ℓ, there are $2\ell + 1$ possible m_ℓ values (m_ℓ can be any integer from 1 to ℓ, from -1 to $-\ell$, or zero), and two possible m_s values. There can be, therefore, at most $2(2\ell + 1)$ electrons in any ℓ subshell. For example, for $\ell = 2$, five m_ℓ values are possible $(2, 1, 0, -1, -2)$, and for each of these, m_s can be $+\frac{1}{2}$ or $-\frac{1}{2}$ for a total of $2(5) = 10$ states. Table 28–3 lists the maximum number of electrons that can occupy each subshell.

Because the energy levels depend almost entirely on the values of n and ℓ, it is customary to specify the electron configuration simply by giving the n value and the appropriate letter for ℓ, with the number of electrons in each subshell given as a superscript. The ground-state configuration of sodium, for example, is written as $1s^2 2s^2 2p^6 3s^1$. This is simplified in the Periodic Table by specifying the configuration only of the outermost electrons and any other nonfilled subshells (see Table 28–4 here, and the Periodic Table inside the back cover).

CONCEPTUAL EXAMPLE 28–6 **Electron configurations.** Which of the following electron configurations are possible, and which are not: (*a*) $1s^2 2s^2 2p^6 3s^3$; (*b*) $1s^2 2s^2 2p^6 3s^2 3p^5 4s^2$; (*c*) $1s^2 2s^2 2p^6 2d^1$?

RESPONSE (*a*) This is not allowed, because too many electrons (three) are shown in the s subshell of the M ($n = 3$) shell. The s subshell has $m_\ell = 0$, with two slots only, for "spin up" and "spin down" electrons.

(*b*) This is allowed, but it is an excited state. One of the electrons from the $3p$ subshell has jumped up to the $4s$ subshell. Since there are 19 electrons, the element is potassium.

(*c*) This is not allowed, because there is no d ($\ell = 2$) subshell in the $n = 2$ shell (Table 28–1). The outermost electron will have to be (at least) in the $n = 3$ shell.

EXERCISE F Write the complete ground-state configuration for gallium, with its 31 electrons.

The grouping of atoms in the Periodic Table is according to increasing atomic number, Z. It was designed to also show regularity according to chemical properties. Although this is treated in chemistry textbooks, we discuss it here briefly because it is a result of quantum mechanics. See the Periodic Table inside the back cover.

All the **noble gases** (in column VIII of the Periodic Table) have completely filled shells or subshells. That is, their outermost subshell is completely full, and the electron distribution is spherically symmetric. With such full spherical symmetry, other electrons are not attracted nor are electrons readily lost (ionization energy is high). This is why the noble gases are chemically inert (more on this when we discuss molecules and bonding in Chapter 29). Column VII contains the **halogens**, which lack one electron from a filled shell. Because of the shapes of the orbits (see Section 29–1), an additional electron can be accepted from another atom, and hence these elements are quite reactive. They have a valence of -1, meaning that when an extra electron is acquired, the resulting ion has a net charge of $-1e$. Column I of the Periodic Table contains the **alkali metals**, all of which have a single outer s electron. This electron spends most of its time outside the inner closed shells and subshells which shield it from most of the nuclear charge. Indeed, it is relatively far from the nucleus and is attracted to it by a net charge of only about $+1e$, because of the shielding effect of the other electrons. Hence this outer electron is easily removed and can spend much of its time around another atom, forming a molecule. This is why the alkali metals are very chemically reactive and have a valence of $+1$. The other columns of the Periodic Table can be treated similarly.

The presence of the **transition elements** in the center of the Periodic Table, as well as the lanthanides (rare earths) and actinides below, is a result of incomplete inner shells. For the lowest Z elements, the subshells are filled in a simple order: first $1s$, then $2s$, followed by $2p$, $3s$, and $3p$. You might expect that $3d$ ($n = 3$, $\ell = 2$) would be filled next, but it isn't. Instead, the $4s$ level actually has a slightly lower energy than the $3d$ (due to electrons interacting with each other), so it fills first (K and Ca). Only then does the $3d$ shell start to fill up, beginning with Sc, as can be seen in Table 28–4. (The $4s$ and $3d$ levels are close, so some elements have only one $4s$ electron, such as Cr.) Most of the chemical properties of these transition elements are governed by the relatively loosely held $4s$ electrons, and hence they usually have valences of $+1$ or $+2$. A similar effect is responsible for the *lanthanides* and *actinides*, which are shown at the bottom of the Periodic Table for convenience. All have very similar chemical properties, which are determined by their two outer $6s$ or $7s$ electrons, whereas the different numbers of electrons in the unfilled inner shells have little effect.

CAUTION
Subshells are not always filled in "order"

*28–9 X-Ray Spectra and Atomic Number

The line spectra of atoms in the visible, UV, and IR regions of the EM spectrum are mainly due to transitions between states of the outer electrons. Much of the positive charge of the nucleus is shielded from these electrons by the negative charge on the inner electrons. But the innermost electrons in the $n = 1$ shell "see" the full charge of the nucleus. Since the energy of a level is proportional to Z^2 (see Eq. 27–15), for an atom with $Z = 50$, we would expect wavelengths about $50^2 = 2500$ times shorter than those found in the Lyman series of hydrogen (around 100 nm), or $(100\text{ nm})/(2500) \approx 10^{-2}$ to 10^{-1} nm. Such short wavelengths lie in the X-ray region of the spectrum.

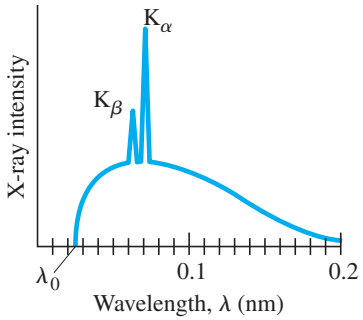
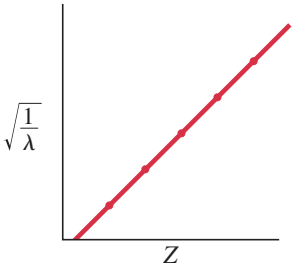


FIGURE 28–11 Spectrum of X-rays emitted from a molybdenum target in an X-ray tube operated at 50 kV.

X-rays are produced when electrons accelerated by a high voltage strike the metal target inside an X-ray tube (Section 25–11). If we look at the spectrum of wavelengths emitted by an X-ray tube, we see that the spectrum consists of two parts: a continuous spectrum with a cutoff at some λ_0 which depends only on the voltage across the tube, and a series of peaks superimposed. A typical example is shown in Fig. 28–11. The smooth curve and the cutoff wavelength λ_0 move to the left as the voltage across the tube increases. The sharp lines or peaks (labeled K_α and K_β in Fig. 28–11), however, remain at the same wavelength when the voltage is changed, although they are located at different wavelengths when different target materials are used. This observation suggests that the peaks are characteristic of the target material used. Indeed, we can explain the peaks by imagining that the electrons accelerated by the high voltage of the tube can reach sufficient energies that, when they collide with the atoms of the target, they can knock out one of the very tightly held inner electrons. Then we explain these **characteristic X-rays** (the peaks in Fig. 28–11) as photons emitted when an electron in an upper state drops down to fill the vacated lower state. The K lines result from transitions *into* the K shell $(n = 1)$. The K_α line consists of photons emitted in a transition that originates from the $n = 2$ (L) shell and drops to the $n = 1$ (K) shell. On the other hand, the K_β line reflects a transition from the $n = 3$ (M) shell down to the K shell. An L line is due to a transition into the L shell, and so on.

FIGURE 28–12 Plot of $\sqrt{1/\lambda}$ vs. Z for K_α X-ray lines.



Measurement of the characteristic X-ray spectra has allowed a determination of the inner energy levels of atoms. It has also allowed the determination of Z values for many atoms, because (as we have seen) the wavelength of the shortest characteristic X-rays emitted will be inversely proportional to Z^2. Actually, for an electron jumping from, say, the $n = 2$ to the $n = 1$ level (K_α line), the wavelength is inversely proportional to $(Z - 1)^2$ because the nucleus is shielded by the one electron that still remains in the 1*s* level. In 1914, H. G. J. Moseley (1887–1915) found that a plot of $\sqrt{1/\lambda}$ vs. Z produced a straight line, Fig. 28–12, where λ is the wavelength of the K_α line. The Z values of a number of elements were determined by fitting them to such a **Moseley plot**. The work of Moseley put the concept of atomic number on a firm experimental basis.

EXAMPLE 28–7 **X-ray wavelength.** Estimate the wavelength for an $n = 2$ to $n = 1$ transition in molybdenum $(Z = 42)$. What is the energy of such a photon?

APPROACH We use the Bohr formula, Eq. 27–16 for $1/\lambda$, with Z^2 replaced by $(Z - 1)^2 = (41)^2$.

SOLUTION Equation 27–16 gives

$$\frac{1}{\lambda} = \left(\frac{2\pi^2 e^4 m k^2}{h^3 c}\right)(Z - 1)^2\left(\frac{1}{n'^2} - \frac{1}{n^2}\right)$$

where $n = 2$, $n' = 1$, and $k = 8.99 \times 10^9\,\text{N}\cdot\text{m}^2/c^2$. We substitute in values:

$$\frac{1}{\lambda} = (1.097 \times 10^7\,\text{m}^{-1})(41)^2\left(\frac{1}{1} - \frac{1}{4}\right) = 1.38 \times 10^{10}\,\text{m}^{-1}.$$

So

$$\lambda = \frac{1}{1.38 \times 10^{10}\,\text{m}^{-1}} = 0.072\,\text{nm}.$$

This is close to the measured value (Fig. 28–11) of 0.071 nm. Each of these photons would have energy (in eV) of:

$$E = hf = \frac{hc}{\lambda} = \frac{(6.63 \times 10^{-34}\,\text{J}\cdot\text{s})(3.00 \times 10^8\,\text{m/s})}{(7.2 \times 10^{-11}\,\text{m})(1.60 \times 10^{-19}\,\text{J/eV})} = 17\,\text{keV}.$$

The denominator includes the conversion factor from joules to eV.

EXAMPLE 28–8 **Determining atomic number.** High-energy electrons are used to bombard an unknown material. The strongest peak is found for X-rays emitted with an energy of 7.5 keV. Guess what the material is.

APPROACH The highest intensity X-rays are generally for the K_α line (see Fig. 28–11) which occurs when high-energy electrons knock out K shell electrons (the innermost orbit, $n = 1$) and their place is taken by electrons from the L shell ($n = 2$). We use the Bohr model, and assume the electrons of the unknown atoms (Z) "see" a nuclear charge of $Z - 1$ (screened by one electron).

SOLUTION The hydrogen transition $n = 2$ to $n = 1$ would yield $E_H = 13.6\,\text{eV} - 3.4\,\text{eV} = 10.2\,\text{eV}$ (see Fig. 27–29 or Example 27–13). Energy of our unknown E_Z is proportional to Z^2 (Eq. 27–15), or rather $(Z - 1)^2$ because the nucleus is shielded by the one electron in a 1s state (see above), so we can use ratios:

$$\frac{E_Z}{E_H} = \frac{(Z-1)^2}{1^2} = \frac{7500\,\text{eV}}{10.2\,\text{eV}} = 735,$$

so $Z - 1 = \sqrt{735} = 27$, and $Z = 28$, which makes it cobalt.

Now we briefly analyze the continuous part of an X-ray spectrum (Fig. 28–11) based on the photon theory of light. When electrons strike the target, they collide with atoms of the material and give up most of their energy as heat (about 99%, so X-ray tubes must be cooled). Electrons can also give up energy by emitting a photon of light: an electron decelerated by interaction with atoms of the target (Fig. 28–13) emits radiation because of its deceleration (Chapter 22), and in this case it is called **bremsstrahlung** (German for "braking radiation"). Because energy is conserved, the energy of the emitted photon, hf, equals the loss of kinetic energy of the electron, $\Delta\text{KE} = \text{KE} - \text{KE}'$, so

$$hf = \Delta\text{KE}.$$

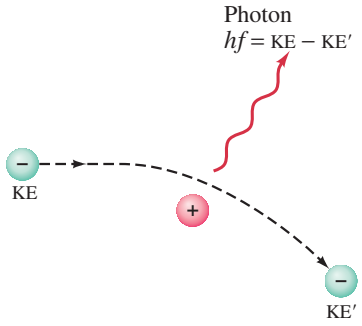


FIGURE 28–13 Bremsstrahlung photon produced by an electron decelerated by interaction with a target atom.

An electron may lose all or a part of its energy in such a collision. The continuous X-ray spectrum (Fig. 28–11) is explained as being due to such bremsstrahlung collisions in which varying amounts of energy are lost by the electrons. The shortest-wavelength X-ray (the highest frequency) must be due to an electron that gives up *all* its kinetic energy to produce one photon in a single collision. Since the initial kinetic energy of an electron is equal to the energy given it by the accelerating voltage, V, then $\text{KE} = eV$. In a single collision in which the electron is brought to rest ($\text{KE}' = 0$), then $\Delta\text{KE} = eV$ and

$$hf_0 = eV.$$

We set $f_0 = c/\lambda_0$ where λ_0 is the cutoff wavelength (Fig. 28–11) and find

$$\lambda_0 = \frac{hc}{eV}. \qquad \textbf{(28–4)}$$

This prediction for λ_0 corresponds precisely with that observed experimentally. This result is further evidence that X-rays are a form of electromagnetic radiation (light) and that the photon theory of light is valid.

EXAMPLE 28–9 **Cutoff wavelength.** What is the shortest-wavelength X-ray photon emitted in an X-ray tube subjected to 50 kV?

APPROACH The electrons striking the target will have a KE of 50 keV. The shortest-wavelength photons are due to collisions in which all of the electron's KE is given to the photon so $\text{KE} = eV = hf_0$.

SOLUTION From Eq. 28–4,

$$\lambda_0 = \frac{hc}{eV} = \frac{(6.63\times10^{-34}\,\text{J}\cdot\text{s})(3.0\times10^{8}\,\text{m/s})}{(1.6\times10^{-19}\,\text{C})(5.0\times10^{4}\,\text{V})} = 2.5\times10^{-11}\,\text{m},$$

or 0.025 nm.

NOTE This result agrees well with experiment, Fig. 28–11.

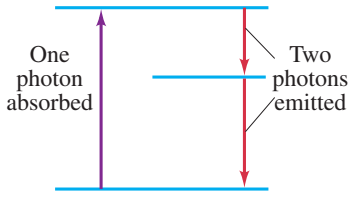
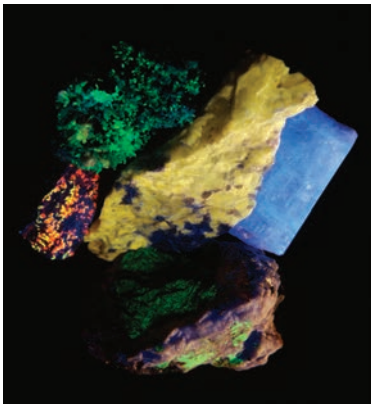


FIGURE 28–14 Fluorescence.

PHYSICS APPLIED

Fluorescence analysis and fluorescent lightbulbs

FIGURE 28–15 When UV light (a range of wavelengths) illuminates these various "fluorescent" rocks, they fluoresce in the visible region of the spectrum.



*28–10 Fluorescence and Phosphorescence

When an atom is excited from one energy state to a higher one by the absorption of a photon, it may return to the lower level in a series of two (or more) transitions if there is at least one energy level in between (Fig. 28–14). The photons emitted will consequently have lower energy and frequency than the absorbed photon. When the absorbed photon is in the UV and the emitted photons are in the visible region of the spectrum, this phenomenon is called **fluorescence** (Fig. 28–15).

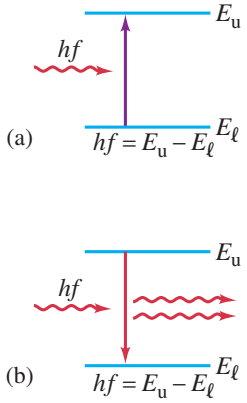
The wavelength for which fluorescence will occur depends on the energy levels of the particular atoms. Because the frequencies are different for different substances, and because many substances fluoresce readily, fluorescence is a powerful tool for identification of compounds. It is also used for assaying—determining how much of a substance is present—and for following substances along a natural *metabolic pathway* in biological organisms. For detection of a given compound, the stimulating light must be monochromatic, and solvents or other materials present must not fluoresce in the same region of the spectrum. Sometimes the observation of fluorescent light being emitted is sufficient to detect a compound. In other cases, spectrometers are used to measure the wavelengths and intensities of the emitted light.

Fluorescent lightbulbs work in a two-step process. The applied voltage accelerates electrons that strike atoms of the gas in the tube and cause them to be excited. When the excited atoms jump down to their normal levels, they emit UV photons which strike a fluorescent coating on the inside of the tube. The light we see is a result of this material fluorescing in response to the UV light striking it.

Materials such as those used for luminous watch dials, and other glow-in-the-dark products, are said to be **phosphorescent**. When an atom is raised to a normal excited state, it drops back down within about 10^{-8} s. In phosphorescent substances, atoms can be excited by photon absorption to energy levels called **metastable**, which are states that last much longer because to jump down is a "forbidden" transition (Section 28–6). Metastable states can last even a few seconds or longer. In a collection of such atoms, many of the atoms will descend to the lower state fairly soon, but many will remain in the excited state for over an hour. Hence light will be emitted even after long periods. When you put a luminous watch dial close to a bright lamp, many atoms are excited to metastable states, and you can see the glow for a long time afterward.

28–11 Lasers

FIGURE 28–16 (a) Absorption of a photon. (b) Stimulated emission. E_u and E_ℓ refer to "upper" and "lower" energy states.



A **laser** is a device that can produce a very narrow intense beam of monochromatic coherent light. (By **coherent**, we mean that across any cross section of the beam, all parts have the same phase.[†]) The emitted beam is a nearly perfect plane wave. An ordinary light source, on the other hand, emits light in all directions (so the intensity decreases rapidly with distance), and the emitted light is incoherent (the different parts of the beam are not in phase with each other). The excited atoms that emit the light in an ordinary lightbulb act independently, so each photon emitted can be considered as a short wave train lasting about 10^{-8} s. Different wave trains bear no phase relation to one another. Just the opposite is true of lasers.

The action of a laser is based on quantum theory. We have seen that a photon can be absorbed by an atom if (and only if) the photon energy hf corresponds to the energy difference between an occupied energy level of the atom and an available excited state, Fig. 28–16a. If the atom is already in the excited state, it may jump down spontaneously (i.e., no stimulus) to the lower state with the emission of a photon. However, if a photon with this same energy strikes the excited atom, it can stimulate the atom to make the transition sooner to the lower state, Fig. 28–16b. This phenomenon is called **stimulated emission**: not only do we still have the original photon, but also a second one of the same frequency as a result

[†]See also Section 24–3.

of the atom's transition. These two photons are exactly *in phase*, and they are moving in the same direction. This is how coherent light is produced in a laser. The name "laser" is an acronym for **L**ight **A**mplification by **S**timulated **E**mission of **R**adiation.

Normally, most atoms are in the lower state, so the majority of incident photons will be absorbed. To obtain the coherent light from stimulated emission, two conditions must be satisfied. First, the atoms must be excited to the higher state so that an **inverted population** is produced in which more atoms are in the upper state than in the lower one (Fig. 28–17). Then *emission* of photons will dominate over absorption. And second, the higher state must be a **metastable state**—a state in which the electrons remain longer than usual† so that the transition to the lower state occurs by stimulated emission rather than spontaneously.

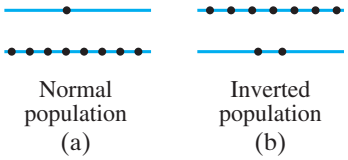


FIGURE 28–17 Two energy levels for a collection of atoms. Each dot represents the energy state of one atom. (a) A normal situation; (b) an inverted population.

Figure 28–18 is a schematic diagram of a laser: the "lasing" material is placed in a long narrow tube at the ends of which are two mirrors, one of which is partially transparent (transmitting perhaps 1 or 2%). Some of the excited atoms drop down fairly soon after being excited. One of these is the blue atom shown on the far left in Fig. 28–18. If the emitted photon strikes another atom in the excited state, it stimulates this atom to emit a photon of the *same* frequency, moving in the *same* direction, and *in phase* with it. These two photons then move on to strike other atoms causing more stimulated emission. As the process continues, the number of photons multiplies. When the photons strike the end mirrors, most are reflected back, and as they move in the opposite direction, they continue to stimulate more atoms to emit photons. As the photons move back and forth between the mirrors, a small percentage passes through the partially transparent mirror at one end. These photons make up the narrow coherent external laser beam. (Inside the tube, some spontaneously emitted photons will be emitted at an angle to the axis, and these will merely go out the side of the tube and not affect the narrow width of the main beam.)

CAUTION

Laser: photons have same frequency and direction, and are in phase

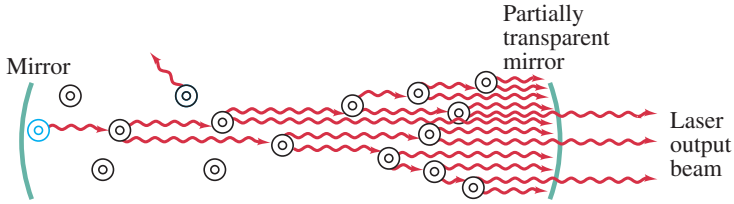


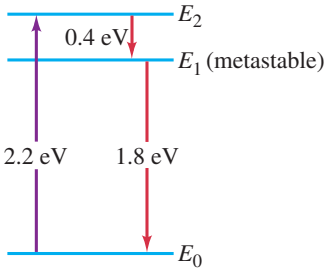
FIGURE 28–18 Laser diagram, showing excited atoms stimulated to emit light.

In a well-designed laser, the spreading of the beam is limited only by diffraction, so the angular spread is $\approx \lambda/D$ (see Eq. 24–3 or 25–7) where D is the diameter of the end mirror. The diffraction spreading can be incredibly small. The light energy, instead of spreading out in space as it does for an ordinary light source, can be a pencil-thin beam.

Creating an Inverted Population

The excitation of the atoms in a laser can be done in several ways to produce the necessary inverted population. In a **ruby laser**, the lasing material is a ruby rod consisting of Al_2O_3 with a small percentage of aluminum (Al) atoms replaced by chromium (Cr) atoms. The Cr atoms are the ones involved in lasing. In a process called **optical pumping**, the atoms are excited by strong flashes of light of wavelength 550 nm, which corresponds to a photon energy of 2.2 eV. As shown in Fig. 28–19, the atoms are excited from state E_0 to state E_2. The atoms quickly decay either back to E_0 or to the intermediate state E_1, which is metastable with a lifetime of about 3×10^{-3} s (compared to 10^{-8} s for ordinary levels). With strong pumping action, more atoms can be found in the E_1 state than are in the E_0 state. Thus we have the inverted population needed for lasing. As soon as a few atoms in the E_1 state jump down to E_0, they emit photons that produce stimulated emission of the other atoms, and the lasing action begins. A ruby laser thus emits a beam whose photons have energy 1.8 eV and a wavelength of 694.3 nm (or "ruby-red" light).

FIGURE 28–19 Energy levels of chromium in a ruby crystal. Photons of energy 2.2 eV "pump" atoms from E_0 to E_2, which then decay to metastable state E_1. Lasing action occurs by stimulated emission of photons in transition from E_1 to E_0.



†An excited atom may land in such a state and can jump to a lower state only by a so-called forbidden transition (Section 28–6), which is why its lifetime is longer than normal.

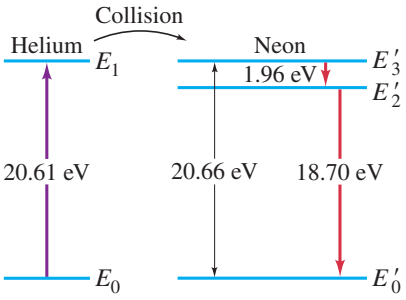


FIGURE 28–20 Energy levels for He and Ne. He is excited in the electric discharge to the E_1 state. This energy is transferred to the E'_3 level of the Ne by collision. E'_3 is metastable and decays to E'_2 by stimulated emission.

In a **helium–neon laser** (He–Ne), the lasing material is a gas, a mixture of about 85% He and 15% Ne. The atoms are excited by applying a high voltage to the tube so that an electric discharge takes place within the gas. In the process, some of the He atoms are raised to the metastable state E_1 shown in Fig. 28–20, which corresponds to a jump of 20.61 eV, almost exactly equal to an excited state in neon, 20.66 eV. The He atoms do not quickly return to the ground state by spontaneous emission, but instead often give their excess energy to a Ne atom when they collide—see Fig. 28–20. In such a collision, the He drops to the ground state and the Ne atom is excited to the state E'_3 (the prime refers to neon states). The slight difference in energy (0.05 eV) is supplied by the kinetic energy of the moving atoms. In this manner, the E'_3 state in Ne—which is metastable—becomes more populated than the E'_2 level. This inverted population between E'_3 and E'_2 is what is needed for lasing.

Very common now are **semiconductor diode lasers**, also called **pn junction lasers**, which utilize an inverted population of electrons between the conduction band and the lower-energy valence band (Section 29–9). When an electron jumps down, a photon can be emitted, which in turn can stimulate another electron to make the transition and emit another photon, in phase. The needed mirrors (as in Fig. 28–18) are made by the polished ends of the *pn* crystal. Semiconductor lasers are used in CD and DVD players (see below), and in many other applications.

Other types of laser include: *chemical lasers*, in which the energy input comes from the chemical reaction of highly reactive gases; *dye lasers*, whose frequency is tunable; CO_2 *gas lasers*, capable of high power output in the infrared; and *rare-earth solid-state lasers* such as the high-power Nd:YAG laser.

The excitation of the atoms in a laser can be done continuously or in pulses. In a **pulsed laser**, the atoms are excited by periodic inputs of energy. In a **continuous laser**, the energy input is continuous: as atoms are stimulated to jump down to the lower level, they are soon excited back up to the upper level so the output is a continuous laser beam.

No laser is a source of energy. Energy must be put in, and the laser converts a part of it into an intense narrow beam output.

CAUTION
Laser is not an energy source

*Applications

PHYSICS APPLIED
DVD and CD players, bar codes

The unique feature of light from a laser, that it is a coherent narrow beam, has found many applications. In everyday life, lasers are used as bar-code readers (at store checkout stands) and in compact disc (CD) and digital video disc (DVD) players. The laser beam reflects off the stripes and spaces of a bar code, or off the tiny pits of a CD or DVD as shown in Fig. 28–21a. The recorded information on a CD or DVD is a series of pits and spaces representing 0s and 1s (or "off" and "on") of a binary code (Section 17–10) that is decoded electronically before being sent to the audio or video system. The laser of a CD player starts reading at the inside of the disc which rotates at about 500 rpm at the start. As the disc rotates, the laser follows the spiral track (Fig. 28–21b), and as it moves outward the disc must slow down because each successive circumference ($C = 2\pi r$) is slightly longer as r increases; at the outer edge, the disc is rotating about 200 rpm. A 1-hour CD has a track roughly 5 km long; the track width is about 1600 nm (= 1.6 μm) and the distance between pits is about 800 nm. DVDs contain much more information.

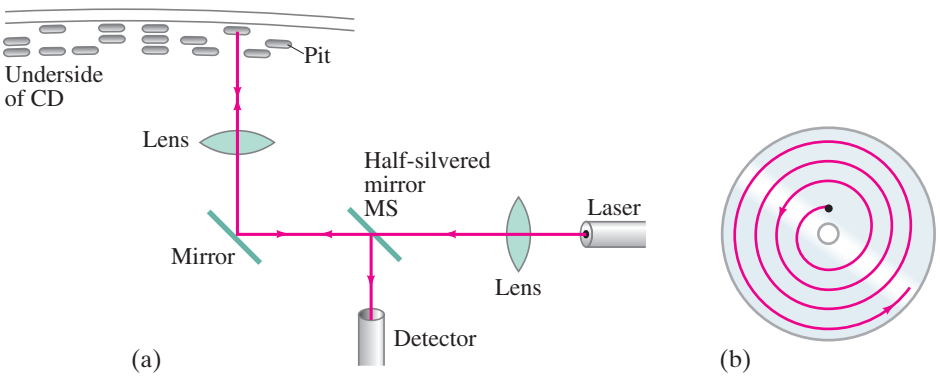


FIGURE 28–21 (a) Reading a CD (or DVD). The fine beam of a laser, focused even more finely with lenses, is directed at the undersurface of a rotating compact disc. The beam is reflected back from the areas between pits but reflects much less from pits. The reflected light is detected as shown, reflected by a half-reflecting mirror MS. The strong and weak reflections correspond to the 0s and 1s of the binary code representing the audio or video signal. (b) A laser follows the CD track which starts near the center and spirals outward.

Standard DVDs use a thinner track (0.7 μm) and shorter pit length (400 nm). Blu-ray discs use a "blue" laser with a short wavelength (405 nm) and narrower beam, allowing a narrower track (0.3 μm) that can store much more data for high definition. DVDs can also have two layers, one below the other. When the laser focuses on the second layer, the light passes through the semitransparent surface layer. The second layer may start reading at the outer edge instead of inside. DVDs can also have a single or double layer on *both* surfaces of the disc.

Lasers are a useful surgical tool. The narrow intense beam can be used to destroy tissue in a localized area, or to break up gallstones and kidney stones. Because of the heat produced, a laser beam can be used to "weld" broken tissue, such as a detached retina, Fig. 28–22, or to mold the cornea of the eye (by vaporizing tiny bits of material) to correct myopia and other eye defects (LASIK surgery). The laser beam can be carried by an optical fiber (Section 23–6) to the surgical point, sometimes as an additional fiber-optic path on an endoscope (again Section 23–6). An example is the removal of plaque clogging human arteries. Lasers have been used to destroy tiny organelles within a living cell by researchers studying how the absence of that organelle affects the behavior of the cell. Laser beams are used to destroy cancerous and precancerous cells; and the heat seals off capillaries and lymph vessels, thus "cauterizing" the wound to prevent spread of the disease.

The intense heat produced in a small area by a laser beam is used for welding and machining metals and for drilling tiny holes in hard materials. Because a laser beam is coherent, monochromatic, narrow, and essentially parallel, lenses can be used to focus the light into even smaller areas. The precise straightness of a laser beam is also useful to surveyors for lining up equipment accurately, especially in inaccessible places.

PHYSICS APPLIED
Medical and other uses of lasers

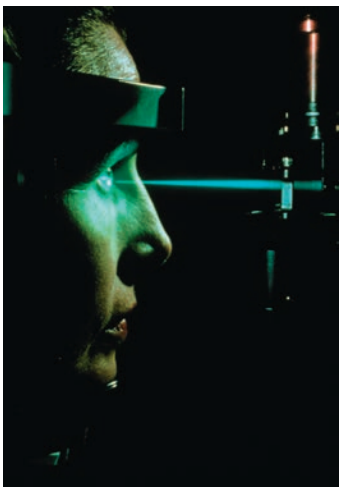


FIGURE 28–22 Laser being used in eye surgery.

*28–12 Holography

One of the most interesting applications of laser light is the production of three-dimensional images called **holograms** (see Fig. 28–23). In an ordinary photograph, the film simply records the intensity of light reaching it at each point. When the photograph or transparency is viewed, light reflecting from it or passing through it gives us a two-dimensional picture. In holography, the images are formed by interference, without lenses. A laser hologram is typically made on a photographic emulsion (film). A broadened laser beam is split into two parts by a half-silvered mirror, Fig. 28–24. One part goes directly to the film; the rest passes to the object to be photographed, from which it is reflected to the film. Light from every point on the object reaches each point on the film, and the interference of the two beams allows the film to record both the intensity and relative phase of the light at each point. It is crucial that the incident light be coherent—that is, in phase at all points—which is why a laser is used. After the film is developed, it is placed again in a laser beam and a three-dimensional image of the object is created. You can walk around such an image and see it from different sides as if it were the original object. Yet, if you try to touch it with your hand, there will be nothing material there.

PHYSICS APPLIED
Holography



FIGURE 28–23 An athlete (race-car driver) puts his hand on, or through, a holographic image of himself.

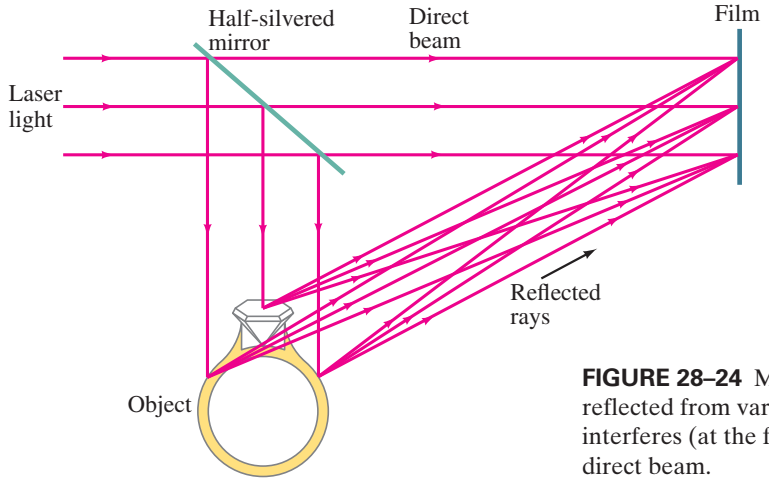


FIGURE 28–24 Making a hologram. Light reflected from various points on the object interferes (at the film) with light from the direct beam.

Volume or **white-light holograms** do not require a laser to see the image, but can be viewed with ordinary white light (preferably a nearly point source, such as the Sun or a clear bulb with a small bright filament). Such holograms must be made, however, with a laser. They are made not on thin film, but on a *thick* emulsion. The interference pattern in the film emulsion can be thought of as an array of bands or ribbons where constructive interference occurred. This array, and the reconstruction of the image, can be compared to Bragg scattering of X-rays from the atoms in a crystal (see Section 25–11). White light can reconstruct the image because the Bragg condition $(m\lambda = 2d \sin\theta)$ selects out the appropriate single wavelength. If the hologram is originally produced by lasers emitting the three additive primary colors (red, green, and blue), the three-dimensional image can be seen in full color when viewed with white light.

Summary

In 1925, Schrödinger and Heisenberg separately worked out a new theory, **quantum mechanics**, which is now considered to be the fundamental theory at the atomic level. It is a statistical theory rather than a deterministic one.

An important aspect of quantum mechanics is the Heisenberg **uncertainty principle**. It results from the wave–particle duality and the unavoidable interaction between an observed object and the observer.

One form of the uncertainty principle states that the position x and momentum p_x of an object cannot both be measured precisely at the same time. The products of the uncertainties, $(\Delta x)(\Delta p_x)$, can be no less than $\hbar$ $(= h/2\pi)$:

$$(\Delta p_x)(\Delta x) \gtrsim \hbar. \qquad \textbf{(28–1)}$$

Another form of the uncertainty principle states that the energy can be uncertain (or nonconserved) by an amount ΔE for a time Δt, where

$$(\Delta E)(\Delta t) \gtrsim \hbar. \qquad \textbf{(28–2)}$$

According to quantum mechanics, the state of an electron in an atom is specified by four **quantum numbers**: n, ℓ, m_ℓ, and m_s:

(1) n, the **principal quantum number**, can take on any integer value $(1, 2, 3, \cdots)$ and corresponds to the quantum number of the old Bohr model;

(2) ℓ, the **orbital quantum number**, can take on values from 0 up to $n - 1$;

(3) m_ℓ, the **magnetic quantum number**, can take on integer values from $-\ell$ to $+\ell$;

(4) m_s, the **spin quantum number**, can be $+\frac{1}{2}$ or $-\frac{1}{2}$.

The energy levels in the hydrogen atom depend on n, whereas in other atoms they depend on n and ℓ.

The orbital angular momentum of an atom has magnitude $L = \sqrt{\ell(\ell + 1)}\ \hbar$ and z component $L_z = m_\ell \hbar$.

When an external magnetic field is applied, the spectral lines are split (the **Zeeman effect**), indicating that the energy depends also on m_ℓ in this case.

Even in the absence of a magnetic field, precise measurements of spectral lines show a tiny splitting of the lines called **fine structure**, whose explanation is that the energy depends very slightly on m_ℓ and m_s.

Transitions between states that obey the **selection rule** $\Delta\ell = \pm 1$ are far more probable than other so-called **forbidden transitions**.

The arrangement of electrons in multi-electron atoms is governed by the Pauli **exclusion principle**, which states that no two electrons can occupy the same quantum state—that is, they cannot have the same set of quantum numbers n, ℓ, m_ℓ, and m_s.

As a result, electrons in multi-electron atoms are grouped into **shells** (according to the value of n) and **subshells** (according to ℓ).

Electron configurations are specified using the numerical values of n, and using letters for ℓ: s, p, d, f, etc., for $\ell = 0, 1, 2, 3$, and so on, plus a superscript for the number of electrons in that subshell. Thus, the ground state of hydrogen is $1s^1$, whereas that for oxygen is $1s^2 2s^2 2p^4$.

In the **Periodic Table**, the elements are arranged in horizontal rows according to increasing **atomic number** (= number of electrons in the neutral atom). The shell structure gives rise to a periodicity in the properties of the elements, so that each vertical column can contain elements with similar chemical properties.

X-rays, which are a form of electromagnetic radiation of very short wavelength, are produced when high-speed electrons strike a target. The spectrum of X-rays so produced consists of two parts, a *continuous* spectrum produced when the electrons are decelerated by atoms of the target, and *peaks* representing photons emitted by atoms of the target after being excited by collision with the high-speed electrons. Measurement of these peaks allows determination of inner energy levels of atoms and determination of atomic number Z.

[*__Fluorescence__ occurs when absorbed UV photons are followed by emission of visible light, due to the special arrangement of energy levels of atoms of the material. **Phosphorescent** materials have **metastable** states (long-lived) that emit light seconds or minutes after absorption of light.]

Lasers produce a narrow beam of monochromatic coherent light (light waves *in phase*).

[*__Holograms__ are images with a 3-dimensional quality, formed by interference of laser light.]

Questions

1. Compare a matter wave Ψ to (*a*) a wave on a string, (*b*) an EM wave. Discuss similarities and differences.

2. Explain why Bohr's theory of the atom is not compatible with quantum mechanics, particularly the uncertainty principle.

3. Explain why it is that the more massive an object is, the easier it becomes to predict its future position.

4. In view of the uncertainty principle, why does a baseball seem to have a well-defined position and speed, whereas an electron does not?

5. Would it ever be possible to balance a very sharp needle precisely on its point? Explain.

6. A cold thermometer is placed in a hot bowl of soup. Will the temperature reading of the thermometer be the same as the temperature of the hot soup before the measurement was made? Explain.

7. Does the uncertainty principle set a limit to how well you can make any single measurement of position? Explain.

8. If you knew the position of an object precisely, with no uncertainty, how well would you know its momentum?

9. When you check the pressure in a tire, doesn't some air inevitably escape? Is it possible to avoid this escape of air altogether? What is the relation to the uncertainty principle?

10. It has been said that the ground-state energy in the hydrogen atom can be precisely known but the excited states have some uncertainty in their values (an "energy width"). Is this consistent with the uncertainty principle in its energy form? Explain.

11. Which model of the hydrogen atom, the Bohr model or the quantum-mechanical model, predicts that the electron spends more time near the nucleus? Explain.

12. The size of atoms varies by only a factor of three or so, from largest to smallest, yet the number of electrons varies from one to over 100. Explain.

13. Excited hydrogen and excited helium atoms both radiate light as they jump down to the $n = 1$, $\ell = 0$, $m_\ell = 0$ state. Why do the two elements have very different emission spectra?

14. How would the Periodic Table look if there were no electron spin but otherwise quantum mechanics were valid? Consider the first 20 elements or so.

15. Which of the following electron configurations are not allowed: (*a*) $1s^2 2s^2 2p^4 3s^2 4p^2$; (*b*) $1s^2 2s^2 2p^8 3s^1$; (*c*) $1s^2 2s^2 2p^6 3s^2 3p^5 4s^2 4d^5 4f^1$? If not allowed, explain why.

16. In what column of the Periodic Table would you expect to find the atom with each of the following configurations: (*a*) $1s^2 2s^2 2p^5$; (*b*) $1s^2 2s^2 2p^6 3s^2$; (*c*) $1s^2 2s^2 2p^6 3s^2 3p^6$; (*d*) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$?

17. Why do chlorine and iodine exhibit similar properties?

18. Explain why potassium and sodium exhibit similar properties.

19. Why are the chemical properties of the rare earths so similar? [*Hint*: Examine the Periodic Table.]

20. The ionization energy for neon ($Z = 10$) is 21.6 eV, and that for sodium ($Z = 11$) is 5.1 eV. Explain the large difference.

21. Why do we expect electron transitions deep within an atom to produce shorter wavelengths than transitions by outer electrons?

22. Does the Bohr model of the atom violate the uncertainty principle? Explain.

23. Briefly explain why noble gases are nonreactive and why alkali metals are highly reactive. (See Section 28–8.)

24. Compare spontaneous emission to stimulated emission.

25. How does laser light differ from ordinary light? How is it the same?

26. Explain how a 0.0005-W laser beam, photographed at a distance, can seem much stronger than a 1000-W street lamp at the same distance.

27. Does the intensity of light from a laser fall off as the inverse square of the distance? Explain.

***28.** Why does the cutoff wavelength in Fig. 28–11 imply a photon nature for light?

***29.** Why do we not expect perfect agreement between measured values of characteristic X-ray line wavelengths and those calculated using the Bohr model, as in Example 28–7?

***30.** How would you figure out which lines in an X-ray spectrum correspond to K_α, K_β, L, etc., transitions?

MisConceptual Questions

1. An atom has the electron configuration $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$. How many electrons does this atom have?
(*a*) 15. (*b*) 19. (*c*) 30. (*d*) 46.

2. For the electron configuration of MisConceptual Question 1, what orbital quantum numbers do the electrons have?
(*a*) 0.
(*b*) 0 and 1.
(*c*) 0 and 1 and 2.
(*d*) 0 and 1 and 2 and 3.
(*e*) 0 and 1 and 2 and 3 and 4.

3. If a beam of electrons is fired through a slit,
(*a*) the electrons can be deflected because of their wave properties.
(*b*) only electrons that hit the edge of the slit are deflected.
(*c*) electrons can interact with electromagnetic waves in the slit, forming a diffraction pattern.
(*d*) the probability of an electron making it through the slit depends on the uncertainty principle.

4. What is meant by the ground state of an atom?
(*a*) All of the quantum numbers have their lowest values $(n = 1,\ \ell = m_\ell = 0)$.
(*b*) The principal quantum number of the electrons in the outer shell is 1.
(*c*) All of the electrons are in the lowest energy state, consistent with the exclusion principle.
(*d*) The electrons are in the lowest state allowed by the uncertainty principle.

5. The Pauli exclusion principle applies to all electrons
(*a*) in the same shell, but not electrons in different shells.
(*b*) in the same container of atoms.
(*c*) in the same column of the Periodic Table.
(*d*) in incomplete shells.
(*e*) in the same atom.

6. Which of the following is the best paraphrasing of the Heisenberg uncertainty principle?
(*a*) Only if you know the exact position of a particle can you know the exact momentum of the particle.
(*b*) The larger the momentum of a particle, the smaller the position of the particle.
(*c*) The more precisely you know the position of a particle, the less well you can know the momentum of the particle.
(*d*) The better you know the position of a particle, the better you can know the momentum of the particle.
(*e*) How well you can determine the position and momentum of a particle depends on the particle's quantum numbers.

7. Which of the following is required by the Pauli exclusion principle?
(*a*) No electron in an atom can have the same set of quantum numbers as any other electron in that atom.
(*b*) Each electron in an atom must have the same n value.
(*c*) Each electron in an atom must have different m_ℓ values.
(*d*) Only two electrons can be in any particular shell of an atom.
(*e*) No two electrons in a collection of atoms can have the exact same set of quantum numbers.

8. Under what condition(s) can the exact location and velocity of an electron be measured at the same time?
(*a*) The electron is in the ground state of the atom.
(*b*) The electron is in an excited state of the atom.
(*c*) The electron is free (not bound to an atom).
(*d*) Both (*a*) and (*b*).
(*e*) Never.

9. According to the uncertainty principle,
(*a*) there is always an uncertainty in a measurement of the position of a particle.
(*b*) there is always an uncertainty in a measurement of the momentum of a particle.
(*c*) there is always an uncertainty in a simultaneous measurement of both the position and momentum of a particle.
(*d*) All of the above.

10. Which of the following is *not* always a property of lasers?
(*a*) All of the photons in laser light have the same phase.
(*b*) All laser photons have nearly identical frequencies.
(*c*) Laser light moves as a beam, spreading out very slowly.
(*d*) Laser light is always brighter than other sources of light.
(*e*) Lasers depend on an inverted population of atoms where more atoms occupy a higher energy state than some lower energy state.

For assigned homework and other learning materials, go to the MasteringPhysics website.

Problems

28–2 Wave Function, Double-Slit

1. (II) The neutrons in a parallel beam, each having kinetic energy 0.025 eV, are directed through two slits 0.40 mm apart. How far apart will the interference peaks be on a screen 1.0 m away? [*Hint*: First find the wavelength of the neutron.]
2. (II) Pellets of mass 2.0 g are fired in parallel paths with speeds of 120 m/s through a hole 3.0 mm in diameter. How far from the hole must you be to detect a 1.0-cm-diameter spread in the beam of pellets?

28–3 Uncertainty Principle

3. (I) A proton is traveling with a speed of $(8.660 \pm 0.012) \times 10^5$ m/s. With what maximum precision can its position be ascertained? [*Hint*: $\Delta p = m\,\Delta v$.]
4. (I) If an electron's position can be measured to a precision of 2.4×10^{-8} m, how precisely can its speed be known?
5. (I) An electron remains in an excited state of an atom for typically 10^{-8} s. What is the minimum uncertainty in the energy of the state (in eV)?
6. (II) The Z^0 boson, discovered in 1985, is the mediator of the weak nuclear force, and it typically decays very quickly. Its average rest energy is 91.19 GeV, but its short lifetime shows up as an intrinsic width of 2.5 GeV. What is the lifetime of this particle? [*Hint*: See Example 28–3.]
7. (II) What is the uncertainty in the mass of a muon $(m = 105.7\ \text{MeV}/c^2)$, specified in eV/c^2, given its lifetime of 2.20 μs?
8. (II) A free neutron $(m = 1.67 \times 10^{-27}\ \text{kg})$ has a mean life of 880 s. What is the uncertainty in its mass (in kg)?
9. (II) An electron and a 140-g baseball are each traveling 120 m/s measured to a precision of 0.065%. Calculate and compare the uncertainty in position of each.
10. (II) A radioactive element undergoes an alpha decay with a lifetime of 12 μs. If alpha particles are emitted with 5.5-MeV kinetic energy, find the percent uncertainty $\Delta E/E$ in the particle energy.
11. (II) If an electron's position can be measured to a precision of 15 nm, what is the uncertainty in its speed? Assuming the minimum speed must be at least equal to its uncertainty, what is the electron's minimum kinetic energy?
12. (II) Estimate the lowest possible energy of a neutron contained in a typical nucleus of radius 1.2×10^{-15} m. [*Hint*: Assume a particle can have an energy as large as its uncertainty.]
13. (III) How precisely can the position of a 5.00-keV electron be measured assuming its energy is known to 1.00%?
14. (III) Use the uncertainty principle to show that if an electron were present in the nucleus $(r \approx 10^{-15}\ \text{m})$, its kinetic energy (use relativity) would be hundreds of MeV. (Since such electron energies are not observed, we conclude that electrons are not present in the nucleus.) [*Hint*: Assume a particle can have an energy as large as its uncertainty.]

28–6 to 28–8 Quantum Numbers, Exclusion Principle

15. (I) For $n = 6$, what values can ℓ have?

16. (I) For $n = 6$, $\ell = 3$, what are the possible values of m_ℓ and m_s?

17. (I) How many electrons can be in the $n = 5$, $\ell = 3$ subshell?

18. (I) How many different states are possible for an electron whose principal quantum number is $n = 4$? Write down the quantum numbers for each state.

19. (I) List the quantum numbers for each electron in the ground state of (*a*) carbon ($Z = 6$), (*b*) aluminum ($Z = 13$).

20. (I) List the quantum numbers for each electron in the ground state of oxygen ($Z = 8$).

21. (I) Calculate the magnitude of the angular momentum of an electron in the $n = 5$, $\ell = 3$ state of hydrogen.

22. (I) If a hydrogen atom has $\ell = 4$, what are the possible values for n, m_ℓ, and m_s?

23. (II) If a hydrogen atom has $m_\ell = -3$, what are the possible values of n, ℓ, and m_s?

24. (II) Show that there can be 18 electrons in a "*g*" subshell.

25. (II) What is the full electron configuration in the ground state for elements with Z equal to (*a*) 26, (*b*) 34, (*c*) 38? [*Hint*: See the Periodic Table inside the back cover.]

26. (II) What is the full electron configuration for (*a*) silver (Ag), (*b*) gold (Au), (*c*) uranium (U)? [*Hint*: See the Periodic Table inside the back cover.]

27. (II) A hydrogen atom is in the 5*d* state. Determine (*a*) the principal quantum number, (*b*) the energy of the state, (*c*) the orbital angular momentum and its quantum number ℓ, and (*d*) the possible values for the magnetic quantum number.

28. (II) Estimate the binding energy of the third electron in lithium using the Bohr model. [*Hint*: This electron has $n = 2$ and "sees" a net charge of approximately $+1e$.] The measured value is 5.36 eV.

29. (II) Show that the total angular momentum is zero for a filled subshell.

30. (II) For each of the following atomic transitions, state whether the transition is *allowed* or *forbidden*, and why: (*a*) $4p \rightarrow 3p$; (*b*) $3p \rightarrow 1s$; (*c*) $4d \rightarrow 2d$; (*d*) $5d \rightarrow 3s$; (*e*) $4s \rightarrow 2p$.

31. (II) An electron has $m_\ell = 2$ and is in its lowest possible energy state. What are the values of n and ℓ for this electron?

*28–9 X-Rays

***32.** (I) What are the shortest-wavelength X-rays emitted by electrons striking the face of a 28.5-kV TV picture tube? What are the longest wavelengths?

***33.** (I) If the shortest-wavelength bremsstrahlung X-rays emitted from an X-ray tube have $\lambda = 0.035$ nm, what is the voltage across the tube?

***34.** (I) Show that the cutoff wavelength λ_0 in an X-ray spectrum is given by

$$\lambda_0 = \frac{1240 \text{ nm}}{V},$$

where V is the X-ray tube voltage in volts.

***35.** (I) For the spectrum of X-rays emitted from a molybdenum target (discussed relative to Fig. 28–11), determine the maximum and minimum energy.

***36.** (II) Estimate the wavelength for an $n = 3$ to $n = 2$ transition in iron ($Z = 26$).

***37.** (II) Use the Bohr model to estimate the wavelength for an $n = 3$ to $n = 1$ transition in molybdenum ($Z = 42$). The measured value is 0.063 nm. Why do we not expect perfect agreement?

***38.** (II) A mixture of iron and an unknown material is bombarded with electrons. The wavelengths of the K_α lines are 194 pm for iron and 229 pm for the unknown. What is the unknown material?

28–11 Lasers

39. (II) A laser used to weld detached retinas puts out 25-ms-long pulses of 640-nm light which average 0.68-W output during a pulse. How much energy can be deposited per pulse and how many photons does each pulse contain? [*Hint*: See Example 27–4.]

40. (II) A low-power laser used in a physics lab might have a power of 0.50 mW and a beam diameter of 3.0 mm. Calculate (*a*) the average light intensity of the laser beam, and (*b*) compare it to the intensity of a very powerful light bulb emitting 100-W of light as viewed from 2.0 m.

41. (II) Calculate the wavelength of the He–Ne laser (see Fig. 28–20).

42. (II) Estimate the angular spread of a laser beam due to diffraction if the beam emerges through a 3.0-mm-diameter mirror. Assume that $\lambda = 694$ nm. What would be the diameter of this beam if it struck (*a*) a satellite 340 km above the Earth, or (*b*) the Moon? [*Hint*: See Sections 24–5 and 25–7.]

General Problems

43. An electron in the $n = 2$ state of hydrogen remains there on average about 10^{-8} s before jumping to the $n = 1$ state. (*a*) Estimate the uncertainty in the energy of the $n = 2$ state. (*b*) What fraction of the transition energy is this? (*c*) What is the wavelength, and width (in nm), of this line in the spectrum of hydrogen?

44. What are the largest and smallest possible values for the angular momentum L of an electron in the $n = 6$ shell?

45. If an electron's position can be measured to a precision of 2.0×10^{-8} m, what is the uncertainty in its momentum? Assuming its momentum must be at least equal to its uncertainty, estimate the electron's wavelength.

46. The ionization (binding) energy of the outermost electron in boron is 8.26 eV. (*a*) Use the Bohr model to estimate the "effective charge," Z_{eff}, seen by this electron. (*b*) Estimate the average orbital radius.

47. Using the Bohr formula for the radius of an electron orbit, estimate the average distance from the nucleus for an electron in the innermost ($n = 1$) orbit of a uranium atom ($Z = 92$). Approximately how much energy would be required to remove this innermost electron?

48. Protons are accelerated from rest across 480 V. They are then directed at two slits 0.70 mm apart. How far apart will the interference peaks be on a screen 28 m away?

49. How many electrons can there be in an "h" subshell?

50. (*a*) Show that the number of different states possible for a given value of ℓ is equal to $2(2\ell + 1)$. (*b*) What is this number for $\ell = 0, 1, 2, 3, 4, 5,$ and 6?

51. Show that the number of different electron states possible for a given value of n is $2n^2$. (See Problem 50.)

52. A beam of electrons with kinetic energy 45 keV is shot through two narrow slits in a barrier. The slits are a distance 2.0×10^{-6} m apart. If a screen is placed 45.0 cm behind the barrier, calculate the spacing between the "bright" fringes of the interference pattern produced on the screen.

53. The angular momentum in the hydrogen atom is given both by the Bohr model and by quantum mechanics. Compare the results for $n = 2$.

54. The lifetime of a typical excited state in an atom is about 10 ns. Suppose an atom falls from one such excited state to a lower one, and emits a photon of wavelength about 500 nm. Find the fractional energy uncertainty $\Delta E/E$ and wavelength uncertainty $\Delta\lambda/\lambda$ of this photon.

55. A 1300-kg car is traveling with a speed of (22 ± 0.22) m/s. With what maximum precision can its position be determined?

56. An atomic spectrum contains a line with a wavelength centered at 488 nm. Careful measurements show the line is really spread out between 487 and 489 nm. Estimate the lifetime of the excited state that produced this line.

57. An electron and a proton, each initially at rest, are accelerated across the same voltage. Assuming that the uncertainty in their position is given by their de Broglie wavelength, find the ratio of the uncertainty in their momentum.

58. If the principal quantum number n were limited to the range from 1 to 6, how many elements would we find in nature?

59. If your de Broglie wavelength were 0.50 m, how fast would you be moving if your mass is 68.0 kg? Would you notice diffraction effects as you walk through a doorway? Approximately how long would it take you to walk through the doorway?

60. Suppose that the spectrum of an unknown element shows a series of lines with one out of every four matching a line from the Lyman series of hydrogen. Assuming that the unknown element is an ion with Z protons and one electron, determine Z and the element in question.

*61. Photons of wavelength 0.154 nm are emitted from the surface of a certain metal when it is bombarded with high-energy radiation. If this photon wavelength corresponds to the K_α line, what is the element?

Search and Learn

1. Use the uncertainty principle to estimate the position uncertainty for the electron in the ground state of the hydrogen atom. [*Hint*: Determine the momentum using the Bohr model of Section 27–12 and assume the momentum can be anywhere between this value and zero.] How does this result compare to the Bohr radius?

2. On what factors does the periodicity of the Periodic Table depend? Consider the exclusion principle, quantization of angular momentum, spin, and any others you can think of.

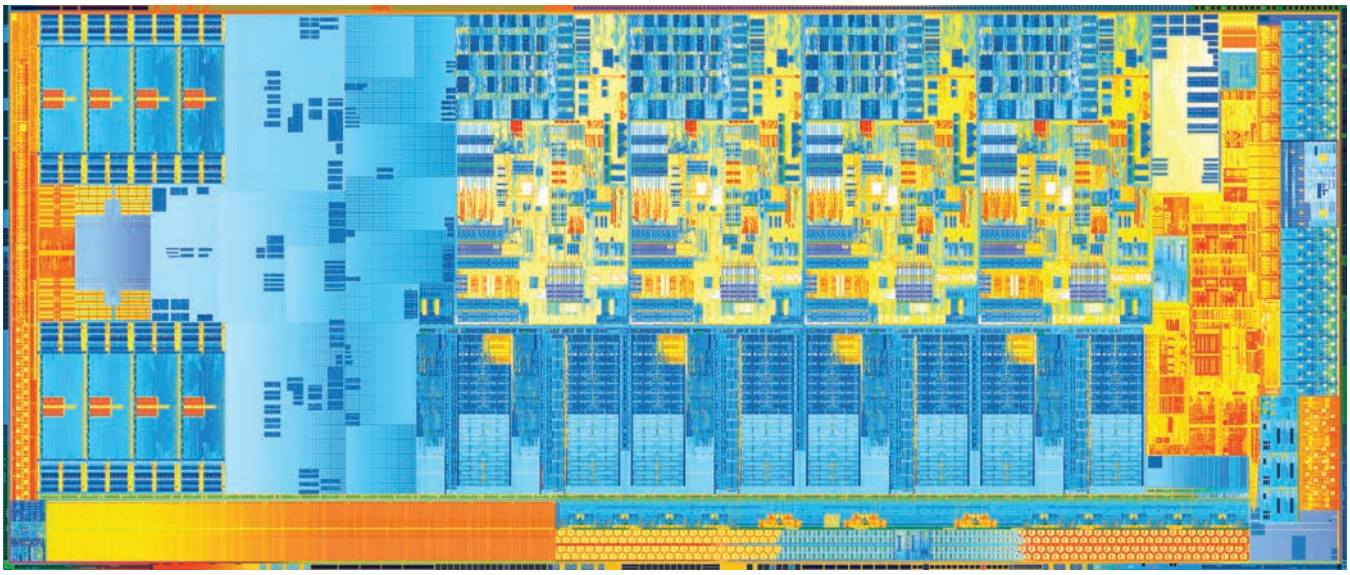
3. Estimate (*a*) the quantum number ℓ for the orbital angular momentum of the Earth about the Sun, and (*b*) the number of possible orientations for the plane of Earth's orbit.

4. Show that the diffraction spread of a laser beam, $\approx \lambda/D$ (Section 28–11), is precisely what you might expect from the uncertainty principle. See also Chapters 24 and 25. [*Hint*: Since the beam's width is constrained by the dimension of the aperture D, the component of the light's momentum perpendicular to the laser axis is uncertain.]

5. For noble gases, the halogens, and the alkali metals, explain the atomic structure that is common to each group and how that structure explains a common property of the group. (See Section 28–8.)

ANSWERS TO EXERCISES

A: (*c*).
B: 2.1×10^{-24} kg·m/s, 2.3×10^{6} m/s.
C: 2, 1, 0, −1, −2.
D: −0.38 eV, $\sqrt{20}\,\hbar$.
E: Add one line to Li in Table 28–2: 2, 0, 0, $-\frac{1}{2}$.
F: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^1$.



This computer processor chip contains over 1.4 billion transistors, plus diodes and other semiconductor electronic elements, all in a space of about 1 cm^2. It uses 22-nm technology, meaning the "wires" (conducting lines) are 22 nm wide.

Before discussing semiconductors and their applications, we study the quantum theory description of bonding between atoms to form molecules, and how it explains molecular behavior. We then examine how atoms and molecules form solids, with emphasis on metals as well as on semiconductors and their use in electronics.

CHAPTER 29

Molecules and Solids

CHAPTER-OPENING QUESTION—Guess now!

What holds a solid together?

(a) Gravitational forces.
(b) Magnetic forces.
(c) Electric forces.
(d) Glue.
(e) Nuclear forces.

Since its development in the 1920s, quantum mechanics has had a profound influence on our lives, both intellectually and technologically. Even the way we view the world has changed, as we have seen in the last few Chapters. Now we discuss how quantum mechanics has given us an understanding of the structure of molecules and matter in bulk, as well as a number of important applications including semiconductor devices and applications to biology. Semiconductor devices, like transistors, now may be only a few atoms thick, which is the realm of quantum mechanics.

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*29–11 Integrated Circuits, 22-nm Technology

*29–1 Bonding in Molecules

One of the great successes of quantum mechanics was to give scientists, at last, an understanding of the nature of chemical bonds. Because it is based in physics, and because this understanding is so important in many fields, we discuss it here.

By a molecule, we mean a group of two or more atoms that are strongly held together so as to function as a single unit. When atoms make such an attachment, we say that a chemical **bond** has been formed. There are two main types of strong chemical bond: covalent and ionic. Many bonds are actually intermediate between these two types.

*Covalent Bonds

To understand how *covalent bonds* are formed, we take the simplest case, the bond that holds two hydrogen atoms together to form the hydrogen molecule, H_2. The mechanism is basically the same for other covalent bonds. As two H atoms approach each other, the electron clouds begin to overlap, and the electrons from each atom can "orbit" both nuclei. (This is sometimes called **sharing** electrons.) If both electrons are in the ground state $(n = 1)$ of their respective atoms, there are two possibilities: their spins can be parallel (both up or both down), in which case the total spin is $S = \frac{1}{2} + \frac{1}{2} = 1$; or their spins can be opposite ($m_s = +\frac{1}{2}$ for one, and $m_s = -\frac{1}{2}$ for the other), so that the total spin $S = 0$. We shall now see that a bond is formed only for the $S = 0$ state, when the spins are opposite.

First we consider the $S = 1$ state, for which the spins are the same. The two electrons cannot both be in the lowest energy state and be attached to the same atom, for then they would have identical quantum numbers in violation of the exclusion principle. The exclusion principle tells us that, because no two electrons can occupy the same quantum state, if two electrons have the same quantum numbers, they must be different in some other way—namely, by being in different places in space (for example, attached to different atoms). Thus, for $S = 1$, when the two atoms approach each other, the electrons will stay away from each other as shown by the probability distribution of Fig. 29–1. The electrons spend very little time between the two nuclei, so the positively charged nuclei repel each other and no bond is formed.

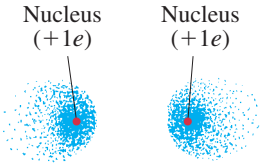


FIGURE 29–1 Electron probability distribution (electron cloud) for two H atoms when the spins are the same: $S = \frac{1}{2} + \frac{1}{2} = 1$.

For the $S = 0$ state, on the other hand, the spins are opposite and the two electrons are consequently in different quantum states (m_s is different, $+\frac{1}{2}$ for one, $-\frac{1}{2}$ for the other). Hence the two electrons can come close together, and the probability distribution looks like Fig. 29–2: the electrons can spend much of their time between the two nuclei. The two positively charged nuclei are attracted to the negatively charged electron cloud between them, and this is the attraction that holds the two hydrogen atoms together to form a hydrogen molecule. This is a **covalent bond**.

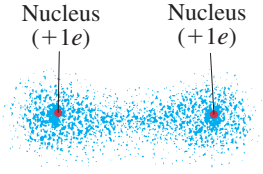


FIGURE 29–2 Electron probability distribution (cloud) around two H atoms when the spins are opposite ($S = 0$): in this case, a bond is formed because the positive nuclei are attracted to the concentration of the electron cloud's negative charge between them. This is a hydrogen molecule, H_2.

The probability distributions of Figs. 29–1 and 29–2 can perhaps be better understood on the basis of waves. What the exclusion principle requires is that when the spins are the same, there is destructive interference of the electron wave functions in the region between the two atoms. But when the spins are opposite, constructive interference occurs in the region between the two atoms, resulting in a large amount of negative charge there. Thus a covalent bond can be said to be the result of constructive interference of the electron wave functions in the space between the two atoms, and of the electrostatic attraction of the two positive nuclei for the negative charge concentration between them.

Why a bond is formed can also be understood from the energy point of view. When the two H atoms approach close to one another, if the spins of their electrons are opposite, the electrons can occupy the same space, as discussed above. This means that each electron can now move about in the space of two atoms instead of in the volume of only one. Because each electron now occupies more space, it is less well localized. From the uncertainty principle with Δx larger, we see that Δp and the minimum momentum can be less. With less momentum, each electron has less energy when the two atoms combine than when they are separate. That is, the molecule has less energy than the two separate atoms, and so is more stable. An energy input is required to break the H_2 molecule into two separate H atoms, so the H_2 molecule is a stable entity. This is what we mean by a *bond*. The energy required to break a bond is called the **bond energy**, the **binding energy**, or the **dissociation energy**. For the hydrogen molecule, H_2, the bond energy is 4.5 eV.

*Ionic Bonds

An *ionic bond* is, in a sense, a special case of the covalent bond. Instead of the electrons being shared equally, they are shared unequally. For example, in sodium chloride (NaCl), the outer electron of the sodium spends nearly all its time around the chlorine (Fig. 29–3). The chlorine atom acquires a net negative charge as a result of the extra electron, whereas the sodium atom is left with a net positive charge. The electrostatic attraction between these two charged atoms holds them together. The resulting bond is called an **ionic bond** because it is created by the attraction between the two ions (Na^+ and Cl^-). But to understand the ionic bond, we must understand why the extra electron from the sodium spends so much of its time around the chlorine. After all, the chlorine atom is neutral; why should it attract another electron?

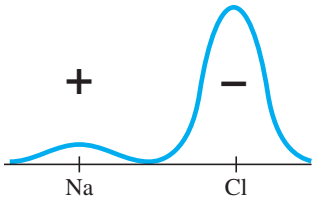
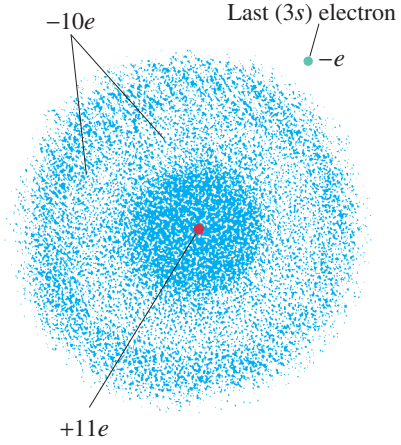


FIGURE 29–3 Probability distribution for the outermost electron of Na in NaCl.

The answer lies in the probability distributions of the electrons in the two neutral atoms. Sodium contains 11 electrons, 10 of which are in spherically symmetric closed shells (Fig. 29–4). The last electron spends most of its time beyond these closed shells. Because the closed shells have a total charge of $-10e$ and the nucleus has charge $+11e$, the outermost electron in sodium "feels" a net attraction due to $+1e$. It is not held very strongly. On the other hand, 12 of chlorine's 17 electrons form closed shells, or subshells (corresponding to $1s^2 2s^2 2p^6 3s^2$). These 12 electrons form a spherically symmetric shield around the nucleus. The other five electrons are in $3p$ states whose probability distributions are not spherically symmetric and have a form similar to those for the $2p$ states in hydrogen shown in Figs. 28–9b and c. Four of these $3p$ electrons can have "doughnut-shaped" distributions symmetric about the z axis, as shown in Fig. 29–5. The fifth can have a "barbell-shaped" distribution (as for $m_\ell = 0$ in Fig. 28–9b), which in Fig. 29–5 is shown only in dashed outline because it is half empty. That is, the exclusion principle allows one more electron to be in this state (it will have spin opposite to that of the electron already there). If an extra electron—say from a Na atom—happens to be in the vicinity, it can be in this state, perhaps at point x in Fig. 29–5. It could experience an attraction due to as much as $+5e$ because the $+17e$ of the nucleus is partly shielded at this point by the 12 inner electrons. Thus, the outer electron of a sodium atom will be more strongly attracted by the $+5e$ of the chlorine atom than by the $+1e$ of its own atom. This, combined with the strong attraction between the two ions when the extra electron stays with the Cl^-, produces the charge distribution of Fig. 29–3, and hence the ionic bond.

FIGURE 29–4 In a neutral sodium atom, the 10 inner electrons shield the nucleus, so the single outer electron is attracted by a net charge of $+1e$.



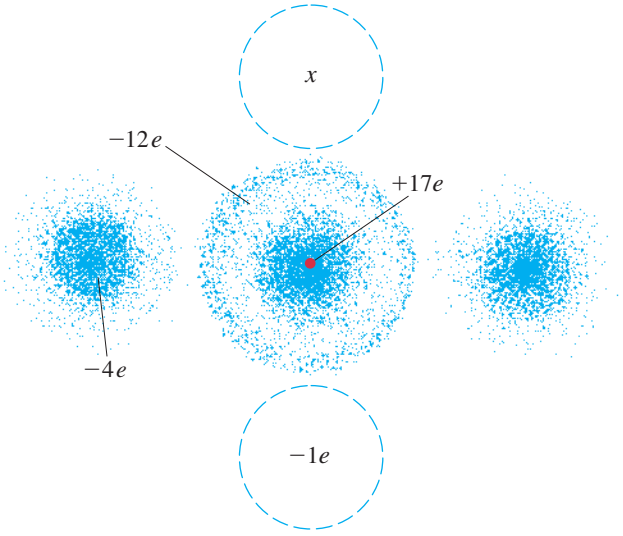


FIGURE 29–5 Neutral chlorine atom. The $+17e$ of the nucleus is shielded by the 12 electrons in the inner shells and subshells. Four of the five $3p$ electrons are shown in doughnut-shaped clouds (seen in cross section at left and right), and the fifth is in the dashed-line cloud concentrated about the z axis (vertical). An extra electron at x will be attracted by a net charge that can be as much as $+5e$.

*Partial Ionic Character of Covalent Bonds

A pure covalent bond in which the electrons are shared equally occurs mainly in symmetrical molecules such as H_2, O_2, and Cl_2. When the atoms involved are different from each other, usually the shared electrons are more likely to be in the vicinity of one atom than the other. The extreme case is an ionic bond. In intermediate cases the *covalent bond* is said to have a **partial ionic character**.

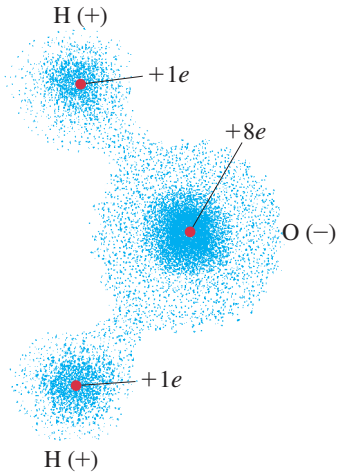


FIGURE 29–6 The water molecule H_2O is polar.

The molecules themselves are **polar**—that is, one part (or parts) of the molecule has a net positive charge and other parts a net negative charge. An example is the water molecule, H_2O (Fig. 29–6). Covalent bonds have shared electrons, which in H_2O are more likely to be found around the oxygen atom than around the two hydrogens. The reason is similar to that discussed above in connection with ionic bonds. Oxygen has eight electrons $(1s^2 2s^2 2p^4)$, of which four form a spherically symmetric core and the other four could have, for example, a doughnut-shaped distribution. The barbell-shaped distribution on the z axis (like that shown dashed in Fig. 29–5) could be empty, so electrons from hydrogen atoms can be attracted by a net charge of $+4e$. They are also attracted by the H nuclei, so they partly orbit the H atoms as well as the O atom. The net effect is that there is a net positive charge on each H atom (less than $+1e$), because the electrons spend only part of their time there. And, there is a net negative charge on the O atom.

*29–2 Potential-Energy Diagrams for Molecules

It is useful to analyze the interaction between two objects—say, between two atoms or molecules—with the use of a potential-energy diagram, which is a plot of the potential energy versus the separation distance.

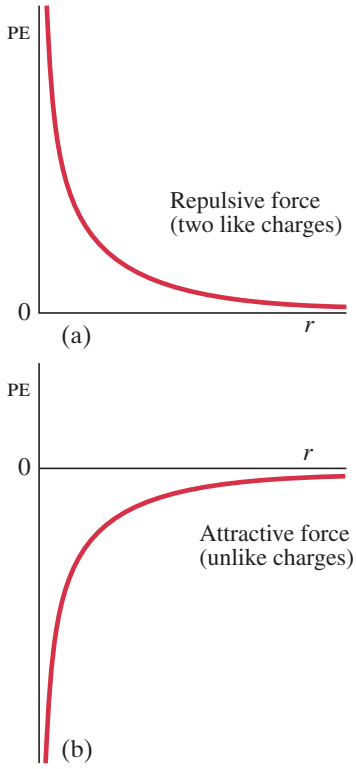
For the simple case of two point charges, q_1 and q_2, the potential energy PE is given by (we combine Eqs. 17–2a and 17–5)

$$\text{PE} = k\frac{q_1 q_2}{r},$$

where r is the distance between the charges, and the constant $k\,(= 1/4\pi\epsilon_0)$ is equal to $9.0 \times 10^9\ \text{N}\cdot\text{m}^2/\text{C}^2$. If the two charges have the same sign, the potential energy is positive for all values of r, and a graph of PE versus r in this case is shown in Fig. 29–7a. The force is repulsive (the charges have the *same* sign) and the curve rises as r decreases; this makes sense because if one particle moves freely toward the other (r getting smaller), the repulsion slows it down so its KE gets smaller, meaning PE gets larger. If, on the other hand, the two charges are of the *opposite* sign, the potential energy is negative because the product $q_1 q_2$ is negative. The force is attractive in this case, and the graph of PE $(\propto -1/r)$ versus r looks like Fig. 29–7b. The potential energy becomes more *negative* as r decreases.

Now let us look at the potential-energy diagram for the formation of a covalent bond, such as for the hydrogen molecule, H_2. The potential energy PE of one H atom in the presence of the other is plotted in Fig. 29–8. Starting at large r, the PE decreases as the atoms approach, because the electrons concentrate between the two nuclei (Fig. 29–2), so attraction occurs. However, at very short distances, the electrons would be "squeezed out"—there is no room for them between the two nuclei. Without the electrons between them, each nucleus would feel a repulsive force due to the other, so the curve rises as r decreases further.

FIGURE 29–7 Potential energy PE as a function of separation r for two point charges of (a) like sign and (b) opposite sign.



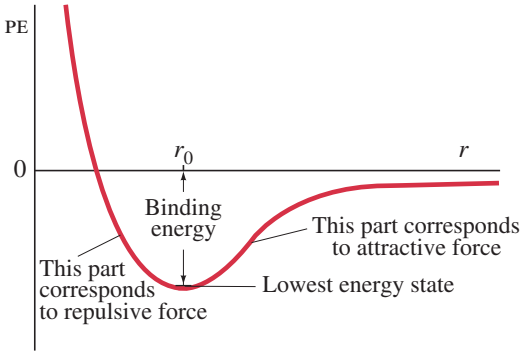


FIGURE 29–8 Potential-energy diagram for the H_2 molecule; r is the separation of the two H atoms. The binding energy (the energy difference between PE = 0 and the lowest energy state near the bottom of the well) is 4.5 eV, and $r_0 = 0.074$ nm.

There is an optimum separation of the atoms, r_0 in Fig. 29–8, at which the energy is lowest. This is the point of greatest stability for the hydrogen molecule, and r_0 is the average separation of atoms in the H_2 molecule. The depth of this "well" is the *binding energy*,† as shown. This is how much energy must be put into the system to separate the two atoms to infinity, where the PE = 0. For the H_2 molecule, the binding energy is about 4.5 eV and $r_0 = 0.074$ nm.

For many bonds, the potential-energy curve has the shape shown in Fig. 29–9. There is still an optimum distance r_0 at which the molecule is stable. But when the atoms approach from a large distance, the force is initially repulsive rather than attractive. The atoms thus do not form a bond spontaneously. Some additional energy must be injected into the system to get it over the "hump" (or barrier) in the potential-energy diagram. This required energy is called the **activation energy**.

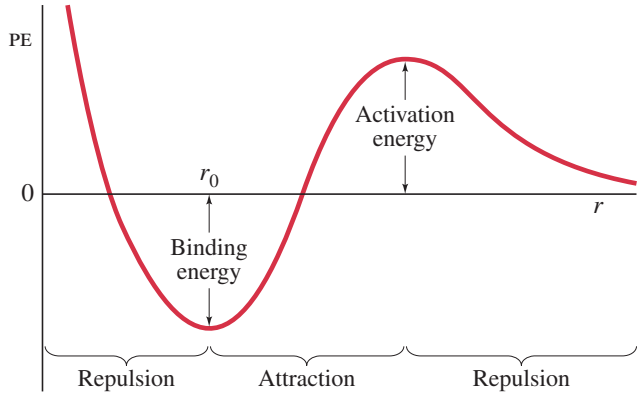


FIGURE 29–9 Potential-energy diagram for a bond requiring an activation energy.

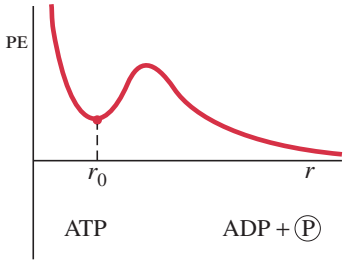
The curve of Fig. 29–9 is much more common than that of Fig. 29–8. The activation energy often reflects a need to break other bonds, before the one under discussion can be made. For example, to make water from O_2 and H_2, the H_2 and O_2 molecules must first be broken into H and O atoms by an input of energy; this is what the activation energy represents. Then the H and O atoms can combine to form H_2O with the release of a great deal more energy than was put in initially. The initial activation energy can be provided by applying an electric spark to a mixture of H_2 and O_2, breaking a few of these molecules into H and O atoms. When these atoms combine to form H_2O, a lot of energy is released (the ground state is near the bottom of the well) which provides the activation energy needed for further reactions: additional H_2 and O_2 molecules are broken up and recombined to form H_2O.

The potential-energy diagrams for ionic bonds, such as NaCl, may be more like Fig. 29–8: the Na^+ and Cl^- ions attract each other at distances a bit larger than some r_0, but at shorter distances the overlapping of inner electron shells gives rise to repulsion. The two atoms thus are most stable at some intermediate separation, r_0. For partially ionic bonds, there is usually an activation energy, Fig. 29–9.

Sometimes the potential energy of a bond looks like that of Fig. 29–10. In this case, the energy of the bonded molecule, at a separation r_0, is greater than when there is no bond ($r = \infty$). That is, an energy *input* is required to make the bond (hence the binding energy is negative), and there is energy release when the bond is broken. Such a bond is stable only because there is the barrier of the activation energy. This type of bond is important in living cells, for it is in such bonds that energy can be stored efficiently in certain molecules, particularly ATP (adenosine triphosphate). The bond that connects the last phosphate group (designated Ⓟ in Fig. 29–10) to the rest of the molecule (ADP, meaning adenosine diphosphate, since it contains only two phosphates) has potential energy of the shape shown in Fig. 29–10. Energy is stored in this bond. When the bond is broken (ATP → ADP + Ⓟ), energy is released and this energy can be used to make other chemical reactions "go."

PHYSICS APPLIED
ATP and energy in the cell

FIGURE 29–10 Potential-energy diagram for the formation of ATP from ADP and phosphate (Ⓟ).



†The binding energy corresponds not quite to the bottom of the potential-energy curve, but to the lowest quantum energy state, slightly above the bottom, as shown in Fig. 29–8.

In living cells, many chemical reactions have activation energies that are often on the order of several eV. Such energy barriers are not easy to overcome in the cell. This is where enzymes come in. They act as *catalysts*, which means they act to lower the activation energy so that reactions can occur that otherwise would not. Enzymes act via the electrostatic force to distort the bonding electron clouds, so that the initial bonds are easily broken.

*29–3 Weak (van der Waals) Bonds

Once a bond between two atoms or ions is made, energy must normally be supplied to break the bond and separate the atoms. As mentioned in Section 29–1, this energy is called the *bond energy* or *binding energy*. The binding energy for covalent and ionic bonds is typically 2 to 5 eV. These bonds, which hold atoms together to *form* molecules, are often called **strong bonds** to distinguish them from so-called "weak bonds." The term **weak bond**, as we use it here, refers to an attachment *between* molecules due to simple electrostatic attraction—such as *between* polar molecules (and not *within* a polar molecule, which is a strong bond). The strength of the attachment is much less than for the strong bonds. Binding energies are typically in the range 0.04 to 0.3 eV—hence their name "weak bonds."

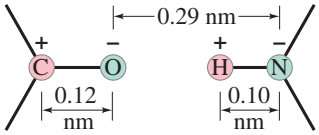


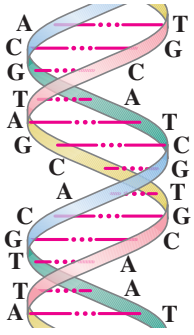
FIGURE 29–11 The C^+—O^- and H^+—N^- dipoles attract each other. (These dipoles may be part of, for example, the nucleotide bases cytosine and guanine in DNA molecules. See Fig. 29–12.) The + and − charges typically have magnitudes of a fraction of e.

Weak bonds are generally the result of attraction between dipoles. (A pair of equal point charges q of opposite sign, separated by a distance ℓ, is called an **electric dipole**, as we saw in Chapter 17.) For example, Fig. 29–11 shows two molecules, which have permanent dipole moments, attracting one another. Besides such **dipole–dipole bonds**, there can also be **dipole–induced dipole bonds**, in which a polar molecule with a permanent dipole moment can induce a dipole moment in an otherwise electrically balanced (nonpolar) molecule, just as a single charge can induce a separation of charge in a nearby object (see Fig. 16–7). There can even be an attraction between two nonpolar molecules, because their electrons are moving about: at any instant there may be a transient separation of charge, creating a brief dipole moment and weak attraction. All these weak bonds are referred to as **van der Waals bonds**, and the forces involved **van der Waals forces**. The potential energy has the general shape shown in Fig. 29–8, with the attractive van der Waals potential energy varying as $1/r^6$. The force decreases greatly with increased distance.

When one of the atoms in a dipole–dipole bond is hydrogen, as in Fig. 29–11, it is called a **hydrogen bond**. A hydrogen bond is generally the strongest of the weak bonds, because the hydrogen atom is the smallest atom and can be approached more closely. Hydrogen bonds also have a partial "covalent" character: that is, electrons between the two dipoles may be shared to a small extent, making a stronger, more lasting bond.

PHYSICS APPLIED
DNA

Weak bonds are very important for understanding the activities of cells, such as the double helix shape of DNA (Fig. 29–12), and DNA replication



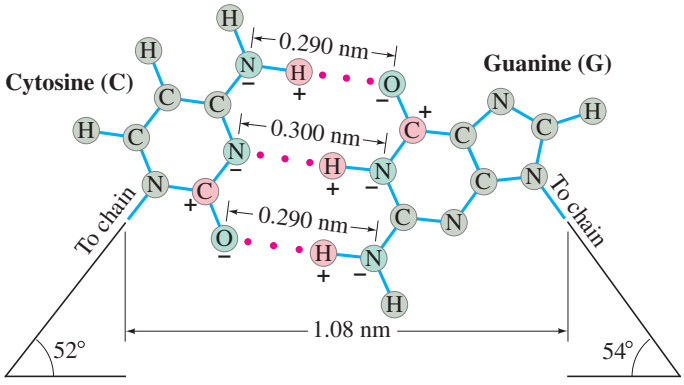


FIGURE 29–12 (a) Model of part of a DNA double helix. The red dots represent hydrogen bonds between the two strands. (b) "Close-up" view: cytosine (C) and guanine (G) molecules on separate strands of a DNA double helix are held together by the hydrogen bonds (red dots) involving an H^+ on one molecule attracted to an N^- or C^+—O^- of a molecule on the adjacent chain. See also Section 16–10 and Figs. 16–39 and 16–40.

(see Section 16–10). The average kinetic energy of molecules in a living cell at normal temperatures ($T \approx 300\,\text{K}$) is around $\frac{3}{2}kT \approx 0.04\,\text{eV}$ (kinetic theory, Chapter 13), about the magnitude of weak bonds. This means that a weak bond can readily be broken just by a molecular collision. Hence weak bonds are not very permanent—they are, instead, brief attachments. This helps them play particular roles in the cell. On the other hand, strong bonds—those that hold molecules together—are almost never broken simply by molecular collision because their binding energies are much higher ($\approx$2 to 5 eV). Thus they are relatively permanent. They can be broken by chemical action (the making of even stronger bonds), and this usually happens in the cell with the aid of an enzyme, which is a protein molecule.

EXAMPLE 29–1 **Nucleotide energy.** Calculate the potential energy between a C^+-O^- dipole of the nucleotide base cytosine and the nearby H^+-N^- dipole of guanine, assuming that the two dipoles are lined up as shown in Fig. 29–11. Dipole moment ($= q\ell$) measurements (see Table 17–2 and Fig. 29–11) give

$$q_\text{H} = -q_\text{N} = \frac{3.0 \times 10^{-30}\,\text{C}\cdot\text{m}}{0.10 \times 10^{-9}\,\text{m}} = 3.0 \times 10^{-20}\,\text{C} = 0.19e,$$

and

$$q_\text{C} = -q_\text{O} = \frac{8.0 \times 10^{-30}\,\text{C}\cdot\text{m}}{0.12 \times 10^{-9}\,\text{m}} = 6.7 \times 10^{-20}\,\text{C} = 0.42e.$$

APPROACH We want to find the potential energy of the two charges in one dipole due to the two charges in the other, because this will be equal to the work needed to pull the two dipoles infinitely far apart. The potential energy of a charge q_1 in the presence of a charge q_2 is $\text{PE} = k(q_1 q_2/r_{12})$ where $k = 9.0 \times 10^9\,\text{N}\cdot\text{m}^2/\text{C}^2$ and r_{12} is the distance between the two charges. (See Eqs. 17–2 and 17–5.)

SOLUTION The potential energy consists of four terms:

$$\text{PE} = \text{PE}_\text{CH} + \text{PE}_\text{CN} + \text{PE}_\text{OH} + \text{PE}_\text{ON}$$

where PE_CH means the potential energy of C in the presence of H, and similarly for the other terms. We do not have terms corresponding to C and O, or N and H, because the two dipoles are assumed to be stable entities. Then, using the distances shown in Fig. 29–11, we get:

$$\begin{aligned}
\text{PE} &= k\left[\frac{q_\text{C}q_\text{H}}{r_\text{CH}} + \frac{q_\text{C}q_\text{N}}{r_\text{CN}} + \frac{q_\text{O}q_\text{H}}{r_\text{OH}} + \frac{q_\text{O}q_\text{N}}{r_\text{ON}}\right] \\
&= (9.0 \times 10^9\,\text{N}\cdot\text{m}^2/\text{C}^2)(6.7 \times 10^{-20}\,\text{C})(3.0 \times 10^{-20}\,\text{C})\left(\frac{1}{r_\text{CH}} - \frac{1}{r_\text{CN}} - \frac{1}{r_\text{OH}} + \frac{1}{r_\text{ON}}\right) \\
&= (9.0 \times 10^9\,\text{N}\cdot\text{m}^2/\text{C}^2)(6.7)(3.0)\frac{(10^{-20}\,\text{C})^2}{(10^{-9}\,\text{m})}\left(\frac{1}{0.31} - \frac{1}{0.41} - \frac{1}{0.19} + \frac{1}{0.29}\right) \\
&= -1.86 \times 10^{-20}\,\text{J} = -0.12\,\text{eV}.
\end{aligned}$$

The potential energy is negative, meaning 0.12 eV of work (or energy input) is required to separate the dipoles. That is, the binding energy of this "weak" or hydrogen bond is 0.12 eV. This is only an estimate, of course, since other charges in the vicinity would have an influence too.

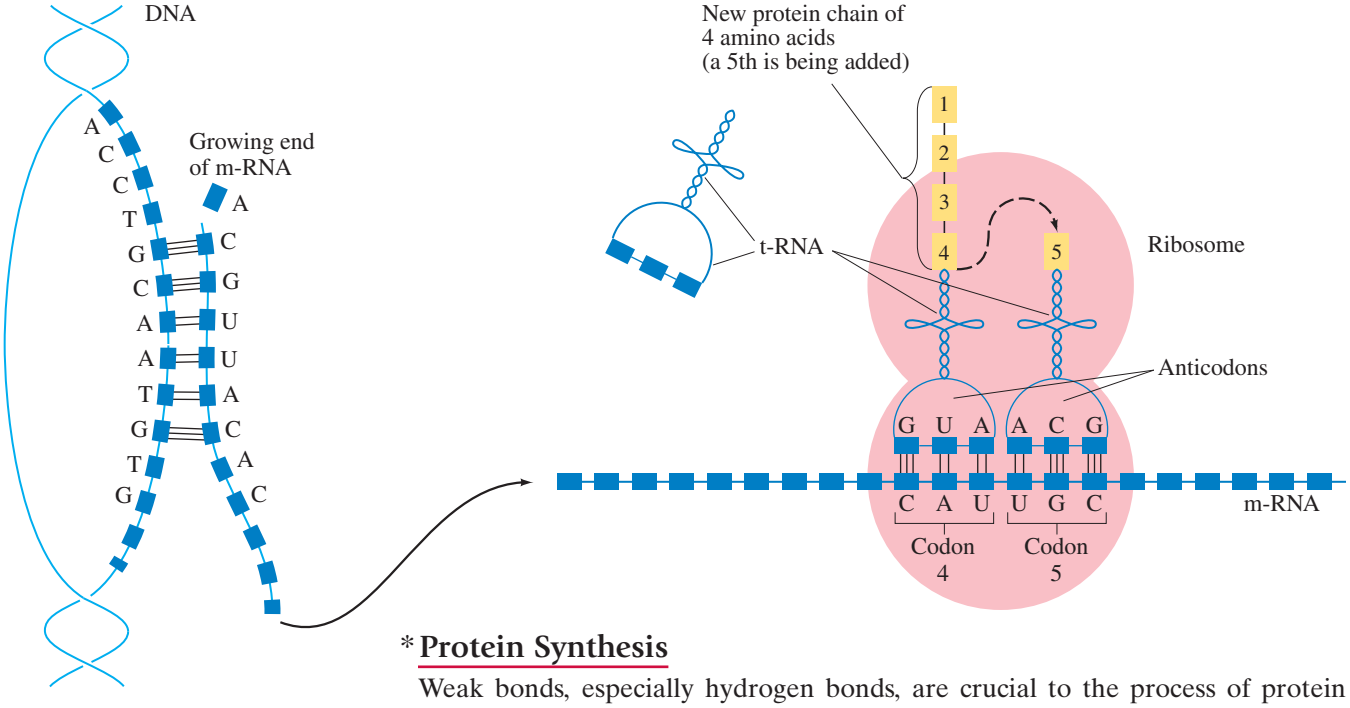


FIGURE 29–13 Protein synthesis. The yellow rectangles represent amino acids. See text for details.

PHYSICS APPLIED
Protein synthesis

* Protein Synthesis

Weak bonds, especially hydrogen bonds, are crucial to the process of protein synthesis. Proteins serve as structural parts of the cell and as enzymes to catalyze chemical reactions needed for the growth and survival of the organism. A protein molecule consists of one or more chains of small molecules known as *amino acids*. There are 20 different amino acids, and a single protein chain may contain hundreds of them in a specific order. The standard model for how amino acids are connected together in the correct order to form a protein molecule is shown schematically in Fig. 29–13.

We begin at the DNA double helix: each gene on a chromosome contains the information for producing one protein. The ordering of the four bases, A, C, G, and T, provides the "code," the **genetic code**, for the order of amino acids in the protein. First, the DNA double helix unwinds and a new molecule called *messenger*-RNA (m-RNA) is synthesized using one strand of the DNA as a "template." m-RNA is a chain molecule containing four different bases, like those of DNA (Section 16–10) except that thymine (T) is replaced by the similar uracil molecule (U). Near the top left in Fig. 29–13, a C has just been added to the growing m-RNA chain in much the same way that DNA replicates (Fig. 16–40); and an A, attracted and held close to the T on the DNA chain by the electrostatic force, will soon be attached to the C by an enzyme. The order of the bases, and thus the genetic information, is preserved in the m-RNA because the shapes of the molecules only allow the "proper" one to get close enough so the electrostatic force can act to form weak bonds.

Next, the m-RNA is buffeted about in the cell (recall kinetic theory, Chapter 13) until it gets close to a tiny organelle known as a *ribosome*, to which it can attach by electrostatic attraction (on the right in Fig. 29–13), because their shapes allow the charged parts to get close enough to form weak bonds. (Recall that force decreases greatly with separation distance.) Also held by the electrostatic force to the ribosome are one or two *transfer*-RNA (t-RNA) molecules. These t-RNA molecules "translate" the genetic code of nucleotide bases into amino acids in the following way. There is a different t-RNA molecule for each amino acid and each combination of three bases. On one end of a t-RNA molecule is an amino acid. On the other end of the t-RNA molecule is the appropriate "anticodon," a set of three nucleotide bases that "code" for that amino acid. If all three bases of an anticodon match the three bases of the "codon" on the m-RNA (in the sense of G to C and A to U), the anticodon is attracted electrostatically to the m-RNA codon and that t-RNA molecule is held there briefly. The

ribosome has two particular attachment sites which hold two t-RNA molecules while enzymes bond the two amino acids together to lengthen the amino acid chain (yellow in Fig. 29–13). As each amino acid is connected by an enzyme (four are already connected in Fig. 29–13, top right, and a fifth is about to be connected), the old t-RNA molecule is removed—perhaps by a random collision with some molecule in the cellular fluid. A new one soon becomes attracted as the ribosome moves along the m-RNA.

This process of protein synthesis is often presented as if it occurred in clockwork fashion—as if each molecule knew its role and went to its assigned place. But this is not the case. The forces of attraction between the electric charges of the molecules are rather weak and become significant only when the molecules can come close together, and when several weak bonds can be made. Indeed, if the shapes are not just right, the electrostatic attraction is nearly zero, which is why there are few mistakes. The fact that weak bonds are weak is very important. If they were strong, collisions with other molecules would not allow a t-RNA molecule to be released from the ribosome, or the m-RNA to be released from the DNA. If they were not temporary encounters, metabolism would grind to a halt.

As each amino acid is added to the next, the protein molecule grows in length until it is complete. Even as it is being made, this chain is being buffeted about in the cell—we might think of a wiggling worm. But a protein molecule has electrically charged polar groups along its length. And as it takes on various shapes, the electric forces of attraction between different parts of the molecule will eventually lead to a particular shape of the protein which is quite stable. Each type of protein has its own special shape, depending on the location of charged atoms. In the last analysis, the final shape depends on the order of the amino acids.

*29–4 Molecular Spectra

When atoms combine to form molecules, the probability distributions of the outer electrons overlap and this interaction alters the energy levels. Nonetheless, molecules can undergo transitions between electron energy levels just as atoms do. For example, the H_2 molecule can absorb a photon of just the right frequency to excite one of its ground-state electrons to an excited state. The excited electron can then return to the ground state, emitting a photon. The energy of photons emitted by molecules can be of the same order of magnitude as for atoms, typically 1 to 10 eV, or less.

Additional energy levels become possible for molecules (but not for atoms) because the molecule as a whole can rotate, and the atoms of the molecule can vibrate relative to each other. The energy levels for both rotational and vibrational levels are quantized, and are generally spaced much more closely (10^{-3} to 10^{-1} eV) than the electronic levels. Each atomic energy level thus becomes a set of closely spaced levels corresponding to the vibrational and rotational motions, Fig. 29–14. Transitions from one level to another appear as many very closely spaced lines. In fact, the lines are not always distinguishable, and these spectra are called **band spectra**. Each type of molecule has its own characteristic spectrum, which can be used for identification and for determination of structure. We now look in more detail at rotational and vibrational states in molecules.

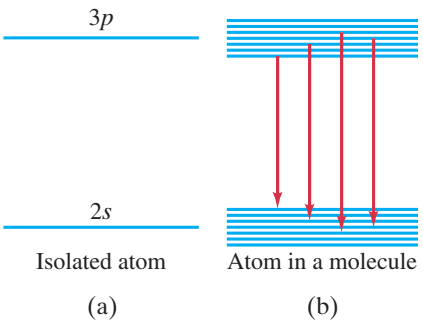


FIGURE 29–14 (a) The individual energy levels of an isolated atom become (b) bands of closely spaced levels in molecules, as well as in solids and liquids.

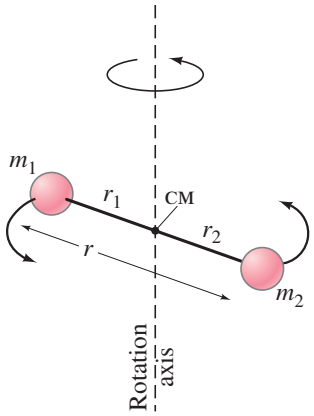
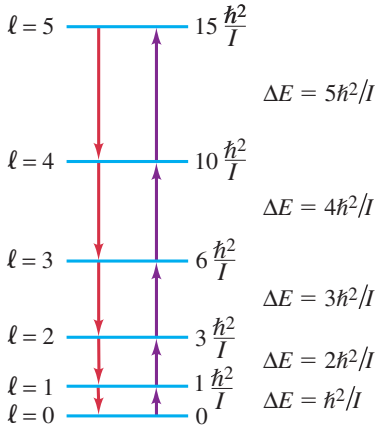


FIGURE 29–15 Diatomic molecule rotating about a vertical axis.

FIGURE 29–16 Rotational energy levels and allowed transitions (emission and absorption) for a diatomic molecule. Upward-pointing arrows represent absorption of a photon, and downward arrows represent emission of a photon.



* Rotational Energy Levels in Molecules

We consider only diatomic molecules, although the analysis can be extended to polyatomic molecules. When a diatomic molecule rotates about its center of mass as shown in Fig. 29–15, its kinetic energy of rotation (see Section 8–7) is

$$E_{\text{rot}} = \frac{1}{2} I\omega^2 = \frac{(I\omega)^2}{2I},$$

where $I\omega$ is the angular momentum (Section 8–8). Quantum mechanics predicts quantization of angular momentum just as in atoms (see Eq. 28–3):

$$I\omega = \sqrt{\ell(\ell+1)}\,\hbar, \qquad \ell = 0,\ 1,\ 2, \cdots,$$

where ℓ is an integer called the **rotational angular momentum quantum number**. Thus the rotational energy is quantized:

$$E_{\text{rot}} = \frac{(I\omega)^2}{2I} = \ell(\ell+1)\frac{\hbar^2}{2I}. \qquad \ell = 0,\ 1,\ 2, \cdots. \qquad \textbf{(29–1)}$$

Transitions between rotational energy levels are subject to the **selection rule** (as in Section 28–6):

$$\Delta\ell = \pm 1.$$

The energy of a photon emitted or absorbed for a transition between rotational states with angular momentum quantum number ℓ and $\ell - 1$ will be

$$\Delta E_{\text{rot}} = E_\ell - E_{\ell-1} = \frac{\hbar^2}{2I}\ell(\ell+1) - \frac{\hbar^2}{2I}(\ell-1)(\ell)$$
$$= \frac{\hbar^2}{I}\ell. \qquad \left[\begin{matrix}\ell \text{ is for upper} \\ \text{energy state}\end{matrix}\right] \qquad \textbf{(29–2)}$$

We see that the transition energy increases directly with ℓ. Figure 29–16 shows some of the allowed rotational energy levels and transitions. Measured absorption lines fall in the microwave or far-infrared regions of the spectrum (energies $\approx 10^{-3}$ eV), and their frequencies are generally 2, 3, 4, $\cdots$ times higher than the lowest one, as predicted by Eq. 29–2.

EXERCISE A Determine the three lowest rotational energy states (in eV) for a nitrogen molecule which has a moment of inertia $I = 1.39 \times 10^{-46}\,\text{kg}\cdot\text{m}^2$.

EXAMPLE 29–2 Rotational transition. A rotational transition $\ell = 1$ to $\ell = 0$ for the molecule CO has a measured absorption wavelength $\lambda_1 = 2.60$ mm (microwave region). Use this to calculate (*a*) the moment of inertia of the CO molecule, and (*b*) the CO bond length, r.

APPROACH The absorption wavelength is used to find the energy of the absorbed photon, and we can then calculate the moment of inertia, I, from Eq. 29–2. The moment of inertia is related to the CO separation (bond length r).

SOLUTION (*a*) The photon energy, $E = hf = hc/\lambda$, equals the rotational energy level difference, ΔE_{rot}. From Eq. 29–2, we can write

$$\frac{\hbar^2}{I}\ell = \Delta E_{\text{rot}} = hf = \frac{hc}{\lambda_1}.$$

With $\ell = 1$ (the upper state) in this case, we solve for I:

$$I = \frac{\hbar^2\ell}{hc}\lambda_1 = \frac{h\lambda_1}{4\pi^2 c} = \frac{(6.63 \times 10^{-34}\,\text{J}\cdot\text{s})(2.60 \times 10^{-3}\,\text{m})}{4\pi^2(3.00 \times 10^8\,\text{m/s})}$$
$$= 1.46 \times 10^{-46}\,\text{kg}\cdot\text{m}^2.$$

(*b*) The molecule rotates about its center of mass (CM) as shown in Fig. 29–15. Let m_1 be the mass of the C atom, $m_1 = 12$ u, and let m_2 be the mass of the O, $m_2 = 16$ u. The distance of the CM from the C atom, which is r_1 in Fig. 29–15, is given by the CM formula, Eq. 7–9:

$$r_1 = \frac{0 + m_2 r}{m_1 + m_2} = \frac{16}{12 + 16}r = 0.57r.$$

The O atom is a distance $r_2 = r - r_1 = 0.43r$ from the CM. The moment of

inertia of the CO molecule about its CM is then (see Example 8–9)

$$\begin{aligned} I &= m_1 r_1^2 + m_2 r_2^2 \\ &= \left[(12\,\text{u})(0.57r)^2 + (16\,\text{u})(0.43r)^2\right]\left[1.66 \times 10^{-27}\,\text{kg/u}\right] \\ &= \left(1.14 \times 10^{-26}\,\text{kg}\right) r^2. \end{aligned}$$

We solve for r and use the result of part (a) for I:

$$r = \sqrt{\frac{1.46 \times 10^{-46}\,\text{kg}\cdot\text{m}^2}{1.14 \times 10^{-26}\,\text{kg}}} = 1.13 \times 10^{-10}\,\text{m} = 0.113\,\text{nm} \approx 0.11\,\text{nm}.$$

EXERCISE B What are the wavelengths of the next three rotational transitions for CO?

*Vibrational Energy Levels in Molecules

The potential energy of the two atoms in a typical diatomic molecule has the shape shown in Fig. 29–8 or 29–9, and Fig. 29–17 again shows the PE for the H_2 molecule (solid curve). This PE curve, at least in the vicinity of the equilibrium separation r_0, closely resembles the potential energy of a harmonic oscillator, $\text{PE} = \frac{1}{2}kx^2$, which is shown superposed in dashed lines. Thus, for small displacements from r_0, each atom experiences a restoring force approximately proportional to the displacement, and the molecule vibrates as a simple harmonic oscillator (SHO)—see Chapter 11. According to quantum mechanics, the possible quantized energy levels are

$$E_{\text{vib}} = \left(\nu + \tfrac{1}{2}\right)hf, \qquad \nu = 0, 1, 2, \cdots, \tag{29–3}$$

where f is the classical frequency (see Chapter 11—f depends on the mass of the atoms and on the bond strength or "stiffness") and ν is an integer called the **vibrational quantum number**. The lowest energy state ($\nu = 0$) is not zero (as for rotation), but has $E = \frac{1}{2}hf$. This is called the **zero-point energy**. Higher states have energy $\frac{3}{2}hf$, $\frac{5}{2}hf$, and so on, as shown in Fig. 29–18. Transitions between vibrational energy levels are subject to the **selection rule**

$$\Delta\nu = \pm 1,$$

so allowed transitions occur only between adjacent states†, and all give off (or absorb) photons of energy

$$\Delta E_{\text{vib}} = hf. \tag{29–4}$$

This is very close to experimental values for small ν. But for higher energies, the PE curve (Fig. 29–17) begins to deviate from a perfect SHO curve, which affects the wavelengths and frequencies of the transitions. Typical transition energies are on the order of 10^{-1} eV, roughly 10 to 100 times larger than for rotational transitions, with wavelengths in the infrared region of the spectrum ($\approx 10^{-5}$ m).

EXAMPLE 29–3 **Vibrational energy levels in hydrogen.** Hydrogen molecule vibrations emit infrared radiation of wavelength around 2300 nm. (a) What is the separation in energy between adjacent vibrational levels? (b) What is the lowest vibrational energy state?

APPROACH The energy separation between adjacent vibrational levels is (Eq. 29–4) $\Delta E_{\text{vib}} = hf = hc/\lambda$. The lowest energy (Eq. 29–3) has $\nu = 0$.

SOLUTION

(a) $$\Delta E_{\text{vib}} = hf = \frac{hc}{\lambda} = \frac{(6.63 \times 10^{-34}\,\text{J}\cdot\text{s})(3.00 \times 10^{8}\,\text{m/s})}{(2300 \times 10^{-9}\,\text{m})(1.60 \times 10^{-19}\,\text{J/eV})} = 0.54\,\text{eV},$$

where the denominator includes the conversion factor from joules to eV.

(b) The lowest vibrational energy has $\nu = 0$ in Eq. 29–3:

$$E_{\text{vib}} = \left(\nu + \tfrac{1}{2}\right)hf = \tfrac{1}{2}hf = 0.27\,\text{eV}.$$

EXERCISE C What is the energy of the first vibrational state above the ground state in the hydrogen molecule?

†Forbidden transitions with $\Delta\nu = 2$ are emitted with much lower probability, but their observation can be important in some cases, such as in astronomy.

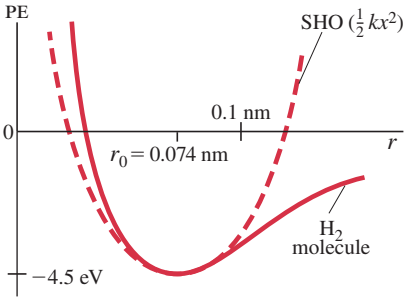
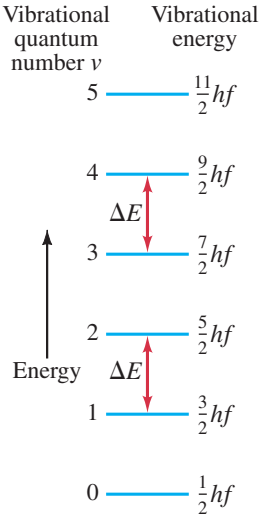
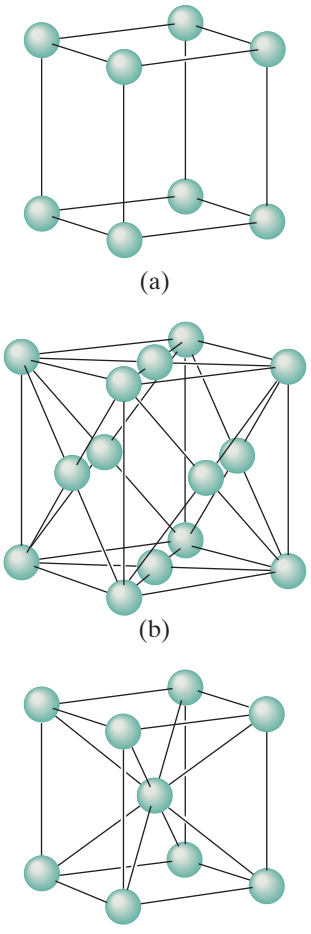


FIGURE 29–17 Potential energy for the H_2 molecule and for a simple harmonic oscillator ($\text{PE} = \frac{1}{2}kx^2$, with $|x| = |r - r_0|$).

FIGURE 29–18 Allowed vibrational energies for a diatomic molecule, where f is the fundamental frequency of vibration (see Chapter 11). The energy levels are equally spaced. Transitions are allowed only between adjacent levels ($\Delta\nu = \pm 1$).

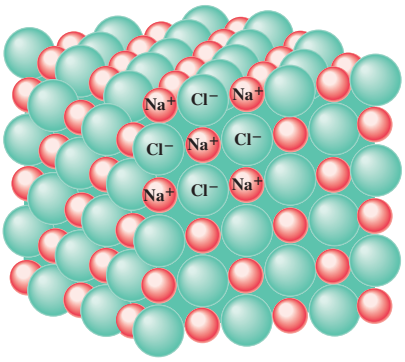




(a) (b) (c)

FIGURE 29–19 Arrangement of atoms in (a) a simple cubic crystal, (b) face-centered cubic crystal (note the atom at the center of each face), and (c) body-centered cubic crystal. Each of these "cells" is repeated in three dimensions to the edges of the macroscopic crystal.

FIGURE 29–20 Diagram of an NaCl crystal, showing the "packing" of atoms.



*29–5 Bonding in Solids

Quantum mechanics has been a great tool for understanding the structure of solids. This active field of research today is called **solid-state physics**, or **condensed-matter physics** so as to include liquids as well. The rest of this Chapter is devoted to this subject, and we begin with a brief look at the structure of solids and the bonds that hold them together.

Although some solid materials are *amorphous* in structure (such as glass), in that the atoms and molecules show no long-range order, we are interested here in the large class of *crystalline* substances whose atoms, ions, or molecules are generally accepted to form an orderly array known as a **lattice**. Figure 29–19 shows three of the possible arrangements of atoms in a crystal: simple cubic, face-centered cubic, and body-centered cubic. The NaCl crystal lattice is shown in Fig. 29–20.

The molecules of a solid are held together in a number of ways. The most common are by *covalent* bonding (such as between the carbon atoms of the diamond crystal) and by *ionic* bonding (as in a NaCl crystal). Often the bonds are partially covalent and partially ionic. Our discussion of these bonds earlier in this Chapter for molecules applies equally well to solids.

Let us look for a moment at the NaCl crystal of Fig. 29–20. Each Na^+ ion feels an attractive Coulomb potential due to each of the six "nearest neighbor" Cl^- ions surrounding it. Note that one Na^+ does not "belong" exclusively to one Cl^-, so we must not think of ionic solids as consisting of individual molecules. Each Na^+ also feels a repulsive Coulomb potential due to other Na^+ ions, although this is weaker since the Na^+ ions are farther away.

A different type of bond occurs in metals. Metal atoms have relatively loosely held outer electrons. **Metallic bond** theories propose that in a metallic solid, these outer electrons roam rather freely among all the metal atoms which, without their outer electrons, act like positive ions. According to the theory, the electrostatic attraction between the metal ions and this negative electron "gas" is responsible, at least in part, for holding the solid together. The binding energy of metal bonds is typically 1 to 3 eV, somewhat weaker than ionic or covalent bonds (5 to 10 eV in solids). The "free electrons" are responsible for the high electrical and thermal conductivity of metals. This theory also nicely accounts for the shininess of smooth metal surfaces: the free electrons can vibrate at any frequency, so when light of a range of frequencies falls on a metal, the electrons can vibrate in response and re-emit light of those same frequencies. Hence, the reflected light will consist largely of the same frequencies as the incident light. Compare this to nonmetallic materials that have a distinct color—the atomic electrons exist only in certain energy states, and when white light falls on them, the atoms absorb at certain frequencies, and reflect other frequencies which make up the color we see.

Here is a brief comparison of important strong bonds:

- ionic: an electron is "grabbed" from one atom by another;
- covalent: electrons are shared by atoms within a single molecule;
- metallic: electrons are shared by all atoms in the metal.

The atoms or molecules of some materials, such as the noble gases, can form only **weak bonds** with each other. As we saw in Section 29–3, weak bonds have very low binding energies and would not be expected to hold atoms together as a liquid or solid at room temperature. The noble gases condense only at very low temperatures, where the atomic (thermal) kinetic energy is small and the weak attraction can then hold the atoms together.

EXERCISE D Return to the Chapter-Opening Question, page 829, and answer it again now. Try to explain why you may have answered differently the first time.

*29–6 Free-Electron Theory of Metals; Fermi Energy

The free-electron theory of metals considers electrons in a metal as being in constant motion like an ideal gas, which we discussed in Chapter 13. For a classical ideal gas, at very low temperatures near absolute zero, $T = 0\,\text{K}$, all the particles would be in the lowest state, with zero kinetic energy $\left(= \frac{3}{2}kT = 0\right)$. But the situation is vastly different for an electron gas because, according to quantum mechanics, electrons obey the exclusion principle and can be only in certain possible energy levels or states. Electrons also obey a quantum statistics called **Fermi–Dirac statistics**† that takes into account the exclusion principle. All particles that have spin $\frac{1}{2}$ (or other half-integral spin: $\frac{3}{2}$, $\frac{5}{2}$, etc.), such as electrons, protons, and neutrons, obey Fermi–Dirac statistics and are referred to as **fermions** (see Section 28–7). The electron gas in a metal is often called a **Fermi gas**. According to the exclusion principle, no two electrons in the metal can have the same set of quantum numbers. Therefore, in each of the energy states available for the electrons in our "gas," there can be at most two electrons: one with spin up $\left(m_s = +\frac{1}{2}\right)$ and one with spin down $\left(m_s = -\frac{1}{2}\right)$. Thus, at $T = 0\,\text{K}$, the possible energy levels will be filled, two electrons each, up to a maximum level called the **Fermi level**. This is shown in Fig. 29–21, where the vertical axis is the "density of occupied states," whose meaning is similar to the Maxwell distribution for a classical gas (Section 13–10). The energy of the state at the Fermi level is called the **Fermi energy**, E_F. For copper, $E_F = 7.0\,\text{eV}$. This is very much greater than the energy of thermal motion at room temperature ($\overline{\text{KE}} = \frac{3}{2}kT \approx 0.04\,\text{eV}$, Eq. 13–8). Clearly, all motion does not stop at absolute zero.

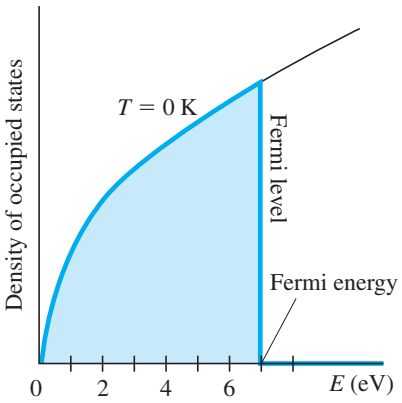


FIGURE 29–21 At $T = 0\,\text{K}$, all states up to energy E_F, called the Fermi energy, are filled. (Shown here for copper.)

At $T = 0$, all states with energy below E_F are occupied, and all states above E_F are empty. What happens for $T > 0$? We expect that at least some of the electrons will increase in energy due to thermal motion. Figure 29–22 shows the density of occupied states for $T = 1200\,\text{K}$, a temperature at which a metal is so hot it would glow. We see that the distribution differs very little from that at $T = 0$. We see also that the changes that do occur are concentrated about the Fermi level. A few electrons from slightly below the Fermi level move to energy states slightly above it. The average energy of the electrons increases only very slightly when the temperature is increased from $T = 0\,\text{K}$ to $T = 1200\,\text{K}$. This is very different from the behavior of an ideal gas, for which kinetic energy increases directly with T. Nonetheless, this behavior is readily understood as follows. Energy of thermal motion at $T = 1200\,\text{K}$ is about $\frac{3}{2}kT \approx 0.1\,\text{eV}$. The Fermi level, on the other hand, is on the order of several eV: for copper it is $E_F \approx 7.0\,\text{eV}$. An electron at $T = 1200\,\text{K}$ may have 7 eV of energy, but it can acquire at most only a few times 0.1 eV of energy by a (thermal) collision with the lattice. Only electrons very near the Fermi level would find vacant states close enough to make such a transition. Essentially none of the electrons could increase in energy by, say, 3 eV, so electrons farther down in the electron gas are unaffected. Only electrons near the top of the energy distribution can be thermally excited to higher states. And their new energy is on the average only slightly higher than their old energy. This model of free electrons in a metal as a "gas," though incomplete, provides good explanations for the thermal and electrical conductivity of metals.

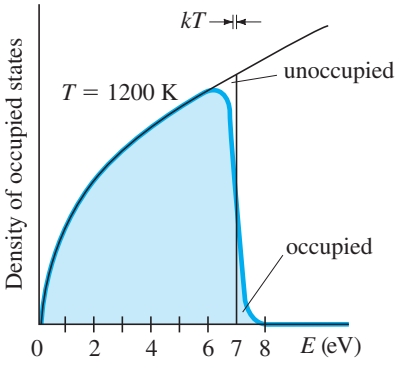


FIGURE 29–22 The density of occupied states for the electron gas in copper. The width kT shown above the graph represents thermal energy at $T = 1200\,\text{K}$.

†Developed independently by Enrico Fermi (Figs. 1–13, 27–11, 28–2, 30–7) in early 1926 and by P. A. M. Dirac a few months later. See Section 28–7.

*29–7 Band Theory of Solids

We saw in Section 29–1 that when two hydrogen atoms approach each other, the wave functions overlap, and the two 1*s* states (one for each atom) divide into two states of different energy. (As we saw, only one of these states, $S = 0$, has low enough energy to give a bound H_2 molecule.) Figure 29–23a shows this situation for 1*s* and 2*s* states for two atoms: as the two atoms get closer (toward the left in Fig. 29–23a), the 1*s* and 2*s* states split into two levels. If six atoms come together, as in Fig. 29–23b, each of the states splits into six levels. If a large number of atoms come together to form a solid, then each of the original atomic levels becomes a **band** as shown in Fig. 29–23c. The energy levels are so close together in each band that they seem essentially continuous. This is why the spectrum of heated solids (Section 27–2) appears continuous. (See also Fig. 29–14 and its discussion at start of Section 29–4.)

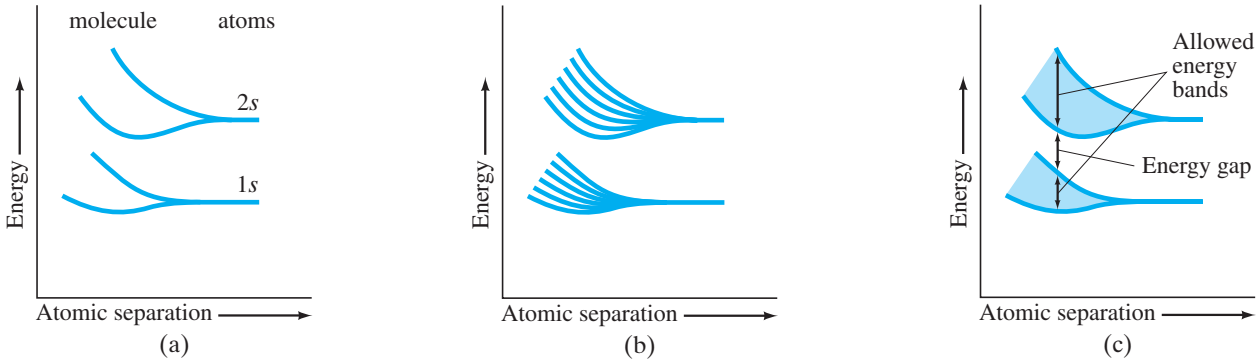
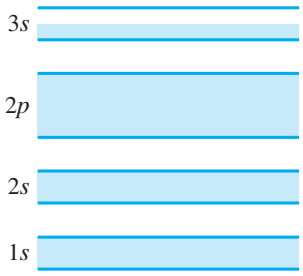


FIGURE 29–23 The splitting of 1*s* and 2*s* atomic energy levels as (a) two atoms approach each other (the atomic separation decreases toward the left on the graph); (b) the same for six atoms, and (c) for many atoms when they come together to form a solid.

FIGURE 29–24 Energy bands for sodium (Na).



The crucial aspect of a good **conductor** is that the highest energy band containing electrons is only partially filled. Consider sodium metal, for example, whose energy bands are shown in Fig. 29–24. The 1*s*, 2*s*, and 2*p* bands are full (just as in a sodium atom) and don't concern us. The 3*s* band, however, is only half full. To see why, recall that the exclusion principle stipulates that in an atom, only two electrons can be in the 3*s* state, one with spin up and one with spin down. These two states have slightly different energy. For a solid consisting of N atoms, the 3*s* band will contain $2N$ possible energy states. A sodium atom has a single 3*s* electron, so in a sample of sodium metal containing N atoms, there are N electrons in the 3*s* band, and N unoccupied states. When a potential difference is applied across the metal, electrons can respond by accelerating and increasing their energy, since there are plenty of unoccupied states of slightly higher energy available. Hence, a current flows readily and sodium is a good conductor. The characteristic of all good conductors is that the highest energy band is only partially filled, or two bands overlap so that unoccupied states are available. An example of the latter is magnesium, which has two 3*s* electrons, so its 3*s* band is filled. But the unfilled 3*p* band overlaps the 3*s* band in energy, so there are lots of available states for the electrons to move into. Thus magnesium, too, is a good conductor.

In a material that is a good **insulator**, on the other hand, the highest band containing electrons, called the **valence band**, is completely filled. The next highest energy band, called the **conduction band**, is separated from the valence band by a "forbidden" **energy gap** (or **band gap**), E_g, of typically 5 to 10 eV. So at room temperature (300 K), where thermal energies (that is, average kinetic energy—see Chapter 13) are on the order of $\frac{3}{2}kT \approx 0.04$ eV, almost no electrons can acquire the 5 eV needed to reach the conduction band. When a potential difference is applied across the material, no available states are accessible to the electrons, and no current flows. Hence, the material is a good insulator.

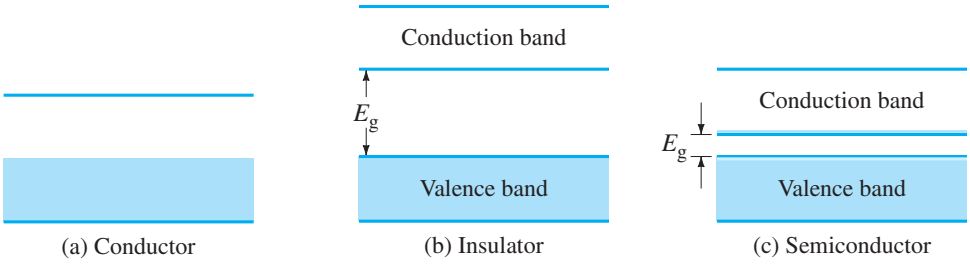


FIGURE 29–25 Energy bands for (a) a conductor, (b) an insulator, which has a large energy gap E_g, and (c) a semiconductor, which has a small energy gap E_g. Shading represents occupied states. Pale shading in (c) represents electrons that can pass from the top of the valence band to the bottom of the conduction band due to thermal agitation at room temperature (exaggerated).

Figure 29–25 compares the relevant energy bands (a) for conductors, (b) for insulators, and also (c) for the important class of materials known as **semiconductors**. The bands for a pure (or **intrinsic**) semiconductor, such as silicon or germanium, are like those for an insulator, except that the unfilled conduction band is separated from the filled valence band by a much smaller energy gap, E_g, which for silicon is $E_g = 1.12$ eV. At room temperature, electrons are moving about with varying amounts of kinetic energy $\left(\overline{\text{KE}} = \frac{3}{2}kT\right)$, according to kinetic theory, Chapter 13. A few electrons can acquire enough thermal energy to reach the conduction band, and so a very small current may flow when a voltage is applied. At higher temperatures, more electrons have enough energy to jump the gap (top end of thermal distribution—see Fig. 13–20). Often this effect can more than offset the effects of more frequent collisions due to increased disorder at higher temperature, so the resistivity of semiconductors can *decrease* with increasing temperature (see Table 18–1). But this is not the whole story of semiconductor conduction. When a potential difference is applied to a semiconductor, the few electrons in the conduction band move toward the positive electrode. Electrons in the valence band try to do the same thing, and a few can because there are a small number of unoccupied states which were left empty by the electrons reaching the conduction band. Such unfilled electron states are called **holes**. Each electron in the valence band that fills a hole in this way as it moves toward the positive electrode leaves behind its own hole, so the holes migrate toward the negative electrode. As the electrons tend to accumulate at one side of the material, the holes tend to accumulate on the opposite side. We will look at this phenomenon in more detail in the next Section.

EXAMPLE 29–4 **Calculating the energy gap.** It is found that the conductivity of a certain semiconductor increases when light of wavelength 345 nm or shorter strikes it, suggesting that electrons are being promoted from the valence band to the conduction band. What is the energy gap, E_g, for this semiconductor?

APPROACH The longest wavelength (lowest energy) photon to cause an increase in conductivity has $\lambda = 345$ nm, and its energy $(= hf)$ equals the energy gap.

SOLUTION The gap energy equals the energy of a $\lambda = 345$-nm photon:

$$E_g = hf = \frac{hc}{\lambda} = \frac{(6.63 \times 10^{-34}\,\text{J}\cdot\text{s})(3.00 \times 10^{8}\,\text{m/s})}{(345 \times 10^{-9}\,\text{m})(1.60 \times 10^{-19}\,\text{J/eV})} = 3.6\,\text{eV}.$$

CONCEPTUAL EXAMPLE 29–5 **Which is transparent?** The energy gap for silicon is 1.12 eV at room temperature, whereas that of zinc sulfide (ZnS) is 3.6 eV. Which one of these is opaque to visible light, and which is transparent?

PHYSICS APPLIED
Transparency

RESPONSE Visible-light photons span energies from roughly 1.8 eV to 3.1 eV. ($E = hf = hc/\lambda$ where $\lambda = 400$ nm to 700 nm and $1\,\text{eV} = 1.6 \times 10^{-19}$ J.) Light is absorbed by the electrons in a material. Silicon's energy gap is small enough to absorb these photons, thus bumping electrons well up into the conduction band, so silicon is opaque. On the other hand, zinc sulfide's energy gap is so large that no visible-light photons would be absorbed; they would pass right through the material which would thus be transparent.

*29–8 Semiconductors and Doping

Nearly all electronic devices today use semiconductors—mainly silicon (Si), although the first transistor (1948) was made with germanium (Ge). An atom of silicon has four outer electrons (group IV of the Periodic Table) that act to hold the atoms in the regular lattice structure of the crystal, shown schematically in Fig. 29–26a. Silicon acquires properties useful for electronics when a tiny amount of impurity is introduced into the crystal structure (perhaps 1 part in 10^6 or 10^7). This is called **doping** the semiconductor. Two kinds of doped semiconductor can be made, depending on the type of impurity used. The impurity can be an element whose atoms have five outer electrons (group V in the Periodic Table), such as arsenic. Then we have the situation shown in Fig. 29–26b, with a few arsenic atoms holding positions in the crystal lattice where normally silicon atoms are. Only four of arsenic's electrons fit into the bonding structure. The fifth does not fit in and can move relatively freely, somewhat like the electrons in a conductor. Because of this small number of extra electrons, a doped semiconductor becomes slightly conducting. The density of conduction electrons in an **intrinsic** (= undoped) semiconductor at room temperature is very low, usually less than 1 per 10^9 atoms. With an impurity concentration of 1 in 10^6 or 10^7 when doped, the conductivity will be much higher and it can be controlled with great precision. An arsenic-doped silicon crystal is an ***n*-type semiconductor** because *negative* charges (electrons) carry the electric current.

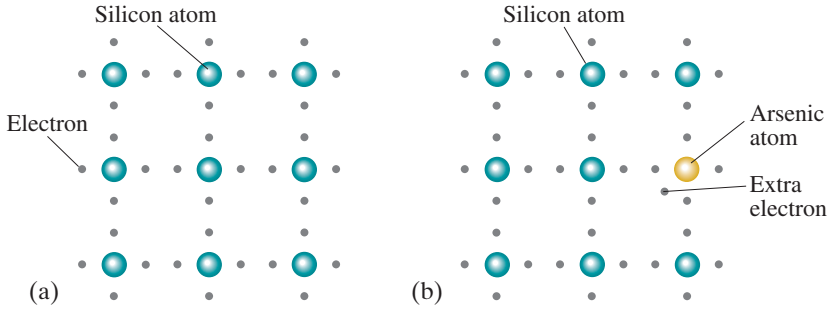


FIGURE 29–26 Two-dimensional representation of a silicon crystal. (a) Four (outer) electrons surround each silicon atom. (b) Silicon crystal doped with a small percentage of arsenic atoms: the extra electron doesn't fit into the crystal lattice and so is free to move about. This is an *n*-type semiconductor.

In a ***p*-type semiconductor**, a small percentage of semiconductor atoms are replaced by atoms with three outer electrons (group III in the Periodic Table), such as boron. As shown in Fig. 29–27a, there is a **hole** in the lattice structure near a boron atom because it has only three outer electrons. An electron from a nearby silicon atom can jump into this hole and fill it. But this leaves a hole where that electron had previously been, Fig. 29–27b. The vast majority of atoms are silicon, so holes are almost always next to a silicon atom. Since silicon atoms require four outer electrons to be neutral, this means there is a net positive charge at the hole. Whenever an electron moves to fill a hole, the positive hole is then at the previous position of that electron. Another electron can then fill this hole, and the hole thus moves to a new location; and so on. This type of semiconductor is called ***p*-type** because it is the positive holes that carry the electric current.[†] Note, however, that both *p-type* and *n-type* semiconductors have *no net charge* on them.

CAUTION

p-type semiconductors act as though + charges move—but electrons actually do the moving

[†]Each electron that fills a hole moves a very short distance ($\sim$1 atom $<$ 1 nm) whereas holes move much larger distances and so are the real carriers of the current. We can tell the current is carried by positive charges (holes) by using the Hall effect, Section 20–4.

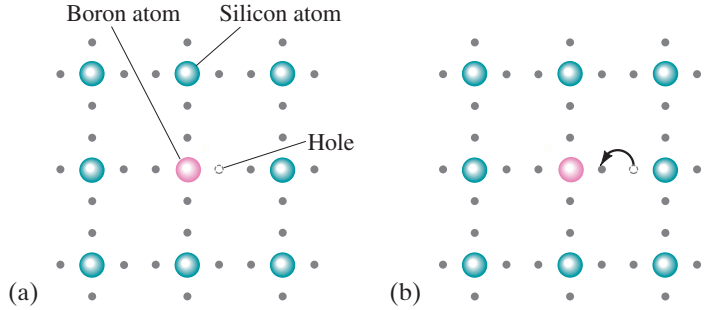


FIGURE 29–27 A *p*-type semiconductor, boron-doped silicon. (a) Boron has only three outer electrons, so there is an empty spot, or *hole* in the structure. (b) An electron from a nearby silicon atom can jump into the hole and fill it. As a result, the hole moves to a new location (to the right in this diagram), to where the electron used to be.

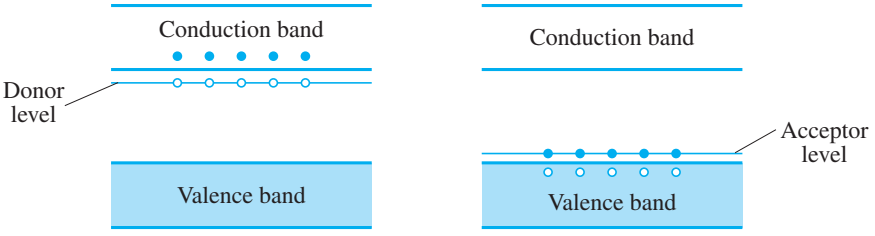


FIGURE 29–28 Impurity energy levels in doped semiconductors.

According to the band theory (Section 29–7), in a doped semiconductor the impurity provides additional energy states between the bands as shown in Fig. 29–28. In an *n*-type semiconductor, the impurity energy level lies just below the conduction band, Fig. 29–28a. Electrons in this energy level need only about 0.05 eV in Si to reach the conduction band which is on the order of the thermal energy, $\frac{3}{2}kT$ ($\approx$ 0.04 eV at 300 K). At room temperature, the tiny % of electrons in this donor level (roughly 1 in 10^6) can readily make the transition upward. This energy level can thus supply electrons to the conduction band, so it is called a **donor** level. In *p*-type semiconductors, the impurity energy level is just above the valence band (Fig. 29–28b). It is called an **acceptor** level because electrons from the valence band can jump into it with only average thermal energy. Positive holes are left behind in the valence band, and as other electrons move into these holes, the holes move as discussed earlier.

EXERCISE E Which of the following impurity atoms in silicon would produce a *p*-type semiconductor? (*a*) Ge; (*b*) Ne; (*c*) Al; (*d*) As; (*e*) Ga; (*f*) none of the above.

*29–9 Semiconductor Diodes, LEDs, OLEDs

Semiconductor diodes and transistors are essential components of modern electronic devices. The miniaturization achieved today allows many billions of diodes, transistors, resistors, etc., to be fabricated (adding doping atoms) on a single *chip* less than a millimeter on a side.

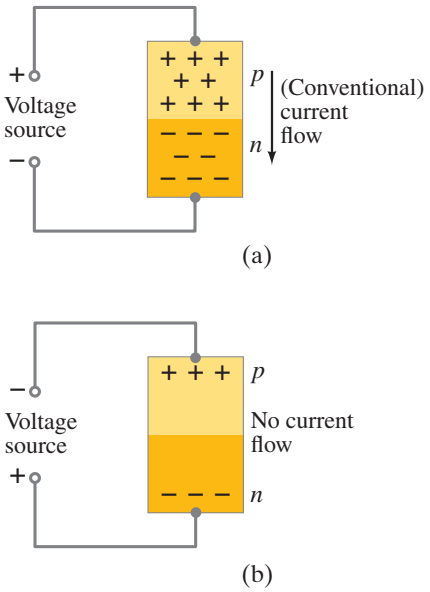
At the interface between an *n*-type and a *p*-type semiconductor, a ***pn* junction diode** is formed. Separately, the two semiconductors are electrically neutral. But near the junction, a few electrons diffuse from the *n*-type into the *p*-type semiconductor, where they fill a few of the holes. The *n*-type is left with a positive charge, and the *p*-type acquires a net negative charge. Thus an "intrinsic" *potential difference* is established, with the *n* side positive relative to the *p* side, and this prevents further diffusion of electrons. The "junction" is actually a very thin layer between the charged *n* and *p* semiconductors where all holes are filled with electrons. This junction region is called the **depletion layer** (depleted of electrons and holes).[†]

If a battery is connected to a diode with the positive terminal to the *p* side and the negative terminal to the *n* side as in Fig. 29–29a, the externally applied voltage opposes the intrinsic potential difference and the diode is said to be **forward biased**. If the voltage is great enough, about 0.6 V for Si at room temperature, it overcomes that intrinsic potential difference and a large current can flow. The positive holes in the *p*-type semiconductor are repelled by the positive terminal of the battery, and the electrons in the *n*-type are repelled by the negative terminal of the battery. The holes and electrons meet at the junction, and the electrons cross over and fill the holes. A current is flowing. The positive terminal of the battery is continually pulling electrons off the *p* end, forming new holes, and electrons are being supplied by the negative terminal at the *n* end.

When the diode is **reverse biased**, as in Fig. 29–29b, the holes in the *p* end are attracted to the battery's negative terminal and the electrons in the *n* end are attracted to the positive terminal. Almost no current carriers meet near the junction and, ideally, no current flows.

[†]One way to form the *pn* boundary at the **nanometer** thicknesses on chips is to implant (or diffuse) *n*-type donor atoms into the surface of a *p*-type semiconductor, converting a layer of the *p*-type semiconductor into *n*-type.

FIGURE 29–29 Schematic diagram showing how a semiconductor diode operates. Current flows when the voltage is connected in forward bias, as in (a), but not when connected in reverse bias, as in (b).



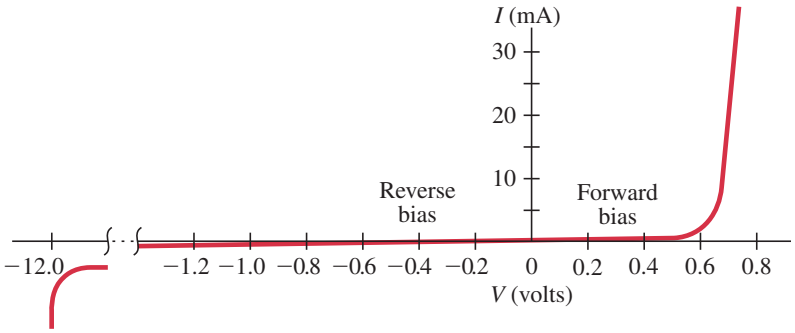


FIGURE 29–30 Current through a silicon *pn* diode as a function of applied voltage.

A graph of current versus voltage for a typical diode is shown in Fig. 29–30. A forward bias greater than 0.6 V allows a large current to flow. In reverse bias, a real diode allows a small amount of reverse current to flow; for most practical purposes, it is negligible.†

The symbol for a diode is



[diode]

where the arrow represents the direction conventional (+) current flows readily.

EXAMPLE 29–6 **A diode.** The diode whose current–voltage characteristics are shown in Fig. 29–30 is connected in series with a 4.0-V battery in forward bias and a resistor. If a current of 15 mA is to pass through the diode, what resistance must the resistor have?

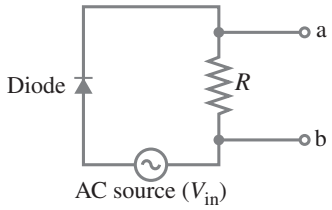
APPROACH We use Fig. 29–30, where we see that the voltage drop across the diode is about 0.7 V when the current is 15 mA. Then we use simple circuit analysis and Ohm's law (Chapters 18 and 19).

SOLUTION The voltage drop across the resistor is $4.0\,\text{V} - 0.7\,\text{V} = 3.3\,\text{V}$, so $R = V/I = (3.3\,\text{V})/(1.5 \times 10^{-2}\,\text{A}) = 220\,\Omega$.

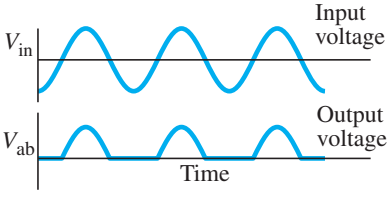
If the voltage across a diode connected in reverse bias is increased greatly, breakdown occurs. The electric field across the junction becomes so large that ionization of atoms results. The electrons thus pulled off their atoms contribute to a larger and larger current as breakdown continues. The voltage remains constant over a wide range of currents. This is shown on the far left in Fig. 29–30. This property of diodes can be used to accurately regulate a voltage supply. A diode designed for this purpose is called a **zener diode**. When placed across the output of an unregulated power supply, a zener diode can maintain the voltage at its own breakdown voltage as long as the supply voltage is always above this point. Zener diodes can be obtained corresponding to voltages of a few volts to hundreds of volts.

A diode is called a **nonlinear device** because the current is not proportional to the voltage. That is, a graph of current versus voltage (Fig. 29–30) is not a straight line, as it is for a resistor (which ideally *is* linear).

FIGURE 29–31 (a) A simple (half-wave) rectifier circuit using a semiconductor diode. (b) AC source input voltage, and output voltage across R, as functions of time.



(a)



(b)

* Rectifiers

Since a *pn* junction diode allows current to flow only in one direction (as long as the voltage is not too high), it can serve as a **rectifier**—to change ac into dc. A simple rectifier circuit is shown in Fig. 29–31a. The ac source applies a voltage across the diode alternately positive and negative. Only during half of each cycle will a current pass through the diode; only then is there a current through the resistor R. Hence, a graph of the voltage V_{ab} across R as a function of time looks like the output voltage shown in Fig. 29–31b. This **half-wave rectification** is not exactly dc, but it is unidirectional. More useful is a **full-wave rectifier** circuit, which uses two diodes (or sometimes four) as shown in Fig. 29–32a (top of next page). At any given instant, either one diode or the other will conduct current to the right.

†At room temperature, the reverse current is a few pA in Si; but it increases rapidly with temperature, and may render a diode ineffective above 200°C.

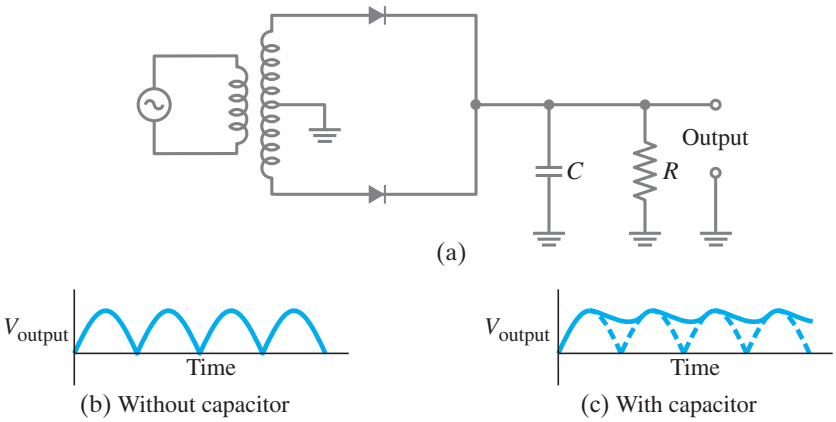


FIGURE 29–32 (a) Full-wave rectifier circuit (including a transformer so the magnitude of the voltage can be changed). (b) Output voltage in the absence of capacitor *C*. (c) Output voltage with the capacitor in the circuit.

Therefore, the output across the load resistor *R* will be as shown in Fig. 29–32b. Actually this is the voltage if the capacitor *C* were not in the circuit. The capacitor tends to store charge and, if the time constant *RC* is sufficiently long, helps to smooth out the current as shown in Fig. 29–32c. (The variation in output shown in Fig. 29–32c is called **ripple voltage**.)

Rectifier circuits are important because most line voltage in buildings is ac, and most electronic devices require a dc voltage for their operation. Hence, diodes are found in nearly all electronic devices including radios, TV sets, computers, and chargers for cell phones and other devices.

*Photovoltaic Cells

Solar cells, also called **photovoltaic cells**, are rather heavily doped *pn* junction diodes used to convert sunlight into electric energy. Photons are absorbed, creating electron–hole pairs if the photon energy is greater than the band gap energy, E_g (see Figs. 29–25c and 29–28). That is, the absorbed photon excites an electron from the valence band up to the conduction band, leaving behind a hole in the valence band. The created electrons and holes produce a current that, when connected to an external circuit, becomes a source of emf and power. A typical silicon *pn* junction may produce about 0.6 V. Many are connected in series to produce a higher voltage. Such series strings are connected in parallel within a **photovoltaic panel**. Research includes experimenting with combinations of semiconductors. A good photovoltaic panel can have an output of perhaps 50 W/m^2, averaged over day and night, sunny and cloudy. The world's total electricity demand is on the order of 10^{12} W, which could be met with solar cells covering an area of only about 200 km × 200 km of Earth's surface.[†]

Photodiodes (Section 27–3) and **semiconductor particle detectors** (Section 30–13) operate similarly.

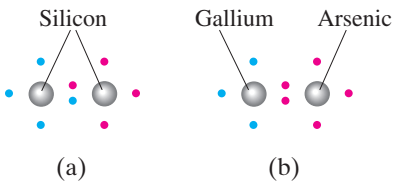
*LEDs

A **light-emitting diode** (**LED**) is sort of the reverse of a photovoltaic cell. When a *pn* junction is forward biased, a current begins to flow. Electrons cross from the *n*-region into the *p*-region, recombining with holes, and a photon can be emitted with an energy about equal to the band gap energy, E_g. This does not work well with silicon diodes.[‡] But high light-emission is achieved with **compound semiconductors**, typically involving a group III and a group V element such as gallium and arsenic (= gallium arsenide = GaAs). Remarkably, GaAs has a crystal structure very similar to Si. See Fig. 29–33. For doping of GaAs, group VI atoms (like Se) can serve as donors, and group II atoms (valence +2, such as Zn) as acceptors. The energy gap for GaAs is E_g = 1.42 eV, corresponding to near-infrared photons with wavelength 870 nm (almost visible). Such infrared LEDs are suitable for use in remote-control devices for TVs, DVD players, stereos, car door locks, and so on.

The first visible-light LED, developed in the early 1960s, was made of a semiconductor compound of gallium, arsenic, and phosphorus (= GaAsP) which emitted red light. The red LED soon found use as the familiar indicator lights (on–off) on electronic devices, and as the bright red read-out on calculators and

PHYSICS APPLIED
LEDs and applications
Car safety (brakes)

FIGURE 29–33 (a) Two Si atoms forming the covalent bond showing the electrons in different colors for each of the two separate atoms. (In Fig. 29–26a we showed each atom separately to emphasize the four outer electrons in each.) (b) A gallium–arsenic pair, also covalently bonded.



[†]Electricity makes up about 5% of total global energy use.

[‡]Electron-hole recombination in silicon results mostly in heat, as lattice vibrations called **phonons**.

digital clocks (brighter than the dimmer LCD readouts). Further development led to LEDs with higher E_g and shorter wavelengths: first yellow, then finally in 1995, blue (InGaN). A blue LED was important because it gave the possibility of a **white-light LED**. White light can be approximated by LEDs in two ways: (1) using a red, a yellow–green, and a blue LED; (2) using a blue LED with coatings of "powders" or "phosphors" that are fluorescent (Section 28–10). For the latter, the high-energy blue LED photons are themselves emitted, plus they can excite the various phosphors to excited states which decay in two or more steps, emitting light of lower energy and longer wavelengths. Figure 29–34 shows typical spectra of both types.

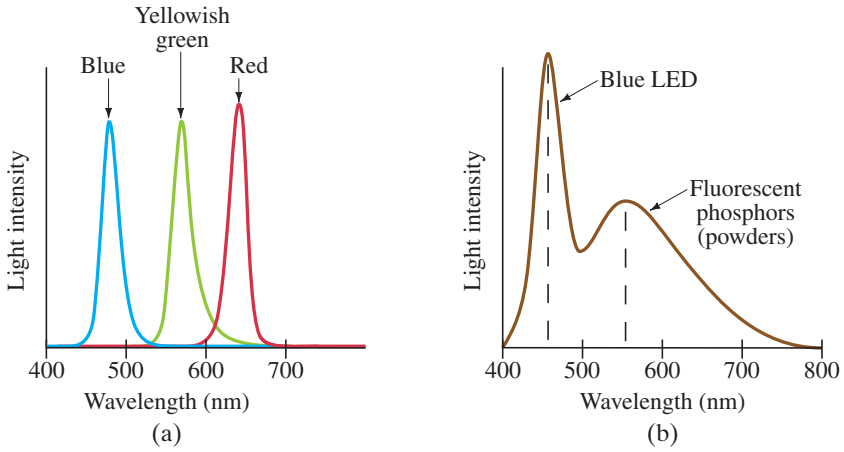


FIGURE 29–34 (a) A combination of three LEDs of three different colors gives a sort of white color, but there are large wavelength gaps, so some colors would not be reflected and would appear black; this type is rarely used now. (b) A blue LED with fluorescent phosphors or powders gives a better approximation of white light. (Thanks to M. Vannoni and G. Molesini for (b).)

FIGURE 29–35 LED flashlights. Note the tiny LEDs, each maybe $\frac{1}{2}$ cm in diameter.



LED "bulbs" are available to replace other types of lighting in applications such as flashlights (Fig. 29–35), street lighting, traffic signals, car brake lights, billboards, backlighting for LCD screens, and large display screens at stadiums. LED lights, sometimes called **solid-state** lighting, are longer-lived (50,000 hours vs. 1000–2000 for ordinary bulbs), more efficient (up to 5 times), and rugged. A small town in Italy, Torraca, was the first to have all its street lighting be LED (2007). LEDs can be as small as 1 or 2 mm wide, and are individual units with wires connected directly to them. They can be used for large TV screens in stadiums, but a home TV would require much smaller LED size, meaning fabrication of many on a crystalline semiconductor, and the pixels would be addressed as discussed in Section 17–11 for LCD screens.

* Pulse Oximeter

A **pulse oximeter** uses two LEDs to measure the % oxygen (O_2) saturation in your blood. One LED is red, 660 nm, and the other IR (900–940 nm). The LED beams pass through a finger (Fig. 29–36) or earlobe and are detected by a photodiode. Oxygenated red blood cells absorb less red and more infrared light than deoxygenated cells. A ratio of absorbed light (red/IR) of 0.5 corresponds to nearly 100% O_2 saturation; a ratio of 1.0 is about 85% and 2.0 corresponds to about 50% (bad). The LED measures during complete pulses, including blood surges, and the device can also count your heartbeat rate.

FIGURE 29–36 A pulse oximeter.



* *pn* Diode Lasers

Diode lasers, using a *pn*-junction in forward bias like an LED, are the most compact of lasers and are very common: they read CDs and DVDs and are used as pointers and in laser printers. They emit photons like an LED but, like all lasers (Section 28–11), need to have an *inverted population* of states for the lasing frequency. This is achieved by applying a high forward-bias voltage. The large current brings many electrons into the conduction band at the junction layer, and holes into the valence band, and before the electrons have time to combine with holes, they form an inverted population. When one electron drops down into a hole and emits a photon, that photon stimulates other electrons to drop down as well, *in phase*, creating coherent laser light. Opposite ends of the crystal are made parallel and very smooth so they act as the mirrors needed for lasing, as shown in our laser diagram, Fig. 28–18.

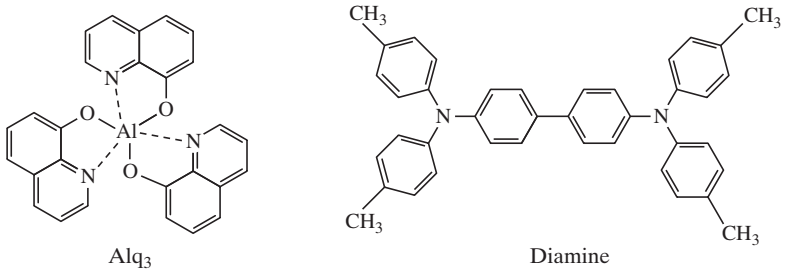


FIGURE 29–37 These two organic molecules were used in the first OLEDs (1987). The hexagons have carbon at each corner, and an attached hydrogen, unless otherwise noted.

*OLED (Organic LED)

Many organic compounds have semiconductor properties. Useful ones can have mobile electrons and holes. A practical organic **electroluminescent** (**EL**) device, an **organic light-emitting diode** (**OLED**) was first described in the late 1980s.

Organic compounds contain carbon (C), hydrogen (H), often nitrogen and oxygen, and sometimes other atoms. We usually think of them as coming from life—plants and animals. They are also found in petroleum, and some can be synthesized in the lab. Organic compounds can be complex, and often contain the familiar hexagonal "benzene ring" with C atoms at all (or most) of the six corners. The two organic compounds shown in Fig. 29–37 were used as *n*-type and *p*-type layers in the earliest useful OLED. **Polymers**, long organic molecules with repeating structural units, can also be used for an OLED.

The simplest OLED consists of two organic layers, the **emissive layer** and the **conductive layer**, each 20 to 50 nm thick, sandwiched between two electrodes, Fig. 29–38. The anode is typically transparent, to let the light out. It can be made of a very thin layer of indium–tin oxide (**ITO**), which is transparent and conductive, coated on a glass slab. The cathode is often metallic, but could also be made of transparent material.

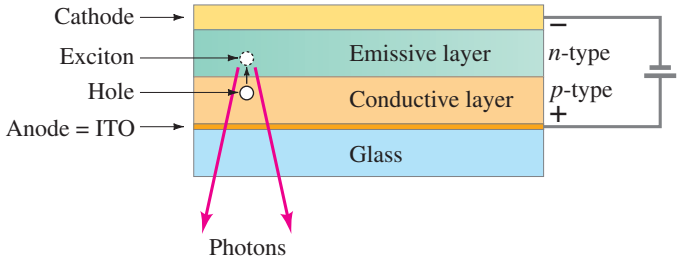


FIGURE 29–38 An OLED with two organic layers. Hole–electron recombination into an exciton (dashed circle) occurs in the emissive layer, followed by photon emission. Photons emitted in the wrong direction (upward in the diagram) reduce efficiency.

OLEDs can be smaller and thinner than ordinary inorganic LEDs. They can be more easily constructed as a unit for a screen display (i.e., more cheaply, but still quite expensive) than for inorganic LEDs. Their use as screens on cell phones, cameras, and TVs produces brighter light and greater contrast, and they need less power (important for battery life of portable devices) than LCD screens. Why? They need no backlight (like LCDs) because they emit the light themselves. OLEDs can be fabricated as a matrix, usually active matrix (**AMOLED**), using the same type of addressing described in Section 17–11 for LCDs. OLED displays are much thinner than LCDs and retain brightness at larger viewing angles. They can even be fabricated on curved or flexible substrates—try the windshield of your car (Fig. 29–39). The array may be RGBG (similar to a Bayer mosaic, Fig. 25–2) or RGBW where W = white is meant to give greater brightness. The subpixels can also be stacked, one above the other (similar to the Foveon, Fig. 25–3).

FIGURE 29–39 Head up displays on curved windshields can use curved OLEDs to show, for example, your speed without having to look down at the speedometer.



*OLED Functioning (advanced)

According to band theory, when a voltage is applied (≈ 2 to 5 V), electrons are "injected" (engineering term) into energy states of the **lowest unoccupied molecular orbitals** (**LUMO**) of the emissive layer. At the same time, electrons are withdrawn from the **highest occupied molecular orbitals** (**HOMO**) of the conductive layer at the cathode—which is equivalent to **holes** being "injected" into the conductive layer. The LUMO and HOMO energy levels are analogous to the conduction and valence bands of inorganic silicon diodes (Fig. 29–28). Holes travel in the HOMO, electrons in the LUMO. ("Orbital" is a chemistry word for the states occupied by the electrons in a molecule.)

When electrons and holes meet near the junction (Fig. 29–38), they can form a sort of bound state (like in the hydrogen atom) known as an **exciton**. An exciton has a small binding energy (0.1 to 1 eV), and a very short lifetime on the order of nanoseconds. When an exciton "decays" (the negative electron and positive hole combine), a photon is emitted. These photons are the useful output.

The energy hf of the photon, and its frequency corresponding to the color, depends on the energy structure of the exciton. The energy gap, LUMO–HOMO, sets an upper limit on hf, but the vibrational energy levels of the molecules reduce that by varying amounts, as does the binding energy of the exciton. The spectrum has a peak, like those in Fig. 29–34a, but is wider, 100–200 nm at half maximum. The organic molecules are chosen so that the photons have frequencies in the color range desired, say for a display subpixel: bluish (B), greenish (G), or red (R).

The conductive layer is also called the **hole transport layer** (**HTL**), which name expresses its purpose. The emissive layer, on the other hand (Fig. 29–38), serves two purposes: (1) it serves to transport electrons toward the junction, and (2) it is in this layer (near the junction) that holes meet electrons to form excitons and then combine and emit light. These two functions can be divided in a more sophisticated OLED that has three layers: Adjacent to the cathode is the **electron transport layer** (**ETL**), plus there is an **emissive layer** (**EML**) sandwiched between the ETL and the HTL. The emissive layer can be complex, containing a **host** material plus a **guest** compound in small concentration—a kind of doping—to fine-tune energy levels and efficiency.

*29–10 Transistors: Bipolar and MOSFETs

The **bipolar junction transistor** was invented in 1948 by J. Bardeen, W. Shockley, and W. Brattain. It consists of a crystal of one type of doped semiconductor sandwiched between two of the opposite type. Both *npn* and *pnp* transistors can be made, and they are shown schematically in Fig. 29–40a. The three semiconductors are given the names **collector**, **base**, and **emitter**. The symbols for *npn* and *pnp* transistors are shown in Fig. 29–40b. The arrow is always placed on the emitter and indicates the direction of (conventional) current flow in normal operation.

The operation of an *npn* transistor as an **amplifier** is shown in Fig. 29–41. A dc voltage V_{CE} is maintained between the collector and emitter by battery $\mathscr{E}_C$. The voltage applied to the base is called the *base bias voltage*, V_{BE}. If V_{BE} is positive, conduction electrons in the emitter are attracted into the base. The base region is very thin, much less than 1 μm, so most of these electrons flow right across into the collector which is maintained at a positive voltage. A large current, I_C, flows between collector and emitter and a much smaller current, I_B, through the base. In the steady state, I_B and I_C can be considered dc. But a small variation (= ac) in the base voltage due to an input signal attracts (or repels)

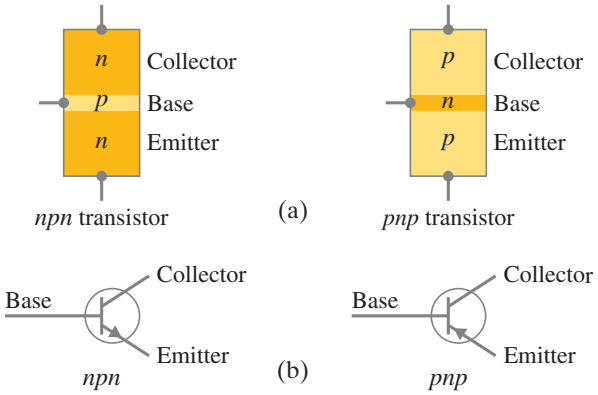


FIGURE 29–40 (a) Schematic diagram of *npn* and *pnp* transistors. (b) Symbols for *npn* and *pnp* transistors.

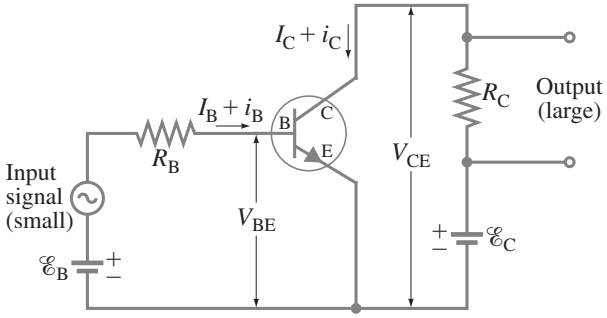


FIGURE 29–41 An *npn* transistor used as an amplifier. I_B is the current produced by $\mathscr{E}_B$ (in the absence of a signal), i_B is the ac signal current (= change in I_B).

charge that passes through into the collector and thus can cause a large *change* in the collector current and a large change in the voltage drop across the output resistor R_C. Hence a transistor can *amplify* a small signal into a larger one.

Typically a small ac signal (call it i_B) is to be amplified, and when added to the base bias current (and voltage) causes the current and voltage at the collector to vary at the same rate but magnified. Thus, what is important for amplification is the *change* in collector current for a given input *change* in base current. We label these ac signal currents (= changes in I_C and I_B) as i_C and i_B. The **current gain** is defined as the ratio

$$\beta_I = \frac{\text{output (collector) ac current}}{\text{input (base) ac current}} = \frac{i_C}{i_B}.$$

β_I may be on the order of 10 to 100. Similarly, the **voltage gain** is

$$\beta_V = \frac{\text{output (collector) ac voltage}}{\text{input (base) ac voltage}}.$$

Transistors are the basic elements in modern electronic amplifiers of all sorts.

A *pnp* transistor operates like an *npn*, except that holes move instead of electrons. The collector voltage is negative, and so is the base voltage in normal operation.

Another kind of transistor, very important, is the **MOSFET** (metal-oxide semiconductor field-effect transistor) common in **digital circuits** as a type of switch. Its construction is shown in Fig. 29–42a, and its symbol in Fig. 29–42b. What is called the emitter in a bipolar transistor is called the **source** in a MOSFET, and the collector is called the **drain**. The base is called the **gate**. The gate acts to let a current flow, or not, from the source to the drain, depending on the electric field it (the gate) provides across an insulator that separates it from the *p*-type semiconductor below, Fig. 29–42a. Hence the name "field-effect transistor" (FET).[†] MOSFETs are often used like switches, on or off, which in digital circuits can allow the storage of a binary bit, a "1" or a "0". We discussed uses of MOSFETs relative to digital TV (Section 17–11) and computer memory storage (Section 21–8).

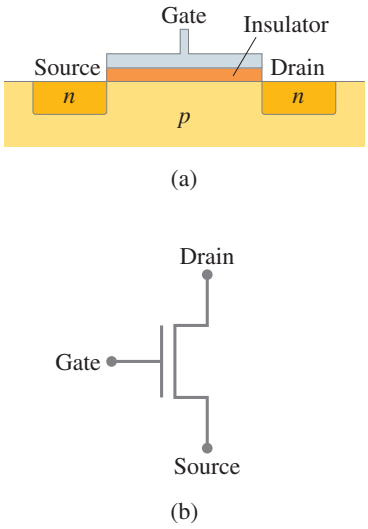


FIGURE 29–42 (a) Construction of a MOSFET of *n*- and *p*-type semiconductors and a gate of metal or heavily doped silicon (= a good conductor). (b) Symbol for a MOSFET which suggests its function.

*29–11 Integrated Circuits, 22-nm Technology

Although individual transistors are very small compared to the once-used vacuum tubes, they are huge compared to **integrated circuits** or **chips** (photo at start of this Chapter), invented in 1959 independently by Jack Kilby and Robert Noyce. Tiny amounts of impurities can be inserted or injected at particular locations within a single silicon crystal or wafer. These can be arranged to form diodes, transistors, resistors (undoped semiconductors), and very thin connecting "wires" (= conductors) which are heavily doped thin lines. Capacitors and inductors can also be formed, but also can be connected separately. Integrated circuits are the heart of computers, televisions, calculators, cameras, and the electronic instruments that control aircraft, space vehicles, and automobiles.

A tiny chip, a few millimeters on a side, may contain billions of transistors and other circuit elements. The number of elements/mm^2 has been doubling every 2 or 3 years. We often hear of the **technology generation**, which is a number that refers to the minimum width of a conducting line ("wire"). The gate of a MOSFET may be even smaller. Since 2003 we have passed from 90-nm technology to 65-nm, to 45-nm, to 32-nm, to 22-nm, every 2 to 3 years, and now 16-nm technology which—being only a few atoms wide—may involve new structures and quantum-mechanical effects. Smaller means more diodes and transistors per mm^2 and therefore greater speed (faster response time) because the distance signals have to travel is less. Smaller also means lower power consumption. Size, speed, and power have all been improved 10 to 100 million times in the last 40 years.

[†]The "MOS" comes from a version with a **M**etal gate, silicon di**O**xide insulator, and a **S**emiconductor (*p*-type shown in Fig. 29–42a). The gate can also be heavily doped silicon (= good conductor).

* Summary

Quantum mechanics explains the bonding together of atoms to form **molecules**. In a **covalent bond**, the atoms share electrons. The electron clouds of two or more atoms overlap because of constructive interference between the electron waves. The positive nuclei are attracted to this concentration of negative charge between them, forming the bond.

An **ionic bond** is an extreme case of a covalent bond in which one or more electrons from one atom spend much more time around the other atom than around their own. The atoms then act as oppositely charged ions that attract each other, forming the bond.

These **strong bonds** hold molecules together, and also hold atoms and molecules together in solids. Also important are **weak bonds** (or **van der Waals bonds**), which are generally dipole attractions between molecules.

When atoms combine to form molecules, the energy levels of the outer electrons are altered because they now interact with each other. Additional energy levels also become possible because the atoms can vibrate with respect to each other, and the molecule as a whole can rotate. The energy levels for both vibrational and rotational motion are quantized, and are very close together (typically, 10^{-1} eV to 10^{-3} eV apart). Each atomic energy level thus becomes a set of closely spaced levels corresponding to the vibrational and rotational motions. Transitions from one level to another appear as many very closely spaced lines. The resulting spectra are called **band spectra**.

The quantized rotational energy levels are given by

$$E_{\text{rot}} = \ell(\ell + 1)\frac{\hbar^2}{2I}, \qquad \ell = 0, 1, 2, \cdots, \qquad \textbf{(29–1)}$$

where I is the moment of inertia of the molecule.

The energy levels for vibrational motion are given by

$$E_{\text{vib}} = \left(v + \tfrac{1}{2}\right)hf, \qquad v = 0, 1, 2, \cdots, \qquad \textbf{(29–3)}$$

where f is the classical natural frequency of vibration for the molecule. Transitions between energy levels are subject to the selection rules $\Delta\ell = \pm 1$ and $\Delta v = \pm 1$.

Some **solids** are bound together by covalent and ionic bonds, just as molecules are. In metals, the electrostatic force between free electrons and positive ions helps form the **metallic bond**.

In the free-electron theory of metals, electrons occupy the possible energy states according to the exclusion principle. At $T = 0$ K, all possible states are filled up to a maximum energy level called the **Fermi energy**, E_F, the magnitude of which is typically a few eV. All states above E_F are vacant at $T = 0$ K.

In a crystalline solid, the possible energy states for electrons are arranged in **bands**. Within each band the levels are very close together, but between the bands there may be forbidden **energy gaps**. Good conductors are characterized by the highest occupied band (the **conduction band**) being only partially full, so lots of states are available to electrons to move about and accelerate when a voltage is applied. In a good insulator, the highest occupied energy band (the **valence band**) is completely full, and there is a large energy gap (5 to 10 eV) to the next highest band, the *conduction band*. At room temperature, molecular kinetic energy (thermal energy) available due to collisions is only about 0.04 eV, so almost no electrons can jump from the valence to the conduction band in an insulator. In a **semiconductor**, the gap between valence and conduction bands is much smaller, on the order of 1 eV, so a few electrons can make the transition from the essentially full valence band to the nearly empty conduction band, allowing a small amount of conductivity.

In a **doped** semiconductor, a small percentage of impurity atoms with five or three valence electrons replace a few of the normal silicon atoms with their four valence electrons. A five-electron impurity produces an ***n*-type** semiconductor with negative electrons as carriers of current. A three-electron impurity produces a ***p*-type** semiconductor in which positive **holes** carry the current. The energy level of impurity atoms lies slightly below the conduction band in an *n*-type semiconductor, and acts as a **donor** from which electrons readily pass into the conduction band. The energy level of impurity atoms in a *p*-type semiconductor lies slightly above the valence band and acts as an **acceptor** level, since electrons from the valence band easily reach it, leaving holes behind to act as charge carriers.

A semiconductor **diode** consists of a ***pn* junction** and allows current to flow in one direction only; *pn* junction diodes are used as **rectifiers** to change ac to dc, as photovoltaic cells to produce electricity from sunlight, and as lasers. **Light-emitting diodes** (**LED**) use compound semiconductors which can emit light when a forward-bias voltage is applied; uses include read-outs, infrared remote controls, visible lighting (flashlights, street lights), and very large TV screens. LEDs using organic molecules or polymers (**OLED**) are used as screens on cell phones and other displays. Common **transistors** consist of three semiconductor sections, either as ***pnp*** or ***npn***. Transistors can amplify electrical signals and in computers serve as switches or **gates** for the 1s and 0s of digital bits. An integrated circuit consists of a tiny semiconductor crystal or **chip** on which many transistors, diodes, resistors, and other circuit elements are constructed by placement of impurities.

Questions

1. What type of bond would you expect for (*a*) the N_2 molecule, (*b*) the HCl molecule, (*c*) Fe atoms in a solid?
2. Describe how the molecule $CaCl_2$ could be formed.
3. Does the H_2 molecule have a permanent dipole moment? Does O_2? Does H_2O? Explain.
4. Although the molecule H_3 is not stable, the ion H_3^+ is. Explain, using the Pauli exclusion principle.
5. Would you expect the molecule H_2^+ to be stable? If so, where would the single electron spend most of its time?
6. Explain why the carbon atom ($Z = 6$) usually forms four bonds with hydrogen-like atoms.
7. The energy of a molecule can be divided into four categories. What are they?
8. If conduction electrons are free to roam about in a metal, why don't they leave the metal entirely?
9. Explain why the resistivity of metals increases with increasing temperature whereas the resistivity of semiconductors may decrease with increasing temperature.
10. Compare the resistance of a *pn* junction diode connected in forward bias to its resistance when connected in reverse bias.
11. Explain how a transistor can be used as a switch.

12. Figure 29–43 shows a "bridge-type" full-wave rectifier. Explain how the current is rectified and how current flows during each half cycle.

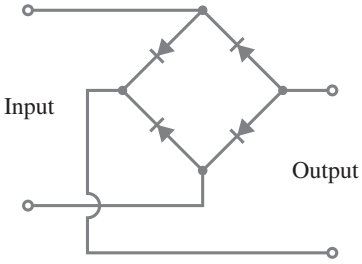


FIGURE 29–43
Question 12.

13. What is the main difference between *n*-type and *p*-type semiconductors?

14. Explain on the basis of energy bands why the sodium chloride crystal is a good insulator. [*Hint*: Consider the shells of Na^+ and Cl^- ions.]

15. In a transistor, the base–emitter junction and the base–collector junction are essentially diodes. Are these junctions reverse-biased or forward-biased in the application shown in Fig. 29–41?

16. A transistor can amplify an electronic signal, meaning it can increase the power of an input signal. Where does it get the energy to increase the power?

17. A silicon semiconductor is doped with phosphorus. Will these atoms be donors or acceptors? What type of semiconductor will this be?

18. Do diodes and transistors obey Ohm's law? Explain.

19. Can a diode be used to amplify a signal? Explain.

MisConceptual Questions

1. What holds molecules together?
(*a*) Gravitational forces.
(*b*) Magnetic forces.
(*c*) Electric forces.
(*d*) Glue.
(*e*) Nuclear forces.

2. Which of the following is true for covalently bound diatomic molecules such as H_2?
(*a*) All electrons in the two atoms have identical quantum numbers.
(*b*) The molecule has fewer electrons than the two separate atoms do.
(*c*) The molecule has less energy than two separate atoms.
(*d*) The energy of the molecule is greatest when the atoms are separated by one bond length.

3. A hydrogen atom ($Z = 1$) is bonded to a lithium atom ($Z = 3$) in lithium hydride, LiH. Which of the following are possible spin states of the two shared electrons?
(*a*) $+\frac{1}{2}, +\frac{1}{2}$.
(*b*) $-\frac{1}{2}, -\frac{1}{2}$.
(*c*) $+\frac{1}{2}, -\frac{1}{2}$.
(*d*) Both (*a*) and (*b*).
(*e*) Any of the above.

4. Ionic bonding is related to
(*a*) magnetic dipole interactions.
(*b*) the transfer of one or more electrons from one atom to another.
(*c*) the sharing of electrons between atoms.
(*d*) the transfer of electrons to the solid.
(*e*) oscillation dipoles.

5. Consider Fig. 29–10. As the last phosphate group approaches and then bonds to the ADP molecule, which of the following is true? Choose all that apply.
(*a*) The phosphate group is first repelled and then attracted to the ADP molecule.
(*b*) The phosphate group is always attracted to the ADP molecule.
(*c*) The phosphate group is always repelled by the ADP molecule.
(*d*) The system first loses and then stores potential energy.
(*e*) Both binding energy and activation energy are negative.
(*f*) Both binding energy and activation energy are positive.

6. Which type of bond holds the molecules of the DNA double helix together?
(*a*) Covalent bond.
(*b*) Ionic bond.
(*c*) Einstein bond.
(*d*) Van der Waals bond.

7. In a *p*-type semiconductor, a hole is
(*a*) a region in the molecular structure where an atom is missing.
(*b*) an extra electron from one of the donor atoms.
(*c*) an extra positively charged particle in the molecular structure.
(*d*) a region missing an electron relative to the rest of the molecular structure.

8. The electrical resistance of a semiconductor may decrease with increasing temperature because, at elevated temperature, more electrons
(*a*) collide with the crystal lattice.
(*b*) move faster.
(*c*) are able to jump across the energy gap.
(*d*) form weak van der Waals bonds.

9. Which of the following would *not* be used as an impurity in doping silicon?
(*a*) Germanium.
(*b*) Gallium.
(*c*) Boron.
(*d*) Phosphorus.
(*e*) Arsenic.

10. Why are metals good conductors?
(*a*) Gaining a tiny bit of energy allows their electrons to move.
(*b*) They have more electrons than protons, so some of the electrons are extra and free to move.
(*c*) They have more protons than electrons, so some of the protons are extra and free to move.
(*d*) Gaining a tiny bit of energy allows their protons to move.
(*e*) Electrons are tightly bound to their atoms.

For assigned homework and other learning materials, go to the MasteringPhysics website.

Problems

*29–1 to 29–3 Molecular Bonds

1. (I) Estimate the binding energy of a KCl molecule by calculating the electrostatic potential energy when the K^+ and Cl^- ions are at their stable separation of 0.28 nm. Assume each has a charge of magnitude $1.0e$.

2. (II) The measured binding energy of KCl is 4.43 eV. From the result of Problem 1, estimate the contribution to the binding energy of the repelling electron clouds at the equilibrium distance $r_0 = 0.28$ nm.

3. (II) The equilibrium distance r_0 between two atoms in a molecule is called the **bond length**. Using the bond lengths of homogeneous molecules (like H_2, O_2, and N_2), one can estimate the bond length of heterogeneous molecules (like CO, CN, and NO). This is done by summing half of each bond length of the homogenous molecules to estimate that of the heterogeneous molecule. Given the following bond lengths: H_2 (= 74 pm), N_2 (= 145 pm), O_2 (= 121 pm), C_2 (= 154 pm), estimate the bond lengths for: HN, CN, and NO.

4. (II) Binding energies are often measured experimentally in kcal per mole, and then the binding energy in eV per molecule is calculated from that result. What is the conversion factor in going from kcal per mole to eV per molecule? What is the binding energy of KCl (= 4.43 eV) in kcal per mole?

5. (III) Estimate the binding energy of the H_2 molecule, assuming the two H nuclei are 0.074 nm apart and the two electrons spend 33% of their time midway between them.

6. (III) (*a*) Apply reasoning similar to that in the text for the $S = 0$ and $S = 1$ states in the formation of the H_2 molecule to show why the molecule He_2 is *not* formed. (*b*) Explain why the He_2^+ molecular ion *could* form. (Experiment shows it has a binding energy of 3.1 eV at $r_0 = 0.11$ nm.)

*29–4 Molecular Spectra

7. (I) Show that the quantity $\hbar^2/I$ has units of energy.

8. (II) (*a*) Calculate the "characteristic rotational energy," $\hbar^2/2I$, for the O_2 molecule whose bond length is 0.121 nm. (*b*) What are the energy and wavelength of photons emitted in an $\ell = 3$ to $\ell = 2$ transition?

9. (II) The "characteristic rotational energy," $\hbar^2/2I$, for N_2 is 2.48×10^{-4} eV. Calculate the N_2 bond length.

10. (II) The equilibrium separation of H atoms in the H_2 molecule is 0.074 nm (Fig. 29–8). Calculate the energies and wavelengths of photons for the rotational transitions (*a*) $\ell = 1$ to $\ell = 0$, (*b*) $\ell = 2$ to $\ell = 1$, and (*c*) $\ell = 3$ to $\ell = 2$.

11. (II) Determine the wavelength of the photon emitted when the CO molecule makes the rotational transition $\ell = 5$ to $\ell = 4$. [*Hint*: See Example 29–2.]

12. (II) Calculate the bond length for the NaCl molecule given that three successive wavelengths for rotational transitions are 23.1 mm, 11.6 mm, and 7.71 mm.

13. (II) (*a*) Use the curve of Fig. 29–17 to estimate the stiffness constant k for the H_2 molecule. (Recall that $\text{PE} = \frac{1}{2}kx^2$.) (*b*) Then estimate the fundamental wavelength for vibrational transitions using the classical formula (Chapter 11), but use only $\frac{1}{2}$ the mass of an H atom (because both H atoms move).

*29–5 Bonding in Solids

14. (II) Common salt, NaCl, has a density of 2.165 g/cm^3. The molecular weight of NaCl is 58.44. Estimate the distance between nearest neighbor Na and Cl ions. [*Hint: Each* ion can be considered to be at the corner of a cube.]

15. (II) Repeat Problem 14 for KCl whose density is 1.99 g/cm^3.

16. (II) The spacing between "nearest neighbor" Na and Cl ions in a NaCl crystal is 0.24 nm. What is the spacing between two nearest neighbor Na ions?

*29–7 Band Theory of Solids

17. (I) A semiconductor is struck by light of slowly increasing frequency and begins to conduct when the wavelength of the light is 620 nm. Estimate the energy gap E_g.

18. (I) Calculate the longest-wavelength photon that can cause an electron in silicon $(E_g = 1.12\,\text{eV})$ to jump from the valence band to the conduction band.

19. (II) The energy gap between valence and conduction bands in germanium is 0.72 eV. What range of wavelengths can a photon have to excite an electron from the top of the valence band into the conduction band?

20. (II) The band gap of silicon is 1.12 eV. (*a*) For what range of wavelengths will silicon be transparent? (See Example 29–5.) In what region of the electromagnetic spectrum does this transparent range begin? (*b*) If window glass is transparent for all visible wavelengths, what is the minimum possible band gap value for glass (assume $\lambda = 400$ nm to 700 nm)? [*Hint*: If the photon has less energy than the band gap, the photon will pass through the solid without being absorbed.]

21. (II) The energy gap E_g in germanium is 0.72 eV. When used as a photon detector, roughly how many electrons can be made to jump from the valence to the conduction band by the passage of an 830-keV photon that loses all its energy in this fashion?

22. (III) We saw that there are $2N$ possible electron states in the 3*s* band of Na, where N is the total number of atoms. How many possible electron states are there in the (*a*) 2*s* band, (*b*) 2*p* band, and (*c*) 3*p* band? (*d*) State a general formula for the total number of possible states in any given electron band.

*29–8 Semiconductors and Doping

23. (III) Suppose that a silicon semiconductor is doped with phosphorus so that one silicon atom in 1.5×10^6 is replaced by a phosphorus atom. Assuming that the "extra" electron in every phosphorus atom is donated to the conduction band, by what factor is the density of conduction electrons increased? The density of silicon is 2330 kg/m^3, and the density of conduction electrons in pure silicon is about 10^{16} m^{-3} at room temperature.

*29–9 Diodes

24. (I) At what wavelength will an LED radiate if made from a material with an energy gap $E_g = 1.3$ eV?

25. (I) If an LED emits light of wavelength $\lambda = 730$ nm, what is the energy gap (in eV) between valence and conduction bands?

26. (I) A semiconductor diode laser emits 1.3-μm light. Assuming that the light comes from electrons and holes recombining, what is the band gap in this laser material?

27. (II) A silicon diode, whose current–voltage characteristics are given in Fig. 29–30, is connected in series with a battery and a 960-Ω resistor. What battery voltage is needed to produce a 14-mA current?

28. (II) An ac voltage of 120-V rms is to be rectified. Estimate very roughly the average current in the output resistor R ($= 31\,\text{k}\Omega$) for (*a*) a half-wave rectifier (Fig. 29–31), and (*b*) a full-wave rectifier (Fig. 29–32) without capacitor.

29. (III) Suppose that the diode of Fig. 29–30 is connected in series to a 180-Ω resistor and a 2.0-V battery. What current flows in the circuit? [*Hint*: Draw a line on Fig. 29–30 representing the current in the resistor as a function of the voltage across the diode; the intersection of this line with the characteristic curve will give the answer.]

30. (III) Sketch the resistance as a function of current, for $V > 0$, for the diode shown in Fig. 29–30.

*29–10 Transistors

31. (I) From Fig. 29–41, write an equation for the relationship between the base current (I_B), the collector current (I_C), and the emitter current (I_E, not labeled in Fig. 29–41). Assume $i_B = i_C = 0$.

32. (I) Draw a circuit diagram showing how a *pnp* transistor can operate as an amplifier, similar to Fig. 29–41 showing polarities, etc.

33. (II) If the current gain of the transistor amplifier in Fig. 29–41 is $\beta = i_C/i_B = 95$, what value must R_C have if a 1.0-μA ac base current is to produce an ac output voltage of 0.42 V?

34. (II) Suppose that the current gain of the transistor in Fig. 29–41 is $\beta = i_C/i_B = 85$. If $R_C = 3.8\,\text{k}\Omega$, calculate the ac output voltage for an ac input current of 2.0 μA.

35. (II) An amplifier has a voltage gain of 75 and a 25-kΩ load (output) resistance. What is the peak output current through the load resistor if the input voltage is an ac signal with a peak of 0.080 V?

36. (II) A transistor, whose current gain $\beta = i_C/i_B = 65$, is connected as in Fig. 29–41 with $R_B = 3.8\,\text{k}\Omega$ and $R_C = 7.8\,\text{k}\Omega$. Calculate (*a*) the voltage gain, and (*b*) the power amplification.

General Problems

37. Use the uncertainty principle to estimate the binding energy of the H_2 molecule by calculating the difference in kinetic energy of the electrons between (i) when they are in separate atoms and (ii) when they are in the molecule. Take Δx for the electrons in the separated atoms to be the radius of the first Bohr orbit, 0.053 nm, and for the molecule take Δx to be the separation of the nuclei, 0.074 nm. [*Hint*: Let $\Delta p \approx \Delta p_x$.]

38. The average translational kinetic energy of an atom or molecule is about $\text{KE} = \frac{3}{2}kT$ (see Section 13–9), where $k = 1.38 \times 10^{-23}\,\text{J/K}$ is Boltzmann's constant. At what temperature T will KE be on the order of the bond energy (and hence the bond easily broken by thermal motion) for (*a*) a covalent bond (say H_2) of binding energy 4.0 eV, and (*b*) a "weak" hydrogen bond of binding energy 0.12 eV?

39. A diatomic molecule is found to have an activation energy of 1.3 eV. When the molecule is disassociated, 1.6 eV of energy is released. Draw a potential energy curve for this molecule.

40. The rotational absorption spectrum of a molecule displays peaks about 8.9×10^{11} Hz apart. Determine the moment of inertia of this molecule.

41. For O_2 with a bond length of 0.121 nm, what is the moment of inertia about an axis through the center of mass perpendicular to the line joining the 2 atoms?

42. Must we consider quantum effects for everyday rotating objects? Estimate the differences between rotational energy levels for a spinning baton compared to the energy of the baton. Assume the baton consists of a uniform 32-cm-long bar with a mass of 230 g and two small end masses, each of mass 380 g, and it rotates at 1.8 rev/s about the bar's center.

43. When EM radiation is incident on diamond, it is found that light with wavelengths shorter than 226 nm will cause the diamond to conduct. What is the energy gap between the valence band and the conduction band for diamond?

44. The energy gap between valence and conduction bands in zinc sulfide is 3.6 eV. What range of wavelengths can a photon have to excite an electron from the top of the valence band into the conduction band?

45. Most of the Sun's radiation has wavelengths shorter than 1100 nm. For a solar cell to absorb all this, what energy gap ought the material have?

46. A TV remote control emits IR light. If the detector on the TV set is *not* to react to visible light, could it make use of silicon as a "window" with its energy gap $E_g = 1.12$ eV? What is the shortest-wavelength light that can strike silicon without causing electrons to jump from the valence band to the conduction band?

47. Green and blue LEDs became available many years after red LEDs were first developed. Approximately what energy gaps would you expect to find in green (525 nm) and in blue (465 nm) LEDs?

48. Consider a monatomic solid with a weakly bound cubic lattice, with each atom connected to six neighbors, each bond having a binding energy of 3.4×10^{-3} eV. When this solid melts, its latent heat of fusion goes directly into breaking the bonds between the atoms. Estimate the latent heat of fusion for this solid, in J/mol. [*Hint*: Show that in a simple cubic lattice (Fig. 29–44), there are *three* times as many bonds as there are atoms, when the number of atoms is large.]

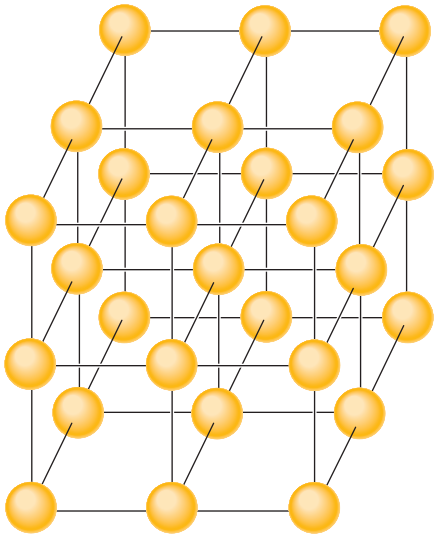


FIGURE 29–44
Problem 48.

Search and Learn

1. Explain why metals are shiny. (See Section 29–5.)

2. (*a*) Why are weak bonds important in cells? (*b*) Explain why heating proteins too much may cause them to denature—that is, lose the specific shape they need to function. (See Section 29–3.) (*c*) What is the strongest weak bond, and why? (*d*) If this bond, and the other weak bonds, were stronger (that is, too strong), what would be the consequence for protein synthesis?

3. Assume conduction electrons in a semiconductor behave as an ideal gas. (This is not true for conduction electrons in a metal.) (*a*) Taking mass $m = 9 \times 10^{-31}$ kg and temperature $T = 300$ K, determine the de Broglie wavelength of a semiconductor's conduction electrons. (*b*) Given that the spacing between atoms in a semiconductor's atomic lattice is on the order of 0.3 nm, would you expect room-temperature conduction electrons to travel in straight lines or diffract when traveling through this lattice? Explain.

4. A strip of silicon 1.6 cm wide and 1.0 mm thick is immersed in a magnetic field of strength 1.5 T perpendicular to the strip (Fig. 29–45). When a current of 0.28 mA is run through the strip, there is a resulting Hall effect voltage of 18 mV across the strip (Section 20–4). How many electrons per silicon atom are in the conduction band? The density of silicon is 2330 kg/m^3.

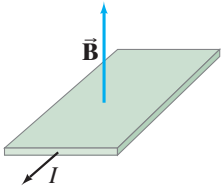
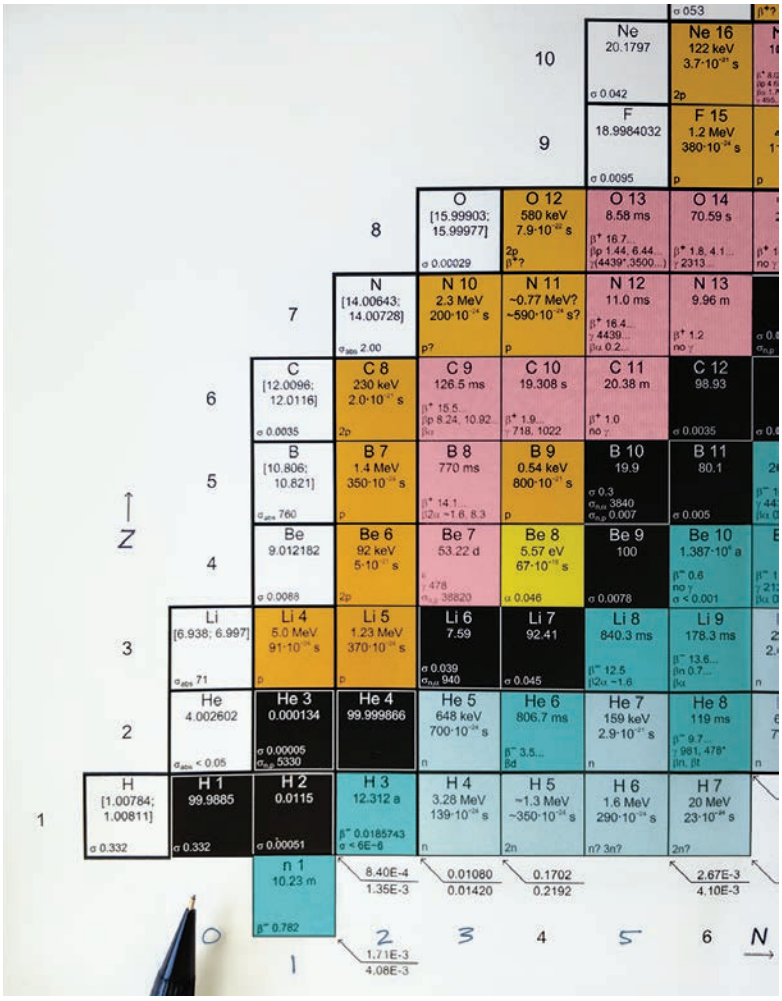


FIGURE 29–45
Search and Learn 4.

5. For an arsenic donor atom in a doped silicon semiconductor, assume that the "extra" electron moves in a Bohr orbit about the arsenic ion. For this electron in the ground state, take into account the dielectric constant $K = 12$ of the Si lattice (which represents the weakening of the Coulomb force due to all the other atoms or ions in the lattice), and estimate (*a*) the binding energy, and (*b*) the orbit radius for this extra electron. [*Hint*: Substitute $\epsilon = K\epsilon_0$ in Coulomb's law; see Section 17–8 and also 27–12.]

ANSWERS TO EXERCISES

A: 0; 5.00×10^{-4} eV; 1.50×10^{-3} eV.
B: 1.30 mm, 0.87 mm, 0.65 mm.
C: 0.81 eV.
D: (*c*).
E: (*c*), (*e*).



In this Chapter we begin our discussion of nuclear physics. We study the properties of nuclei, the various forms of radioactivity, and how radioactive decay can be used in a variety of fields to determine the age of old objects, from bones and trees to rocks and other mineral substances, and obtain information on the history of the Earth.

Shown is one version of a **Chart of the Nuclides**. Each horizontal row has a square for each known isotope (nuclide) of one element with a particular Z value (= number of electrons in the neutral atom = number of protons in the nucleus). At the far left is a white box with the average atomic weight (or a range if uncertain) of the naturally occurring isotopes of that element. Each vertical column contains nuclides with the same neutron number N. For $N = 1$ (to right of pencil), starting at the bottom, there is a lone neutron, then above it ${}^{2}_{1}\text{H}$, then ${}^{3}_{2}\text{He}$ and ${}^{4}_{3}\text{Li}$. Each square is color coded: black means a stable nuclide. Radioactive nuclides are blue green for β^- decay, pink for β^+ decay or electron capture (ε) such as ${}^{7}_{4}\text{Be}$, yellow for α decay, and so on. Thus ${}^{1}_{1}\text{H}$ and ${}^{2}_{1}\text{H}$ are stable but ${}^{3}_{1}\text{H}$ (tritium) undergoes β^- decay with half-life = 12.3 years ("a" is for Latin "anno" = year). The squares contain the atomic mass of that isotope, or half-life and energy released if radioactive. Other details may be alternate decay modes and certain cross sections (σ).

CHAPTER 30

Nuclear Physics and Radioactivity

CHAPTER-OPENING QUESTION—Guess now!

If half of an 80-μg sample of ${}^{60}_{27}\text{Co}$ decays in 5.3 years, how much ${}^{60}_{27}\text{Co}$ is left in 10.6 years?

(a) 10 μg.
(b) 20 μg.
(c) 30 μg.
(d) 40 μg.
(e) 0 μg.

In the early part of the twentieth century, Rutherford's experiments (Section 27–10) led to the idea that at the center of an atom there is a tiny but massive nucleus with a positive charge. At the same time that the quantum theory was being developed and scientists were attempting to understand the structure of the atom and its electrons, investigations into the nucleus itself had also begun. In this Chapter and the next, we take a brief look at *nuclear physics*.

CONTENTS

30–1 Structure and Properties of the Nucleus
30–2 Binding Energy and Nuclear Forces
30–3 Radioactivity
30–4 Alpha Decay
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30–7 Conservation of Nucleon Number and Other Conservation Laws
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30–9 Calculations Involving Decay Rates and Half-Life
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30–13 Detection of Particles

30–1 Structure and Properties of the Nucleus

An important question for physicists was whether the nucleus had a structure, and what that structure might be. By the early 1930s, a model of the nucleus had been developed that is still useful. According to this model, a nucleus is made up of two types of particles: protons and neutrons. [These "particles" also have wave properties, but for ease of visualization and language, we usually refer to them simply as "particles."] A **proton** is the nucleus of the simplest atom, hydrogen. The proton has a positive charge ($= +e = +1.60 \times 10^{-19}$ C, the same magnitude as for the electron) and its mass is measured to be

$$m_p = 1.67262 \times 10^{-27}\,\text{kg}.$$

The **neutron**, whose existence was ascertained in 1932 by the English physicist James Chadwick (1891–1974), is electrically neutral ($q = 0$), as its name implies. Its mass is very slightly larger than that of the proton:

$$m_n = 1.67493 \times 10^{-27}\,\text{kg}.$$

These two constituents of a nucleus, neutrons and protons, are referred to collectively as **nucleons**.

Although a normal hydrogen nucleus consists of a single proton alone, the nuclei of all other elements consist of both neutrons and protons. The different nuclei are often referred to as **nuclides**. The number of protons in a nucleus (or nuclide) is called the **atomic number** and is designated by the symbol Z. The total number of nucleons, neutrons plus protons, is designated by the symbol A and is called the **atomic mass number**, or sometimes simply **mass number**. This name is used since the mass of a nucleus is very closely A times the mass of one nucleon. A nuclide with 7 protons and 8 neutrons thus has $Z = 7$ and $A = 15$. The **neutron number** N is $N = A - Z$.

To specify a given nuclide, we need give only A and Z. A special symbol is commonly used which takes the form

$${}^{A}_{Z}\text{X},$$

where X is the chemical symbol for the element (see Appendix B, and the Periodic Table inside the back cover), A is the atomic mass number, and Z is the atomic number. For example, ${}^{15}_{7}\text{N}$ means a nitrogen nucleus containing 7 protons and 8 neutrons for a total of 15 nucleons. In a neutral atom, the number of electrons orbiting the nucleus is equal to the atomic number Z (since the charge on an electron has the same magnitude but opposite sign to that of a proton). The main properties of an atom, and how it interacts with other atoms, are largely determined by the number of electrons. Hence Z determines what kind of atom it is: carbon, oxygen, gold, or whatever. It is redundant to specify both the symbol of a nucleus and its atomic number Z as described above. If the nucleus is nitrogen, for example, we know immediately that $Z = 7$. The subscript Z is thus sometimes dropped and ${}^{15}_{7}\text{N}$ is then written simply ${}^{15}\text{N}$; in words we say "nitrogen fifteen."

For a particular type of atom (say, carbon), nuclei are found to contain different numbers of neutrons, although they all have the same number of protons. For example, carbon nuclei always have 6 protons, but they may have 5, 6, 7, 8, 9, or 10 neutrons. Nuclei that contain the same number of protons but different numbers of neutrons are called **isotopes**. Thus, ${}^{11}_{6}\text{C}$, ${}^{12}_{6}\text{C}$, ${}^{13}_{6}\text{C}$, ${}^{14}_{6}\text{C}$, ${}^{15}_{6}\text{C}$, and ${}^{16}_{6}\text{C}$ are all isotopes of carbon. The isotopes of a given element are not all equally common. For example, 98.9% of naturally occurring carbon (on Earth) is the isotope ${}^{12}_{6}\text{C}$, and about 1.1% is ${}^{13}_{6}\text{C}$. These percentages are referred to as the **natural abundances**.[†] Even hydrogen has isotopes: 99.99% of natural hydrogen is ${}^{1}_{1}\text{H}$, a simple proton, as the nucleus; there are also ${}^{2}_{1}\text{H}$, called **deuterium**, and ${}^{3}_{1}\text{H}$, **tritium**, which besides the proton contain 1 or 2 neutrons. (The bare nucleus in each case is called the **deuteron** and **triton**.)

[†]The mass value for each element as given in the Periodic Table (inside back cover) is an average weighted according to the natural abundances of its isotopes.

Many isotopes that do not occur naturally can be produced in the laboratory by means of nuclear reactions (more on this later). Indeed, all elements beyond uranium ($Z > 92$) do not occur naturally on Earth and are only produced artificially (in the laboratory), as are many nuclides with $Z \leq 92$.

The approximate size of nuclei was determined originally by Rutherford from the scattering of charged particles by thin metal foils. We cannot speak about a definite size for nuclei because of the wave–particle duality (Section 27–7): their spatial extent must remain somewhat fuzzy. Nonetheless a rough "size" can be measured by scattering high-speed electrons off nuclei. It is found that nuclei have a roughly spherical shape with a radius that increases with A according to the approximate formula

$$r \approx (1.2 \times 10^{-15}\,\text{m})(A^{\frac{1}{3}}). \qquad \textbf{(30–1)}$$

Since the volume of a sphere is $V = \frac{4}{3}\pi r^3$, we see that the volume of a nucleus is approximately proportional to the number of nucleons, $V \propto A$ (because $(A^{\frac{1}{3}})^3 = A$). This is what we would expect if nucleons were like impenetrable billiard balls: if you double the number of balls, you double the total volume. Hence, all nuclei have nearly the same density, and it is enormous (see Example 30–2).

The metric abbreviation for 10^{-15} m is the fermi (after Enrico Fermi, Fig. 30–7) or the femtometer, fm (see Table 1–4 or inside the front cover). Thus 1.2×10^{-15} m $= 1.2$ fm or 1.2 fermis.

EXAMPLE 30–1 | ESTIMATE **Nuclear sizes.** Estimate the diameter of the smallest and largest naturally occurring nuclei: (*a*) ${}^{1}_{1}\text{H}$, (*b*) ${}^{238}_{92}\text{U}$.

APPROACH The radius r of a nucleus is related to its number of nucleons A by Eq. 30–1. The diameter $d = 2r$.

SOLUTION (*a*) For hydrogen, $A = 1$, Eq. 30–1 gives

$$d = \text{diameter} = 2r \approx 2(1.2 \times 10^{-15}\,\text{m})(A^{\frac{1}{3}}) = 2.4 \times 10^{-15}\,\text{m}$$

since $A^{\frac{1}{3}} = 1^{\frac{1}{3}} = 1$.

(*b*) For uranium $d \approx (2.4 \times 10^{-15}\,\text{m})(238)^{\frac{1}{3}} = 15 \times 10^{-15}\,\text{m}$.

The range of nuclear diameters is only from 2.4 fm to 15 fm.

NOTE Because nuclear radii vary as $A^{\frac{1}{3}}$, the largest nuclei (such as uranium with $A = 238$) have a radius only about $\sqrt[3]{238} \approx 6$ times that of the smallest, hydrogen ($A = 1$).

EXAMPLE 30–2 | ESTIMATE **Nuclear and atomic densities.** Compare the density of nuclear matter to the density of normal solids.

APPROACH The density of normal liquids and solids is on the order of 10^3 to $10^4\,\text{kg/m}^3$ (see Table 10–1), and because the atoms are close packed, atoms have about this density too. We therefore compare the density (mass per volume) of a nucleus to that of its atom as a whole.

SOLUTION The mass of a proton is greater than the mass of an electron by a factor

$$\frac{1.67 \times 10^{-27}\,\text{kg}}{9.1 \times 10^{-31}\,\text{kg}} \approx 2000.$$

Thus, over 99.9% of the mass of an atom is in the nucleus, and for our estimate we can say the mass of the atom equals the mass of the nucleus, $m_{\text{nucl}}/m_{\text{atom}} = 1$. Atoms have a radius of about 10^{-10} m (Chapter 27) and nuclei on the order of 10^{-15} m (Eq. 30–1). Thus the ratio of nuclear density to atomic density is about

$$\frac{\rho_{\text{nucl}}}{\rho_{\text{atom}}} = \frac{(m_{\text{nucl}}/V_{\text{nucl}})}{(m_{\text{atom}}/V_{\text{atom}})} = \left(\frac{m_{\text{nucl}}}{m_{\text{atom}}}\right)\frac{\frac{4}{3}\pi r_{\text{atom}}^3}{\frac{4}{3}\pi r_{\text{nucl}}^3} \approx (1)\frac{(10^{-10})^3}{(10^{-15})^3} = 10^{15}.$$

The nucleus is 10^{15} times more dense than ordinary matter.

The masses of nuclei can be determined from the radius of curvature of fast-moving nuclei (as ions) in a known magnetic field using a mass spectrometer, as discussed in Section 20–11. Indeed the existence of different isotopes of the same element (different number of neutrons) was discovered using this device.

CAUTION
Masses are for neutral atom (nucleus plus electrons)

Nuclear masses can be specified in **unified atomic mass units** (u). On this scale, a neutral $^{12}_{6}C$ atom is given the exact value 12.000000 u. A neutron then has a measured mass of 1.008665 u, a proton 1.007276 u, and a neutral hydrogen atom $^{1}_{1}H$ (proton plus electron) 1.007825 u. The masses of many nuclides are given in Appendix B. It should be noted that the masses in this Table, as is customary, are for the *neutral atom* (including electrons), and not for a bare nucleus.

Masses may be specified using the electron-volt energy unit, $1\text{ eV} = 1.6022 \times 10^{-19}\text{ J}$ (Section 17–4). This can be done because mass and energy are related, and the precise relationship is given by Einstein's equation $E = mc^2$ (Chapter 26). Since the mass of a proton is $1.67262 \times 10^{-27}\text{ kg}$, or 1.007276 u, then 1 u is equal to

$$1.0000\text{ u} = (1.0000\text{ u})\left(\frac{1.67262 \times 10^{-27}\text{ kg}}{1.007276\text{ u}}\right) = 1.66054 \times 10^{-27}\text{ kg};$$

this is equivalent to an energy (see Table inside front cover) in MeV ($= 10^6\text{ eV}$) of

$$E = mc^2 = \frac{(1.66054 \times 10^{-27}\text{ kg})(2.9979 \times 10^8\text{ m/s})^2}{(1.6022 \times 10^{-19}\text{ J/eV})} = 931.5\text{ MeV}.$$

Thus,

$$1\text{ u} = 1.6605 \times 10^{-27}\text{ kg} = 931.5\text{ MeV}/c^2.$$

The rest masses of some of the basic particles are given in Table 30–1. As a rule of thumb, to remember, the masses of neutron and proton are about $1\text{ GeV}/c^2$ ($= 1000\text{ MeV}/c^2$) which is about 2000 times the mass of an electron ($\approx \frac{1}{2}\text{MeV}/c^2$).

TABLE 30–1
Rest Masses in Kilograms, Unified Atomic Mass Units, and MeV/c^2

	Mass		
Object	**kg**	**u**	**MeV/c^2**
Electron	9.1094×10^{-31}	0.00054858	0.51100
Proton	1.67262×10^{-27}	1.007276	938.27
$^{1}_{1}H$ atom	1.67353×10^{-27}	1.007825	938.78
Neutron	1.67493×10^{-27}	1.008665	939.57

Just as an electron has intrinsic spin and angular momentum quantum numbers, so too do nuclei and their constituents, the proton and neutron. Both the proton and the neutron are spin $\frac{1}{2}$ particles, just like the electron. A nucleus, made up of protons and neutrons, has a **nuclear spin** quantum number, I, that can be either integer or half integer, depending on whether it is made up of an even or an odd number of nucleons.

30–2 Binding Energy and Nuclear Forces

Binding Energies

The total mass of a stable nucleus is always less than the sum of the masses of its separate protons and neutrons, as the following Example shows.

EXAMPLE 30–3 **$^{4}_{2}$He mass compared to its constituents.** Compare the mass of a $^{4}_{2}$He atom to the total mass of its constituent particles.

PROBLEM SOLVING
Keep track of electron masses

APPROACH The $^{4}_{2}$He nucleus contains 2 protons and 2 neutrons. Tables normally give the masses of neutral atoms—that is, nucleus plus its Z electrons. We must therefore be sure to balance out the electrons when we compare masses. Thus we use the mass of $^{1}_{1}H$ rather than that of a proton alone. We look up the mass of the $^{4}_{2}$He atom in Appendix B (it includes the mass of 2 electrons), as well as the mass for the 2 neutrons and 2 hydrogen atoms ($= 2$ protons $+ 2$ electrons).

SOLUTION The mass of a neutral $^4_2$He atom, from Appendix B, is 4.002603 u. The mass of two neutrons and two H atoms (2 protons including the 2 electrons) is

$$\begin{aligned} 2m_n &= 2(1.008665\text{ u}) = 2.017330\text{ u} \\ 2m(^1_1\text{H}) &= 2(1.007825\text{ u}) = \underline{2.015650\text{ u}} \\ \text{sum} &= 4.032980\text{ u}. \end{aligned}$$

Thus the mass of $^4_2$He is measured to be less than the masses of its constituents by an amount $4.032980\text{ u} - 4.002603\text{ u} = 0.030377\text{ u}$.

Where has this lost mass of 0.030377 u disappeared to? It must be $E = mc^2$.

If the four nucleons suddenly came together to form a $^4_2$He nucleus, the mass "loss" would appear as energy of another kind (such as radiation, or kinetic energy). The mass (or energy) difference in the case of $^4_2$He, given in energy units, is $(0.030377\text{ u})(931.5\text{ MeV/u}) = 28.30\text{ MeV}$. This difference is referred to as the **total binding energy** of the nucleus. The total binding energy represents the amount of energy that must be put *into* a nucleus in order to break it apart into its constituents. If the mass of, say, a $^4_2$He nucleus were exactly equal to the mass of two neutrons plus two protons, the nucleus could fall apart without any input of energy. To be stable, the mass of a nucleus *must* be less than that of its constituent nucleons, so that energy input *is* needed to break it apart.

CAUTION
Mass of nucleus must be less *than mass of constituents*

Binding energy is not something a nucleus has—it is energy it "lacks" relative to the total mass of its separate constituents.

[As a comparison, we saw in Chapter 27 that the binding energy of the one electron in the hydrogen atom is 13.6 eV; so the mass of a $^1_1$H atom is less than that of a single proton plus a single electron by $13.6\text{ eV}/c^2$. The binding energies of nuclei are on the order of MeV, so the eV binding energies of electrons can be ignored. Nuclear binding energies, compared to nuclear masses, are on the order of $(28\text{ MeV}/4000\text{ MeV}) \approx 1 \times 10^{-2}$, where we used helium's binding energy of 28.3 MeV (see above) and mass $\approx 4 \times 940\text{ MeV} \approx 4000\text{ MeV}$.]

EXERCISE A Determine how much less the mass of the $^7_3$Li nucleus is compared to that of its constituents. See Appendix B.

The **binding energy per nucleon** is defined as the total binding energy of a nucleus divided by A, the total number of nucleons. We calculated above that the binding energy of $^4_2$He is 28.3 MeV, so its binding energy per nucleon is $28.3\text{ MeV}/4 = 7.1\text{ MeV}$. Figure 30–1 shows the measured binding energy per nucleon as a function of A for stable nuclei. The curve rises as A increases and reaches a plateau at about 8.7 MeV per nucleon above $A \approx 40$. Beyond $A \approx 80$, the curve decreases slowly, indicating that larger nuclei are held together less tightly than those in the middle of the Periodic Table. We will see later that these characteristics allow the release of nuclear energy in the processes of fission and fusion.

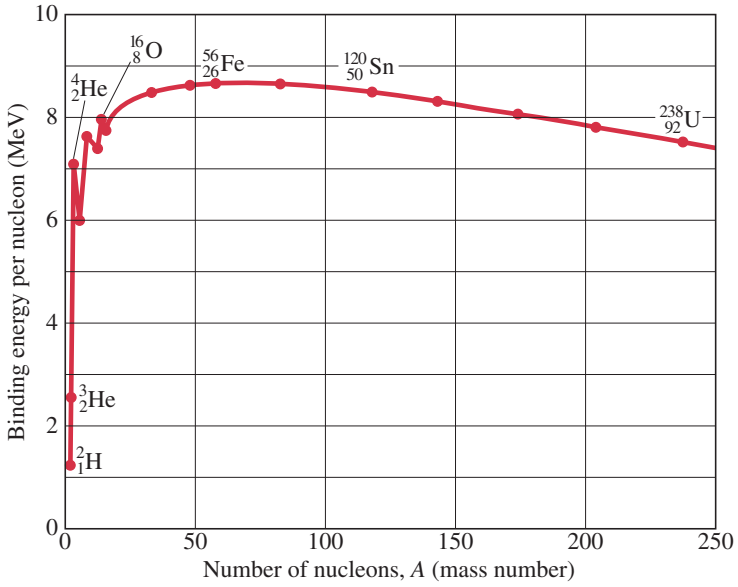


FIGURE 30–1 Binding energy per nucleon for the more stable nuclides as a function of mass number A.

EXAMPLE 30–4 **Binding energy for iron.** Calculate the total binding energy and the binding energy per nucleon for $^{56}_{26}\text{Fe}$, the most common stable isotope of iron.

APPROACH We subtract the mass of a $^{56}_{26}\text{Fe}$ atom from the total mass of 26 hydrogen atoms and 30 neutrons, all found in Appendix B. Then we convert mass units to energy units; finally we divide by $A = 56$, the total number of nucleons.

SOLUTION $^{56}_{26}\text{Fe}$ has 26 protons and 30 neutrons whose separate masses are

$$\begin{aligned} 26m\left(^1_1\text{H}\right) &= (26)(1.007825\ \text{u}) = 26.20345\ \text{u} \quad \text{(includes 26 electrons)} \\ 30m_\text{n} &= (30)(1.008665\ \text{u}) = 30.25995\ \text{u} \\ \text{sum} &= 56.46340\ \text{u}. \\ \text{Subtract mass of } ^{56}_{26}\text{Fe}: &= -55.93494\ \text{u} \quad \text{(Appendix B)} \\ \Delta m &= 0.52846\ \text{u}. \end{aligned}$$

The total binding energy is thus

$$(0.52846\ \text{u})(931.5\ \text{MeV/u}) = 492.26\ \text{MeV}$$

and the binding energy per nucleon is

$$\frac{492.26\ \text{MeV}}{56\ \text{nucleons}} = 8.79\ \text{MeV}.$$

NOTE The binding energy per nucleon graph (Fig. 30–1) peaks about here, for iron. So the iron nucleus, and its neighbors, are the most stable of nuclei.

EXERCISE B Determine the binding energy per nucleon for $^{16}_{8}\text{O}$.

EXAMPLE 30–5 **Binding energy of last neutron.** What is the binding energy of the last neutron in $^{13}_{6}\text{C}$?

APPROACH If $^{13}_{6}\text{C}$ lost one neutron, it would be $^{12}_{6}\text{C}$. We subtract the mass of $^{13}_{6}\text{C}$ from the masses of $^{12}_{6}\text{C}$ and a free neutron.

SOLUTION Obtaining the masses from Appendix B, we have

$$\begin{aligned} \text{Mass } ^{12}_{6}\text{C} &= 12.000000\ \text{u} \\ \text{Mass } ^{1}_{0}\text{n} &= 1.008665\ \text{u} \\ \text{Total} &= 13.008665\ \text{u}. \\ \text{Subtract mass of } ^{13}_{6}\text{C}: &\quad -13.003355\ \text{u} \\ \Delta m &= 0.005310\ \text{u} \end{aligned}$$

which in energy is $(931.5\ \text{MeV/u})(0.005310\ \text{u}) = 4.95\ \text{MeV}$. That is, it would require 4.95 MeV input of energy to remove one neutron from $^{13}_{6}\text{C}$.

Nuclear Forces

We can analyze nuclei not only from the point of view of energy, but also from the point of view of the forces that hold them together. We might not expect a collection of protons and neutrons to come together spontaneously, since protons are all positively charged and thus exert repulsive electric forces on each other. Since stable nuclei *do* stay together, another force must be acting. This new force has to be stronger than the electric force in order to hold the nucleus together, and is called the **strong nuclear force**. The strong nuclear force acts as an attractive force between all nucleons, protons and neutrons alike. Thus protons attract each other via the strong nuclear force at the same time they repel each other via the electric force. Neutrons, because they are electrically neutral, only attract other neutrons or protons via the strong nuclear force.

The strong nuclear force turns out to be far more complicated than the gravitational and electromagnetic forces. One important aspect of the strong nuclear force is that it is a **short-range** force: it acts only over a very short distance.

It is very strong between two nucleons if they are less than about 10^{-15} m apart, but it is essentially zero if they are separated by a distance greater than this. Compare this to electric and gravitational forces, which decrease as $1/r^2$ but continue acting over any distances and are therefore called **long-range** forces.

The strong nuclear force has some strange features. For example, if a nuclide contains too many or too few neutrons relative to the number of protons, the binding of the nucleons is reduced; nuclides that are too unbalanced in this regard are unstable. As shown in Fig. 30–2, stable nuclei tend to have the same number of protons as neutrons $(N = Z)$ up to about $A = 30$. Beyond this, stable nuclei contain more neutrons than protons. This makes sense since, as Z increases, the electrical repulsion increases, so a greater number of neutrons—which exert only the attractive strong nuclear force—are required to maintain stability. For very large Z, no number of neutrons can overcome the greatly increased electric repulsion. Indeed, there are no completely stable nuclides above $Z = 82$.

What we mean by a *stable nucleus* is one that stays together indefinitely. What then is an *unstable nucleus*? It is one that comes apart; and this results in radioactive decay. Before we discuss the important subject of radioactivity (next Section), we note that there is a second type of nuclear force that is much weaker than the strong nuclear force. It is called the **weak nuclear force**, and we are aware of its existence only because it shows itself in certain types of radioactive decay. These two nuclear forces, the strong and the weak, together with the gravitational and electromagnetic forces, comprise the four fundamental types of force in nature.

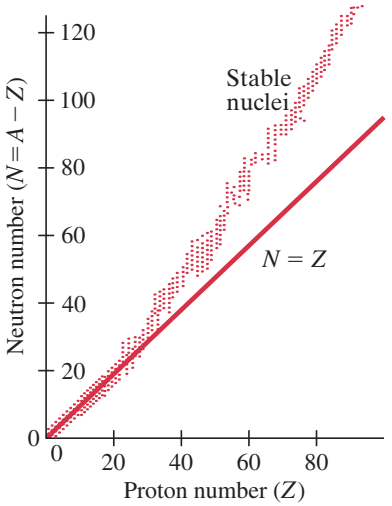


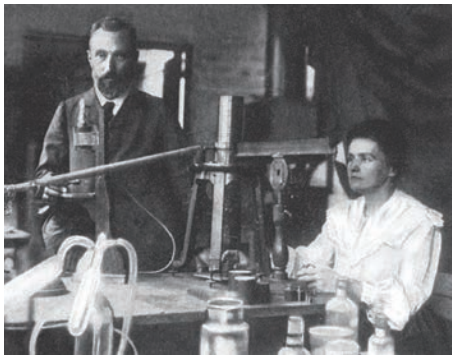
FIGURE 30–2 Number of neutrons versus number of protons for stable nuclides, which are represented by dots. The straight line represents $N = Z$.

30–3 Radioactivity

Nuclear physics had its beginnings in 1896. In that year, Henri Becquerel (1852–1908) made an important discovery: in his studies of phosphorescence, he found that a certain mineral (which happened to contain uranium) would darken a photographic plate even when the plate was wrapped to exclude light. It was clear that the mineral emitted some new kind of radiation that, unlike X-rays (Section 25–11), occurred without any external stimulus. This new phenomenon eventually came to be called **radioactivity**.

Soon after Becquerel's discovery, Marie Curie (1867–1934) and her husband, Pierre Curie (1859–1906), isolated two previously unknown elements that were very highly radioactive (Fig. 30–3). These were named polonium and radium. Other radioactive elements were soon discovered as well. The radioactivity was found in every case to be unaffected by the strongest physical and chemical treatments, including strong heating or cooling or the action of strong chemicals. It was suspected that the source of radioactivity must be deep within the atom, coming from the nucleus. It became apparent that radioactivity is the result of the **disintegration** or **decay** of an unstable nucleus. Certain isotopes are not stable, and they decay with the emission of some type of radiation or "rays."

FIGURE 30–3 Marie and Pierre Curie in their laboratory (about 1906) where radium was discovered.



Many unstable isotopes occur in nature, and such radioactivity is called "natural radioactivity." Other unstable isotopes can be produced in the laboratory by nuclear reactions (Section 31–1); these are said to be produced "artificially" and to have "artificial radioactivity." Radioactive isotopes are sometimes referred to as **radioisotopes** or **radionuclides**.

Rutherford and others began studying the nature of the rays emitted in radioactivity about 1898. They classified the rays into three distinct types according to their penetrating power. One type of radiation could barely penetrate a piece of paper. The second type could pass through as much as 3 mm of aluminum. The third was extremely penetrating: it could pass through several centimeters of lead and still be detected on the other side. They named these three types of radiation alpha (α), beta (β), and gamma (γ), respectively, after the first three letters of the Greek alphabet.

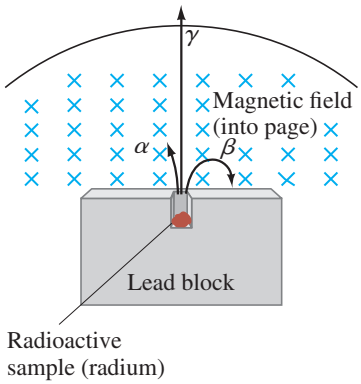


FIGURE 30–4 Alpha and beta rays are bent in opposite directions by a magnetic field, whereas gamma rays are not bent at all.

Each type of ray was found to have a different charge and hence is bent differently in a magnetic field, Fig. 30–4; α rays are positively charged, β rays are negatively charged, and γ rays are neutral. It was soon found that all three types of radiation consisted of familiar kinds of particles. Gamma rays are very high-energy *photons* whose energy is even higher than that of X-rays. Beta rays were found to be identical to *electrons* that orbit the nucleus, but they are created within the nucleus itself. Alpha rays (or α particles) are simply the nuclei of *helium* atoms, ${}^4_2\text{He}$; that is, an α ray consists of two protons and two neutrons bound together.

We now discuss each of these three types of radioactivity, or decay, in more detail.

30–4 Alpha Decay

Experiments show that when nuclei decay, the number of nucleons (= mass number A) is conserved, as well as electric charge (= Ze). When a nucleus emits an α particle (${}^4_2\text{He}$), the remaining nucleus will be different from the original: it has lost two protons and two neutrons. Radium 226 (${}^{226}_{88}\text{Ra}$), for example, is an α emitter. It decays to a nucleus with $Z = 88 - 2 = 86$ and $A = 226 - 4 = 222$. The nucleus with $Z = 86$ is radon (Rn)—see Appendix B or the Periodic Table. Thus radium decays to radon with the emission of an α particle. This is written

$$ {}^{226}_{88}\text{Ra} \rightarrow {}^{222}_{86}\text{Rn} + {}^4_2\text{He}. $$

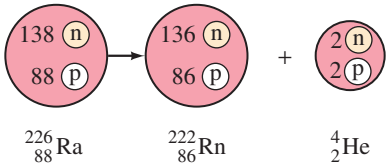


FIGURE 30–5 Radioactive decay of radium to radon with emission of an alpha particle.

See Fig. 30–5.

When α decay occurs, a different element is formed. The **daughter** nucleus (${}^{222}_{86}\text{Rn}$ in this case) is different from the **parent** nucleus (${}^{226}_{88}\text{Ra}$ in this case). This changing of one element into another is called **transmutation** of the elements.

Alpha decay can be written in general as

$$ {}^A_Z\text{N} \rightarrow {}^{A-4}_{Z-2}\text{N}' + {}^4_2\text{He} \qquad [\alpha \text{ decay}] $$

where N is the parent, N' the daughter, and Z and A are the atomic number and atomic mass number, respectively, of the parent.

EXERCISE C ${}^{154}_{66}\text{Dy}$ decays by α emission to what element? (*a*) Pb, (*b*) Gd, (*c*) Sm, (*d*) Er, (*e*) Yb.

Alpha decay occurs because the strong nuclear force is unable to hold very large nuclei together. The nuclear force is a short-range force: it acts only between neighboring nucleons. But the electric force acts all the way across a large nucleus. For very large nuclei, the large Z means the repulsive electric force becomes so large (Coulomb's law) that the strong nuclear force is unable to hold the nucleus together.

We can express the instability of the parent nucleus in terms of energy (or mass): the mass of the parent nucleus is greater than the mass of the daughter nucleus plus the mass of the α particle. The mass difference appears as kinetic energy, which is carried away by the α particle and the recoiling daughter nucleus. The total energy released is called the **disintegration energy**, Q, or the ***Q*-value** of the decay. From conservation of energy,

$$M_P c^2 = M_D c^2 + m_\alpha c^2 + Q,$$

where Q equals the kinetic energy of the daughter and α particle, and M_P, M_D, and m_α are the masses of the parent, daughter, and α particle, respectively. Thus

$$Q = M_P c^2 - (M_D + m_\alpha)c^2. \qquad \textbf{(30–2)}$$

If the parent had *less* mass than the daughter plus the α particle (so $Q < 0$), the decay would violate conservation of energy. Such decays have never been observed, another confirmation of this great conservation law.

EXAMPLE 30–6 **Uranium decay energy release.** Calculate the disintegration energy when $^{232}_{92}$U (mass = 232.037156 u) decays to $^{228}_{90}$Th (228.028741 u) with the emission of an α particle. (As always, masses given are for neutral atoms.)

APPROACH We use conservation of energy as expressed in Eq. 30–2. $^{232}_{92}$U is the parent, $^{228}_{90}$Th is the daughter.

SOLUTION Since the mass of the $^4_2$He is 4.002603 u (Appendix B), the total mass in the final state $(m_{Th} + m_{He})$ is

$$228.028741\text{ u} + 4.002603\text{ u} = 232.031344\text{ u}.$$

The mass lost when the $^{232}_{92}$U decays $(m_U - m_{Th} - m_{He})$ is

$$232.037156\text{ u} - 232.031344\text{ u} = 0.005812\text{ u}.$$

Because 1 u = 931.5 MeV, the energy Q released is

$$Q = (0.005812\text{ u})(931.5\text{ MeV/u}) = 5.4\text{ MeV}$$

and this energy appears as kinetic energy of the α particle and the daughter nucleus.

Additional Example

EXAMPLE 30–7 **Kinetic energy of the α in $^{232}_{92}$U decay.** For the $^{232}_{92}$U decay of Example 30–6, how much of the 5.4-MeV disintegration energy will be carried off by the α particle?

APPROACH In any reaction, momentum must be conserved as well as energy.

SOLUTION Before disintegration, the nucleus can be assumed to be at rest, so the total momentum was zero. After disintegration, the total vector momentum must still be zero so the magnitude of the α particle's momentum must equal the magnitude of the daughter's momentum (Fig. 30–6):

$$m_\alpha v_\alpha = m_D v_D.$$

Thus $v_\alpha = m_D v_D / m_\alpha$ and the α's kinetic energy is

$$\text{KE}_\alpha = \tfrac{1}{2} m_\alpha v_\alpha^2 = \tfrac{1}{2} m_\alpha \left(\frac{m_D v_D}{m_\alpha}\right)^2 = \tfrac{1}{2} m_D v_D^2 \left(\frac{m_D}{m_\alpha}\right) = \left(\frac{m_D}{m_\alpha}\right)\text{KE}_D$$

$$= \left(\frac{228.028741\text{ u}}{4.002603\text{ u}}\right)\text{KE}_D = 57\,\text{KE}_D.$$

The total disintegration energy is $Q = \text{KE}_\alpha + \text{KE}_D = 57\,\text{KE}_D + \text{KE}_D = 58\,\text{KE}_D$. Hence

$$\text{KE}_\alpha = 57\,\text{KE}_D = \frac{57}{58} Q = 5.3\text{ MeV}.$$

The lighter α particle carries off (57/58) or 98% of the total kinetic energy. The total energy released is 5.4 MeV, so the daughter nucleus, which recoils in the opposite direction, carries off only 0.1 MeV.

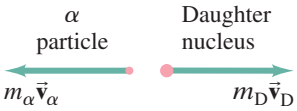


FIGURE 30–6 Momentum conservation in Example 30–7.

Why α Particles?

Why, you may wonder, do nuclei emit this combination of four nucleons called an α particle? Why not just four separate nucleons, or even one? The answer is that the α particle is very strongly bound, so that its mass is significantly less than that of four separate nucleons. That helps the final state in α decay to have less total mass, thus allowing certain nuclides to decay which could not decay to, say, 2 protons plus 2 neutrons. For example, $^{232}_{92}\text{U}$ could not decay to 2p + 2n because the masses of the daughter $^{228}_{90}\text{Th}$ plus four separate nucleons is $228.028741\ \text{u} + 2(1.007825\ \text{u}) + 2(1.008665\ \text{u}) = 232.061721\ \text{u}$, which is greater than the mass of the $^{232}_{92}\text{U}$ parent (232.037156 u). Such a decay would violate the conservation of energy. Indeed, we have never seen $^{232}_{92}\text{U} \rightarrow {}^{228}_{90}\text{Th} + 2\text{p} + 2\text{n}$. Similarly, it is almost always true that the emission of a single nucleon is energetically not possible; see Example 30–5.

Smoke Detectors—An Application

PHYSICS APPLIED
Smoke detector

One widespread application of nuclear physics is present in nearly every home in the form of an ordinary **smoke detector**. One type of smoke detector contains about 0.2 mg of the radioactive americium isotope, $^{241}_{95}\text{Am}$, in the form of AmO_2. The radiation continually ionizes the nitrogen and oxygen molecules in the air space between two oppositely charged plates. The resulting conductivity allows a small steady electric current. If smoke enters, the radiation is absorbed by the smoke particles rather than by the air molecules, thus reducing the current. The current drop is detected by the device's electronics and sets off the alarm. The radiation dose that escapes from an intact americium smoke detector is much less than the natural radioactive background, and so can be considered relatively harmless. There is no question that smoke detectors save lives and reduce property damage.

30–5 Beta Decay

β^- Decay

Transmutation of elements also occurs when a nucleus decays by β decay—that is, with the emission of an electron or β^- particle. The nucleus $^{14}_{6}\text{C}$, for example, emits an electron when it decays:

$$^{14}_{6}\text{C} \rightarrow {}^{14}_{7}\text{N} + \text{e}^- + \text{neutrino},$$

where e^- is the symbol for the electron. The particle known as the neutrino has charge $q = 0$ and a very small mass, long thought to be zero. It was not initially detected and was only later hypothesized to exist, as we shall discuss later in this Section. No nucleons are lost when an electron is emitted, and the total number of nucleons, A, is the same in the daughter nucleus as in the parent. But because an electron has been emitted from the nucleus itself, the charge on the daughter nucleus is $+1e$ greater than that on the parent. The parent nucleus in the decay written above had $Z = +6$, so from charge conservation the nucleus remaining behind must have a charge of $+7e$. So the daughter nucleus has $Z = 7$, which is nitrogen.

CAUTION
β-decay e^- comes from nucleus (it is not an orbital electron)

It must be carefully noted that the electron emitted in β decay is *not* an orbital electron. Instead, the electron is created *within the nucleus itself*. What happens is that one of the neutrons changes to a proton and in the process (to conserve charge) emits an electron. Indeed, free neutrons actually do decay in this fashion:

$$\text{n} \rightarrow \text{p} + \text{e}^- + \text{neutrino}.$$

To remind us of their origin in the nucleus, the electrons emitted in β decay are often referred to as "β particles." They are, nonetheless, indistinguishable from orbital electrons.

EXAMPLE 30–8 **Energy release in $^{14}_{6}$C decay.** How much energy is released when $^{14}_{6}$C decays to $^{14}_{7}$N by β emission?

APPROACH We find the mass difference before and after decay, Δm. The energy released is $E = (\Delta m)c^2$. The masses given in Appendix B are those of the neutral atom, and we have to keep track of the electrons involved. Assume the parent nucleus has six orbiting electrons so it is neutral; its mass is 14.003242 u. The daughter in this decay, $^{14}_{7}$N, is not neutral because it has the same six orbital electrons circling it but the nucleus has a charge of $+7e$. However, the mass of this daughter with its six electrons, plus the mass of the emitted electron (which makes a total of seven electrons), is just the mass of a neutral nitrogen atom.

SOLUTION The total mass in the final state is

$$(\text{mass of } ^{14}_{7}\text{N nucleus} + 6 \text{ electrons}) + (\text{mass of 1 electron}),$$

and this is equal to

$$\text{mass of neutral } ^{14}_{7}\text{N (includes 7 electrons)},$$

which from Appendix B is a mass of 14.003074 u. So the mass difference is $14.003242\text{ u} - 14.003074\text{ u} = 0.000168\text{ u}$, which is equivalent to an energy change $\Delta m\,c^2 = (0.000168\text{ u})(931.5\text{ MeV/u}) = 0.156\text{ MeV}$ or 156 keV.

NOTE The neutrino doesn't contribute to either the mass or charge balance because it has $q = 0$ and $m \approx 0$.

CAUTION
Be careful with atomic and electron masses in β decay

According to Example 30–8, we would expect the emitted electron to have a kinetic energy of 156 keV. (The daughter nucleus, because its mass is very much larger than that of the electron, recoils with very low velocity and hence gets very little of the kinetic energy—see Example 30–7.) Indeed, very careful measurements indicate that a few emitted β particles do have kinetic energy close to this calculated value. But the vast majority of emitted electrons have somewhat less energy. In fact, the energy of the emitted electron can be anywhere from zero up to the maximum value as calculated above. This range of electron kinetic energy was found for any β decay. It was as if the law of conservation of energy was being violated, and Bohr actually considered this possibility. Careful experiments indicated that linear momentum and angular momentum also did not seem to be conserved. Physicists were troubled at the prospect of giving up these laws, which had worked so well in all previous situations.

In 1930, Wolfgang Pauli proposed an alternate solution: perhaps a new particle that was very difficult to detect was emitted during β decay in addition to the electron. This hypothesized particle could be carrying off the energy, momentum, and angular momentum required to maintain the conservation laws. This new particle was named the **neutrino**—meaning "little neutral one"—by the great Italian physicist Enrico Fermi (1901–1954; Fig. 30–7), who in 1934 worked out a detailed theory of β decay. (It was Fermi who, in this theory, postulated the existence of the fourth force in nature which we call the *weak nuclear force*.) The neutrino has zero charge, spin of $\frac{1}{2}\hbar$, and was long thought to have zero mass, although today we are quite sure that it has a very tiny mass $(< 0.14\text{ eV}/c^2)$. If its mass were zero, it would be much like a photon in that it is neutral and would travel at the speed of light. But the neutrino is very difficult to detect. In 1956, complex experiments produced further evidence for the existence of the neutrino; but by then, most physicists had already accepted its existence.

The symbol for the neutrino is the Greek letter nu (ν). The correct way of writing the decay of $^{14}_{6}$C is then

$$^{14}_{6}\text{C} \rightarrow {}^{14}_{7}\text{N} + \text{e}^- + \bar{\nu}.$$

The bar $(^-)$ over the neutrino symbol is to indicate that it is an "antineutrino." (Why this is called an antineutrino rather than simply a neutrino is discussed in Chapter 32.)



FIGURE 30–7 Enrico Fermi, as portrayed on a US postage stamp. Fermi contributed significantly to both theoretical and experimental physics, a feat almost unique in modern times: statistical theory of identical particles that obey the exclusion principle (= fermions); theory of the weak interaction and β decay; neutron physics; induced radioactivity and new elements; first nuclear reactor; first resonance of particle physics; led and inspired a vast amount of other nuclear research.

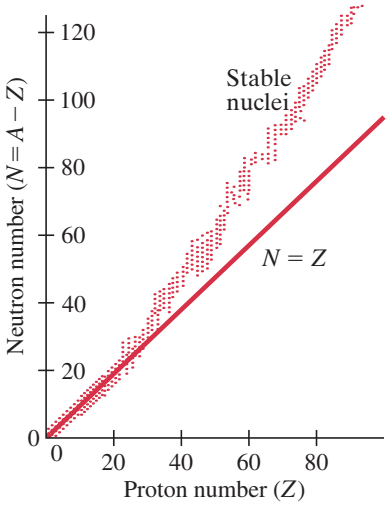


FIGURE 30–2 (Repeated.) Number of neutrons versus number of protons for stable nuclides, which are represented by dots. The straight line represents $N = Z$.

In β decay, it is the weak nuclear force that plays the crucial role. The neutrino is unique in that it interacts with matter only via the weak force, which is why it is so hard to detect.

β^+ Decay

Many isotopes decay by electron emission. They are always isotopes that have too many neutrons compared to the number of protons. That is, they are isotopes that lie above the stable isotopes plotted in Fig. 30–2. But what about unstable isotopes that have too few neutrons compared to their number of protons—those that fall below the stable isotopes of Fig. 30–2? These, it turns out, decay by emitting a **positron** instead of an electron. A positron (sometimes called an e^+ or β^+ particle) has the same mass as the electron, but it has a positive charge of $+1e$. Because it is so like an electron, except for its charge, the positron is called the **antiparticle**[†] to the electron. An example of a β^+ decay is that of ${}^{19}_{10}\text{Ne}$:

$$ {}^{19}_{10}\text{Ne} \rightarrow {}^{19}_{9}\text{F} + e^+ + \nu, $$

where e^+ stands for a positron. Note that the ν emitted here is a neutrino, whereas that emitted in β^- decay is called an antineutrino. Thus an antielectron (= positron) is emitted with a neutrino, whereas an antineutrino is emitted with an electron; this gives a certain balance as discussed in Chapter 32.

We can write β^- and β^+ decay, in general, as follows:

$$ {}^{A}_{Z}\text{N} \rightarrow {}^{A}_{Z+1}\text{N}' + e^- + \bar{\nu} \qquad [\beta^- \text{ decay}] $$
$$ {}^{A}_{Z}\text{N} \rightarrow {}^{A}_{Z-1}\text{N}' + e^+ + \nu, \qquad [\beta^+ \text{ decay}] $$

where N is the parent nucleus and N′ is the daughter.

Electron Capture

Besides β^- and β^+ emission, there is a third related process. This is **electron capture** (abbreviated EC in Appendix B) and occurs when a nucleus absorbs one of its orbiting electrons. An example is ${}^{7}_{4}\text{Be}$, which as a result becomes ${}^{7}_{3}\text{Li}$. The process is written

$$ {}^{7}_{4}\text{Be} + e^- \rightarrow {}^{7}_{3}\text{Li} + \nu, $$

or, in general,

$$ {}^{A}_{Z}\text{N} + e^- \rightarrow {}^{A}_{Z-1}\text{N}' + \nu. \qquad \text{[electron capture]} $$

Usually it is an electron in the innermost (K) shell that is captured, in which case the process is called **K-capture**. The electron disappears in the process, and a proton in the nucleus becomes a neutron; a neutrino is emitted as a result. This process is inferred experimentally by detection of emitted X-rays (due to other electrons jumping down to fill the state of the captured e^-).

30–6 Gamma Decay

Gamma rays are photons having very high energy. They have their origin in the decay of a nucleus, much like emission of photons by excited atoms. Like an atom, a nucleus itself can be in an excited state. When it jumps down to a lower energy state, or to the ground state, it emits a photon which we call a γ ray. The possible states of a nucleus are much farther apart in energy than those of an atom: on the order of keV or MeV, as compared to a few eV for electrons in an atom. Hence, the emitted photons have energies that can range from a few keV to several MeV. For a given decay, the γ ray always has the same energy. Since a γ ray carries no charge, there is no change in the element as a result of a γ decay.

How does a nucleus get into an excited state? It may occur because of a violent collision with another particle. More commonly, the nucleus remaining after a previous radioactive decay may be in an excited state. A typical example is shown

[†]Discussed in Chapter 32. Briefly, an antiparticle has the same mass as its corresponding particle, but opposite charge. A particle and its antiparticle can quickly annihilate each other, releasing energy in the form of two γ rays: $e^+ + e^- \rightarrow 2\gamma$.

in the energy-level diagram of Fig. 30–8. $^{12}_{5}$B can decay by β decay directly to the ground state of $^{12}_{6}$C; or it can go by β decay to an excited state of $^{12}_{6}$C, written $^{12}_{6}$C*, which itself decays by emission of a 4.4-MeV γ ray to the ground state of $^{12}_{6}$C.

We can write γ decay as

$$^{A}_{Z}\mathrm{N}^{*} \rightarrow {}^{A}_{Z}\mathrm{N} + \gamma, \qquad [\gamma \text{ decay}]$$

where the asterisk means "excited state" of that nucleus.

What, you may wonder, is the difference between a γ ray and an X-ray? They both are electromagnetic radiation (photons) and, though γ rays usually have higher energy than X-rays, their range of energies overlap to some extent. The difference is not intrinsic. We use the term X-ray if the photon is produced by an electron–atom interaction, and γ ray if the photon is produced in a nuclear process.

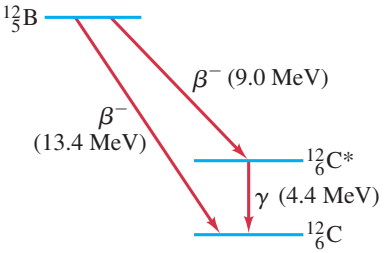


FIGURE 30–8 Energy-level diagram showing how $^{12}_{5}$B can decay to the ground state of $^{12}_{6}$C by β decay (total energy released = 13.4 MeV), or can instead β decay to an excited state of $^{12}_{6}$C (indicated by *), which subsequently decays to its ground state by emitting a 4.4-MeV γ ray.

*Isomers; Internal Conversion

In some cases, a nucleus may remain in an excited state for some time before it emits a γ ray. The nucleus is then said to be in a **metastable state** and is called an **isomer**.

An excited nucleus can sometimes return to the ground state by another process known as **internal conversion** with no γ ray emitted. In this process, the excited nucleus interacts with one of the orbital electrons and ejects this electron from the atom with the same kinetic energy (minus the binding energy of the electron) that an emitted γ ray would have had.

30–7 Conservation of Nucleon Number and Other Conservation Laws

In all three types of radioactive decay, the classical conservation laws hold. Energy, linear momentum, angular momentum, and electric charge are all conserved. These quantities are the same before the decay as after. But a new conservation law is also revealed, the **law of conservation of nucleon number**. According to this law, the total number of nucleons (A) remains constant in any process, although one type can change into the other type (protons into neutrons or vice versa). This law holds in all three types of decay. [In Chapter 32 we will generalize this and call it conservation of baryon number.]

Table 30–2 gives a summary of α, β, and γ decay.

TABLE 30–2 The Three Types of Radioactive Decay

α decay:
$^{A}_{Z}\mathrm{N} \rightarrow {}^{A-4}_{Z-2}\mathrm{N}' + {}^{4}_{2}\mathrm{He}$
β decay:
$^{A}_{Z}\mathrm{N} \rightarrow {}^{A}_{Z+1}\mathrm{N}' + \mathrm{e}^{-} + \bar{\nu}$
$^{A}_{Z}\mathrm{N} \rightarrow {}^{A}_{Z-1}\mathrm{N}' + \mathrm{e}^{+} + \nu$
$^{A}_{Z}\mathrm{N} + \mathrm{e}^{-} \rightarrow {}^{A}_{Z-1}\mathrm{N}' + \nu$ [EC]†
γ decay:
$^{A}_{Z}\mathrm{N}^{*} \rightarrow {}^{A}_{Z}\mathrm{N} + \gamma$

†Electron capture.
*Indicates the excited state of a nucleus.

30–8 Half-Life and Rate of Decay

A macroscopic sample of any radioactive isotope consists of a vast number of radioactive nuclei. These nuclei do not all decay at one time. Rather, they decay one by one over a period of time. This is a random process: we can not predict exactly when a given nucleus will decay. But we can determine, on a probabilistic basis, approximately how many nuclei in a sample will decay over a given time period, by assuming that each nucleus has the same probability of decaying in each second that it exists.

The number of decays ΔN that occur in a very short time interval Δt is then proportional to Δt and to the total number N of radioactive (parent) nuclei present:

$$\Delta N = -\lambda N\, \Delta t \qquad \textbf{(30–3a)}$$

where the minus sign means N is decreasing. We rewrite this to get the **rate of decay** (number of decays per second):

$$\frac{\Delta N}{\Delta t} = -\lambda N. \qquad \textbf{(30–3b)}$$

In these equations, λ is a measurable constant called the **decay constant**, which is different for different isotopes. The greater λ is, the greater the rate of decay, $\Delta N/\Delta t$, and the more "radioactive" that isotope is said to be.

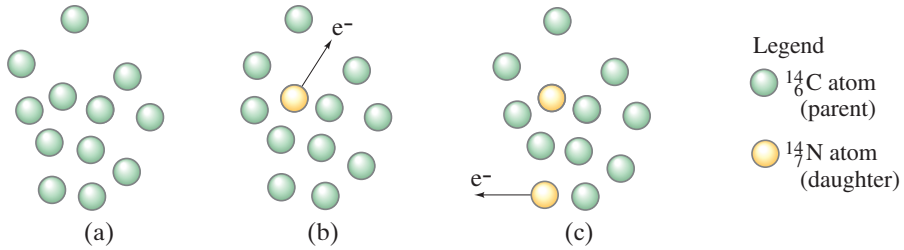


FIGURE 30–9 Radioactive nuclei decay one by one. Hence, the number of parent nuclei in a sample is continually decreasing. When a $^{14}_{6}$C nucleus emits an electron (b), the nucleus becomes a $^{14}_{7}$N nucleus. Another decays in (c).

The number of decays that occur in the short time interval Δt is designated ΔN because each decay that occurs corresponds to a decrease by one in the number N of parent nuclei present. That is, radioactive decay is a "one-shot" process, Fig. 30–9. Once a particular parent nucleus decays into its daughter, it cannot do it again.

Exponential Decay

Equation 30–3a or b can be solved for N (using calculus) and the result is

$$N = N_0 e^{-\lambda t}, \tag{30–4}$$

where N_0 is the number of parent nuclei present at any chosen time $t = 0$, and N is the number remaining after a time t. The symbol e is the natural exponential (encountered earlier in Sections 19–6 and 21–12) whose value is $e = 2.718\cdots$. Thus the number of parent nuclei in a sample decreases exponentially in time. This is shown in Fig. 30–10a for the decay of $^{14}_{6}$C. Equation 30–4 is called the **radioactive decay law**.

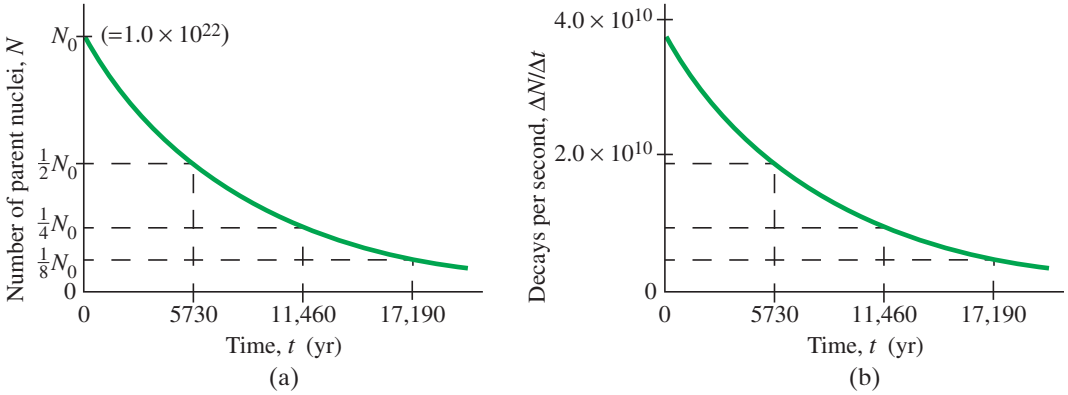


FIGURE 30–10 (a) The number N of parent nuclei in a given sample of $^{14}_{6}$C decreases exponentially. We assume a sample that has $N_0 = 1.00 \times 10^{22}$ nuclei. (b) The number of decays per second also decreases exponentially. The half-life of $^{14}_{6}$C is 5730 yr, which means that the number of parent nuclei, N, and the rate of decay, $\Delta N/\Delta t$, decrease by half every 5730 yr.

The number of decays per second, or decay rate R, is the magnitude of $\Delta N/\Delta t$, and is also called the **activity** of the sample. The magnitude (always positive) of a quantity is often indicated using vertical lines. The magnitude of $\Delta N/\Delta t$ is written $|\Delta N/\Delta t|$ and it is proportional to N (see Eq. 30–3b). So it too decreases exponentially in time at the same rate (Fig. 30–10b). The activity of a pure sample at time t is

$$R = \left|\frac{\Delta N}{\Delta t}\right| = R_0 e^{-\lambda t}, \tag{30–5}$$

where $R_0 = |\Delta N/\Delta t|_0$ is the activity at $t = 0$.

Equation 30–5 is also referred to as the **radioactive decay law** (as is Eq. 30–4).

Half-Life

The rate of decay of any isotope is often specified by giving its "half-life" rather than the decay constant λ. The **half-life** of an isotope is defined as the time it takes for half the original amount of parent isotope in a given sample to decay.

For example, the half-life of $^{14}_{6}C$ is about 5730 years. If at some time a piece of petrified wood contains, say, 1.00×10^{22} nuclei of $^{14}_{6}C$, then 5730 years later it will contain half as many, 0.50×10^{22} nuclei. After another 5730 years it will contain 0.25×10^{22} nuclei, and so on. This is shown in Fig. 30–10a. Since the rate of decay $\Delta N/\Delta t$ is proportional to N, it, too, decreases by a factor of 2 every half-life (Fig. 30–10b).

The half-lives of known radioactive isotopes vary from very short ($\approx 10^{-22}$ s) to more than 10^{23} yr ($> 10^{30}$ s). The half-lives of many isotopes are given in Appendix B. It should be clear that the half-life (which we designate $T_{\frac{1}{2}}$) bears an inverse relationship to the decay constant. The longer the half-life of an isotope, the more slowly it decays, and hence λ is smaller. Conversely, very active isotopes (large λ) have very short half-lives. The precise relationship between half-life and decay constant is

$$T_{\frac{1}{2}} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}. \qquad \textbf{(30–6)}$$

We derive this in the next (optional) subsection.

EXERCISE D The half-life of $^{22}_{11}Na$ is 2.6 years. How much $^{22}_{11}Na$ will be left of a pure 1.0-μg sample after 7.8 yr? (*a*) None. (*b*) $\frac{1}{8}$ μg. (*c*) $\frac{1}{4}$ μg. (*d*) $\frac{1}{2}$ μg. (*e*) 0.693 μg.

EXERCISE E Return to the Chapter-Opening Question, page 857, and answer it again now. Try to explain why you may have answered differently the first time.

*Deriving the Half-Life Formula

We can derive Eq. 30–6 starting from Eq. 30–4 by setting $N = N_0/2$ at $t = T_{\frac{1}{2}}$:

$$\frac{N_0}{2} = N_0 e^{-\lambda T_{\frac{1}{2}}}$$

so

$$\frac{1}{2} = e^{-\lambda T_{\frac{1}{2}}}$$

and

$$e^{\lambda T_{\frac{1}{2}}} = 2.$$

We take natural logs of both sides ("ln" and "e" are inverse operations, meaning $\ln(e^x) = x$) and find

$$\ln\left(e^{\lambda T_{\frac{1}{2}}}\right) = \ln 2,$$

so

$$\lambda T_{\frac{1}{2}} = \ln 2 = 0.693$$

and

$$T_{\frac{1}{2}} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda},$$

which is Eq. 30–6.

*Mean Life

Sometimes the **mean life** τ of an isotope is quoted, which is defined as $\tau = 1/\lambda$. Then Eq. 30–4 can be written $N = N_0 e^{-t/\tau}$, just as for RC and LR circuits (Chapters 19 and 21 where τ was called the time constant). The mean life of an isotope is then given by (see also Eq. 30–6)

$$\tau = \frac{1}{\lambda} = \frac{T_{\frac{1}{2}}}{0.693}. \qquad \text{[mean life]} \quad \textbf{(30–7)}$$

The mean life and half-life differ by a factor of 0.693, so confusing them can cause serious error (and has). The radioactive decay law, Eq. 30–5, can then be written as $R = R_0 e^{-t/\tau}$.

CAUTION
Do not confuse half-life and mean life

30–9 Calculations Involving Decay Rates and Half-Life

Let us now consider Examples of what we can determine about a sample of radioactive material if we know the half-life.

EXAMPLE 30–9 **Sample activity.** The isotope $^{14}_{6}C$ has a half-life of 5730 yr. If a sample contains 1.00×10^{22} carbon-14 nuclei, what is the activity of the sample?

APPROACH We first use the half-life to find the decay constant (Eq. 30–6), and use that to find the activity, Eq. 30–3b. The number of seconds in a year is $(60)(60)(24)(365\frac{1}{4}) = 3.156 \times 10^7$ s.

SOLUTION The decay constant λ from Eq. 30–6 is

$$\lambda = \frac{0.693}{T_{\frac{1}{2}}} = \frac{0.693}{(5730\text{ yr})(3.156 \times 10^7\text{ s/yr})} = 3.83 \times 10^{-12}\,\text{s}^{-1}.$$

From Eqs. 30–3b and 30–5, the activity or rate of decay is

$$R = \left|\frac{\Delta N}{\Delta t}\right| = \lambda N = (3.83 \times 10^{-12}\,\text{s}^{-1})(1.00 \times 10^{22}) = 3.83 \times 10^{10}\text{ decays/s}.$$

Notice that the graph of Fig. 30–10b starts at this value, corresponding to the original value of $N = 1.0 \times 10^{22}$ nuclei in Fig. 30–10a.

NOTE The unit "decays/s" is often written simply as s^{-1} since "decays" is not a unit but refers only to the number. This simple unit of activity is called the becquerel: 1 Bq = 1 decay/s, as discussed in Chapter 31.

CONCEPTUAL EXAMPLE 30–10 **Safety: Activity versus half-life.** One might think that a short half-life material is safer than a long half-life material because it will not last as long. Is that true?

RESPONSE No. A shorter half-life means the activity is higher and thus more "radioactive" and can cause more biological damage. In contrast, a longer half-life for the same sample size N means a lower activity but we have to worry about it for longer and find safe storage until it reaches a safe (low) level of activity.

EXAMPLE 30–11 **A sample of radioactive $^{13}_{7}N$.** A laboratory has 1.49 μg of pure $^{13}_{7}N$, which has a half-life of 10.0 min (600 s). (*a*) How many nuclei are present initially? (*b*) What is the rate of decay (activity) initially? (*c*) What is the activity after 1.00 h? (*d*) After approximately how long will the activity drop to less than one per second $(= 1\,\text{s}^{-1})$?

APPROACH We use the definition of the mole and Avogadro's number (Sections 13–6 and 13–8) to find (*a*) the number of nuclei. For (*b*) we get λ from the given half-life and use Eq. 30–3b for the rate of decay. For (*c*) and (*d*) we use Eq. 30–5.

SOLUTION (*a*) The atomic mass is 13.0, so 13.0 g will contain 6.02×10^{23} nuclei (Avogadro's number). We have only 1.49×10^{-6} g, so the number of nuclei N_0 that we have initially is given by the ratio

$$\frac{N_0}{6.02 \times 10^{23}} = \frac{1.49 \times 10^{-6}\,\text{g}}{13.0\,\text{g}}.$$

Solving for N_0, we find $N_0 = 6.90 \times 10^{16}$ nuclei.

(*b*) From Eq. 30–6,

$$\lambda = 0.693/T_{\frac{1}{2}} = (0.693)/(600\,\text{s}) = 1.155 \times 10^{-3}\,\text{s}^{-1}.$$

Then, at $t = 0$ (see Eqs. 30–3b and 30–5)

$$R_0 = \left|\frac{\Delta N}{\Delta t}\right|_0 = \lambda N_0 = (1.155 \times 10^{-3}\,\text{s}^{-1})(6.90 \times 10^{16}) = 7.97 \times 10^{13}\text{ decays/s}.$$

(*c*) After 1.00 h = 3600 s, the magnitude of the activity will be (Eq. 30–5)

$$R = R_0 e^{-\lambda t} = (7.97 \times 10^{13}\,\text{s}^{-1}) e^{-(1.155\times10^{-3}\,\text{s}^{-1})(3600\,\text{s})} = 1.25 \times 10^{12}\,\text{s}^{-1}.$$

(*d*) We want to determine the time t when $R = 1.00\,\text{s}^{-1}$. From Eq. 30–5, we have

$$e^{-\lambda t} = \frac{R}{R_0} = \frac{1.00\,\text{s}^{-1}}{7.97 \times 10^{13}\,\text{s}^{-1}} = 1.25 \times 10^{-14}.$$

We take the natural log (ln) of both sides $(\ln e^{-\lambda t} = -\lambda t)$ and divide by λ to find

$$t = -\frac{\ln(1.25 \times 10^{-14})}{\lambda} = 2.77 \times 10^4\,\text{s} = 7.70\,\text{h}.$$

Easy Alternate Solution to (c) 1.00 h = 60.0 minutes is 6 half-lives, so the activity will decrease to $(\frac{1}{2})(\frac{1}{2})(\frac{1}{2})(\frac{1}{2})(\frac{1}{2})(\frac{1}{2}) = (\frac{1}{2})^6 = \frac{1}{64}$ of its original value, or $(7.97 \times 10^{13})/(64) = 1.25 \times 10^{12}$ per second.

30–10 Decay Series

It is often the case that one radioactive isotope decays to another isotope that is also radioactive. Sometimes this daughter decays to yet a third isotope which also is radioactive. Such successive decays are said to form a **decay series**. An important example is illustrated in Fig. 30–11. As can be seen, $^{238}_{92}$U decays by α emission to $^{234}_{90}$Th, which in turn decays by β decay to $^{234}_{91}$Pa. The series continues as shown, with several possible branches near the bottom, ending at the stable lead isotope, $^{206}_{82}$Pb. The two last decays can be

$$^{206}_{81}\text{Tl} \rightarrow {}^{206}_{82}\text{Pb} + e^- + \bar{\nu}, \qquad (T_{\frac{1}{2}} = 4.2\,\text{min})$$

or

$$^{210}_{84}\text{Po} \rightarrow {}^{206}_{82}\text{Pb} + \alpha. \qquad (T_{\frac{1}{2}} = 138\,\text{days})$$

Other radioactive series also exist.

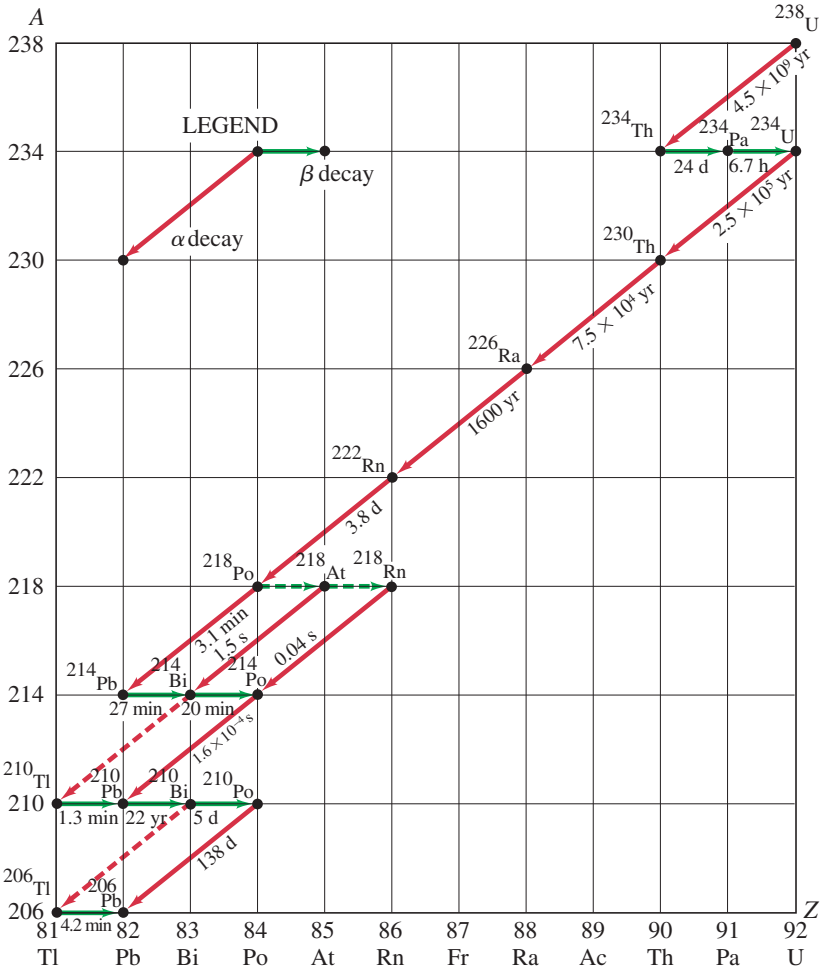


FIGURE 30–11 Decay series beginning with $^{238}_{92}$U. Nuclei in the series are specified by a dot representing A and Z values. Half-lives are given in seconds (s), minutes (min), hours (h), days (d), or years (yr). Note that a horizontal arrow represents β decay (A does not change), whereas a diagonal line represents α decay (A changes by 4, Z changes by 2). For the four nuclides shown that can decay by both α and β decay, the more prominent decay (in these four cases, >99.9%) is shown as a solid arrow and the less common decay (< 0.1%) as a dashed arrow.

Because of such decay series, certain radioactive elements are found in nature that otherwise would not be. When the solar system (including Earth) was formed about 5 billion years ago, it is believed that nearly all nuclides were present, having been formed (by fusion and neutron capture, Sections 31–3 and 33–2) in a nearby supernova explosion (Section 33–2). Many isotopes with short half-lives decayed quickly and no longer are detected in nature today. But long-lived isotopes, such as $^{238}_{92}\text{U}$ with a half-life of 4.5×10^9 yr, still do exist in nature today. Indeed, about half of the original $^{238}_{92}\text{U}$ still remains. We might expect, however, that radium ($^{226}_{88}\text{Ra}$), with a half-life of 1600 yr, would have disappeared from the Earth long ago. Indeed, the original $^{226}_{88}\text{Ra}$ nuclei must by now have all decayed. However, because $^{238}_{92}\text{U}$ decays (in several steps, Fig. 30–11) to $^{226}_{88}\text{Ra}$, the supply of $^{226}_{88}\text{Ra}$ is continually replenished, which is why it is still found on Earth today. The same can be said for many other radioactive nuclides.

CONCEPTUAL EXAMPLE 30–12 **Decay chain.** In the decay chain of Fig. 30–11, if we look at the decay of $^{234}_{92}\text{U}$, we see four successive nuclides with half-lives of 250,000 yr, 75,000 yr, 1600 yr, and a little under 4 days. Each decay in the chain has an alpha particle of a characteristic energy, and so we can monitor the radioactive decay rate of each nuclide. Given a sample that was pure $^{234}_{92}\text{U}$ a million years ago, which alpha decay would you expect to have the highest activity rate in the sample?

RESPONSE The first instinct is to say that the process with the shortest half-life would show the highest activity. Surprisingly, perhaps, the activities of the four nuclides in this sample are all the same. The reason is that in each case the decay of the parent acts as a bottleneck to the decay of the daughter. Compared to the 1600-yr half-life of $^{226}_{88}\text{Ra}$, for example, its daughter $^{222}_{86}\text{Rn}$ decays almost immediately, but it cannot decay until it is made. (This is like an automobile assembly line: if worker A takes 20 minutes to do a task and then worker B takes only 1 minute to do the next task, worker B still does only one car every 20 minutes.)

30–11 Radioactive Dating

Radioactive decay has many interesting applications. One is the technique of *radioactive dating* by which the age of ancient materials can be determined.

PHYSICS APPLIED
Carbon-14 dating

The age of any object made from once-living matter, such as wood, can be determined using the natural radioactivity of $^{14}_{6}\text{C}$. All living plants absorb carbon dioxide (CO_2) from the air and use it to synthesize organic molecules. The vast majority of these carbon atoms are $^{12}_{6}\text{C}$, but a small fraction, about 1.3×10^{-12}, is the radioactive isotope $^{14}_{6}\text{C}$. The ratio of $^{14}_{6}\text{C}$ to $^{12}_{6}\text{C}$ in the atmosphere has remained roughly constant over many thousands of years, in spite of the fact that $^{14}_{6}\text{C}$ decays with a half-life of about 5730 yr. This is because energetic nuclei in the cosmic radiation, which impinges on the Earth from outer space, strike nuclei of atoms in the atmosphere and break those nuclei into pieces, releasing free neutrons. Those neutrons can collide with nitrogen nuclei in the atmosphere to produce the nuclear transformation $\text{n} + {}^{14}_{7}\text{N} \rightarrow {}^{14}_{6}\text{C} + \text{p}$. That is, a neutron strikes and is absorbed by a $^{14}_{7}\text{N}$ nucleus, and a proton is knocked out in the process. The remaining nucleus is $^{14}_{6}\text{C}$. This continual production of $^{14}_{6}\text{C}$ in the atmosphere roughly balances the loss of $^{14}_{6}\text{C}$ by radioactive decay.

As long as a plant or tree is alive, it continually uses the carbon from carbon dioxide in the air to build new tissue and to replace old. Animals eat plants, so they too are continually receiving a fresh supply of carbon for their tissues.

Organisms cannot distinguish[†] $^{14}_{6}C$ from $^{12}_{6}C$, and because the ratio of $^{14}_{6}C$ to $^{12}_{6}C$ in the atmosphere remains nearly constant, the ratio of the two isotopes within the living organism remains nearly constant as well. When an organism dies, carbon dioxide is no longer taken in and utilized. Because the $^{14}_{6}C$ decays radioactively, the ratio of $^{14}_{6}C$ to $^{12}_{6}C$ in a dead organism decreases over time. The half-life of $^{14}_{6}C$ is about 5730 yr, so the $^{14}_{6}C/^{12}_{6}C$ ratio decreases by half every 5730 yr. If, for example, the $^{14}_{6}C/^{12}_{6}C$ ratio of an ancient wooden tool is half of what it is in living trees, then the object must have been made from a tree that was felled about 5730 years ago.

Actually, corrections must be made for the fact that the $^{14}_{6}C/^{12}_{6}C$ ratio in the atmosphere has not remained precisely constant over time. The determination of what this ratio has been over the centuries has required techniques such as comparing the expected ratio to the actual ratio for objects whose age is known, such as very old trees whose annual rings can be counted reasonably accurately.

EXAMPLE 30–13 **An ancient animal.** The mass of carbon in an animal bone fragment found in an archeological site is 200 g. If the bone registers an activity of 16 decays/s, what is its age?

PHYSICS APPLIED
Archeological dating

APPROACH First we determine how many $^{14}_{6}C$ atoms there were in our 200-g sample when the animal was alive, given the known fraction of $^{14}_{6}C$ to $^{12}_{6}C$, 1.3×10^{-12}. Then we use Eq. 30–3b to find the activity back then, and Eq. 30–5 to find out how long ago that was by solving for the time t.

SOLUTION The 200 g of carbon is nearly all $^{12}_{6}C$; 12.0 g of $^{12}_{6}C$ contains 6.02×10^{23} atoms, so 200 g contains

$$\left(\frac{6.02 \times 10^{23}\ \text{atoms/mol}}{12.0\ \text{g/mol}}\right)(200\ \text{g}) = 1.00 \times 10^{25}\ \text{atoms}.$$

When the animal was alive, the ratio of $^{14}_{6}C$ to $^{12}_{6}C$ in the bone was 1.3×10^{-12}. The number of $^{14}_{6}C$ nuclei at that time was

$$N_0 = (1.00 \times 10^{25}\ \text{atoms})(1.3 \times 10^{-12}) = 1.3 \times 10^{13}\ \text{atoms}.$$

From Eq. 30–3b with $\lambda = 3.83 \times 10^{-12}\ \text{s}^{-1}$ (Example 30–9) the magnitude of the activity when the animal was still alive ($t = 0$) was

$$R_0 = \left|\frac{\Delta N}{\Delta t}\right|_0 = \lambda N_0 = (3.83 \times 10^{-12}\ \text{s}^{-1})(1.3 \times 10^{13}) = 50\ \text{s}^{-1}.$$

From Eq. 30–5

$$R = R_0 e^{-\lambda t}$$

where R, its activity now, is given as $16\ \text{s}^{-1}$. Then

$$16\ \text{s}^{-1} = (50\ \text{s}^{-1})e^{-\lambda t}$$

or

$$e^{\lambda t} = \frac{50}{16}.$$

We take the natural logs of both sides (and divide by λ) to get

$$t = \frac{1}{\lambda}\ln\left(\frac{50}{16}\right) = \frac{1}{3.83 \times 10^{-12}\ \text{s}^{-1}}\ln\left(\frac{50}{16}\right)$$
$$= 2.98 \times 10^{11}\ \text{s} = 9400\ \text{yr},$$

which is the time elapsed since the death of the animal.

[†]Organisms operate almost exclusively via chemical reactions—which involve only the outer orbital electrons of the atom; extra neutrons in the nucleus have essentially no effect.

Geological Time Scale Dating

Carbon dating is useful only for determining the age of objects less than about 60,000 years old. The amount of $^{14}_{6}C$ remaining in objects older than that is usually too small to measure accurately, although new techniques are allowing detection of even smaller amounts of $^{14}_{6}C$, pushing the time frame further back. On the other hand, radioactive isotopes with longer half-lives can be used in certain circumstances to obtain the age of older objects. For example, the decay of $^{238}_{92}U$, because of its long half-life of 4.5×10^9 years, is useful in determining the ages of rocks on a geologic time scale. When molten material on Earth long ago solidified into rock as the temperature dropped, different compounds solidified according to the melting points, and thus different compounds separated to some extent. Uranium present in a material became fixed in position and the daughter nuclei that result from the decay of uranium were also fixed in that position. Thus, by measuring the amount of $^{238}_{92}U$ remaining in the material relative to the amount of daughter nuclei, the time when the rock solidified can be determined.

PHYSICS APPLIED
Geological dating

PHYSICS APPLIED
Oldest Earth rocks and earliest life

Radioactive dating methods using $^{238}_{92}U$ and other isotopes have shown the age of the oldest Earth rocks to be about 4×10^9 yr. The age of rocks in which the oldest fossilized organisms are embedded indicates that life appeared more than $3\frac{1}{2}$ billion years ago. The earliest fossilized remains of mammals are found in rocks 200 million years old, and humanlike creatures seem to have appeared more than 2 million years ago. Radioactive dating has been indispensable for the reconstruction of Earth's history.

*30–12 Stability and Tunneling

Radioactive decay occurs only if the mass of the parent nucleus is greater than the sum of the masses of the daughter nucleus and all particles emitted. For example, $^{238}_{92}U$ can decay to $^{234}_{90}Th$ because the mass of $^{238}_{92}U$ is greater than the mass of the $^{234}_{90}Th$ plus the mass of the α particle. Because systems tend to go in the direction that reduces their internal or potential energy (a ball rolls downhill, a positive charge moves toward a negative charge), you may wonder why an unstable nucleus doesn't fall apart immediately. In other words, why do $^{238}_{92}U$ nuclei $\left(T_{\frac{1}{2}} = 4.5 \times 10^9 \text{ yr}\right)$ and other isotopes have such long half-lives? Why don't parent nuclei all decay at once?

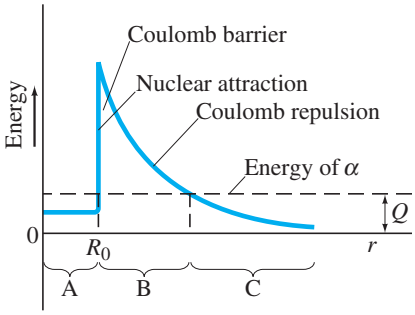


FIGURE 30–12 Potential energy for alpha particle and nucleus, showing the "Coulomb barrier" through which the α particle must tunnel to escape. The Q-value of the reaction is also indicated.

The answer has to do with quantum theory and the nature of the forces involved. One way to view the situation is with the aid of a potential-energy diagram, as in Fig. 30–12. Let us consider the particular case of the decay $^{238}_{92}U \rightarrow {}^{234}_{90}Th + {}^{4}_{2}He$. The blue line represents the potential energy, including rest mass, where we imagine the α particle as a separate entity within the $^{238}_{92}U$ nucleus. The region labeled A in Fig. 30–12 represents the PE of the α particle when it is held within the uranium nucleus by the strong nuclear force (R_0 is the nuclear radius). Region C represents the PE when the α particle is free of the nucleus. The downward-curving PE (proportional to $1/r$) represents the electrical repulsion (Coulomb's law) between the positively charged α and the $^{234}_{90}Th$ nucleus. To get to region C, the α particle has to get through the "**Coulomb barrier**" shown. Since the PE just beyond $r = R_0$ (region B) is greater than the energy of the alpha particle (dashed line), the α particle could not escape the nucleus if it were governed by classical physics. It could escape only if there were an input of energy equal to the height of the barrier. Nuclei decay spontaneously, however, without any input of energy. How, then, does the α particle get from region A to region C? It actually passes through the barrier in a process known as quantum-mechanical **tunneling**. Classically, this could not happen, because an α particle in region B (within the barrier) would be violating the conservation-of-energy principle.[†]

[†]The total energy E (dashed line in Fig. 30–12) would be less than the PE; because $\text{KE} = \frac{1}{2}mv^2 > 0$, then classically, $E = \text{KE} + \text{PE}$ could not be less than the PE.

The uncertainty principle, however, tells us that energy conservation can be violated by an amount ΔE for a length of time Δt given by

$$(\Delta E)(\Delta t) \approx \frac{h}{2\pi}.$$

We saw in Section 28–3 that this is a result of the wave–particle duality. Thus quantum mechanics allows conservation of energy to be violated for brief periods that may be long enough for an α particle to "tunnel" through the barrier. ΔE would represent the energy difference between the average barrier height and the particle's energy, and Δt the time to pass through the barrier. The higher and wider the barrier, the less time the α particle has to escape and the less likely it is to do so. It is therefore the height and width of this barrier that control the rate of decay and half-life of an isotope.

30–13 Detection of Particles

Individual particles such as electrons, protons, α particles, neutrons, and γ rays are not detected directly by our senses. Consequently, a variety of instruments have been developed to detect them.

Counters

One of the most common detectors is the **Geiger counter**. As shown in Fig. 30–13, it consists of a cylindrical metal tube filled with a certain type of gas. A long wire runs down the center and is kept at a high positive voltage ($\approx 10^3$ V) with respect to the outer cylinder. The voltage is just slightly less than that required to ionize the gas atoms. When a charged particle enters through the thin "window" at one end of the tube, it ionizes a few atoms of the gas. The freed electrons are attracted toward the positive wire, and as they are accelerated they strike and ionize additional atoms. An "avalanche" of electrons is quickly produced, and when it reaches the wire anode, it produces a voltage pulse. The pulse, after being amplified, can be sent to an electronic counter, which counts how many particles have been detected. Or the pulses can be sent to a loudspeaker and each detection of a particle is heard as a "click." Only a fraction of the radiation emitted by a sample is detected by any detector.

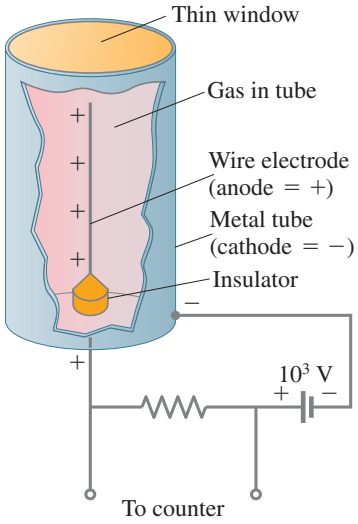
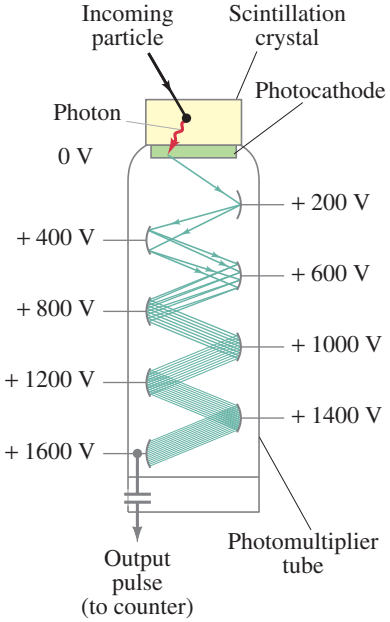


FIGURE 30–13 Diagram of a Geiger counter.

A **scintillation counter** makes use of a solid, liquid, or gas known as a **scintillator** or **phosphor**. The atoms of a scintillator are easily excited when struck by an incoming particle and emit visible light when they return to their ground states. Typical scintillators are crystals of NaI and certain plastics. One face of a solid scintillator is cemented to a photomultiplier tube, and the whole is wrapped with opaque material to keep it light-tight (in the dark) or is placed within a light-tight container. The **photomultiplier** (PM) **tube** converts the energy of the scintillator-emitted photon(s) into an electric signal. A PM tube is a vacuum tube containing several electrodes (typically 8 to 14), called *dynodes*, which are maintained at successively higher voltages as shown in Fig. 30–14. At its top surface is a photoelectric surface, called the *photocathode*, whose work function (Section 27–3) is low enough that an electron is easily released when struck by a photon from the scintillator. Such an electron is accelerated toward the positive voltage of the first dynode. When it strikes the first dynode, the electron has acquired sufficient kinetic energy so that it can eject two to five more electrons. These, in turn, are accelerated toward the higher voltage second dynode, and a multiplication process begins. The number of electrons striking the last dynode may be 10^6 or more. Thus the passage of a particle through the scintillator results in an electric signal at the output of the PM tube that can be sent to an electronic counter just as for a Geiger tube. Solid scintillators are much more dense than the gas of a Geiger counter, and so are much more efficient detectors—especially for γ rays, which interact less with matter than do α or β particles. Scintillators that can measure the total energy deposited are much used today and are called **calorimeters**.

FIGURE 30–14 Scintillation counter with a photomultiplier tube.



In tracer work (Section 31–7), **liquid scintillators** are often used. Radioactive samples taken at different times or from different parts of an organism are placed directly in small bottles containing the liquid scintillator. This is particularly convenient for detection of β rays from ${}^{3}_{1}\mathrm{H}$ and ${}^{14}_{6}\mathrm{C}$, which have very low energies and have difficulty passing through the outer covering of a crystal scintillator or Geiger tube. A PM tube is still used to produce the electric signal from the liquid scintillator.

A **semiconductor detector** consists of a reverse-biased *pn* junction diode (Section 29–9). A charged particle passing through the junction can excite electrons into the conduction band, leaving holes in the valence band. The freed charges produce a short electric current pulse that can be counted as for Geiger and scintillation counters.

Hospital workers and others who work around radiation may carry *film badges* which detect the accumulation of radiation exposure. The film inside is periodically replaced and developed, the darkness of the developed film being related to total exposure (see Section 31–5).

Visualization

The devices discussed so far are used for counting the number of particles (or decays of a radioactive isotope). There are also devices that allow the track of charged particles to be *seen*. Very important are semiconductor detectors. **Silicon wafer semiconductors** have their surface etched into separate tiny pixels, each providing particle position information. They are much used in elementary particle physics (Chapter 32) to track the positions of particles produced and to determine their point of origin and/or their momentum (with the help of a magnetic field). The pixel arrangement can be CCD or CMOS (Section 25–1), the latter able to incorporate electronics inside, allowing fast readout.

One of the oldest tracking devices is the **photographic emulsion**, which can be small and portable, used now particularly for cosmic-ray studies from balloons. A charged particle passing through an emulsion ionizes the atoms along its path. These points undergo a chemical change, and when the emulsion is developed (like film) the particle's path is revealed.

In a **cloud chamber**, used in the early days of nuclear physics, a gas is cooled to a temperature slightly below its usual condensation point ("supercooled"). Tiny droplets form around ions produced when a charged particle passes through (Fig. 30–15). Light scattering from these droplets reveals the track of the particle.

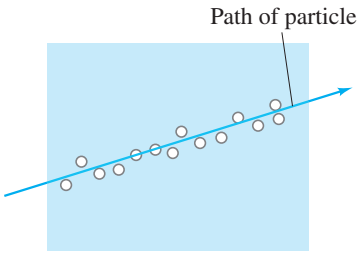
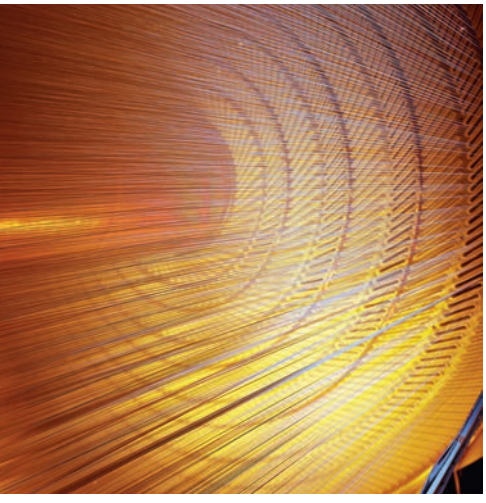


FIGURE 30–15 In a cloud chamber or bubble chamber, droplets or bubbles are formed around ions produced by the passage of a charged particle.

The **bubble chamber**, invented in 1952 by Donald A. Glaser (1926–2013), makes use of a superheated liquid kept close to its normal boiling point. Bubbles characteristic of boiling form around ions produced by the passage of a charged particle, revealing paths of particles that recently passed through. Because a bubble chamber uses a liquid, often liquid hydrogen, many more interactions can occur than in a cloud chamber. A magnetic field applied across the chamber makes charged particle paths curve (Chapter 20) and allows the momentum of charged particles to be determined from the radius of curvature of their paths.

FIGURE 30–16 Multiwire chamber inside the Collider Detector at Fermilab (CDF). Figure 32–15 in Chapter 32 was made with this detector.



A **multiwire**[†] **chamber** consists of a set of closely spaced fine wires immersed in a gas (Fig. 30–16). Many wires are grounded, and the others between are kept at very high voltage. A charged particle passing through produces ions in the gas. Freed electrons drift toward the nearest high voltage wire, creating an "avalanche" of many more ions, and producing an electric pulse or signal at that wire. The positions of the particles are determined electronically by the position of the wire and by the time it takes the pulses to reach "readout" electronics at the ends of the wires. The paths of the particles are reconstructed electronically by computers which can "draw" a picture of the tracks, as shown in Fig. 32–15, Chapter 32. An external magnetic field curves the paths, allowing the momentum of the particles to be measured.

Many detectors are also **calorimeters** which measure the energy of the particles.

[†]Also called *wire drift chamber* or *wire proportional chamber*.

Summary

Nuclear physics is the study of atomic nuclei. Nuclei contain **protons** and **neutrons**, which are collectively known as **nucleons**. The total number of nucleons, A, is the nucleus's **atomic mass number**. The number of protons, Z, is the **atomic number**. The number of neutrons equals $A - Z$. **Isotopes** are nuclei with the same Z, but with different numbers of neutrons. For an element X, an isotope of given Z and A is represented by

$$^{A}_{Z}\mathrm{X}.$$

The nuclear radius is approximately proportional to $A^{\frac{1}{3}}$, indicating that all nuclei have about the same density. Nuclear masses are specified in **unified atomic mass units** (u), where the mass of $^{12}_{6}\mathrm{C}$ (including its 6 electrons) is defined as exactly 12.000000 u. In terms of the energy equivalent (because $E = mc^2$),

$$1\,\mathrm{u} = 931.5\,\mathrm{MeV}/c^2 = 1.66 \times 10^{-27}\,\mathrm{kg}.$$

The mass of a stable nucleus is less than the sum of the masses of its constituent nucleons. The difference in mass (times c^2) is the **total binding energy**. It represents the energy needed to break the nucleus into its constituent nucleons. The **binding energy per nucleon** averages about 8 MeV per nucleon, and is lowest for low mass and high mass nuclei.

Unstable nuclei undergo **radioactive decay**; they change into other nuclei with the emission of an α, β, or γ particle. An α particle is a $^{4}_{2}\mathrm{He}$ nucleus; a β particle is an electron or positron; and a γ ray is a high-energy photon. In β decay, a **neutrino** is also emitted. The transformation of **parent** nuclei into **daughter** nuclei is called **transmutation** of the elements. Radioactive decay occurs spontaneously only when the mass of the products is less than the mass of the parent nucleus. The loss in mass appears as kinetic energy of the products.

Nuclei are held together by the **strong nuclear force**. The **weak nuclear force** makes itself apparent in β decay. These two forces, plus the gravitational and electromagnetic forces, are the four known types of force.

Electric charge, linear and angular momentum, mass–energy, and **nucleon number** are **conserved** in all decays.

Radioactive decay is a statistical process. For a given type of radioactive nucleus, the number of nuclei that decay (ΔN) in a time Δt is proportional to the number N of parent nuclei present:

$$\Delta N = -\lambda N\,\Delta t; \qquad \textbf{(30–3a)}$$

the minus sign means N *decreases* in time.

The proportionality constant λ is called the **decay constant** and is characteristic of the given nucleus. The number N of nuclei remaining after a time t decreases exponentially,

$$N = N_0 e^{-\lambda t}, \qquad \textbf{(30–4)}$$

as does the **activity**, R = magnitude of $\Delta N/\Delta t$:

$$R = \left|\frac{\Delta N}{\Delta t}\right|_0 e^{-\lambda t}. \qquad \textbf{(30–5)}$$

The **half-life**, $T_{\frac{1}{2}}$, is the time required for half the nuclei of a radioactive sample to decay. It is related to the decay constant by

$$T_{\frac{1}{2}} = \frac{0.693}{\lambda}. \qquad \textbf{(30–6)}$$

Radioactive dating is the use of radioactive decay to determine the age of certain objects, such as carbon dating.

[*Alpha decay occurs via a purely quantum-mechanical process called **tunneling** through a barrier.]

Particle **detectors** include **Geiger counters**, **scintillators** with attached **photomultiplier tubes**, and **semiconductor detectors**. Detectors that can image particle tracks include **semiconductors**, photographic **emulsions, bubble chambers**, and **multiwire chambers**.

Questions

1. What do different isotopes of a given element have in common? How are they different?

2. What are the elements represented by the X in the following: (*a*) $^{232}_{92}\mathrm{X}$; (*b*) $^{18}_{7}\mathrm{X}$; (*c*) $^{1}_{1}\mathrm{X}$; (*d*) $^{86}_{38}\mathrm{X}$; (*e*) $^{252}_{100}\mathrm{X}$?

3. How many protons and how many neutrons do each of the isotopes in Question 2 have?

4. Identify the element that has 87 nucleons and 50 neutrons.

5. Why are the atomic masses of many elements (see the Periodic Table) not close to whole numbers?

6. Why are atoms much more likely to emit an alpha particle than to emit separate neutrons and protons?

7. What are the similarities and the differences between the strong nuclear force and the electric force?

8. What is the experimental evidence in favor of radioactivity being a nuclear process?

9. The isotope $^{64}_{29}\mathrm{Cu}$ is unusual in that it can decay by γ, β^-, and β^+ emission. What is the resulting nuclide for each case?

10. A $^{238}_{92}\mathrm{U}$ nucleus decays via α decay to a nucleus containing how many neutrons?

11. Describe, in as many ways as you can, the difference between α, β, and γ rays.

12. Fill in the missing particle or nucleus:
(*a*) $^{45}_{20}\mathrm{Ca} \rightarrow\ ? + \mathrm{e}^- + \bar{\nu}$
(*b*) $^{58}_{29}\mathrm{Cu}^* \rightarrow\ ? + \gamma$
(*c*) $^{46}_{24}\mathrm{Cr} \rightarrow {}^{46}_{23}\mathrm{V} + ?$
(*d*) $^{234}_{94}\mathrm{Pu} \rightarrow\ ? + \alpha$
(*e*) $^{239}_{93}\mathrm{Np} \rightarrow {}^{239}_{94}\mathrm{Pu} + ?$

13. Immediately after a $^{238}_{92}\mathrm{U}$ nucleus decays to $^{234}_{90}\mathrm{Th} + {}^{4}_{2}\mathrm{He}$, the daughter thorium nucleus may still have 92 electrons circling it. Since thorium normally holds only 90 electrons, what do you suppose happens to the two extra ones?

14. When a nucleus undergoes either β^- or β^+ decay, what happens to the energy levels of the atomic electrons? What is likely to happen to these electrons following the decay?

15. The alpha particles from a given alpha-emitting nuclide are generally monoenergetic; that is, they all have the same kinetic energy. But the beta particles from a beta-emitting nuclide have a spectrum of energies. Explain the difference between these two cases.

16. Do isotopes that undergo electron capture generally lie above or below the stable nuclides in Fig. 30–2?

17. Can hydrogen or deuterium emit an α particle? Explain.

18. Why are many artificially produced radioactive isotopes rare in nature?

19. An isotope has a half-life of one month. After two months, will a given sample of this isotope have completely decayed? If not, how much remains?

20. Why are none of the elements with $Z > 92$ stable?

21. A proton strikes a ${}^{6}_{3}\text{Li}$ nucleus. As a result, an α particle and another particle are released. What is the other particle?

22. Can ${}^{14}_{6}\text{C}$ dating be used to measure the age of stone walls and tablets of ancient civilizations? Explain.

23. Explain the absence of β^+ emitters in the radioactive decay series of Fig. 30–11.

24. As ${}^{222}_{86}\text{Rn}$ decays into ${}^{206}_{82}\text{Pb}$, how many alpha and beta particles are emitted? Does it matter which path in the decay series is chosen? Why or why not?

25. A ${}^{238}\text{U}$ nucleus (initially at rest) decays into a ${}^{234}\text{Th}$ nucleus and an alpha particle. Which has the greater (i) momentum, (ii) velocity, (iii) kinetic energy? Explain.
(*a*) The ${}^{234}\text{Th}$ nucleus.
(*b*) The alpha particle.
(*c*) Both the same.

MisConceptual Questions

1. Elements of the Periodic Table are distinguished by
(*a*) the number of protons in the nucleus.
(*b*) the number of neutrons in the nucleus.
(*c*) the number of electrons in the atom.
(*d*) Both (*a*) and (*b*).
(*e*) (*a*), (*b*), and (*c*).

2. A nucleus has
(*a*) more energy than its component neutrons and protons have.
(*b*) less energy than its component neutrons and protons have.
(*c*) the same energy as its component neutrons and protons have.
(*d*) more energy than its component neutrons and protons have when the nucleus is at rest but less energy than when it is moving.

3. Which of the following will generally create a more stable nucleus?
(*a*) Having more nucleons.
(*b*) Having more protons than neutrons.
(*c*) Having a larger binding energy per nucleon.
(*d*) Having the same number of electrons as protons.
(*e*) Having a larger total binding energy.

4. There are 82 protons in a lead nucleus. Why doesn't the lead nucleus burst apart?
(*a*) Coulomb repulsive force doesn't act inside the nucleus.
(*b*) Gravity overpowers the Coulomb repulsive force inside the nucleus.
(*c*) The negatively charged neutrons balance the positively charged protons.
(*d*) Protons lose their positive charge inside the nucleus.
(*e*) The strong nuclear force holds the nucleus together.

5. The half-life of a radioactive nucleus is
(*a*) half the time it takes for the entire substance to decay.
(*b*) the time it takes for half of the substance to decay.
(*c*) the same as the decay constant.
(*d*) Both (*a*) and (*b*) (they are the same).
(*e*) All of the above.

6. As a radioactive sample decays,
(*a*) the half-life increases.
(*b*) the half-life decreases.
(*c*) the activity remains the same.
(*d*) the number of radioactive nuclei increases.
(*e*) None of the above.

7. If the half-life of a radioactive sample is 10 years, then it should take _____ years for the sample to decay completely.
(*a*) 10. (*b*) 20. (*c*) 40.
(*d*) Cannot be determined.

8. A sample's half-life is 1 day. What fraction of the original sample will have decayed after 3 days?
(*a*) $\frac{1}{8}\cdot$ (*b*) $\frac{1}{4}\cdot$ (*c*) $\frac{1}{2}\cdot$ (*d*) $\frac{3}{4}\cdot$ (*e*) $\frac{7}{8}\cdot$ (*f*) All of it.

9. After three half-lives, what fraction of the original radioactive material is left?
(*a*) None. (*b*) $\frac{1}{16}\cdot$ (*c*) $\frac{1}{8}\cdot$ (*d*) $\frac{1}{4}\cdot$ (*e*) $\frac{3}{4}\cdot$ (*f*) $\frac{7}{8}\cdot$

10. Technetium ${}^{98}_{43}\text{Tc}$ has a half-life of 4.2×10^6 yr. Strontium ${}^{90}_{38}\text{Sr}$ has a half-life of 28.79 yr. Which statements are true?
(*a*) The decay constant of Sr is greater than the decay constant of Tc.
(*b*) The activity of 100 g of Sr is less than the activity of 100 g of Tc.
(*c*) The long half-life of Tc means that it decays by alpha decay.
(*d*) A Tc atom has a higher probability of decaying in 1 yr than a Sr atom.
(*e*) 28.79 g of Sr has the same activity as 4.2×10^6 g of Tc.

11. A material having which decay constant would have the shortest half-life?
(*a*) 100/second.
(*b*) 5/year.
(*c*) 8/century.
(*d*) 10^9/day.

12. Uranium-238 decays to lead-206 through a series of
(*a*) alpha decays.
(*b*) beta decays.
(*c*) gamma decays.
(*d*) some combination of alpha, beta, and gamma decays.

13. Carbon dating is useful only for determining the age of objects less than about _____ years old.
(*a*) 4.5 million.
(*b*) 1.2 million.
(*c*) 600,000.
(*d*) 60,000.
(*e*) 6000.

14. Radon has a half-life of about 1600 years. The Earth is several billion years old, so why do we still find radon on this planet?
(*a*) Ice-age temperatures preserved some of it.
(*b*) Heavier unstable isotopes decay into it.
(*c*) It is created in lightning strikes.
(*d*) It is replenished by cosmic rays.
(*e*) Its half-life has increased over time.
(*f*) Its half-life has decreased over time.

15. How does an atom's nucleus stay together and remain stable?
(*a*) The attractive gravitational force between the protons and neutrons overcomes the repulsive electrostatic force between the protons.
(*b*) Having just the right number of neutrons overcomes the electrostatic force between the protons.
(*c*) A strong covalent bond develops between the neutrons and protons, because they are so close to each other.
(*d*) None of the above.

16. What has greater mass?
(*a*) A neutron and a proton that are far from each other (unbound).
(*b*) A neutron and a proton that are bound together in a hydrogen (deuterium) nucleus.
(*c*) Both the same.

For assigned homework and other learning materials, go to the MasteringPhysics website.

Problems

[See Appendix B for masses]

30–1 Nuclear Properties

1. (I) A pi meson has a mass of 139 MeV/c^2. What is this in atomic mass units?

2. (I) What is the approximate radius of an α particle ($^4_2$He)?

3. (I) By what % is the radius of $^{238}_{92}$U greater than the radius of $^{232}_{92}$U?

4. (II) (*a*) What is the approximate radius of a $^{112}_{48}$Cd nucleus? (*b*) Approximately what is the value of A for a nucleus whose radius is 3.7×10^{-15} m?

5. (II) What is the mass of a bare α particle (without electrons) in MeV/c^2?

6. (II) Suppose two alpha particles were held together so they were just "touching" (use Eq. 30–1). Estimate the electrostatic repulsive force each would exert on the other. What would be the acceleration of an alpha particle subjected to this force?

7. (II) (*a*) What would be the radius of the Earth if it had its actual mass but had the density of nuclei? (*b*) By what factor would the radius of a $^{238}_{92}$U nucleus increase if it had the Earth's density?

8. (II) What stable nucleus has approximately half the radius of a uranium nucleus? [*Hint*: Find A and use Appendix B to get Z.]

9. (II) If an alpha particle were released from rest near the surface of a $^{257}_{100}$Fm nucleus, what would its kinetic energy be when far away?

10. (II) (*a*) What is the fraction of the hydrogen atom's mass ($^1_1$H) that is in the nucleus? (*b*) What is the fraction of the hydrogen atom's volume that is occupied by the nucleus?

11. (II) Approximately how many nucleons are there in a 1.0-kg object? Does it matter what the object is made of? Why or why not?

12. (III) How much kinetic energy, in MeV, must an α particle have to just "touch" the surface of a $^{232}_{92}$U nucleus?

30–2 Binding Energy

13. (I) Estimate the total binding energy for $^{63}_{29}$Cu, using Fig. 30–1.

14. (I) Use Fig. 30–1 to estimate the total binding energy of (*a*) $^{238}_{92}$U, and (*b*) $^{84}_{36}$Kr.

15. (II) Calculate the binding energy per nucleon for a $^{15}_{7}$N nucleus, using Appendix B.

16. (II) Use Appendix B to calculate the binding energy of $^2_1$H (deuterium).

17. (II) Determine the binding energy of the last neutron in a $^{23}_{11}$Na nucleus.

18. (II) Calculate the total binding energy, and the binding energy per nucleon, for (*a*) $^7_3$Li, (*b*) $^{195}_{78}$Pt. Use Appendix B.

19. (II) Compare the average binding energy of a nucleon in $^{23}_{11}$Na to that in $^{24}_{11}$Na, using Appendix B.

20. (III) How much energy is required to remove (*a*) a proton, (*b*) a neutron, from $^{15}_{7}$N? Explain the difference in your answers.

21. (III) (*a*) Show that the nucleus $^8_4$Be (mass = 8.005305 u) is unstable and will decay into two α particles. (*b*) Is $^{12}_{6}$C stable against decay into three α particles? Show why or why not.

30–3 to 30–7 Radioactive Decay

22. (I) The $^7_3$Li nucleus has an excited state 0.48 MeV above the ground state. What wavelength gamma photon is emitted when the nucleus decays from the excited state to the ground state?

23. (II) Show that the decay $^{11}_{6}\text{C} \rightarrow {}^{10}_{5}\text{B} + \text{p}$ is not possible because energy would not be conserved.

24. (II) Calculate the energy released when tritium, $^3_1$H, decays by β^- emission.

25. (II) What is the maximum kinetic energy of an electron emitted in the β decay of a free neutron?

26. (II) Give the result of a calculation that shows whether or not the following decays are possible:
(*a*) $^{233}_{92}\text{U} \rightarrow {}^{232}_{92}\text{U} + \text{n}$;
(*b*) $^{14}_{7}\text{N} \rightarrow {}^{13}_{7}\text{N} + \text{n}$;
(*c*) $^{40}_{19}\text{K} \rightarrow {}^{39}_{19}\text{K} + \text{n}$.

27. (II) $^{24}_{11}$Na is radioactive. (*a*) Is it a β^- or β^+ emitter? (*b*) Write down the decay reaction, and estimate the maximum kinetic energy of the emitted β.

28. (II) A $^{238}_{92}$U nucleus emits an α particle with kinetic energy = 4.20 MeV. (*a*) What is the daughter nucleus, and (*b*) what is the approximate atomic mass (in u) of the daughter atom? Ignore recoil of the daughter nucleus.

29. (II) Calculate the maximum kinetic energy of the β particle emitted during the decay of $^{60}_{27}$Co.

30. (II) How much energy is released in electron capture by beryllium: $^{7}_{4}\text{Be} + e^- \rightarrow {}^{7}_{3}\text{Li} + \nu$?

31. (II) The isotope $^{218}_{84}$Po can decay by either α or β^- emission. What is the energy release in each case? The mass of $^{218}_{84}$Po is 218.008973 u.

32. (II) A photon with a wavelength of 1.15×10^{-13} m is ejected from an atom. Calculate its energy and explain why it is a γ ray from the nucleus or a photon from the atom.

33. (II) How much recoil energy does a $^{40}_{19}$K nucleus get when it emits a 1.46-MeV gamma ray?

34. (II) Determine the maximum kinetic energy of β^+ particles released when $^{11}_{6}$C decays to $^{11}_{5}$B. What is the maximum energy the neutrino can have? What is the minimum energy of each?

35. (III) Show that when a nucleus decays by β^+ decay, the total energy released is equal to

$$(M_P - M_D - 2m_e)c^2,$$

where M_P and M_D are the masses of the parent and daughter atoms (neutral), and m_e is the mass of an electron or positron.

36. (III) When $^{238}_{92}$U decays, the α particle emitted has 4.20 MeV of kinetic energy. Calculate the recoil kinetic energy of the daughter nucleus and the Q-value of the decay.

30–8 to 30–11 Half-Life, Decay Rates, Decay Series, Dating

37. (I) (*a*) What is the decay constant of $^{238}_{92}$U whose half-life is 4.5×10^9 yr? (*b*) The decay constant of a given nucleus is $3.2 \times 10^{-5}\ \text{s}^{-1}$. What is its half-life?

38. (I) A radioactive material produces 1120 decays per minute at one time, and 3.6 h later produces 140 decays per minute. What is its half-life?

39. (I) What fraction of a sample of $^{68}_{32}$Ge, whose half-life is about 9 months, will remain after 2.5 yr?

40. (I) What is the activity of a sample of $^{14}_{6}$C that contains 6.5×10^{20} nuclei?

41. (I) What fraction of a radioactive sample is left after exactly 5 half-lives?

42. (II) The iodine isotope $^{131}_{53}$I is used in hospitals for diagnosis of thyroid function. If 782 μg are ingested by a patient, determine the activity (*a*) immediately, (*b*) 1.50 h later when the thyroid is being tested, and (*c*) 3.0 months later. Use Appendix B.

43. (II) How many nuclei of $^{238}_{92}$U remain in a rock if the activity registers 420 decays per second?

44. (II) In a series of decays, the nuclide $^{235}_{92}$U becomes $^{207}_{82}$Pb. How many α and β^- particles are emitted in this series?

45. (II) $^{124}_{55}$Cs has a half-life of 30.8 s. (*a*) If we have 8.7 μg initially, how many Cs nuclei are present? (*b*) How many are present 2.6 min later? (*c*) What is the activity at this time? (*d*) After how much time will the activity drop to less than about 1 per second?

46. (II) Calculate the mass of a sample of pure $^{40}_{19}$K with an initial decay rate of $2.4 \times 10^5\ \text{s}^{-1}$. The half-life of $^{40}_{19}$K is 1.248×10^9 yr.

47. (II) Calculate the activity of a pure 6.7-μg sample of $^{32}_{15}$P $\left(T_{\frac{1}{2}} = 1.23 \times 10^6\ \text{s}\right)$.

48. (II) A sample of $^{233}_{92}$U $\left(T_{\frac{1}{2}} = 1.59 \times 10^5\ \text{yr}\right)$ contains 4.50×10^{18} nuclei. (*a*) What is the decay constant? (*b*) Approximately how many disintegrations will occur per minute?

49. (II) The activity of a sample drops by a factor of 6.0 in 9.4 minutes. What is its half-life?

50. (II) **Rubidium–strontium dating**. The rubidium isotope $^{87}_{37}$Rb, a β emitter with a half-life of 4.75×10^{10} yr, is used to determine the age of rocks and fossils. Rocks containing fossils of ancient animals contain a ratio of $^{87}_{38}$Sr to $^{87}_{37}$Rb of 0.0260. Assuming that there was no $^{87}_{38}$Sr present when the rocks were formed, estimate the age of these fossils.

51. (II) Two of the naturally occurring radioactive decay sequences start with $^{232}_{90}$Th and with $^{235}_{92}$U. The first five decays of these two sequences are:

$$\alpha,\ \beta,\ \beta,\ \alpha,\ \alpha$$

and

$$\alpha,\ \beta,\ \alpha,\ \beta,\ \alpha.$$

Determine the resulting intermediate daughter nuclei in each case.

52. (II) An ancient wooden club is found that contains 73 g of carbon and has an activity of 7.0 decays per second. Determine its age assuming that in living trees the ratio of $^{14}\text{C}/^{12}\text{C}$ atoms is about 1.3×10^{-12}.

53. (II) Use Fig. 30–11 and calculate the relative decay rates for α decay of $^{218}_{84}$Po and $^{214}_{84}$Po.

54. (III) The activity of a radioactive source decreases by 5.5% in 31.0 hours. What is the half-life of this source?

55. (III) $^{7}_{4}$Be decays with a half-life of about 53 d. It is produced in the upper atmosphere, and filters down onto the Earth's surface. If a plant leaf is detected to have 350 decays/s of $^{7}_{4}$Be, (*a*) how long do we have to wait for the decay rate to drop to 25 per second? (*b*) Estimate the initial mass of $^{7}_{4}$Be on the leaf.

56. (III) At $t = 0$, a pure sample of radioactive nuclei contains N_0 nuclei whose decay constant is λ. Determine a formula for the number of daughter nuclei, N_D, as a function of time; assume the daughter is stable and that $N_D = 0$ at $t = 0$.

General Problems

57. Which radioactive isotope of lead is being produced if the measured activity of a sample drops to 1.050% of its original activity in 4.00 h?

58. An old wooden tool is found to contain only 4.5% of the $^{14}_{6}$C that an equal mass of fresh wood would. How old is the tool?

59. A neutron star consists of neutrons at approximately nuclear density. Estimate, for a 10-km-diameter neutron star, (*a*) its mass number, (*b*) its mass (kg), and (*c*) the acceleration of gravity at its surface.

60. **Tritium dating**. The $^{3}_{1}$H isotope of hydrogen, which is called *tritium* (because it contains three nucleons), has a half-life of 12.3 yr. It can be used to measure the age of objects up to about 100 yr. It is produced in the upper atmosphere by cosmic rays and brought to Earth by rain. As an application, determine approximately the age of a bottle of wine whose $^{3}_{1}$H radiation is about $\frac{1}{10}$ that present in new wine.

61. Some elementary particle theories (Section 32–11) suggest that the proton may be unstable, with a half-life $\geq 10^{33}$ yr. (*a*) How long would you expect to wait for one proton in your body to decay (approximate your body as all water)? (*b*) Of the roughly 7 billion people on Earth, about how many would have a proton in their body decay in a 70 yr lifetime?

62. How long must you wait (in half-lives) for a radioactive sample to drop to 2.00% of its original activity?

63. If the potassium isotope $^{40}_{19}$K gives 42 decays/s in a liter of milk, estimate how much $^{40}_{19}$K and regular $^{39}_{19}$K are in a liter of milk. Use Appendix B.

64. Strontium-90 is produced as a nuclear fission product of uranium in both reactors and atomic bombs. Look at its location in the Periodic Table to see what other elements it might be similar to chemically, and tell why you think it might be dangerous to ingest. It has too many neutrons to be stable, and it decays with a half-life of about 29 yr. How long will we have to wait for the amount of $^{90}_{38}$Sr on the Earth's surface to reach 1% of its current level, assuming no new material is scattered about? Write down the decay reaction, including the daughter nucleus. The daughter is radioactive: write down its decay.

65. The activity of a sample of $^{35}_{16}$S $\left(T_{\frac{1}{2}} = 87.37 \text{ days}\right)$ is 4.28×10^{4} decays per second. What is the mass of the sample?

66. The nuclide $^{191}_{76}$Os decays with β^- energy of 0.14 MeV accompanied by γ rays of energy 0.042 MeV and 0.129 MeV. (*a*) What is the daughter nucleus? (*b*) Draw an energy-level diagram showing the ground states of the parent and daughter and excited states of the daughter. (*c*) To which of the daughter states does β^- decay of $^{191}_{76}$Os occur?

67. Determine the activities of (*a*) 1.0 g of $^{131}_{53}$I $\left(T_{\frac{1}{2}} = 8.02 \text{ days}\right)$ and (*b*) 1.0 g of $^{238}_{92}$U $\left(T_{\frac{1}{2}} = 4.47 \times 10^{9} \text{ yr}\right)$.

68. Use Fig. 30–1 to estimate the total binding energy for copper and then estimate the energy, in joules, needed to break a 3.0-g copper penny into its constituent nucleons.

69. Instead of giving atomic masses for nuclides as in Appendix B, some Tables give the **mass excess**, Δ, defined as $\Delta = M - A$, where A is the atomic mass number and M is the mass in u. Determine the mass excess, in u and in MeV/c^2, for: (*a*) $^{4}_{2}$He; (*b*) $^{12}_{6}$C; (*c*) $^{86}_{38}$Sr; (*d*) $^{235}_{92}$U. (*e*) From a glance at Appendix B, can you make a generalization about the sign of Δ as a function of Z or A?

70. When water is placed near an intense neutron source, the neutrons can be slowed down to almost zero speed by collisions with the water molecules, and are eventually captured by a hydrogen nucleus to form the stable isotope called **deuterium**, $^{2}_{1}$H, giving off a gamma ray. What is the energy of the gamma ray?

71. The practical limit for carbon-14 dating is about 60,000 years. If a bone contains 1.0 kg of carbon, and the animal died 60,000 years ago, what is the activity today?

72. Using Section 30–2 and Appendix B, determine the energy required to remove one neutron from $^{4}_{2}$He. How many times greater is this energy than the binding energy of the last neutron in $^{13}_{6}$C?

73. (*a*) If all of the atoms of the Earth were to collapse and simply become nuclei, what would be the Earth's new radius? (*b*) If all of the atoms of the Sun were to collapse and simply become nuclei, what would be the Sun's new radius?

74. (*a*) A 72-gram sample of natural carbon contains the usual fraction of $^{14}_{6}$C. Estimate roughly how long it will take before there is only one $^{14}_{6}$C nucleus left. (*b*) How does the answer in (*a*) change if the sample is 340 grams? What does this tell you about the limits of carbon dating?

75. If the mass of the proton were just a little closer to the mass of the neutron, the following reaction would be possible even at low collision energies:

$$e^- + p \rightarrow n + \nu.$$

(*a*) Why would this situation be catastrophic? (See last paragraph of Chapter 33.) (*b*) By what percentage would the proton's mass have to be increased to make this reaction possible?

76. What is the ratio of the kinetic energies for an alpha particle and a beta particle if both make tracks with the same radius of curvature in a magnetic field, oriented perpendicular to the paths of the particles?

77. Almost all of naturally occurring uranium is $^{238}_{92}$U with a half-life of 4.468×10^{9} yr. Most of the rest of natural uranium is $^{235}_{92}$U with a half-life of 7.04×10^{8} yr. Today a sample contains 0.720% $^{235}_{92}$U. (*a*) What was this percentage 1.0 billion years ago? (*b*) What percentage of uranium will be $^{235}_{92}$U 100 million years from now?

78. A banana contains about 420 mg of potassium, of which a small fraction is the radioactive isotope $^{40}_{19}$K (Appendix B). Estimate the activity of an average banana due to $^{40}_{19}$K.

79. When $^{23}_{10}$Ne (mass = 22.9947 u) decays to $^{23}_{11}$Na (mass = 22.9898 u), what is the maximum kinetic energy of the emitted electron? What is its minimum energy? What is the energy of the neutrino in each case? Ignore recoil of the daughter nucleus.

80. (*a*) In α decay of, say, a $^{226}_{88}$Ra nucleus, show that the nucleus carries away a fraction $1/(1 + \frac{1}{4}A_D)$ of the total energy available, where A_D is the mass number of the daughter nucleus. [*Hint*: Use conservation of momentum as well as conservation of energy.] (*b*) Approximately what percentage of the energy available is thus carried off by the α particle when $^{226}_{88}$Ra decays?

Search and Learn

1. Describe in detail why we think there is a strong nuclear force.

2. (*a*) Under what circumstances could a fermium nucleus decay into an einsteinium nucleus? (*b*) What about the reverse, an Es nucleus decaying into Fm?

3. Using the uncertainty principle and the radius of a nucleus, estimate the minimum possible kinetic energy of a nucleon in, say, iron. Ignore relativistic corrections. [*Hint*: A particle can have a momentum at least as large as its momentum uncertainty.]

4. In Fig. 30–17, a nucleus decays and emits a particle that enters a region with a uniform magnetic field of 0.012 T directed into the page. The path of the detected particle is shown. (*a*) What type of radioactive decay is this? (*b*) If the radius of the circular arc is 4.7 mm, what is the velocity of the particle?

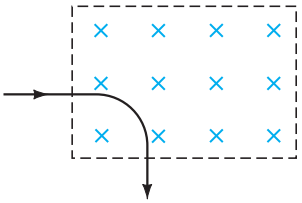


FIGURE 30–17
Search and Learn 4.

5. In both internal conversion and β decay, an electron is emitted. How could you determine which decay process occurred?

6. In 1991, the frozen remains of a Neolithic-age man, nicknamed Otzi, were found in the Italian Alps by hikers. The body was well preserved, as were his bow, arrows, knife, axe, other tools, and clothing. The date of his death can be determined using carbon-14 dating. (*a*) What is the decay constant for $^{14}_{6}$C? (*b*) How many $^{14}_{6}$C atoms per gram of $^{12}_{6}$C are there in a living organism? (*c*) What is the activity per gram in naturally occurring carbon for a living organism? (*d*) For Otzi, the activity per gram of carbon was measured to be 0.121. How long ago did he live?

7. Some radioactive isotopes have half-lives that are greater than the age of the universe (like gadolinium or samarium). The only way to determine these half-lives is to monitor the decay rate of a sample that contains these isotopes. For example, suppose we find an asteroid that currently contains about 15,000 kg of $^{152}_{64}$Gd (gadolinium) and we detect an activity of 1 decay/s. Estimate the half-life of gadolinium (in years).

ANSWERS TO EXERCISES

A: 0.042130 u.
B: 7.98 MeV/nucleon.
C: (*b*) Gd.
D: (*b*) $\frac{1}{8}\,\mu$g.
E: (*b*) 20 μg.



Technicians are looking at an MRI image of sections through a patient's body. MRI is one of several powerful types of medical imaging based on physics used by doctors to diagnose illnesses.

This Chapter opens with basic and important physics topics of nuclear reactions, nuclear fission, and nuclear fusion, and how we obtain nuclear energy. Then we examine the health aspects of radiation—dosimetry, therapy, and imaging: MRI, PET, and SPECT.

CHAPTER 31

Nuclear Energy; Effects and Uses of Radiation

CHAPTER-OPENING QUESTION—Guess now!

The Sun is powered by

(a) nuclear alpha decay.
(b) nuclear beta decay.
(c) nuclear gamma decay.
(d) nuclear fission.
(e) nuclear fusion.

CONTENTS

31–1 Nuclear Reactions and the Transmutation of Elements
31–2 Nuclear Fission; Nuclear Reactors
31–3 Nuclear Fusion
31–4 Passage of Radiation Through Matter; Biological Damage
31–5 Measurement of Radiation—Dosimetry
*31–6 Radiation Therapy
*31–7 Tracers in Research and Medicine
*31–8 Emission Tomography: PET and SPECT
31–9 Nuclear Magnetic Resonance (NMR) and Magnetic Resonance Imaging (MRI)

We continue our study of nuclear physics in this Chapter. We begin with a discussion of nuclear reactions, and then we examine the important huge energy-releasing processes of fission and fusion. We also deal with the effects of nuclear radiation passing through matter, particularly biological matter, and how radiation is used medically for therapy, diagnosis, and imaging techniques.

31–1 Nuclear Reactions and the Transmutation of Elements

When a nucleus undergoes α or β decay, the daughter nucleus is a different element from the parent. The transformation of one element into another, called **transmutation**, also occurs via nuclear reactions. A **nuclear reaction** is said to occur when a nucleus is struck by another nucleus, or by a simpler particle such as a γ ray, neutron, or proton, and an interaction takes place. Ernest Rutherford was the first to report seeing a nuclear reaction. In 1919 he observed that some of the α particles passing through nitrogen gas were absorbed and protons emitted. He concluded that nitrogen nuclei had been transformed into oxygen nuclei via the reaction

$$^{4}_{2}\text{He} + {}^{14}_{7}\text{N} \rightarrow {}^{17}_{8}\text{O} + {}^{1}_{1}\text{H},$$

where $^{4}_{2}\text{He}$ is an α particle, and $^{1}_{1}\text{H}$ is a proton.

Since then, a great many nuclear reactions have been observed. Indeed, many of the radioactive isotopes used in the laboratory are made by means of nuclear reactions. Nuclear reactions can be made to occur in the laboratory, but they also occur regularly in nature. In Chapter 30 we saw an example: ${}^{14}_{6}\text{C}$ is continually being made in the atmosphere via the reaction $\text{n} + {}^{14}_{7}\text{N} \rightarrow {}^{14}_{6}\text{C} + \text{p}$.

Nuclear reactions are sometimes written in a shortened form: for example,

$$\text{n} + {}^{14}_{7}\text{N} \rightarrow {}^{14}_{6}\text{C} + \text{p}$$

can be written

$${}^{14}_{7}\text{N}\ (\text{n},\ \text{p})\ {}^{14}_{6}\text{C}.$$

The symbols outside the parentheses on the left and right represent the initial and final nuclei, respectively. The symbols inside the parentheses represent the bombarding particle (first) and the emitted small particle (second).

In any nuclear reaction, both electric charge and nucleon number are conserved. These conservation laws are often useful, as the following Example shows.

CONCEPTUAL EXAMPLE 31–1 **Deuterium reaction.** A neutron is observed to strike an ${}^{16}_{8}\text{O}$ nucleus, and a deuteron is given off. (A **deuteron**, or **deuterium**, is the isotope of hydrogen containing one proton and one neutron, ${}^{2}_{1}\text{H}$; it is sometimes given the symbol d or D.) What is the nucleus that results?

RESPONSE We have the reaction $\text{n} + {}^{16}_{8}\text{O} \rightarrow ? + {}^{2}_{1}\text{H}$. The total number of nucleons initially is $1 + 16 = 17$, and the total charge is $0 + 8 = 8$. The same totals apply after the reaction. Hence the product nucleus must have $Z = 7$ and $A = 15$. From the Periodic Table, we find that it is nitrogen that has $Z = 7$, so the nucleus produced is ${}^{15}_{7}\text{N}$.

EXERCISE A Determine the resulting nucleus in the reaction $\text{n} + {}^{137}_{56}\text{Ba} \rightarrow ? + \gamma$.

Energy and momentum are also conserved in nuclear reactions, and can be used to determine whether or not a given reaction can occur. For example, if the total mass of the final products is less than the total mass of the initial particles, this decrease in mass (recall $\Delta E = \Delta m\, c^2$) is converted to kinetic energy (KE) of the outgoing particles. But if the total mass of the products is greater than the total mass of the initial reactants, the reaction requires energy. The reaction will then not occur unless the bombarding particle has sufficient kinetic energy. Consider a nuclear reaction of the general form

$$\text{a} + \text{X} \rightarrow \text{Y} + \text{b}, \tag{31–1}$$

where particle a is a moving projectile particle (or small nucleus) that strikes nucleus X, producing nucleus Y and particle b (typically, p, n, α, γ). We define the **reaction energy**, or ***Q*-value**, in terms of the masses involved, as

$$Q = (M_\text{a} + M_\text{X} - M_\text{b} - M_\text{Y})c^2. \tag{31–2a}$$

For a γ ray, $M = 0$. If energy is released by the reaction, $Q > 0$. If energy is required, $Q < 0$.

Because energy is conserved, Q has to be equal to the change in kinetic energy (final minus initial):

$$Q = \text{KE}_\text{b} + \text{KE}_\text{Y} - \text{KE}_\text{a} - \text{KE}_\text{X}. \tag{31–2b}$$

If X is a target nucleus at rest (or nearly so) struck by incoming particle a, then $\text{KE}_\text{X} = 0$. For $Q > 0$, the reaction is said to be *exothermic* or *exoergic*; energy is released in the reaction, so the total kinetic energy is greater after the reaction than before. If Q is negative, the reaction is said to be *endothermic* or *endoergic*: an energy input is required to make the reaction happen. The energy input comes from the kinetic energy of the initial colliding particles (a and X).

EXAMPLE 31–2 **A slow-neutron reaction.** The nuclear reaction

$$\text{n} + {}^{10}_{5}\text{B} \rightarrow {}^{7}_{3}\text{Li} + {}^{4}_{2}\text{He}$$

is observed to occur even when very slow-moving neutrons (mass $M_\text{n} = 1.0087\ \text{u}$) strike boron atoms at rest. For a particular reaction in which $\text{KE}_\text{n} \approx 0$, the outgoing helium ($M_\text{He} = 4.0026\ \text{u}$) is observed to have a speed of $9.30 \times 10^6\ \text{m/s}$. Determine (*a*) the kinetic energy of the lithium ($M_\text{Li} = 7.0160\ \text{u}$), and (*b*) the Q-value of the reaction.

APPROACH Since the neutron and boron are both essentially at rest, the total momentum before the reaction is zero; momentum is conserved and so must be zero afterward as well. Thus,

$$M_{\text{Li}}\, v_{\text{Li}} = M_{\text{He}}\, v_{\text{He}}.$$

We solve this for v_{Li} and substitute it into the equation for kinetic energy. In (*b*) we use Eq. 31–2b.

SOLUTION (*a*) We can use classical kinetic energy with little error, rather than relativistic formulas, because $v_{\text{He}} = 9.30 \times 10^6\ \text{m/s}$ is not close to the speed of light *c*. And v_{Li} will be even less because $M_{\text{Li}} > M_{\text{He}}$. Thus we can write the KE of the lithium, using the momentum equation just above, as

$$\text{KE}_{\text{Li}} = \frac{1}{2} M_{\text{Li}} v_{\text{Li}}^2 = \frac{1}{2} M_{\text{Li}} \left(\frac{M_{\text{He}}\, v_{\text{He}}}{M_{\text{Li}}} \right)^2 = \frac{M_{\text{He}}^2\, v_{\text{He}}^2}{2 M_{\text{Li}}}.$$

We put in numbers, changing the mass in u to kg and recall that $1.60 \times 10^{-13}\ \text{J} = 1\ \text{MeV}$:

$$\text{KE}_{\text{Li}} = \frac{(4.0026\ \text{u})^2 (1.66 \times 10^{-27}\ \text{kg/u})^2 (9.30 \times 10^6\ \text{m/s})^2}{2(7.0160\ \text{u})(1.66 \times 10^{-27}\ \text{kg/u})}$$

$$= 1.64 \times 10^{-13}\ \text{J} = 1.02\ \text{MeV}.$$

(*b*) We are given the data $\text{KE}_{\text{a}} = \text{KE}_{\text{X}} = 0$ in Eq. 31–2b, so $Q = \text{KE}_{\text{Li}} + \text{KE}_{\text{He}}$, where

$$\text{KE}_{\text{He}} = \tfrac{1}{2} M_{\text{He}}\, v_{\text{He}}^2 = \tfrac{1}{2} (4.0026\ \text{u})(1.66 \times 10^{-27}\ \text{kg/u})(9.30 \times 10^6\ \text{m/s})^2$$

$$= 2.87 \times 10^{-13}\ \text{J} = 1.80\ \text{MeV}.$$

Hence, $Q = 1.02\ \text{MeV} + 1.80\ \text{MeV} = 2.82\ \text{MeV}$.

EXAMPLE 31–3 Will the reaction "go"? Can the reaction

$$\text{p} + {}^{13}_{6}\text{C} \rightarrow {}^{13}_{7}\text{N} + \text{n}$$

occur when ${}^{13}_{6}\text{C}$ is bombarded by 2.0-MeV protons?

APPROACH The reaction will "go" if the reaction is exothermic ($Q > 0$) and even if $Q < 0$ if the input momentum and kinetic energy are sufficient. First we calculate Q from the difference between final and initial masses using Eq. 31–2a, and look up the masses in Appendix B.

SOLUTION The total masses before and after the reaction are:

Before		After	
$M({}^{13}_{6}\text{C})$	13.003355	$M({}^{13}_{7}\text{N})$	13.005739
$M({}^{1}_{1}\text{H})$	1.007825	$M(\text{n})$	1.008665
	14.011180		14.014404

(We must use the mass of the ${}^{1}_{1}\text{H}$ atom rather than that of the bare proton because the masses of ${}^{13}_{6}\text{C}$ and ${}^{13}_{7}\text{N}$ include the electrons, and we must include an equal number of electron masses on each side of the equation.) The products have an excess mass of

$$(14.014404 - 14.011180)\text{u} = 0.003224\ \text{u} \times 931.5\ \text{MeV/u} = 3.00\ \text{MeV}.$$

Thus $Q = -3.00\ \text{MeV}$, and the reaction is endothermic. This reaction requires energy, and the 2.0-MeV protons do not have enough to make it go.

NOTE The incoming proton in this Example would need more than 3.00 MeV of kinetic energy to make this reaction go; 3.00 MeV would be enough to conserve energy, but a proton of this energy would produce the ${}^{13}_{7}\text{N}$ and n with no kinetic energy and hence no momentum. Since an incident 3.0-MeV proton has momentum, conservation of momentum would be violated. A calculation using conservation of energy *and* of momentum, as we did in Examples 30–7 and 31–2, shows that the minimum proton energy, called the **threshold energy**, is 3.23 MeV in this case.

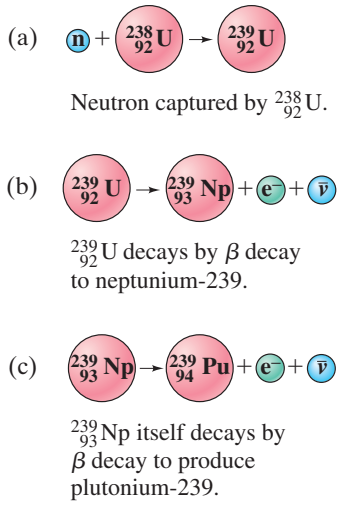


FIGURE 31–1 Neptunium and plutonium are produced in this series of reactions, after bombardment of $^{238}_{92}\text{U}$ by neutrons.

Neutron Physics

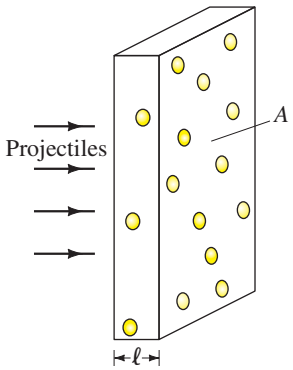
The artificial transmutation of elements took a great leap forward in the 1930s when Enrico Fermi realized that neutrons would be the most effective projectiles for causing nuclear reactions and in particular for producing new elements. Because neutrons have no net electric charge, they are not repelled by positively charged nuclei as are protons or alpha particles. Hence the probability of a neutron reaching the nucleus and causing a reaction is much greater than for charged projectiles,[†] particularly at low energies. Between 1934 and 1936, Fermi and his co-workers in Rome produced many previously unknown isotopes by bombarding different elements with neutrons. Fermi realized that if the heaviest known element, uranium, is bombarded with neutrons, it might be possible to produce new elements with atomic numbers greater than that of uranium. After several years of hard work, it was suspected that two new elements had been produced, neptunium ($Z = 93$) and plutonium ($Z = 94$). The full confirmation that such "transuranic" elements could be produced came several years later at the University of California, Berkeley. The reactions are shown in Fig. 31–1.

It was soon shown that what Fermi had actually observed when he bombarded uranium was an even stranger process—one that was destined to play an extraordinary role in the world at large. We discuss it in Section 31–2.

*Cross Section

Some reactions have a higher probability of occurring than others. The reaction probability is specified by a quantity called the collision **cross section**. Although the size of a nucleus, like that of an atom, is not a clearly defined quantity since the edges are not distinct like those of a tennis ball or baseball, we can nonetheless define a *cross section* for nuclei undergoing collisions by using an analogy. Suppose that projectile particles strike a stationary target of total area A and thickness ℓ, as shown in Fig. 31–2. Assume also that the target is made up of identical objects (such as marbles or nuclei), each of which has a cross-sectional area σ, and we assume the incoming projectiles are small by comparison. We assume that the target objects are fairly far apart and the thickness ℓ is so small that we don't have to worry about overlapping. This is often a reasonable assumption because nuclei have diameters on the order of 10^{-14} m but are at least 10^{-10} m (atomic size) apart even in solids. If there are n nuclei per unit volume, the total cross-sectional area of all these tiny targets is

FIGURE 31–2 Projectile particles strike a target of area A and thickness ℓ made up of n nuclei per unit volume.



$$A' = nA\ell\sigma$$

since $nA\ell = (n)(\text{volume})$ is the total number of targets and σ is the cross-sectional area of each. If $A' \ll A$, most of the incident projectile particles will pass through the target without colliding. If R_0 is the rate at which the projectile particles strike the target (number/second), the rate at which collisions occur, R, is

$$R = R_0 \frac{A'}{A} = R_0 \frac{nA\ell\sigma}{A}$$

so

$$R = R_0 n\ell\sigma.$$

Thus, by measuring the collision rate, R, we can determine σ:

$$\sigma = \frac{R}{R_0 n\ell}.$$

The cross section σ is an "effective" target area. It is a *measure of the probability of a collision or of a particular reaction occurring* per target nucleus, independent of the dimensions of the entire target. The concept of cross section is useful

[†]That is, positively charged particles. Electrons rarely cause nuclear reactions because they do not interact via the strong nuclear force.

because σ depends only on the properties of the interacting particles, whereas R depends on the thickness and area of the physical (macroscopic) target, on the number of particles in the incident beam, and so on.

31–2 Nuclear Fission; Nuclear Reactors

In 1938, the German scientists Otto Hahn and Fritz Strassmann made an amazing discovery. Following up on Fermi's work, they found that uranium bombarded by neutrons sometimes produced smaller nuclei that were roughly half the size of the original uranium nucleus. Lise Meitner and Otto Frisch quickly realized what had happened: the uranium nucleus, after absorbing a neutron, actually had split into two roughly equal pieces. This was startling, for until then the known nuclear reactions involved knocking out only a tiny fragment (for example, n, p, or α) from a nucleus.

Nuclear Fission and Chain Reactions

This new phenomenon was named **nuclear fission** because of its resemblance to biological fission (cell division). It occurs much more readily for $^{235}_{92}\text{U}$ than for the more common $^{238}_{92}\text{U}$. The process can be visualized by imagining the uranium nucleus to be like a liquid drop. According to this **liquid-drop model**, the neutron absorbed by the $^{235}_{92}\text{U}$ nucleus (Fig. 31–3a) gives the nucleus extra internal energy (like heating a drop of water). This intermediate state, or **compound nucleus**, is $^{236}_{92}\text{U}$ (because of the absorbed neutron), Fig. 31–3b. The extra energy of this nucleus—it is in an excited state—appears as increased motion of the individual nucleons inside, which causes the nucleus to take on abnormal elongated shapes. When the nucleus elongates (in this model) into the shape shown in Fig. 31–3c, the attraction of the two ends via the short-range nuclear force is greatly weakened by the increased separation distance. Then the electric repulsive force becomes dominant, and the nucleus splits in two (Fig. 31–3d). The two resulting nuclei, X_1 and X_2, are called **fission fragments**, and in the process a number of neutrons (typically two or three) are also given off. The reaction can be written

$$\text{n} + {}^{235}_{92}\text{U} \rightarrow {}^{236}_{92}\text{U} \rightarrow \text{X}_1 + \text{X}_2 + \text{neutrons}. \qquad \textbf{(31–3)}$$

The compound nucleus, $^{236}_{92}\text{U}$, exists for less than 10^{-12} s, so the process occurs very quickly. The two fission fragments, X_1 and X_2, rarely split the original uranium mass precisely half and half, but more often as about 40%–60%. A typical fission reaction is

$$\text{n} + {}^{235}_{92}\text{U} \rightarrow {}^{141}_{56}\text{Ba} + {}^{92}_{36}\text{Kr} + 3\text{n}, \qquad \textbf{(31–4)}$$

although many others also occur.

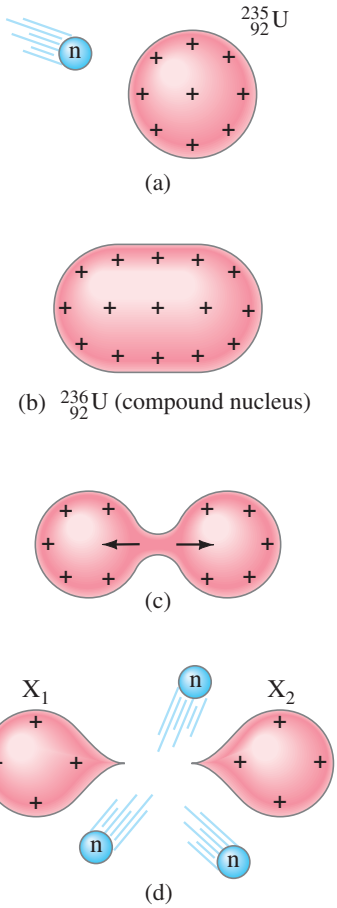


FIGURE 31–3 Fission of a $^{235}_{92}\text{U}$ nucleus after capture of a neutron, according to the liquid-drop model.

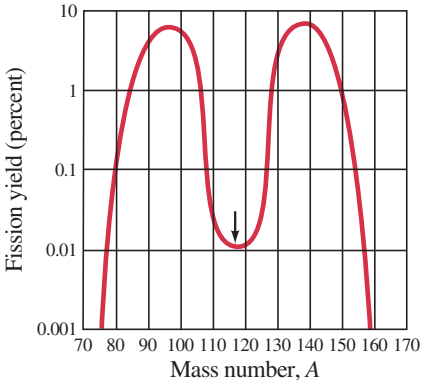
CONCEPTUAL EXAMPLE 31–4 **Counting nucleons.** Identify the element X in the fission reaction $\text{n} + {}^{235}_{92}\text{U} \rightarrow {}^{A}_{Z}\text{X} + {}^{93}_{38}\text{Sr} + 2\text{n}$.

RESPONSE The number of nucleons is conserved (Section 30–7). The uranium nucleus with 235 nucleons plus the incoming neutron make $235 + 1 = 236$ nucleons. So there must be 236 nucleons after the reaction. The Sr has 93 nucleons, and the two neutrons make 95 nucleons, so X has $A = 236 - 95 = 141$. Electric charge is also conserved: before the reaction, the total charge is $92e$. After the reaction the total charge is $(Z + 38)e$ and must equal $92e$. Thus $Z = 92 - 38 = 54$. The element with $Z = 54$ is xenon (see Appendix B or the Periodic Table inside the back cover), so the isotope is $^{141}_{54}\text{Xe}$.

EXERCISE B In the fission reaction $\text{n} + {}^{235}_{92}\text{U} \rightarrow {}^{137}_{53}\text{I} + {}^{96}_{39}\text{Y} + \text{neutrons}$, how many neutrons are produced?

Figure 31–4 shows the measured distribution of $^{235}_{92}\text{U}$ fission fragments according to mass. Only rarely (about 1 in 10^4) does a fission result in equal mass fragments (arrow in Fig. 31–4).

FIGURE 31–4 Mass distribution of fission fragments from $^{235}_{92}\text{U} + \text{n}$. The small arrow indicates equal mass fragments ($\frac{1}{2} \times (236 - 2) = 117$, assuming 2 neutrons are liberated). Note that the vertical scale is logarithmic.



A tremendous amount of energy is released in a fission reaction because the mass of $^{235}_{92}U$ is considerably greater than the total mass of the fission fragments plus released neutrons. This can be seen from the binding-energy-per-nucleon curve of Fig. 30–1; the binding energy per nucleon for uranium is about 7.6 MeV/nucleon, but for fission fragments that have intermediate mass (in the center portion of the graph, $A \approx 100$), the average binding energy per nucleon is about 8.5 MeV/nucleon. Since the fission fragments are more tightly bound, the sum of their masses is less than the mass of the uranium. The difference in mass, or energy, between the original uranium nucleus and the fission fragments is about $8.5 - 7.6 = 0.9$ MeV per nucleon. Because there are 236 nucleons involved in each fission, the total energy released per fission is

$$(0.9\,\text{MeV/nucleon})(236\,\text{nucleons}) \approx 200\,\text{MeV}. \qquad \textbf{(31–5)}$$

This is an enormous amount of energy for one single nuclear event. At a practical level, the energy from one fission is tiny. But if many such fissions could occur in a short time, an enormous amount of energy at the macroscopic level would be available. A number of physicists, including Fermi, recognized that the neutrons released in each fission (Eqs. 31–3 and 31–4) could be used to create a **chain reaction**. That is, one neutron initially causes one fission of a uranium nucleus; the two or three neutrons released can go on to cause additional fissions, so the process multiplies as shown schematically in Fig. 31–5.

If a **self-sustaining chain reaction** was actually possible in practice, the enormous energy available in fission could be released on a larger scale. Fermi and his co-workers (at the University of Chicago) showed it was possible by constructing the first **nuclear reactor** in 1942 (Fig. 31–6).

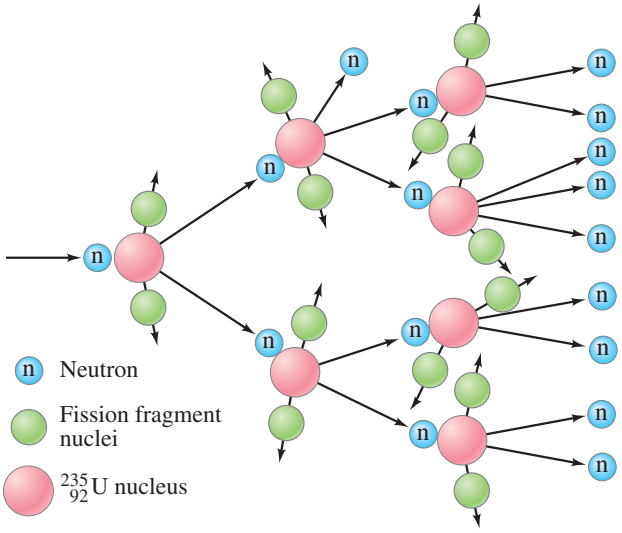


FIGURE 31–5 Chain reaction.

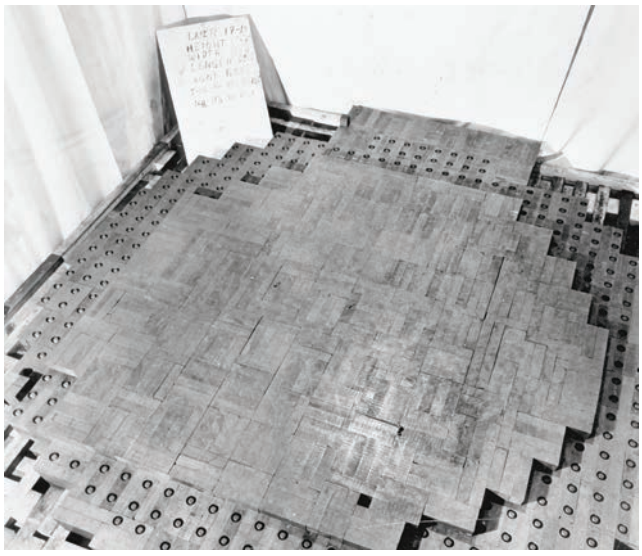


FIGURE 31–6 This is the only photograph of the first nuclear reactor, built by Fermi under the grandstand of Stagg Field at the University of Chicago. It is shown here under construction as a layer of graphite (used as moderator) was being placed over a layer of natural uranium. On December 2, 1942, Fermi slowly withdrew the cadmium control rods and the reactor went critical. This first self-sustaining chain reaction was announced to Washington, via telephone, by Arthur Compton who witnessed the event and reported: "The Italian navigator has just landed in the new world."

Nuclear Reactors

Several problems have to be overcome to make any nuclear reactor function. First, the probability that a $^{235}_{92}U$ nucleus will absorb a neutron is large only for slow neutrons, but the neutrons emitted during a fission (which are needed to sustain a chain reaction) are moving very fast. A substance known as a **moderator** must be used to slow down the neutrons. The most effective moderator will consist of atoms whose mass is as close as possible to that of the neutrons. (To see why this is true, recall from Chapter 7 that a billiard ball striking an equal mass ball at rest can itself be stopped in one collision; but a billiard ball striking a heavy object bounces off with nearly unchanged speed.) The best moderator would thus contain $^{1}_{1}H$ atoms. Unfortunately, $^{1}_{1}H$ tends to absorb neutrons. But the isotope of hydrogen called *deuterium*, $^{2}_{1}H$, does not absorb many neutrons and is thus an almost ideal moderator. Either $^{1}_{1}H$ or $^{2}_{1}H$ can be used in the form of water. In the latter case, it is **heavy water**, in which the hydrogen atoms have been replaced by deuterium. Another common moderator is *graphite*, which consists of $^{12}_{6}C$ atoms.

A second problem is that the neutrons produced in one fission may be absorbed and produce other nuclear reactions with other nuclei in the reactor, rather than produce further fissions. In a "light-water" reactor, the $^{1}_{1}H$ nuclei absorb neutrons, as does $^{238}_{92}U$ to form $^{239}_{92}U$ in the reaction $n + {}^{238}_{92}U \rightarrow {}^{239}_{92}U + \gamma$. Naturally occurring uranium[†] contains 99.3% $^{238}_{92}U$ and only 0.7% fissionable $^{235}_{92}U$. To increase the probability of fission of $^{235}_{92}U$ nuclei, natural uranium can be **enriched** to increase the percentage of $^{235}_{92}U$ by using processes such as diffusion or centrifugation. Enrichment is not usually necessary for reactors using heavy water as moderator because heavy water doesn't absorb neutrons.[‡]

The third problem is that some neutrons will escape through the surface of the reactor core before they can cause further fissions (Fig. 31–7). Thus the mass of fuel must be sufficiently large for a self-sustaining chain reaction to take place. The minimum mass of uranium needed is called the **critical mass**. The value of the critical mass depends on the moderator, the fuel ($^{239}_{94}Pu$ may be used instead of $^{235}_{92}U$), and how much the fuel is enriched, if at all. Typical values are on the order of a few kilograms (that is, neither grams nor thousands of kilograms). Critical mass depends also on the average number of neutrons released per fission: 2.5 for $^{235}_{92}U$, 2.9 for $^{239}_{94}Pu$ so the critical mass for $^{239}_{94}Pu$ is smaller.

To have a self-sustaining chain reaction, on average at least one neutron produced in each fission must go on to produce another fission. The average number of neutrons per fission that do go on to produce further fissions is called the **neutron multiplication factor**, f. For a self-sustaining chain reaction, we must have $f \geq 1$. If $f < 1$, the reactor is "subcritical." If $f > 1$, it is "supercritical" (and could become dangerously explosive). Reactors are equipped with movable **control rods** (good neutron absorbers like cadmium or boron), whose function is to absorb neutrons and maintain the reactor at just barely "critical," $f = 1$.

The release of neutrons and subsequent fissions occur so quickly that manipulation of the control rods to maintain $f = 1$ would not be possible if it weren't for the small percentage ($\approx 1\%$) of so-called **delayed neutrons**. They come from the decay of neutron-rich fission fragments (or their daughters) having lifetimes on the order of seconds—sufficient to allow enough reaction time to operate the control rods and maintain $f = 1$.

Nuclear reactors have been built for use in research and to produce electric power. Fission produces many neutrons and a "research reactor" is basically an intense source of neutrons. These neutrons can be used as projectiles in nuclear reactions to produce nuclides not found in nature, including isotopes used as tracers and for therapy. A "power reactor" is used to produce electric power.

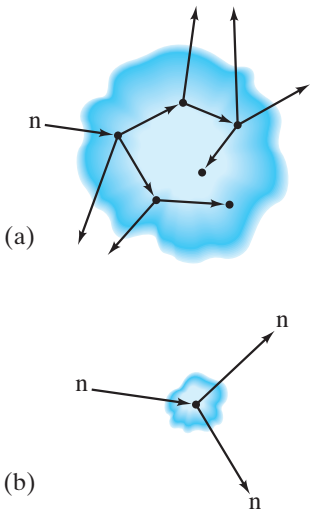


FIGURE 31–7 If the amount of uranium exceeds the critical mass, as in (a), a sustained chain reaction is possible. If the mass is less than critical, as in (b), too many neutrons escape before additional fissions occur, and the chain reaction is not sustained.

[†]$^{238}_{92}U$ will fission, but only with fast neutrons ($^{238}_{92}U$ is more stable than $^{235}_{92}U$). The probability of absorbing a fast neutron and producing a fission is too low to produce a self-sustaining chain reaction.

[‡]Other types of power or research reactors need 3% to 20% of $^{235}_{92}U$. Atom bonds need greater enrichment: $\gtrsim 80\%$.

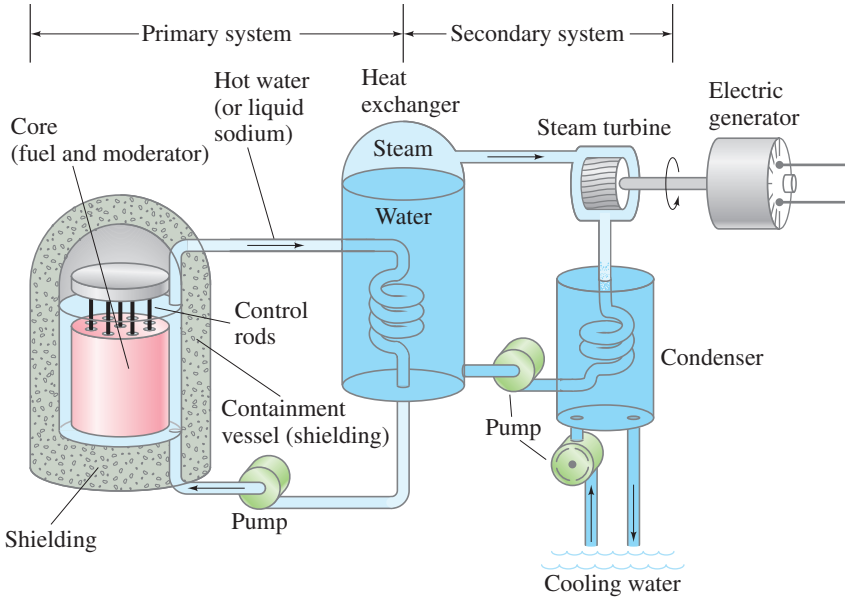


FIGURE 31–8 A nuclear reactor. The heat generated by the fission process in the fuel rods is carried off by hot water or liquid sodium and is used to boil water to steam in the heat exchanger. The steam drives a turbine to generate electricity and is then cooled in the condenser (to reduce pressure on the back side of the turbine blades).

The energy released in the fission process appears as heat, which is used to boil water and produce steam to drive a turbine connected to an electric generator (Fig. 31–8). The **core** of a nuclear reactor consists of the fuel and a moderator (water in most U.S. commercial reactors). The fuel is usually uranium enriched so that it contains 2 to 4 percent $^{235}_{92}\text{U}$. Water at high pressure or other liquid (such as liquid sodium) is allowed to flow through the core. The thermal energy it absorbs is used to produce steam in the heat exchanger, so the fissionable fuel acts as the heat input for a heat engine (Chapter 15).

There are problems associated with nuclear power plants. Besides the usual thermal pollution associated with any heat engine (Section 15–11), there is the serious problem of disposal of the radioactive fission fragments produced in the reactor, plus radioactive nuclides produced by neutrons interacting with the structural parts of the reactor. Fission fragments, like their uranium or plutonium parents, have about 50% more neutrons than protons. Nuclei with atomic number in the typical range for fission fragments ($Z \approx 30$ to 60) are stable only if they have more nearly equal numbers of protons and neutrons (see Fig. 30–2). Hence the highly neutron-rich fission fragments are very unstable and decay radioactively. The accidental release of highly radioactive fission fragments into the atmosphere poses a serious threat to human health (Section 31–4), as does possible leakage of the radioactive wastes when they are disposed of. The accidents at Three Mile Island, Pennsylvania (1979), at Chernobyl, Russia (1986), and at Fukushima, Japan (2011), have illustrated some of the dangers and have shown that nuclear plants must be located, constructed, maintained, and operated with great care and precision (Fig. 31–9).

FIGURE 31–9 Smoke rising from Fukushima, Japan, after the nuclear power plant meltdown in 2011.



Finally, the lifetime of nuclear power plants is limited to 30-some years, due to buildup of radioactivity and the fact that the structural materials themselves are weakened by the intense conditions inside. The cost of "decommissioning" a power plant is very great.

So-called **breeder reactors** were proposed as a solution to the problem of limited supplies of fissionable uranium, $^{235}_{92}\text{U}$. A breeder reactor is one in which some of the neutrons produced in the fission of $^{235}_{92}\text{U}$ are absorbed by $^{238}_{92}\text{U}$, and $^{239}_{94}\text{Pu}$ is produced via the set of reactions shown in Fig. 31–1. $^{239}_{94}\text{Pu}$ is fissionable with slow neutrons, so after separation it can be used as a fuel in a nuclear reactor. Thus a breeder reactor "breeds" new fuel[†] ($^{239}_{94}\text{Pu}$) from otherwise useless $^{238}_{92}\text{U}$. Natural uranium is 99.3 percent $^{238}_{92}\text{U}$, which in a breeder becomes useful fissionable $^{239}_{94}\text{Pu}$, thus increasing the supply of fissionable fuel by more than a factor of 100. But breeder reactors have the same problems as other reactors, plus other serious problems. Not only is plutonium a serious health hazard in itself (radioactive with a half-life of 24,000 years), but plutonium produced in a reactor can readily be used in a bomb, increasing the danger of nuclear proliferation and theft of fuel to produce a bomb.

[†]A breeder reactor does *not* produce more fuel than it uses.

Nuclear power presents risks. Other large-scale energy-conversion methods, such as conventional oil and coal-burning steam plants, also present health and environmental hazards; some of them were discussed in Section 15–11, and include air pollution, oil spills, and the release of CO_2 gas which can trap heat as in a greenhouse to raise the Earth's temperature. The solution to the world's needs for energy is not only technological, but also economic and political. A major factor surely is to "conserve"—to minimize our energy use. "Reduce, reuse, recycle."

EXAMPLE 31–5 **Uranium fuel amount.** Estimate the minimum amount of $^{235}_{92}\text{U}$ that needs to undergo fission in order to run a 1000-MW power reactor per year of continuous operation. Assume an efficiency (Chapter 15) of about 33%.

APPROACH At 33% efficiency, we need $3 \times 1000\,\text{MW} = 3000 \times 10^6\,\text{J/s}$ input. Each fission releases about 200 MeV (Eq. 31–5), so we divide the energy for a year by 200 MeV to get the number of fissions needed per year. Then we multiply by the mass of one uranium atom.

SOLUTION For 1000 MW output, the total power generation needs to be 3000 MW, of which 2000 MW is dumped as "waste" heat. Thus the total energy release in 1 yr $(3 \times 10^7\,\text{s})$ from fission needs to be about

$$(3 \times 10^9\,\text{J/s})(3 \times 10^7\,\text{s}) \approx 10^{17}\,\text{J}.$$

If each fission releases 200 MeV of energy, the number of fissions required for a year is

$$\frac{(10^{17}\,\text{J})}{(2 \times 10^8\,\text{eV/fission})(1.6 \times 10^{-19}\,\text{J/eV})} \approx 3 \times 10^{27}\,\text{fissions}.$$

The mass of a single uranium atom is about $(235\,\text{u})(1.66 \times 10^{-27}\,\text{kg/u}) \approx 4 \times 10^{-25}\,\text{kg}$, so the total uranium mass needed is

$$(4 \times 10^{-25}\,\text{kg/fission})(3 \times 10^{27}\,\text{fissions}) \approx 1000\,\text{kg},$$

or about a ton of $^{235}_{92}\text{U}$.

NOTE Because $^{235}_{92}\text{U}$ makes up only 0.7% of natural uranium, the yearly requirement for uranium is on the order of a hundred tons. This is orders of magnitude less than coal, both in mass and volume. Coal releases $2.8 \times 10^7\,\text{J/kg}$, whereas $^{235}_{92}\text{U}$ can release $10^{17}\,\text{J}$ per ton, as we just calculated, or $10^{17}\,\text{J}/10^3\,\text{kg} = 10^{14}\,\text{J/kg}$. For natural uranium, the figure is 100 times less, $10^{12}\,\text{J/kg}$.

EXERCISE C A nuclear-powered submarine needs 6000-kW input power. How many $^{235}_{92}\text{U}$ fissions is this per second?

Atom Bomb

The first use of fission, however, was not to produce electric power. Instead, it was first used as a fission bomb (called the "atomic bomb"). In early 1940, with Europe already at war, Germany's leader, Adolf Hitler, banned the sale of uranium from the Czech mines he had recently taken over. Research into the fission process suddenly was enshrouded in secrecy. Physicists in the United States were alarmed. A group of them approached Einstein—a man whose name was a household word—to send a letter to President Franklin Roosevelt about the possibilities of using nuclear fission for a bomb far more powerful than any previously known, and inform him that Germany might already have begun development of such a bomb. Roosevelt responded by authorizing the program known as the Manhattan Project, to see if a bomb could be built. Work began in earnest after Fermi's demonstration in 1942 that a sustained chain reaction was possible. A new secret laboratory was developed on an isolated mesa in New Mexico known as Los Alamos. Under the direction of J. Robert Oppenheimer (1904–1967; Fig. 31–10), it became the home of famous scientists from all over Europe and the United States.



FIGURE 31–10 J. Robert Oppenheimer, on the left, with General Leslie Groves, who was the administrative head of Los Alamos during World War II. The photograph was taken at the Trinity site in the New Mexico desert, where the first atomic bomb was exploded.



FIGURE 31–11 Photo taken a month after the bomb was dropped on Nagasaki. The shacks were constructed afterwards from debris in the ruins. The bombs dropped on Hiroshima and Nagasaki were each equivalent to about 20,000 tons of the common explosive TNT ($\sim 10^{14}$ J).

To build a bomb that was subcritical during transport but that could be made supercritical (to produce a chain reaction) at just the right moment, two pieces of uranium were used, each less than the critical mass but together greater than the critical mass. The two masses, kept separate until the moment of detonation, were then forced together quickly by a kind of gun, and a chain reaction of explosive proportions occurred. An alternate bomb detonated conventional explosives (TNT) surrounding a plutonium sphere to compress it by implosion to double its density, making it more than critical and causing a nuclear explosion. The first fission bomb was tested in the New Mexico desert in July 1945. It was successful. In early August, a fission bomb using uranium was dropped on Hiroshima and a second, using plutonium, was dropped on Nagasaki (Fig. 31–11), both in Japan. World War II ended shortly thereafter.

Besides its destructive power, a fission bomb produces many highly radioactive fission fragments, as does a nuclear reactor. When a fission bomb explodes, these radioactive isotopes are released into the atmosphere as **radioactive fallout**.

Testing of nuclear bombs in the atmosphere after World War II was a cause of concern, because the movement of air masses spread the fallout all over the globe. Radioactive fallout eventually settles to the Earth, particularly in rainfall, and is absorbed by plants and grasses and enters the food chain. This is a far more serious problem than the same radioactivity on the exterior of our bodies, because α and β particles are largely absorbed by clothing and the outer (dead) layer of skin. But inside our bodies as food, the isotopes are in contact with living cells. One particularly dangerous radioactive isotope is $^{90}_{38}\text{Sr}$, which is chemically much like calcium and becomes concentrated in bone, where it causes bone cancer and destroys bone marrow. The 1963 treaty signed by over 100 nations that bans nuclear weapons testing in the atmosphere was motivated because of the hazards of fallout.

31–3 Nuclear Fusion

The mass of every stable nucleus is less than the sum of the masses of its constituent protons and neutrons. For example, the mass of the helium isotope $^{4}_{2}\text{He}$ is less than the mass of two protons plus two neutrons, Example 30–3. If two protons and two neutrons were to come together to form a helium nucleus, there would be a loss of mass. This mass loss is manifested in the release of energy.

Nuclear Fusion; Stars

The process of building up nuclei by bringing together individual protons and neutrons, or building larger nuclei by combining small nuclei, is called **nuclear fusion**. In Fig. 31–12 (same as Fig. 30–1), we can see why small nuclei can combine to form larger ones with the release of energy: it is because the binding energy per nucleon is less for light nuclei than it is for heavier nuclei (up to about $A \approx 60$).

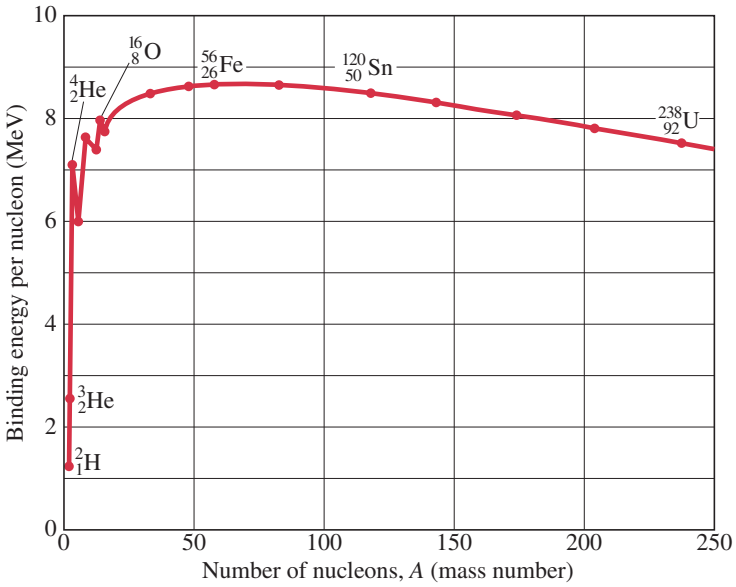


FIGURE 31–12 Average binding energy per nucleon as a function of mass number A for stable nuclei. Same as Fig. 30–1.

For two positively charged nuclei to get close enough to fuse, they must have very high kinetic energy to overcome the electric repulsion. It is believed that many of the elements in the universe were originally formed through the process of fusion in stars (see Chapter 33) where the temperature is extremely high, corresponding to high KE (Eq. 13–8). Today fusion is still producing the prodigious amounts of light energy (EM waves) stars emit, including our Sun.

EXAMPLE 31–6 **Fusion energy release.** One of the simplest fusion reactions involves the production of deuterium, ${}^{2}_{1}\mathrm{H}$, from a neutron and a proton: ${}^{1}_{1}\mathrm{H} + \mathrm{n} \rightarrow {}^{2}_{1}\mathrm{H} + \gamma$. How much energy is released in this reaction?

APPROACH The energy released equals the difference in mass (times c^2) between the initial and final masses.

SOLUTION From Appendix B, the initial mass is

$$1.007825\ \mathrm{u} + 1.008665\ \mathrm{u} = 2.016490\ \mathrm{u},$$

and after the reaction the mass is that of the ${}^{2}_{1}\mathrm{H}$, namely 2.014102 u (the γ is massless). The mass difference is

$$2.016490\ \mathrm{u} - 2.014102\ \mathrm{u} = 0.002388\ \mathrm{u},$$

so the energy released is

$$(\Delta m)c^2 = (0.002388\ \mathrm{u})(931.5\ \mathrm{MeV/u}) = 2.22\ \mathrm{MeV},$$

and it is carried off by the ${}^{2}_{1}\mathrm{H}$ nucleus and the γ ray.

The energy output of our Sun is believed to be due principally to the following sequence of fusion reactions:

Proton–proton chain

$${}^{1}_{1}\mathrm{H} + {}^{1}_{1}\mathrm{H} \rightarrow {}^{2}_{1}\mathrm{H} + \mathrm{e}^{+} + \nu \qquad (0.42\ \mathrm{MeV}) \qquad \textbf{(31–6a)}$$

$${}^{1}_{1}\mathrm{H} + {}^{2}_{1}\mathrm{H} \rightarrow {}^{3}_{2}\mathrm{He} + \gamma \qquad (5.49\ \mathrm{MeV}) \qquad \textbf{(31–6b)}$$

$${}^{3}_{2}\mathrm{He} + {}^{3}_{2}\mathrm{He} \rightarrow {}^{4}_{2}\mathrm{He} + {}^{1}_{1}\mathrm{H} + {}^{1}_{1}\mathrm{H}. \qquad (12.86\ \mathrm{MeV}) \qquad \textbf{(31–6c)}$$

where the energy released (Q-value) for each reaction is given in parentheses. (They include keeping track of atomic electrons as for β-decay, Section 30–5.) These reactions are between nuclei (without electrons at these very high temperatures); the first reaction can be written as

$$\mathrm{p} + \mathrm{p} \rightarrow \mathrm{d} + \mathrm{e}^{+} + \nu$$

where p = proton and d = deuteron. The net effect of this sequence, which is called the **proton–proton chain**, is for four protons to combine to form one ${}^{4}_{2}\mathrm{He}$ nucleus plus two positrons, two neutrinos, and two gamma rays:

$$4\,{}^{1}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + 2\mathrm{e}^{+} + 2\nu + 2\gamma. \qquad \textbf{(31–7)}$$

Note that it takes two of each of the first two reactions (Eqs. 31–6a and b) to produce the two ${}^{3}_{2}\mathrm{He}$ for the third reaction. Also, each of the two e^{+} (Eq. 31–6a) quickly annihilates with an electron to produce 2 γ rays (Section 27–6) with total energy $2m_{\mathrm{e}}c^2 = 1.02\ \mathrm{MeV}$. So the total energy release for the net reaction, Eq. 31–7, is

$$2(0.42\ \mathrm{MeV}) + 2(1.02\ \mathrm{MeV}) + 2(5.49\ \mathrm{MeV}) + 12.86\ \mathrm{MeV} = 26.7\ \mathrm{MeV}.$$

The first reaction, the formation of deuterium from two protons (Eq. 31–6a), has a very low probability, and so limits the rate at which the Sun produces energy. (Thank goodness; this is why the Sun and other stars have long lifetimes and are still shining brightly.)

EXERCISE D Return to the Chapter-Opening Question, page 885, and answer it again now. Try to explain why you may have answered it differently the first time.

EXAMPLE 31–7 **ESTIMATE** **Estimating fusion energy.** Estimate the energy released if the following reaction occurred:

$${}^{2}_{1}\mathrm{H} + {}^{2}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He}.$$

APPROACH We use Fig. 31–12 for a quick estimate.

SOLUTION We see in Fig. 31–12 that each ${}^{2}_{1}\mathrm{H}$ has a binding energy of about $1\frac{1}{4}$ MeV/nucleon, which for 2 nuclei of mass 2 is $4 \times (1\frac{1}{4}) \approx 5$ MeV. The ${}^{4}_{2}\mathrm{He}$ has a binding energy per nucleon (Fig. 31–12) of about 7 MeV for a total of $4 \times 7\ \mathrm{MeV} \approx 28$ MeV. Hence the energy release is about $28\ \mathrm{MeV} - 5\ \mathrm{MeV} \approx 23\ \mathrm{MeV}$.

In stars hotter than the Sun, it is more likely that the energy output comes principally from the **carbon** (or **CNO**) **cycle**, which comprises the following sequence of reactions:

Carbon cycle

$$\begin{aligned}
{}^{12}_{6}\mathrm{C} + {}^{1}_{1}\mathrm{H} &\rightarrow {}^{13}_{7}\mathrm{N} + \gamma \\
{}^{13}_{7}\mathrm{N} &\rightarrow {}^{13}_{6}\mathrm{C} + \mathrm{e}^{+} + \nu \\
{}^{13}_{6}\mathrm{C} + {}^{1}_{1}\mathrm{H} &\rightarrow {}^{14}_{7}\mathrm{N} + \gamma \\
{}^{14}_{7}\mathrm{N} + {}^{1}_{1}\mathrm{H} &\rightarrow {}^{15}_{8}\mathrm{O} + \gamma \\
{}^{15}_{8}\mathrm{O} &\rightarrow {}^{15}_{7}\mathrm{N} + \mathrm{e}^{+} + \nu \\
{}^{15}_{7}\mathrm{N} + {}^{1}_{1}\mathrm{H} &\rightarrow {}^{12}_{6}\mathrm{C} + {}^{4}_{2}\mathrm{He}.
\end{aligned}$$

No net carbon is consumed in this cycle and the net effect is the same as the proton–proton chain, Eq. 31–7 (plus one extra γ). The theory of the proton–proton chain and of the carbon cycle as the source of energy for the Sun and stars was first worked out by Hans Bethe (1906–2005) in 1939.

CONCEPTUAL EXAMPLE 31–8 **Stellar fusion.** What is the heaviest element likely to be produced in fusion processes in stars?

RESPONSE Fusion is possible if the final products have more binding energy (less mass) than the reactants, because then there is a net release of energy. Since the binding energy curve in Fig. 31–12 (or Fig. 30–1) peaks near $A \approx 56$ to 58 which corresponds to iron or nickel, it would not be energetically favorable to produce elements heavier than that. Nevertheless, in the center of massive stars or in supernova explosions, there is enough initial kinetic energy available to drive endothermic reactions that produce heavier elements as well.

EXERCISE E If the Sun is generating a constant amount of energy via fusion, the mass of the Sun must be (*a*) increasing, (*b*) decreasing, (*c*) constant, (*d*) irregular.

Possible Fusion Reactors

PHYSICS APPLIED
Fusion energy reactors

The possibility of utilizing the energy released in fusion to make a power reactor is very attractive. The fusion reactions most likely to succeed in a reactor involve the isotopes of hydrogen, ${}^{2}_{1}\mathrm{H}$ (deuterium) and ${}^{3}_{1}\mathrm{H}$ (tritium), and are as follows, with the energy released given in parentheses:

$$\begin{aligned}
{}^{2}_{1}\mathrm{H} + {}^{2}_{1}\mathrm{H} &\rightarrow {}^{3}_{1}\mathrm{H} + {}^{1}_{1}\mathrm{H} \qquad (4.03\ \mathrm{MeV}) \qquad \textbf{(31–8a)} \\
{}^{2}_{1}\mathrm{H} + {}^{2}_{1}\mathrm{H} &\rightarrow {}^{3}_{2}\mathrm{He} + \mathrm{n} \qquad (3.27\ \mathrm{MeV}) \qquad \textbf{(31–8b)} \\
{}^{2}_{1}\mathrm{H} + {}^{3}_{1}\mathrm{H} &\rightarrow {}^{4}_{2}\mathrm{He} + \mathrm{n}. \qquad (17.59\ \mathrm{MeV}) \qquad \textbf{(31–8c)}
\end{aligned}$$

Comparing these energy yields with that for the fission of ${}^{235}_{92}\mathrm{U}$, we can see that the energy released in fusion reactions can be greater for a given mass of fuel than in fission. Furthermore, as fuel, a fusion reactor could use deuterium, which is very plentiful in the water of the oceans (the natural abundance of ${}^{2}_{1}\mathrm{H}$ is 0.0115% on average, or about 1 g of deuterium per 80 L of water). The simple proton–proton reaction of Eq. 31–6a, which could use a much more plentiful source of fuel, ${}^{1}_{1}\mathrm{H}$, has such a small probability of occurring that it cannot be considered a possibility on Earth.

Although a useful fusion reactor has not yet been achieved, considerable progress has been made in overcoming the inherent difficulties. The problems are associated with the fact that all nuclei have a positive charge and repel each other. However, if they can be brought close enough together so that the short-range attractive strong nuclear force can come into play, it can pull the nuclei together and fusion can occur. For the nuclei to get close enough together, they must have large kinetic energy to overcome the electric repulsion. High kinetic energies are readily attainable with particle accelerators (Chapter 32), but the number of particles involved is too small. To produce realistic amounts of energy, we must deal with matter in bulk, for which high kinetic energy means higher temperatures.

Indeed, very high temperatures are required for sustained fusion to occur, and fusion devices are often referred to as **thermonuclear devices**. The interiors of the Sun and other stars are very hot, many millions of degrees, so the nuclei are moving fast enough for fusion to take place, and the energy released keeps the temperature high so that further fusion reactions can occur. The Sun and the stars represent huge self-sustaining thermonuclear reactors that stay together because of their great gravitational mass. But on Earth, containment of the fast-moving nuclei at the high temperatures and densities required has proven difficult.

It was realized after World War II that the temperature produced within a fission (or "atomic") bomb was close to 10^8 K. This suggested that a fission bomb could be used to ignite a fusion bomb (popularly known as a thermonuclear or hydrogen bomb) to release the vast energy of fusion. The uncontrollable release of fusion energy in an H-bomb (in 1952) was relatively easy to obtain. But to realize usable energy from fusion at a slow and controlled rate has turned out to be a serious challenge.

EXAMPLE 31–9 | ESTIMATE **Temperature needed for d–t fusion.** Estimate the temperature required for deuterium–tritium fusion (d–t) to occur.

APPROACH We assume the nuclei approach head-on, each with kinetic energy KE, and that the nuclear force comes into play when the distance between their centers equals the sum of their nuclear radii. The electrostatic potential energy (Chapter 17) of the two particles at this distance equals the minimum total kinetic energy of the two particles when far apart. The average kinetic energy is related to Kelvin temperature by Eq. 13–8.

SOLUTION The radii of the two nuclei ($A_d = 2$ and $A_t = 3$) are given by Eq. 30–1: $r_d \approx 1.5\,\text{fm}$, $r_t \approx 1.7\,\text{fm}$, so $r_d + r_t = 3.2 \times 10^{-15}\,\text{m}$. We equate the kinetic energy of the two initial particles to the potential energy when at this distance:

$$2\text{KE} \approx \frac{1}{4\pi\epsilon_0}\frac{e^2}{(r_d + r_t)}$$

$$\approx \left(9.0 \times 10^9 \frac{\text{N}\cdot\text{m}^2}{\text{C}^2}\right)\frac{(1.6 \times 10^{-19}\,\text{C})^2}{(3.2 \times 10^{-15}\,\text{m})(1.6 \times 10^{-19}\,\text{J/eV})} \approx 0.45\,\text{MeV}.$$

Thus, KE ≈ 0.22 MeV, and if we ask that the average kinetic energy be this high, then from Eq. 13–8, $\frac{3}{2}kT = \overline{\text{KE}}$, we have a temperature of

$$T = \frac{2\overline{\text{KE}}}{3k} = \frac{2(0.22\,\text{MeV})(1.6 \times 10^{-13}\,\text{J/MeV})}{3(1.38 \times 10^{-23}\,\text{J/K})} \approx 2 \times 10^9\,\text{K}.$$

NOTE More careful calculations show that the temperature required for fusion is actually about an order of magnitude less than this rough estimate, partly because it is not necessary that the *average* kinetic energy be 0.22 MeV—a small percentage of nuclei with this much energy (in the high-energy tail of the Maxwell distribution, Fig. 13–20) would be sufficient. Reasonable estimates for a usable fusion reactor are in the range $T \gtrsim 1$ to 4×10^8 K.

A high temperature is required for a fusion reactor. But there must also be a high density of nuclei to ensure a sufficiently high collision rate. A real difficulty with controlled fusion is to contain nuclei long enough and at a high enough density for sufficient reactions to occur so that a usable amount of energy is obtained. At the temperatures needed for fusion, the atoms are ionized, and the resulting collection of nuclei and electrons is referred to as a **plasma**. Ordinary materials vaporize at a few thousand degrees at most, and hence cannot be used to contain a high-temperature plasma. Two major containment techniques are *magnetic confinement* and *inertial confinement*.

In **magnetic confinement**, magnetic fields are used to try to contain the hot plasma. A simple approach is the "magnetic bottle" shown in Fig. 31–13. The paths of the charged particles in the plasma are bent by the magnetic field; where magnetic field lines are close together, the force on the particles reflects them

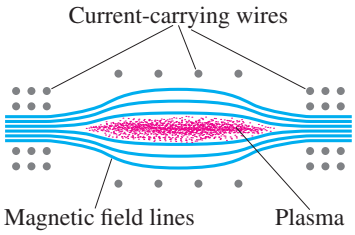


FIGURE 31–13 "Magnetic bottle" used to confine a plasma.

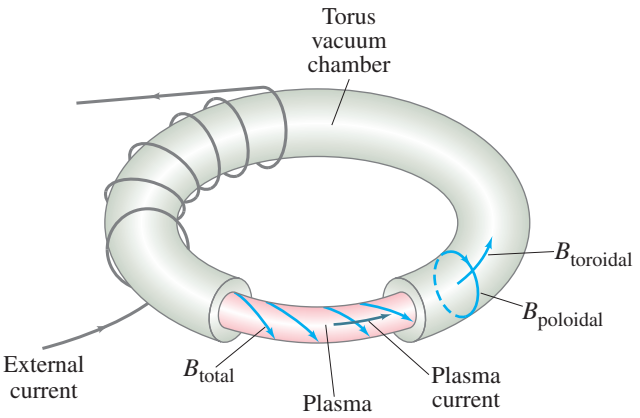
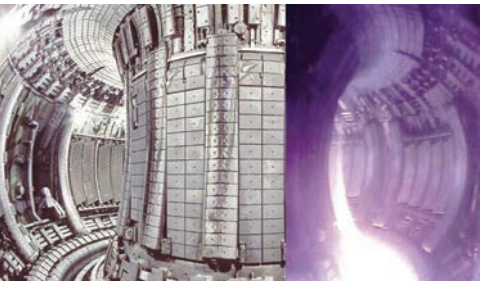
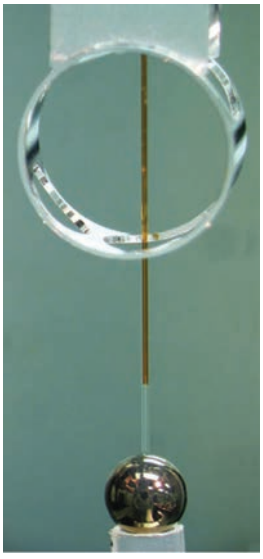


FIGURE 31–14 Tokamak configuration, showing the total $\vec{\mathbf{B}}$ field due to external current plus current in the plasma itself.



(a)



(b)

FIGURE 31–15 (a) Tokamak: split image view of the Joint European Torus (JET) located near Oxford, England. Interior, on the left, and an actual plasma in there ($T \approx 1 \times 10^8$ K) on the right. (b) A 2-mm-diameter round d–t (deuterium–tritium) inertial target, being filled through a thin glass tube from above, at the National Ignition Facility (NIF), Lawrence Livermore National Laboratory, California.

back toward the center. Unfortunately, magnetic bottles develop "leaks" and the charged particles slip out before sufficient fusion takes place. The most promising design today is the **tokamak**, first developed in Russia. A tokamak (Fig. 31–14) is toroid-shaped (a torus, which is like a donut) and involves complicated magnetic fields: current-carrying conductors produce a magnetic field directed along the axis of the toroid ("toroidal" field); an additional field is produced by currents within the plasma itself ("poloidal" field). The combination produces a helical field as shown in Fig. 31–14, confining the plasma, at least briefly, so it doesn't touch the vacuum chamber's metal walls (Fig. 31–15a).

In 1957, J. D. Lawson showed that the product of ion density n (= ions/m^3) and confinement time τ must exceed a minimum value of approximately

$$n\tau \gtrsim 3 \times 10^{20}\ \text{s/m}^3.$$

This **Lawson criterion** must be reached to produce **ignition**, meaning fusion that continues after all external heating is turned off. Practically, it is expected to be achieved with $n \approx 1$ to $3 \times 10^{20}\ \text{m}^{-3}$ and $\tau \approx 1$ to 3 s. To reach **break-even**, the point at which the energy output due to fusion is equal to the energy input to heat the plasma, requires an $n\tau$ about an order of magnitude less. The break-even point was very closely approached in the 1990s at the Tokamak Fusion Test Reactor (TFTR) at Princeton, and the very high temperature needed for ignition $(4 \times 10^8\ \text{K})$ was exceeded—although not both of these at the same time.

Magnetic confinement fusion research continues throughout the world. This research will help us in developing the huge multinational test device (European Union, India, Japan, South Korea, Russia, China, and the U.S.), called ITER (International Thermonuclear Experimental Reactor), situated in France. It is expected that ITER will produce temperatures above 10^8 K for extended periods (minutes or hours) and to begin running by 2020s, with an expected power output of about 500 MW, 10 times the input energy. ITER is planned to be the final research step before building a working reactor.

The second method for containing the fuel for fusion is **inertial confinement fusion** (ICF): a small pellet or capsule of deuterium and tritium (Fig. 31–15b) is struck simultaneously from hundreds of directions by very intense laser beams. The intense influx of energy heats and ionizes the pellet into a plasma, compressing it and heating it to temperatures at which fusion can occur $(> 10^8\ \text{K})$. The confinement time is on the order of 10^{-11} to 10^{-9} s, during which time the ions do not move appreciably because of their own inertia, and fusion can take place.

31–4 Passage of Radiation Through Matter; Biological Damage

When we speak of *radiation*, we include α, β, γ, and X-rays, as well as protons, neutrons, and other particles such as pions (see Chapter 32). Because charged particles can ionize the atoms or molecules of any material they pass through, they are referred to as **ionizing radiation**. And because radiation produces ionization, it can cause considerable damage to materials, particularly to biological tissue.

Charged particles, such as α and β rays and protons, cause ionization because of electric forces. That is, when they pass through a material, they can attract or repel electrons strongly enough to remove them from the atoms of the material. Since the α and β rays emitted by radioactive substances have energies on the order of 1 MeV (10^4 to 10^7 eV), whereas ionization of atoms and molecules requires on the order of 10 eV (Chapter 27), we see that a single α or β particle can cause thousands of ionizations.

Neutral particles also give rise to ionization when they pass through materials. For example, X-ray and γ-ray photons can ionize atoms by knocking out electrons by means of the photoelectric and Compton effects (Chapter 27). Furthermore, if a γ ray has sufficient energy (greater than 1.02 MeV), it can undergo pair production: an electron and a positron are produced (Section 27–6). The charged particles produced in all of these processes can themselves go on to produce further ionization. Neutrons, on the other hand, interact with matter mainly by collisions with nuclei, with which they interact strongly. Often the nucleus is broken apart by such a collision, altering the molecule of which it was a part. The fragments produced can in turn cause ionization.

Radiation passing through matter can do considerable damage. Metals and other structural materials become brittle and their strength can be weakened if the radiation is very intense, as in nuclear reactor power plants and for space vehicles that must pass through areas of intense cosmic radiation.

Biological Damage

PHYSICS APPLIED
Biological radiation damage

The radiation damage produced in biological organisms is due primarily to ionization produced in cells. Several related processes can occur. Ions or radicals are produced that are highly reactive and take part in chemical reactions that interfere with the normal operation of the cell. All forms of radiation can ionize atoms by knocking out electrons. If these are bonding electrons, the molecule may break apart, or its structure may be altered so that it does not perform its normal function or may perform a harmful function. In the case of proteins, the loss of one molecule is not serious if there are other copies of the protein in the cell and additional copies can be made from the gene that codes for it. However, large doses of radiation may damage so many molecules that new copies cannot be made quickly enough, and the cell dies.

Damage to the DNA is more serious, since a cell may have only one copy. Each alteration in the DNA can affect a gene and alter the molecule that gene codes for (Section 29–3), so that needed proteins or other molecules may not be made at all. Again the cell may die. The death of a single cell is not normally a problem, since the body can replace it with a new one. (There are exceptions, such as neurons, which are mostly not replaceable, so their loss is serious.) But if many cells die, the organism may not be able to recover. On the other hand, a cell may survive but be defective. It may go on dividing and produce many more defective cells, to the detriment of the whole organism. Thus radiation can cause cancer—the rapid uncontrolled production of cells.

The possible damage done by the medical use of X-rays and other radiation must be balanced against the medical benefits and prolongation of life as a result of their diagnostic use.

31–5 Measurement of Radiation—Dosimetry

Although the passage of ionizing radiation through the human body can cause considerable damage, radiation can also be used to treat certain diseases, particularly cancer, often by using very narrow beams directed at a cancerous tumor in order to destroy it (Section 31–6). It is therefore important to be able to quantify the amount, or **dose**, of radiation. This is the subject of **dosimetry**.

PHYSICS APPLIED
Dosimetry

The strength of a source can be specified at a given time by stating the **source activity**: how many nuclear decays (or disintegrations) occur per second. The traditional unit is the **curie** (Ci), defined as

$$1\ \text{Ci} \ = \ 3.70 \times 10^{10}\ \text{decays per second.}$$

(This number comes from the original definition as the activity of exactly one gram of radium.) Although the curie is still in common use, the SI unit for source activity is the **becquerel** (Bq), defined as

$$1\ \text{Bq} \ = \ 1\ \text{decay/s.}$$

CAUTION

In the lab, activity will be less than written on the bottle—note the date

Commercial suppliers of **radionuclides** (radioactive nuclides) used as tracers specify the activity at a given time. Because the activity decreases over time, more so for short-lived isotopes, it is important to take this decrease into account.

The magnitude of the source activity, $\Delta N/\Delta t$, is related to the number of radioactive nuclei present, N, and to the half-life, $T_{\frac{1}{2}}$, by (see Section 30–8):

$$\frac{\Delta N}{\Delta t} \ = \ \lambda N \ = \ \frac{0.693}{T_{\frac{1}{2}}} N.$$

EXAMPLE 31–10 **Radioactivity taken up by cells.** In a certain experiment, 0.016 μCi of $^{32}_{15}$P is injected into a medium containing a culture of bacteria. After 1.0 h the cells are washed and a 70% efficient detector (counts 70% of emitted β rays) records 720 counts per minute from the cells. What percentage of the original $^{32}_{15}$P was taken up by the cells?

APPROACH The half-life of $^{32}_{15}$P is about 14 days (Appendix B), so we can ignore any loss of activity over 1 hour. From the given activity, we find how many β rays are emitted. We can compare 70% of this to the $(720/\text{min})/(60\ \text{s/min}) = 12$ per second detected.

SOLUTION The total number of decays per second originally was $(0.016 \times 10^{-6})(3.7 \times 10^{10}) = 590$. The counter could be expected to count 70% of this, or 410 per second. Since it counted $720/60 = 12$ per second, then $12/410 = 0.029$ or 2.9% was incorporated into the cells.

Another type of measurement is the exposure or **absorbed dose**—that is, the *effect* the radiation has on the absorbing material. The earliest unit of dosage was the **roentgen** (R), defined in terms of the amount of ionization produced by the radiation ($1\ \text{R} = 1.6 \times 10^{12}$ ion pairs per gram of dry air at standard conditions). Today, 1 R is defined as the amount of X-ray or γ radiation that deposits 0.878×10^{-2} J of energy per kilogram of air. The roentgen was largely superseded by another unit of absorbed dose applicable to any type of radiation, the **rad**: *1 rad is that amount of radiation which deposits energy per unit mass of* 1.00×10^{-2} *J/kg in any absorbing material.* (This is quite close to the roentgen for X- and γ rays.) The proper SI unit for absorbed dose is the **gray** (Gy):

$$1\ \text{Gy} \ = \ 1\ \text{J/kg} \ = \ 100\ \text{rad.} \qquad \textbf{(31–9)}$$

The absorbed dose depends not only on the energy per particle and on the strength of a given source or of a radiation beam (number of particles per second), but also on the type of material that is absorbing the radiation. Bone, for example, absorbs more of X-ray or γ radiation normally used than does flesh, so the same beam passing through a human body deposits a greater dose (in rads or grays) in bone than in flesh.

The gray and the rad are physical units of dose—the energy deposited per unit mass of material. They are, however, not the most meaningful units for measuring the biological damage produced by radiation because equal doses of different types of radiation cause differing amounts of damage. For example, 1 rad of α radiation does 10 to 20 times the amount of damage as 1 rad of β or γ rays. This difference arises largely because α rays (and other heavy particles such as protons and neutrons) move much more slowly than β and γ rays of equal energy due to their greater mass. Hence, ionizing collisions occur closer together,

so more irreparable damage can be done. The **relative biological effectiveness** (RBE) of a given type of radiation is defined as the number of rads of X-ray or γ radiation that produces the same biological damage as 1 rad of the given radiation. For example, 1 rad of slow neutrons does the same damage as 5 rads of X-rays. Table 31–1 gives the RBE for several types of radiation. The numbers are approximate because they depend somewhat on the energy of the particles and on the type of damage that is used as the criterion.

The **effective dose** can be given as the product of the dose in rads and the RBE, and this unit is known as the **rem** (which stands for *rad equivalent man*):

$$\text{effective dose (in rem)} = \text{dose (in rad)} \times \text{RBE}. \qquad \textbf{(31–10a)}$$

This unit is being replaced by the SI unit for "effective dose," the **sievert** (Sv):

$$\text{effective dose (Sv)} = \text{dose (Gy)} \times \text{RBE} \qquad \textbf{(31–10b)}$$

so

$$1\ \text{Sv} = 100\ \text{rem} \qquad \text{or} \qquad 1\ \text{rem} = 10\ \text{mSv}.$$

By these definitions, 1 rem (or 1 Sv) of any type of radiation does approximately the same amount of biological damage. For example, 50 rem of fast neutrons does the same damage as 50 rem of γ rays. But note that 50 rem of fast neutrons is only 5 rads, whereas 50 rem of γ rays is 50 rads.

TABLE 31–1 Relative Biological Effectiveness (RBE)

Type	RBE
X- and γ rays	1
β (electrons)	1
Protons	2
Slow neutrons	5
Fast neutrons	≈ 10
α particles and heavy ions	≈ 20

Human Exposure to Radiation

PHYSICS APPLIED
Radon

We are constantly exposed to low-level radiation from natural sources: cosmic rays, natural radioactivity in rocks and soil, and naturally occurring radioactive isotopes in our food, such as $^{40}_{19}\text{K}$. **Radon**, $^{222}_{86}\text{Rn}$, is of considerable concern today. It is the product of radium decay and is an intermediate in the decay series from uranium (see Fig. 30–11). Most intermediates remain in the rocks where formed, but radon is a gas that can escape from rock (and from building material like concrete) to enter the air we breathe, and damage the interior of the lung.

PHYSICS APPLIED
Human radiation exposure

The **natural radioactive background** averages about 0.30 rem (300 mrem) per year per person in the U.S., although there are large variations. From medical X-rays and scans, the average person receives about 50 to 60 mrem per year, giving an average total dose of about 360 mrem (3.6 mSv) per person. U.S. government regulators suggest an upper limit of allowed radiation for an individual in the general populace at about 100 mrem (1 mSv) per year in addition to natural background. It is believed that even low doses of radiation increase the chances of cancer or genetic defects; there is *no safe level* or threshold of radiation exposure.

PHYSICS APPLIED
Radiation worker exposure
Film badge

The upper limit for people who work around radiation—in hospitals, in power plants, in research—has been set higher, a maximum of 20 mSv (2 rem) whole-body dose, averaged over some years (a maximum of 50 mSv (5 rem/yr) in any one year). To monitor exposure, those people who work around radiation generally carry some type of dosimeter, one common type being a **radiation film badge** which is a piece of film wrapped in light-tight material. The passage of ionizing radiation through the film changes it so that the film is darkened upon development, and thus indicates the received dose. Newer types include the *thermoluminescent dosimeter* (TLD). Dosimeters and badges do not protect the worker, but high levels detected suggest reassignment or modified work practices to reduce radiation exposure to acceptable levels.

PHYSICS APPLIED
Radiation sickness

Large doses of radiation can cause unpleasant symptoms such as nausea, fatigue, and loss of body hair, because of cellular damage. Such effects are sometimes referred to as **radiation sickness**. Very large doses can be fatal, although the time span of the dose is important. A brief dose of 10 Sv (1000 rem) is nearly always fatal. A 3-Sv (300-rem) dose in a short period of time is fatal in about 50% of patients within a month. However, the body possesses remarkable repair processes, so that a 3-Sv dose spread over several weeks is usually not fatal. It will, nonetheless, cause considerable damage to the body.

The effects of low doses over a long time are difficult to determine and are not well known as yet.

CONCEPTUAL EXAMPLE 31–11 **Limiting the dose.** A worker in an environment with a radioactive source is warned that she is accumulating a dose too quickly and will have to lower her exposure by a factor of ten to continue working for the rest of the year. If the worker is able to work farther away from the source, how much farther away is necessary?

RESPONSE If the energy is radiated uniformly in all directions, then the intensity (dose/area) should decrease as the distance squared, just as it does for sound and light. If she can work four times farther away, the exposure lowers by a factor of sixteen, enough to make her safe.

EXAMPLE 31–12 **Whole-body dose.** What whole-body dose is received by a 70-kg laboratory worker exposed to a 40-mCi $^{60}_{27}$Co source, assuming the person's body has cross-sectional area 1.5 m^2 and is normally about 4.0 m from the source for 4.0 h per day? $^{60}_{27}$Co emits γ rays of energy 1.33 MeV and 1.17 MeV in quick succession. Approximately 50% of the γ rays interact in the body and deposit all their energy. (The rest pass through.)

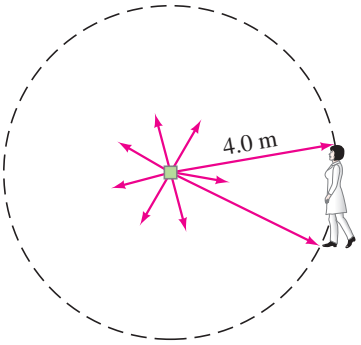


FIGURE 31–16 Radiation spreads out in all directions. A person 4.0 m away intercepts only a fraction: her cross-sectional area divided by the area of a sphere of radius 4.0 m. Example 31–12.

APPROACH Of the given energy emitted, only a fraction passes through the worker, equal to *her* area divided by the total area (or $4\pi r^2$) over a full sphere of radius $r = 4.0$ m (Fig. 31–16).

SOLUTION The total γ-ray energy per decay is $(1.33 + 1.17)$ MeV $= 2.50$ MeV, so the total energy emitted by the source per second is

$$(0.040\ \text{Ci})(3.7 \times 10^{10}\ \text{decays/Ci}\cdot\text{s})(2.50\ \text{MeV}) = 3.7 \times 10^{9}\ \text{MeV/s}.$$

The proportion of this energy intercepted by the body is its 1.5-m^2 area divided by the area of a sphere of radius 4.0 m (Fig. 31–16):

$$\frac{1.5\ \text{m}^2}{4\pi r^2} = \frac{1.5\ \text{m}^2}{4\pi(4.0\ \text{m})^2} = 7.5 \times 10^{-3}.$$

So the rate energy is deposited in the body (remembering that only 50% of the γ rays interact in the body) is

$$E = \left(\tfrac{1}{2}\right)(7.5 \times 10^{-3})(3.7 \times 10^{9}\ \text{MeV/s})(1.6 \times 10^{-13}\ \text{J/MeV}) = 2.2 \times 10^{-6}\ \text{J/s}.$$

Since 1 Gy = 1 J/kg, the whole-body dose rate for this 70-kg person is $(2.2 \times 10^{-6}\ \text{J/s})/(70\ \text{kg}) = 3.1 \times 10^{-8}$ Gy/s. In 4.0 h, this amounts to a dose of

$$(4.0\ \text{h})(3600\ \text{s/h})(3.1 \times 10^{-8}\ \text{Gy/s}) = 4.5 \times 10^{-4}\ \text{Gy}.$$

RBE ≈ 1 for gammas, so the effective dose is 450 μSv (Eqs. 31–10b and 31–9) or:

$$(100\ \text{rad/Gy})(4.5 \times 10^{-4}\ \text{Gy})(1\ \text{rem/rad}) = 45\ \text{mrem} = 0.45\ \text{mSv}.$$

NOTE This 45-mrem effective dose is almost 50% of the normal allowed dose for a whole year (100 mrem/yr), or 1% of the maximum one-year allowance for radiation workers. This worker should not receive such a large dose every day and should seek ways to reduce it (shield the source, vary the work, work farther from the source, work less time this close to source, etc.).

We have assumed that the intensity of radiation decreases as the square of the distance. It actually falls off faster than $1/r^2$ because of absorption in the air, so our answers are a slight overestimate of dose received.

PHYSICS APPLIED
Radon exposure

EXAMPLE 31–13 **Radon exposure.** In the U.S., yearly deaths from radon exposure (the second leading cause of lung cancer) are estimated to be on the order of 20,000 and maybe much more. The Environmental Protection Agency recommends taking action to reduce the radon concentration in living areas if it exceeds 4 pCi/L of air. In some areas 50% of houses exceed this level from naturally occurring radon in the soil. Estimate (*a*) the number of decays/s in 1.0 m^3 of air and (*b*) the mass of radon that emits 4.0 pCi of $^{222}_{86}$Rn radiation.

APPROACH We can use the definition of the curie to determine how many decays per second correspond to 4 pCi, then Eq. 30–3b to determine how many nuclei of radon it takes to have this activity $\Delta N/\Delta t$.

SOLUTION (*a*) We saw at the start of this Section that $1\,\text{Ci} = 3.70 \times 10^{10}$ decays/s. Thus

$$\frac{\Delta N}{\Delta t} = 4.0\,\text{pCi} = (4.0 \times 10^{-12}\,\text{Ci})(3.70 \times 10^{10}\,\text{decays/s/Ci})$$
$$= 0.148\,\text{s}^{-1}$$

per liter of air. In $1\,\text{m}^3$ of air ($1\,\text{m}^3 = 10^6\,\text{cm}^3 = 10^3\,\text{L}$) there would be $(0.148\,\text{s}^{-1})(1000) = 150$ decays/s.

(*b*) From Eqs. 30–3b and 30–6

$$\frac{\Delta N}{\Delta t} = \lambda N = \frac{0.693}{T_{\frac{1}{2}}} N.$$

Appendix B tells us $T_{\frac{1}{2}} = 3.8235$ days for radon, so

$$N = \left(\frac{\Delta N}{\Delta t}\right)\frac{T_{\frac{1}{2}}}{0.693}$$
$$= (0.148\,\text{s}^{-1})\,\frac{(3.8235\,\text{days})(8.64 \times 10^4\,\text{s/day})}{0.693}$$
$$= 7.06 \times 10^4\,\text{atoms of radon-222.}$$

The molar mass (222 u) and Avogadro's number are used to find the mass:

$$m = \frac{(7.06 \times 10^4\,\text{atoms})(222\,\text{g/mol})}{6.02 \times 10^{23}\,\text{atoms/mol}} = 2.6 \times 10^{-17}\,\text{g}$$

or 26 attograms in 1 L of air at the limit of 4 pCi/L. This 2.6×10^{-17} g/L is 2.6×10^{-14} grams of radon per m^3 of air.

NOTE Each radon atom emits 4 α particles and 4 β particles, each one capable of causing many harmful ionizations, before the sequence of decays reaches a stable element.

*31–6 Radiation Therapy

The medical application of radioactivity and radiation to human beings involves two basic aspects: (1) **radiation therapy**—the treatment of disease (mainly cancer)—which we discuss in this Section; and (2) the *diagnosis* of disease, which we discuss in the following Sections of this Chapter.

Radiation can cause cancer. It can also be used to treat it. Rapidly growing cancer cells are especially susceptible to destruction by radiation. Nonetheless, large doses are needed to kill the cancer cells, and some of the surrounding normal cells are inevitably killed as well. It is for this reason that cancer patients receiving radiation therapy often suffer side effects characteristic of radiation sickness. To minimize the destruction of normal cells, a narrow beam of γ or X-rays is often used when a cancerous tumor is well localized. The beam is directed at the tumor, and the source (or body) is rotated so that the beam passes through various parts of the body to keep the dose at any one place as low as possible—except at the tumor and its immediate surroundings, where the beam passes at all times (Fig. 31–17). The radiation may be from a radioactive source such as $^{60}_{27}$Co, or it may be from an X-ray machine that produces photons in the range 200 keV to 5 MeV. Protons, neutrons, electrons, and pions, which are produced in particle accelerators (Section 32–1), are also being used in cancer therapy.

PHYSICS APPLIED
Radiation therapy

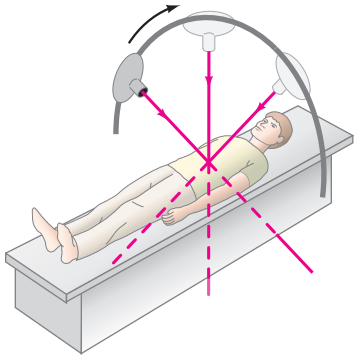


FIGURE 31–17 Radiation source rotates so that the beam always passes through the diseased tissue, but minimizes the dose in the rest of the body.

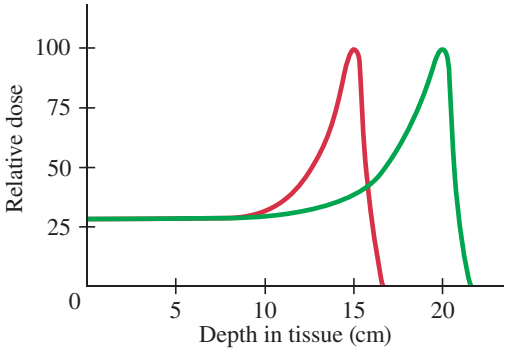


FIGURE 31–18 Energy deposited in tissue as a function of depth for 170-MeV protons (red curve) and 190-MeV protons (green). The peak of each curve is often called the Bragg peak.

PHYSICS APPLIED
Proton therapy

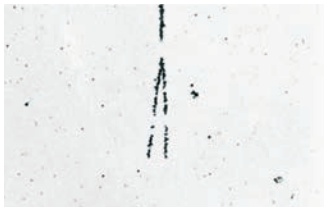
Protons used to kill tumors have a special property that makes them particularly useful. As shown in Fig. 31–18, when protons enter tissue, most of their energy is deposited at the end of their path. The protons' initial kinetic energy can be chosen so that most of the energy is deposited at the depth of the tumor itself, to destroy it. The incoming protons deposit only a small amount of energy in the tissue in front of the tumor, and none at all behind the tumor, thus having less negative effect on healthy tissue than X- or γ rays. Because tumors have physical size, even several centimeters in diameter, a range of proton energies is often used. Heavier ions, such as α particles or carbon ions, are similarly useful. This **proton therapy** technique is more than a half century old, but the necessity of having a large accelerator has meant that few hospitals have used the technique until now. Many such "proton centers" are now being built.

Another form of treatment is to insert a tiny radioactive source directly inside a tumor, which will eventually kill the majority of the cells. A similar technique is used to treat cancer of the thyroid with the radioactive isotope $^{131}_{53}\text{I}$. The thyroid gland concentrates iodine present in the bloodstream, particularly in any area where abnormal growth is taking place. Its intense radioactivity can destroy the defective cells.

Another application of radiation is for sterilizing bandages, surgical equipment, and even packaged foods such as ground beef, chicken, and produce, because bacteria and viruses can be killed or deactivated by large doses of radiation.



(a)



(b)

FIGURE 31–19 (a) Autoradiograph of a leaf exposed for 30 s to $^{14}CO_2$. Only the tissue where the CO_2 has been taken up, to be used in photosynthesis (Example 27–7), has become radioactive. The non-metabolizing tissue of the veins is free of $^{14}_{6}\text{C}$ and does not blacken the X-ray sheet. (b) Autoradiograph of chromosomal DNA. The dashed arrays of film grains show the Y-shaped growing point of replicating DNA.

*31–7 Tracers in Research and Medicine

Radioactive isotopes are used in biological and medical research as **tracers**. A given compound is artificially synthesized incorporating a radioactive isotope such as $^{14}_{6}\text{C}$ or $^{3}_{1}\text{H}$. Such "tagged" molecules can then be traced as they move through an organism or as they undergo chemical reactions. The presence of these tagged molecules (or parts of them, if they undergo chemical change) can be detected by a Geiger or scintillation counter, which detects emitted radiation (see Section 30–13). How food molecules are digested, and to what parts of the body they are diverted, can be traced in this way.

Radioactive tracers have been used to determine how amino acids and other essential compounds are synthesized by organisms. The permeability of cell walls to various molecules and ions can be determined using radioactive tracers: the tagged molecule or ion is injected into the extracellular fluid, and the radioactivity present inside and outside the cells is measured as a function of time.

In a technique known as **autoradiography**, the position of the radioactive isotopes is detected on film. For example, the distribution of carbohydrates produced in the leaves of plants from absorbed CO_2 can be observed by keeping the plant in an atmosphere where the carbon atom in the CO_2 is $^{14}_{6}\text{C}$. After a time, a leaf is placed firmly on a photographic plate and the emitted radiation darkens the film most strongly where the isotope is most strongly concentrated (Fig. 31–19a). Autoradiography using labeled nucleotides (components of DNA) has revealed much about the details of DNA replication (Fig. 31–19b). Today gamma cameras are used in a similar way—see next page.

For medical diagnosis, the radionuclide commonly used today is $^{99m}_{43}$Tc, a long-lived excited state of technetium-99 (the "m" in the symbol stands for "metastable" state). It is formed when $^{99}_{42}$Mo decays. The great usefulness of $^{99m}_{43}$Tc derives from its convenient half-life of 6 h (short, but not too short) and the fact that it can combine with a large variety of compounds. The compound to be labeled with the radionuclide is so chosen because it concentrates in the organ or region of the anatomy to be studied. Detectors outside the body then record, or image, the distribution of the radioactively labeled compound. The detection could be done by a single detector (Fig. 31–20a) which is moved across the body, measuring the intensity of radioactivity at a large number of points. The image represents the relative intensity of radioactivity at each point. The relative radioactivity is a diagnostic tool. For example, high or low radioactivity may represent overactivity or underactivity of an organ or part of an organ, or in another case may represent a lesion or tumor. More complex **gamma cameras** make use of many detectors which simultaneously record the radioactivity at many points. The measured intensities can be displayed on a TV or computer monitor. The image is sometimes called a scintigram (after scintillator), Fig. 31–20b. Gamma cameras are relatively inexpensive, but their resolution is limited—by non-perfect collimation†. Yet they allow "dynamic" studies: images that change in time, like a movie.

PHYSICS APPLIED
Medical diagnosis

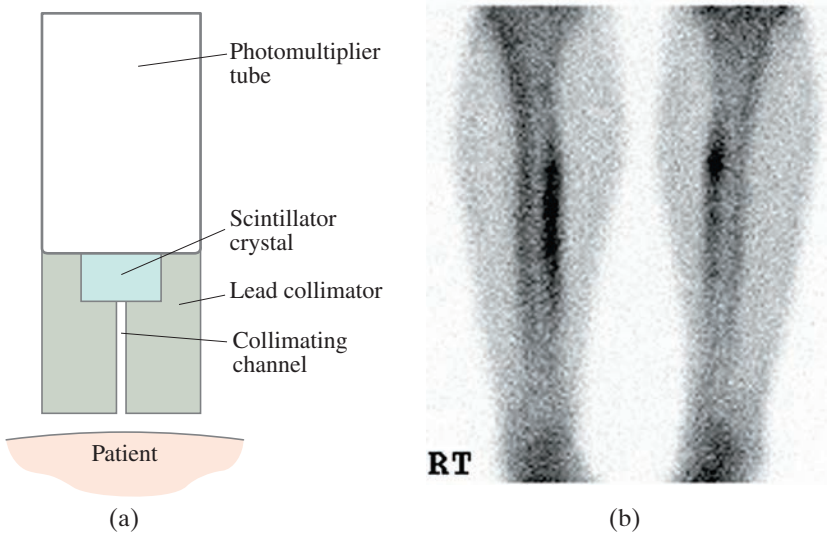


FIGURE 31–20 (a) Collimated gamma-ray detector for scanning (moving) over a patient. The collimator selects γ rays that come in a (nearly) straight line from the patient. Without the collimator, γ rays from all parts of the body could strike the scintillator, producing a poor image. Detectors today usually have many collimator tubes and are called *gamma cameras*. (b) Gamma camera image (scintigram), of both legs of a patient with shin splints, detecting γs from $^{99m}_{43}$Tc.

*31–8 Emission Tomography: PET and SPECT

The images formed using the standard techniques of nuclear medicine, as briefly discussed in the previous Section, are produced from radioactive tracer sources within the *volume* of the body. It is also possible to image the radioactive emissions from a single plane or slice through the body using the computed tomography techniques discussed in Section 25–12. A gamma camera measures the radioactive intensity from the tracer at many points and angles around the patient. The data are processed in much the same way as for X-ray CT scans (Section 25–12). This technique is referred to as **single photon emission computed tomography** (SPECT), or simply SPET (single photon emission tomography).

PHYSICS APPLIED
Medical imaging

Another important technique is **positron emission tomography** (PET), which makes use of positron emitters such as $^{11}_{6}$C, $^{13}_{7}$N, $^{15}_{8}$O, and $^{18}_{9}$F whose half-lives are short. These isotopes are incorporated into molecules that, when inhaled or injected, accumulate in the organ or region of the body to be studied.

PHYSICS APPLIED
Emission tomography (SPECT, PET)

†To "collimate" means to "make parallel," usually by blocking non-parallel rays with a narrow tube inside lead, as in Fig. 31–20a.

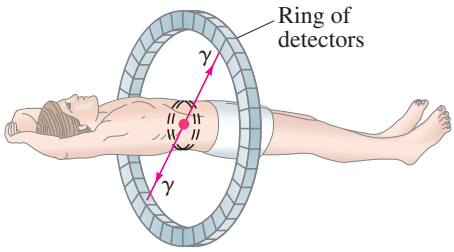


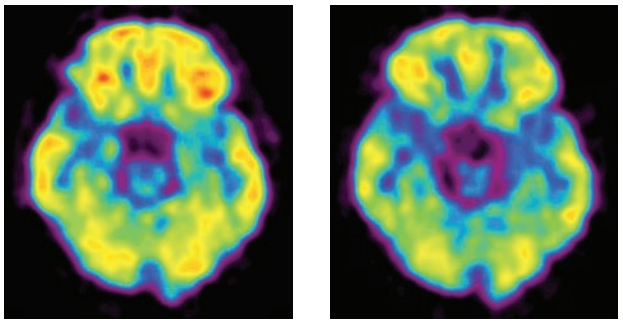
FIGURE 31–21 Positron emission tomography (PET) system showing a ring of detectors to detect the two annihilation γ rays $(e^+ + e^- \rightarrow 2\gamma)$ emitted at 180° to each other.

When such a nuclide undergoes β^+ decays, the emitted positron travels at most a few millimeters before it collides with a normal electron. In this collision, the positron and electron are annihilated, producing two γ rays $(e^+ + e^- \rightarrow 2\gamma)$, each having an energy of 511 keV $(= m_e c^2)$. The two γ rays fly off in opposite directions $(180° \pm 0.25°)$ since they must have almost exactly equal and opposite momenta to conserve momentum (the momenta of the initial e^+ and e^- are essentially zero compared to the momenta afterward of the γ rays). Because the photons travel along the same line in opposite directions, their detection in coincidence by rings of detectors surrounding the patient (Fig. 31–21) readily establishes the line along which the emission took place. If the difference in time of arrival of the two photons could be determined accurately, the actual position of the emitting nuclide along that line could be calculated. Present-day electronics can measure times to at best ± 300 ps, so at the γ ray's speed $(c = 3 \times 10^8 \text{ m/s})$, the actual position could be determined to an accuracy on the order of about $d = vt \approx (3 \times 10^8 \text{ m/s})(300 \times 10^{-12} \text{ s}) \approx 10$ cm, which is not very useful. Although there may be future potential for *time-of-flight* measurements to determine position, today computed tomography techniques are used instead, similar to those for X-ray CT, which can reconstruct PET images with a resolution on the order of 2–5 mm. One big advantage of PET is that no collimators are needed (as for detection of a single photon—see Fig. 31–20a). Thus, fewer photons are "wasted" and lower doses can be administered to the patient with PET.

Both PET and SPECT systems can give images that relate to biochemistry, metabolism, and function. This is to be compared to X-ray CT scans, whose images reflect shape and structure—that is, the anatomy of the imaged region.

Figure 31–22 shows PET scans of the same person's brain (a) when using a cell phone near the ear and (b) with the cell phone off. The bright red spots in (a) indicate a higher rate of glucose metabolism, suggesting excitability of brain tissue (the glucose was tagged with a radioactive tracer). Emfs from the cell phone antenna thus seem to affect metabolism and may be harmful to us!

The colors shown here are faked (only visible light has colors). The original images are various shades of gray, representing intensity (or counts).



(a) (b)

FIGURE 31–22 False-color PET scans of a horizontal section through a brain showing glucose metabolism rates (red is high) by a person (a) using a cell phone near the ear, and (b) with the cell phone off.

31–9 Nuclear Magnetic Resonance (NMR) and Magnetic Resonance Imaging (MRI)

Nuclear magnetic resonance (NMR) is a phenomenon which soon after its discovery in 1946 became a powerful research tool in a variety of fields from physics to chemistry and biochemistry. It is also an important medical imaging technique. We first briefly discuss the phenomenon, and then look at its applications.

* Nuclear Magnetic Resonance (NMR)

We saw in Chapter 28 (Section 28–6) that when atoms are placed in a magnetic field, atomic energy levels split into several closely spaced levels (see Fig. 28–8). Nuclei, too, exhibit these magnetic properties. We examine only the simplest, the hydrogen (H) nucleus, since it is the one most used, even for medical imaging.

The $^1_1$H nucleus consists of a single proton. Its spin angular momentum (and its magnetic moment), like that of the electron, can take on only two values when placed in a magnetic field: we call these "spin up" (parallel to the field) and "spin down" (antiparallel to the field), as suggested in Fig. 31–23. When a magnetic field is present, the energy of the nucleus splits into two levels as shown in Fig. 31–24, with the spin up (parallel to field) having the lower energy. (This is like the Zeeman effect for atomic levels, Fig. 28–8.) The difference in energy ΔE between these two levels is proportional to the total magnetic field B_T at the nucleus:

$$\Delta E = kB_T,$$

where k is a proportionality constant that is different for different nuclides.

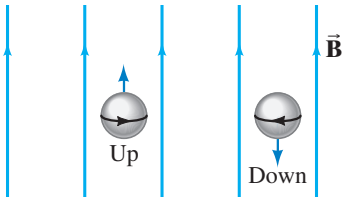


FIGURE 31–23 Schematic picture of a proton in a magnetic field $\vec{\mathbf{B}}$ (pointing upward) with the two possible states of proton spin, up and down.

In a standard **nuclear magnetic resonance** (NMR) setup, the sample to be examined is placed in a static magnetic field. A radiofrequency (RF) pulse of electromagnetic radiation (that is, photons) is applied to the sample. If the frequency, f, of this pulse corresponds precisely to the energy difference between the two energy levels (Fig. 31–24), so that

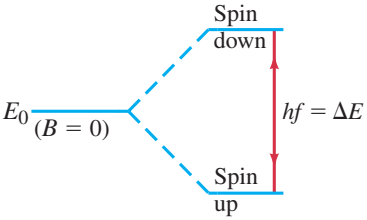
$$hf = \Delta E = kB_T, \qquad \textbf{(31–11)}$$

then the photons of the RF beam will be absorbed, exciting many of the nuclei from the lower state to the upper state. This is a resonance phenomenon because there is significant absorption only if f is very near $f = kB_T/h$. Hence the name "nuclear magnetic resonance." For free $^1_1$H nuclei, the frequency is 42.58 MHz for a magnetic field $B_T = 1.0\,\text{T}$. If the H atoms are bound in a molecule the total magnetic field B_T at the H nuclei will be the sum of the external applied field (B_{ext}) plus the local magnetic field (B_{local}) due to electrons and nuclei of neighboring atoms. Since f is proportional to B_T, the value of f for a given external field will be slightly different for bound H atoms than for free atoms:

$$hf = k(B_{ext} + B_{local}).$$

This small change in frequency can be measured, and is called the "chemical shift." A great deal has been learned about the structure of molecules and bonds using this NMR technique.

FIGURE 31–24 Energy E_0 in the absence of a magnetic field splits into two levels in the presence of a magnetic field.



* Magnetic Resonance Imaging (MRI)

PHYSICS APPLIED
NMR imaging (MRI)

For producing medically useful NMR images—now commonly called MRI, or **magnetic resonance imaging**—the element most used is hydrogen since it is the commonest element in the human body and gives the strongest NMR signals. The experimental apparatus is shown in Fig. 31–25. The large coils set up the static magnetic field, and the RF coils produce the RF pulse of electromagnetic waves (photons) that cause the nuclei to jump from the lower state to the upper one (Fig. 31–24). These same coils (or another coil) can detect the absorption of energy or the emitted radiation (also of frequency $f = \Delta E/h$, Eq. 31–11) when the nuclei jump back down to the lower state.

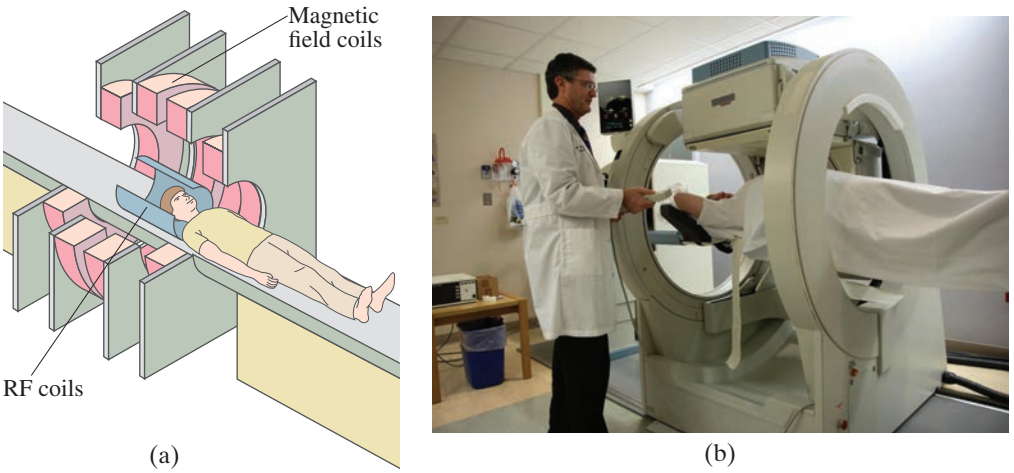


FIGURE 31–25 NMR imaging setup: (a) diagram; (b) photograph.

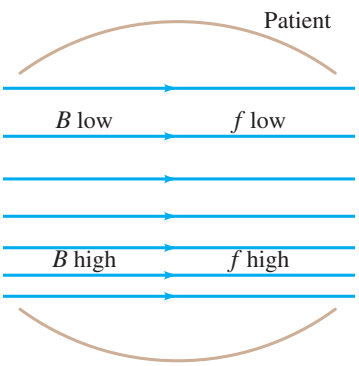
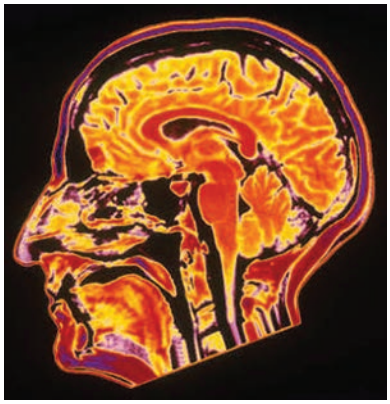


FIGURE 31–26 A static field that is stronger at the bottom than at the top. The frequency of absorbed or emitted radiation is proportional to B in NMR.

The formation of a two-dimensional or three-dimensional image can be done using techniques similar to those for computed tomography (Section 25–12). The simplest thing to measure for creating an image is the intensity of absorbed and/or reemitted radiation from many different points of the body, and this would be a measure of the density of H atoms at each point. But how do we determine from what part of the body a given photon comes? One technique is to give the static magnetic field a gradient; that is, instead of applying a uniform magnetic field, B_T, the field is made to vary with position across the width of the sample (or patient), Fig. 31–26. Because the frequency absorbed by the H nuclei is proportional to B_T (Eq. 31–11), only one plane within the body will have the proper value of B_T to absorb photons of a particular frequency f. By varying f, absorption by different planes can be measured. Alternately, if the field gradient is applied *after* the RF pulse, the frequency of the emitted photons will be a measure of where they were emitted. If a magnetic field gradient in one direction is applied during excitation (absorption of photons) and photons of a single frequency are transmitted, only H nuclei in one thin slice will be excited. By applying a gradient during reemission in a direction perpendicular to the first, the frequency f of the reemitted radiation will represent depth in that slice. Other ways of varying the magnetic field throughout the volume of the body can be used in order to correlate NMR frequency with position.

A reconstructed image based on the density of H atoms (that is, the intensity of absorbed or emitted radiation) is not very interesting. More useful are images based on the rate at which the nuclei decay back to the ground state, and such images can produce resolution of 1 mm or better. This NMR technique (sometimes called **spin-echo**) produces images of great diagnostic value, both in the delineation of structure (anatomy) and in the study of metabolic processes. An NMR image is shown in Fig. 31–27.

FIGURE 31–27 False-color NMR image (MRI) through the head showing structures in the brain.



NMR imaging is considered to be noninvasive. We can calculate the energy of the photons involved: as mentioned above, in a 1.0-T magnetic field, $f = 42.58\,\text{MHz}$ for ${}^{1}_{1}\text{H}$. This corresponds to an energy of $hf = (6.6 \times 10^{-34}\,\text{J}\cdot\text{s})(43 \times 10^{6}\,\text{Hz}) \approx 3 \times 10^{-26}\,\text{J}$ or about $10^{-7}\,\text{eV}$. Since molecular bonds are on the order of 1 eV, the RF photons can cause little cellular disruption. This should be compared to X- or γ rays, whose energies are 10^4 to 10^6 eV and thus can cause significant damage. The static magnetic fields, though often large (as high as 1.0 to 1.5 T), are believed to be harmless (except for people who wear heart pacemakers).

TABLE 31–2 Medical Imaging Techniques

Technique	Where Discussed in This Book	Optimal Resolution
Conventional X-ray	Section 25–12	$\frac{1}{2}$ mm
CT scan, X-ray	Section 25–12	$\frac{1}{2}$ mm
Nuclear medicine (tracers)	Section 31–7	1 cm
SPECT (single photon emission)	Section 31–8	1 cm
PET (positron emission)	Section 31–8	2–5 mm
MRI (NMR)	Section 31–9	$\frac{1}{2}$–1 mm
Ultrasound	Section 12–9	0.3–2 mm

Table 31–2 lists the major techniques we have discussed for imaging the interior of the human body, along with the optimum resolution attainable today. Resolution is only one factor that must be considered, because the different imaging techniques provide different types of information that are useful for different types of diagnosis.

Summary

A **nuclear reaction** occurs when two nuclei collide and two or more other nuclei (or particles) are produced. In this process, as in radioactivity, **transmutation** (change) of elements occurs.

The **reaction energy** or ***Q*-value** of a reaction $a + X \rightarrow Y + b$ is

$$Q = (M_a + M_X - M_b - M_Y)c^2 \qquad \textbf{(31–2a)}$$

$$= KE_b + KE_Y - KE_a - KE_X. \qquad \textbf{(31–2b)}$$

In **fission**, a heavy nucleus such as uranium splits into two intermediate-sized nuclei after being struck by a neutron. $^{235}_{92}U$ is fissionable by slow neutrons, whereas some fissionable nuclei require fast neutrons. Much energy is released in fission (≈ 200 MeV per fission) because the binding energy per nucleon is lower for heavy nuclei than it is for intermediate-sized nuclei, so the mass of a heavy nucleus is greater than the total mass of its fission products. The fission process releases neutrons, so that a **chain reaction** is possible. The **critical mass** is the minimum mass of fuel needed so that enough emitted neutrons go on to produce more fissions and sustain a chain reaction. In a **nuclear reactor** or nuclear weapon, a **moderator** is used to slow down the released neutrons.

The **fusion** process, in which small nuclei combine to form larger ones, also releases energy. The energy from our Sun originates in the fusion reactions known as the **proton–proton chain** in which four protons fuse to form a ^4_2He nucleus producing 25 MeV of energy. A useful fusion reactor for power generation has not yet proved possible because of the difficulty in containing the fuel (e.g., deuterium) long enough at the extremely high temperature required ($\approx 10^8$K). Nonetheless, progress has been made in confining the collection of charged ions known as a **plasma**. The two main methods are **magnetic confinement**, using a magnetic field in a device such as the donut-shaped **tokamak**, and **inertial confinement** in which intense laser beams compress a fuel pellet of deuterium and tritium.

Radiation can cause damage to materials, including biological tissue. Quantifying amounts of radiation is the subject of **dosimetry**. The **curie** (Ci) and the **becquerel** (Bq) are units that measure the **source activity** or rate of decay of a sample: $1\text{ Ci} = 3.70 \times 10^{10}$ decays per second, whereas 1 Bq = 1 decay/s. The **absorbed dose**, often specified in **rads**, measures the amount of energy deposited per unit mass of absorbing material: 1 rad is the amount of radiation that deposits energy at the rate of 10^{-2} J/kg of material. The SI unit of absorbed dose is the **gray**: 1 Gy = 1 J/kg = 100 rad. The **effective dose** is often specified by the **rem** = rad × RBE, where RBE is the "relative biological effectiveness" of a given type of radiation; 1 rem of any type of radiation does approximately the same amount of biological damage. The average dose received per person per year in the United States is about 360 mrem. The SI unit for effective dose is the **sievert**: 1 Sv = 100 rem.

[*Nuclear radiation is used in medicine for cancer therapy, and for imaging of biological structure and processes. Tomographic imaging of the human body, which can provide 3-dimensional detail, includes several types: PET, SPET (= SPECT), MRI, and CT scans (discussed in Chapter 25). MRI makes use of **nuclear magnetic resonance** (NMR).]

Questions

1. Fill in the missing particles or nuclei:
(a) $n + {}^{232}_{90}Th \rightarrow ? + \gamma$;
(b) $n + {}^{137}_{56}Ba \rightarrow {}^{137}_{55}Cs + ?$;
(c) $d + {}^{2}_{1}H \rightarrow {}^{4}_{2}He + ?$;
(d) $\alpha + {}^{197}_{79}Au \rightarrow ? + d$
where d stands for deuterium.

2. When $^{22}_{11}Na$ is bombarded by deuterons $\left({}^{2}_{1}H\right)$, an α particle is emitted. What is the resulting nuclide? Write down the reaction equation.

3. Why are neutrons such good projectiles for producing nuclear reactions?

4. What is the Q-value for radioactive decay reactions?
(a) $Q < 0$. (b) $Q > 0$. (c) $Q = 0$.
(d) The sign of Q depends on the nucleus.

5. The energy from nuclear fission appears in the form of thermal energy—but the thermal energy of what?

6. (a) If $^{235}_{92}U$ released only 1.5 neutrons per fission on average (instead of 2.5), would a chain reaction be possible? (b) If so, how would the chain reaction be different than if 3 neutrons were released per fission?

7. Why can't uranium be enriched by chemical means?

8. How can a neutron, with practically no kinetic energy, excite a nucleus to the extent shown in Fig. 31–3?

9. Why would a porous block of uranium be more likely to explode if kept under water rather than in air?

10. A reactor that uses highly enriched uranium can use ordinary water (instead of heavy water) as a moderator and still have a self-sustaining chain reaction. Explain.

11. Why must the fission process release neutrons if it is to be useful?

12. Why are neutrons released in a fission reaction?

13. What is the reason for the "secondary system" in a nuclear reactor, Fig. 31–8? That is, why is the water heated by the fuel in a nuclear reactor not used directly to drive the turbines?

14. What is the basic difference between fission and fusion?

15. Discuss the relative merits and disadvantages, including pollution and safety, of power generation by fossil fuels, nuclear fission, and nuclear fusion.

16. Why do gamma particles penetrate matter more easily than beta particles do?

17. Light energy emitted by the Sun and stars comes from the fusion process. What conditions in the interior of stars make this possible?

18. How do stars, and our Sun, maintain confinement of the plasma for fusion?

19. People who work around metals that emit alpha particles are trained that there is little danger from proximity or touching the material, but they must take extreme precautions against ingesting it. Why? (Eating and drinking while working are forbidden.)

20. What is the difference between absorbed dose and effective dose? What are the SI units for each?

21. Radiation is sometimes used to sterilize medical supplies and even food. Explain how it works.

***22.** How might radioactive tracers be used to find a leak in a pipe?

MisConceptual Questions

1. In a nuclear reaction, which of the following is *not* conserved?
(*a*) Energy.
(*b*) Momentum.
(*c*) Electric charge.
(*d*) Nucleon number.
(*e*) None of the above.

2. Fission fragments are typically
(*a*) β^+ emitters.
(*b*) β^- emitters.
(*c*) Both.
(*d*) Neither.

3. Which of the following properties would decrease the critical mass needed to sustain a nuclear chain reaction?
(*a*) Low boiling point.
(*b*) High melting point.
(*c*) More neutrons released per fission.
(*d*) Low nuclear density.
(*e*) Filled valence shell.
(*f*) All of the above.

4. Rather than having a maximum at about $A \approx 60$, as shown in Fig. 31–12, suppose the average binding energy per nucleon continually increased with increasing mass number. Then,
(*a*) fission would still be possible, but not fusion.
(*b*) fusion would still be possible, but not fission.
(*c*) both fission and fusion would still be possible.
(*d*) neither fission nor fusion would be possible.

5. Why is a moderator needed in a normal uranium fission reactor?
(*a*) To increase the rate of neutron capture by uranium-235.
(*b*) To increase the rate of neutron capture by uranium-238.
(*c*) To increase the rate of production of plutonium-239.
(*d*) To increase the critical mass of the fission fuel.
(*e*) To provide more neutrons for the reaction.
(*f*) All of the above.

6. What is the difference between nuclear fission and nuclear fusion?
(*a*) Nuclear fission is used for bombs; nuclear fusion is used in power plants.
(*b*) There is no difference. Fission and fusion are different names for the same physical phenomenon.
(*c*) Nuclear fission refers to using deuterium to create a nuclear reaction.
(*d*) Nuclear fusion occurs spontaneously, as happens to the C^{14} used in carbon dating.
(*e*) In nuclear fission, a nucleus splits; in nuclear fusion, nucleons or nuclei and nucleons join to form a new nucleus.

7. A primary difficulty in energy production by fusion is
(*a*) the scarcity of necessary fuel.
(*b*) the disposal of radioactive by-products produced.
(*c*) the high temperatures necessary to overcome the electrical repulsion of protons.
(*d*) the fact that it is possible in volcanic regions only.

8. If two hydrogen nuclei, ${}^2_1\text{H}$, each of mass m_H, fuse together and form a helium nucleus of mass m_He,
(*a*) $m_\text{He} < 2m_\text{H}$.
(*b*) $m_\text{He} = 2m_\text{H}$.
(*c*) $m_\text{He} > 2m_\text{H}$.
(*d*) All of the above are possible.

9. Which radiation induces the most biological damage for a given amount of energy deposited in tissue?
(*a*) Alpha particles.
(*b*) Gamma radiation.
(*c*) Beta radiation.
(*d*) All do the same damage for the same deposited energy.

10. Which would produce the most energy in a single reaction?
(*a*) The fission reaction associated with uranium-235.
(*b*) The fusion reaction of the Sun (two hydrogen nuclei fused to one helium nucleus).
(*c*) Both (*a*) and (*b*) are about the same.
(*d*) Need more information.

11. The fuel necessary for fusion-produced energy could be derived from
(*a*) water.
(*b*) superconductors.
(*c*) uranium.
(*d*) helium.
(*e*) sunlight.

12. Which of the following is true?
(*a*) Any amount of radiation is harmful to living tissue.
(*b*) Radiation is a natural part of the environment.
(*c*) All forms of radiation will penetrate deep into living tissue.
(*d*) None of the above is true.

13. Which of the following would reduce the cell damage due to radiation for a lab technician who works with radioactive isotopes in a hospital or lab?
(*a*) Increase the worker's distance from the radiation source.
(*b*) Decrease the time the worker is exposed to the radiation.
(*c*) Use shielding to reduce the amount of radiation that strikes the worker.
(*d*) Have the worker wear a radiation badge when working with the radioactive isotopes.
(*e*) All of the above.

14. If the same dose of each type of radiation was provided over the same amount of time, which type would be most harmful?
(*a*) X-rays.
(*b*) γ rays.
(*c*) β rays.
(*d*) α particles.

15. ${}^{235}_{92}\text{U}$ releases an average of 2.5 neutrons per fission compared to 2.9 for ${}^{239}_{94}\text{Pu}$. Which has the smaller critical mass?
(*a*) ${}^{235}_{92}\text{U}$.
(*b*) ${}^{239}_{94}\text{Pu}$.
(*c*) Both the same.

For assigned homework and other learning materials, go to the MasteringPhysics website.

Problems

(NOTE: Masses are found in Appendix B.)

31–1 Nuclear Reactions, Transmutation

1. (I) Natural aluminum is all $^{27}_{13}\text{Al}$. If it absorbs a neutron, what does it become? Does it decay by β^+ or β^-? What will be the product nucleus?
2. (I) Determine whether the reaction $^{2}_{1}\text{H} + ^{2}_{1}\text{H} \rightarrow ^{3}_{2}\text{He} + \text{n}$ requires a threshold energy, and why.
3. (I) Is the reaction $\text{n} + ^{238}_{92}\text{U} \rightarrow ^{239}_{92}\text{U} + \gamma$ possible with slow neutrons? Explain.
4. (II) (*a*) Complete the following nuclear reaction, $\text{p} + ? \rightarrow ^{32}_{16}\text{S} + \gamma$. (*b*) What is the *Q*-value?
5. (II) The reaction $\text{p} + ^{18}_{8}\text{O} \rightarrow ^{18}_{9}\text{F} + \text{n}$ requires an input of energy equal to 2.438 MeV. What is the mass of $^{18}_{9}\text{F}$?
6. (II) (*a*) Can the reaction $\text{n} + ^{24}_{12}\text{Mg} \rightarrow ^{23}_{11}\text{Na} + \text{d}$ occur if the bombarding particles have 18.00 MeV of kinetic energy? (d stands for deuterium, $^{2}_{1}\text{H}$.) (*b*) If so, how much energy is released? If not, what kinetic energy is needed?
7. (II) (*a*) Can the reaction $\text{p} + ^{7}_{3}\text{Li} \rightarrow ^{4}_{2}\text{He} + \alpha$ occur if the incident proton has kinetic energy = 3100 keV? (*b*) If so, what is the total kinetic energy of the products? If not, what kinetic energy is needed?
8. (II) In the reaction $\alpha + ^{14}_{7}\text{N} \rightarrow ^{17}_{8}\text{O} + \text{p}$, the incident α particles have 9.85 MeV of kinetic energy. The mass of $^{17}_{8}\text{O}$ is 16.999132 u. (*a*) Can this reaction occur? (*b*) If so, what is the total kinetic energy of the products? If not, what kinetic energy is needed?
9. (II) Calculate the *Q*-value for the "capture" reaction $\alpha + ^{16}_{8}\text{O} \rightarrow ^{20}_{10}\text{Ne} + \gamma$.
10. (II) Calculate the total kinetic energy of the products of the reaction $\text{d} + ^{13}_{6}\text{C} \rightarrow ^{14}_{7}\text{N} + \text{n}$ if the incoming deuteron has kinetic energy KE = 41.4 MeV.
11. (II) Radioactive $^{14}_{6}\text{C}$ is produced in the atmosphere when a neutron is absorbed by $^{14}_{7}\text{N}$. Write the reaction and find its *Q*-value.
12. (II) An example of a **stripping** nuclear reaction is $\text{d} + ^{6}_{3}\text{Li} \rightarrow \text{X} + \text{p}$. (*a*) What is X, the resulting nucleus? (*b*) Why is it called a "stripping" reaction? (*c*) What is the *Q*-value of this reaction? Is the reaction endothermic or exothermic?
13. (II) An example of a **pick-up** nuclear reaction is $^{3}_{2}\text{He} + ^{12}_{6}\text{C} \rightarrow \text{X} + \alpha$. (*a*) Why is it called a "pick-up" reaction? (*b*) What is the resulting nucleus? (*c*) What is the *Q*-value of this reaction? Is the reaction endothermic or exothermic?
14. (II) Does the reaction $\text{p} + ^{7}_{3}\text{Li} \rightarrow ^{4}_{2}\text{He} + \alpha$ require energy, or does it release energy? How much energy?
15. (II) Calculate the energy released (or energy input required) for the reaction $\alpha + ^{9}_{4}\text{Be} \rightarrow ^{12}_{6}\text{C} + \text{n}$.

31–2 Nuclear Fission

16. (I) What is the energy released in the fission reaction of Eq. 31–4? (The masses of $^{141}_{56}\text{Ba}$ and $^{92}_{36}\text{Kr}$ are 140.914411 u and 91.926156 u, respectively.)
17. (I) Calculate the energy released in the fission reaction $\text{n} + ^{235}_{92}\text{U} \rightarrow ^{88}_{38}\text{Sr} + ^{136}_{54}\text{Xe} + 12\text{n}$. Use Appendix B, and assume the initial kinetic energy of the neutron is very small.
18. (I) How many fissions take place per second in a 240-MW reactor? Assume 200 MeV is released per fission.
19. (I) The energy produced by a fission reactor is about 200 MeV per fission. What fraction of the mass of a $^{235}_{92}\text{U}$ nucleus is this?
20. (II) Suppose that the average electric power consumption, day and night, in a typical house is 960 W. What initial mass of $^{235}_{92}\text{U}$ would have to undergo fission to supply the electrical needs of such a house for a year? (Assume 200 MeV is released per fission, as well as 100% efficiency.)
21. (II) Consider the fission reaction

$$^{235}_{92}\text{U} + \text{n} \rightarrow ^{133}_{51}\text{Sb} + ^{98}_{41}\text{Nb} + ?\text{n}.$$

(*a*) How many neutrons are produced in this reaction? (*b*) Calculate the energy release. The atomic masses for Sb and Nb isotopes are 132.915250 u and 97.910328 u, respectively.

22. (II) How much mass of $^{235}_{92}\text{U}$ is required to produce the same amount of energy as burning 1.0 kg of coal (about 3×10^7 J)?
23. (II) What initial mass of $^{235}_{92}\text{U}$ is required to operate a 950-MW reactor for 1 yr? Assume 34% efficiency.
24. (II) If a 1.0-MeV neutron emitted in a fission reaction loses one-half of its kinetic energy in each collision with moderator nuclei, how many collisions must it make to reach thermal energy $(\frac{3}{2}kT = 0.040\text{ eV})$?
25. (II) Assuming a fission of $^{236}_{92}\text{U}$ into two roughly equal fragments, estimate the electric potential energy just as the fragments separate from each other. Assume that the fragments are spherical (see Eq. 30–1) and compare your calculation to the nuclear fission energy released, about 200 MeV.
26. (III) Suppose that the neutron multiplication factor is 1.0004. If the average time between successive fissions in a chain of reactions is 1.0 ms, by what factor will the reaction rate increase in 1.0 s?

31–3 Nuclear Fusion

27. (I) What is the average kinetic energy of protons at the center of a star where the temperature is 2×10^7 K? [*Hint*: See Eq. 13–8.]
28. (II) Show that the energy released in the fusion reaction $^{2}_{1}\text{H} + ^{3}_{1}\text{H} \rightarrow ^{4}_{2}\text{He} + \text{n}$ is 17.59 MeV.
29. (II) Show that the energy released when two deuterium nuclei fuse to form $^{3}_{2}\text{He}$ with the release of a neutron is 3.27 MeV (Eq. 31–8b).

30. (II) Verify the Q-value stated for each of the reactions of Eqs. 31–6. [*Hint*: Use Appendix B; be careful with electrons (included in mass values except for p, d, t).]

31. (II) (*a*) Calculate the energy release per gram of fuel for the reactions of Eqs. 31–8a, b, and c. (*b*) Calculate the energy release per gram of uranium $^{235}_{92}$U in fission, and give its ratio to each reaction in (*a*).

32. (II) How much energy is released when $^{238}_{92}$U absorbs a slow neutron (kinetic energy ≈ 0) and becomes $^{239}_{92}$U?

33. (II) If a typical house requires 960 W of electric power on average, what minimum amount of deuterium fuel would have to be used in a year to supply these electrical needs? Assume the reaction of Eq. 31–8b.

34. (II) Suppose a fusion reactor ran on "d–d" reactions, Eqs. 31–8a and b in equal amounts. Estimate how much natural water, for fuel, would be needed per hour to run a 1150-MW reactor, assuming 33% efficiency.

35. (III) Show that the energies carried off by the $^{4}_{2}$He nucleus and the neutron for the reaction of Eq. 31–8c are about 3.5 MeV and 14 MeV, respectively. Are these fixed values, independent of the plasma temperature?

36. (III) How much energy (J) is contained in 1.00 kg of water if its natural deuterium is used in the fusion reaction of Eq. 31–8a? Compare to the energy obtained from the burning of 1.0 kg of gasoline, about 5×10^7 J.

37. (III) (*a*) Give the ratio of the energy needed for the first reaction of the *carbon cycle* to the energy needed for a deuterium–tritium reaction (Example 31–9). (*b*) If a deuterium–tritium reaction actually requires a temperature $T \approx 3 \times 10^8$ K, estimate the temperature needed for the first carbon-cycle reaction.

31–5 Dosimetry

38. (I) 350 rads of α-particle radiation is equivalent to how many rads of X-rays in terms of biological damage?

39. (I) How many rads of slow neutrons will do as much biological damage as 72 rads of fast neutrons?

40. (II) How much energy is deposited in the body of a 65-kg adult exposed to a 2.5-Gy dose?

41. (II) A cancer patient is undergoing radiation therapy in which protons with an energy of 1.2 MeV are incident on a 0.20-kg tumor. (*a*) If the patient receives an effective dose of 1.0 rem, what is the absorbed dose? (*b*) How many protons are absorbed by the tumor? Assume RBE ≈ 1.

42. (II) A 0.035-μCi sample of $^{32}_{15}$P is injected into an animal for tracer studies. If a Geiger counter intercepts 35% of the emitted β particles, what will be the counting rate, assumed 85% efficient?

43. (II) About 35 eV is required to produce one ion pair in air. Show that this is consistent with the two definitions of the roentgen given in the text.

44. (II) A 1.6-mCi source of $^{32}_{15}$P (in $NaHPO_4$), a β emitter, is implanted in a tumor where it is to administer 32 Gy. The half-life of $^{32}_{15}$P is 14.3 days, and 1.0 mCi delivers about 10 mGy/min. Approximately how long should the source remain implanted?

45. (II) What is the mass of a 2.50-μCi $^{14}_{6}$C source?

46. (II) $^{57}_{27}$Co emits 122-keV γ rays. If a 65-kg person swallowed 1.55 μCi of $^{57}_{27}$Co, what would be the dose rate (Gy/day) averaged over the whole body? Assume that 50% of the γ-ray energy is deposited in the body. [*Hint*: Determine the rate of energy deposited in the body and use the definition of the gray.]

47. (II) Ionizing radiation can be used on meat products to reduce the levels of microbial pathogens. Refrigerated meat is limited to 4.5 kGy. If 1.6-MeV electrons irradiate 5 kg of beef, how many electrons would it take to reach the allowable limit?

48. (III) Huge amounts of radioactive $^{131}_{53}$I were released in the accident at Chernobyl in 1986. Chemically, iodine goes to the human thyroid. (It can be used for diagnosis and treatment of thyroid problems.) In a normal thyroid, $^{131}_{53}$I absorption can cause damage to the thyroid. (*a*) Write down the reaction for the decay of $^{131}_{53}$I. (*b*) Its half-life is 8.0 d; how long would it take for ingested $^{131}_{53}$I to become 5.0% of the initial value? (*c*) Absorbing 1 mCi of $^{131}_{53}$I can be harmful; what mass of iodine is this?

49. (III) Assume a liter of milk typically has an activity of 2000 pCi due to $^{40}_{19}$K. If a person drinks two glasses (0.5 L) per day, estimate the total effective dose (in Sv and in rem) received in a year. As a crude model, assume the milk stays in the stomach 12 hr and is then released. Assume also that roughly 10% of the 1.5 MeV released per decay is absorbed by the body. Compare your result to the normal allowed dose of 100 mrem per year. Make your estimate for (*a*) a 60-kg adult, and (*b*) a 6-kg baby.

50. (III) Radon gas, $^{222}_{86}$Rn, is considered a serious health hazard (see discussion in text). It decays by α-emission. (*a*) What is the daughter nucleus? (*b*) Is the daughter nucleus stable or radioactive? If the latter, how does it decay, and what is its half-life? (See Fig. 30–11.) (*c*) Is the daughter nucleus also a noble gas, or is it chemically reactive? (*d*) Suppose 1.4 ng of $^{222}_{86}$Rn seeps into a basement. What will be its activity? If the basement is then sealed, what will be the activity 1 month later?

31–9 NMR

51. (II) Calculate the wavelength of photons needed to produce NMR transitions in free protons in a 1.000-T field. In what region of the spectrum is this wavelength?

General Problems

52. Consider a system of nuclear power plants that produce 2100 MW. (*a*) What total mass of $^{235}_{92}$U fuel would be required to operate these plants for 1 yr, assuming that 200 MeV is released per fission? (*b*) Typically 6% of the $^{235}_{92}$U nuclei that fission produce strontium–90, $^{90}_{38}$Sr, a β^- emitter with a half-life of 29 yr. What is the total radioactivity of the $^{90}_{38}$Sr, in curies, produced in 1 yr? (Neglect the fact that some of it decays during the 1-yr period.)

53. J. Chadwick discovered the neutron by bombarding $^{9}_{4}$Be with the popular projectile of the day, alpha particles. (*a*) If one of the reaction products was the then unknown neutron, what was the other product? (*b*) What is the *Q*-value of this reaction?

54. Fusion temperatures are often given in keV. Determine the conversion factor from kelvins to keV using, as is common in this field, $\overline{\text{KE}} = kT$ without the factor $\frac{3}{2}$.

55. One means of enriching uranium is by diffusion of the gas UF_6. Calculate the ratio of the speeds of molecules of this gas containing $^{235}_{92}$U and $^{238}_{92}$U, on which this process depends.

56. (*a*) What mass of $^{235}_{92}$U was actually fissioned in the first atomic bomb, whose energy was the equivalent of about 20 kilotons of TNT (1 kiloton of TNT releases 5×10^{12} J)? (*b*) What was the actual mass transformed to energy?

57. The average yearly background radiation in a certain town consists of 32 mrad of X-rays and γ rays plus 3.4 mrad of particles having a RBE of 10. How many rem will a person receive per year on average?

58. A shielded γ-ray source yields a dose rate of 0.048 rad/h at a distance of 1.0 m for an average-sized person. If workers are allowed a maximum dose of 5.0 rem in 1 year, how close to the source may they operate, assuming a 35-h work week? Assume that the intensity of radiation falls off as the square of the distance. (It actually falls off more rapidly than $1/r^2$ because of absorption in the air, so your answer will give a better-than-permissible value.)

59. In the net reaction, Eq. 31–7, for the proton–proton chain in the Sun, the neutrinos escape from the Sun with energy of about 0.5 MeV. The remaining energy, 26.2 MeV, is available to heat the Sun. Use this value to calculate the "heat of combustion" per kilogram of hydrogen fuel and compare it to the heat of combustion of coal, about 3×10^7 J/kg.

60. Energy reaches Earth from the Sun at a rate of about 1300 W/m^2. Calculate (*a*) the total power output of the Sun, and (*b*) the number of protons consumed per second in the reaction of Eq. 31–7, assuming that this is the source of all the Sun's energy. (*c*) Assuming that the Sun's mass of 2.0×10^{30} kg was originally all protons and that all could be involved in nuclear reactions in the Sun's core, how long would you expect the Sun to "glow" at its present rate? See Problem 59. [*Hint*: Use $1/r^2$ law.]

61. Radon gas, $^{222}_{86}$Rn, is formed by α decay. (*a*) Write the decay equation. (*b*) Ignoring the kinetic energy of the daughter nucleus (it's so massive), estimate the kinetic energy of the α particle produced. (*c*) Estimate the momentum of the alpha and of the daughter nucleus. (*d*) Estimate the kinetic energy of the daughter, and show that your approximation in (*b*) was valid.

62. Estimate how many solar neutrinos pass through a 180-m^2 ceiling of a room, at latitude 44°, for an hour around midnight on midsummer night. [*Hint*: See Problems 59 and 60.]

63. Estimate how much total energy would be released via fission if 2.0 kg of uranium were enriched to 5% of the isotope $^{235}_{92}$U.

64. Some stars, in a later stage of evolution, may begin to fuse two $^{12}_{6}$C nuclei into one $^{24}_{12}$Mg nucleus. (*a*) How much energy would be released in such a reaction? (*b*) What kinetic energy must two carbon nuclei each have when far apart, if they can then approach each other to within 6.0 fm, center-to-center? (*c*) Approximately what temperature would this require?

65. An average adult body contains about 0.10 μCi of $^{40}_{19}$K, which comes from food. (*a*) How many decays occur per second? (*b*) The potassium decay produces beta particles with energies of around 1.4 MeV. Estimate the dose per year in sieverts for a 65-kg adult. Is this a significant fraction of the 3.6-mSv/yr background rate?

66. When the nuclear reactor accident occurred at Chernobyl in 1986, 2.0×10^7 Ci were released into the atmosphere. Assuming that this radiation was distributed uniformly over the surface of the Earth, what was the activity per square meter? (The actual activity was not uniform; even within Europe wet areas received more radioactivity from rainfall.)

67. A star with a large helium abundance can burn helium in the reaction $^{4}_{2}\text{He} + {}^{4}_{2}\text{He} + {}^{4}_{2}\text{He} \rightarrow {}^{12}_{6}\text{C}$. What is the *Q*-value for this reaction?

68. A 1.2-μCi $^{137}_{55}$Cs source is used for 1.4 hours by a 62-kg worker. Radioactive $^{137}_{55}$Cs decays by β^- decay with a half-life of 30 yr. The average energy of the emitted betas is about 190 keV per decay. The β decay is quickly followed by a γ with an energy of 660 keV. Assuming the person absorbs *all* emitted energy, what effective dose (in rem) is received?

69. If a 65-kg power plant worker has been exposed to the maximum slow-neutron radiation for a given year, how much total energy (in J) has that worker absorbed? What if he were exposed to fast protons?

70. A large amount of $^{90}_{38}$Sr was released during the Chernobyl nuclear reactor accident in 1986. The $^{90}_{38}$Sr enters the body through the food chain. How long will it take for 85% of the $^{90}_{38}$Sr released during the accident to decay? See Appendix B.

71. Three radioactive sources have the same activity, 35 mCi. Source A emits 1.0-MeV γ rays, source B emits 2.0-MeV γ rays, and source C emits 2.0-MeV alphas. What is the relative danger of these sources?

72. A 55-kg patient is to be given a medical test involving the ingestion of $^{99m}_{43}$Tc (Section 31–7) which decays by emitting a 140-keV gamma. The half-life for this decay is 6 hours. Assuming that about half the gamma photons exit the body without interacting with anything, what must be the initial activity of the Tc sample if the whole-body dose cannot exceed 50 mrem? Make the rough approximation that biological elimination of Tc can be ignored.

Search and Learn

1. Referring to Section 31–3, (*a*) why can small nuclei combine to form larger ones, releasing energy in the process? (*b*) Why does the first reaction in the proton–proton chain limit the rate at which the Sun produces energy? (*c*) What are the heaviest elements for which energy is released if the elements are created by fusion of lighter elements? (*d*) What keeps the Sun and stars together, allowing them to sustain fusion? (*e*) What two methods are currently being investigated to contain high-temperature plasmas on the Earth to create fusion in the laboratory?

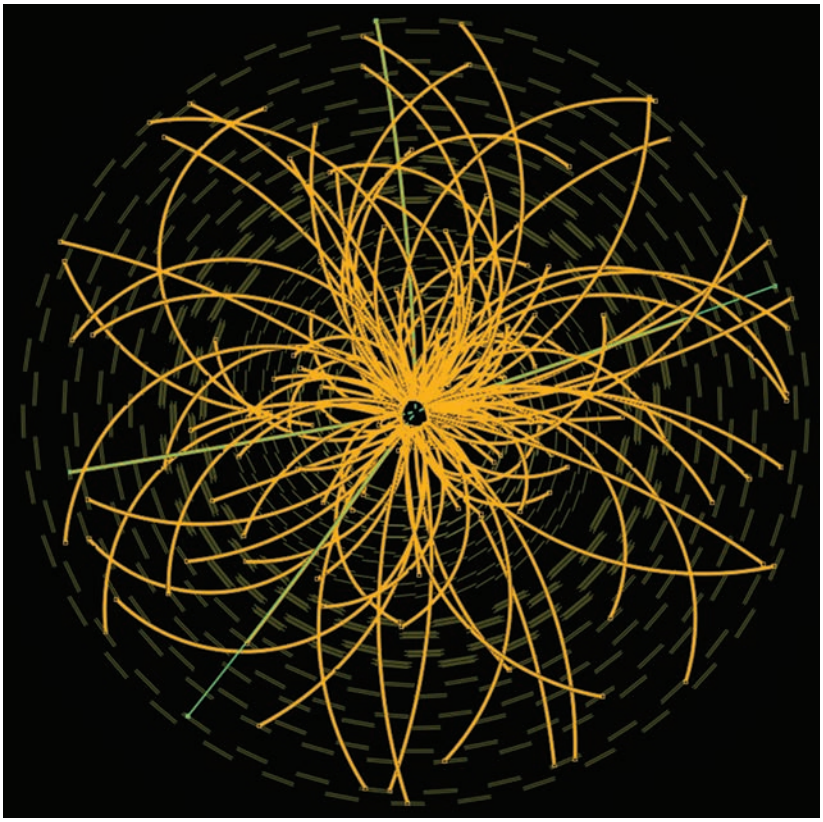
2. Deuterium makes up 0.0115% of natural hydrogen on average. Make a rough estimate of the total deuterium in the Earth's oceans and estimate the total energy released if all of it were used in fusion reactors.

3. The energy output of massive stars is believed to be due to the *carbon cycle* (see text). (*a*) Show that no carbon is consumed in this cycle and that the net effect is the same as for the proton–proton chain. (*b*) What is the total energy release? (*c*) Determine the energy output for each reaction and decay. (*d*) Why might the carbon cycle require a higher temperature $(\approx 2 \times 10^7\ \text{K})$ than the proton–proton chain $(\approx 1.5 \times 10^7\ \text{K})$?

4. (*a*) Explain how each of the following can cause damage to materials: beta particles, alpha particles, energetic neutrons, and gamma rays. (*b*) How might metals be damaged? (*c*) How can the damage affect living cells?

ANSWERS TO EXERCISES

A: $^{138}_{56}$Ba.
B: 3 neutrons.
C: 2×10^{17}.
D: (*e*).
E: (*b*).



This photo is a computer reconstruction of particles produced due to a 7 TeV proton–proton collision at the Large Hadron Collider (LHC). It is a candidate for having produced the long-sought Higgs boson (plus other particles). The Higgs in this case could have decayed (very quickly $\sim 10^{-22}$ s) into two Z bosons (which are carriers of the weak force):

$$H^0 \rightarrow Z^0 + Z^0.$$

We don't see the tracks of the Z^0 particles because (1) they are neutral and (2) they decay too quickly ($\sim 10^{-24}$ s), in this case:

$$Z^0 \rightarrow e^- + e^+.$$

The tracks of the 2 electrons and 2 positrons are shown as green lines. The Higgs is thought to play a fundamental role in the Standard Model of particle physics, importantly providing mass to fundamental particles.

The CMS detector of this photo uses a combination of the detector types discussed in Section 30–13. A magnetic field causes particles to move in curved paths so the momentum of each can be measured (Section 20–4). Tracks of particles with very large momentum, such as our electrons here, are barely curved.

In this Chapter we will study elementary particle physics from its beginnings until today, including antiparticles, neutrinos, quarks, the Standard Model, and theories that go beyond. We start with the great machines that accelerate particles so they can collide at high energies.

CHAPTER 32

Elementary Particles

CHAPTER-OPENING QUESTIONS—Guess now!

1. Physicists reserve the term "fundamental particle" for particles with a special property. What do you think that special property is?

(a) Particles that are massless.
(b) Particles that possess the minimum allowable electric charge.
(c) Particles that have no internal structure.
(d) Particles that produce no force on other objects.

2. The fundamental particles as we see them today, besides the long-sought-for Higgs boson, are

(a) atoms and electrons.
(b) protons, neutrons, and electrons.
(c) protons, neutrons, electrons, and photons.
(d) quarks, leptons, and gauge bosons (carriers of force).
(e) hadrons, leptons, and gauge bosons.

In the final two Chapters of this book we discuss two of the most exciting areas of contemporary physics: elementary particles in this Chapter, and cosmology and astrophysics in Chapter 33. These are subjects at the forefront of knowledge—elementary particles treats the smallest objects in the universe; cosmology treats the largest (and oldest) aspects of the universe. The reader who wants an understanding of the great beauties of present-day science (and its limits) will want to read these Chapters. So will those who want to be good citizens, even if there is not time to cover them in a physics course.

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32–1 High-Energy Particles and Accelerators
32–2 Beginnings of Elementary Particle Physics—Particle Exchange
32–3 Particles and Antiparticles
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32–7 Particle Stability and Resonances
32–8 Strangeness? Charm? Towards a New Model
32–9 Quarks
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In this penultimate Chapter we discuss *elementary particle* physics, which represents the human endeavor to understand the basic building blocks of all matter, and the fundamental forces that govern their interactions.

Almost a century ago, by the 1930s, it was accepted that all atoms can be considered to be made up of neutrons, protons, and electrons. The basic constituents of the universe were no longer considered to be atoms (as they had been for 2000 years) but rather the proton, neutron, and electron. Besides these three "elementary particles," several others were also known: the positron (a positive electron), the neutrino, and the γ particle (or photon), for a total of six elementary particles.

By the 1950s and 1960s many new types of particles similar to the neutron and proton were discovered, as well as many "midsized" particles called *mesons* whose masses were mostly less than nucleon masses but more than the electron mass. (Other mesons, found later, have masses greater than nucleons.) Physicists felt that these particles could not all be fundamental, and must be made up of even smaller constituents (later confirmed by experiment), which were given the name *quarks*.

By the term **fundamental particle**, we mean a particle that is so simple, so basic, that it has no internal structure[†] (is not made up of smaller subunits)—see Chapter-Opening Question 1.

Today, the fundamental constituents of matter are considered to be **quarks** (they make up protons and neutrons as well as mesons) and **leptons** (a class that includes electrons, positrons, and neutrinos). There are also the "carriers of force" known as **gauge bosons**, including the photon, gluons, and W and Z bosons. In addition there is the elusive **Higgs** boson, predicted in the 1960s but whose first suggestions of experimental detection came only in 2011–2013. The theory that describes our present view is called the **Standard Model**. How we came to our present understanding of elementary particles is the subject of this Chapter.

One of the exciting developments of the last few years is an emerging synthesis between the study of elementary particles and astrophysics (Chapter 33). In fact, recent observations in astrophysics have led to the conclusion that the greater part of the mass–energy content of the universe is not ordinary matter but two mysterious and invisible forms known as "dark matter" and "dark energy" which cannot be explained by the Standard Model in its present form.

Indeed, we are now aware that the Standard Model is not sufficient. There are problems and important questions still unanswered, and we will mention some of them in this Chapter and how we hope to answer them.

32–1 High-Energy Particles and Accelerators

In the late 1940s, after World War II, it was found that if the incoming particle in a nuclear reaction (Section 31–1) has sufficient energy, new types of particles can be produced. The earliest experiments used **cosmic rays**—particles that impinge on the Earth from space. In the laboratory, various types of particle accelerators have been constructed to accelerate protons or electrons to high energies so they can collide with other particles—often protons (the hydrogen nucleus). Heavy ions, up to lead (Pb), have also been accelerated. These **high-energy accelerators** have been used to probe more deeply into matter, to produce and study new particles, and to give us information about the basic forces and constituents of nature. The particles produced in high-energy collisions can be detected by a variety of special detectors, discussed in Section 30–13, including scintillation counters, bubble chambers, multiwire chambers, and semiconductors. The rate of production of

[†]Recall from Section 13–1 that the word "atom" comes from the Greek meaning "indivisible." Atoms have a substructure (protons, neutrons) so are not fundamental. Yet an atom is still the smallest "piece" of an element that has the characteristics of that material.

any group of particles is quantified using the concept of *cross section*, Section 31–1. Because the projectile particles are at high energy, this field is sometimes called **high-energy physics**.

Wavelength and Resolution

Particles accelerated to high energy can probe the interior of nuclei and nucleons or other particles they strike. An important factor is that faster-moving projectiles can reveal more detail. The wavelength of projectile particles is given by de Broglie's wavelength formula (Eq. 27–8),

$$\lambda = \frac{h}{p}, \tag{32–1}$$

showing that the greater the momentum p of the bombarding particle, the shorter its wavelength. As discussed in Chapter 25 on optical instruments, resolution of details in images is limited by the wavelength: the shorter the wavelength, the finer the detail that can be obtained. This is one reason why particle accelerators of higher and higher energy have been built in recent years: to probe ever deeper into the structure of matter, to smaller and smaller size.

EXAMPLE 32–1 **High resolution with electrons.** What is the wavelength, and hence the expected resolution, for 1.3-GeV electrons?

APPROACH Because 1.3 GeV is much larger than the electron mass, we must be dealing with relativistic speeds. The momentum of the electrons is found from Eq. 26–9, and the wavelength is $\lambda = h/p$.

SOLUTION Each electron has KE = 1.3 GeV = 1300 MeV, which is about 2500 times the rest energy of the electron $(mc^2 = 0.51\,\text{MeV})$. Thus we can ignore the term $(mc^2)^2$ in Eq. 26–9, $E^2 = p^2c^2 + m^2c^4$, and we solve for p:

$$p = \sqrt{\frac{E^2 - m^2c^4}{c^2}} \approx \sqrt{\frac{E^2}{c^2}} = \frac{E}{c}.$$

Therefore the de Broglie wavelength is

$$\lambda = \frac{h}{p} = \frac{hc}{E},$$

where $E = 1.3\,\text{GeV}$. Hence

$$\lambda = \frac{(6.63 \times 10^{-34}\,\text{J}\cdot\text{s})(3.0 \times 10^{8}\,\text{m/s})}{(1.3 \times 10^{9}\,\text{eV})(1.6 \times 10^{-19}\,\text{J/eV})} = 0.96 \times 10^{-15}\,\text{m},$$

or 0.96 fm. This resolution of about 1 fm is on the order of the size of nuclei (see Eq. 30–1).

NOTE The maximum possible resolution of this beam of electrons is far greater than for a light beam in a light microscope $(\lambda \approx 500\,\text{nm})$.

EXERCISE A What is the wavelength of a proton with KE = 1.00 TeV?

A major reason today for building high-energy accelerators is that new particles of greater mass can be produced at higher collision energies, transforming the kinetic energy of the colliding particles into massive particles by $E = mc^2$, as we will discuss shortly. Now we look at particle accelerators.

Cyclotron

The cyclotron was developed in 1930 by E. O. Lawrence (1901–1958; Fig. 32–1) at the University of California, Berkeley. It uses a magnetic field to maintain charged ions—usually protons—in nearly circular paths. Although particle physicists no longer use simple cyclotrons, they are used in medicine for treating cancer, and their operating principles are useful for understanding modern accelerators.



FIGURE 32–1 Ernest O. Lawrence, left, with Donald Cooksey and the "dees" of an early cyclotron.

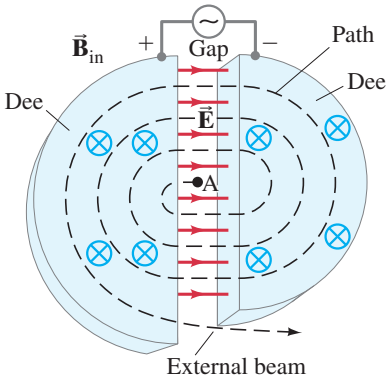


FIGURE 32–2 Diagram of a cyclotron. The magnetic field, applied by a large electromagnet, points into the page. The protons start at A, the ion source. The red electric field lines shown are for the alternating electric field in the gap at a certain moment.

The protons move in a vacuum inside two D-shaped cavities, as shown in Fig. 32–2. Each time they pass into the gap between the "dees," a voltage accelerates them (the electric force), increasing their speed and increasing the radius of curvature of their path in the magnetic field. After many revolutions, the protons acquire high kinetic energy and reach the outer edge of the cyclotron where they strike a target. The protons speed up only when they are in the gap *between* the dees, and the voltage must be alternating. When protons are moving to the right across the gap in Fig. 32–2, the right dee must be electrically negative and the left one positive. A half-cycle later, the protons are moving to the left, so the left dee must be negative in order to accelerate them.

The frequency, f, of the applied voltage must be equal to the frequency of the circulating protons. When ions of charge q are circulating *within* the hollow dees, the net force F on each ion is due to the magnetic field B, so $F = qvB$, where v is the speed of the ion at a given moment (Eq. 20–4). The magnetic force is perpendicular to both $\vec{\mathbf{v}}$ and $\vec{\mathbf{B}}$, and does not speed up the ions but causes them to move in circles; the acceleration within the dees is centripetal and equals v^2/r, where r is the radius of the ion's path at a given moment. We use Newton's second law, $F = ma$, and find that

$$F = ma$$
$$qvB = \frac{mv^2}{r}$$

when the protons are within the dees (not the gap), so their (constant) speed at radius r is

$$v = \frac{qBr}{m}.$$

The time required for a complete revolution is the period T and is equal to

$$T = \frac{\text{distance}}{\text{speed}} = \frac{2\pi r}{qBr/m} = \frac{2\pi m}{qB}.$$

Hence the frequency of revolution f is

$$f = \frac{1}{T} = \frac{qB}{2\pi m}. \qquad \textbf{(32–2)}$$

This is known as the **cyclotron frequency**.

EXAMPLE 32–2 Cyclotron. A small cyclotron of maximum radius $R = 0.25\,\text{m}$ accelerates protons in a 1.7 T magnetic field. Calculate (*a*) the frequency needed for the applied alternating voltage, and (*b*) the kinetic energy of protons when they leave the cyclotron.

APPROACH The frequency of the protons revolving within the dees (Eq. 32–2) must equal the frequency of the voltage applied across the gap if the protons are going to increase in speed.

SOLUTION (*a*) From Eq. 32–2,

$$f = \frac{qB}{2\pi m} = \frac{(1.6 \times 10^{-19}\,\text{C})(1.7\,\text{T})}{(6.28)(1.67 \times 10^{-27}\,\text{kg})} = 2.6 \times 10^7\,\text{Hz} = 26\,\text{MHz},$$

which is in the radio-wave region of the EM spectrum (Fig. 22–8).
(*b*) The protons leave the cyclotron at $r = R = 0.25\,\text{m}$. From $qvB = mv^2/r$ (see above), we have $v = qBr/m$, so their kinetic energy is

$$\text{KE} = \frac{1}{2}mv^2 = \frac{1}{2}m\frac{q^2B^2R^2}{m^2} = \frac{q^2B^2R^2}{2m}$$
$$= \frac{(1.6 \times 10^{-19}\,\text{C})^2(1.7\,\text{T})^2(0.25\,\text{m})^2}{(2)(1.67 \times 10^{-27}\,\text{kg})} = 1.4 \times 10^{-12}\,\text{J} = 8.7\,\text{MeV}.$$

NOTE The kinetic energy is much less than the rest energy of the proton (938 MeV), so relativity is not needed.

NOTE The magnitude of the voltage applied to the dees does not appear in the formula for KE, and so does not affect the final energy. But the higher this voltage, the fewer the revolutions required to bring the protons to full energy.

An important aspect of the cyclotron is that the frequency of the applied voltage, as given by Eq. 32–2, does not depend on the radius r of the particle's path. Thus the frequency does not have to be changed as the protons or ions start from the source and are accelerated to paths of larger and larger radii. But this is only true at nonrelativistic energies. At higher speeds, the momentum (Eq. 26–4) is $p = \gamma mv = mv/\sqrt{1 - v^2/c^2}$, so m in Eq. 32–2 has to be replaced by γm and the cyclotron frequency f (Eq. 32–2) depends on speed v because γ does. To keep the particles in sync, machines called **synchrocyclotrons** reduce the frequency in time to correspond to the increase of γm (in Eq. 32–2) as a packet of charged particles increases in speed more slowly at larger orbits.

Synchrotron

Another way to accelerate relativistic particles is to increase the magnetic field B in time so as to keep f (Eq. 32–2) constant as the particles speed up. Such devices are called **synchrotrons**; the particles move in a circle of fixed radius, which can be very large. The larger the radius, the greater the KE of the particles can be for a given magnetic field strength (see argument on previous page). The biggest synchrotron of all is at the European Center for Nuclear Research (CERN) in Geneva, Switzerland, the Large Hadron Collider (LHC). It is 4.3 km in radius, and 27 km in circumference, and accelerates protons to 4 TeV (soon to be 7 TeV).

The *Tevatron* accelerator at Fermilab (Fermi National Accelerator Laboratory, near Chicago, Illinois, has a radius of 1.0 km.[†] The Tevatron accelerated protons to about 1000 GeV = 1 TeV (hence its name, 1 TeV = 10^{12} eV). It was shut down in 2011.

These large synchrotrons use a narrow ring of magnets (see Fig. 32–3) with each magnet placed at the same radius from the center of the circle. The magnets are interrupted by gaps where high voltage accelerates the particles to higher speeds. Another way to describe the acceleration is to say the particles "surf" on a traveling electromagnetic wave within radiofrequency (RF) cavities. (The particles are first given considerable energy in smaller accelerators, "injectors," before being injected into the large ring of the large synchrotron.)

One problem of any accelerator is that accelerating electric charges radiate electromagnetic energy (see Chapter 22). Since ions or electrons are accelerated in an accelerator, we can expect considerable energy to be lost by radiation. The effect increases with energy and is especially important in circular machines where centripetal acceleration is present, such as synchrotrons, and hence is called **synchrotron radiation**. Synchrotron radiation can be useful, however. Intense beams of photons (γ rays) are sometimes needed, and they are often obtained from an electron synchrotron. Strong sources of such photons are referred to as **light sources**.



FIGURE 32–3 The interior of the tunnel of the main accelerator at Fermilab, showing (red) the ring of superconducting magnets used to keep particles moving in a circular path at the 1-TeV Tevatron.

[†]Robert Wilson, who helped design the Tevatron, and founded the field of proton therapy (Section 31–6), expressed his views on accelerators and national security in this exchange with Senator John Pastore during testimony before a Congressional Committee in 1969:

> Pastore: "Is there anything connected with the hopes of this accelerator [the Tevatron] that in any way involves the security of the country?"
>
> Robert Wilson: "No sir, I don't believe so."
>
> Pastore: "Nothing at all?"
>
> Wilson: "Nothing at all. … "
>
> Pastore: "It has no value in that respect?"
>
> Wilson: "It has only to do with the respect with which we regard one another, the dignity of men, our love of culture. … It has to do with are we good painters, good sculptors, great poets? I mean all the things we really venerate in our country and are patriotic about … it has nothing to do directly with defending our country except to make it worth defending."

Linear Accelerators

In a **linear accelerator** (linac), electrons or ions are accelerated along a straight-line path, Fig. 32–4, passing through a series of tubular conductors. Voltage applied to the tubes is alternating so that when electrons (say) reach a gap, the tube in front of them is positive and the one they just left is negative. At low speeds, the particles cover less distance in the same amount of time, so the tubes are shorter at first. Electrons, with their small mass, get close to the speed of light quickly, $v \approx c$, and the tubes are nearly equal in length. Linear accelerators are particularly important for accelerating electrons to avoid loss of energy due to synchrotron radiation. The largest electron linear accelerator has been at Stanford University (Stanford Linear Accelerator Center, or SLAC), about 3 km (2 mi) long, accelerating electrons to 50 GeV. Linacs accelerating protons are used as injectors into circular machines to provide initial kinetic energy. Many hospitals have 10-MeV electron linacs that strike a metal foil to produce γ ray photons to irradiate tumors.

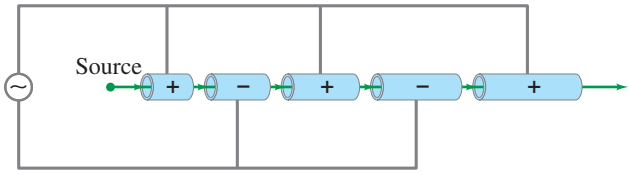


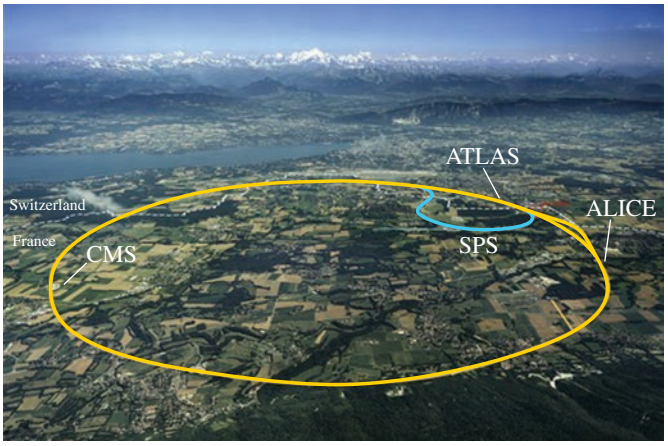
FIGURE 32–4 Diagram of a simple linear accelerator.

Colliding Beams

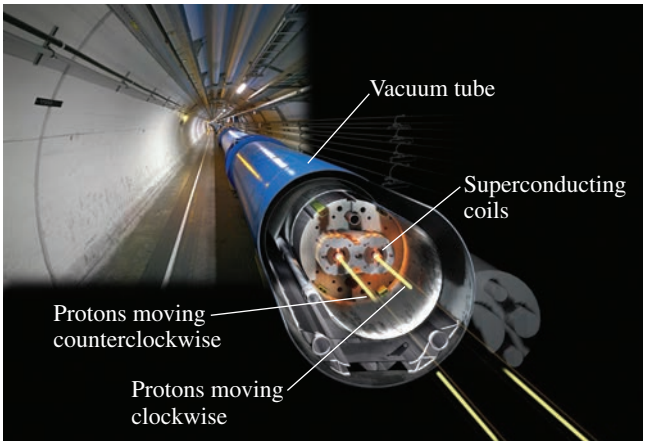
High-energy physics experiments were once done by aiming a beam of particles from an accelerator at a stationary target. But to obtain the maximum possible collision energy from a given accelerator, two beams of particles are now accelerated to very high energy and are steered so that they collide head-on. One way to accomplish such **colliding beams** with a single accelerator is through the use of **storage rings**, in which oppositely circulating beams can be repeatedly brought into collision with one another at particular points. For example, in the experiments that provided strong evidence for the top quark (Section 32–9 and Fig. 32–15), the Fermilab Tevatron accelerated protons and antiprotons each to 900 GeV, so that the combined energy of head-on collisions was 1.8 TeV.

The largest collider is the Large Hadron Collider (LHC) at CERN, with a circumference of 26.7 km (Fig. 32–5). The two colliding beams are designed to each carry 7-TeV protons for a total interaction energy of 14 TeV. For the experiments in 2011 and 2012 the total interaction energy was 7 TeV and 8 TeV. The protons for each of the beams, moving in opposite directions, are accelerated in several stages. The penultimate is SPS (Super Proton Synchrotron), seen in Fig. 32–5a which accelerates protons from 28 GeV to the 450 GeV at which they are injected into the LHC itself.

FIGURE 32–5 (a) The large circle represents the position of the tunnel, about 100 m below the ground at CERN (near Geneva) on the French–Swiss border, which houses the LHC. The smaller circle shows the position of the Super Proton Synchrotron used for accelerating protons prior to injection into the LHC. (b) Circulating proton beams, in opposite directions, inside the vacuum tube within the LHC tunnel.



(a)



(b)

Figure 32–6 shows part of one of the detectors (ATLAS) as it was being constructed at the LHC. The detectors within ATLAS include silicon semiconductor detectors with huge numbers of pixels used to track particle paths and find their point of interaction, and to measure their radius of curvature in a magnetic field and thus determine their momentum (Section 20–4). Their energy is determined in "calorimeters" utilizing plastic, liquid, or dense metal compound crystal scintillators (Section 30–13).

In the planning stage is the International Linear Collider (ILC) which would have colliding beams of e^- and e^+ at around 0.3 to 1 TeV. It would utilize semiconductor detectors using CMOS (Section 25–1) with embedded transistors to allow fast readout.

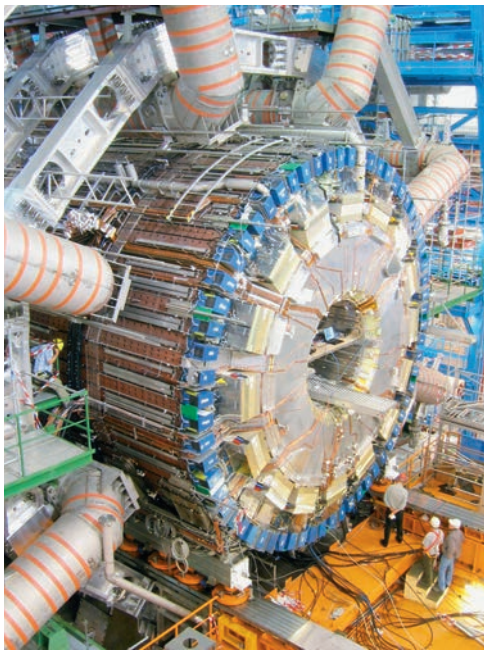


FIGURE 32–6 ATLAS, one of the large complex detectors at the LHC, is shown here as it was being built. In 2012 it was used to provide evidence for the Higgs boson. Note the people near the bottom. From the outside, the CMS detector at the LHC looks similar.

EXAMPLE 32–3 **Protons at relativistic speeds.** Determine the energy required to accelerate a proton in a high-energy accelerator (*a*) from rest to $v = 0.900c$, and (*b*) from $v = 0.900c$ to $v = 0.999c$. (*c*) What is the kinetic energy achieved by the proton in each case?

APPROACH We use the work-energy principle, which is still valid relativistically as mentioned in Section 26–9: $W = \Delta\text{KE}$.

SOLUTION The kinetic energy of a proton of mass m is given by Eq. 26–5,

$$\text{KE} = (\gamma - 1)mc^2,$$

where the relativistic factor γ is

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}.$$

The work-energy theorem becomes

$$W = \Delta\text{KE} = (\gamma_2 - 1)mc^2 - (\gamma_1 - 1)mc^2 = (\gamma_2 - \gamma_1)mc^2$$

where γ_1 and γ_2 are for the initial and final speeds, $v_1 = 0$, $v_2 = 0.900c$.
(*a*) For $v = v_1 = 0$, $\gamma_1 = 1$; and for $v_2 = 0.900c$

$$\gamma_2 = \frac{1}{\sqrt{1 - (0.900)^2}} = 2.29.$$

For a proton, $mc^2 = 938\,\text{MeV}$, so the work (or energy) needed to accelerate it from rest to $v_2 = 0.900c$ is

$$W = \Delta\text{KE} = (\gamma_2 - \gamma_1)mc^2$$
$$= (2.29 - 1.00)(938\,\text{MeV}) = 1.21\,\text{GeV}.$$

(*b*) To go from $v_2 = 0.900c$ to $v_3 = 0.999c$, we need

$$\gamma_3 = \frac{1}{\sqrt{1 - (0.999)^2}} = 22.4.$$

So the work needed to accelerate a proton from $0.900c$ to $0.999c$ is

$$W = \Delta\text{KE} = (\gamma_3 - \gamma_2)mc^2$$
$$= (22.4 - 2.29)(938\,\text{MeV}) = 18.9\,\text{GeV},$$

which is 15 times as much.
(*c*) The kinetic energy reached by the proton in (*a*) is just equal to the work done on it, $\text{KE} = 1.21\,\text{GeV}$. The final kinetic energy of the proton in (*b*), moving at $v_3 = 0.999c$, is

$$\text{KE} = (\gamma_3 - 1)mc^2 = (21.4)(938\,\text{MeV}) = 20.1\,\text{GeV}.$$

NOTE This result makes sense because, starting from rest, we did work

$$W = 1.21\,\text{GeV} + 18.9\,\text{GeV} = 20.1\,\text{GeV}$$

on it.

32–2 Beginnings of Elementary Particle Physics—Particle Exchange

The accepted model for elementary particles today views *quarks* and *leptons* as the fundamental constituents of ordinary matter. To understand our present-day view of elementary particles, it is necessary to understand the ideas leading up to its formulation.

Elementary particle physics might be said to have begun in 1935 when the Japanese physicist Hideki Yukawa (1907–1981) predicted the existence of a new particle that would in some way mediate the strong nuclear force. To understand Yukawa's idea, we first consider the electromagnetic force. When we first discussed electricity, we saw that the electric force acts over a distance, without contact. To better perceive how a force can act over a distance, we used the idea of a **field**. The force that one charged particle exerts on a second can be said to be due to the electric field set up by the first. Similarly, the magnetic field can be said to carry the magnetic force. Later (Chapter 22), we saw that electromagnetic (EM) fields can travel through space as waves. Finally, in Chapter 27, we saw that electromagnetic radiation (light) can be considered as either a wave or as a collection of particles called *photons*. Because of this wave–particle duality, it is possible to imagine that the electromagnetic force between charged particles is due to

(1) the EM field set up by one charged particle and felt by the other, or

(2) an exchange of photons (γ particles) between them.



(a) Repulsive force (children throwing pillows)



(b) Attractive force (children grabbing pillows from each other's hands)

FIGURE 32–7 Forces equivalent to particle exchange. (a) Repulsive force (children on roller skates throwing pillows at each other). (b) Attractive force (children grabbing pillows from each other's hands).

It is (2) that we want to concentrate on here, and a crude analogy for how an exchange of particles could give rise to a force is suggested in Fig. 32–7. In part (a), two children start throwing heavy pillows at each other; each throw and each catch results in the child being pushed backward by the impulse. This is the equivalent of a repulsive force. On the other hand, if the two children exchange pillows by grabbing them out of the other person's hand, they will be pulled toward each other, as when an attractive force acts.

For the electromagnetic force, it is photons exchanged between two charged particles that give rise to the force between them. A simple diagram describing this photon exchange is shown in Fig. 32–8. Such a diagram, called a **Feynman diagram** after its inventor, the American physicist Richard Feynman (1918–1988), is based on the theory of **quantum electrodynamics** (QED).

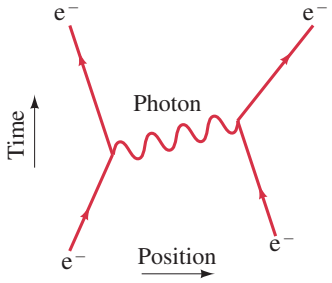


FIGURE 32–8 Feynman diagram showing a photon acting as the carrier of the electromagnetic force between two electrons. This is sort of an x vs. t graph, with t increasing upward. Starting at the bottom, two electrons approach each other. As they get close, momentum and energy get transferred from one to the other, carried by a photon (or more than one), and the two electrons bounce apart.

Figure 32–8 represents the simplest case in QED, in which a single photon is exchanged. One of the charged particles emits the photon and recoils somewhat as a result; and the second particle absorbs the photon. In such a collision or *interaction*, energy and momentum are transferred from one charged particle to the other, carried by the photon. The photon is absorbed by the second particle after it is emitted by the first particle and is not observable. Hence the photon is referred to as a **virtual** photon, in contrast to one that is free and can be detected by instruments. The photon is said to **mediate**, or **carry**, the electromagnetic force.

By analogy with photon exchange that mediates the electromagnetic force, Yukawa argued that there ought to be a particle that mediates the strong nuclear force—the force that holds nucleons together in the nucleus. Yukawa called this predicted particle a **meson** (meaning "medium mass"). Figure 32–9 is a Feynman diagram showing the original model of meson exchange: a meson carrying the strong force between a neutron and a proton.

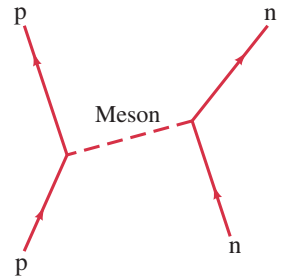


FIGURE 32–9 Early model showing meson exchange when a proton and neutron interact via the strong nuclear force. (Today, as we shall see shortly, we view the strong force as carried by gluons between quarks.)

A rough estimate of the mass of the meson can be made as follows. Suppose the proton on the left in Fig. 32–9 is at rest. For it to emit a meson would require energy (to make the meson's mass) which, coming from nowhere, would violate conservation of energy. But the uncertainty principle allows nonconservation of energy by an amount ΔE if it occurs only for a time Δt given by $(\Delta E)(\Delta t) \approx h/2\pi$. We set ΔE equal to the energy needed to create the mass m of the meson: $\Delta E = mc^2$. Conservation of energy is violated only as long as the meson exists, which is the time Δt required for the meson to pass from one nucleon to the other, where it is absorbed and disappears. If we assume the meson travels at relativistic speed, close to the speed of light c, then Δt need be at most about $\Delta t = d/c$, where d is the maximum distance that can separate the interacting nucleons. Thus we can write

$$\Delta E\,\Delta t \approx \frac{h}{2\pi}$$
$$mc^2\left(\frac{d}{c}\right) \approx \frac{h}{2\pi}$$

or

$$mc^2 \approx \frac{hc}{2\pi d}. \qquad \textbf{(32–3)}$$

The range of the strong nuclear force (the maximum distance away it can be felt) is small—not much more than the size of a nucleon or small nucleus (see Eq. 30–1)—so let us take $d \approx 1.5 \times 10^{-15}\,\text{m}$. Then from Eq. 32–3,

$$mc^2 \approx \frac{hc}{2\pi d} = \frac{(6.6\times10^{-34}\,\text{J}\cdot\text{s})(3.0\times10^{8}\,\text{m/s})}{(6.28)(1.5\times10^{-15}\,\text{m})} \approx 2.1\times10^{-11}\,\text{J} = 130\,\text{MeV}.$$

The mass of the predicted meson, roughly $130\,\text{MeV}/c^2$, is about 250 times the electron mass of $0.51\,\text{MeV}/c^2$.

EXERCISE B What effect does an increase in the mass of the virtual exchange particle have on the range of the force it mediates? (*a*) Decreases it; (*b*) increases it; (*c*) has no appreciable effect; (*d*) decreases the range for charged particles and increases the range for neutral particles.

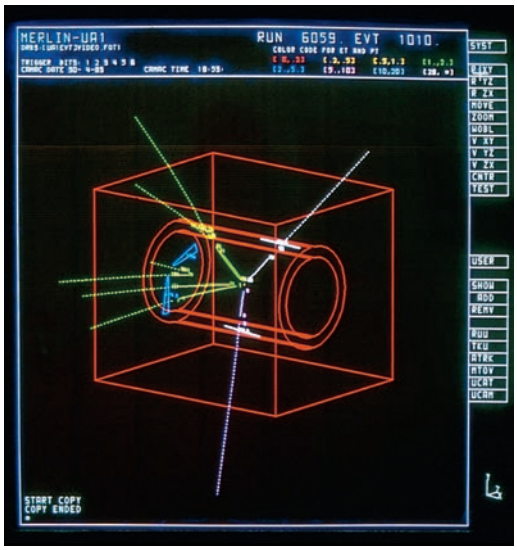
Note that since the electromagnetic force has infinite range, Eq. 32–3 with $d = \infty$ tells us that the exchanged particle for the electromagnetic force, the photon, will have zero mass, which it does.

The particle predicted by Yukawa was discovered in cosmic rays by C. F. Powell and G. Occhialini in 1947, and is called the "π" or pi meson, or simply the **pion**. It comes in three charge states: $+e$, $-e$, or 0, where $e = 1.6 \times 10^{-19}\,\text{C}$. The π^+ and π^- have mass of $139.6\,\text{MeV}/c^2$ and the π^0 a mass of $135.0\,\text{MeV}/c^2$, all close to Yukawa's prediction. All three interact strongly with matter. Reactions observed in the laboratory, using a particle accelerator, include

$$\text{p} + \text{p} \rightarrow \text{p} + \text{p} + \pi^0,$$
$$\text{p} + \text{p} \rightarrow \text{p} + \text{n} + \pi^+. \qquad \textbf{(32–4)}$$

The incident proton from the accelerator must have sufficient energy to produce the additional mass of the free pion.

Yukawa's theory of pion exchange as carrier of the strong force has been superseded by *quantum chromodynamics* in which protons, neutrons, and other strongly interacting particles are made up of basic entities called *quarks*, and the basic carriers of the strong force are *gluons*, as we shall discuss shortly. But the basic idea of the earlier theory, that forces can be understood as the exchange of particles, remains valid.

(a) 

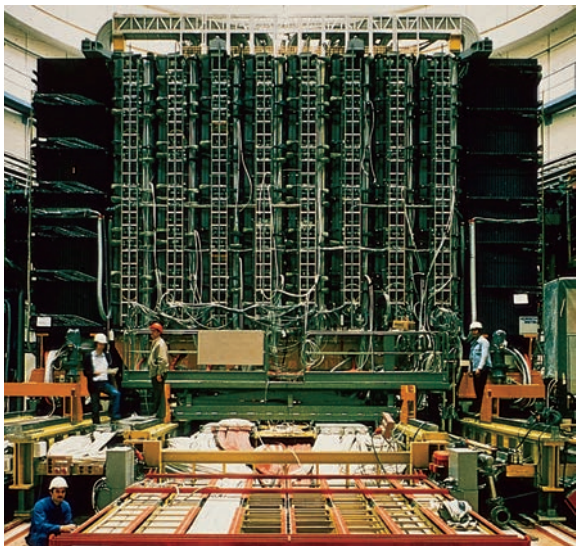
(b) 

FIGURE 32–10 (a) Computer reconstruction of a Z-particle decay into an electron and a positron $(Z^0 \rightarrow e^+ + e^-)$ whose tracks are shown in white, which took place in the UA1 detector at CERN. (b) Photo of the UA1 detector at CERN as it was being built. 1980s.

There are four known types of force—or interactions—in nature. The electromagnetic force is carried by the photon, the strong force by gluons. What about the other two: the weak force and gravity? These too are believed to be mediated by particles. The particles that transmit the weak force are referred to as the W^+, W^-, and Z^0, and were detected in 1983 (Fig. 32–10). The quantum (or carrier) of the gravitational force has been named the **graviton**, but its existence has not been detected and it may not be detectable.

A comparison of the four forces is given in Table 32–1, where they are listed according to their (approximate) relative strengths. Although gravity may be the most obvious force in daily life (because of the huge mass of the Earth), on a nuclear scale gravity is by far the weakest of the four forces and its effect at the particle level can nearly always be ignored.

TABLE 32–1 The Four Forces in Nature

Type	Relative Strength (approx., for 2 protons in nucleus)	Field Particle
Strong	1	Gluons (= g)
Electromagnetic	10^{-2}	Photon (= γ)
Weak	10^{-6}	$W^{\pm}$ and Z^0
Gravitational	10^{-38}	Graviton (?)

32–3 Particles and Antiparticles

The positron, as we discussed in Sections 27–6 (pair production) and 30–5 (β^+ decay), is basically a positive electron. That is, many of its properties are the same as for the electron, such as mass, but it has the opposite electric charge $(+e)$. Other quantum numbers that we will discuss shortly are also reversed. The positron is said to be the **antiparticle** to the electron.

The positron was first detected as a curved path in a cloud chamber in a magnetic field by Carl Anderson in 1932. It was predicted that other particles also would have antiparticles. It was decades before another type was found.

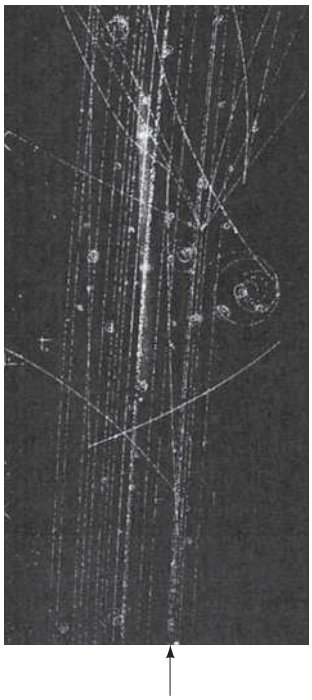


FIGURE 32–11 Liquid-hydrogen bubble-chamber photograph of an antiproton ($\bar{p}$) colliding with a proton at rest, producing a Xi—anti-Xi pair ($\bar{p} + p \rightarrow \Xi^- + \Xi^+$) that subsequently decay into other particles. The drawing indicates the assignment of particles to each track, which is based on how or if that particle decays, and on mass values estimated from measurement of momentum (curvature of track in magnetic field) and energy (thickness of track, for example). Neutral particle paths are shown by dashed lines since neutral particles rarely ionize atoms, around which bubbles form, and hence leave no tracks. 1950s.

Finally, in 1955 the antiparticle to the proton, the **antiproton** ($\bar{p}$), which carries a negative charge (Fig. 32–11), was discovered at the University of California, Berkeley, by Emilio Segrè (1905–1989, Fig. 32–12) and Owen Chamberlain (1920–2006). A bar, such as over the p, is used to indicate the antiparticle ($\bar{p}$). Soon after, the antineutron ($\bar{n}$) was found. All particles have antiparticles. But a few, like the photon, the π^0, and the Higgs, do not have distinct antiparticles—we say that they are their own antiparticles.

Antiparticles are produced in nuclear reactions when there is sufficient energy available to produce the required mass, and they do not live very long in the presence of matter. For example, a positron is stable when by itself but rarely survives for long; as soon as it encounters an electron, the two annihilate each other. The energy of their vanished mass, plus any kinetic energy they possessed, is usually converted into the energy of two γ rays. Annihilation also occurs for all other particle–antiparticle pairs.

Antimatter is a term referring to material that would be made up of "antiatoms" in which antiprotons and antineutrons would form the nucleus around which positrons (antielectrons) would move. The term is also used for antiparticles in general. If there were pockets of antimatter in the universe, a huge explosion would occur if it should encounter normal matter. It is believed that antimatter was prevalent in the very early universe (Section 33–7).

FIGURE 32–12 Emilio Segrè: he worked with Fermi in the 1930s, later discovered the first "man-made" element, technetium, and other elements, and then the antiproton. The inscription below the photo is from a book by Segrè given to this book's author.



Emilio Segrè
Feb. 20 1985
To enlist the help of the subjects of this book for the next edition of Giancoli's texts

* Negative Sea of Electrons; Vacuum State

The original idea for antiparticles came from a relativistic wave equation developed in 1928 by the Englishman P. A. M. Dirac (1902–1984). Recall that, as we saw in Chapter 26, the total energy E of a particle with mass m and momentum p and zero potential energy is given by Eq. 26–9, $E^2 = p^2c^2 + m^2c^4$. Thus

$$E = \pm\sqrt{p^2c^2 + m^2c^4}.$$

Dirac applied his new equation and found that it included solutions with both + and − signs. He could not ignore the solution with the negative sign, which we might have thought unphysical. If those negative energy states are real, then we would expect normal free electrons to drop down into those states, emitting

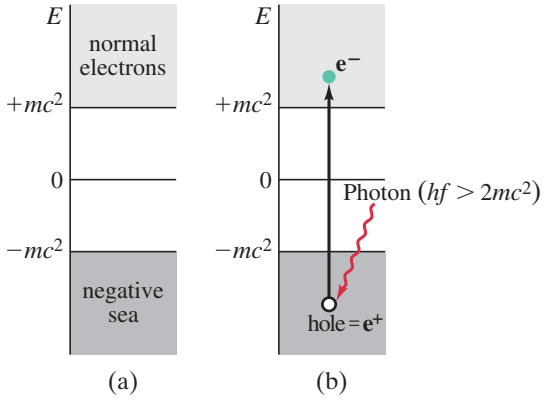


FIGURE 32–13 (a) Possible energy states for an electron. Note the vast sea of fully occupied electron states at $E < -mc^2$. (b) An electron in the negative sea is hit by a photon $(E > 2mc^2)$ and knocks it up to a normal positive energy state. The positive "hole" left behind acts like a positive electron—it is a positron.

photons—never experimentally seen. To deal with this difficulty, Dirac postulated that all those negative energy states are *normally occupied*. That is, what we thought was the **vacuum** is instead a vast **sea of electrons** in negative energy states (Fig. 32–13a). These electrons are not normally observable. But if a photon strikes one of these negative energy electrons, that electron can be knocked up to a normal $(E > mc^2)$ energy state as shown in Fig. 32–13b. Note in Fig. 32–13 that there are no energy states between $E = -mc^2$ and $E = +mc^2$ (because p^2 cannot be negative in the equation $E = \pm\sqrt{p^2c^2 + m^2c^4}$). The photon that knocks an e^- from the negative sea up to a normal state (Fig. 32–13b) must have an energy greater than $2mc^2$. What is left behind is a hole (as in semiconductors, Sections 29–7 and 29–8) with positive charge. We call that "hole" a **positron**, and it can move around as a free particle with positive energy. Thus Fig. 32–13b represents (Section 27–6) **pair production**: $\gamma \rightarrow e^-e^+$.

The vast sea of electrons with negative energy in Fig. 32–13 is the vacuum (or **vacuum state**). According to quantum mechanics, the vacuum is not empty, but contains electrons and other particles as well. The uncertainty principle allows a particle to jump briefly up to a normal energy, thus creating a **particle–antiparticle** pair. It is possible that they could be the source of the recently discovered *dark energy* that fills the universe (Chapter 33). We still have a lot to learn.

32–4 Particle Interactions and Conservation Laws

One of the important uses of high-energy accelerators and colliders is to study the interactions of elementary particles with each other. As a means of ordering this subnuclear world, the conservation laws are indispensable. The laws of conservation of energy, of momentum, of angular momentum, and of electric charge are found to hold precisely in all particle interactions.

A study of particle interactions has revealed a number of new conservation laws which (just like the old ones) are ordering principles: they help to explain why some reactions occur and others do not. For example, the following reaction has never been observed:

$$p + n \nrightarrow p + p + \bar{p}$$

even though charge, energy, and so on, are conserved ($\nrightarrow$ means the reaction does not occur). To understand why such a reaction does not occur, physicists hypothesized a new conservation law, the conservation of **baryon number**. (Baryon number is a generalization of nucleon number, which we saw earlier is conserved in nuclear reactions and decays.) All nucleons are defined to have baryon number $B = +1$, and all antinucleons (antiprotons, antineutrons) have $B = -1$. All other types of particles, such as photons, mesons, and electrons and

other leptons, have $B = 0$. The reaction shown at the start of this paragraph does not conserve baryon number since the left side has $B = (+1) + (+1) = +2$, and the right has $B = (+1) + (+1) + (-1) = +1$. On the other hand, the following reaction does conserve B and *does* occur if the incoming proton has sufficient energy:

$$\begin{aligned} \mathrm{p} + \mathrm{p} &\rightarrow \mathrm{p} + \mathrm{p} + \bar{\mathrm{p}} + \mathrm{p}, \\ B = +1 + 1 &= +1 + 1 - 1 + 1. \end{aligned}$$

As indicated, $B = +2$ on both sides of this equation. From these and other reactions, the **conservation of baryon number** has been established as a basic principle of physics.

Also useful are conservation laws for the three **lepton numbers**, associated with weak interactions including decays. In ordinary β decay, an electron or positron is emitted along with a neutrino or antineutrino. In another type of decay, a particle known as a "μ" or **muon**, can be emitted instead of an electron. The muon (discovered in 1937) seems to be much like an electron, except its mass is 207 times larger $(106\,\mathrm{MeV}/c^2)$. The neutrino (ν_e) that accompanies an emitted electron is found to be different from the neutrino (ν_μ) that accompanies an emitted muon. Each of these neutrinos has an antiparticle: $\bar{\nu}_\mathrm{e}$ and $\bar{\nu}_\mu$. In ordinary β decay we have, for example,

$$\mathrm{n} \rightarrow \mathrm{p} + \mathrm{e}^- + \bar{\nu}_\mathrm{e}$$

but not $\mathrm{n} \nrightarrow \mathrm{p} + \mathrm{e}^- + \bar{\nu}_\mu$. To explain why these do not occur, the concept of **electron lepton number**, L_e, was invented. If the electron (e^-) and the electron neutrino (ν_e) are assigned $L_\mathrm{e} = +1$, and e^+ and $\bar{\nu}_\mathrm{e}$ are assigned $L_\mathrm{e} = -1$, whereas all other particles have $L_\mathrm{e} = 0$, then all observed decays conserve L_e. For example, in $\mathrm{n} \rightarrow \mathrm{p} + \mathrm{e}^- + \bar{\nu}_\mathrm{e}$, initially $L_\mathrm{e} = 0$, and afterward $L_\mathrm{e} = 0 + (+1) + (-1) = 0$. Decays that do not conserve L_e, even though they would obey the other conservation laws, are not observed to occur.

In a decay involving muons, such as

$$\pi^+ \rightarrow \mu^+ + \nu_\mu,$$

a second quantum number, **muon lepton number** (L_μ), is conserved. The μ^- and ν_μ are assigned $L_\mu = +1$, and their antiparticles μ^+ and $\bar{\nu}_\mu$ have $L_\mu = -1$, whereas all other particles have $L_\mu = 0$. L_μ too is conserved in interactions and decays. Similar assignments can be made for the **tau lepton number**, L_τ, associated with the τ lepton (discovered in 1976 with mass more than 3000 times the electron mass) and its neutrino, ν_τ.

Antiparticles have not only opposite electric charge from their particles, but also opposite B, L_e, L_μ, and L_τ. For example, a neutron has $B = +1$, an antineutron has $B = -1$ (and all the L's are zero).

CAUTION
The different types of neutrinos are not identical

CONCEPTUAL EXAMPLE 32–4 **Lepton number in muon decay.** Which of the following decay schemes is possible for muon decay: (*a*) $\mu^- \rightarrow \mathrm{e}^- + \bar{\nu}_\mathrm{e}$; (*b*) $\mu^- \rightarrow \mathrm{e}^- + \bar{\nu}_\mathrm{e} + \nu_\mu$; (*c*) $\mu^- \rightarrow \mathrm{e}^- + \nu_\mathrm{e}$? All of these particles have $L_\tau = 0$.

RESPONSE A μ^- has $L_\mu = +1$ and $L_\mathrm{e} = 0$. This is the initial state for all decays given, and the final state must also have $L_\mu = +1$, $L_\mathrm{e} = 0$. In (*a*), the final state has $L_\mu = 0 + 0 = 0$, and $L_\mathrm{e} = +1 - 1 = 0$; L_μ would not be conserved and indeed this decay is not observed to occur. The final state of (*b*) has $L_\mu = 0 + 0 + 1 = +1$ and $L_\mathrm{e} = +1 - 1 + 0 = 0$, so both L_μ and L_e are conserved. This is in fact the most common decay mode of the μ^-. Lastly, (*c*) does not occur because L_e $(= +2$ in the final state$)$ is not conserved, nor is L_μ.

EXAMPLE 32–5 **Energy and momentum are conserved.** In addition to the "number" conservation laws which help explain the decay schemes of particles, we can also apply the laws of conservation of energy and momentum. The decay of a Σ^+ particle at rest with mass 1189 MeV/c^2 (Table 32–2 in Section 32–6) can yield a proton ($m_p = 938\ \text{MeV}/c^2$) and a neutral pion, π^0 ($m_{\pi^0} = 135\ \text{MeV}/c^2$):

$$\Sigma^+ \rightarrow p + \pi^0.$$

Determine the kinetic energies of the proton and π^0.

APPROACH We find the energy release from the change in mass ($E = mc^2$) as we did for nuclear processes (Eq. 30–2 or 31–2a). Then we apply conservation of energy and momentum, using relativistic formulas as the energies are large.

SOLUTION The energy released, $Q = \text{KE}_p + \text{KE}_{\pi^0}$, is the change in mass $\times\ c^2$:

$$Q = [m_{\Sigma^+} - (m_p + m_{\pi^0})]c^2 = [1189 - (938 + 135)]\ \text{MeV} = 116\ \text{MeV}.$$

Next we apply conservation of momentum: the initial particle Σ^+ is at rest, so the π^0 and p have opposite momentum but are equal in magnitude: $p_{\pi^0} = p_p$. We square this equation, $p_{\pi^0}^2 = p_p^2$, which becomes, using Eq. 26–9 ($p^2c^2 = E^2 - m^2c^4$),

$$E_{\pi^0}^2 - m_{\pi^0}^2c^4 = E_p^2 - m_p^2c^4.$$

Solving for $E_{\pi^0}^2$:

$$E_{\pi^0}^2 = E_p^2 - m_p^2c^4 + m_{\pi^0}^2c^4.$$

We substitute Eq. 26–6a, $E = \text{KE} + mc^2$, for both the π^0 and the p:

$$(\text{KE}_{\pi^0} + m_{\pi^0}c^2)^2 = (\text{KE}_p + m_pc^2)^2 - m_p^2c^4 + m_{\pi^0}^2c^4$$

$$\text{KE}_{\pi^0}^2 + 2\text{KE}_{\pi^0}m_{\pi^0}c^2 + \cancel{m_{\pi^0}^2c^4} = \text{KE}_p^2 + 2\text{KE}_p m_pc^2 + \cancel{m_p^2c^4} - \cancel{m_p^2c^4} + \cancel{m_{\pi^0}^2c^4}.$$

Next (after cancelling as shown) we substitute $\text{KE}_p = Q - \text{KE}_{\pi^0}$:

$$\cancel{\text{KE}_{\pi^0}^2} + 2\text{KE}_{\pi^0}m_{\pi^0}c^2 = Q^2 - 2Q\text{KE}_{\pi^0} + \cancel{\text{KE}_{\pi^0}^2} + 2Qm_pc^2 - 2\text{KE}_{\pi^0}m_pc^2.$$

After cancelling as shown, we solve for KE_{π^0}:

$$\text{KE}_{\pi^0} = \frac{Q^2 + 2Qm_pc^2}{2m_{\pi^0}c^2 + 2Q + 2m_pc^2} = \frac{(116\ \text{MeV})^2 + 2(116\ \text{MeV})(938\ \text{MeV})}{2(135\ \text{MeV}) + 2(116\ \text{MeV}) + 2(938\ \text{MeV})}$$

which gives $\text{KE}_{\pi^0} = 97\ \text{MeV}$. Then $\text{KE}_p = 116\ \text{MeV} - 97\ \text{MeV} = 19\ \text{MeV}$.

32–5 Neutrinos

We first met neutrinos with regard to β decay in Section 30–5. The study of neutrinos is a "hot" subject today. Experiments are being carried out in deep underground laboratories, sometimes in deep mine shafts. The thick layer of earth above is meant to filter out all other "background" particles, leaving mainly the very weakly interacting neutrinos to arrive at the detectors.

Some very important results have come to the fore in recent years. First there was the **solar neutrino problem**. The energy output of the Sun is believed to be due to the nuclear fusion reactions discussed in Chapter 31, Eqs. 31–6 and 31–7. The neutrinos emitted in these reactions are all ν_e (accompanied by e^+). But the rate at which ν_e arrive at Earth was measured starting in the late 1960s to be much less than expected based on the power output of the Sun. It was then proposed that, on the long trip between Sun and Earth, ν_e might turn into ν_μ or ν_τ. Subsequent experiments, definitive only in 2001, confirmed this hypothesis. Thus the three neutrinos, ν_e, ν_μ, ν_τ, can change into one another in certain circumstances, a phenomenon called **neutrino flavor oscillation**[†]. (Each of the three neutrino types is called, whimsically, a different "flavor.") This result suggests that the lepton numbers L_e, L_μ, and L_τ are not perfectly conserved. But the sum, $L_e + L_\mu + L_\tau$, is believed to be always conserved.

[†]Neutrino oscillations had first been proposed in 1957 by Bruno Pontecorvo. He also proposed that the electron and muon neutrinos are different species; and he also suggested a way to confirm the existence of neutrinos by detecting $\bar{\nu}_e$ emitted in huge numbers by a nuclear reactor, an experiment carried out by Frederick Reines and Clyde Cowan in the 1950s. The experimentalists who confirmed these two predictions were awarded the Nobel Prize, but not the theorist who proposed them.

The second exceptional result has long been speculated on: are neutrinos massless as originally thought, or do they have a nonzero mass? Rough upper limits on the masses have been made. Today astrophysical experiments show that the sum of all three neutrino masses combined is less than about $0.14\ \text{eV}/c^2$. But can all the masses be zero? Not if there are the flavor oscillations discussed above. It seems that at most, one type could have zero mass, and it is likely that at least one neutrino type has a mass of at least $0.04\ \text{eV}/c^2$.

As a result of neutrino oscillations, the three types of neutrino may not be exactly what we thought they were (e, μ, τ). If not, the three basic neutrinos, called 1, 2, and 3, are combinations of ν_e, ν_μ, and ν_τ.

Another outstanding question is whether or not neutrinos are in the category called **Majorana particles**,[†] meaning they would be their own antiparticles, like γ, π^0, and Higgs. If so, a lot of other questions (and answers) would appear.

* Neutrino Mass Estimate from a Supernova

The explosion of a supernova in the outer parts of our Galaxy in 1987 (Section 33–2) released lots of neutrinos and offered an opportunity to estimate electron neutrino mass. If neutrinos do have mass, then their speed would be less than c, and neutrinos of different energy would take different times to travel the 170,000 light-years from the supernova to Earth. To get an idea of how such a measurement could be done, suppose two neutrinos from "SN1987A" were emitted at the same time and were actually detected on Earth (via the reaction $\bar{\nu}_e + p \rightarrow n + e^+$) 10 seconds apart, with measured kinetic energies of about 20 MeV and 10 MeV. From other laboratory measurements we expect the neutrino mass to be less than 100 eV; and since our neutrinos have kinetic energy of 20 MeV and 10 MeV, we can make the approximation $m_\nu c^2 \ll E$, so that E (the total energy) is essentially equal to the kinetic energy. We use Eq. 26–6b, which tells us

$$E = \frac{m_\nu c^2}{\sqrt{1 - v^2/c^2}}.$$

We solve this for v, the velocity of a neutrino with energy E:

$$v = c\left(1 - \frac{m_\nu^2 c^4}{E^2}\right)^{\frac{1}{2}} = c\left(1 - \frac{m_\nu^2 c^4}{2E^2} + \cdots\right),$$

where we have used the binomial expansion $(1 - x)^{\frac{1}{2}} = 1 - \frac{1}{2}x + \cdots$, and we ignore higher-order terms since $m_\nu^2 c^4 \ll E^2$. The time t for a neutrino to travel a distance d $(= 170{,}000\ \text{ly})$ is

$$t = \frac{d}{v} = \frac{d}{c\left(1 - \dfrac{m_\nu^2 c^4}{2E^2}\right)} \approx \frac{d}{c}\left(1 + \frac{m_\nu^2 c^4}{2E^2}\right),$$

where again we used the binomial expansion $\left[(1 - x)^{-1} = 1 + x + \cdots\right]$. The difference in arrival times for our two neutrinos of energies $E_1 = 20\ \text{MeV}$ and $E_2 = 10\ \text{MeV}$ is

$$t_2 - t_1 = \frac{d}{c}\frac{m_\nu^2 c^4}{2}\left(\frac{1}{E_2^2} - \frac{1}{E_1^2}\right).$$

We solve this for $m_\nu c^2$ and set $t_2 - t_1 = 10\ \text{s}$:

$$m_\nu c^2 = \left[\frac{2c(t_2 - t_1)}{d}\frac{E_1^2 E_2^2}{E_1^2 - E_2^2}\right]^{\frac{1}{2}} = 22 \times 10^{-6}\ \text{MeV} = 22\ \text{eV}.$$

This calculation, with its optimistic assumptions, estimates the mass of the neutrino to be $22\ \text{eV}/c^2$. But there would be experimental uncertainties, and even worse there is the unwarranted assumption that the two neutrinos were emitted at the same time.

[†]The brilliant young physicist Ettore Majorana (1906–1938) disappeared from a ship under mysterious circumstances in 1938 at the age of 31.

Theoretical models of supernova explosions suggest that the neutrinos are emitted in a burst that lasts from a second or two up to perhaps 10 s. If we assume the neutrinos are not emitted simultaneously but rather at any time over a 10-s interval, then that 10-s difference in arrival times could be due to a 10-s difference in their emission time. In this case the data would be consistent with zero neutrino mass, and it put an approximate *upper limit* of 22 eV/c^2.

The actual detection of these neutrinos was brilliant—it was a rare event that allowed us to detect something other than EM radiation from beyond the solar system, and was an exceptional confirmation of theory. In the experiments, the most sensitive detector consisted of several thousand tons of water in an underground chamber. It detected 11 events in 12 seconds, probably via the reaction $\bar{\nu}_e + p \rightarrow n + e^+$. There was not a clear correlation between energy and time of arrival. Nonetheless, a careful analysis of that experiment set a rough upper limit on the electron antineutrino mass of about 4 eV/c^2. The more recent results mentioned above are much more definitive—they provide evidence that mass is much smaller, and that it is *not zero*—but precise neutrino masses still elude us.

32–6 Particle Classification

In the decades after the discovery of the π meson in the late 1940s, hundreds of other subnuclear particles were discovered. One way to categorize the particles is according to their interactions, since not all particles interact via all four of the forces known in nature (though all interact via gravity). Table 32–2 (next page) lists some of the more common particles classified in this way along with many of their properties. At the top of Table 32–2 are the so-called "fundamental" particles which we believe have no internal structure. Below them are some of the "composite" particles which are made up of quarks, according to the Standard Model.

The **fundamental particles** include the **gauge bosons** (so-named after the theory that describes them, *gauge theory*), which include the gluons, the photon, and the W and Z particles; these are the particles that mediate (or "carry") the strong, electromagnetic, and weak interactions, respectively.

Also fundamental are the **leptons**, which are particles that do not interact via the strong force but do interact via the weak nuclear force. Leptons that carry electric charge also interact via the electromagnetic force. The leptons include the electron, the muon, and the tau, and three types of neutrino: the electron neutrino (ν_e), the muon neutrino (ν_μ), and the tau neutrino (ν_τ). Each lepton has an antiparticle. Finally, the recently detected Higgs boson is also considered to be fundamental, with no internal structure.

The second category of particle in Table 32–2 is the **hadrons**, which are **composite** particles (made up of quarks as we will discuss shortly). Hadrons are particles that interact via the strong nuclear force and are said to be **strongly interacting particles**. They also interact via the other forces, but the strong force predominates at short distances. The hadrons include the proton, neutron, pion, and many other particles. They are divided into two subgroups: **baryons**, which are particles that have baryon number $+1$ (or -1 in the case of their antiparticles) and, as we shall see, are each made up of three quarks; and **mesons**, which have baryon number $= 0$, and are made up of a quark and an antiquark.

Only a few of the hundreds of hadrons (a veritable "zoo") are included in Table 32–2. Notice that the baryons Λ, Σ, Ξ, and Ω all decay to lighter-mass baryons, and eventually to a proton or neutron. All these processes conserve baryon number. Since there is no particle lighter than the proton with $B = +1$, if baryon number is strictly conserved, the proton itself cannot decay and is stable. (But see Section 32–11.) Note that Table 32–2 gives the **mean life** (τ) of each particle (as is done in particle physics), not the half-life $(T_{\frac{1}{2}})$. Recall that they differ by a factor 0.693: $\tau = T_{\frac{1}{2}}/\ln 2 = T_{\frac{1}{2}}/0.693$, Eq. 30–7. The term **lifetime** in particle physics means the mean life τ $(=$ mean lifetime$)$.

The baryon and lepton numbers $(B,\ L_e, L_\mu, L_\tau)$, as well as strangeness S (Section 32–8), as given in Table 32–2 are for particles; their antiparticles have opposite sign for these numbers.

TABLE 32–2 Particles (selected)†

Category	Forces involved	Particle name	Symbol	Anti-particle	Spin	Mass (MeV/c^2)	B	L_e	L_μ	L_τ	S	Mean life (s)	Principal Decay Modes
							[antiparticles have opposite sign]						
Fundamental													
Gauge bosons (force carriers)	str	Gluons	g	Self	1	0	0	0	0	0	0	Stable	
	em	Photon	γ	Self	1	0	0	0	0	0	0	Stable	
	w, em	W	W^+	W^-	1	80.385×10^3	0	0	0	0	0	3×10^{-25}	$e\nu_e$, $\mu\nu_\mu$, $\tau\nu_\tau$, hadrons
	w	Z	Z^0	Self	1	91.19×10^3	0	0	0	0	0	3×10^{-25}	e^+e^-, $\mu^+\mu^-$, $\tau^+\tau^-$, hadrons
Higgs boson	w, str	Higgs	H^0	Self	0	125×10^3	0	0	0	0	0	1.6×10^{-22}	$b\bar{b}$, Z^0Z^0, W^+W^-, $g\bar{g}$, $\tau\bar{\tau}$, $\gamma\gamma$
Leptons	w, em‡	Electron	e^-	e^+	$\frac{1}{2}$	0.511	0	+1	0	0	0	Stable	
		Neutrino (e)	ν_e	$\bar{\nu}_e$	$\frac{1}{2}$	0 ($<0.14\ eV/c^2$)‡	0	+1	0	0	0	Stable	
		Muon	μ^-	μ^+	$\frac{1}{2}$	105.7	0	0	+1	0	0	2.20×10^{-6}	$e^-\bar{\nu}_e\nu_\mu$
		Neutrino (μ)	ν_μ	$\bar{\nu}_\mu$	$\frac{1}{2}$	0 ($<0.14\ eV/c^2$)‡	0	0	+1	0	0	Stable	
		Tau	τ^-	τ^+	$\frac{1}{2}$	1777	0	0	0	+1	0	2.91×10^{-13}	$\mu^-\bar{\nu}_\mu\nu_\tau$, $e^-\bar{\nu}_e\nu_\tau$, hadrons $+\nu_\tau$
		Neutrino (τ)	ν_τ	$\bar{\nu}_\tau$	$\frac{1}{2}$	0 ($<0.14\ eV/c^2$)‡	0	0	0	+1	0	Stable	
Quarks	w, em, str	(see Table 32–3)											
Hadrons (composite), selected													
Mesons (quark–antiquark)	str, em, w	Pion	π^+	π^-	0	139.6	0	0	0	0	0	2.60×10^{-8}	$\mu^+\nu_\mu$
			π^0	Self	0	135.0	0	0	0	0	0	0.85×10^{-16}	2γ
		Kaon	K^+	K^-	0	493.7	0	0	0	0	+1	1.24×10^{-8}	$\mu^+\nu_\mu$, $\pi^+\pi^0$
			K^0_S	$\overline{K}^0_S$	0	497.6	0	0	0	0	+1	0.895×10^{-10}	$\pi^+\pi^-$, $2\pi^0$
			K^0_L	$\overline{K}^0_L$	0	497.6	0	0	0	0	+1	5.12×10^{-8}	$\pi^\pm e^\mp \overset{(-)}{\nu}_e$, $\pi^\pm \mu^\mp \overset{(-)}{\nu}_\mu$, 3π
		Eta	η^0	Self	0	547.9	0	0	0	0	0	5.1×10^{-19}	2γ, $3\pi^0$, $\pi^+\pi^-\pi^0$
		Rho	ρ^0	Self	1	775	0	0	0	0	0	4.4×10^{-24}	$\pi^+\pi^-$, $2\pi^0$
			ρ^+	ρ^-	1	775	0	0	0	0	0	4.4×10^{-24}	$\pi^+\pi^0$
		and others											
Baryons (3 quarks)	str, em, w	Proton	p	$\overline{p}$	$\frac{1}{2}$	938.3	+1	0	0	0	0	Stable	
		Neutron	n	$\overline{n}$	$\frac{1}{2}$	939.6	+1	0	0	0	0	882	$pe^-\bar{\nu}_e$
		Lambda	Λ^0	$\overline{\Lambda}^0$	$\frac{1}{2}$	1115.7	+1	0	0	0	−1	2.63×10^{-10}	$p\pi^-$, $n\pi^0$
		Sigma	Σ^+	$\overline{\Sigma}^-$	$\frac{1}{2}$	1189.4	+1	0	0	0	−1	0.80×10^{-10}	$p\pi^0$, $n\pi^+$
			Σ^0	$\overline{\Sigma}^0$	$\frac{1}{2}$	1192.6	+1	0	0	0	−1	7.4×10^{-20}	$\Lambda^0\gamma$
			Σ^-	$\overline{\Sigma}^+$	$\frac{1}{2}$	1197.4	+1	0	0	0	−1	1.48×10^{-10}	$n\pi^-$
		Xi	Ξ^0	$\overline{\Xi}^0$	$\frac{1}{2}$	1314.9	+1	0	0	0	−2	2.90×10^{-10}	$\Lambda^0\pi^0$
			Ξ^-	$\overline{\Xi}^+$	$\frac{1}{2}$	1321.7	+1	0	0	0	−2	1.64×10^{-10}	$\Lambda^0\pi^-$
		Omega	Ω^-	Ω^+	$\frac{3}{2}$	1672.5	+1	0	0	0	−3	0.82×10^{-10}	$\Xi^0\pi^-$, Λ^0K^-, $\Xi^-\pi^0$
		and others											

†See also Table 32–4 for particles with charm and bottom. S in this Table stands for "strangeness" (see Section 32–8). More detail online at: pdg.lbl.gov.
‡Neutrinos partake only in the weak interaction. Experimental upper limits on neutrino masses are given in parentheses, as obtained mainly from the WMAP survey (Chapter 33). Detection of neutrino oscillations suggests that at least one type of neutrino has a nonzero mass greater than $0.04\ eV/c^2$.

EXAMPLE 32–6 **Baryon decay.** Show that the decay modes of the Σ^+ baryon given in Table 32–2 do not violate the conservation laws we have studied up to now: energy, charge, baryon number, lepton numbers.

APPROACH Table 32–2 shows two possible decay modes, (*a*) $\Sigma^+ \rightarrow p + \pi^0$, (*b*) $\Sigma^+ \rightarrow n + \pi^+$. All the particles have lepton numbers equal to zero.

SOLUTION (*a*) Energy: for $\Sigma^+ \rightarrow p + \pi^0$ the change in mass-energy is

$$\begin{aligned}\Delta(Mc^2) &= m_\Sigma c^2 - m_p c^2 - m_{\pi^0} c^2 \\ &= 1189.4\ \text{MeV} - 938.3\ \text{MeV} - 135.0\ \text{MeV} = +116.1\ \text{MeV},\end{aligned}$$

so energy can be conserved with the resulting particles having kinetic energy.

Charge: $+e = +e + 0$, so charge is conserved.

Baryon number: $+1 = +1 + 0$, so baryon number is conserved.

(*b*) Energy: for $\Sigma^+ \rightarrow n + \pi^+$, the mass-energy change is

$$\begin{aligned}\Delta(Mc^2) &= m_\Sigma c^2 - m_n c^2 - m_{\pi^+} c^2 \\ &= 1189.4\ \text{MeV} - 939.6\ \text{MeV} - 139.6\ \text{MeV} = +110.2\ \text{MeV}.\end{aligned}$$

This reaction releases 110.2 MeV of energy as kinetic energy of the products.

Charge: $+e = 0 + e$, so charge is conserved.

Baryon number: $+1 = +1 + 0$, so baryon number is conserved.

32–7 Particle Stability and Resonances

Many particles listed in Table 32–2 are unstable. The lifetime of an unstable particle depends on which force is most active in causing the decay. When a stronger force influences a decay, that decay occurs more quickly. Decays caused by the weak force typically have lifetimes of 10^{-13} s or longer (W and Z decay directly and more quickly). Decays via the electromagnetic force have much shorter lifetimes, typically about 10^{-16} to 10^{-19} s, and normally involve a γ (photon). Most of the unstable particles included in Table 32–2 decay either via the weak or the electromagnetic interaction.

Many particles have been found that decay via the strong interaction, with very short lifetimes, typically about 10^{-23} s. Their lifetimes are so short they do not travel far enough to be detected before decaying. The existence of such short-lived particles is inferred from their decay products. Consider the first such particle discovered (by Fermi), using a beam of π^+ particles with varying amounts of energy directed through a hydrogen target (protons). The number of interactions (π^+ scattered) plotted versus the pion's kinetic energy is shown in Fig. 32–14. The large number of interactions around 200 MeV led Fermi to conclude that the π^+ and proton combined momentarily to form a short-lived particle before coming apart again, or at least that they resonated together for a short time. Indeed, the large peak in Fig. 32–14 resembles a resonance curve (see Figs. 11–18 and 21–46), and this new "particle"—now called the Δ—is referred to as a **resonance**. Hundreds of other resonances have been found, and are regarded as excited states of lighter mass particles such as a nucleon.

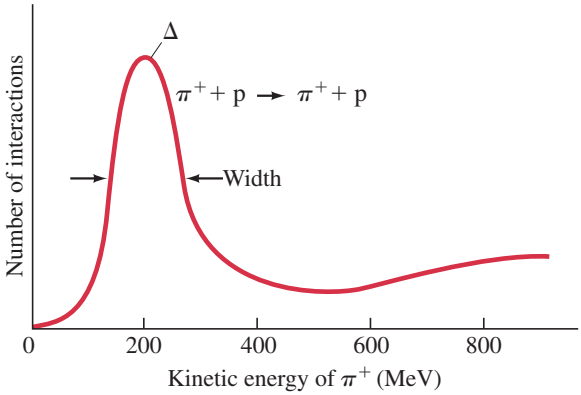


FIGURE 32–14 Number of π^+ particles scattered elastically by a proton target as a function of the incident π^+ kinetic energy. The resonance shape represents the formation of a short-lived particle, the Δ, which has a charge in this case of $+2e$ (Δ^{++}).

The **width** of a resonance—in Fig. 32–14 the full width of the Δ peak at half the maximum is on the order of 100 MeV—is an interesting application of the uncertainty principle. If a particle lives only 10^{-23} s, then its mass (i.e., its rest energy) will be uncertain by an amount

$$\Delta E \approx \frac{h}{2\pi\,\Delta t} \approx \frac{(6.6 \times 10^{-34}\,\text{J}\cdot\text{s})}{(6)(10^{-23}\,\text{s})} \approx 10^{-11}\,\text{J} \approx 100\,\text{MeV},$$

which is what is observed. Actually, the lifetimes of $\approx 10^{-23}$ s for such resonances are inferred by the reverse process: from the measured width being ≈ 100 MeV.

32–8 Strangeness? Charm? Towards a New Model

In the early 1950s, the newly found particles K, Λ, and Σ were found to behave rather strangely in two ways. First, they were always produced in pairs. For example, the reaction

$$\pi^- + \text{p} \rightarrow \text{K}^0 + \Lambda^0$$

occurred with high probability, but the similar reaction $\pi^- + \text{p} \nrightarrow \text{K}^0 + \text{n}$ was never observed to occur even though it did not violate any known conservation law. The second feature of these **strange particles**, as they came to be called, was that they were produced via the strong interaction (that is, at a high interaction rate),

but did not decay at a fast rate characteristic of the strong interaction (even though they decayed into strongly interacting particles).

To explain these observations, a new quantum number, **strangeness**, and a new conservation law, **conservation of strangeness**, were introduced. By assigning the strangeness numbers (S) indicated in Table 32–2, the production of strange particles in pairs was explained. Antiparticles were assigned opposite strangeness from their particles. For example, in the reaction $\pi^- + p \rightarrow K^0 + \Lambda^0$, the initial state has strangeness $S = 0 + 0 = 0$, and the final state has $S = +1 - 1 = 0$, so strangeness is conserved. But for $\pi^- + p \not\rightarrow K^0 + n$, the initial state has $S = 0$ and the final state has $S = +1 + 0 = +1$, so strangeness would not be conserved; and this reaction is not observed.

CAUTION
Partially conserved quantities

To explain the decay of strange particles, it is assumed that strangeness is conserved in the strong interaction but is *not conserved in the weak interaction*. Thus, strange particles were forbidden by strangeness conservation to decay to nonstrange particles of lower mass via the strong interaction, but could decay by means of the weak interaction at the observed longer lifetimes of 10^{-10} to 10^{-8} s.

The conservation of strangeness was the first example of a **partially conserved** quantity. In this case, the quantity strangeness is conserved by strong interactions but not by weak.

CONCEPTUAL EXAMPLE 32–7 **Guess the missing particle.** Using the conservation laws for particle interactions, determine the possibilities for the missing particle in the reaction

$$\pi^- + p \rightarrow K^0 + ?$$

in addition to $K^0 + \Lambda^0$ mentioned above.

RESPONSE We write equations for the conserved numbers in this reaction, with B, L_e, S, and Q as unknowns whose determination will reveal what the possible particle might be:

Baryon number:	$0 + 1 = 0 + B$
Lepton number:	$0 + 0 = 0 + L_e$
Charge:	$-1 + 1 = 0 + Q$
Strangeness:	$0 + 0 = 1 + S$.

The unknown product particle would have to have these characteristics:

$$B = +1 \qquad L_e = 0 \qquad Q = 0 \qquad S = -1.$$

In addition to Λ^0, a neutral sigma particle, Σ^0, is also consistent with these numbers.

In the next Section we will discuss another partially conserved quantity which was given the name **charm**. The discovery in 1974 of a particle with charm helped solidify a new theory involving quarks, which we now discuss.

32–9 Quarks

One difference between leptons and hadrons is that the hadrons interact via the strong interaction, whereas the leptons do not. There is an even more fundamental difference. The six leptons $(e^-, \mu^-, \tau^-, \nu_e, \nu_\mu, \nu_\tau)$ are considered to be truly fundamental particles because they do not show any internal structure, and have no measurable size. (Attempts to determine the size of leptons have put an upper limit of about 10^{-18} m.) On the other hand, there are hundreds of hadrons, and experiments indicate they do have an internal structure. When an electron collides with another electron, it scatters off as per Coulomb's law. But electrons scattering off a proton reveal a more complex pattern, implying internal parts within the proton (= quarks).

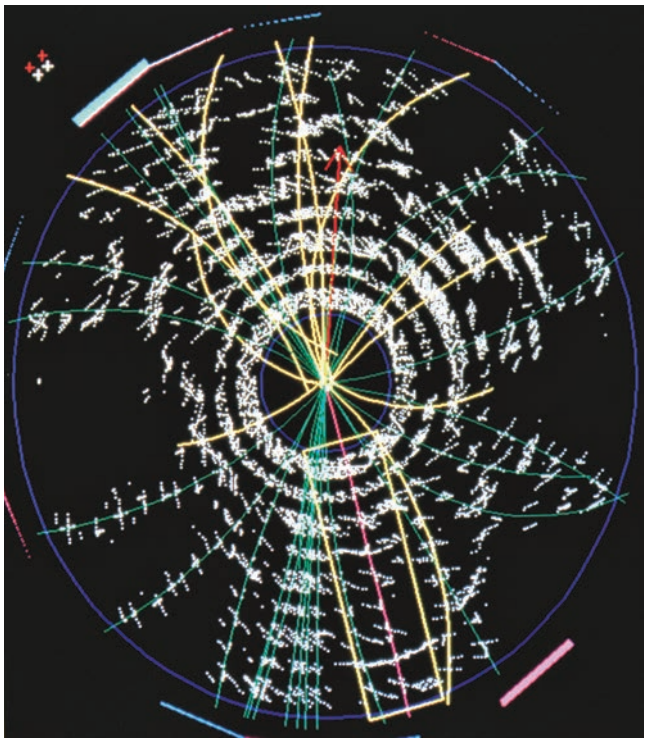


FIGURE 32–15 This computer-generated reconstruction of a proton–antiproton collision at Fermilab occurred at an energy of nearly 2 TeV. It is one of the events that provided evidence for the top quark (1995). The multiwire chamber (Section 30–13) is in a magnetic field, and the radius of curvature of the charged particle tracks is a measure of each particle's momentum (Chapter 20). The white dots represent signals seen on the electric wires of the multiwire chamber. The colored lines are particle paths. The top quark (t) has too brief a lifetime $(\approx 10^{-23}\,\text{s})$ to be detected itself, so we look for its possible decay products. Analysis indicates the following interaction and subsequent decays:

$$
\begin{aligned}
&p + \bar{p} \longrightarrow t + \bar{t}\\
&\quad\quad\quad

TABLE 32–4 Partial List of Heavy Hadrons, with Charm and Bottom ($L_e = L_\mu = L_\tau = 0$)

Category	Particle	Anti-particle	Spin	Mass (MeV/c^2)	Baryon Number B	Strangeness S	Charm c	Bottom b	Mean life (s)	Principal Decay Modes
Mesons	D^+	D^-	0	1869.6	0	0	+1	0	10.4×10^{-13}	K + others, e + others
	D^0	$\overline{D}^0$	0	1864.9	0	0	+1	0	4.1×10^{-13}	K + others, μ or e + others
	D_S^+	D_S^-	0	1968.5	0	+1	+1	0	5.0×10^{-13}	K + others
	J/ψ (3097)	Self	1	3096.9	0	0	0	0	0.71×10^{-20}	Hadrons, e^+e^-, $\mu^+\mu^-$
	Y (9460)	Self	1	9460.3	0	0	0	0	1.2×10^{-20}	Hadrons, $\mu^+\mu^-$, e^+e^-, $\tau^+\tau^-$
	B^-	B^+	0	5279.3	0	0	0	−1	1.6×10^{-12}	D^0 + others
	B^0	$\overline{B}^0$	0	5279.6	0	0	0	−1	1.5×10^{-12}	D^0 + others
Baryons	Λ_c^+	Λ_c^-	$\frac{1}{2}$	2286	+1	0	+1	0	2.0×10^{-13}	Hadrons (e.g., Λ + others)
	Σ_c^{++}	Σ_c^{--}	$\frac{1}{2}$	2454	+1	0	+1	0	2.9×10^{-22}	$\Lambda_c^+\pi^+$
	Σ_c^+	Σ_c^-	$\frac{1}{2}$	2453	+1	0	+1	0	$>1.4 \times 10^{-22}$	$\Lambda_c^+\pi^0$
	Σ_c^0	$\overline{\Sigma}_c^0$	$\frac{1}{2}$	2454	+1	0	+1	0	3.0×10^{-22}	$\Lambda_c^+\pi^-$
	Λ_b^0	$\overline{\Lambda}_b^0$	$\frac{1}{2}$	5619	+1	0	0	−1	1.4×10^{-12}	$J/\psi\Lambda^0$, $pD^0\pi^-$, $\Lambda_c^+\pi^+\pi^-\pi^-$

All hadrons are considered to be made up of combinations of quarks (plus the gluons that hold them together), and their properties are described by looking at their quark content. Mesons consist of a quark–antiquark pair. For example, a π^+ meson is a $u\bar{d}$ combination: note that for the $u\bar{d}$ pair (Table 32–3), $Q = \frac{2}{3}e + \frac{1}{3}e = +1e$, $B = \frac{1}{3} - \frac{1}{3} = 0$, $S = 0 + 0 = 0$, as they must for a π^+; and a $K^+ = u\bar{s}$, with $Q = +1$, $B = 0$, $S = +1$. A π^0 can be made of $u\bar{u}$ or $d\bar{d}$.

Baryons, on the other hand, consist of three quarks. For example, a neutron is n = ddu, whereas an antiproton is $\bar{p} = \bar{u}\,\bar{u}\,\bar{d}$. See Fig. 32–16. Strange particles all contain an s or $\bar{s}$ quark, whereas charm particles contain a c or $\bar{c}$ quark. A few of these hadrons are listed in Table 32–4.

Current models suggest that quarks may be so tightly bound together that they may not ever exist singly in the free state. But quarks can be detected indirectly when they turn into narrow **jets** of other particles, as in Fig. 32–15. Also, observations of very high energy electrons scattered off protons suggest that protons are indeed made up of constituents.

Today, the truly **fundamental particles** are considered to be the six quarks, the six leptons, the gauge bosons that carry the fundamental forces, and the Higgs (page 939). See Table 32–5, where the quarks and leptons are arranged in three "families" or "generations." Ordinary matter—atoms made of protons, neutrons, and electrons—is contained in the "first generation." The others are thought to have existed in the very early universe, but are seen by us today only at powerful colliders or in cosmic rays. All of the hundreds of hadrons can be accounted for by combinations of the six quarks and six antiquarks.

EXERCISE C Return to the Chapter-Opening Questions, page 915, and answer them again now. Try to explain why you may have answered differently the first time.

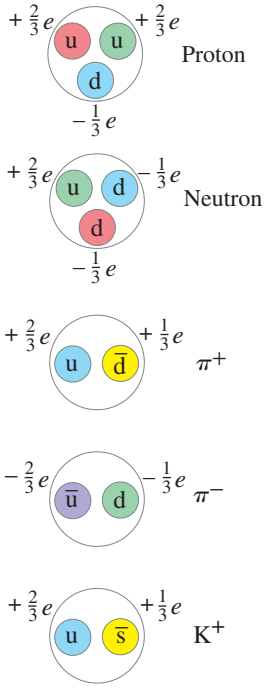


FIGURE 32–16 Quark compositions for several particles.

TABLE 32–5 The Fundamental Particles† of the Standard Model

Gauge bosons	γ	g	Z^0	$W^\pm$
Higgs boson	H^0			
	Quarks		and Leptons	
First generation	u	d	e	ν_e
Second generation	s	c	μ	ν_μ
Third generation	b	t	τ	ν_τ

†The graviton (G^0) has been hypothesized but not detected and may not be detectable. It is not part of the Standard Model.

CONCEPTUAL EXAMPLE 32–8 **Quark combinations.** Find the baryon number, charge, and strangeness for the following quark combinations, and identify the hadron particle that is made up of these quark combinations: (*a*) udd, (*b*) $u\bar{u}$, (*c*) uss, (*d*) sdd, and (*e*) $b\bar{u}$.

RESPONSE We use Table 32–3 to get the properties of the quarks, then Table 32–2 or 32–4 to find the particle that has these properties.

(*a*) udd has

$$\begin{aligned} Q &= +\tfrac{2}{3}e - \tfrac{1}{3}e - \tfrac{1}{3}e = 0, \\ B &= \tfrac{1}{3} + \tfrac{1}{3} + \tfrac{1}{3} = 1, \\ S &= 0 + 0 + 0 = 0, \end{aligned}$$

as well as $c = 0$, bottom $= 0$, top $= 0$. The only baryon ($B = +1$) that has $Q = 0$, $S = 0$, etc., is the neutron (Table 32–2).

(*b*) $u\bar{u}$ has $Q = \frac{2}{3}e - \frac{2}{3}e = 0$, $B = \frac{1}{3} - \frac{1}{3} = 0$, and all other quantum numbers $= 0$. Sounds like a π^0 ($d\bar{d}$ also gives a π^0).

(*c*) uss has $Q = 0$, $B = +1$, $S = -2$, others $= 0$. This is a Ξ^0.

(*d*) sdd has $Q = -1$, $B = +1$, $S = -1$, so must be a Σ^-.

(*e*) $b\bar{u}$ has $Q = -1$, $B = 0$, $S = 0$, $c = 0$, bottom $= -1$, top $= 0$. This must be a B^- meson (Table 32–4).

EXERCISE D What is the quark composition of a K^- meson?

32–10 The Standard Model: QCD and Electroweak Theory

Not long after the quark theory was proposed, it was suggested that quarks have another property (or quality) called **color**, or "color charge" (analogous to electric charge). The distinction between the six types of quark (u, d, s, c, b, t) was referred to as **flavor**. According to theory, each flavor of quark can have one of three colors, usually designated red, green, and blue. (These are the three primary colors which, when added together in appropriate amounts, as on a TV screen, produce white.) Note that the names "color" and "flavor" have nothing to do with our senses, but are purely whimsical—as are other names, such as charm, in this new field. (We did, however, "color" the quarks in Fig. 32–16.) The antiquarks are colored antired, antigreen, and antiblue. Baryons are made up of three quarks, one of each color. Mesons consist of a quark–antiquark pair of a particular color and its anticolor. Both baryons and mesons are thus colorless or white.

Originally, the idea of quark color was proposed to preserve the Pauli exclusion principle (Section 28–7). Not all particles obey the exclusion principle. Those that do, such as electrons, protons, and neutrons, are called **fermions**. Those that don't are called **bosons**. These two categories are distinguished also in their spin: bosons have integer spin (0, 1, 2, etc.) whereas fermions have half-integer spin, usually $\frac{1}{2}$ as for electrons and nucleons, but other fermions have spin $\frac{3}{2}, \frac{5}{2}$, etc. Matter is made up mainly of fermions, but the carriers of the forces (γ, W, Z, and gluons) are all bosons. Quarks are fermions (they have spin $\frac{1}{2}$) and therefore should obey the exclusion principle. Yet for three particular baryons (uuu, ddd, and sss), all three quarks would have the same quantum numbers, and at least two quarks have their spin in the same direction (since there are only two choices, spin up $\left[m_s = +\frac{1}{2}\right]$ or spin down $\left[m_s = -\frac{1}{2}\right]$). This would seem to violate the exclusion principle; but if quarks have that additional quantum number *color*, which is different for each quark, it would serve to distinguish them and allow the exclusion principle to hold. Although quark color,

and the resulting threefold increase in the number of quarks, was originally an *ad hoc* idea, it also served to bring the theory into better agreement with experiment, such as predicting the correct lifetime of the π^0 meson, and the measured rate of hadron production in observed e^+e^- collisions at accelerators. The idea of color soon became a central feature of the theory as determining the force binding quarks together in a hadron.

Each quark is assumed to carry a *color charge*, analogous to electric charge, and the strong force between quarks is referred to as the **color force**. This theory of the strong force is called **quantum chromodynamics** (*chroma* = color in Greek), or **QCD**, to indicate that the force acts between color charges (and not between, say, electric charges). The strong force between two hadrons is considered to be a force between the quarks that make them up, as suggested in Fig. 32–17.

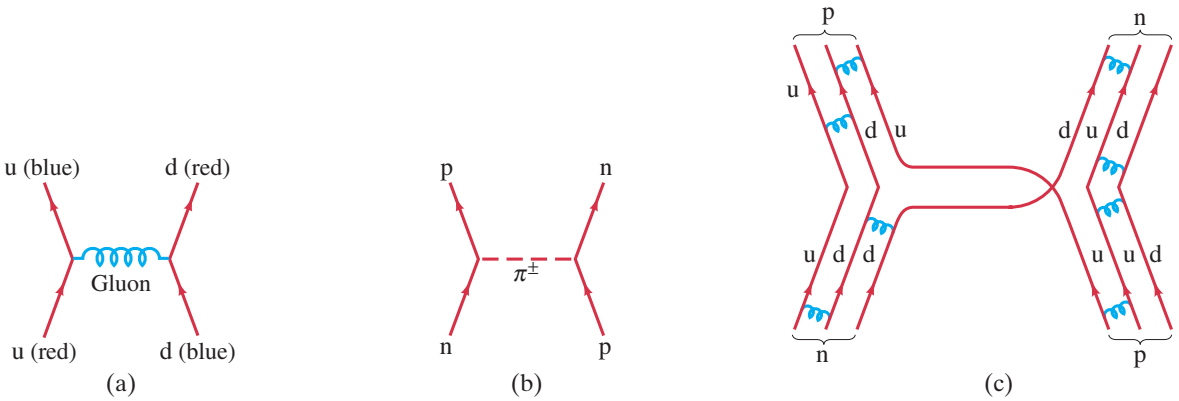


FIGURE 32–17 (a) The force between two quarks holding them together as part of a proton, for example, is carried by a gluon, which in this case involves a change in color. (b) Strong interaction $n + p \rightarrow n + p$ with the exchange of a charged π meson (+ or −, depending on whether it is considered moving to the left or to the right). (c) Quark representation of the same interaction $n + p \rightarrow n + p$. The blue coiled lines between quarks represent gluon exchanges holding the hadrons together. (The exchanged meson may be regarded as $\bar{u}d$ emitted by the n and absorbed by the p, or as $u\bar{d}$ emitted by p and absorbed by n, because a u (or d) quark going to the left in the diagram is equivalent to a $\bar{u}$ (or $\bar{d}$) going to the right.)

The particles that transmit the color force (analogous to photons for the EM force) are called **gluons** (a play on "glue"). They are included in Tables 32–2 and 32–5. There are eight gluons, according to the theory, all massless and all have color charge.[†]

You might ask what would happen if we try to see a single quark with color by reaching deep inside a hadron and extracting a single quark. Quarks are so tightly bound to other quarks that extracting one would require a tremendous amount of energy, so much that it would be sufficient to create more quarks $(E = mc^2)$. Indeed, such experiments are done at modern particle colliders and all we get is more hadrons (quark–antiquark pairs, or triplets, which we observe as mesons or baryons), never an isolated quark. This property of quarks, that they are always bound in groups that are colorless, is called **confinement**.

The color force has the interesting property that, as two quarks approach each other very closely (equivalently, have high energy), the force between them becomes small. This aspect is referred to as **asymptotic freedom**.

The weak force, as we have seen, is thought to be mediated by the W^+, W^-, and Z^0 particles. It acts between the "weak charges" that each particle has. Each fundamental particle can thus have electric charge, weak charge, color charge, and gravitational mass, although one or more of these could be zero. For example, all leptons have color charge of zero, so they do not interact via the strong force.

[†]Compare to the EM interaction, where the photon has no electric charge. Because gluons have color charge, they could attract each other and form composite particles (photons cannot). Such "glueballs" are being searched for.

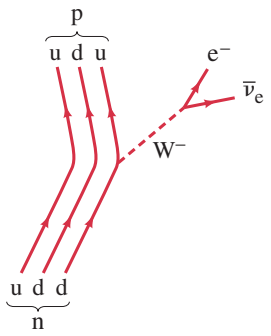


FIGURE 32–18 Quark representation of the Feynman diagram for β decay of a neutron into a proton. Example 32–9.

CONCEPTUAL EXAMPLE 32–9 **Beta decay.** Draw a Feynman diagram, showing what happens in beta decay using quarks.

RESPONSE Beta decay is a result of the weak interaction, and the mediator is either a $W^{\pm}$ or Z^0 particle. What happens, in part, is that a neutron (udd quarks) decays into a proton (uud). Apparently a d quark (charge $-\frac{1}{3}e$) has turned into a u quark (charge $+\frac{2}{3}e$). Charge conservation means that a negatively charged particle, namely a W^-, was emitted by the d quark. Since an electron and an antineutrino appear in the final state, they must have come from the decay of the virtual W^-, as shown in Fig. 32–18.

To summarize, the Standard Model says that the truly fundamental particles (Table 32–5) are the leptons, the quarks, the gauge bosons (photon, W and Z, and the gluons), and the Higgs boson. The photon, leptons, W^+, W^-, and Z^0 have all been observed in experiments. But only combinations of quarks (baryons and mesons) have been observed in the free state, and it seems likely that free quarks and gluons cannot be observed in isolation.

One important aspect of theoretical work is the attempt to find a **unified** basis for the different forces in nature. This was a long-held hope of Einstein, which he was never able to fulfill. A so-called **gauge theory** that unifies the weak and electromagnetic interactions was put forward in the 1960s by S. Weinberg, S. Glashow, and A. Salam. In this **electroweak theory**, the weak and electromagnetic forces are seen as two different manifestations of a single, more fundamental, *electroweak* interaction. The electroweak theory has had many successes, including the prediction of the $W^{\pm}$ particles as carriers of the weak force, with masses of $80.38 \pm 0.02\,\text{GeV}/c^2$ in excellent agreement with the measured values of $80.385 \pm 0.015\,\text{GeV}/c^2$ (and similar accuracy for the Z^0).

The combination of electroweak theory plus QCD for the strong interaction is referred to today as the **Standard Model** (**SM**) of fundamental particles.

EXAMPLE 32–10 | **ESTIMATE** **Range of weak force.** The weak nuclear force is of very short range, meaning it acts over only a very short distance. Estimate its range using the masses (Table 32–2) of the $W^{\pm}$ and Z: $m \approx 80$ or $90\,\text{GeV}/c^2 \approx 10^2\,\text{GeV}/c^2$.

APPROACH We assume the $W^{\pm}$ or Z^0 exchange particles can exist for a time Δt given by the uncertainty principle (Section 28–3), $\Delta t \approx \hbar/\Delta E$, where $\Delta E \approx mc^2$ is the energy needed to create the virtual particle $(W^{\pm}, Z)$ that carries the weak force.

SOLUTION Let Δx be the distance the virtual W or Z can move before it must be reabsorbed within the time $\Delta t \approx \hbar/\Delta E$. To find an upper limit on Δx, and hence the maximum range of the weak force, we let the W or Z travel close to the speed of light, so $\Delta x \lesssim c\,\Delta t$. Recalling that $1\,\text{GeV} = 1.6 \times 10^{-10}\,\text{J}$, then

$$\Delta x \lesssim c\,\Delta t \approx \frac{c\hbar}{\Delta E} \approx \frac{(3 \times 10^8\,\text{m/s})(10^{-34}\,\text{J}\cdot\text{s})}{(10^2\,\text{GeV})(1.6 \times 10^{-10}\,\text{J/GeV})} \approx 10^{-18}\,\text{m}.$$

This is indeed a very small range.

NOTE Compare this to the range of the electromagnetic force whose range is infinite ($1/r^2$ never becomes zero for any finite r), which makes sense because the mass of its virtual exchange particle, the photon, is zero (in the denominator of the above equation).

[We did a similar calculation for the strong force in Section 32–2, estimating the mass of the π meson as exchange particle. In our deeper view of the strong force, namely the color force between quarks within a nucleon, the gluons have zero mass, which implies infinite range (see formula in Example 32–10). We might have expected a range of about 10^{-15} m (nuclear size). But according to the Standard Model, the color force is weak at very close distances and increases greatly with distance (causing quark confinement). Thus its range could be infinite.]

Theoreticians have wondered why the W and Z have large masses rather than being massless like the photon. Peter Higgs and others in 1964 used electroweak theory to suggest an explanation by means of a **Higgs field** and its particle, the **Higgs boson**, which interact with the W and Z to "slow them down." In being forced to go slower than the speed of light, they would have to have mass ($m = 0$ only if $v = c$). Indeed, the Higgs field is thought to permeate the vacuum ("empty space") and to perhaps confer mass on particles that now have mass by slowing them down. In 2012 strong evidence was announced at CERN's Large Hadron Collider (Section 32–1) for a particle of mass 125 GeV/c^2 that is thought to be the long-sought Higgs boson of the Standard Model. But intense research continues, not only to better understand this particle, but to search for additional Higgs-like particles suggested by theories that go beyond the Standard Model such as *supersymmetry* (Section 32–12).

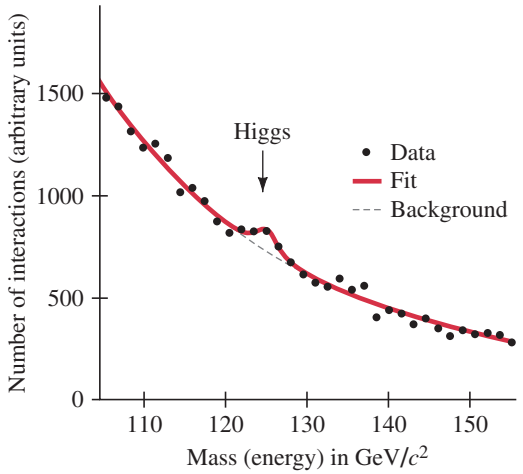


FIGURE 32–19 Evidence for the Higgs boson.

FIGURE 32–20 Fabiola Gianotti, leader of the ATLAS team (3000 physicists), at the LHC with theorist Peter Higgs, July 4, 2012, when the long hoped-for boson was announced.



Figure 32–19 shows the "resonance" bump (Section 32–7) that represents the Higgs boson as detected by the CMS team at the LHC. A second experiment, ATLAS, came up with the same mass. ATLAS is considered the largest scientific experiment ever (see Fig. 32–20).

There is no way to know if the Chapter-Opening Photo of a possible Higgs event, page 915, is actually a Higgs or is a background event. As can be seen in Fig. 32–19, there are many more background events around 125 GeV than there are in the resonance bump representing the Higgs boson.

32–11 Grand Unified Theories

The Standard Model, for all its success, cannot explain some important issues—such as why the charge on the electron has *exactly* the same magnitude as the charge on the proton. This is crucial, because if the charge magnitudes were even a little different, atoms would not be neutral and the resulting large electric forces would surely have made life impossible. Indeed, the Standard Model is now considered to be a low-energy approximation to a more complete theory.

With the success of unified electroweak theory, theorists are trying to incorporate it and QCD for the strong (color) force into a so-called **grand unified theory** (**GUT**).

One type of such a grand unified theory of the electromagnetic, weak, and strong forces has been proposed in which there is only one class of particle—leptons and quarks belong to the same family and are able to change freely from one type to the other—and the three forces are different aspects of a single underlying force. The unity is predicted to occur, however, only on a scale of less than about 10^{-31} m, corresponding to a typical particle energy of about 10^{16} GeV.

If two elementary particles (leptons or quarks) approach each other to within this **unification scale**, the apparently fundamental distinction between them would not exist at this level, and a quark could readily change to a lepton, or vice versa. Baryon and lepton numbers would not be conserved. The weak, electromagnetic, and strong (color) force would blend to a force of a single strength.

What happens between the unification distance of 10^{-31} m and more normal (larger) distances is referred to as **symmetry breaking**. As an analogy, consider an atom in a crystal. Deep within the atom, there is much symmetry—in the innermost regions the electron cloud is spherically symmetric (Chapter 28). Farther out, this symmetry breaks down—the electron clouds are distributed preferentially along the lines (bonds) joining the atoms in the crystal. In a similar way, at 10^{-31} m the force between elementary particles is theorized to be a single force—it is symmetrical and does not single out one type of "charge" over another. But at larger distances, that symmetry is broken and we see three distinct forces. (In the "Standard Model" of electroweak interactions, Section 32–10, the symmetry breaking between the electromagnetic and the weak interactions occurs at about 10^{-18} m.)

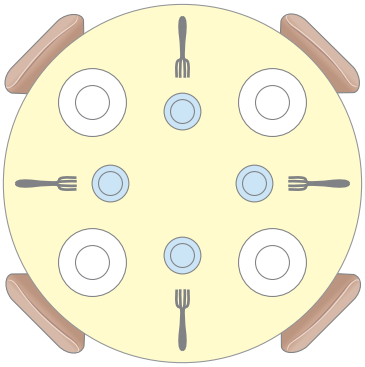


FIGURE 32–21 Symmetry around a table. Example 32–11.

CONCEPTUAL EXAMPLE 32–11 **Symmetry.** The table in Fig. 32–21 has four identical place settings. Four people sit down to eat. Describe the symmetry of this table and what happens to it when someone starts the meal.

RESPONSE The table has several kinds of symmetry. It is symmetric to rotations of 90°: that is, the table will look the same if everyone moved one chair to the left or to the right. It is also north–south symmetric and east–west symmetric, so that swaps across the table don't affect the way the table looks. It also doesn't matter whether any person picks up the fork to the left of the plate or the fork to the right. But once that first person picks up either fork, the choice is set for all the rest at the table as well. The symmetry has been *broken*. The underlying symmetry is still there—the blue glasses could still be chosen either way—but some choice must get made and at that moment the symmetry of the diners is broken.

Another example of symmetry breaking is a pencil standing on its point before falling. Standing, it looks the same from any horizontal direction. From above, it is a tiny circle. But when it falls to the table, it points in one particular direction—the symmetry is broken.

Proton Decay

Since unification is thought to occur at such tiny distances and huge energies, the theory is difficult to test experimentally. But it is not completely impossible. One testable prediction is the idea that the proton might decay (via, for example, $p \rightarrow \pi^0 + e^+$) and violate conservation of baryon number. This could happen if two quarks within a proton approached to within 10^{-31} m of each other. But it is very unlikely at normal temperature and energy, so the decay of a proton can only be an unlikely process. In the simplest form of GUT, the theoretical estimate of the proton mean life for the decay mode $p \rightarrow \pi^0 + e^+$ is about 10^{31} yr, and this is now within the realm of testability.[†] Proton decays have still not been seen, and experiments put the lower limit on the proton mean life for the above mode to be about 10^{33} yr, somewhat greater than this prediction. This may seem a disappointment, but on the other hand, it presents a challenge. Indeed more complex GUTs may resolve this conflict.

[†]This is much larger than the age of the universe ($\approx 14 \times 10^9$ yr). But we don't have to wait 10^{31} yr to see. Instead we can wait for one decay among 10^{31} protons over a year (see Eqs. 30–3a and 30–7, $\Delta N = \lambda N \, \Delta t = N \, \Delta t/\tau$, and Example 32–12).

EXAMPLE 32–12 **ESTIMATE** **Proton decay.** An experiment uses 3300 tons of water waiting to see a proton decay of the type $p \rightarrow \pi^0 + e^+$. If the experiment is run for 4 years without detecting a decay, estimate the lower limit on the proton mean life.

APPROACH As with radioactive decay, the number of decays is proportional to the number of parent species (N), the time interval (Δt), and the decay constant (λ) which is related to the mean life τ by (see Eqs. 30–3 and 30–7):

$$\Delta N = -\lambda N \, \Delta t = -\frac{N \, \Delta t}{\tau}.$$

SOLUTION Dealing only with magnitudes, we solve for τ:

$$\tau = \frac{N \, \Delta t}{\Delta N}.$$

Thus for $\Delta N < 1$ (we don't see even one decay) over the four-year trial,

$$\tau > N\,(4\,\text{yr}),$$

where N is the number of protons in 3300 tons of water. To determine N, we note that each molecule of H_2O contains $2 + 8 = 10$ protons. So one mole of water $(18\,\text{g},\ 6 \times 10^{23}\,\text{molecules})$ contains $10 \times 6 \times 10^{23} = 6 \times 10^{24}$ protons in 18 g of water ($= 18\,\text{g}/1000\,\text{g} = 1/56$ of a kg), or about 3×10^{26} protons per kilogram. One ton is 10^3 kg, so the 3300 tons contains $(3.3 \times 10^6\,\text{kg})(3 \times 10^{26}\,\text{protons/kg}) \approx 1 \times 10^{33}$ protons. Then our very rough estimate for a lower limit on the proton mean life is $\tau > (10^{33})(4\,\text{yr}) \approx 4 \times 10^{33}\,\text{yr}$.

*GUT and Cosmology

An interesting prediction of unified theories relates to cosmology (Chapter 33). It was thought by many theorists that during the first 10^{-35} s after the theorized Big Bang that created the universe, the temperature was so extremely high that particles had energies corresponding to the unification scale. Baryon number would not have been conserved then, perhaps allowing an imbalance that might account for the observed predominance of matter ($B > 0$) over antimatter ($B < 0$) in the universe. The fact that we are surrounded by matter, with no significant antimatter in sight, is considered a problem in search of an explanation (not given by the Standard Model). We call this the **matter–antimatter problem**. To understand it may require still undiscovered phenomena—perhaps related to quarks or neutrinos, or the Higgs boson or supersymmetry (next Section).

Many theorists no longer think the Big Bang was sufficiently hot to create unification. Nonetheless we see that there is a deep connection between investigations at either end of the size scale: theories about the tiniest objects (elementary particles) have a strong bearing on the understanding of the universe on a large scale. We look at this more in the next Chapter.

Figure 32–22 is a rough diagram indicating how the four fundamental forces in nature might have "condensed out" (a symmetry was broken) as time went on after the Big Bang (Chapter 33), and as the mean temperature of the universe and the typical particle energy decreased.

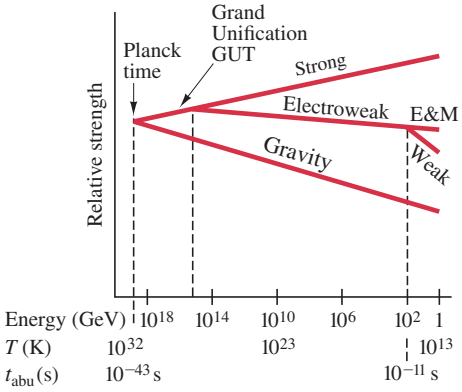


FIGURE 32–22 Time and energy plot of the four fundamental forces, perhaps unified at the "Planck time" (10^{-43} s after the birth of the universe), and how each condensed out, assuming a very hot Big Bang. The symbol t_{abu} means time after the birth of the universe. Note that the typical particle energy (and average temperature of the universe) decreases to the right, as time increases.

32–12 Strings and Supersymmetry

We have seen that the Standard Model is unable to address important experimental issues, and that theoreticians are attacking the problem as experimenters search for new data, new particles, new concepts.

Even more ambitious than grand unified theories are attempts to also incorporate gravity, and thus unify all four forces in nature into a single theory. (Such theories are sometimes referred to misleadingly as **theories of everything**.) A major attempt to unify all four forces is called **string theory**, introduced by Gabriele Veneziano in 1968: Each fundamental particle (Table 32–5) is imagined not as a point but as a one-dimensional **string**, perhaps 10^{-35} m long, which vibrates in a particular standing wave pattern. (You might say each particle is a different note on a tiny stretched string.) More sophisticated theories propose the fundamental entities as being multidimensional **branes** (after 2-D membranes).

A related idea that also goes way beyond the Standard Model is **supersymmetry**, which applied to strings is known as **superstring theory**. Supersymmetry, developed by Bruno Zumino (1923–) and Julius Wess (1934–2007), predicts that interactions exist that would change fermions into bosons and vice versa, and that each known fermion would have a supersymmetric boson partner. Thus, for each quark (a fermion), there would be a **squark** (a boson) or "supersymmetric" quark. For every lepton there would be a **slepton**. Likewise, for every known boson (photons and gluons, for example), there would be a supersymmetric fermion (**photinos** and **gluinos**). Supersymmetry predicts also that a *graviton*, which transmits the gravity force, has a partner, the **gravitino**. Supersymmetry (often abbreviated SUSY) offers solutions to a number of important theoretical problems. Supersymmetric particles are a candidate for the "dark matter" of the universe (discussed in Chapter 33). But why hasn't this "missing part" of the universe ever been detected? The best guess is that supersymmetric particles might be heavier than their conventional counterparts, perhaps too heavy to have been produced in today's accelerators. A search for supersymmetric particles is already being done at CERN's new Large Hadron Collider.

Versions of these theories predict other interesting properties, such as that space has 11 dimensions, but 7 of them are "coiled up" so we normally only notice the 4-D of space–time. We would like to know if and how many extra dimensions there are, and how and why they are hidden. We hope to have some answers from the new LHC (Section 32–1).

Some theorists think SUSY and other theories are approximations to a more fundamental, still undiscovered, **M-theory**. Edward Witten coined the term when proposing an 11 dimensional approximation, but never said what "M" stands for.

The world of elementary particles is opening new vistas. What happens in the future is bound to be exciting.

Summary

Particle accelerators are used to accelerate charged particles, such as electrons and protons, so they can have very high energy collisions with other particles. High-energy particles have short wavelength and so can be used to probe the structure of matter in great detail (very small distances). High kinetic energy also allows the creation of new particles through collisions (via $E = mc^2$).

Cyclotrons and **synchrotrons** use a magnetic field to keep the particles in a circular path and accelerate them at intervals by high voltage. **Linear accelerators** accelerate particles along a line. **Colliding beams** allow higher interaction energy.

An **antiparticle** has the same mass as a particle but opposite charge. Certain other properties may also be opposite: for example, the antiproton has **baryon number** (nucleon number) opposite $(B = -1)$ to that for the proton $(B = +1)$.

In all nuclear and particle reactions, the following conservation laws hold: momentum, angular momentum, mass–energy, electric charge, baryon number, and **lepton numbers**.

Certain particles have a property called **strangeness**, which is conserved by the strong force but not by the weak force. The properties **charm**, **bottom**, and **top** also are conserved by the strong force but not by the weak force.

Just as the electromagnetic force can be said to be due to an exchange of photons, the strong nuclear force is carried by massless **gluons**. The W and Z particles carry the weak force. These fundamental force carriers (photon, W and Z, gluons) are called **gauge bosons**.

Other particles can be classified as either *leptons* or *hadrons*. **Leptons** participate only in gravity, the weak, and the electromagnetic interactions. **Hadrons**, which today are considered **composite** particles, are made up of **quarks**, and participate in all four interactions, including the strong interaction. The hadrons can be classified as **mesons**, with baryon number zero, and **baryons**, with nonzero baryon number.

Most particles, except for the photon, electron, neutrinos, and proton, decay with measurable mean lives varying from 10^{-25} s to 10^{3} s. The mean life depends on which force is predominant. Weak decays usually have mean lives greater than about 10^{-13} s. Electromagnetic decays typically have mean lives on the order of 10^{-16} to 10^{-19} s. The shortest lived particles, called **resonances**, decay via the strong interaction and live typically for only about 10^{-23} s.

Today's **Standard Model** of elementary particles considers **quarks** as the basic building blocks of the hadrons. The six quark "flavors" are called **up**, **down**, **strange**, **charm**, **bottom**, and **top**. It is expected that there are the same number of quarks as leptons (six of each), and that quarks and leptons are truly fundamental particles along with the gauge bosons (γ, W, Z, gluons) and the Higgs boson.

Quarks are said to have **color**, and, according to **quantum chromodynamics** (QCD), the strong color force acts between their color charges and is transmitted by **gluons**. **Electroweak theory** views the weak and electromagnetic forces as two aspects of a single underlying interaction. QCD plus the electroweak theory are referred to as the *Standard Model* of the fundamental particles.

Grand unified theories of forces suggest that at very short distance $(10^{-31}\,\text{m})$ and very high energy, the weak, electromagnetic, and strong forces would appear as a single force, and the fundamental difference between quarks and leptons would disappear.

According to **string theory**, the fundamental particles may be tiny strings, 10^{-35} m long, distinguished by their standing wave pattern. **Supersymmetry** predicts that each fermion (or boson) has a corresponding boson (or fermion) partner.

Questions

1. Give a reaction between two nucleons, similar to Eq. 32–4, that could produce a π^-.
2. If a proton is moving at very high speed, so that its kinetic energy is much greater than its rest energy (mc^2), can it then decay via $\text{p} \rightarrow \text{n} + \pi^+$?
3. What would an "antiatom," made up of the antiparticles to the constituents of normal atoms, consist of? What might happen if *antimatter*, made of such antiatoms, came in contact with our normal world of matter?
4. What particle in a decay signals the electromagnetic interaction?
5. (*a*) Does the presence of a neutrino among the decay products of a particle necessarily mean that the decay occurs via the weak interaction? (*b*) Do all decays via the weak interaction produce a neutrino? Explain.
6. Why is it that a neutron decays via the weak interaction even though the neutron and one of its decay products (proton) are strongly interacting?
7. Which of the four interactions (strong, electromagnetic, weak, gravitational) does an electron take part in? A neutrino? A proton?
8. Verify that charge and baryon number are conserved in each of the decays shown in Table 32–2.
9. Which of the particle decays listed in Table 32–2 occur via the electromagnetic interaction?
10. Which of the particle decays listed in Table 32–2 occur by the weak interaction?
11. The Δ baryon has spin $\frac{3}{2}$, baryon number 1, and charge $Q = +2, +1, 0,$ or -1. Why is there no charge state $Q = -2$?
12. Which of the particle decays in Table 32–4 occur via the electromagnetic interaction?
13. Which of the particle decays in Table 32–4 occur by the weak interaction?
14. Quarks have spin $\frac{1}{2}$. How do you account for the fact that baryons have spin $\frac{1}{2}$ or $\frac{3}{2}$, and mesons have spin 0 or 1?
15. Suppose there were a kind of "neutrinolet" that was massless, had no color charge or electrical charge, and did not feel the weak force. Could you say that this particle even exists?
16. Is it possible for a particle to be both (*a*) a lepton and a baryon? (*b*) a baryon and a hadron? (*c*) a meson and a quark? (*d*) a hadron and a lepton? Explain.
17. Using the ideas of quantum chromodynamics, would it be possible to find particles made up of two quarks and no antiquarks? What about two quarks and two antiquarks?
18. Why can neutrons decay when they are free, but not when they are inside a stable nucleus?
19. Is the reaction $\text{e}^- + \text{p} \rightarrow \text{n} + \bar{\nu}_\text{e}$ possible? Explain.
20. Occasionally, the Λ will decay by the following reaction: $\Lambda^0 \rightarrow \text{p}^+ + \text{e}^- + \bar{\nu}_\text{e}$. Which of the four forces in nature is responsible for this decay? How do you know?

MisConceptual Questions

1. There are six kinds (= flavors) of quarks: up, down, strange, charm, bottom, and top. Which flavors make up most of the known matter in the universe?
   (*a*) Up and down quarks.
   (*b*) Strange and charm quarks.
   (*c*) Bottom and top quarks.
   (*d*) All of the above.
2. Which of the following particles can not be composed of quarks?
   (*a*) Proton.
   (*b*) Electron.
   (*c*) π meson.
   (*d*) Neutron.
   (*e*) Higgs boson.

3. If gravity is the weakest force, why is it the one we notice most?
(*a*) Our bodies are not sensitive to the other forces.
(*b*) The other forces act only within atoms and therefore have no effect on us.
(*c*) Gravity may be "very weak" but always attractive, and the Earth has enormous mass. The strong and weak nuclear forces have very short range. The electromagnetic force has a long range, but most matter is electrically neutral.
(*d*) At long distances, the gravitational force is actually stronger than the other forces.
(*e*) The other forces act only on elementary particles, not on objects our size.

4. Is it possible for a tau lepton (whose mass is almost twice that of a proton) to decay into only hadrons?
(*a*) Yes, because it is so massive it could decay into a proton and pions.
(*b*) Yes, it could decay into pions and nothing else.
(*c*) No, such a decay would violate lepton number; all of its decay products must be leptons.
(*d*) No, its decay products must include a tau neutrino but could include hadrons such as pions.
(*e*) No, the tau lepton is too massive to decay.

5. Many particle accelerators are circular because:
(*a*) particles accelerate faster around circles.
(*b*) in order to move in a circle, acceleration is required.
(*c*) a circular accelerator has a shorter length than a square one.
(*d*) the particles can be accelerated through the same potential difference many times, making the accelerator more compact.
(*e*) a particle moving in a circle needs more energy than a particle moving in a straight line.

6. Which of the following are today considered fundamental particles (that is, not composed of smaller components)? Choose as many as apply.
(*a*) Atoms. (*b*) Electrons. (*c*) Protons. (*d*) Neutrons.
(*e*) Quarks. (*f*) Photon. (*g*) Higgs boson.

7. The electron's antiparticle is called the positron. Which of the following properties, if any, are the same for electrons and positrons?
(*a*) Mass.
(*b*) Charge.
(*c*) Lepton number.
(*d*) None of the above.

8. The strong nuclear force between a neutron and a proton is due to
(*a*) the exchange of π mesons between the neutron and the proton.
(*b*) the conservation of baryon number.
(*c*) the beta decay of the neutron into the proton.
(*d*) the exchange of gluons between the quarks within the neutron and the proton.
(*e*) Both (*a*) and (*d*) at different scales.

9. Electrons are still considered fundamental particles (in the group called leptons). But protons and neutrons are no longer considered fundamental; they have substructure and are made up of
(*a*) pions. (*b*) leptons. (*c*) quarks. (*d*) bosons. (*e*) photons.

10. Which of the following will interact via the weak nuclear force *only*?
(*a*) Quarks. (*b*) Gluons. (*c*) Neutrons. (*d*) Neutrinos.
(*e*) Electrons. (*f*) Muons. (*g*) Higgs boson.

For assigned homework and other learning materials, go to the MasteringPhysics website. 

Problems

32–1 Particles and Accelerators

1. (I) What is the total energy of a proton whose kinetic energy is 4.65 GeV?

2. (I) Calculate the wavelength of 28-GeV electrons.

3. (I) If α particles are accelerated by the cyclotron of Example 32–2, what must be the frequency of the voltage applied to the dees?

4. (I) What is the time for one complete revolution for a very high-energy proton in the 1.0-km-radius Fermilab accelerator?

5. (II) What strength of magnetic field is used in a cyclotron in which protons make 3.1×10^7 revolutions per second?

6. (II) (*a*) If the cyclotron of Example 32–2 accelerated α particles, what maximum energy could they attain? What would their speed be? (*b*) Repeat for deuterons $\left({}^{2}_{1}\mathrm{H}\right)$. (*c*) In each case, what frequency of voltage is required?

7. (II) Which is better for resolving details of the nucleus: 25-MeV alpha particles or 25-MeV protons? Compare each of their wavelengths with the size of a nucleon in a nucleus.

8. (II) What is the wavelength (= minimum resolvable size) of 7.0-TeV protons at the LHC?

9. (II) The 1.0-km radius Fermilab Tevatron took about 20 seconds to bring the energies of the stored protons from 150 GeV to 1.0 TeV. The acceleration was done once per turn. Estimate the energy given to the protons on each turn. (You can assume that the speed of the protons is essentially c the whole time.)

10. (II) A cyclotron with a radius of 1.0 m is to accelerate deuterons $\left({}^{2}_{1}\mathrm{H}\right)$ to an energy of 12 MeV. (*a*) What is the required magnetic field? (*b*) What frequency is needed for the voltage between the dees? (*c*) If the potential difference between the dees averages 22 kV, how many revolutions will the particles make before exiting? (*d*) How much time does it take for one deuteron to go from start to exit? (*e*) Estimate how far it travels during this time.

11. (III) Show that the energy of a particle (charge e) in a synchrotron, in the relativistic limit ($v \approx c$), is given by E (in eV) $= Brc$, where B is the magnetic field and r is the radius of the orbit (SI units).

32–2 to 32–6 Particle Interactions, Particle Exchange

12. (I) About how much energy is released when a Λ^0 decays to $\mathrm{n} + \pi^0$? (See Table 32–2.)

13. (I) How much energy is released in the decay

$$\pi^+ \rightarrow \mu^+ + \nu_\mu ?$$

See Table 32–2.

14. (I) Estimate the range of the strong force if the mediating particle were the kaon instead of a pion.

15. (I) How much energy is required to produce a neutron–antineutron pair?

16. (II) Determine the total energy released when Σ^0 decays to Λ^0 and then to a proton.

17. (II) Two protons are heading toward each other with equal speeds. What minimum kinetic energy must each have if a π^0 meson is to be created in the process? (See Table 32–2.)

18. (II) What minimum kinetic energy must a proton and an antiproton each have if they are traveling at the same speed toward each other, collide, and produce a K^+K^- pair in addition to themselves? (See Table 32–2.)

19. (II) What are the wavelengths of the two photons produced when a proton and antiproton at rest annihilate?

20. (II) The Λ^0 cannot decay by the following reactions. What conservation laws are violated in each of the reactions?
(*a*) $\Lambda^0 \not\rightarrow n + \pi^-$
(*b*) $\Lambda^0 \not\rightarrow p + K^-$
(*c*) $\Lambda^0 \not\rightarrow \pi^+ + \pi^-$

21. (II) What would be the wavelengths of the two photons produced when an electron and a positron, each with 420 keV of kinetic energy, annihilate in a head-on collision?

22. (II) Which of the following reactions and decays are possible? For those forbidden, explain what laws are violated.
(*a*) $\pi^- + p \rightarrow n + \eta^0$
(*b*) $\pi^+ + p \rightarrow n + \pi^0$
(*c*) $\pi^+ + p \rightarrow p + e^+$
(*d*) $p \rightarrow e^+ + \nu_e$
(*e*) $\mu^+ \rightarrow e^+ + \bar{\nu}_\mu$
(*f*) $p \rightarrow n + e^+ + \nu_e$

23. (III) In the rare decay $\pi^+ \rightarrow e^+ + \nu_e$, what is the kinetic energy of the positron? Assume the π^+ decays from rest and $m_\nu = 0$.

24. (III) For the decay $\Lambda^0 \rightarrow p + \pi^-$, calculate (*a*) the Q-value (energy released), and (*b*) the kinetic energy of the p and π^-, assuming the Λ^0 decays from rest. (Use relativistic formulas.)

32–7 to 32–11 Resonances, Standard Model, Quarks, QCD, GUT

25. (I) The mean life of the Σ^0 particle is 7×10^{-20} s. What is the uncertainty in its rest energy? Express your answer in MeV.

26. (I) The measured width of the ψ (3686) meson is about 300 keV. Estimate its mean life.

27. (I) The measured width of the J/ψ meson is 88 keV. Estimate its mean life.

28. (I) The B^- meson is a $b\bar{u}$ quark combination. (*a*) Show that this is consistent for all quantum numbers. (*b*) What are the quark combinations for B^+, B^0, $\bar{B}^0$?

29. (I) What is the energy width (or uncertainty) of (*a*) η^0, and (*b*) ρ^+? See Table 32–2.

30. (II) Which of the following decays are possible? For those that are forbidden, explain which laws are violated.
(*a*) $\Xi^0 \rightarrow \Sigma^+ + \pi^-$
(*b*) $\Omega^- \rightarrow \Sigma^0 + \pi^- + \nu$
(*c*) $\Sigma^0 \rightarrow \Lambda^0 + \gamma + \gamma$

31. (II) In ordinary radioactive decay, a W particle may be created even though the decaying particle has less mass than the W particle. If you assume $\Delta E \approx$ mass of the virtual W, what is the expected lifetime of the W?

32. (II) What quark combinations produce (*a*) a Ξ^0 baryon and (*b*) a Ξ^- baryon?

33. (II) What are the quark combinations that can form (*a*) a neutron, (*b*) an antineutron, (*c*) a Λ^0, (*d*) a $\bar{\Sigma}^0$?

34. (II) What particles do the following quark combinations produce: (*a*) uud, (*b*) $\bar{u}\,\bar{u}\,\bar{s}$, (*c*) $\bar{u}s$, (*d*) $d\bar{u}$, (*e*) $\bar{c}s$?

35. (II) What is the quark combination needed to produce a D^0 meson ($Q = B = S = 0$, $c = +1$)?

36. (II) The D_S^+ meson has $S = c = +1$, $B = 0$. What quark combination would produce it?

37. (II) Draw a possible Feynman diagram using quarks (as in Fig. 32–17c) for the reaction $\pi^- + p \rightarrow \pi^0 + n$.

38. (II) Draw a Feynman diagram for the reaction $n + \nu_\mu \rightarrow p + \mu^-$.

General Problems

39. What is the total energy of a proton whose kinetic energy is 15 GeV? What is its wavelength?

40. The mean lifetimes listed in Table 32–2 are in terms of *proper time*, measured in a reference frame where the particle is at rest. If a tau lepton is created with a kinetic energy of 950 MeV, how long would its track be as measured in the lab, on average, ignoring any collisions?

41. (*a*) How much energy is released when an electron and a positron annihilate each other? (*b*) How much energy is released when a proton and an antiproton annihilate each other? (All particles have KE $\approx$ 0.)

42. If 2×10^{14} protons moving at $v \approx c$, with KE = 4.0 TeV, are stored in the 4.3-km-radius ring of the LHC, (*a*) how much current (amperes) is carried by this beam? (*b*) How fast would a 1500-kg car have to move to carry the same kinetic energy as this beam?

43. Protons are injected into the 4.3-km-radius Large Hadron Collider with an energy of 450 GeV. If they are accelerated by 8.0 MV each revolution, how far do they travel and approximately how much time does it take for them to reach 4.0 TeV?

44. Which of the following reactions are possible, and by what interaction could they occur? For those forbidden, explain why.
(*a*) $\pi^- + p \rightarrow K^0 + p + \pi^0$
(*b*) $K^- + p \rightarrow \Lambda^0 + \pi^0$
(*c*) $K^+ + n \rightarrow \Sigma^+ + \pi^0 + \gamma$
(*d*) $K^+ \rightarrow \pi^0 + \pi^0 + \pi^+$
(*e*) $\pi^+ \rightarrow e^+ + \nu_e$

45. Which of the following reactions are possible, and by what interaction could they occur? For those forbidden, explain why.
(*a*) $\pi^- + p \rightarrow K^+ + \Sigma^-$
(*b*) $\pi^+ + p \rightarrow K^+ + \Sigma^+$
(*c*) $\pi^- + p \rightarrow \Lambda^0 + K^0 + \pi^0$
(*d*) $\pi^+ + p \rightarrow \Sigma^0 + \pi^0$
(*e*) $\pi^- + p \rightarrow p + e^- + \bar{\nu}_e$

46. One decay mode for a π^+ is $\pi^+ \rightarrow \mu^+ + \nu_\mu$. What would be the equivalent decay for a π^-? Check conservation laws.

47. Symmetry breaking occurs in the electroweak theory at about 10^{-18} m. Show that this corresponds to an energy that is on the order of the mass of the $W^\pm$.

48. Calculate the Q-value for each of the reactions, Eq. 32–4, for producing a pion.

49. How many fundamental fermions are there in a water molecule?

50. The mass of a π^0 can be measured by observing the reaction $\pi^- + p \rightarrow \pi^0 + n$ with initial kinetic energies near zero. The neutron is observed to be emitted with a kinetic energy of 0.60 MeV. Use conservation of energy and momentum to determine the π^0 mass.

51. (*a*) Show that the so-called unification distance of 10^{-31} m in grand unified theory is equivalent to an energy of about 10^{16} GeV. Use the uncertainty principle, and also de Broglie's wavelength formula, and explain how they apply. (*b*) Calculate the temperature corresponding to 10^{16} GeV.

52. Calculate the Q-value for the reaction $\pi^- + p \rightarrow \Lambda^0 + K^0$, when negative pions strike stationary protons. Estimate the minimum pion kinetic energy needed to produce this reaction. [*Hint*: Assume Λ^0 and K^0 move off with the same velocity.]

53. A proton and an antiproton annihilate each other at rest and produce two pions, π^- and π^+. What is the kinetic energy of each pion?

54. For the reaction $p + p \rightarrow 3p + \bar{p}$, where one of the initial protons is at rest, use relativistic formulas to show that the threshold energy is $6m_p c^2$, equal to three times the magnitude of the Q-value of the reaction, where m_p is the proton mass. [*Hint*: Assume all final particles have the same velocity.]

55. At about what kinetic energy (in eV) can the rest energy of a proton be ignored when calculating its wavelength, if the wavelength is to be within 1.0% of its true value? What are the corresponding wavelength and speed of the proton?

56. Use the quark model to describe the reaction

$$\bar{p} + n \rightarrow \pi^- + \pi^0.$$

57. Identify the missing particle in the following reactions.
(*a*) $p + p \rightarrow p + n + \pi^+ + ?$ (*b*) $p + ? \rightarrow n + \mu^+$

58. What fraction of the speed of light c is the speed of a 7.0-TeV proton?

59. Using the information in Section 32–1, show that the Large Hadron Collider's two colliding proton beams can resolve details that are less than 1/10,000 the size of a nucleus.

60. Searches are underway for a process called **neutrinoless double beta decay**, in which a nucleus decays by emitting two electrons. (*a*) If the parent nucleus is $^{96}_{40}Zr$, what would the daughter nucleus be? (*b*) What conservation laws would be violated during this decay? (*c*) How could $^{96}_{40}Zr$ decay to the same daughter nucleus without violating any conservation laws?

61. Estimate the lifetime of the Higgs boson from the width of the "bump" in Fig. 32–19, using the uncertainty principle. [Note: This is not a realistic estimate because the underlying processes are very complicated.]

Search and Learn

1. (*a*) Show, by conserving momentum and energy, that it is impossible for an isolated electron to radiate only a single photon. (*b*) With this result in mind, how can you defend the photon exchange diagram in Fig. 32–8?

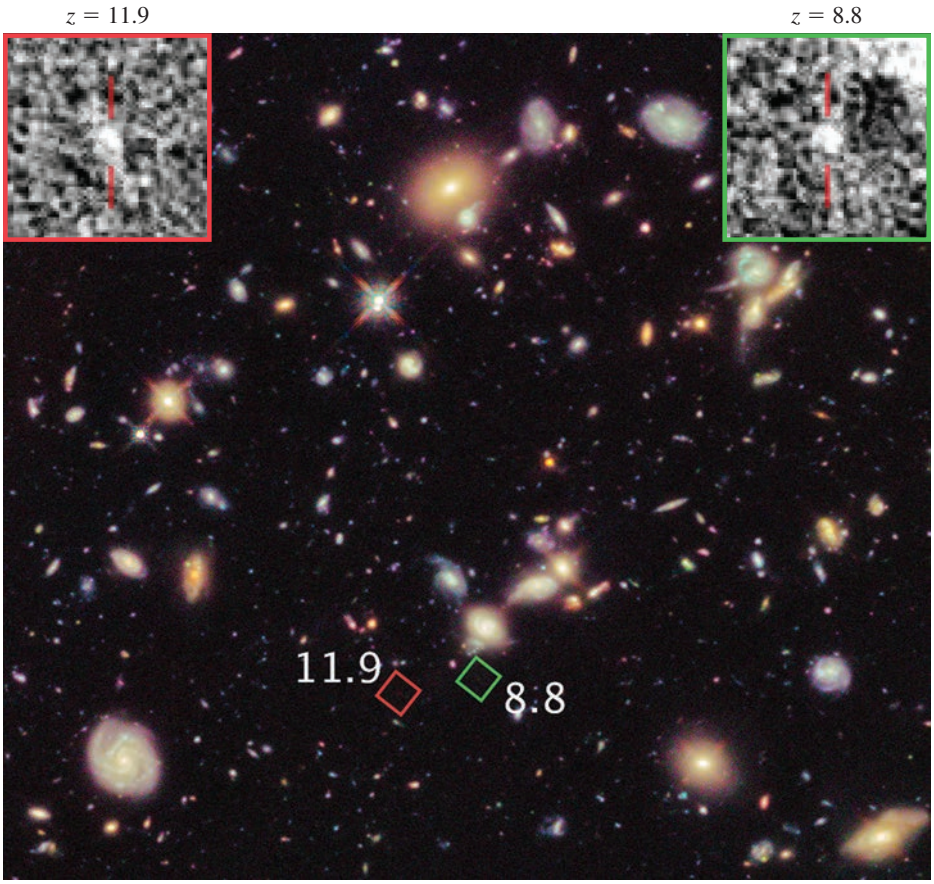
2. (*a*) What property characterizes all hadrons? (*b*) What property characterizes all baryons? (*c*) What property characterizes all mesons?

3. Show that all conservation laws hold for all the decays described in Fig. 32–15 for the decays of the top quark.

4. The Higgs boson, Section 32–10, has very probably been detected at the CERN LHC. (*a*) If a Higgs boson at rest decays into two tau leptons, what is the kinetic energy of each tau? Follow the analysis of Example 32–5. See Table 32–2. (*b*) What are the signs of the electric charges of the two tau leptons? (*c*) Could a Higgs boson decay into two Z bosons (Table 32–2)?

ANSWERS TO EXERCISES

A: 1.24×10^{-18} m = 1.24 am.
B: (*a*).
C: (*c*); (*d*).
D: $s\bar{u}$.



This Hubble eXtreme Deep Field (XDF) photograph is of a very small part of the sky. It includes what may be the most distant galaxies observable by us (small red and green squares, and shown enlarged in the corners), with $z \approx 8.8$ and 11.9, that already existed when the universe was about 0.4 billion years old. We see these galaxies as they appeared then, 13.4 billion years ago, which is when they emitted this light. The most distant galaxies were young and small and grew to become large galaxies by colliding and merging with other small galaxies.

We examine the latest theories on how stars and galaxies form and evolve, including the role of nucleosynthesis, as well as Einstein's general theory of relativity which deals with gravity and curvature of space. We take a thorough look at the evidence for the expansion of the universe, and the Standard Model of the universe evolving from an initial Big Bang. We point out some unsolved problems, including the nature of dark matter and dark energy that make up most of our universe.

CHAPTER 33

Astrophysics and Cosmology

CHAPTER-OPENING QUESTIONS—Guess now!

1. Until recently, astronomers expected the expansion rate of the universe would be decreasing. Why?

(a) Friction.
(b) The second law of thermodynamics.
(c) Gravity.
(d) The electromagnetic force.

2. The universe began expanding right at the beginning. How long will it continue to expand?

(a) Until it runs out of room.
(b) Until friction slows it down and brings it to a stop.
(c) Until all galaxies are moving at the speed of light relative to the center.
(d) Possibly forever.

In the previous Chapter, we studied the tiniest objects in the universe—the elementary particles. Now we leap to the grandest objects in the universe—stars, galaxies, and clusters of galaxies—plus the history and structure of the universe itself. These two extreme realms, elementary particles and the cosmos, are among the most intriguing and exciting subjects in science. And, surprisingly, these two extreme realms are related in a fundamental way, as was already hinted in Chapter 32.

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33–2 Stellar Evolution: Birth and Death of Stars, Nucleosynthesis
33–3 Distance Measurements
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33–6 The Big Bang and the Cosmic Microwave Background
33–7 The Standard Cosmological Model: Early History of the Universe
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33–9 Dark Matter and Dark Energy
33–10 Large-Scale Structure of the Universe
33–11 Finally …

Use of the techniques and ideas of physics to study the night sky is often referred to as **astrophysics**. Central to our present theoretical understanding of the universe (or cosmos) is Einstein's *general theory of relativity* which represents our most complete understanding of gravitation. Many other aspects of physics are involved, from electromagnetism and thermodynamics to atomic and nuclear physics as well as elementary particles. General Relativity serves also as the foundation for modern **cosmology**, which is the study of the universe as a whole. Cosmology deals especially with the search for a theoretical framework to understand the observed universe, its origin, and its future. The questions posed by cosmology are profound and difficult; the possible answers stretch the imagination. They are questions like "Has the universe always existed, or did it have a beginning in time?" Either alternative is difficult to imagine: time going back indefinitely into the past, or an actual moment when the universe began (but, then, what was there before?). And what about the size of the universe? Is it infinite in size? It is hard to imagine infinity. Or is it finite in size? This is also hard to imagine, for if the universe is finite, it does not make sense to ask what is beyond it, because the universe is all there is.

In the last 10 to 20 years, so much progress has occurred in astrophysics and cosmology that many scientists are calling recent work a "Golden Age" for cosmology. Our survey will be qualitative, but we will nonetheless touch on the major ideas. We begin with a look at what can be seen beyond the Earth.



(a)



(b)

FIGURE 33–1 Sections of the Milky Way. In (a), the thin line is the trail of an artificial Earth satellite in this long time exposure. The dark diagonal area is due to dust absorption of visible light, blocking the view. In (b) the view is toward the center of the Galaxy (taken in summer from Arizona).

33–1 Stars and Galaxies

According to the ancients, the stars, except for the few that seemed to move relative to the others (the planets), were fixed on a sphere beyond the last planet. The universe was neatly self-contained, and we on Earth were at or near its center. But in the centuries following Galileo's first telescopic observations of the night sky in 1609, our view of the universe has changed dramatically. We no longer place ourselves at the center, and we view the universe as vastly larger. The distances involved are so great that we specify them in terms of the time it takes light to travel the given distance: for example,

$$1 \text{ light-second} = (3.0 \times 10^8 \text{ m/s})(1.0 \text{ s}) = 3.0 \times 10^8 \text{ m} = 300{,}000 \text{ km};$$
$$1 \text{ light-minute} = (3.0 \times 10^8 \text{ m/s})(60 \text{ s}) = 18 \times 10^6 \text{ km}.$$

The most common unit is the **light-year** (ly):

$$1 \text{ ly} = (2.998 \times 10^8 \text{ m/s})(3.156 \times 10^7 \text{ s/yr})$$
$$= 9.46 \times 10^{15} \text{ m} \approx 10^{13} \text{ km} \approx 10^{16} \text{ m}.$$

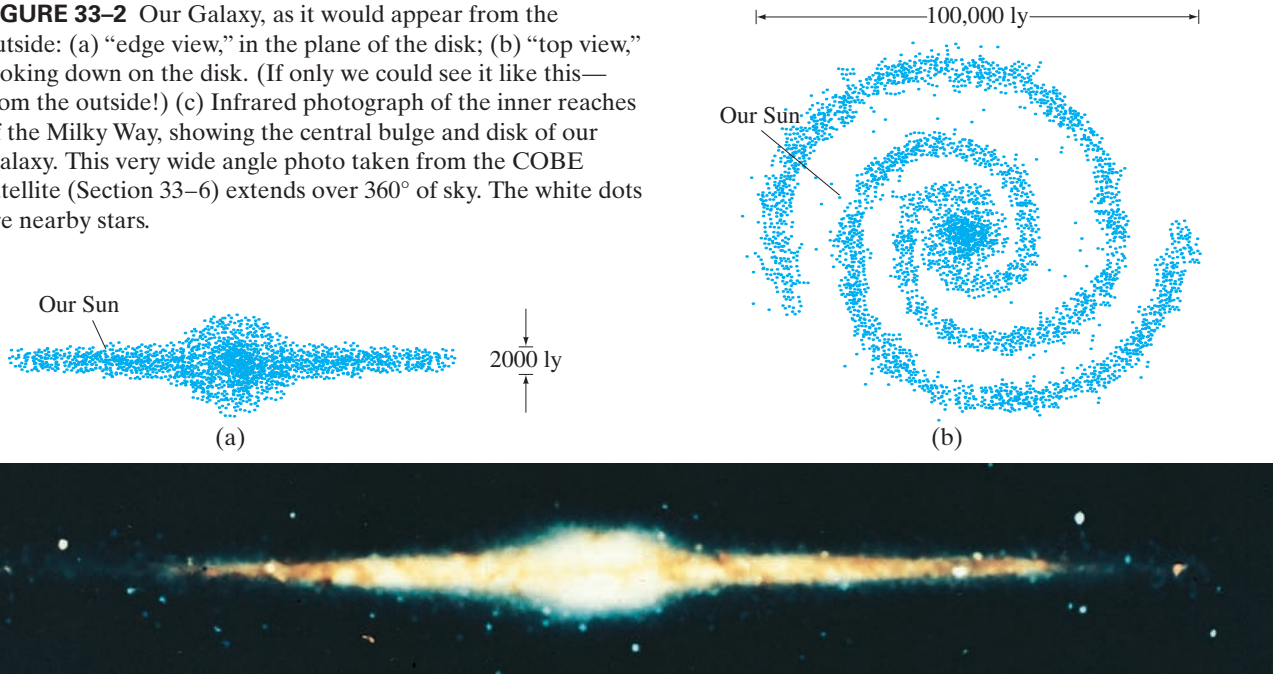
For specifying distances to the Sun and Moon, we usually use meters or kilometers, but we could specify them in terms of light seconds or minutes. The Earth–Moon distance is 384,000 km, which is 1.28 light-seconds. The Earth–Sun distance is 1.50×10^{11} m, or 150,000,000 km; this is equal to 8.3 light-minutes (it takes 8.3 min for light emitted by the Sun to reach us). Far out in our solar system, Pluto is about 6×10^9 km from the Sun, or 6×10^{-4} ly.[†] The nearest star to us, other than the Sun, is Proxima Centauri, about 4.2 ly away.

On a clear moonless night, thousands of stars of varying degrees of brightness can be seen, as well as the long cloudy stripe known as the Milky Way (Fig. 33–1). Galileo first observed, with his telescope, that the Milky Way is comprised of countless individual stars. A century and a half later (about 1750), Thomas Wright suggested that the Milky Way was a flat disk of stars extending to great distances in a plane, which we call the **Galaxy** (Greek for "milky way").

[†]We can also say this is about 5 light-hours.

Our Galaxy has a diameter of almost 100,000 light-years and a thickness of roughly 2000 ly. It has a central bulge and spiral arms (Fig. 33–2). Our Sun, which is a star like many others, is located about halfway from the galactic center to the edge, some 26,000 ly from the center. Our Galaxy contains roughly 400 billion (4×10^{11}) stars. The Sun orbits the galactic center approximately once every 250 million years, so its speed is roughly 200 km/s relative to the center of the Galaxy. The total mass of all the stars in our Galaxy is estimated to be about 4×10^{41} kg of ordinary matter. There is also strong evidence that our Galaxy is permeated and surrounded by a massive invisible "halo" of "dark matter" (Section 33–9).

FIGURE 33–2 Our Galaxy, as it would appear from the outside: (a) "edge view," in the plane of the disk; (b) "top view," looking down on the disk. (If only we could see it like this—from the outside!) (c) Infrared photograph of the inner reaches of the Milky Way, showing the central bulge and disk of our Galaxy. This very wide angle photo taken from the COBE satellite (Section 33–6) extends over 360° of sky. The white dots are nearby stars.



EXAMPLE 33–1 **ESTIMATE** **Our Galaxy's mass.** Estimate the total mass of our Galaxy using the orbital data above for the Sun about the center of the Galaxy. Assume the mass of the Galaxy is concentrated in the central bulge.

APPROACH We assume that the Sun (including our solar system) has total mass m and moves in a circular orbit about the center of the Galaxy (total mass M), and that the mass M can be considered as being located at the center of the Galaxy. We then apply Newton's second law, $F = ma$, with a being the centripetal acceleration, $a = v^2/r$, and for F we use the universal law of gravitation (Chapter 5).

SOLUTION Our Sun and solar system orbit the center of the Galaxy, according to the best measurements as mentioned above, with a speed of about $v = 200$ km/s at a distance from the Galaxy center of about $r = 26{,}000$ ly. We use Newton's second law:

$$F = ma$$

$$G\frac{Mm}{r^2} = m\frac{v^2}{r}$$

where M is the mass of the Galaxy and m is the mass of our Sun and solar system. Solving this, we find

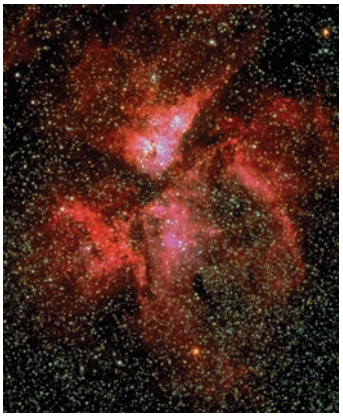
$$M = \frac{rv^2}{G} \approx \frac{(26{,}000\text{ ly})(10^{16}\text{ m/ly})(2 \times 10^5\text{ m/s})^2}{6.67 \times 10^{-11}\text{ N}\cdot\text{m}^2/\text{kg}^2} \approx 2 \times 10^{41}\text{ kg}.$$

NOTE In terms of *numbers* of stars, if they are like our Sun $(m = 2.0 \times 10^{30}\text{ kg})$, there would be about $(2 \times 10^{41}\text{ kg})/(2 \times 10^{30}\text{ kg}) \approx 10^{11}$ or very roughly on the order of 100 billion stars.



FIGURE 33–3 This globular star cluster is located in the constellation Hercules.

FIGURE 33–4 This gaseous nebula, found in the constellation Carina, is about 9000 light-years from us.



In addition to stars both within and outside the Milky Way, we can see by telescope many faint cloudy patches in the sky which were all referred to once as "nebulae" (Latin for "clouds"). A few of these, such as those in the constellations Andromeda and Orion, can actually be discerned with the naked eye on a clear night. Some are **star clusters** (Fig. 33–3), groups of stars that are so numerous they appear to be a cloud. Others are glowing clouds of gas or dust (Fig. 33–4), and it is for these that we now mainly reserve the word **nebula**.

Most fascinating are those that belong to a third category: they often have fairly regular elliptical shapes. Immanuel Kant (about 1755) guessed they are faint because they are a great distance beyond our Galaxy. At first it was not universally accepted that these objects were **extragalactic**—that is, outside our Galaxy. But the very large telescopes constructed in the twentieth century revealed that individual stars could be resolved within these extragalactic objects and that many contain spiral arms. Edwin Hubble (1889–1953) did much of this observational work in the 1920s using the 2.5-m (100-inch) telescope[†] on Mt. Wilson near Los Angeles, California, then the world's largest. Hubble demonstrated that these objects were indeed extragalactic because of their great distances. The distance to our nearest large galaxy,[‡] Andromeda, is over 2 million light-years, a distance 20 times greater than the diameter of our Galaxy. It seemed logical that these nebulae must be **galaxies** similar to ours. (Note that it is usual to capitalize the word "galaxy" only when it refers to our own.) Today it is thought there are roughly 10^{11} galaxies in the observable universe—that is, roughly as many galaxies as there are stars in a galaxy. See Fig. 33–5.

Many galaxies tend to be grouped in **galaxy clusters** held together by their mutual gravitational attraction. There may be anywhere from a few dozen to many thousands of galaxies in each cluster. Furthermore, clusters themselves seem to be organized into even larger aggregates: clusters of clusters of galaxies, or **superclusters**. The farthest detectable galaxies are more than 10^{10} ly distant. See Table 33–1 (top of next page).

[†]2.5 m (= 100 inches) refers to the diameter of the curved objective mirror. The bigger the mirror, the more light it collects (greater brightness) and the less diffraction there is (better resolution), so more and fainter stars can be seen. See Chapter 25. Until recently, photographic films or plates were used to take long time exposures. Now large solid-state CCD or CMOS sensors (Section 25–1) are available containing hundreds of millions of pixels (compared to 10 million pixels in a good-quality digital camera).

[‡]The *Magellanic clouds* are much closer than Andromeda, but are small and are usually considered small satellite galaxies of our own Galaxy.

FIGURE 33–5 Photographs of galaxies. (a) Spiral galaxy in the constellation Hydra. (b) Two galaxies: the larger and more dramatic one is known as the Whirlpool galaxy. (c) An infrared image (given "false" colors) of the same galaxies as in (b), here showing the arms of the spiral as having more substance than in the visible light photo (b); the different colors correspond to different light intensities. Visible light is scattered and absorbed by interstellar dust much more than infrared is, so infrared gives us a clearer image.



(a) (b) (c)

CONCEPTUAL EXAMPLE 33–2 **Looking back in time.** Astronomers often think of their telescopes as time machines, looking back toward the origin of the universe. How far back do they look?

RESPONSE The distance in light-years measures how long in years the light has been traveling to reach us, so Table 33–1 tells us also how far back in time we are looking. For example, if we saw Proxima Centauri explode into a supernova today, then the event would have really occurred about 4.2 years ago. The most distant galaxies emitted the light we see now roughly 13×10^9 years ago. What we see was how they were then, 13×10^9 yr ago.

EXERCISE A Suppose we could place a huge mirror 1 light-year away from us. What would we see in this mirror if it is facing us on Earth? When did what we see in the mirror take place? (This might be called a "time machine.")

Besides the usual stars, clusters of stars, galaxies, and clusters and superclusters of galaxies, the universe contains many other interesting objects. Among these are stars known as *red giants, white dwarfs*, *neutron stars*, exploding stars called *novae* and *supernovae*, and *black holes* whose gravity is so strong that even light cannot escape them. In addition, there is electromagnetic radiation that reaches the Earth but does not come from the bright pointlike objects we call stars: particularly important is the microwave background radiation that arrives nearly uniformly from all directions in the universe.

Finally, there are **active galactic nuclei** (**AGN**), which are very luminous pointlike sources of light in the centers of distant galaxies. The most dramatic examples of AGN are **quasars** ("quasistellar objects" or QSOs), which are so luminous that the surrounding starlight of the galaxy is drowned out. Their luminosity is thought to come from matter falling into a giant black hole at a galaxy's center.

Table 33–1 Astronomical Distances

Object	Approx. Distance from Earth (ly)
Moon	4×10^{-8}
Sun	1.6×10^{-5}
Size of solar system (distance to Pluto)	6×10^{-4}
Nearest star (Proxima Centauri)	4.2
Center of our Galaxy	2.6×10^4
Nearest large galaxy	2.4×10^6
Farthest galaxies	13.4×10^9

33–2 Stellar Evolution: Birth and Death of Stars, Nucleosynthesis

The stars appear unchanging. Night after night the night sky reveals no significant variations. Indeed, on a human time scale, the vast majority of stars change very little (except for novae, supernovae, and certain variable stars). Although stars *seem* fixed in relation to each other, many move sufficiently for the motion to be detected. Speeds of stars relative to neighboring stars can be hundreds of km/s, but at their great distance from us, this motion is detectable only by careful measurement. There is also a great range of brightness among stars, due to differences in the rate stars emit energy and to their different distances from us.

Luminosity and Brightness of Stars

Any star or galaxy has an **intrinsic luminosity**, L (or simply **luminosity**), which is its total power radiated in watts. Also important is the **apparent brightness**, b, defined as the power crossing unit area at the Earth perpendicular to the path of the light. Given that energy is conserved, and ignoring any absorption in space, the total emitted power L when it reaches a distance d from the star will be spread over a sphere of surface area $4\pi d^2$. If d is the distance from the star to the Earth, then L must be equal to $4\pi d^2$ times b (power per unit area at Earth). That is,

$$b = \frac{L}{4\pi d^2}. \qquad \textbf{(33–1)}$$

EXAMPLE 33–3 **Apparent brightness.** Suppose a star has luminosity equal to that of our Sun. If it is 10 ly away from Earth, how much dimmer will it appear?

APPROACH We use the inverse square law in Eq. 33–1 to determine the relative brightness $(b \propto 1/d^2)$ since the luminosity L is the same for both stars.

SOLUTION Using the inverse square law, the star appears dimmer by a factor

$$\frac{b_{\text{star}}}{b_{\text{Sun}}} = \frac{d_{\text{Sun}}^2}{d_{\text{star}}^2} = \frac{(1.5 \times 10^8\,\text{km})^2}{(10\,\text{ly})^2(10^{13}\,\text{km/ly})^2} \approx 2 \times 10^{-12}.$$

Careful study of nearby stars has shown that the luminosity for most stars depends on the mass: *the more massive the star, the greater its luminosity*[†]. Another important parameter of a star is its surface temperature, which can be determined from the spectrum of electromagnetic frequencies it emits. As we saw in Chapter 27, as the temperature of a body increases, the spectrum shifts from predominantly lower frequencies (and longer wavelengths, such as red) to higher frequencies (and shorter wavelengths such as blue). Quantitatively, the relation is given by Wien's law (Eq. 27–2): the wavelength λ_P at the peak of the spectrum of light emitted by a blackbody (we often approximate stars as blackbodies) is inversely proportional to its Kelvin temperature T; that is, $\lambda_P T = 2.90 \times 10^{-3}\,\text{m}\cdot\text{K}$. The surface temperatures of stars typically range from about 3000 K (reddish) to about 50,000 K (UV).

EXAMPLE 33–4 **Determining star temperature and star size.** Suppose that the distances from Earth to two nearby stars can be reasonably estimated, and that their measured apparent brightnesses suggest the two stars have about the same luminosity, L. The spectrum of one of the stars peaks at about 700 nm (so it is reddish). The spectrum of the other peaks at about 350 nm (bluish). Use Wien's law (Eq. 27–2) and the Stefan-Boltzmann equation (Section 14–8) to determine (*a*) the surface temperature of each star, and (*b*) how much larger one star is than the other.

APPROACH We determine the surface temperature T for each star using Wien's law and each star's peak wavelength. Then, using the Stefan-Boltzmann equation (power output or luminosity $\propto AT^4$ where A = surface area of emitter), we can find the surface area ratio and relative sizes of the two stars.

SOLUTION (*a*) Wien's law (Eq. 27–2) states that $\lambda_P T = 2.90 \times 10^{-3}\,\text{m}\cdot\text{K}$. So the temperature of the reddish star is

$$T_r = \frac{2.90 \times 10^{-3}\,\text{m}\cdot\text{K}}{\lambda_P} = \frac{2.90 \times 10^{-3}\,\text{m}\cdot\text{K}}{700 \times 10^{-9}\,\text{m}} = 4140\,\text{K}.$$

The temperature of the bluish star will be double this because its peak wavelength is half (350 nm vs. 700 nm):

$$T_b = 8280\,\text{K}.$$

(*b*) The Stefan-Boltzmann equation, Eq. 14–6, states that the power radiated *per unit area* of surface from a blackbody is proportional to the fourth power of the Kelvin temperature, T^4. The temperature of the bluish star is double that of the reddish star, so the bluish one must radiate $(T_b/T_r)^4 = 2^4 = 16$ times as much energy per unit area. But we are given that they have the same luminosity (the same total power output); so the surface area of the blue star must be $\frac{1}{16}$ that of the red one. The surface area of a sphere is $4\pi r^2$, so the radius of the reddish star is $\sqrt{16} = 4$ times larger than the radius of the bluish star (or $4^3 = 64$ times the volume).

H–R Diagram

An important astronomical discovery, made around 1900, was that for most stars, the color is related to the intrinsic luminosity and therefore to the mass. A useful way to present this relationship is by the so-called Hertzsprung–Russell (H–R) diagram. On the H–R diagram, the horizontal axis shows the surface temperature T and the vertical axis is the luminosity L; each star is represented by a point

[†]Applies to "main-sequence" stars (see next page). The mass of a star can be determined by observing its gravitational effects on other visible objects. Many stars are part of a cluster, the simplest being a binary star in which two stars orbit around each other, allowing their masses to be determined using rotational mechanics.

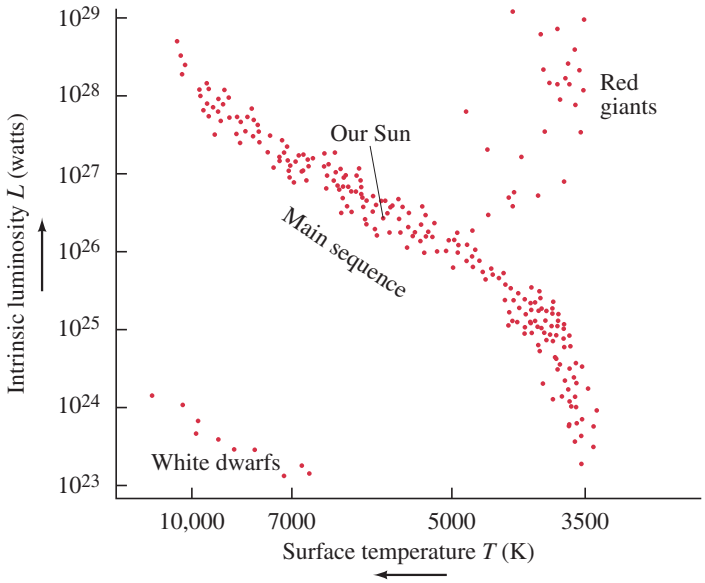


FIGURE 33–6 Hertzsprung–Russell (H–R) diagram is a logarithmic graph of luminosity vs. surface temperature T of stars (note that T increases to the left).

on the diagram, Fig. 33–6. Most stars fall along the diagonal band termed the **main sequence**. Starting at the lower right we find the coolest stars: by Wien's law, $\lambda_P T$ = constant, their light output peaks at long wavelengths, so they are reddish in color. They are also the least luminous and therefore of low mass. Farther up toward the left we find hotter and more luminous stars that are whitish, like our Sun. Still farther up we find even more luminous and more massive stars, bluish in color. Stars that fall on this diagonal band are called *main-sequence stars*. There are also stars that fall outside the main sequence. Above and to the right we find extremely large stars, with high luminosities but with low (reddish) color temperature: these are called **red giants**. At the lower left, there are a few stars of low luminosity but with high temperature: these are the **white dwarfs**.

EXAMPLE 33–5 | ESTIMATE Distance to a star using the H–R diagram and color. Suppose that detailed study of a certain star suggests that it most likely fits on the main sequence of an H–R diagram. Its measured apparent brightness is $b = 1.0 \times 10^{-12}\,\text{W/m}^2$, and the peak wavelength of its spectrum is $\lambda_P \approx 600\,\text{nm}$. Estimate its distance from us.

APPROACH We find the temperature using Wien's law, Eq. 27–2. The luminosity is estimated for a main-sequence star on the H–R diagram of Fig. 33–6, and then the distance is found using the relation between brightness and luminosity, Eq. 33–1.

SOLUTION The star's temperature, from Wien's law (Eq. 27–2), is

$$T \approx \frac{2.90 \times 10^{-3}\,\text{m}\cdot\text{K}}{600 \times 10^{-9}\,\text{m}} \approx 4800\,\text{K}.$$

A star on the main sequence of an H–R diagram at this temperature has luminosity of about $L \approx 1 \times 10^{26}\,\text{W}$, read off of Fig. 33–6. Then, from Eq. 33–1,

$$d = \sqrt{\frac{L}{4\pi b}} \approx \sqrt{\frac{1 \times 10^{26}\,\text{W}}{4(3.14)(1.0 \times 10^{-12}\,\text{W/m}^2)}} \approx 3 \times 10^{18}\,\text{m}.$$

Its distance from us in light-years is

$$d = \frac{3 \times 10^{18}\,\text{m}}{10^{16}\,\text{m/ly}} \approx 300\,\text{ly}.$$

EXERCISE B Estimate the distance to a 6000-K main-sequence star with an apparent brightness of $2.0 \times 10^{-12}\,\text{W/m}^2$.

Stellar Evolution; Nucleosynthesis

Why are there different types of stars, such as red giants and white dwarfs, as well as main-sequence stars? Were they all born this way, in the beginning? Or might each different type represent a different age in the life cycle of a star? Astronomers and astrophysicists today believe the latter is the case. Note, however, that we cannot actually follow any but the tiniest part of the life cycle of any given star because they live for ages vastly greater than ours, on the order of millions or billions of years. Nonetheless, let us follow the process of **stellar evolution** from the birth to the death of a star, as astrophysicists have theoretically reconstructed it today.

Stars are born, it is believed, when gaseous clouds (mostly hydrogen) contract due to the pull of gravity. A huge gas cloud might fragment into numerous contracting masses, each mass centered in an area where the density is only slightly greater than that at nearby points. Once such "globules" form, gravity causes each to contract in toward its center of mass. As the particles of such a *protostar* accelerate inward, their kinetic energy increases. Eventually, when the kinetic energy is sufficiently high, the Coulomb repulsion between the positive charges is not strong enough to keep all the hydrogen nuclei apart, and nuclear fusion can take place.

In a star like our Sun, the fusion of hydrogen (sometimes referred to as "burning")[†] occurs via the *proton–proton chain* (Section 31–3, Eqs. 31–6), in which four protons fuse to form a ${}^4_2\text{He}$ nucleus with the release of γ rays, positrons, and neutrinos: $4\,{}^1_1\text{H} \rightarrow {}^4_2\text{He} + 2\,e^+ + 2\nu_e + 2\gamma$. These reactions require a temperature of about 10^7 K, corresponding to an average kinetic energy ($\approx kT$) of about 1 keV (Eq. 13–8). In more massive stars, the carbon cycle produces the same net effect: four ${}^1_1\text{H}$ produce a ${}^4_2\text{He}$—see Section 31–3. The fusion reactions take place primarily in the core of a star, where T may be on the order of 10^7 to 10^8 K. (The surface temperature is much lower—on the order of a few thousand kelvins.) The tremendous release of energy in these fusion reactions produces an outward pressure sufficient to halt the inward gravitational contraction. Our protostar, now really a young *star*, stabilizes on the *main sequence*. Exactly where the star falls along the main sequence depends on its mass. The more massive the star, the farther up (and to the left) it falls on the H–R diagram of Fig. 33–6. Our Sun required perhaps 30 million years to reach the main sequence, and is expected to remain there about 10 billion years (10^{10} yr). Although most stars are billions of years old, evidence is strong that stars are actually being born at this moment. More massive stars have shorter lives, because they are hotter and the Coulomb repulsion is more easily overcome, so they use up their fuel faster. Our Sun may remain on the main sequence for 10^{10} years, but a star ten times more massive may reside there for only 10^7 years.

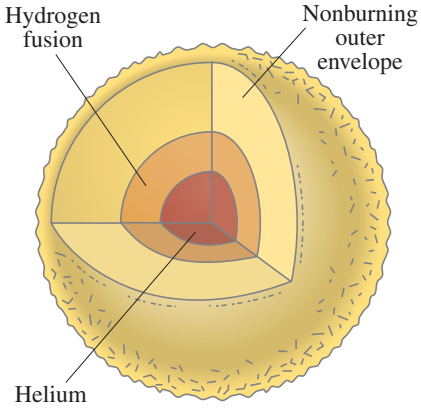


FIGURE 33–7 A shell of "burning" hydrogen (fusing to become helium) surrounds the core where the newly formed helium gravitates.

As hydrogen fuses to form helium, the helium that is formed is denser and tends to accumulate in the central core where it was formed. As the core of helium grows, hydrogen continues to fuse in a shell around it: see Fig. 33–7. When much of the hydrogen within the core has been consumed, the production of energy decreases at the center and is no longer sufficient to prevent the huge gravitational forces from once again causing the core to contract and heat up. The hydrogen in the shell around the core then fuses even more fiercely because of this rise in temperature, allowing the outer envelope of the star to expand and to cool. The surface temperature, thus reduced, produces a spectrum of light that peaks at longer wavelength (reddish).

This process marks a new step in the evolution of a star. The star has become redder, it has grown in size, and it has become more luminous, which means it has left the main sequence. It will have moved to the right and upward on the

[†]The word "burn," meaning fusion, is put in quotation marks because these high-temperature fusion reactions occur via a *nuclear* process, and must not be confused with ordinary burning (of, say, paper, wood, or coal) in air, which is a *chemical* reaction, occurring at the *atomic* level (and at a much lower temperature).

H–R diagram, as shown in Fig. 33–8. As it moves upward, it enters the **red giant** stage. Thus, theory explains the origin of red giants as a natural step in a star's evolution. Our Sun, for example, has been on the main sequence for about $4\frac{1}{2}$ billion years. It will probably remain there another 5 or 6 billion years. When our Sun leaves the main sequence, it is expected to grow in diameter (as it becomes a red giant) by a factor of 100 or more, possibly swallowing up inner planets such as Mercury and possibly Venus and even Earth.

If the star is like our Sun, or larger, further fusion can occur. As the star's outer envelope expands, its core continues to shrink and heat up. When the temperature reaches about 10^8 K, even helium nuclei, in spite of their greater charge and hence greater electrical repulsion, can come close enough to each other to undergo fusion. The reactions are

$$\begin{aligned} {}^4_2\text{He} + {}^4_2\text{He} &\rightarrow {}^8_4\text{Be} \\ {}^4_2\text{He} + {}^8_4\text{Be} &\rightarrow {}^{12}_6\text{C} \end{aligned} \qquad \textbf{(33–2)}$$

with the emission of two γ rays. These two reactions must occur in quick succession (because ${}^8_4$Be is very unstable), and the net effect is

$$3\,{}^4_2\text{He} \rightarrow {}^{12}_6\text{C} + 2\gamma. \qquad (Q = 7.3\,\text{MeV})$$

This fusion of helium causes a change in the star which moves rapidly to the "horizontal branch" on the H–R diagram (Fig. 33–8). Further fusion reactions are possible, with ${}^4_2$He fusing with ${}^{12}_6$C to form ${}^{16}_8$O. In more massive stars, higher Z elements like ${}^{20}_{10}$Ne or ${}^{24}_{12}$Mg can be made. This process of creating heavier nuclei from lighter ones (or by absorption of neutrons which tends to occur at higher Z) is called **nucleosynthesis**.

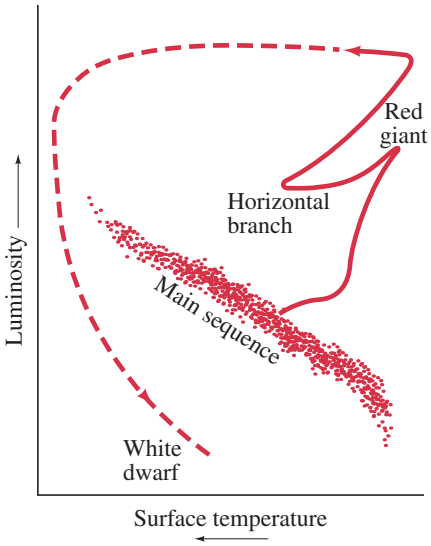


FIGURE 33–8 Evolutionary "track" of a star like our Sun represented on an H–R diagram.

Low Mass Stars—White Dwarfs

The final fate of a star depends on its mass. Stars can lose mass as parts of their outer envelope move off into space. Stars born with a mass less than about 8 solar masses (8× the mass of our Sun) eventually end up with a residual mass less than about 1.4 solar masses. A residual mass of 1.4 solar masses is known as the **Chandrasekhar limit**. For stars smaller than this, no further fusion energy can be obtained because of the large Coulomb repulsion between nuclei. The core of such a "low mass" star (original mass $\lesssim$ 8 solar masses) contracts under gravity. The outer envelope expands again and the star becomes an even brighter and larger red giant, Fig. 33–8. Eventually the outer layers escape into space, and the newly revealed surface is hotter than before. So the star moves to the left in the H–R diagram (horizontal dashed line in Fig. 33–8). Then, as the core shrinks the star cools, and typically follows the downward dashed route shown on the left in Fig. 33–8, becoming a **white dwarf**. A white dwarf with a residual mass equal to that of the Sun would be about the size of the Earth. A white dwarf contracts to the point at which the electrons start to overlap, but no further because, by the Pauli exclusion principle, no two electrons can be in the same quantum state. At this point the star is supported against further collapse by this **electron degeneracy** pressure. A white dwarf continues to lose internal energy by radiation, decreasing in temperature and becoming dimmer until it glows no more. It has then become a cold dark chunk of extremely dense material.

High Mass Stars—Supernovae, Neutron Stars, Black Holes

Stars whose original mass is greater than about 8 solar masses are thought to follow a very different scenario. A star with this great a mass can contract under gravity and heat up even further. At temperatures $T \approx 3$ or 4×10^9 K, nuclei as heavy as ${}^{56}_{26}$Fe and ${}^{56}_{28}$Ni can be made. But here the formation of heavy nuclei from lighter ones, by fusion, ends. As we saw in Fig. 30–1, the average binding energy per nucleon begins to decrease for A greater than about 60. Further fusions would *require* energy, rather than release it.



(a)



(b)

FIGURE 33–9 The star indicated by the arrow in (a) exploded in 1987 as a supernova (SN1987A), as shown in (b). The bright spot in (b) indicates a huge release of energy but does not represent the physical size.

At these extremely high temperatures, well above 10^9 K, high-energy collisions can cause the breaking apart of iron and nickel nuclei into He nuclei, and eventually into protons and neutrons:

$$^{56}_{26}\text{Fe} \rightarrow 13\,^{4}_{2}\text{He} + 4\text{n}$$
$$^{4}_{2}\text{He} \rightarrow 2\text{p} + 2\text{n}.$$

These are energy-requiring (endothermic) reactions, which rob energy from the core, allowing gravitational contraction to begin. This then can force electrons and protons together to form neutrons in **inverse β decay**:

$$\text{e}^- + \text{p} \rightarrow \text{n} + \nu.$$

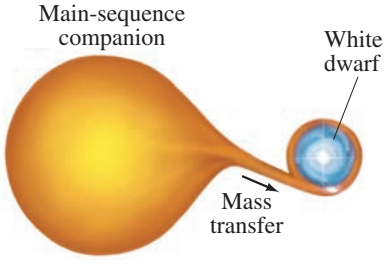
As a result of these reactions, the pressure in the core drops precipitously. As the core collapses under the huge gravitational forces, the tremendous mass becomes essentially an enormous nucleus made up almost exclusively of neutrons. The size of the star is no longer limited by the exclusion principle applied to electrons, but rather by **neutron degeneracy** pressure, and the star contracts rapidly to form an enormously dense **neutron star**. The core of a neutron star contracts to the point at which all neutrons are as close together as they are in an atomic nucleus. That is, the density of a neutron star is on the order of 10^{14} times greater than normal solids and liquids on Earth. A cupful of such dense matter would weigh billions of tons. A neutron star that has a mass 1.5 times that of our Sun would have a diameter of only about 20 km. (Compare this to a white dwarf with 1 solar mass whose diameter would be $\approx 10^4$ km, as mentioned on the previous page.)

The contraction of the core of a massive star would mean a great reduction in gravitational potential energy. Somehow this energy would have to be released. Indeed, it was suggested in the 1930s that the final core collapse to a neutron star could be accompanied by a catastrophic explosion known as a **supernova** (plural = supernovae). The tremendous energy release (Fig. 33–9) could form virtually all elements of the Periodic Table (see below) and blow away the entire outer envelope of the star, spreading its contents into interstellar space. The presence of heavy elements on Earth and in our solar system suggests that our solar system formed from the debris of many such supernova explosions.

The elements heavier than Ni are thought to form mainly by **neutron capture** in these exploding supernovae (rather than by fusion, as for elements up to Ni). Large numbers of free neutrons, resulting from nuclear reactions, are present inside those highly evolved stars and they can readily combine with, say, a $^{56}_{26}$Fe nucleus to form (if three are captured) $^{59}_{26}$Fe, which decays to $^{59}_{27}$Co. The $^{59}_{27}$Co can capture neutrons, also becoming neutron rich and decaying by β^- to the next higher Z element, and so on to the highest Z elements.

The final state of a neutron star depends on its mass. If the final mass is less than about three solar masses, the subsequent evolution of the neutron star is thought to resemble that of a white dwarf. If the mass is greater than this (original mass $\gtrsim$40 solar masses), the neutron star collapses under gravity, overcoming even neutron degeneracy. Gravity would then be so strong that emitted light could not escape—it would be pulled back in by the force of gravity. Since no radiation could escape from such a "star," we could not see it—it would be black. An object may pass by it and be deflected by its gravitational field, but if the object came too close it would be swallowed up, never to escape. This is a **black hole**.

FIGURE 33–10 Hypothetical model for novae and Type Ia supernovae, showing how a white dwarf could pull mass from its normal companion.



Novae and Supernovae

Novae (singular is *nova*, meaning "new" in Latin) are faint stars that have suddenly increased in brightness by as much as a factor of 10^6 and last for a month or two before fading. Novae are thought to be faint white dwarfs that have pulled mass from a nearby companion (they make up a *binary* system), as illustrated in Fig. 33–10. The captured mass of hydrogen suddenly fuses into helium at a high rate for a few weeks. Many novae (maybe all) are *recurrent*—they repeat their bright glow years later.

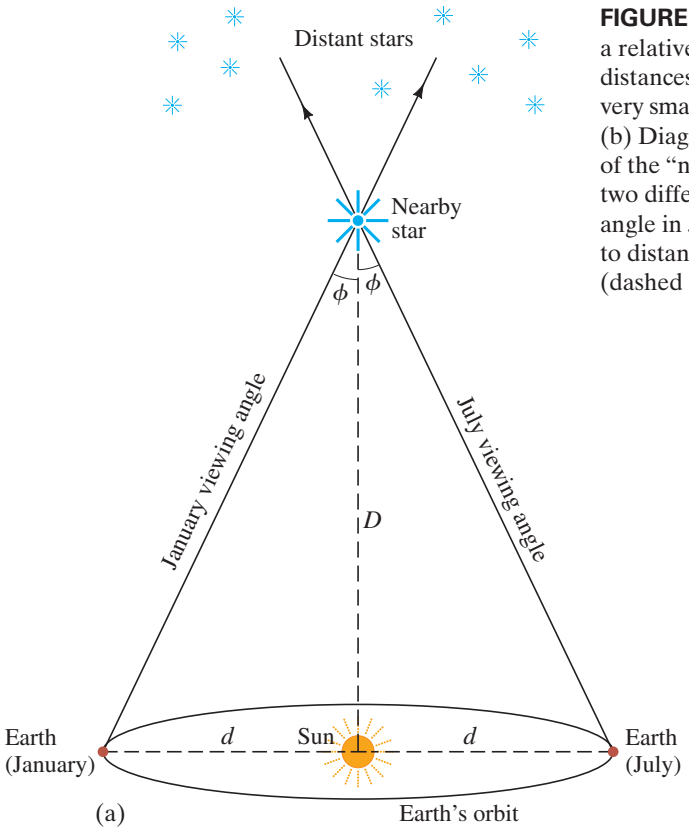
Supernovae are also brief explosive events, but release millions of times more energy than novae, up to 10^{10} times more luminous than our Sun. The peak of brightness may exceed that of the entire galaxy in which they are located, but lasts only a few days or weeks. They slowly fade over a few months. Many supernovae form by core collapse to a neutron star as described above. See Fig. 33–9.

Type Ia supernovae are different. They all seem to have very nearly the same luminosity. They are believed to be binary stars, one of which is a white dwarf that pulls mass from its companion, much like for a nova, Fig. 33–10. The mass is higher, and as mass is captured and the total mass approaches the Chandrasekhar limit of 1.4 solar masses, it explodes as a "white-dwarf" supernova by undergoing a "thermonuclear runaway"—an uncontrolled chain of nuclear reactions that entirely destroys the white dwarf. Type Ia supernovae are useful to us as "standard candles" in the night sky to help us determine distance—see next Section.

33–3 Distance Measurements

Parallax

We have talked about the vast distances of objects in the universe. But how do we measure these distances? One basic technique employs simple geometry to measure the **parallax** of a star. By parallax we mean the apparent motion of a star, against the background of much more distant stars, due to the Earth's motion around the Sun. As shown in Fig. 33–11, we can measure the angle 2ϕ that the star appears to shift, relative to very distant stars, when viewed 6 months apart. If we know the distance d from Earth to Sun, we can reconstruct the right triangles shown in Fig. 33–11 and can then determine the distance D to the star. This is essentially the way the heights of mountains are determined, by "triangulation": see Example 1–8.



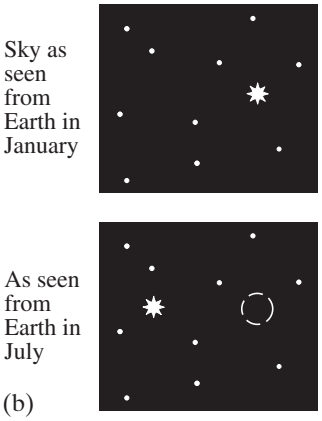


FIGURE 33–11 (a) Determining the distance D to a relatively nearby star using parallax. Horizontal distances are greatly exaggerated: in reality ϕ is a very small angle (less than $\frac{1}{3600}° = 1'' = 1$ second of arc). (b) Diagram of the sky showing the apparent position of the "nearby" star relative to more distant stars, at two different times (January and July). The viewing angle in January puts the star more to the right relative to distant stars, whereas in July it is more to the left (dashed circle shows January location).

EXAMPLE 33–6 **ESTIMATE** **Distance to a star using parallax.** Estimate the distance D to a star if the angle 2ϕ in Fig. 33–11a is measured to be $2\phi = 0.00012°$.

APPROACH From trigonometry, $\tan\phi = d/D$ in Fig. 33–11a. The Sun–Earth distance is $d = 1.5 \times 10^8$ km (inside front cover).

SOLUTION The angle $\phi = 0.00006°$, or about $(0.00006°)(2\pi \text{ rad}/360°) = 1.0 \times 10^{-6}$ radians. We can use $\tan\phi \approx \phi$ because ϕ is very small. We solve for D in $\tan\phi = d/D$. The distance D to the star is

$$D = \frac{d}{\tan\phi} \approx \frac{d}{\phi} = \frac{1.5 \times 10^8 \text{ km}}{1.0 \times 10^{-6} \text{ rad}} = 1.5 \times 10^{14} \text{ km},$$

or about 15 ly.

* Parsec

Distances to stars are often specified in terms of parallax angle (ϕ in Fig. 33–11a) given in seconds of arc: 1 second (1″) is $\frac{1}{60}$ of one minute (1′) of arc, which is $\frac{1}{60}$ of a degree, so $1'' = \frac{1}{3600}$ of a degree. The distance is then specified in **parsecs** (pc) (meaning *par*allax angle in *sec*onds of arc): $D = 1/\phi$ with ϕ in seconds of arc. In Example 33–6, $\phi = (6 \times 10^{-5})°(3600) = 0.22''$ of arc, so we would say the star is at a distance of $1/0.22'' = 4.5$ pc. One parsec is given by (recall $D = d/\phi$, and we set the Sun–Earth distance (Fig. 33–11a) as $d = 1.496 \times 10^{11}$ m):

$$1 \text{ pc} = \frac{d}{1''} = \frac{1.496 \times 10^{11} \text{ m}}{(1'')\left(\frac{1'}{60''}\right)\left(\frac{1°}{60'}\right)\left(\frac{2\pi \text{ rad}}{360°}\right)} = 3.086 \times 10^{16} \text{ m}$$

$$1 \text{ pc} = (3.086 \times 10^{16} \text{ m})\left(\frac{1 \text{ ly}}{9.46 \times 10^{15} \text{ m}}\right) = 3.26 \text{ ly}.$$

Distant Stars and Galaxies

Parallax can be used to determine the distance to stars as far away as about 100 light-years from Earth, and from an orbiting spacecraft perhaps 5 to 10 times farther. Beyond that distance, parallax angles are too small to measure. For greater distances, more subtle techniques must be employed. We might compare the apparent brightnesses of two stars, or two galaxies, and use the *inverse square law* (apparent brightness drops off as the square of the distance) to roughly estimate their relative distances. We can't expect this technique to be very precise because we don't expect any two stars, or two galaxies, to have the same intrinsic luminosity. When comparing galaxies, a perhaps better estimate assumes the brightest stars in all galaxies (or the brightest galaxies in galaxy clusters) are similar and have about the same intrinsic luminosity. Consequently, their *apparent brightness* would be a measure of how far away they were.

Another technique makes use of the H–R diagram. Measurement of a star's surface temperature (from its spectrum) places it at a certain point (within 20%) on the H–R diagram, assuming it is a main-sequence star, and then its luminosity can be estimated from the vertical axis (Fig. 33–6). Its apparent brightness and Eq. 33–1 give its approximate distance; see Example 33–5.

A better estimate comes from comparing *variable stars*, especially *Cepheid variables* whose luminosity varies over time with a period that is found to be related to their average luminosity. Thus, from their period and apparent brightness we get their distance.

Distance via SNIa, Redshift

The largest distances are estimated by comparing the apparent brightnesses of Type Ia supernovae ("SNIa"). Type Ia supernovae all have a similar origin (as described on the previous page and Fig. 33–10), and their brief explosive burst of light is expected to be of nearly the same luminosity. They are thus sometimes referred to as "standard candles."

Another important technique for estimating the distance of very distant galaxies is from the "redshift" in the line spectra of elements and compounds. The redshift is related to the expansion of the universe, as we shall discuss in Section 33–5. It is useful for objects farther than 10^7 to 10^8 ly away.

As we look farther and farther away, measurement techniques are less and less reliable, so there is more uncertainty in the measurements of large distances.

33–4 General Relativity: Gravity and the Curvature of Space

We have seen that the force of gravity plays an important role in the processes that occur in stars. Gravity too is important for the evolution of the universe as a whole. The reasons gravity plays a dominant role in the universe, and not one of the other of the four forces in nature, are (1) it is long-range and (2) it is always attractive. The strong and weak nuclear forces act over very short distances only, on the order of the size of a nucleus; hence they do not act over astronomical distances (they do act between nuclei and nucleons in stars to produce nuclear reactions). The electromagnetic force, like gravity, acts over great distances. But it can be either attractive or repulsive. And since the universe does not seem to contain large areas of net electric charge, a large net force does not occur. But gravity acts only as an *attractive* force between *all* masses, and there are large accumulations of mass in the universe. The force of gravity as Newton described it in his law of universal gravitation was modified by Einstein. In his general theory of relativity, Einstein developed a theory of gravity that now forms the basis of cosmological dynamics.

In the *special theory of relativity* (Chapter 26), Einstein concluded that there is no way for an observer to determine whether a given frame of reference is at rest or is moving at constant velocity in a straight line. Thus the laws of physics must be the same in different inertial reference frames. But what about the more general case of motion where reference frames can be *accelerating*?

Einstein tackled the problem of accelerating reference frames in his **general theory of relativity** and in it also developed a theory of gravity. The mathematics of General Relativity is complex, so our discussion will be mainly qualitative.

We begin with Einstein's **principle of equivalence**, which states that

> **no experiment can be performed that could distinguish between a uniform gravitational field and an equivalent uniform acceleration.**

If observers sensed that they were accelerating (as in a vehicle speeding around a sharp curve), they could not prove by any experiment that in fact they weren't simply experiencing the pull of a gravitational field. Conversely, we might think we are being pulled by gravity when in fact we are undergoing an acceleration having nothing to do with gravity.

As a thought experiment, consider a person in a freely falling elevator near the Earth's surface. If our observer held out a book and let go of it, what would happen? Gravity would pull it downward toward the Earth, but at the same rate $(g = 9.8\,\text{m/s}^2)$ at which the person and elevator were falling. So the book would hover right next to the person's hand (Fig. 33–12). The effect is exactly the same as if this reference frame was at rest and *no* forces were acting. On the other hand, if the elevator was out in space where the gravitational field is essentially zero, the released book would float, just as it does in Fig. 33–12. Next, if the elevator (out in space) is accelerated upward (using rockets) at an acceleration of $9.8\,\text{m/s}^2$, the book as seen by our observer would fall to the floor with an acceleration of $9.8\,\text{m/s}^2$, just as if it were falling due to gravity at the surface of the Earth. According to the principle of equivalence, the observer could not determine whether the book fell because the elevator was accelerating upward, or because a gravitational field was acting downward and the elevator was at rest. The two descriptions are equivalent.

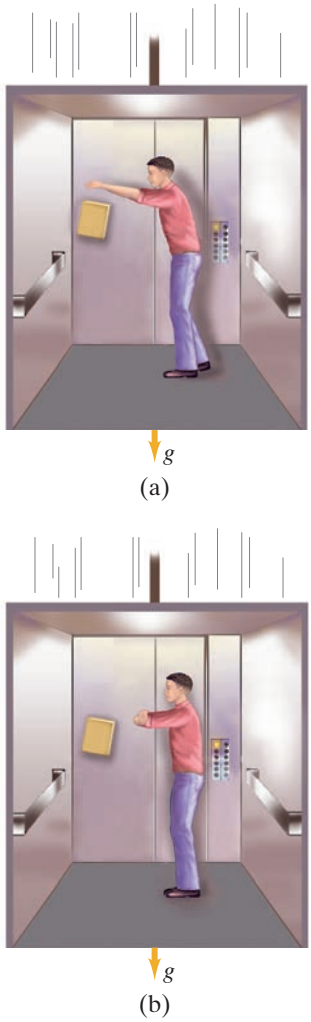
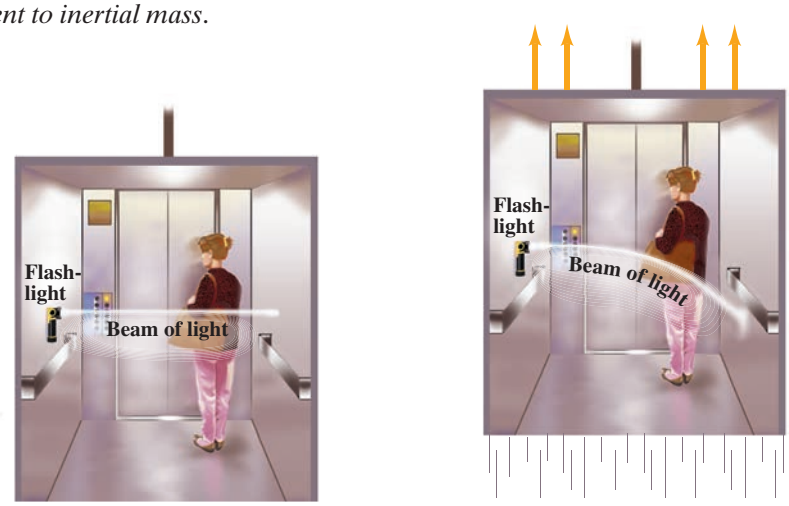


FIGURE 33–12 In an elevator falling freely under gravity, (a) a person releases a book; (b) the released book hovers next to the owner's hand; (b) is a few moments after (a).

The principle of equivalence is related to the concept that there are two types of mass. Newton's second law, $F = ma$, uses **inertial mass**. We might say that inertial mass represents "resistance" to any type of force. The second type of mass is **gravitational mass**. When one object attracts another by the gravitational force (Newton's law of universal gravitation, $F = Gm_1m_2/r^2$, Chapter 5), the strength of the force is proportional to the product of the *gravitational masses* of the two objects. This is much like Coulomb's law for the electric force between two objects which is proportional to the product of their electric charges. The electric charge on an object is not related to its inertial mass; so why should we expect that an object's gravitational mass (call it gravitational charge if you like) be related to its inertial mass? All along we have assumed they were the same. Why? Because no experiment—not even of high precision—has been able to discern any measurable difference between inertial mass and gravitational mass. (For example, in the absence of air resistance, all objects fall at the same acceleration, g, on Earth.) This is another way to state the equivalence principle: *gravitational mass is equivalent to inertial mass*.



(a) (b)

FIGURE 33–13 (a) Light beam goes straight across an elevator which is not accelerating. (b) The light beam bends (exaggerated) according to an observer in an accelerating elevator whose speed increases in the upward direction.

The principle of equivalence can be used to show that light ought to be deflected by the gravitational force due to a massive object. Consider another thought experiment, in which an elevator is in free space where virtually no gravity acts. If a light beam is emitted by a flashlight attached to the side of the elevator, the beam travels straight across the elevator and makes a spot on the opposite side if the elevator is at rest or moving at constant velocity (Fig. 33–13a). If instead the elevator is accelerating upward, as in Fig. 33–13b, the light beam still travels straight across in a reference frame at rest. In the upwardly accelerating elevator, however, the beam is observed to curve downward. Why? Because during the time the light travels from one side of the elevator to the other, the elevator is moving upward at a vertical speed that is increasing relative to the light. Next we note that according to the equivalence principle, an upwardly accelerating reference frame is equivalent to a downward gravitational field. Hence, we can picture the curved light path in Fig. 33–13b as being due to the effect of a gravitational field. Thus, from the principle of equivalence, we expect gravity to exert a force on a beam of light and to bend it out of a straight-line path!

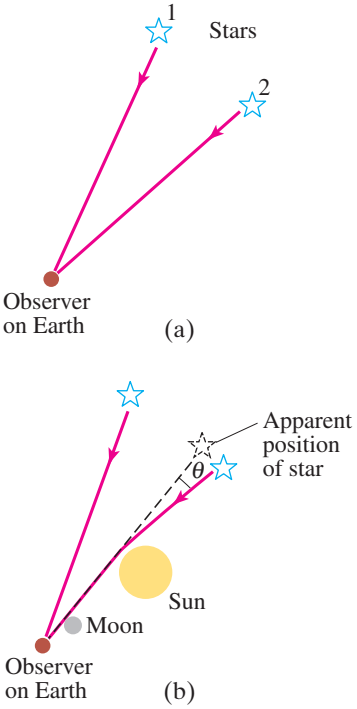


FIGURE 33–14 (a) Two stars in the sky observed from Earth. (b) If the light from one of these stars passes very near the Sun, whose gravity bends the rays, the star will appear higher than it actually is (follow the ray backwards). [Not to scale.]

That light is affected by gravity is an important prediction of Einstein's general theory of relativity. And it can be tested. The amount a light beam would be deflected from a straight-line path must be small even when passing a massive object. (For example, light near the Earth's surface after traveling 1 km is predicted to drop only about 10^{-10} m, which is equal to the diameter of a small atom and not detectable.) The most massive object near us is the Sun, and it was calculated that light from a distant star would be deflected by 1.75″ of arc (tiny but detectable) as it passed by the edge of the Sun (Fig. 33–14). However, such a measurement could be made only during a total eclipse of the Sun, so that the Sun's tremendous brightness would not obscure the starlight passing near its edge.

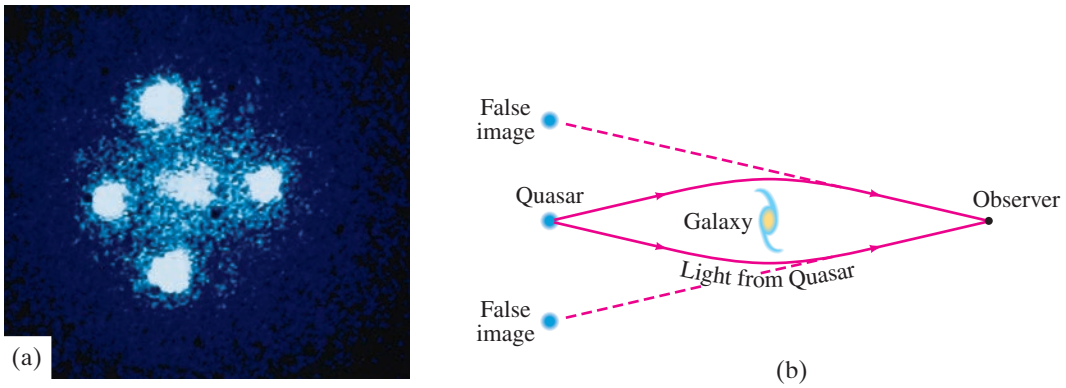


FIGURE 33–15 (a) Hubble Space Telescope photograph of the so-called "Einstein cross," thought to represent "gravitational lensing": the central spot is a relatively nearby galaxy, whereas the four other spots are thought to be images of a single quasar *behind* the galaxy. (b) Diagram showing how the galaxy could bend the light coming from the quasar behind it to produce the four images. See also Fig. 33–14. [If the shape of the nearby galaxy and distant quasar were perfect spheres and perfectly aligned, we would expect the "image" of the distant quasar to be a circular ring or halo instead of the four separate images seen here. Such a ring is called an "Einstein ring."]

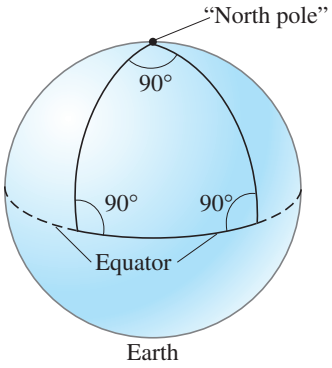
An opportune eclipse occurred in 1919, and scientists journeyed to the South Atlantic to observe it. Their photos of stars just behind the Sun revealed shifts in accordance with Einstein's prediction. Another example of gravitational deflection of light is **gravitational lensing**, as described in Fig. 33–15. The very distant galaxies shown in the XDF photo at the start of this Chapter, page 947, are thought to be visible only because of gravitational lensing (and magnification of their emitted light) by nearer galaxies—as if the nearby galaxies acted as a magnifying glass.

The mathematician Fermat showed in the 1600s that optical phenomena, including reflection, refraction, and effects of lenses, can be derived from a simple principle: that light traveling between two points follows the shortest path in space. Thus if gravity curves the path of light, then gravity must be able to curve space itself. That is, *space itself can be curved*, and it is gravitational mass that causes the curvature. Indeed, the curvature of space—or rather, of four-dimensional space-time—is a basic aspect of Einstein's General Relativity.

What is meant by **curved space**? To understand, recall that our normal method of viewing the world is via Euclidean plane geometry. In Euclidean geometry, there are many axioms and theorems we take for granted, such as that the sum of the angles of any triangle is 180°. Non-Euclidean geometries, which involve curved space, have also been imagined by mathematicians. It is hard enough to imagine three-dimensional curved space, much less curved four-dimensional space-time. So let us try to understand the idea of curved space by using two-dimensional surfaces.

Consider, for example, the two-dimensional surface of a sphere. It is clearly curved, Fig. 33–16, at least to us who view it from the outside—from our three-dimensional world. But how would hypothetical two-dimensional creatures determine whether their two-dimensional space was flat (a plane) or curved? One way would be to measure the sum of the angles of a triangle. If the surface is a plane, the sum of the angles is 180°, as we learn in plane geometry. But if the space is curved, and a sufficiently large triangle is constructed, the sum of the angles will *not* be 180°. To construct a triangle on a curved surface, say the sphere of Fig. 33–16, we must use the equivalent of a straight line: that is, the shortest distance between two points, which is called a **geodesic**. On a sphere, a geodesic is an arc of a great circle (an arc in a plane passing through the center of the sphere) such as the Earth's equator and the Earth's longitude lines. Consider, for example, the large triangle of Fig. 33–16: its sides are two longitude lines passing from the north pole to the equator, and the third side is a section of the equator as shown. The two longitude lines make 90° angles with the equator (look at a world globe to see this more clearly). They make an angle with each other at the north pole, which could be, say, 90° as shown; the sum of these angles is 90° + 90° + 90° = 270°. This is clearly *not* a Euclidean space. Note, however, that if the triangle is small in comparison to the radius of the sphere, the angles will add up to nearly 180°, and the triangle (and space) will seem flat.

FIGURE 33–16 On a two-dimensional curved surface, the sum of the angles of a triangle may not be 180°.



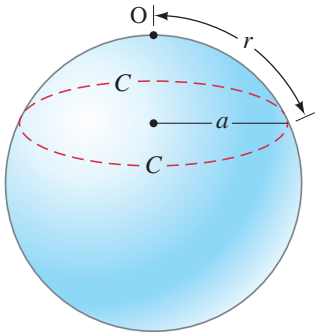


FIGURE 33–17 On a spherical surface (a two-dimensional world) a circle of circumference C is drawn (red) about point O as the center. The radius of the circle (not the sphere) is the distance r along the surface. (Note that in our three-dimensional view, we can tell that $C = 2\pi a$. Since $r > a$, then $C < 2\pi r$.)

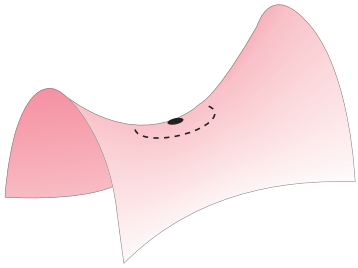


FIGURE 33–18 Example of a two-dimensional surface with negative curvature.

Another way to test the curvature of space is to measure the radius r and circumference C of a large circle. On a plane surface, $C = 2\pi r$. But on a two-dimensional spherical surface, C is *less* than $2\pi r$, as can be seen in Fig. 33–17. The proportionality between C and r is *less* than 2π. Such a surface is said to have *positive curvature*. On the saddlelike surface of Fig. 33–18, the circumference of a circle is greater than $2\pi r$, and the sum of the angles of a triangle is less than 180°. Such a surface is said to have a *negative curvature*.

Curvature of the Universe

What about our universe? On a large scale (not just near a large mass), what is the overall curvature of the universe? Does it have positive curvature, negative curvature, or is it flat (zero curvature)? We perceive our world as Euclidean (flat), but we can not exclude the possibility that space could have a curvature so slight that we don't normally notice it. This is a crucial question in cosmology, and it can be answered only by precise experimentation.

If the universe had a positive curvature, the universe would be *closed*, or *finite* in volume. This would *not* mean that the stars and galaxies extended out to a certain boundary, beyond which there is empty space. There is no boundary or edge in such a universe. The universe is all there is. If a particle were to move in a straight line in a particular direction, it would eventually return to the starting point—perhaps eons of time later.

On the other hand, if the curvature of space was zero or negative, the universe would be *open*. It could just go on forever. An open universe could be *infinite*; but according to recent research, even that may not necessarily be so.

Today the evidence is very strong that the universe on a large scale is very close to being flat. Indeed, it is so close to being flat that we can't tell if it might have very slightly positive or very slightly negative curvature.

Black Holes

According to Einstein's theory of general relativity (sometimes abbreviated GR), space-time is curved near massive objects. We might think of space as being like a thin rubber sheet: if a heavy weight is placed on the sheet, it sags as shown in Fig. 33–19a (top of next page). The weight corresponds to a huge mass that causes space (space itself!) to curve. Thus, in the context of

general relativity[†] we do not speak of the "force" of gravity acting on objects. Instead we say that objects and light rays move as they do because space-time is curved. An object starting at rest or moving slowly near the great mass of Fig. 33–19a would follow a geodesic (the equivalent of a straight line in plane geometry) toward that great mass.

The extreme curvature of space-time shown in Fig. 33–19b could be produced by a **black hole**. A black hole, as we mentioned in Section 33–2, has such strong gravity that even light cannot escape from it. To become a black hole, an object of mass M must undergo **gravitational collapse**, contracting by gravitational self-attraction to within a radius called the **Schwarzschild radius**,

$$R = \frac{2GM}{c^2},$$

where G is the gravitational constant and c the speed of light. If an object collapses to within this radius, it is predicted by general relativity to collapse to a point at $r = 0$, forming an infinitely dense singularity. This prediction is uncertain, however, because in this realm we need to combine quantum mechanics with gravity, a unification of theories not yet achieved (Section 32–12).

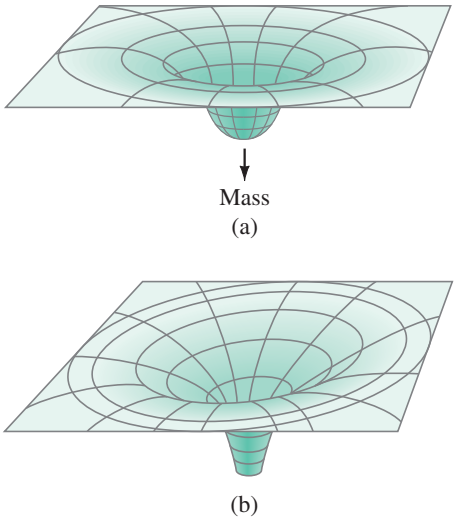


FIGURE 33–19 (a) Rubber-sheet analogy for space-time curved by matter. (b) Same analogy for a black hole, which can "swallow up" objects that pass near.

EXERCISE C What is the Schwarzschild radius for an object with 10 solar masses?

The Schwarzschild radius also represents the event horizon of a black hole. By **event horizon** we mean the surface beyond which no emitted signals can ever reach us, and thus inform us of events that happen beyond that surface. As a star collapses toward a black hole, the light it emits is pulled harder and harder by gravity, but we can still see it. Once the matter passes within the event horizon, the emitted light cannot escape but is pulled back in by gravity (= curvature of space-time).

All we can know about a black hole is its mass, its angular momentum (rotating black holes), and its electric charge. No other information, no details of its structure or the kind of matter it was formed of, can be known because no information can escape.

How might we observe black holes? We cannot see them because no light can escape from them. They would be black objects against a black sky. But they do exert a gravitational force on nearby objects, and also on light rays (or photons) that pass nearby (just like in Fig. 33–15). The black hole believed to be at the center of our Galaxy ($M \approx 4 \times 10^6\, M_{\text{Sun}}$) was discovered by examining the motion of matter in its vicinity. Another technique is to examine stars which appear to move as if they were one member of a *binary system* (two stars rotating about their common center of mass), but without a visible companion. If the unseen star is a black hole, it might be expected to pull off gaseous material from its visible companion (as in Fig. 33–10). As this matter approached the black hole, it would be highly accelerated and should emit X-rays of a characteristic type before plunging inside the event horizon. Such X-rays, plus a sufficiently high mass estimate from the rotational motion, can provide evidence for a black hole. One of the many candidates for a black hole is in the binary-star system Cygnus X-1. It is widely believed that the center of most galaxies is occupied by a black hole with a mass 10^6 to 10^9 times the mass of a typical star like our Sun.

EXERCISE D A black hole has radius R. Its mass is proportional to (*a*) R, (*b*) R^2, (*c*) R^3. Justify your answer.

[†]Alexander Pope (1688–1744) wrote an epitaph for Newton:

"Nature, and Nature's laws lay hid in night:
God said, *Let Newton be!* and all was light."

Sir John Squire (1884–1958), perhaps uncomfortable with Einstein's profound thoughts, added:

"It did not last: the Devil howling '*Ho!*
Let Einstein be!' restored the status quo."

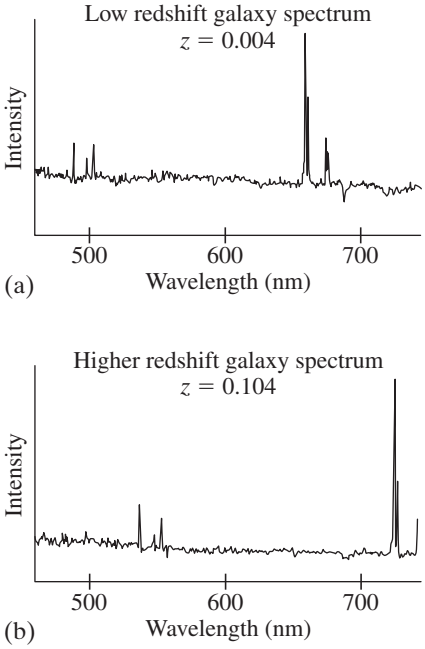


FIGURE 33–20 Atoms and molecules emit and absorb light of particular frequencies depending on the spacing of their energy levels, as we saw in Chapters 27 to 29. (a) The spectrum of light received from a relatively slow-moving galaxy. (b) Spectrum of a galaxy moving away from us at a much higher speed. Note how the peaks (or lines) in the spectrum have moved to longer wavelengths. The redshift is $z = (\lambda_{obs} - \lambda_{rest})/\lambda_{rest}$.

33–5 The Expanding Universe: Redshift and Hubble's Law

We discussed in Section 33–2 how individual stars evolve from their birth to their death as white dwarfs, neutron stars, or black holes. But what about the universe as a whole: is it static, or does it change? One of the most important scientific discoveries of the twentieth century was that distant galaxies are racing away from us, and that the farther they are from us at a given time, the faster they are moving away. How astronomers arrived at this astonishing idea, and what it means for the past history of the universe as well as its future, will occupy us for the remainder of the book.

Observational evidence that the universe is expanding was first put forth by Edwin Hubble in 1929. This idea was based on distance measurements of galaxies (Section 33–3), and determination of their velocities by the Doppler shift of spectral lines in the light received from them (Fig. 33–20). In Chapter 12 we saw how the frequency of sound is higher and the wavelength shorter if the source and observer move toward each other. If the source moves away from the observer, the frequency is lower and the wavelength longer. The **Doppler effect** occurs also for light, but the formula for light is slightly different than for sound and is given by[†]

$$\lambda_{obs} = \lambda_{rest}\sqrt{\frac{1 + v/c}{1 - v/c}}, \qquad \left[\begin{array}{l}\text{source and observer moving}\\ \text{away from each other}\end{array}\right] \qquad \textbf{(33–3)}$$

where λ_{rest} is the emitted wavelength as seen in a reference frame at rest with respect to the source, and λ_{obs} is the wavelength observed in a frame moving with velocity v away from the source along the line of sight. (For relative motion *toward* each other, $v < 0$ in this formula.) When a distant source emits light of a particular wavelength, and the source is moving away from us, the wavelength appears longer to us: the color of the light (if it is visible) is shifted toward the red end of the visible spectrum, an effect known as a **redshift**. (If the source moves toward us, the color shifts toward the blue or shorter wavelength.)

In the spectra of stars in other galaxies, lines are observed that correspond to lines in the known spectra of particular atoms (see Section 27–11 and Figs. 24–28 and 27–23). What Hubble found was that the lines seen in the spectra from distant galaxies were generally *redshifted*, and that the amount of shift seemed to be approximately proportional to the distance of the galaxy from us. That is, the velocity v of a galaxy moving away from us is proportional to its distance d from us:

HUBBLE'S LAW

$$v = H_0 d. \qquad \textbf{(33–4)}$$

This is **Hubble's law**, one of the most fundamental astronomical ideas. It was first suggested, in 1927, by Georges Lemaître, a Belgian physics professor and priest, who also first proposed what later came to be called the Big Bang. The constant H_0 is called the **Hubble parameter**.

The value of H_0 until recently was uncertain by over 20%, and thought to be between 15 and 25 km/s/Mly. But recent measurements now put its value more precisely at

$$H_0 = 21\ \text{km/s/Mly}$$

(that is, 21 km/s per million light-years of distance). The current uncertainty is about 2%, or ± 0.5 km/s/Mly. [H_0 can be written in terms of parsecs (Section 33–3) as $H_0 = 67$ km/s/Mpc (that is, 67 km/s per megaparsec of distance) with an uncertainty of about ± 1.2 km/s/Mpc.]

[†]For light there is no medium and we can make no distinction between motion of the source and motion of the observer (special relativity), as we did for sound which travels in a medium.

Redshift Origins

Galaxies very near us seem to be moving randomly relative to us: some move towards us (blueshifted), others away from us (redshifted); their speeds are on the order of $0.001c$. But for more distant galaxies, the velocity of recession is much greater than the velocity of local random motion, and so is dominant and Hubble's law (Eq. 33–4) holds very well. More distant galaxies have higher recession velocity and a larger redshift, and we call their redshift a **cosmological redshift**. We interpret this redshift today as due to the *expansion of space* itself. We can think of the originally emitted wavelength λ_{rest} as being stretched out (becoming longer) along with the expanding space around it, as suggested in Fig. 33–21. Although Hubble thought of the redshift as a Doppler shift, now we prefer to understand it in this sense of expanding space. (But note that atoms in galaxies do not expand as space expands; they keep their regular size.)

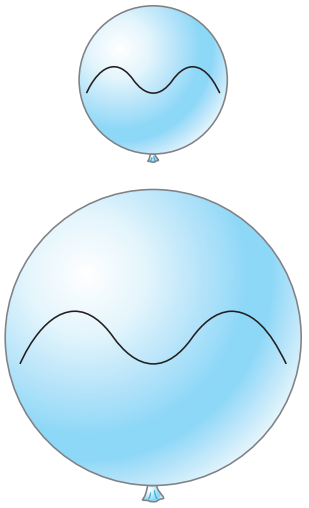


FIGURE 33–21 Simplified model of a 2-dimensional universe, imagined as a balloon. As you blow up the balloon (= expanding universe), the wavelength of a wave on its surface gets longer (redshifted).

There is a third way to produce a redshift, which we mention for completeness: a **gravitational redshift**. Light leaving a massive star is gaining in gravitational potential energy (just like a stone thrown upward from Earth). So the kinetic energy of each photon, hf, must be getting smaller (to conserve energy). A smaller frequency f means a larger (longer) wavelength λ $(= c/f)$, which is a redshift.

The amount of a redshift is specified by the **redshift parameter**, z, defined as

$$z = \frac{\lambda_{\text{obs}} - \lambda_{\text{rest}}}{\lambda_{\text{rest}}} = \frac{\Delta\lambda}{\lambda_{\text{rest}}}, \qquad \textbf{(33–5a)}$$

where λ_{rest} is a wavelength as seen by an observer at rest relative to the source, and λ_{obs} is the wavelength measured by a moving observer. Equation 33–5a can be written as

$$z = \frac{\lambda_{\text{obs}}}{\lambda_{\text{rest}}} - 1 \qquad \textbf{(33–5b)}$$

and

$$z + 1 = \frac{\lambda_{\text{obs}}}{\lambda_{\text{rest}}}. \qquad \textbf{(33–5c)}$$

For low speeds not close to the speed of light $(v \lesssim 0.1\,c)$, the Doppler formula (Eq. 33–3) can be used to show (Problem 31) that z is proportional to the speed of the source toward or away from us:

$$z = \frac{\lambda_{\text{obs}} - \lambda_{\text{rest}}}{\lambda_{\text{rest}}} = \frac{\Delta\lambda}{\lambda_{\text{rest}}} \approx \frac{v}{c}. \qquad [v \ll c] \quad \textbf{(33–6)}$$

But redshifts are not always small, in which case the approximation of Eq. 33–6 is not valid. For high z galaxies, not even Eq. 33–3 applies because the redshift is due to the expansion of space (cosmological redshift), not the Doppler effect. Our Chapter-Opening Photograph, page 947, shows two very distant high z galaxies, $z = 8.8$ and 11.9, which are also shown enlarged.

* Scale Factor (advanced)

The expansion of space can be described as a scaling of the typical distance between two points or objects in the universe. If two distant galaxies are a distance d_0 apart at some initial time, then a time t later they will be separated by a greater distance $d(t)$. The **scale factor** is the same as for light, expressed in Eq. 33–5a:

$$\frac{d(t) - d_0}{d_0} = \frac{\Delta\lambda}{\lambda} = z$$

or

$$\frac{d(t)}{d_0} = 1 + z.$$

Thus, for example, if a galaxy has $z = 3$, then the scale factor is now $(1 + 3) = 4$ times larger than when the light was emitted from that galaxy. That is, the average distance between galaxies has become 4 times larger. Thus the factor by which the wavelength has increased since it was emitted tells us by what factor the universe (or the typical distance between objects) has increased.

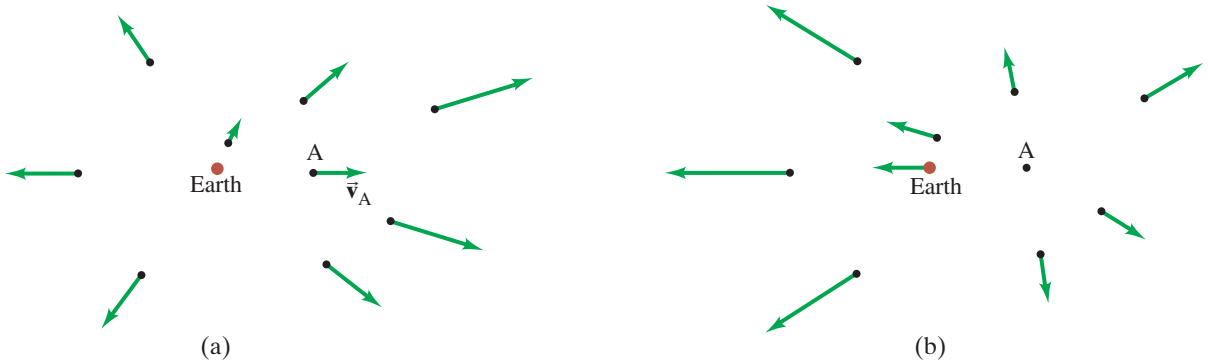


FIGURE 33–22 Expansion of the universe looks the same from any point in the universe. If you are on Earth as shown in part (a), or you are instead at galaxy A (which is at rest in the reference frame shown in (b)), all other galaxies appear to be racing away from you.

Expansion, and the Cosmological Principle

What does it mean that distant galaxies are all moving away from us, and with ever greater speed the farther they are from us? It seems to suggest some kind of explosive expansion that started at some very distant time in the past. And at first sight we seem to be in the middle of it all. But we aren't. The expansion appears the same from any other point in the universe. To understand why, see Fig. 33–22. In Fig. 33–22a we have the view from Earth (or from our Galaxy). The velocities of surrounding galaxies are indicated by arrows, pointing away from us, and the arrows are longer (faster speeds) for galaxies more distant from us. Now, what if we were on the galaxy labeled A in Fig. 33–22a? From Earth, galaxy A appears to be moving to the right at a velocity, call it $\vec{\mathbf{v}}_A$, represented by the arrow pointing to the right. If we were *on* galaxy A, Earth would appear to be moving to the left at velocity $-\vec{\mathbf{v}}_A$. To determine the velocities of other galaxies relative to A, we vectorially add the velocity vector, $-\vec{\mathbf{v}}_A$, to all the velocity arrows shown in Fig. 33–22a. This yields Fig. 33–22b, where we see that the universe is expanding away from galaxy A as well; and the velocities of galaxies receding from A are proportional to their current distance from A. *The universe looks pretty much the same from different points.*

Thus the expansion of the universe can be stated as follows: all galaxies are racing away from *each other* at an average rate of about 21 km/s per million light-years of distance between them. The ramifications of this idea are profound, and we discuss them in a moment.

A basic assumption in cosmology has been that on a large scale, the universe would look the same to observers at different places at the same time. In other words, the universe is both *isotropic* (looks the same in all directions) and *homogeneous* (would look the same if we were located elsewhere, say in another galaxy). This assumption is called the **cosmological principle**. On a local scale, say in our solar system or within our Galaxy, it clearly does not apply (the sky looks different in different directions). But it has long been thought to be valid if we look on a large enough scale, so that the average population density of galaxies and clusters of galaxies ought to be the same in different areas of the sky. This seems to be valid on distances greater than about 700 Mly. The expansion of the universe (Fig. 33–22) is consistent with the cosmological principle; and the near uniformity of the cosmic microwave background radiation (discussed in Section 33–6) supports it. Another way to state the cosmological principle is that *our place in the universe is not special.*

The expansion of the universe, as described by Hubble's law, strongly suggests that galaxies must have been closer together in the past than they are now. This is, in fact, the basis of the *Big Bang* theory of the origin of the universe, which pictures the universe as a relentless expansion starting from a very hot and compressed beginning. We discuss the Big Bang in detail shortly, but first let us see what can be said about the age of the universe.

One way to estimate the age of the universe uses the Hubble parameter. With $H_0 \approx 21\,\text{km/s}$ per 10^6 light-years, the time required for the galaxies to arrive at their present separations would be approximately (starting with $v = d/t$ and using Hubble's law, Eq. 33–4),

$$t = \frac{d}{v} = \frac{d}{H_0 d} = \frac{1}{H_0} \approx \frac{(10^6\,\text{ly})(0.95 \times 10^{13}\,\text{km/ly})}{(21\,\text{km/s})(3.16 \times 10^7\,\text{s/yr})} \approx 14 \times 10^9\,\text{yr},$$

or 14 billion years. The age of the universe calculated in this way is called the *characteristic expansion time* or "Hubble age." It is a very rough estimate and assumes the rate of expansion of the universe was constant (which today we are quite sure is not true). Today's best measurements give the age of the universe as about $13.8 \times 10^9\,\text{yr}$, in remarkable agreement with the rough Hubble age estimate.

* Steady-State Model

Before discussing the Big Bang in detail, we mention one alternative to the Big Bang—the **steady-state model**—which assumed that the universe is infinitely old and on average looks the same now as it always has. (This assumed uniformity in time as well as space was called the *perfect cosmological principle*.) According to the steady-state model, no large-scale changes have taken place in the universe as a whole, particularly no Big Bang. To maintain this view in the face of the recession of galaxies away from each other, matter would need to be created continuously to maintain the assumption of uniformity. The rate of mass creation required is very small—about one nucleon per cubic meter every 10^9 years.

The steady-state model provided the Big Bang model with healthy competition in the mid-twentieth century. But the discovery of the cosmic microwave background radiation (next Section), as well as other observations of the universe, has made the Big Bang model universally accepted.

33–6 The Big Bang and the Cosmic Microwave Background

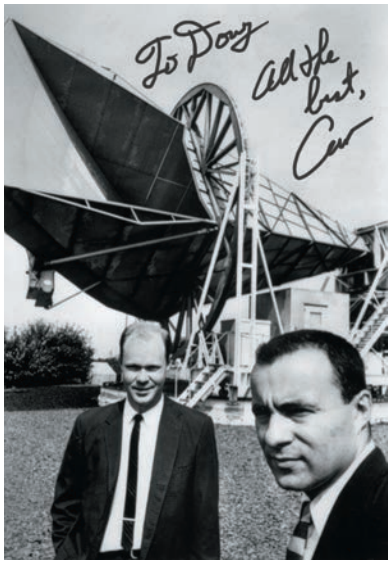
The expansion of the universe suggests that typical objects in the universe were once much closer together than they are now. This is the basis for the idea that the universe began about 14 billion years ago as an expansion from a state of very high density and temperature known affectionately as the **Big Bang**.

The birth of the universe was not an explosion, because an explosion blows pieces out into the surrounding space. Instead, the Big Bang was the start of an expansion of space itself. The observable universe was relatively very small at the start and has been expanding, getting ever larger, ever since. The initial tiny universe of extremely dense matter is not to be thought of as a concentrated mass in the midst of a much larger space around it. The initial tiny but dense universe was the *entire universe*. There wouldn't have been anything else. When we say that the universe was once smaller than it is now, we mean that the average separation between objects (such as electrons or galaxies) was less. The universe may have been infinite in extent even then, and it may still be now (only bigger). The **observable universe** (that which we have the possibility of observing because light has had time to reach us) is, however, finite.

A major piece of evidence supporting the Big Bang is the **cosmic microwave background** radiation (or CMB) whose discovery came about as follows.

In 1964, Arno Penzias and Robert Wilson pointed their horn antenna for detecting radio waves (Fig. 33–23) into the sky. With it they detected widespread emission, and became convinced that it was coming from outside our Galaxy. They made precise measurements at a wavelength $\lambda = 7.35\,\text{cm}$, in the microwave region of the electromagnetic spectrum (Fig. 22–8). The intensity of this radiation was found initially not to vary by day or night or time of year, nor to depend on direction. It came from all directions in the universe with equal intensity, to a precision of better than 1%. It could only be concluded that this radiation came from the universe as a whole.

FIGURE 33–23 Photo of Arno Penzias (right, who signed it "Arno") and Robert Wilson. Behind them their "horn antenna."



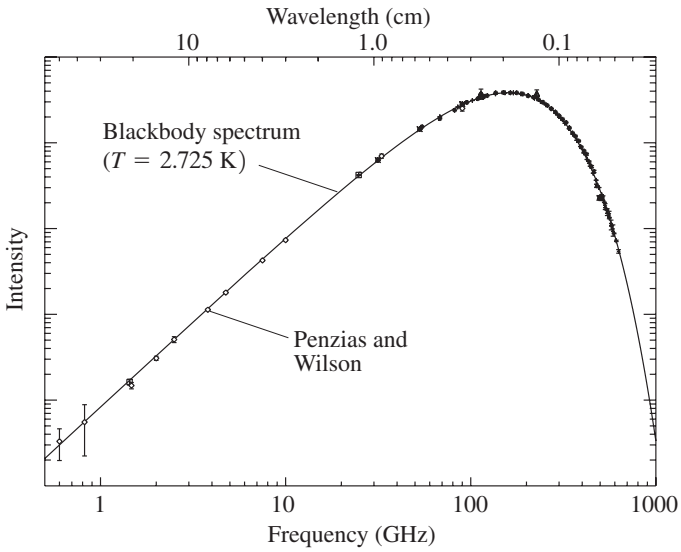


FIGURE 33–24 Spectrum of cosmic microwave background radiation, showing blackbody curve and experimental measurements including at the frequency detected by Penzias and Wilson. (Thanks to G. F. Smoot and D. Scott. The vertical bars represent the most recent experimental uncertainty in a measurement.)

The intensity of this CMB measured at $\lambda = 7.35\,\text{cm}$ corresponds to blackbody radiation (see Section 27–2) at a temperature of about 3 K. When radiation at other wavelengths was measured by the COBE satellite (COsmic Background Explorer), the intensities were found to fall on a nearly perfect blackbody curve as shown in Fig. 33–24, corresponding to a temperature of 2.725 K (± 0.002 K).

The remarkable uniformity of the CMB was in accordance with the cosmological principle. But theorists felt that there needed to be some small inhomogeneities, or "anisotropies," in the CMB that would have provided "seeds" at which galaxy formation could have started. Small areas of slightly higher density, which could have contracted under gravity to form clusters of galaxies, were indeed found. These tiny inhomogeneities in density and temperature were detected first by the COBE satellite experiment in 1992, led by George Smoot and John Mather (Fig. 33–25).

FIGURE 33–25 COBE scientists John Mather (chief scientist and responsible for measuring the blackbody form of the spectrum) and George Smoot (chief investigator for anisotropy experiment) shown here during celebrations for their Dec. 2006 Nobel Prize, given for their discovery of the spectrum and anisotropy of the CMB using the COBE instrument.



This discovery of the **anisotropy** of the CMB ranks with the discovery of the CMB itself in the history of cosmology. The blackbody fit and the anisotropy were the culmination of decades of research by pioneers such as Richard Muller, Paul Richards, and David Wilkinson. Subsequent experiments gave us greater detail in 2003, 2006, and 2012 with the WMAP (Wilkinson Microwave Anisotropy Probe) results, Fig. 33–26, and even more recently with the European Planck satellite results in 2013.

The CMB provides strong evidence in support of the Big Bang, and gives us information about conditions in the very early universe. In fact, in the late 1940s, George Gamow and his collaborators calculated that a Big Bang origin of the universe should have generated just such a microwave background radiation.

To understand why, let us look at what a Big Bang might have been like. (Today we usually use the term "Big Bang" to refer to the *process*, starting from a moment after the birth of the universe through the subsequent expansion.) The temperature must have been extremely high at the start, so high that there could not have been any atoms in the very early stages of the universe (high energy collisions would have broken atoms apart into nuclei and free electrons). Instead, the universe would have consisted solely of radiation (photons) and a plasma of charged electrons and other elementary particles. The universe would have been

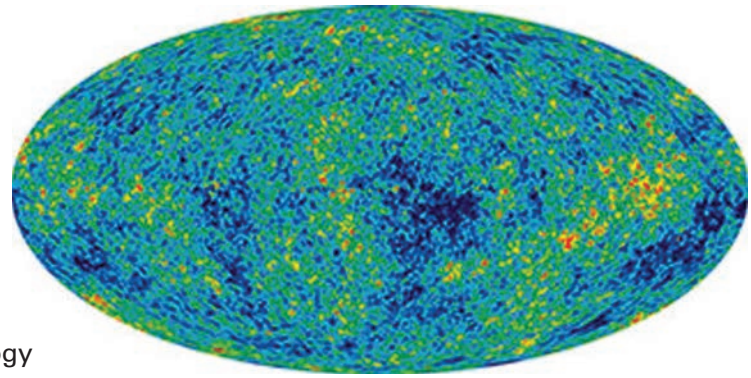


FIGURE 33–26 Measurements of the cosmic microwave background radiation over the entire sky, color-coded to represent differences in temperature from the average 2.725 K: the color scale ranges from $+200\,\mu$K (red) to $-200\,\mu$K (dark blue), representing slightly hotter and colder spots (associated with variations in density). Results are from the WMAP satellite in 2012: the angular resolution is 0.2°.

opaque—the photons in a sense "trapped," traveling very short distances before being scattered again, primarily by electrons. Indeed, the details of the microwave background radiation provide strong evidence that matter and radiation were once in equilibrium at a very high temperature. As the universe expanded, the energy spread out over an increasingly larger volume and the temperature dropped. Not long before the temperature had fallen to ~3000 K, some 380,000 years later, could nuclei and electrons combine together as stable atoms. With the disappearance of free electrons, as they combined with nuclei to form atoms, the radiation would have been freed—**decoupled** from matter, we say. The universe became *transparent* because photons were now free to travel nearly unimpeded straight through the universe.

It is this radiation, from 380,000 years after the birth of the universe, that we now see as the CMB. As the universe expanded, so too the wavelengths of the radiation lengthened, thus redshifting to longer wavelengths that correspond to lower temperature (recall Wien's law, $\lambda_P T = \text{constant}$, Section 27–2), until they would have reached the 2.7-K background radiation we observe today.

Looking Back toward the Big Bang—Lookback Time

Figure 33–27 shows our Earth point of view, looking out in all directions back toward the Big Bang and the brief (380,000-year-long) period when radiation was trapped in the early plasma (yellow band). The time it takes light to reach us from an event is called its **lookback time**. The "close-up" insert in Fig. 33–27 shows a photon scattering repeatedly inside that early plasma and then exiting the plasma in a straight line. No matter what direction we look, our view of the very early universe is blocked by this wall of plasma. It is like trying to look into a very thick fog or into the surface of the Sun—we can see only as far as its surface, called the **surface of last scattering**, but not into it. Wavelengths from there are redshifted by $z \approx 1100$. Time $\Delta t'$ in Fig. 33–27 is the lookback time (not real time that goes forward).

Recall that when we view an object far away, we are seeing it as it was then, when the light was emitted, not as it would appear today.

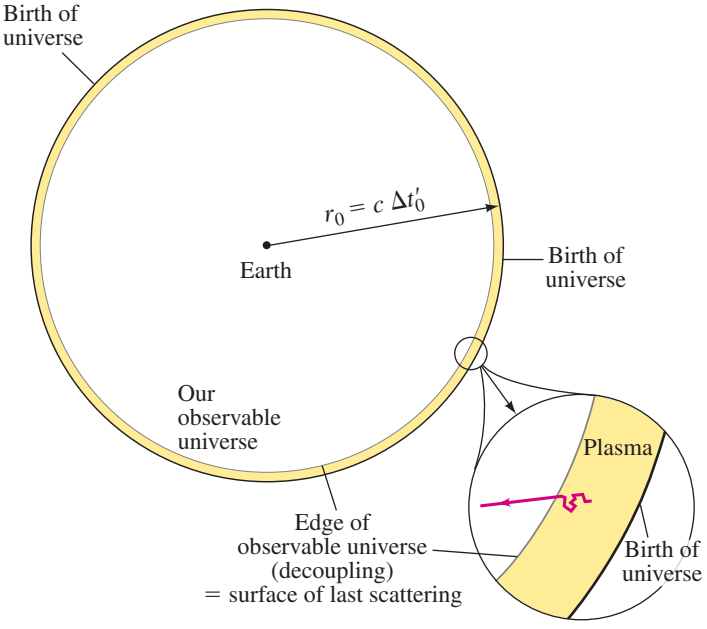


FIGURE 33–27 When we look out from the Earth, we look back in time. Any other observer in the universe would see more or less the same thing. The farther an object is from us, the longer ago the light we see had to have left it. We cannot see quite as far as the Big Bang; we can see only as far as the "surface of last scattering," which radiated the CMB. The insert on the lower right shows the earliest 380,000 years of the universe when it was opaque: a photon is shown scattering many times and then (at decoupling, 380,000 yr after the birth of the universe) becoming free to travel in a straight line. If this photon wasn't heading our way when "liberated," many others were. Galaxies are not shown, but would be concentrated close to Earth in this diagram because they were created relatively recently. *Note:* This diagram is not a normal map. Maps show a section of the world as might be seen all *at a given time*. This diagram shows space (like a map), but each point is *not* at the same time. The light coming from a point a distance r from Earth took a time $\Delta t' = r/c$ to reach Earth, and thus shows an event that took place long ago, a time $\Delta t' = r/c$ in the past, which we call its "lookback time." The universe began $\Delta t'_0 = 13.8\ \text{Gyr}$ ago.

The Observable Universe

Figure 33–27 can easily be misinterpreted: it is not a picture of the universe at a given instant, but is intended to suggest how we look out in all directions from our observation point (the Earth, or near it). Be careful not to think that the birth of the universe took place in a circle or a sphere surrounding us as if Fig. 33–27 were a photo taken at a given moment. What Fig. 33–27 does show is what we can see, the *observable universe*. Better yet, it shows the *most* we could see.

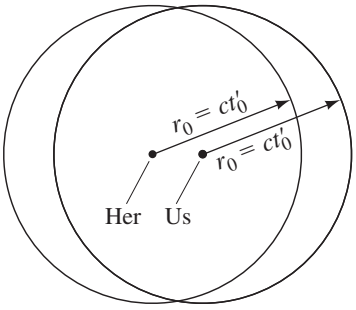


FIGURE 33–28 Two observers, on widely separated galaxies, have different horizons, different observable universes.

We would undoubtedly be arrogant to think that we could see the entire universe. Indeed, theories assume that we cannot see everything, that the **entire universe** is greater than the **observable universe**, which is a sphere of radius $r_0 = ct_0$ centered on the observer, with t_0 being the age of the universe. We can never see further back than the time it takes light to reach us.

Consider, for example, an observer in another galaxy, very far from us, located to the left of our observation point in Fig. 33–27. That observer would not yet have seen light coming from the far right of the large circle in Fig. 33–27 that we see—it will take some time for that light to reach her. But she will have already, some time ago, seen the light coming from the left that we are seeing now. In fact, her observable universe, superimposed on ours, is suggested by Fig. 33–28.

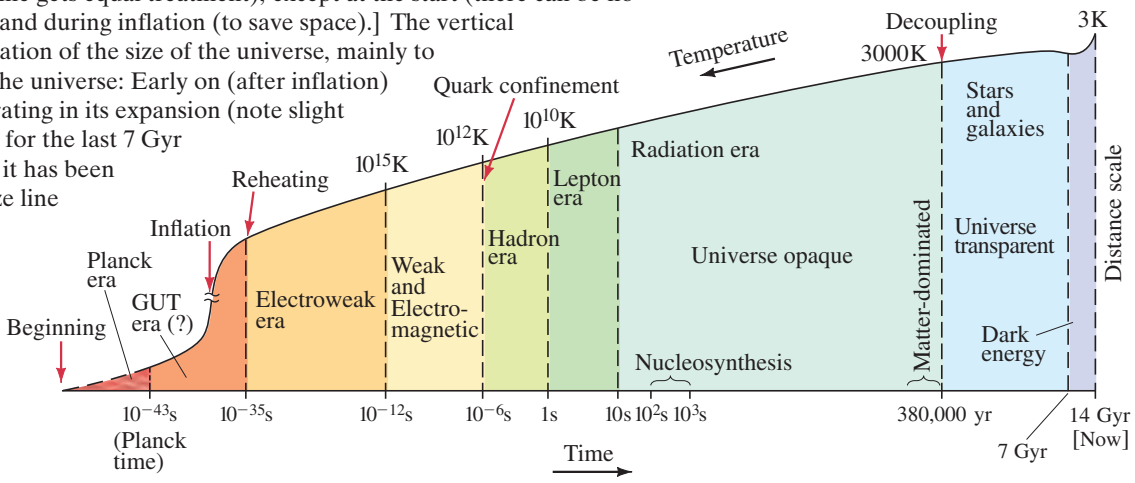
The edge of our observable universe is called the **horizon**. We could, in principle, see as far as the horizon, but not beyond it. An observer in another galaxy, far from us, will have a different horizon.

33–7 The Standard Cosmological Model: Early History of the Universe

In the last decade or two, a convincing theory of the origin and evolution of the universe has been developed, now called the **Standard Cosmological Model**. Part of this theory is based on recent theoretical and experimental advances in elementary particle physics, and part from observations of the universe including COBE, WMAP, and Planck. Indeed, cosmology and elementary particle physics have cross-fertilized to a surprising extent.

Let us go back to the earliest of times—as close as possible to the Big Bang—and follow a Standard Model theoretical scenario of events as the universe expanded and cooled after the Big Bang. Initially we talk of extremely small time intervals as well as extremely high temperatures, far higher than any temperature in the universe today. Figure 33–29 is a compressed graphical representation of the events, and it may be helpful to consult it as we go along.

FIGURE 33–29 Compressed graphical representation of the development of the universe after the Big Bang, according to modern cosmology. [The time scale is mostly logarithmic (each factor of 10 in time gets equal treatment), except at the start (there can be no $t = 0$ on a log scale), and during inflation (to save space).] The vertical height is a rough indication of the size of the universe, mainly to suggest expansion of the universe: Early on (after inflation) the universe is decelerating in its expansion (note slight downward curve); but for the last 7 Gyr (= thin strip on right) it has been accelerating, so the size line on the top curves upward at upper right.



The History

We begin at a time only a minuscule fraction of a second after the "beginning" of the universe, 10^{-43} s. This time (sometimes referred to as the **Planck time**) is an unimaginably short time, and predictions can be only speculative. Earlier, we can say nothing because we do not have a theory of quantum gravity which would be needed for the incredibly high densities and temperatures during this "Planck era."

The first theories of the Big Bang assumed the universe was extremely hot in the beginning, maybe 10^{32} K, and then gradually cooled down while expanding. In those first moments after 10^{-43} s, the four forces of nature were thought to be united—there was only one force (Chapter 32, Fig. 32–22). Then a kind of

"phase transition" would have occurred during which the gravitational force would have "condensed out" as a separate force. This and subsequent phase transitions, as shown in Fig. 32–22, are analogous to phase transitions water undergoes as it cools from a gas condensing into a liquid, and with further cooling freezes into ice.[†] The *symmetry* of the four forces would have been broken leaving the strong, weak, and electromagnetic forces still unified, and the universe would have entered the **grand unified era** (GUT—see Section 32–11).

This scenario of a *hot* Big Bang is now doubted by some important theorists, such as Andreí Linde, whose theories suggest the universe was much cooler at the Planck time. But what happened next to the universe, though very strange, is accepted by most cosmologists: a brilliant idea, suggested by Linde and Alan Guth in the early 1980s, proposed that the universe underwent an incredible exponential expansion, increasing in size by a factor of 10^{30} or maybe much more, in a tiny fraction of a second, perhaps 10^{-35} s or 10^{-32} s. The usefulness of this **inflationary scenario** is that it solved major problems with earlier Big Bang models, such as explaining why the universe is flat, as well as the thermal equilibrium to provide the nearly uniform CMB, as discussed below.

When inflation ended, whatever energy caused it then ended up being transformed into elementary particles with very high kinetic energy, corresponding to very high temperature (Eq. 13–8, $\overline{\text{KE}} = \frac{3}{2}kT$). That process is referred to as **reheating**, and the universe was now a "soup" of leptons, quarks, and other particles. We can think of this "soup" as a plasma of particles and antiparticles, as well as photons—all in roughly equal numbers—colliding with one another frequently and exchanging energy.

The temperature of the universe at the end of inflation was much lower than that expected by the hot Big Bang theory. But it would have been high enough so that the weak and electromagnetic forces were unified into a single force, and this stage of the universe is sometimes called the **electroweak era**. Approximately 10^{-12} s after the Big Bang, the temperature dropped to about 10^{15} K corresponding to randomly moving particles with an average kinetic energy KE of about 100 GeV (see Eq. 13–8):

$$\text{KE} \approx kT \approx \frac{(1.4 \times 10^{-23}\,\text{J/K})(10^{15}\,\text{K})}{1.6 \times 10^{-19}\,\text{J/eV}} \approx 10^{11}\,\text{eV} = 100\,\text{GeV}.$$

(As an estimate, we usually ignore the factor $\frac{3}{2}$ in Eq. 13–8.) At that time, symmetry between weak and electromagnetic forces would have broken down, and the weak force separated from the electromagnetic.

As the universe cooled down to about 10^{12} K (KE $\approx$ 100 MeV), approximately 10^{-6} s after the Big Bang, quarks stop moving freely and begin to "condense" into more normal particles: nucleons and the other hadrons and their antiparticles. With this **confinement of quarks**, the universe entered the **hadron era**. But it did not last long. Very soon the vast majority of hadrons disappeared. To see why, let us focus on the most familiar hadrons: nucleons and their antiparticles. When the average kinetic energy of particles was somewhat higher than 1 GeV, protons, neutrons, and their antiparticles were continually being created out of the energies of collisions involving photons and other particles, such as

$$\begin{aligned}\text{photons} &\rightarrow \text{p} + \bar{\text{p}} \\ &\rightarrow \text{n} + \bar{\text{n}}.\end{aligned}$$

But just as quickly, particles and antiparticles would annihilate: for example

$$\text{p} + \bar{\text{p}} \rightarrow \text{photons or leptons.}$$

So the processes of creation and annihilation of nucleons were in equilibrium. The numbers of nucleons and antinucleons were high—roughly as many as there were electrons, positrons, or photons. But as the universe expanded and cooled, and the average kinetic energy of particles dropped below about 1 GeV, which is the minimum energy needed in a typical collision to create nucleons and antinucleons (about 940 MeV each), the process of nucleon creation could not continue.

[†]It may be interesting to point out that this story of origins here bears some resemblance to ancient accounts (nonscientific) that mention the "void," "formless wasteland" (or "darkness over the deep"), "abyss," "divide the waters" (= a phase transition?), not to mention the sudden appearance of light.

Annihilation could continue, however, with antinucleons annihilating nucleons, until almost no nucleons were left. But not quite zero. Somehow we need to explain our present world of matter (nucleons and electrons) with very little antimatter in sight.

To explain our world of matter, we might suppose that earlier in the universe, after the inflationary period, a slight excess of quarks over antiquarks was formed.[†] This would have resulted in a slight excess of nucleons over antinucleons. And it is these "leftover" nucleons that we are made of today. The excess of nucleons over antinucleons was probably about one part in 10^9. During the hadron era, there should have been about as many nucleons as photons. After it ended, the "leftover" nucleons thus numbered only about one nucleon per 10^9 photons, and this ratio has persisted to this day. Protons, neutrons, and all other heavier particles were thus tremendously reduced in number by about 10^{-6} s after the Big Bang. The lightest hadrons, the pions, soon disappeared, about 10^{-4} s after the Big Bang; because they are the lightest mass hadrons (140 MeV), pions were the last hadrons able to be created as the temperature (and average kinetic energy) dropped. Lighter particles, including electrons and neutrinos, were the dominant form of matter, and the universe entered the **lepton era**.

By the time the first full second had passed (clearly the most eventful second in history!), the universe had cooled to about 10 billion degrees, 10^{10} K. The average kinetic energy was about 1 MeV. This was still sufficient energy to create electrons and positrons and balance their annihilation reactions, since their masses correspond to about 0.5 MeV. So there were about as many e^+ and e^- as there were photons. But within a few more seconds, the temperature had dropped sufficiently so that e^+ and e^- could no longer be formed. Annihilation $(e^+ + e^- \rightarrow \text{photons})$ continued. And, like nucleons before them, electrons and positrons all but disappeared from the universe—except for a slight excess of electrons over positrons (later to join with nuclei to form atoms). Thus, about $t = 10\,\text{s}$ after the Big Bang, the universe entered the **radiation era** (Fig. 33–29). Its major constituents were photons and neutrinos. But the neutrinos, partaking only in the weak force, rarely interacted. So the universe, until then experiencing significant amounts of energy in matter and in radiation, now became **radiation-dominated**: much more energy was contained in radiation than in matter, a situation that would last more than 50,000 years.

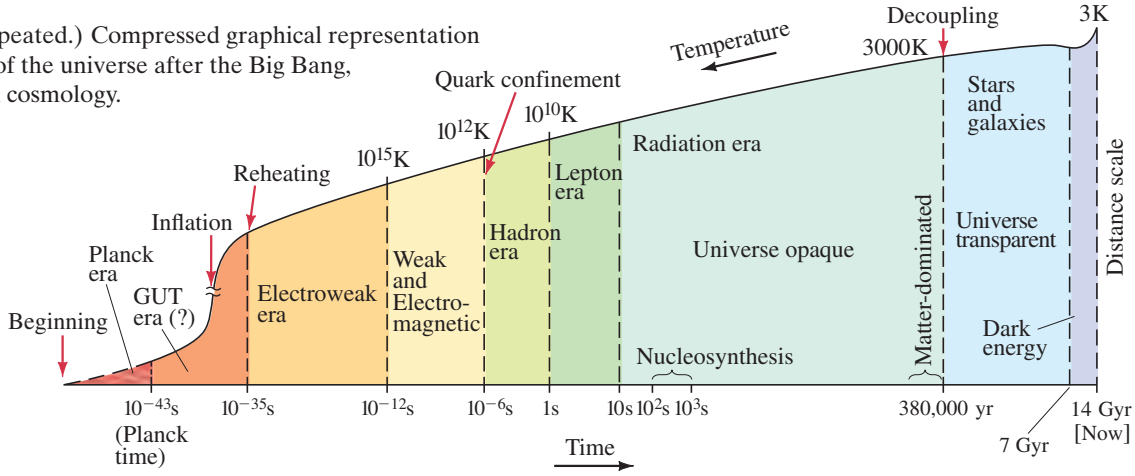


FIGURE 33–29 (Repeated.) Compressed graphical representation of the development of the universe after the Big Bang, according to modern cosmology.

Meanwhile, during the next few minutes, crucial events were taking place. Beginning about 2 or 3 minutes after the Big Bang, nuclear fusion began to occur. The temperature had dropped to about 10^9 K, corresponding to an average kinetic energy $\overline{\text{KE}} \approx 100\,\text{keV}$, where nucleons could strike each other and be able to fuse (Section 31–3), but now cool enough so newly formed nuclei would not be immediately broken apart by subsequent collisions. Deuterium, helium, and very tiny amounts of lithium nuclei were made. But the universe was cooling too quickly, and larger nuclei were not made. After only a few minutes, probably not even a quarter of an hour after the Big Bang, the temperature dropped far enough that nucleosynthesis stopped, not to start again for millions of years (in stars).

[†]Why this could have happened is a question for which we are seeking an answer today.

Thus, after the first quarter hour or so of the universe, matter consisted mainly of bare nuclei of hydrogen (about 75%) and helium (about 25%)[†] as well as electrons. But radiation (photons) continued to dominate.

Our story is almost complete. The next important event is thought to have occurred 380,000 years later. The universe had expanded to about $\frac{1}{1000}$ of its present scale, and the temperature had cooled to about 3000 K. The average kinetic energy of nuclei, electrons, and photons was less than an electron volt. Since ionization energies of atoms are on the order of eV, then as the temperature dropped below this point, electrons could orbit the bare nuclei and remain there (without being ejected by collisions), thus forming atoms. This period is often called the **recombination** epoch (a misnomer since electrons had never before been combined with nuclei to form atoms). With the disappearance of free electrons and the birth of atoms, the photons—which had been continually scattering from the free electrons—now became free to spread throughout the universe. As mentioned in the previous Section, we say that the photons became **decoupled** from matter. Thus *decoupling* occurred at *recombination*. The energy contained in radiation had been decreasing (lengthening in wavelength as the universe expanded); and at about $t = 56{,}000\,\text{yr}$ (even before decoupling) the energy contained in matter became dominant over radiation. The universe was said to have become **matter-dominated** (marked on Fig. 33–29). As the universe continued to expand, the electromagnetic radiation cooled further, to 2.7 K today, forming the cosmic microwave background radiation we detect from everywhere in the universe.

After the birth of atoms, then stars and galaxies could begin to form: by self-gravitation around mass concentrations (inhomogeneities). Stars began to form about 200 million years after the Big Bang, galaxies after almost 10^9 years. The universe continued to evolve until today, some 14 billion years after it started.

* * *

This scenario, like other scientific models, cannot be said to be "proven." Yet this model is remarkably effective in explaining the evolution of the universe we live in, and makes predictions which can be tested against the next generation of observations.

A major event, and something only discovered recently, is that when the universe was about half as old as it is now (about 7 Gyr ago), its expansion began to accelerate. This was a big surprise because it was assumed the expansion of the universe would slow down due to gravitational attraction of all objects toward each other. This acceleration in the expansion of the universe is said to be due to "dark energy," as we discuss in Section 33–9. On the right in Fig. 33–29 is a narrow vertical strip that represents the most recent 7 billion years of the universe, during which *dark energy* seems to have dominated.

33–8 Inflation: Explaining Flatness, Uniformity, and Structure

The idea that the universe underwent a period of exponential inflation early in its life, expanding by a factor of 10^{30} or more (previous Section), was first put forth by Alan Guth and Andreí Linde. Many sophisticated models based on this general idea have since been proposed. The energy required for this wild expansion may have been due to fields somewhat like the Higgs field (Section 32–10). So far, the evidence for inflation is indirect; yet it is a feature of most viable cosmological models because it alone is able to provide natural explanations for several remarkable features of our universe.

[†]This Standard Model prediction of a 25% primordial production of helium agrees with what we observe today—the universe *does* contain about 25% He—and it is strong evidence in support of the Standard Big Bang Model. Furthermore, the theory says that 25% He abundance is fully consistent with there being three neutrino types, which is the number we observe. And it sets an upper limit of four to the maximum number of possible neutrino types. This is a striking example of the powerful connection between particle physics and cosmology.

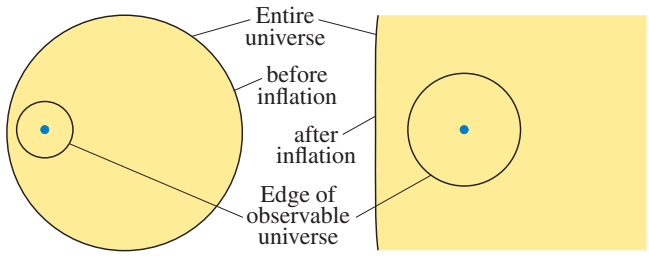


FIGURE 33–30 (a) Simple 2-D model of the entire universe; the observable universe is suggested by the small circle centered on us (blue dot). (b) Edge of entire universe is essentially flat after the 10^{30}-fold expansion during inflation.

Flatness

First of all, our best measurements suggest that the universe is flat, that it has zero curvature. As scientists, we would like some reason for this remarkable result. To see how inflation explains flatness, consider a simple 2-dimensional model of the universe as we did earlier in Figs. 33–16 and 33–21. A circle in this 2-dimensional universe (= surface of a sphere, Fig. 33–30a) represents the *observable* universe as seen by an observer at the blue dot. A possible hypothesis is that inflation occurred over a time interval that very roughly doubled the age of the universe from, let us say, $t = 1 \times 10^{-35}\,\text{s}$ to $t = 2 \times 10^{-35}\,\text{s}$. The size of the *observable* universe $(r = ct)$ would have increased by a factor of two during inflation, while the radius of curvature of the *entire* universe increased by an enormous factor of 10^{30} or more. Thus the edge of our 2-D sphere representing the entire universe would have seemed flat to a high degree of precision, as shown in Fig. 33–30b. Even if the time of inflation was a factor of 10 or 100 (instead of 2), the expansion factor of 10^{30} or more would have blotted out any possibility of observing anything but a flat universe.

CMB Uniformity

Inflation also explains why the CMB is so uniform. Without inflation, the tiny universe at $10^{-35}\,\text{s}$ would not have been small enough for all parts of it to have been in contact and so reach the same temperature (information cannot travel faster than c). To see this, suppose that the currently observable universe came from a region of space about 1 cm in diameter at $t = 10^{-36}\,\text{s}$, as per original Big Bang theory. In that $10^{-36}\,\text{s}$, light could have traveled $d = ct = (3 \times 10^8\,\text{m/s})(10^{-36}\,\text{s}) = 10^{-27}\,\text{m}$, way too small for the opposite sides of a 1-cm-wide "universe" to have been in communication. But if that region had been 10^{30} times smaller $(= 10^{-32}\,\text{m})$, as proposed by the inflation model, there could have been contact and thermal equilibrium to produce the observed nearly uniform CMB. Inflation, by making the very early universe extremely small, assures that all parts of that region which is today's observable universe could have been in thermal equilibrium. And after inflation the universe could be large enough to give us today's observable universe.

Galaxy Seeds, Fluctuations, Magnetic Monopoles

Inflation also gives us a clue as to how the present structure of the universe (galaxies and clusters of galaxies) came about. We saw earlier that, according to the uncertainty principle, energy might be not conserved by an amount ΔE for a time $\Delta t \approx \hbar/\Delta E$. Forces, whether electromagnetic or other types, can undergo such tiny **quantum fluctuations** according to quantum theory, but they are so tiny they are not detectable unless magnified in some way. That is what inflation might have done: it could have magnified those fluctuations perhaps 10^{30} times in size, which would give us the density irregularities seen in the cosmic microwave background (WMAP, Fig. 33–26). That would be very nice, because the density variations we see in the CMB are what we believe were the seeds that later coalesced under gravity into galaxies and galaxy clusters, and our models fit the data extremely well.

Sometimes it is said that the quantum fluctuations occurred in the **vacuum state** or vacuum energy. This could be possible because the vacuum is no longer considered to be empty, as we discussed in Section 32–3 relative to positrons as holes in a negative energy sea of electrons. Indeed, the vacuum is thought to be filled with fields and particles occupying all the possible negative energy states.

Also, the virtual exchange particles that carry the forces, as discussed in Chapter 32, could leave their brief virtual states and actually become real as a result of the 10^{30} magnification of space (according to inflation) and the very short time over which it occurred ($\Delta t = \hbar/\Delta E$).

Inflation helps us too with the puzzle of why **magnetic monopoles** (Section 20–1) have never been observed, yet isolated magnetic poles may well have been copiously produced at the start. After inflation, they would have been so far apart that we have never stumbled on one.

Inflation may solve outstanding problems, but we may need new physics to understand how inflation occurred. Many predictions of inflationary theory have been confirmed by recent cosmological observations.

33–9 Dark Matter and Dark Energy

According to the Standard Big Bang Model, the universe is evolving and changing. Individual stars are being created, evolving, and then dying to become white dwarfs, neutron stars, or black holes. At the same time, the universe as a whole is expanding. One important question is whether the universe will continue to expand forever. Until the late 1990s, the universe was thought to be dominated by matter which interacts by gravity, and the fate of the universe was connected to the curvature of space-time (Section 33–4). If the universe had *negative* curvature, the expansion of the universe would never stop, although the rate of expansion would decrease due to the gravitational attraction of its parts. Such a universe would be *open* and infinite. If the universe is *flat* (no curvature), it would still be open and infinite but its expansion would slowly approach a zero rate. If the universe had *positive* curvature, it would be *closed* and finite; the effect of gravity would be strong enough that the expansion would eventually stop and the universe would begin to contract, collapsing back onto itself in a **big crunch**.

Critical Density

According to the above scenario (which does not include inflation or the recently discovered acceleration of the universe), the fate of the universe would depend on the average mass–energy density in the universe. For an average mass density greater than a critical value known as the **critical density**, estimated to be about

$$\rho_c \approx 10^{-26}\,\text{kg/m}^3$$

(i.e., a few nucleons/m^3 on average throughout the universe), space-time would have a positive curvature and gravity would prevent expansion from continuing forever. Eventually (if $\rho > \rho_c$) gravity would pull the universe back into a big crunch. If instead the actual density was equal to the critical density, $\rho = \rho_c$, the universe would be flat and open, just barely expanding forever. If the actual density was less than the critical density, $\rho < \rho_c$, the universe would have negative curvature and would easily expand forever. See Fig. 33–31. Today we believe the universe is very close to flat. But recent evidence suggests the universe is expanding at an *accelerating* rate, as discussed below.

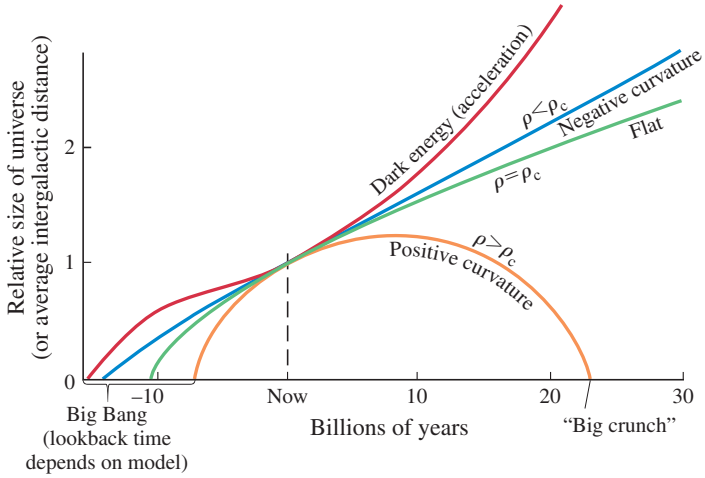


FIGURE 33–31 Three future possibilities for the universe, depending on the density ρ of ordinary matter, plus a fourth possibility that includes dark energy. Note that all curves have been chosen to have the same slope ($= H_0$, the Hubble parameter) right now. Looking back in time, the Big Bang occurs where each curve touches the horizontal (time) axis.

EXERCISE E Return to the Chapter-Opening Questions, page 947, and answer them again. Try to explain why you may have answered differently the first time.

Dark Matter

WMAP and other experiments have convinced scientists that the universe is flat and $\rho = \rho_c$. But this ρ cannot be only normal baryonic matter (atoms are 99.9% baryons—protons and neutrons—by weight). These recent experiments put the amount of normal baryonic matter in the universe at only about 5% of the critical density. What is the other 95%? There is strong evidence for a significant amount of nonluminous matter in the universe referred to as **dark matter**, which acts normally under gravity, but does not absorb or radiate light sufficiently to be visible. For example, observations of the rotation of galaxies suggest that they rotate as if they had considerably more mass than we can see. Recall from Chapter 5, Example 5–12, that for a satellite of mass m revolving around Earth (mass M)

$$m\frac{v^2}{r} = G\frac{mM}{r^2}$$

and hence $v = \sqrt{GM/r}$. If we apply this equation to stars in a galaxy, we see that their speed depends on galactic mass. Observations show that stars farther from the galactic center revolve much faster than expected if there is only the pull of visible matter, suggesting a great deal of invisible matter. Similarly, observations of the motion of galaxies within clusters also suggest that they have considerably more mass than can be seen. Furthermore, theory suggests that without dark matter, galaxies and stars probably would not have formed and would not exist. Dark matter seems to hold the universe together.

What might this nonluminous matter in the universe be? We don't know yet. But we hope to find out soon. It cannot be made of ordinary (baryonic) matter, so it must consist of some other sort of elementary particle, perhaps created at a very early time. Perhaps it is made up of previously undetected *weakly interacting massive particles* (**WIMP**s), possibly supersymmetric particles (Section 32–12) such as neutralinos. We are anxiously awaiting the results of intense searches for such particles, looking both at what arrives from far out in the cosmos with underground detectors[†], and by producing them in particle colliders (the LHC, Section 32–1).

Dark matter makes up roughly 25% of the mass–energy of the universe, according to the latest observations and models. Thus the total mass–energy is 25% dark matter plus 5% baryons for a total of about 30%, which does not bring ρ up to ρ_c. What is the other 70%? We are not sure about that either, but we have given it a name: "dark energy."

Dark Energy—Cosmic Acceleration

In 1998, just before the turn of the millennium, two groups, one led by Saul Perlmutter and the other by Brian Schmidt and Adam Riess (Fig. 33–32), reported a huge surprise. Gravity was assumed to be the predominant force on a large scale in the universe, and it was thought that the expansion of the universe ought to be slowing down in time because gravity acts as an attractive force between objects. But measurements of Type Ia supernovae (our best standard candles—see Section 33–3) unexpectedly showed that very distant (high z) supernovae were dimmer than expected. That is, given their great distance d as determined from their low brightness, their speed v as determined from the measured z was less than expected according to Hubble's law. This result suggests that nearer galaxies are moving away from us relatively faster than those very distant ones, meaning the expansion of the universe in more recent epochs has sped up.



FIGURE 33–32 Saul Perlmutter, center, flanked by Adam G. Riess (left) and Brian P. Schmidt, at the Nobel Prize celebrations, December 2011.

[†]In deep mines and under mountains to block out most other particles.

This **acceleration** in the expansion of the universe (in place of the expected deceleration due to gravitational attraction between masses) seems to have begun roughly 7 billion years ago (7 Gyr, which would be about halfway back to what we call the Big Bang).

What could be causing the universe to accelerate in its expansion, against the attractive force of gravity? Does our understanding of gravity need to be revised? We don't yet know the answers to these questions. There are several speculations. Somehow there seems to be a long-range *repulsive* effect on space, like a negative gravity, causing objects to speed away from each other ever faster. Whatever it is, it has been given the name **dark energy**. Many scientists say dark energy is the biggest mystery facing physical science today.

One idea is a sort of quantum field given the name **quintessence**. Another possibility suggests an energy latent in space itself (**vacuum energy**) and relates to an aspect of General Relativity known as the **cosmological constant** (symbol Λ). When Einstein developed his equations, he found that they offered no solutions for a static universe. In those days (1917) it was thought the universe was static—unchanging and everlasting. Einstein added an arbitrary constant (Λ) to his equations to provide solutions for a static universe.[†] A decade later, when Hubble showed us an expanding universe, Einstein discarded his cosmological constant as no longer needed ($\Lambda = 0$). But today, measurements are consistent with dark energy being due to a nonzero cosmological constant, although further measurements are needed to see subtle differences among theories.

There is increasing evidence that the effects of some form of dark energy are very real. Observations of the CMB, supernovae, and large-scale structure (Section 33–10) agree well with theories and computer models when they input dark energy as providing about 70% of the mass–energy in the universe, and when the total mass–energy density equals the critical density ρ_c.

Today's best estimate of how the mass–energy in the universe is distributed is approximately (see also Fig. 33–33):

70% dark energy

30% matter, subject to the known gravitational force.

Of this 30%, about

25% is dark matter

5% is baryons (what atoms are made of); of this 5% only $\frac{1}{10}$ is readily visible matter—stars and galaxies (that is, 0.5% of the total); the other $\frac{9}{10}$ of ordinary matter, which is not visible, is mainly gaseous plasma.

It is remarkable that only 0.5% of all the mass–energy in the universe is visible as stars and galaxies.

The idea that the universe is dominated by completely unknown forms of matter and energy seems bizarre. Nonetheless, the ability of our present model to precisely explain observations of the CMB anisotropy, cosmic expansion, and large-scale structure (next Section) presents a compelling case.

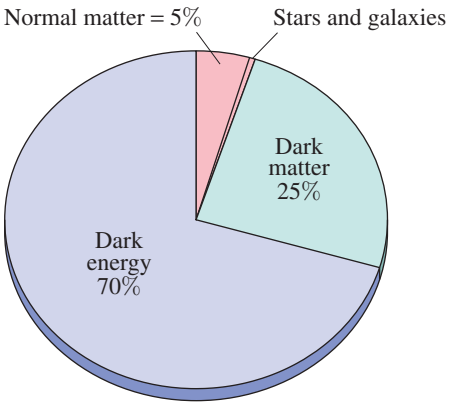


FIGURE 33–33 Portions of total mass–energy in the universe (approximate).

33–10 Large-Scale Structure of the Universe

The beautiful WMAP pictures of the sky (Fig. 33–26) show small but significant inhomogeneities in the temperature of the cosmic microwave background (CMB). These anisotropies reflect compressions and expansions in the primordial plasma just before decoupling (Fig. 33–29), from which galaxies and clusters of galaxies formed. Analyses of the irregularities in the CMB using mammoth computer

[†]It seems strange that Einstein and other scientists believed in a static universe. The ancients, including the Roman Lucretius argued against it: *If there was no birth-time of earth and heaven and they have been from everlasting, why before the* [Trojan] *war . . . . have not other poets as well sung other themes?* [The reference is to Homer being the oldest known writings.] The ancient Hebrews also argued for a beginning (like our Big Bang): see Genesis.

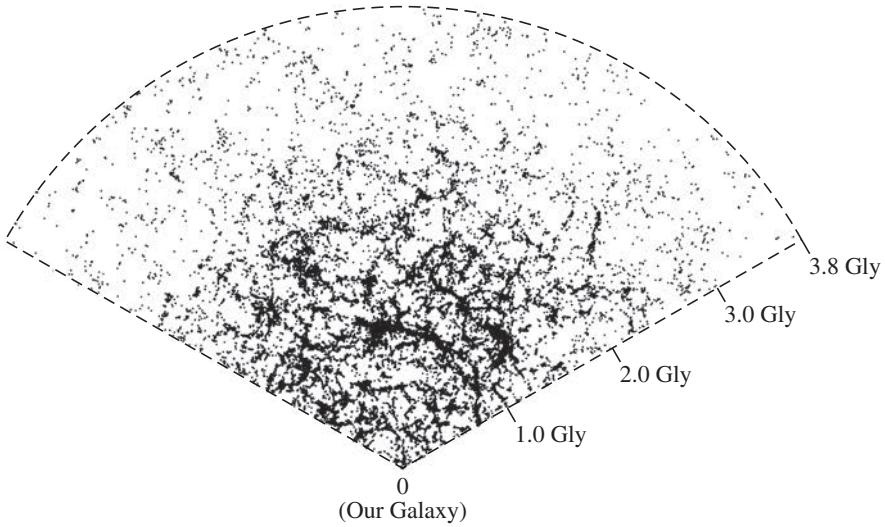


FIGURE 33–34 Distribution of some 50,000 galaxies in a 2.5° slice through almost half of the sky above the equator, as measured by the Sloan Digital Sky Survey (SDSS). Each dot represents a galaxy. The distance from us is obtained from the redshift and Hubble's law, and is given in units of 10^9 light-years (Gly). The point 0 represents us, our observation point. This diagram may seem to put us at the center, but remember that at greater distances, fewer galaxies are bright enough to be detected, thus resulting in an apparent thinning out of galaxies. Note the "walls" and "voids" of galaxies.

simulations predict a large-scale distribution of galaxies very similar to what is seen today (Fig. 33–34). These simulations are very successful if they contain dark energy and dark matter; and the dark matter needs to be *cold* (slow speed—think of Eq. 13–8, $\frac{1}{2}m\overline{v^2} = \frac{3}{2}kT$ where T is temperature), rather than "hot" dark matter such as neutrinos which move at or very near the speed of light. Indeed, the modern **cosmological model** is called the **ΛCDM** model, where lambda (Λ) stands for the cosmological constant, and CDM is **cold dark matter**.

Cosmologists have gained substantial confidence in this cosmological model from such a precise fit between observations and theory. They can also extract very precise values for cosmological parameters which previously were only known with low accuracy. The CMB is such an important cosmological observable that every effort is being made to extract all of the information it contains. A new generation of ground, balloon, and satellite experiments is observing the CMB with greater resolution and sensitivity. They may detect interaction of **gravity waves** (produced in the inflationary epoch) with the CMB and thereby provide direct evidence for cosmic inflation, and also provide information about elementary particle physics at energies far beyond the reach of man-made accelerators and colliders.

33–11 Finally . . .

When we look up into the night sky, we see stars; and with the best telescopes, we see galaxies and the exotic objects we discussed earlier, including rare supernovae. But even with our best instruments we do not see the processes going on inside stars and supernovae that we hypothesized (and believe). We are dependent on brilliant theorists who come up with viable theories and verifiable models. We depend on complicated computer models whose parameters are varied until the outputs compare favorably with our observations and analyses of WMAP and other experiments. And we now have a surprisingly precise idea about some aspects of our universe: it is flat, it is about 14 billion years old, it contains only 5% "normal" baryonic matter (for atoms), and so on.

The questions raised by cosmology are difficult and profound, and may seem removed from everyday "reality." We can always say, "the Sun is shining, it's going to shine on for an unimaginably long time, all is well." Nonetheless, the questions of cosmology are deep ones that fascinate the human intellect. One aspect that is especially intriguing is this: calculations on the formation and evolution of the universe have been performed that deliberately varied the values—just slightly—of certain fundamental physical constants. The result?

A universe in which life as we know it could not exist. [For example, if the difference in mass between a proton and a neutron were zero, or less than the mass of the electron, $0.511\,\text{MeV}/c^2$, there would be no atoms: electrons would be captured by protons to make neutrons.] Such results have contributed to a philosophical idea called the **anthropic principle**, which says that if the universe were even a little different than it is, we could not be here. We physicists are trying to find out if there are some undiscovered fundamental laws that determined those conditions that allowed us to exist. A poet might say that the universe is exquisitely tuned, almost as if to accommodate us.

Summary

The night sky contains myriads of stars including those in the Milky Way, which is a "side view" of our **Galaxy** looking along the plane of the disk. Our Galaxy includes over 10^{11} stars. Beyond our Galaxy are billions of other galaxies.

Astronomical distances are measured in **light-years** $(1\,\text{ly} \approx 10^{13}\,\text{km})$. The nearest star is about 4 ly away and the nearest large galaxy is 2 million ly away. Our Galactic disk has a diameter of about 100,000 ly. [Distances are sometimes specified in **parsecs**, where 1 parsec = 3.26 ly.]

Stars are believed to begin life as collapsing masses of gas (protostars), largely hydrogen. As they contract, they heat up (potential energy is transformed to kinetic energy). When the temperature reaches about 10 million degrees, nuclear fusion begins and forms heavier elements (**nucleosynthesis**), mainly helium at first. The energy released during these reactions heats the gas so its outward pressure balances the inward gravitational force, and the young star stabilizes as a **main-sequence** star. The tremendous luminosity of stars comes from the energy released during these thermonuclear reactions. After billions of years, as helium is collected in the core and hydrogen is used up, the core contracts and heats further. The outer envelope expands and cools, and the star becomes a **red giant** (larger diameter, redder color).

The next stage of stellar evolution depends on the mass of the star, which may have lost much of its original mass as its outer envelope escaped into space. Stars of residual mass less than about 1.4 solar masses cool further and become **white dwarfs**, eventually fading and going out altogether. Heavier stars contract further due to their greater gravity: the density approaches nuclear density, the huge pressure forces electrons to combine with protons to form neutrons, and the star becomes essentially a huge nucleus of neutrons. This is a **neutron star**, and the energy released during its final core collapse is believed to produce **supernova** explosions. If the star is very massive, it may contract even further and form a **black hole**, which is so dense that no matter or light can escape from it.

In the **general theory of relativity**, the **equivalence principle** states that an observer cannot distinguish acceleration from a gravitational field. Said another way, gravitational and inertial masses are the same. The theory predicts gravitational bending of light rays to a degree consistent with experiment. Gravity is treated as a curvature in space and time, the curvature being greater near massive objects. The universe as a whole may be curved. With sufficient mass, the curvature of the universe would be positive, and the universe is *closed* and *finite*; otherwise, it would be *open* and *infinite*. Today we believe the universe is **flat**.

Distant galaxies display a **redshift** in their spectral lines, originally interpreted as a Doppler shift. The universe is observed to be **expanding**, its galaxies racing away from each other at speeds (v) proportional to the distance (d) between them:

$$v = H_0 d, \qquad \textbf{(33–4)}$$

which is known as **Hubble's law** (H_0 is the **Hubble parameter**). This expansion of the universe suggests an explosive origin, the **Big Bang**, which occurred about 13.8 billion years ago. It is not like an ordinary explosion, but rather an expansion of space itself.

The **cosmological principle** assumes that the universe, on a large scale, is homogeneous and isotropic.

Important evidence for the Big Bang model of the universe was the discovery of the **cosmic microwave background** radiation (CMB), which conforms to a blackbody radiation curve at a temperature of 2.725 K.

The **Standard Model** of the Big Bang provides a possible scenario as to how the universe developed as it expanded and cooled after the Big Bang. Starting at 10^{-43} seconds after the Big Bang, according to this model, the universe underwent a brief but rapid exponential expansion, referred to as **inflation**. Shortly thereafter, quarks were **confined** into hadrons (the **hadron era**). About 10^{-4} s after the Big Bang, the majority of hadrons disappeared, having combined with anti-hadrons, producing photons, leptons, and energy, leaving mainly photons and leptons to freely move, thus introducing the **lepton era**. By the time the universe was about 10 s old, the electrons too had mostly disappeared, having combined with their antiparticles; the universe was **radiation-dominated**. A couple of minutes later, nucleosynthesis began, but lasted only a few minutes. It then took almost four hundred thousand years before the universe was cool enough for electrons to combine with nuclei to form atoms (**recombination**). Photons, up to then continually being scattered off of free electrons, could now move freely—they were **decoupled** from matter and the universe became transparent. The background radiation had expanded and cooled so much that its total energy became less than the energy in matter, and **matter dominated** increasingly over radiation. Then stars and galaxies formed, producing a universe not much different than it is today—some 14 billion years later.

Recent observations indicate that the universe is essentially flat, that it contains an as-yet unknown type of **dark matter**, and that it is dominated by a mysterious **dark energy** which exerts a sort of negative gravity causing the expansion of the universe to accelerate. The total contributions of baryonic (normal) matter, dark matter, and dark energy sum up to the **critical density**.

Questions

1. The Milky Way was once thought to be "murky" or "milky" but is now considered to be made up of point sources. Explain.
2. A star is in equilibrium when it radiates at its surface all the energy generated in its core. What happens when it begins to generate more energy than it radiates? Less energy? Explain.
3. Describe a red giant star. List some of its properties.
4. Does the H–R diagram directly reveal anything about the core of a star?
5. Why do some stars end up as white dwarfs, and others as neutron stars or black holes?
6. If you were measuring star parallaxes from the Moon instead of Earth, what corrections would you have to make? What changes would occur if you were measuring parallaxes from Mars?
7. *Cepheid variable* stars change in luminosity with a typical period of several days. The period has been found to have a definite relationship with the average intrinsic luminosity of the star. How could these stars be used to measure the distance to galaxies?
8. What is a geodesic? What is its role in General Relativity?
9. If it were discovered that the redshift of spectral lines of galaxies was due to something other than expansion, how might our view of the universe change? Would there be conflicting evidence? Discuss.
10. Almost all galaxies appear to be moving away from us. Are we therefore at the center of the universe? Explain.
11. If you were located in a galaxy near the boundary of our observable universe, would galaxies in the direction of the Milky Way appear to be approaching you or receding from you? Explain.
12. Compare an explosion on Earth to the Big Bang. Consider such questions as: Would the debris spread at a higher speed for more distant particles, as in the Big Bang? Would the debris come to rest? What type of universe would this correspond to, open or closed?
13. If nothing, not even light, escapes from a black hole, then how can we tell if one is there?
14. The Earth's age is often given as about 4.6 billion years. Find that time on Fig. 33–29. Modern humans have lived on Earth on the order of 200,000 years. Where is that on Fig. 33–29?
15. Why were atoms, as opposed to bare nuclei, unable to exist until hundreds of thousands of years after the Big Bang?
16. (*a*) Why are Type Ia supernovae so useful for determining the distances of galaxies? (*b*) How are their distances actually measured?
17. Under what circumstances would the universe eventually collapse in on itself?
18. (*a*) Why did astronomers expect that the expansion rate of the universe would be decreasing (decelerating) with time? (*b*) How, in principle, could astronomers hope to determine whether the universe used to expand faster than it does now?

MisConceptual Questions

1. Which one of the following is *not* expected to occur on an H–R diagram during the lifetime of a single star?
   (*a*) The star will move off the main sequence toward the upper right of the diagram.
   (*b*) Low-mass stars will become white dwarfs and end up toward the lower left of the diagram.
   (*c*) The star will move along the main sequence from one place to another.
   (*d*) All of the above.
2. When can parallax be used to determine the approximate distance from the Earth to a star?
   (*a*) Only during January and July.
   (*b*) Only when the star's distance is relatively small.
   (*c*) Only when the star's distance is relatively large.
   (*d*) Only when the star appears to move directly toward or away from the Earth.
   (*e*) Only when the star is the Sun.
   (*f*) Always.
   (*g*) Never.
3. Observations show that all galaxies tend to move away from Earth, and that more distant galaxies move away from Earth at faster velocities than do galaxies closer to the Earth. These observations imply that
   (*a*) the Earth is the center of the universe.
   (*b*) the universe is expanding.
   (*c*) the expansion of the universe will eventually stop.
   (*d*) All of the above.
4. Which process results in a tremendous amount of energy being emitted by the Sun?
   (*a*) Hydrogen atoms burn in the presence of oxygen—that is, hydrogen atoms oxidize.
   (*b*) The Sun contracts, decreasing its gravitational potential energy.
   (*c*) Protons in hydrogen atoms fuse, forming helium nuclei.
   (*d*) Radioactive atoms such as uranium, plutonium, and cesium emit gamma rays with high energy.
   (*e*) None of the above.
5. Which of the following methods can be used to find the distance from us to a star outside our galaxy? Choose all that apply.
   (*a*) Parallax.
   (*b*) Using luminosity and temperature from the H–R diagram and measuring the apparent brightness.
   (*c*) Using supernova explosions as a "standard candle."
   (*d*) Redshift in the line spectra of elements and compounds.
6. The history of the universe can be determined by observing astronomical objects at various (large) distances from the Earth. This method of discovery works because
   (*a*) time proceeds at different rates in different regions of the universe.
   (*b*) light travels at a finite speed.
   (*c*) matter warps space.
   (*d*) older galaxies are farther from the Earth than are younger galaxies.

7. Where did the Big Bang occur?
(*a*) Near the Earth.
(*b*) Near the center of the Milky Way Galaxy.
(*c*) Several billion light-years away.
(*d*) Throughout all space.
(*e*) Near the Andromeda Galaxy.

8. When and how were virtually all of the elements of the Periodic Table formed?
(*a*) In the very early universe a few seconds after the Big Bang.
(*b*) At the centers of stars during their main-sequence phases.
(*c*) At the centers of stars during novae.
(*d*) At the centers of stars during supernovae.
(*e*) On the surfaces of planets as they cooled and hardened.

9. We know that there must be dark matter in the universe because
(*a*) we see dark dust clouds.
(*b*) we see that the universe is expanding.
(*c*) we see that stars far from the galactic center are moving faster than can be explained by visible matter.
(*d*) we see that the expansion of the universe is accelerating.

10. Acceleration of the universe's expansion rate is due to
(*a*) the repulsive effect of dark energy.
(*b*) the attractive effect of dark matter.
(*c*) the attractive effect of gravity.
(*d*) the thermal expansion of stellar cores.

For assigned homework and other learning materials, go to the MasteringPhysics website.

Problems

33–1 to 33–3 Stars, Galaxies, Stellar Evolution, Distances

1. (I) The parallax angle of a star is 0.00029°. How far away is the star?
2. (I) A star exhibits a parallax of 0.27 seconds of arc. How far away is it?
3. (I) If one star is twice as far away from us as a second star, will the parallax angle of the farther star be greater or less than that of the nearer star? By what factor?
4. (II) What is the relative brightness of the Sun as seen from Jupiter, as compared to its brightness from Earth? (Jupiter is 5.2 times farther from the Sun than the Earth is.)
5. (II) When our Sun becomes a red giant, what will be its average density if it expands out to the orbit of Mercury (6×10^{10} m from the Sun)?
6. (II) We saw earlier (Chapter 14) that the rate energy reaches the Earth from the Sun (the "solar constant") is about $1.3 \times 10^3\ \text{W/m}^2$. What is (*a*) the apparent brightness *b* of the Sun, and (*b*) the intrinsic luminosity *L* of the Sun?
7. (II) Estimate the angular width that our Galaxy would subtend if observed from the nearest galaxy to us (Table 33–1). Compare to the angular width of the Moon from Earth.
8. (II) Assuming our Galaxy represents a good average for all other galaxies, how many stars are in the observable universe?
9. (II) Calculate the density of a white dwarf whose mass is equal to the Sun's and whose radius is equal to the Earth's. How many times larger than Earth's density is this?
10. (II) A neutron star whose mass is 1.5 solar masses has a radius of about 11 km. Calculate its average density and compare to that for a white dwarf (Problem 9) and to that of nuclear matter.

*11. (II) A star is 56 pc away. What is its parallax angle? State (*a*) in seconds of arc, and (*b*) in degrees.

*12. (II) What is the parallax angle for a star that is 65 ly away? How many parsecs is this?

*13. (II) A star is 85 pc away. How long does it take for its light to reach us?

14. (III) Suppose two stars of the same apparent brightness *b* are also believed to be the same size. The spectrum of one star peaks at 750 nm whereas that of the other peaks at 450 nm. Use Wien's law and the Stefan-Boltzmann equation (Eq. 14–6) to estimate their relative distances from us. [*Hint*: See Examples 33–4 and 33–5.]
15. (III) Stars located in a certain cluster are assumed to be about the same distance from us. Two such stars have spectra that peak at $\lambda_1 = 470$ nm and $\lambda_2 = 720$ nm, and the ratio of their apparent brightness is $b_1/b_2 = 0.091$. Estimate their relative sizes (give ratio of their diameters) using Wien's law and the Stefan-Boltzmann equation, Eq. 14–6.

33–4 General Relativity, Gravity and Curved Space

16. (I) Show that the Schwarzschild radius for Earth is 8.9 mm.
17. (II) What is the Schwarzschild radius for a typical galaxy (like ours)?
18. (II) What mass will give a Schwarzschild radius equal to that of the hydrogen atom in its ground state?
19. (II) What is the maximum sum-of-the-angles for a triangle on a sphere?
20. (II) Describe a triangle, drawn on the surface of a sphere, for which the sum of the angles is (*a*) 359°, and (*b*) 179°.
21. (III) What is the apparent deflection of a light beam in an elevator (Fig. 33–13) which is 2.4 m wide if the elevator is accelerating downward at $9.8\ \text{m/s}^2$?

33–5 Redshift, Hubble's Law

22. (I) The redshift of a galaxy indicates a recession velocity of 1850 km/s. How far away is it?
23. (I) If a galaxy is traveling away from us at 1.5% of the speed of light, roughly how far away is it?
24. (II) A galaxy is moving away from Earth. The "blue" hydrogen line at 434 nm emitted from the galaxy is measured on Earth to be 455 nm. (*a*) How fast is the galaxy moving? (*b*) How far is it from Earth based on Hubble's law?
25. (II) Estimate the wavelength shift for the 656.3-nm line in the Balmer series of hydrogen emitted from a galaxy whose distance from us is (*a*) 7.0×10^6 ly, (*b*) 7.0×10^7 ly.

26. (II) If an absorption line of calcium is normally found at a wavelength of 393.4 nm in a laboratory gas, and you measure it to be at 423.4 nm in the spectrum of a galaxy, what is the approximate distance to the galaxy?

27. (II) What is the speed of a galaxy with $z = 0.060$?

28. (II) What would be the redshift parameter z for a galaxy traveling away from us at $v = 0.075c$?

29. (II) Estimate the distance d from the Earth to a galaxy whose redshift parameter $z = 1$.

30. (II) Estimate the speed of a galaxy, and its distance from us, if the wavelength for the hydrogen line at 434 nm is measured on Earth as being 610 nm.

31. (III) Starting from Eq. 33–3, show that the Doppler shift in wavelength is $\Delta\lambda/\lambda_{\text{rest}} \approx v/c$ (Eq. 33–6) for $v \ll c$. [*Hint*: Use the binomial expansion.]

33–6 to 33–8 The Big Bang, CMB, Universe Expansion

32. (I) Calculate the wavelength at the peak of the blackbody radiation distribution at 2.7 K using Wien's law.

33. (II) Calculate the peak wavelength of the CMB at 1.0 s after the birth of the universe. In what part of the EM spectrum is this radiation?

34. (II) The critical density for closure of the universe is $\rho_c \approx 10^{-26}\,\text{kg/m}^3$. State ρ_c in terms of the average number of nucleons per cubic meter.

35. (II) The scale factor of the universe (average distance between galaxies) at any given time is believed to have been inversely proportional to the absolute temperature. Estimate the size of the universe, compared to today, at (*a*) $t = 10^6\,\text{yr}$, (*b*) $t = 1\,\text{s}$, (*c*) $t = 10^{-6}\,\text{s}$, and (*d*) $t = 10^{-35}\,\text{s}$.

36. (II) At approximately what time had the universe cooled below the threshold temperature for producing (*a*) kaons $(M \approx 500\,\text{MeV}/c^2)$, (*b*) $\Upsilon\,(M \approx 9500\,\text{MeV}/c^2)$, and (*c*) muons $(M \approx 100\,\text{MeV}/c^2)$?

33–9 Dark Matter, Dark Energy

37. (II) Only about 5% of the energy in the universe is composed of baryonic matter. (*a*) Estimate the average density of baryonic matter in the observable universe with a radius of 14 billion light-years that contains 10^{11} galaxies, each with about 10^{11} stars like our Sun. (*b*) Estimate the density of dark matter in the universe.

General Problems

38. Use conservation of angular momentum to estimate the angular velocity of a neutron star which has collapsed to a diameter of 16 km, from a star whose core radius was equal to that of Earth $(6 \times 10^6\,\text{m})$. Assume its mass is 1.5 times that of the Sun, and that it rotated (like our Sun) about once a month.

39. By what factor does the rotational kinetic energy change when the star in Problem 38 collapses to a neutron star?

40. Suppose that three main-sequence stars could undergo the three changes represented by the three arrows, A, B, and C, in the H–R diagram of Fig. 33–35. For each case, describe the changes in temperature, intrinsic luminosity, and size.

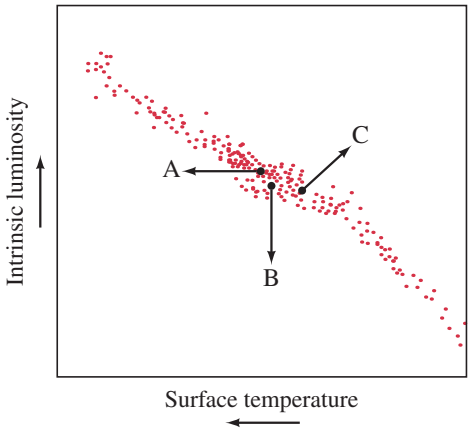


FIGURE 33–35 Problem 40.

41. Assume that the nearest stars to us have an intrinsic luminosity about the same as the Sun's. Their apparent brightness, however, is about 10^{11} times fainter than the Sun. From this, estimate the distance to the nearest stars.

42. A certain pulsar, believed to be a neutron star of mass 1.5 times that of the Sun, with diameter 16 km, is observed to have a rotation speed of 1.0 rev/s. If it loses rotational kinetic energy at the rate of 1 part in 10^9 per day, which is all transformed into radiation, what is the power output of the star?

43. The nearest large galaxy to our Galaxy is about 2×10^6 ly away. If both galaxies have a mass of $4 \times 10^{41}\,\text{kg}$, with what gravitational force does each galaxy attract the other? Ignore dark matter.

44. How large would the Sun be if its density equaled the critical density of the universe, $\rho_c \approx 10^{-26}\,\text{kg/m}^3$? Express your answer in light-years and compare with the Earth–Sun distance and the diameter of our Galaxy.

45. Two stars, whose spectra peak at 660 nm and 480 nm, respectively, both lie on the main sequence. Use Wien's law, the Stefan-Boltzmann equation, and the H–R diagram (Fig. 33–6) to estimate the ratio of their diameters.

46. (*a*) In order to measure distances with parallax at 100 ly, what minimum angular resolution (in degrees) is needed? (*b*) What diameter mirror or lens would be needed?

47. In the later stages of stellar evolution, a star (if massive enough) will begin fusing carbon nuclei to form, for example, magnesium:

$$^{12}_{6}\mathrm{C} + {}^{12}_{6}\mathrm{C} \rightarrow {}^{24}_{12}\mathrm{Mg} + \gamma.$$

(*a*) How much energy is released in this reaction (see Appendix B)? (*b*) How much kinetic energy must each carbon nucleus have (assume equal) in a head-on collision if they are just to "touch" (use Eq. 30–1) so that the strong force can come into play? (*c*) What temperature does this kinetic energy correspond to?

48. Use *dimensional analysis* with the fundamental constants c, G, and $\hbar$ to estimate the value of the so-called *Planck time.* It is thought that physics as we know it can say nothing about the universe before this time.

49. Estimate the mass of our observable universe using the following assumptions: Our universe is spherical in shape, it has been expanding at the speed of light since the Big Bang, and its density is the critical density.

Search and Learn

1. Estimate what neutrino mass (in eV/c^2) would provide the critical density to close the universe. Assume the neutrino density is, like photons, about 10^9 times that of nucleons, and that nucleons make up only (*a*) 2% of the mass needed, or (*b*) 5% of the mass needed.

2. Describe how we can estimate the distance from us to other stars. Which methods can we use for nearby stars, and which can we use for very distant stars? Which method gives the most accurate distance measurements for the most distant stars?

3. The evolution of stars, as discussed in Section 33–2, can lead to a white dwarf, a neutron star, or even a black hole, depending on the mass. (*a*) Referring to Sections 33–2 and 33–4, give the radius of (i) a white dwarf of 1 solar mass, (ii) a neutron star of 1.5 solar masses, and (iii) a black hole of 3 solar masses. (*b*) Express these three radii as ratios ($r_{\mathrm{i}} : r_{\mathrm{ii}} : r_{\mathrm{iii}}$).

4. When stable nuclei first formed, about 3 minutes after the Big Bang, there were about 7 times more protons than neutrons. Explain how this leads to a ratio of the mass of hydrogen to the mass of helium of 3:1. This is about the actual ratio observed in the universe.

5. We cannot use Hubble's law to measure the distances to nearby galaxies, because their random motions are larger than the overall expansion. Indeed, the closest galaxy to us, the Andromeda Galaxy, 2.5 million light-years away, is approaching us at a speed of about 130 km/s. (*a*) What is the shift in wavelength of the 656-nm line of hydrogen emitted from the Andromeda Galaxy, as seen by us? (*b*) Is this a redshift or a blueshift? (*c*) Ignoring the expansion, how soon will it and the Milky Way Galaxy collide?

ANSWERS TO EXERCISES

A: Our Earth and ourselves, 2 years ago.

B: 600 ly (estimating L from Fig. 33–6 as $L \approx 8 \times 10^{26}$ W; note that on a log scale, 6000 K is closer to 7000 K than it is to 5000 K).

C: 30 km.

D: (*a*); not the usual R^3, but R: see formula for the Schwarzschild radius.

E: (*c*); (*d*).

APPENDIX A

Mathematical Review

A–1 Relationships, Proportionality, and Equations

One of the important aspects of physics is the search for relationships between different quantities—that is, determining how one quantity affects another.

As a simple example, the ancients found that if one circle has twice the diameter of a second circle, the first also has twice the circumference. If the diameter is three times as large, the circumference is also three times as large. In other words, an increase in the diameter results in a proportional increase in the circumference. We say that the circumference is **directly proportional** to the diameter. This can be written in symbols as $C \propto D$, where "$\propto$" means "is proportional to," and C and D refer to the circumference and diameter of a circle, respectively. The next step is to change this proportionality to an equation, which will make it possible to link the two quantities numerically. This means inserting a proportionality constant, which in many cases is determined by measurement. The ancients found that the ratio of the circumference to the diameter of any circle was 3.1416 (to keep only the first few decimal places). This number is designated by the Greek letter π. It is the constant of proportionality for the relationship $C \propto D$. To obtain an equation, we insert π into the proportion and change the $\propto$ to $=$. Thus,

$$C = \pi D.$$

Other kinds of proportionality occur as well. For example, the area of a circle is proportional to the *square* of its radius. That is, if the radius is doubled, the area becomes four times as large; and so on. In this case we can write $A \propto r^2$, where A stands for the area and r for the radius of the circle. The constant of proportionality is found to be π again: $A = \pi r^2$.

Sometimes two quantities are related in such a way that an increase in one leads to a proportional *decrease* in the other. This is called **inverse proportion**. For example, the time required to travel a given distance is inversely proportional to the speed of travel. The greater the speed, the less time it takes. We can write this inverse proportion as

$$\text{time} \propto 1/\text{speed}.$$

The larger the denominator of a fraction, the lower the value of the fraction is as a whole. For example, $\frac{1}{4}$ is less than $\frac{1}{2}$. Thus, if the speed is doubled, the time is halved, which is what we want to express by this inverse proportionality relationship.

If you suspect that a relationship exists between two or more quantities, you can try to determine the precise nature of this relationship by varying one of the quantities and measuring how the other varies as a result. Sometimes a given quantity is affected by two or more quantities; for instance, the acceleration of an object is related to both its mass and the applied force. In such a case, only one quantity is varied at a time, while the others are held constant.

When one quantity affects another, we often use the expression **is a function of** to indicate this dependence; for example, we say that the pressure in a tire is a function of the temperature.

Whatever kind of proportion is found to hold, it can be changed to an equality by finding the proper proportionality constant. Quantitative statements or predictions about the physical world can then be made with the equation.

A–2 Exponents

When we write 10^4, we mean that you multiply 10 by itself four times: $10^4 = 10 \times 10 \times 10 \times 10 = 10{,}000$. The superscript 4 is called an **exponent**, and 10 is said to be raised to the fourth power. Any number or symbol can be raised to a power. Special names are used when the exponent is 2 (a^2 is "a squared") or 3 (a^3 is "a cubed"). For any other power, we say a^n is "a to the nth power." If the exponent is 1, it is usually dropped: $a^1 = a$, since no multiplication is involved.

The rules for multiplying numbers expressed as powers are as follows: first,

$$(a^n)(a^m) = a^{n+m}. \qquad \textbf{(A–1)}$$

That is, the exponents are added. To see why, consider the result of the multiplication of 3^3 by 3^4:

$$(3^3)(3^4) = (3)(3)(3) \times (3)(3)(3)(3) = (3)^7.$$

Here the sum of the exponents is $3 + 4 = 7$, so rule A–1 works. Notice that this rule works only if the base numbers (a in Eq. A–1) are the same. Thus we *cannot* use the rule of summing exponents for $(6^3)(5^2)$; these numbers would have to be written out. However, if the base numbers are different but the exponents are the same, we can write a second rule:

$$(a^n)(b^n) = (ab)^n. \qquad \textbf{(A–2)}$$

For example, $(5^3)(6^3) = (30)^3$ since

$$(5)(5)(5)(6)(6)(6) = (30)(30)(30).$$

The third rule involves a power raised to another power: $(a^3)^2$ means $(a^3)(a^3)$, which is equal to $a^{3+3} = a^6$. The general rule is then

$$(a^n)^m = a^{nm}. \qquad \textbf{(A–3)}$$

In this case, the exponents are multiplied.

Negative exponents are used for reciprocals. Thus,

$$\frac{1}{a} = a^{-1}, \qquad \frac{1}{a^3} = a^{-3},$$

and so on. The reason for using negative exponents is to allow us to use the multiplication rules given above. For example, $(a^5)(a^{-3})$ means

$$\frac{(a)(a)(a)(a)(a)}{(a)(a)(a)} = a^2,$$

after canceling 3 of the a's. Rule A–1 gives us the same result:

$$(a^5)(a^{-3}) = a^{5-3} = a^2.$$

What does an exponent of zero mean? That is, what is a^0? Any number raised to the zeroth power is defined as being equal to 1:

$$a^0 = 1.$$

This definition is used because it follows from the rules for adding exponents. For example,

$$a^3a^{-3} = a^{3-3} = a^0 = 1.$$

But *does* a^3a^{-3} actually equal 1? Yes, because

$$a^3a^{-3} = \frac{a^3}{a^3} = 1.$$

Fractional exponents are used to represent *roots*. For example, $a^{\frac{1}{2}}$ means the square root of a; that is, $a^{\frac{1}{2}} = \sqrt{a}$. Similarly, $a^{\frac{1}{3}}$ means the cube root of a, and so on. The fourth root of a means that if you multiply the fourth root of a by itself four times, you again get a:

$$(a^{\frac{1}{4}})^4 = a.$$

This is consistent with rule A–3 since $(a^{\frac{1}{4}})^4 = a^{\frac{4}{4}} = a^1 = a$.

A–3 Powers of 10, or Exponential Notation

Writing out very large and very small numbers such as the distance of Neptune from the Sun, 4,500,000,000 km, or the diameter of a typical atom, 0.00000001 cm, is inconvenient and prone to error. It also leaves in question (see Section 1–4) the number of significant figures. (How many of the zeros are significant in the number 4,500,000,000 km?)

For these reasons we make use of the "powers of 10," or exponential notation. The distance from Neptune to the Sun is then expressed as 4.50×10^9 km (assuming that the value is significant to three digits), and the diameter of an atom 1.0×10^{-8} cm. This way of writing numbers is based on the use of exponents, where a^n signifies a multiplied by itself n times. For example, $10^4 = 10 \times 10 \times 10 \times 10 = 10{,}000$. Thus, $4.50 \times 10^9 = 4.50 \times 1{,}000{,}000{,}000 = 4{,}500{,}000{,}000$. Notice that the exponent (9 in this case) is just the number of places the decimal point is moved to the right to obtain the fully written-out number (4.500,000,000.)

When two numbers are multiplied (or divided), you first multiply (or divide) the simple parts and then the powers of 10. Thus, 2.0×10^3 multiplied by 5.5×10^4 equals $(2.0 \times 5.5) \times (10^3 \times 10^4) = 11 \times 10^7$, where we have used the rule for adding exponents (Appendix A–2). Similarly, 8.2×10^5 divided by 2.0×10^2 equals

$$\frac{8.2 \times 10^5}{2.0 \times 10^2} = \frac{8.2}{2.0} \times \frac{10^5}{10^2} = 4.1 \times 10^3.$$

For numbers less than 1, say 0.01, the exponent power of 10 is written with a negative sign (see previous page): $0.01 = 1/100 = 1/10^2 = 1 \times 10^{-2}$. Similarly, $0.002 = 2 \times 10^{-3}$. The decimal point has again been moved the number of places expressed in the exponent. For example, $0.020 \times 3600 = 72$, or in exponential notation $(2.0 \times 10^{-2}) \times (3.6 \times 10^3) = 7.2 \times 10^1 = 72$.

Notice also that $10^1 \times 10^{-1} = 10 \times 0.1 = 1$, and by the law of exponents, $10^1 \times 10^{-1} = 10^0$. Therefore, $10^0 = 1$.

When writing a number in exponential notation, it is usual to make the simple number be between 1 and 10. Thus it is conventional to write 4.5×10^9 rather than 45×10^8, although they are the same number.[†] This notation also allows the number of *significant figures* to be clearly expressed. We write 4.50×10^9 if this value is accurate to three significant figures, but 4.5×10^9 if it is accurate to only two.

A–4 Algebra

Physical relationships between quantities can be represented as equations involving symbols (usually letters of the Greek or Roman alphabet) that represent the quantities. The manipulation of such equations is the field of algebra, and it is used a great deal in physics. An equation involves an equals sign, which tells us that the quantities on either side of the equals sign have the same value. Examples of equations are

$$\begin{aligned} 3 + 8 &= 11 \\ 2x + 7 &= 15 \\ a^2b + c &= 6. \end{aligned}$$

The first equation involves only numbers, so is called an arithmetic equation. The other two equations are algebraic since they involve symbols. In the third equation, the quantity a^2b means the product of a times a times b: $a^2b = a \times a \times b$.

[†]Another convention used, particularly with computers, is that the simple number be between 0.1 and 1. Thus we could write 4,500,000,000 as 0.450×10^{10}. This is slightly less compact.

Solving for an Unknown

Often we wish to solve for one (or more) symbols, and we treat it as an *unknown*. For example, in the equation $2x + 7 = 15$, x is the unknown; this equation is true, however, only when $x = 4$. Determining what value (or values) the unknown can have to satisfy the equation is called *solving the equation*. To solve an equation, the following rule can be used:

> *An equation will remain true if any operation performed on one side is also performed on the other side.* For example: (*a*) addition or subtraction of a number or symbol; (*b*) multiplication or division by a number or symbol; (*c*) raising each side of the equation to the same power, or taking the same root (such as square root).

EXAMPLE A–1 Solve for x in the equation

$$2x + 7 = 15.$$

APPROACH We perform the same operations on both sides of the equation to isolate x as the only variable on the left side of the equals sign.

SOLUTION We first subtract 7 from both sides:

$$2x + 7 - 7 = 15 - 7$$

or

$$2x = 8.$$

Then we divide both sides by 2 to get

$$\frac{2x}{2} = \frac{8}{2},$$

or, carrying out the divisions,

$$x = 4,$$

and this solves the equation.

EXAMPLE A–2 (*a*) Solve the equation

$$a^2b + c = 24$$

for the unknown a in terms of b and c. (*b*) Solve for a assuming that $b = 2$ and $c = 6$.

APPROACH We perform operations to isolate a as the only variable on the left side of the equals sign.

SOLUTION (*a*) We are trying to solve for a, so we first subtract c from both sides:

$$a^2b = 24 - c,$$

then divide by b:

$$a^2 = \frac{24 - c}{b},$$

and finally take square roots:

$$a = \sqrt{\frac{24 - c}{b}}.$$

(*b*) If we are given that $b = 2$ and $c = 6$, then

$$a = \sqrt{\frac{24 - 6}{2}} = 3.$$

But this is not the only answer. Whenever we take a square root, the number can be either positive or negative. Thus $a = -3$ is also a solution. Why? Because $(-3)^2 = 9$, just as $(+3)^2 = 9$. So we actually get two solutions: $a = +3$ and $a = -3$.

NOTE When an unknown appears squared in an equation, there are generally two solutions for that unknown.

To check a solution, we put it back into the original equation (this is really a check that we did all the manipulations correctly). In the equation

$$a^2b + c = 24,$$

we put in $a = 3,\ b = 2,\ c = 6$ and find

$$(3)^2(2) + (6) \stackrel{?}{=} 24$$
$$24 = 24,$$

which checks.

EXERCISE A Put $a = -3$ into the equation of Example A–2 and show that it works too.

Two or More Unknowns

If we have two or more unknowns, one equation is not sufficient to find them. In general, if there are n unknowns, n independent equations are needed. For example, if there are two unknowns, we need two equations. If the unknowns are called x and y, a typical procedure is to solve one equation for x in terms of y, and substitute this into the second equation.

EXAMPLE A–3 Solve the following pair of equations for x and y:

$$3x - 2y = 19$$
$$x + 4y = -3.$$

APPROACH We have two unknowns and two equations; we can start by solving the second equation for x in terms of y. Then we substitute this result for x into the first equation.

SOLUTION We subtract $4y$ from both sides of the second equation:

$$x = -3 - 4y.$$

We substitute this expression for x into the first equation, and simplify:

$$3(-3 - 4y) - 2y = 19$$
$$-9 - 12y - 2y = 19 \quad \text{(carried out the multiplication by 3)}$$
$$-14y = 28 \quad \text{(added 9 to both sides)}$$
$$y = -2. \quad \text{(divided both sides by } -14\text{)}$$

Now that we know $y = -2$, we substitute this into the expression for x:

$$x = -3 - 4y$$
$$= -3 - 4(-2) = -3 + 8 = 5.$$

Our solution is $x = 5,\ y = -2$. We check this solution by putting these values back into the original equations:

$$3x - 2y \stackrel{?}{=} 19$$
$$3(5) - 2(-2) \stackrel{?}{=} 19$$
$$15 + 4 \stackrel{?}{=} 19$$
$$19 = 19 \quad \text{(it checks)}$$

and

$$x + 4y \stackrel{?}{=} -3$$
$$5 + 4(-2) \stackrel{?}{=} -3$$
$$-3 = -3. \quad \text{(it checks)}$$

Other methods for solving two or more equations, such as the method of determinants, can be found in an algebra textbook.

The Quadratic Formula

We sometimes encounter equations that involve an unknown, say x, that appears not only to the first power, but squared as well. Such a **quadratic equation** can be written in the general form

$$ax^2 + bx + c = 0.$$

The quantities a, b, and c are typically numbers or constants that are given.[†] The general solutions to such an equation are given by the **quadratic formula**:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}. \qquad \textbf{(A–4)}$$

The $\pm$ sign indicates that there are two solutions for x: one where the plus sign is used, the other where the minus sign is used.

EXAMPLE A–4 Find the solutions for x in the equation

$$3x^2 - 5x = 2.$$

APPROACH Here x appears both to the first power and squared, so we use the quadratic equation.

SOLUTION First we write this equation in the standard form

$$ax^2 + bx + c = 0$$

by subtracting 2 from both sides:

$$3x^2 - 5x - 2 = 0.$$

In this case, a, b, and c in the standard formula take the values $a = 3$, $b = -5$, and $c = -2$. The two solutions for x are, using Eq. A–4,

$$x = \frac{+5 + \sqrt{25 - (4)(3)(-2)}}{(2)(3)} = \frac{5 + 7}{6} = 2$$

and

$$x = \frac{+5 - \sqrt{25 - (4)(3)(-2)}}{(2)(3)} = \frac{5 - 7}{6} = -\frac{1}{3}.$$

In this Example, the two solutions are $x = 2$ and $x = -\frac{1}{3}$. In physics problems, it sometimes happens that only one of the solutions corresponds to a real-life situation; in this case, the other solution is discarded. In other cases, both solutions may correspond to physical reality.

Notice, incidentally, that b^2 must be greater than $4ac$, so that $\sqrt{b^2 - 4ac}$ yields a real number. If $(b^2 - 4ac)$ is less than zero (negative), there is no real solution. The square root of a negative number is called **imaginary**.

A second-order equation—one in which the highest power of x is 2—has two solutions; a third-order equation—involving x^3—has three solutions; and so on.

A–5 The Binomial Expansion

Sometimes we end up with a quantity of the form $(1 + x)^n$. That is, the quantity $(1 + x)$ is raised to the nth power. This can be written as an infinite sum of terms known as the **binomial expansion**:

$$(1 + x)^n = 1 + nx + \frac{n(n - 1)}{2}x^2 + \cdots. \qquad \textbf{(A–5)}$$

This formula is useful for us mainly when x is very small compared to one ($x \ll 1$). In this case, each successive term is much smaller than the preceding

[†]Or one or more of them could be variables, in which case additional equations are needed.

term. For example, let $x = 0.01$, and $n = 2$. Then whereas the first term equals 1, the second term is $nx = (2)(0.01) = 0.02$, and the third term is $[(2)(1)/2](0.01)^2 = 0.0001$, and so on. Thus, when x is small, we can ignore all but the first two (or three) terms and can write

$$(1 + x)^n \approx 1 + nx. \qquad \textbf{(A–6)}$$

This approximation often allows us to solve an equation easily that otherwise might be very difficult. Some examples of the binomial expansion are

$$(1 + x)^2 \approx 1 + 2x,$$
$$\frac{1}{1 + x} = (1 + x)^{-1} \approx 1 - x,$$
$$\sqrt{1 + x} = (1 + x)^{\frac{1}{2}} \approx 1 + \tfrac{1}{2}x,$$
$$\frac{1}{\sqrt{1 + x}} = (1 + x)^{-\frac{1}{2}} \approx 1 - \tfrac{1}{2}x,$$

where $x \ll 1$.

As a numerical example, let us evaluate $\sqrt{1.02}$ using the binomial expansion since $x = 0.02$ is much smaller than 1:

$$\sqrt{1.02} = (1.02)^{\frac{1}{2}} = (1 + 0.02)^{\frac{1}{2}} \approx 1 + \tfrac{1}{2}(0.02) = 1.01.$$

You can check with a calculator (and maybe not even more quickly) that $\sqrt{1.02} \approx 1.01$.

A–6 Plane Geometry

We review here a number of theorems involving angles and triangles that are useful in physics.

1. **Equal angles.** Two angles are equal if any of the following conditions are true:
   (*a*) They are vertical angles (Fig. A–1); *or*
   (*b*) the left side of one is parallel to the left side of the other, and the right side of one is parallel to the right side of the other (Fig. A–2; the left and right sides are as seen from the vertex, where the two sides meet); *or*
   (*c*) the left side of one is perpendicular to the left side of the other, and the right sides are likewise perpendicular (Fig. A–3).

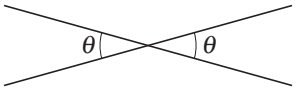


FIGURE A–1

FIGURE A–2

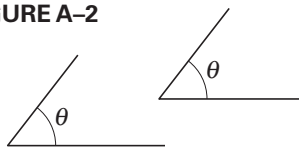


FIGURE A–3

2. *The sum of the angles* in any plane triangle is 180°.

3. **Similar triangles.** Two triangles are said to be similar if all three of their angles are equal (in Fig. A–4, $\theta_1 = \phi_1$, $\theta_2 = \phi_2$, and $\theta_3 = \phi_3$). Similar triangles thus have the same basic shape but may be different sizes and have different orientations. Two useful theorems about similar triangles are:
   (*a*) Two triangles are similar if any two of their angles are equal. (This follows because the third angles must also be equal since the sum of the angles of a triangle is 180°.)
   (*b*) The ratios of corresponding sides of two similar triangles are equal. That is (Fig. A–4),

$$\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3}.$$

4. **Congruent triangles.** Two triangles are congruent if one can be placed precisely on top of the other. That is, they are similar triangles and they have the same size. Two triangles are congruent if any of the following holds:
   (*a*) The three corresponding sides are equal.
   (*b*) Two sides and the enclosed angle are equal ("side-angle-side").
   (*c*) Two angles and the enclosed side are equal ("angle-side-angle").

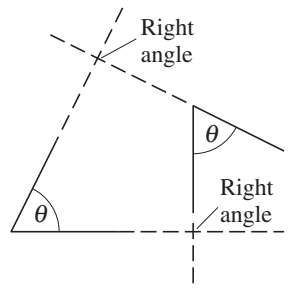


FIGURE A–4

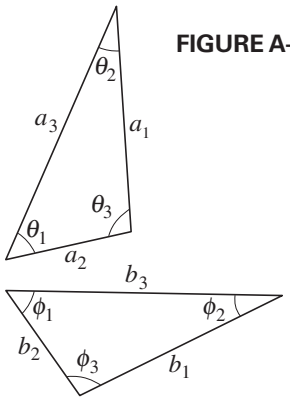
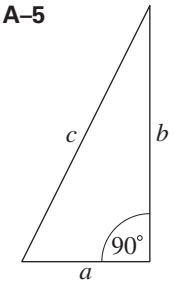


FIGURE A–5



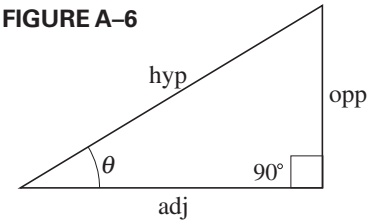
5. **Right triangles.** A right triangle has one angle that is 90° (a **right angle**); that is, the two sides that meet at the right angle are perpendicular (Fig. A–5). The two other (acute) angles in the right triangle add up to 90°.
6. **Pythagorean theorem.** In any right triangle, the square of the length of the hypotenuse (the side opposite the right angle) is equal to the sum of the squares of the lengths of the other two sides. In Fig. A–5,

$$c^2 = a^2 + b^2.$$

A–7 Trigonometric Functions and Identities

Trigonometric functions for any angle θ are defined by constructing a right triangle about that angle as shown in Fig. A–6; opp and adj are the lengths of the sides opposite and adjacent to the angle θ, and hyp is the length of the hypotenuse:

FIGURE A–6



$$\sin\theta = \frac{\text{opp}}{\text{hyp}} \qquad \csc\theta = \frac{1}{\sin\theta} = \frac{\text{hyp}}{\text{opp}}$$

$$\cos\theta = \frac{\text{adj}}{\text{hyp}} \qquad \sec\theta = \frac{1}{\cos\theta} = \frac{\text{hyp}}{\text{adj}}$$

$$\tan\theta = \frac{\text{opp}}{\text{adj}} = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{1}{\tan\theta} = \frac{\text{adj}}{\text{opp}}$$

$$\text{adj}^2 + \text{opp}^2 = \text{hyp}^2 \qquad \text{(Pythagorean theorem).}$$

FIGURE A–7

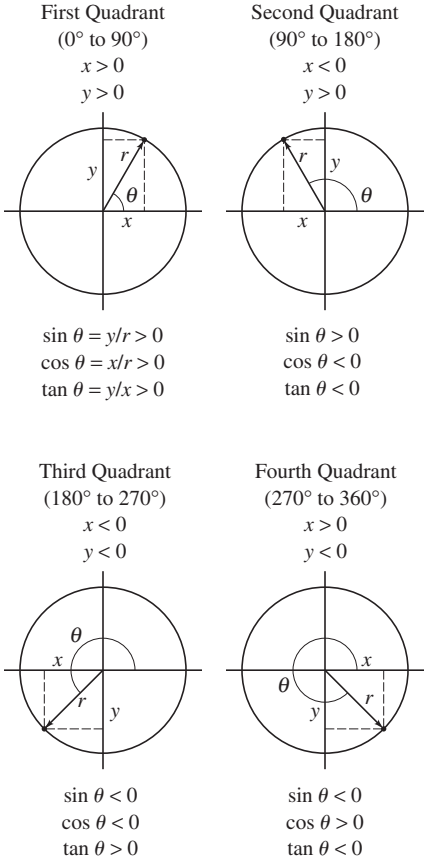


Figure A–7 shows the signs (+ or −) that cosine, sine, and tangent take on for angles θ in the four quadrants (0° to 360°). Note that angles are measured counterclockwise from the x axis as shown; negative angles are measured from *below* the x axis, clockwise: for example, $-30° = +330°$, and so on.

The following are some useful identities among the trigonometric functions:

$$\sin^2\theta + \cos^2\theta = 1$$

$$\sin 2\theta = 2\sin\theta\cos\theta$$

$$\cos 2\theta = \cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta$$

$$\tan 2\theta = \frac{2\tan\theta}{1 - \tan^2\theta}$$

$$\sin(A \pm B) = \sin A\cos B \pm \cos A\sin B$$

$$\cos(A \pm B) = \cos A\cos B \mp \sin A\sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A\tan B}$$

$$\sin(180° - \theta) = \sin\theta$$

$$\cos(180° - \theta) = -\cos\theta$$

$$\sin(90° - \theta) = \cos\theta$$

$$\cos(90° - \theta) = \sin\theta$$

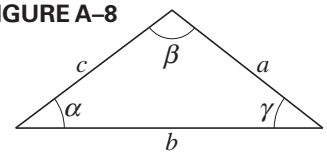
$$\sin\tfrac{1}{2}\theta = \sqrt{\frac{1 - \cos\theta}{2}}$$

$$\cos\tfrac{1}{2}\theta = \sqrt{\frac{1 + \cos\theta}{2}}$$

$$\tan\tfrac{1}{2}\theta = \sqrt{\frac{1 - \cos\theta}{1 + \cos\theta}}$$

$$\sin A \pm \sin B = 2\sin\left(\frac{A \pm B}{2}\right)\cos\left(\frac{A \mp B}{2}\right).$$

FIGURE A–8



For any triangle (see Fig. A–8):

$$\frac{\sin\alpha}{a} = \frac{\sin\beta}{b} = \frac{\sin\gamma}{c} \qquad \text{(law of sines)}$$

$$c^2 = a^2 + b^2 - 2ab\cos\gamma. \qquad \text{(law of cosines)}$$

Trigonometric Table: Numerical Values of Sin, Cos, Tan

Angle in Degrees	Angle in Radians	Sine	Cosine	Tangent	Angle in Degrees	Angle in Radians	Sine	Cosine	Tangent
0°	0.000	0.000	1.000	0.000					
1°	0.017	0.017	1.000	0.017	46°	0.803	0.719	0.695	1.036
2°	0.035	0.035	0.999	0.035	47°	0.820	0.731	0.682	1.072
3°	0.052	0.052	0.999	0.052	48°	0.838	0.743	0.669	1.111
4°	0.070	0.070	0.998	0.070	49°	0.855	0.755	0.656	1.150
5°	0.087	0.087	0.996	0.087	50°	0.873	0.766	0.643	1.192
6°	0.105	0.105	0.995	0.105	51°	0.890	0.777	0.629	1.235
7°	0.122	0.122	0.993	0.123	52°	0.908	0.788	0.616	1.280
8°	0.140	0.139	0.990	0.141	53°	0.925	0.799	0.602	1.327
9°	0.157	0.156	0.988	0.158	54°	0.942	0.809	0.588	1.376
10°	0.175	0.174	0.985	0.176	55°	0.960	0.819	0.574	1.428
11°	0.192	0.191	0.982	0.194	56°	0.977	0.829	0.559	1.483
12°	0.209	0.208	0.978	0.213	57°	0.995	0.839	0.545	1.540
13°	0.227	0.225	0.974	0.231	58°	1.012	0.848	0.530	1.600
14°	0.244	0.242	0.970	0.249	59°	1.030	0.857	0.515	1.664
15°	0.262	0.259	0.966	0.268	60°	1.047	0.866	0.500	1.732
16°	0.279	0.276	0.961	0.287	61°	1.065	0.875	0.485	1.804
17°	0.297	0.292	0.956	0.306	62°	1.082	0.883	0.469	1.881
18°	0.314	0.309	0.951	0.325	63°	1.100	0.891	0.454	1.963
19°	0.332	0.326	0.946	0.344	64°	1.117	0.899	0.438	2.050
20°	0.349	0.342	0.940	0.364	65°	1.134	0.906	0.423	2.145
21°	0.367	0.358	0.934	0.384	66°	1.152	0.914	0.407	2.246
22°	0.384	0.375	0.927	0.404	67°	1.169	0.921	0.391	2.356
23°	0.401	0.391	0.921	0.424	68°	1.187	0.927	0.375	2.475
24°	0.419	0.407	0.914	0.445	69°	1.204	0.934	0.358	2.605
25°	0.436	0.423	0.906	0.466	70°	1.222	0.940	0.342	2.747
26°	0.454	0.438	0.899	0.488	71°	1.239	0.946	0.326	2.904
27°	0.471	0.454	0.891	0.510	72°	1.257	0.951	0.309	3.078
28°	0.489	0.469	0.883	0.532	73°	1.274	0.956	0.292	3.271
29°	0.506	0.485	0.875	0.554	74°	1.292	0.961	0.276	3.487
30°	0.524	0.500	0.866	0.577	75°	1.309	0.966	0.259	3.732
31°	0.541	0.515	0.857	0.601	76°	1.326	0.970	0.242	4.011
32°	0.559	0.530	0.848	0.625	77°	1.344	0.974	0.225	4.331
33°	0.576	0.545	0.839	0.649	78°	1.361	0.978	0.208	4.705
34°	0.593	0.559	0.829	0.675	79°	1.379	0.982	0.191	5.145
35°	0.611	0.574	0.819	0.700	80°	1.396	0.985	0.174	5.671
36°	0.628	0.588	0.809	0.727	81°	1.414	0.988	0.156	6.314
37°	0.646	0.602	0.799	0.754	82°	1.431	0.990	0.139	7.115
38°	0.663	0.616	0.788	0.781	83°	1.449	0.993	0.122	8.144
39°	0.681	0.629	0.777	0.810	84°	1.466	0.995	0.105	9.514
40°	0.698	0.643	0.766	0.839	85°	1.484	0.996	0.087	11.43
41°	0.716	0.656	0.755	0.869	86°	1.501	0.998	0.070	14.301
42°	0.733	0.669	0.743	0.900	87°	1.518	0.999	0.052	19.081
43°	0.750	0.682	0.731	0.933	88°	1.536	0.999	0.035	28.636
44°	0.768	0.695	0.719	0.966	89°	1.553	1.000	0.017	57.290
45°	0.785	0.707	0.707	1.000	90°	1.571	1.000	0.000	∞

A–8 Logarithms

Logarithms are defined in the following way:

$$\text{if } y = A^x, \qquad \text{then } x = \log_A y.$$

That is, the logarithm of a number y to the base A is that number which, as the exponent of A, gives back the number y. For **common logarithms**, the base is 10, so

$$\text{if } y = 10^x, \qquad \text{then } x = \log y.$$

The subscript 10 on $\log_{10}$ is usually omitted when dealing with common logs. Another base sometimes used is the exponential base $e = 2.718\cdots$, a natural number.[†] Such logarithms are called **natural logarithms** and are written "ln". Thus,

$$\text{if } y = e^x, \qquad \text{then } x = \ln y.$$

For any number y, the two types of logarithm are related by

$$\ln y = 2.3026 \log y.$$

Some simple rules for logarithms include:

$$\log (ab) = \log a + \log b. \tag{A–7}$$

This is true because if $a = 10^n$ and $b = 10^m$, then $ab = 10^{n+m}$. From the definition of logarithm, $\log a = n$, $\log b = m$, and $\log (ab) = n + m$; hence, $\log (ab) = n + m = \log a + \log b$. In a similar way, we can show the rules

$$\log\left(\frac{a}{b}\right) = \log a - \log b \tag{A–8}$$

and

$$\log a^n = n \log a. \tag{A–9}$$

These three rules apply not only to common logs but to natural or any other kind of logarithm.

Logs were once used as a technique for simplifying certain types of calculation. Because of the advent of electronic calculators and computers, they are not often used any more for that purpose. However, logs do appear in certain physical equations, so it is helpful to know how to deal with them. If you do not have a calculator that calculates logs, you can use a **log table**, such as the small one shown here (Table A–1). The number N is given to two digits (some tables give N to three or more digits); the first digit is in the vertical column to the left, the second digit is in the horizontal row across the top. For example, the Table tells us that $\log 1.0 = 0.000$, $\log 1.1 = 0.041$, and $\log 4.1 = 0.613$. The Table gives logs for numbers between 1.0 and 9.9; for larger or smaller numbers, we use rule A–7:

$$\log (ab) = \log a + \log b.$$

For example,

$$\log (380) = \log \left(3.8 \times 10^2\right) = \log (3.8) + \log \left(10^2\right).$$

From the Table, $\log 3.8 = 0.580$; and from rule A–9,

$$\log \left(10^2\right) = 2 \log (10) = 2,$$

since $\log (10) = 1$. [This follows from the definition of the logarithm: if

[†]The exponential base e can be written as an infinite series:

$$e = 1 + \frac{1}{1} + \frac{1}{1\cdot 2} + \frac{1}{1\cdot 2\cdot 3} + \frac{1}{1\cdot 2\cdot 3\cdot 4} + \cdots.$$

TABLE A–1 Short Table of Common Logarithms

N	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
1	.000	.041	.079	.114	.146	.176	.204	.230	.255	.279
2	.301	.322	.342	.362	.380	.398	.415	.431	.447	.462
3	.477	.491	.505	.519	.531	.544	.556	.568	.580	.591
4	.602	.613	.623	.633	.643	.653	.663	.672	.681	.690
5	.699	.708	.716	.724	.732	.740	.748	.756	.763	.771
6	.778	.785	.792	.799	.806	.813	.820	.826	.833	.839
7	.845	.851	.857	.863	.869	.875	.881	.886	.892	.898
8	.903	.908	.914	.919	.924	.929	.935	.940	.944	.949
9	.954	.959	.964	.968	.973	.978	.982	.987	.991	.996

$10 = 10^1$, then $1 = \log(10)$.] Thus,

$$\begin{aligned}\log(380) &= \log(3.8) + \log(10^2)\\ &= 0.580 + 2\\ &= 2.580.\end{aligned}$$

Similarly,

$$\begin{aligned}\log(0.081) &= \log(8.1) + \log(10^{-2})\\ &= 0.908 - 2 = -1.092.\end{aligned}$$

Sometimes we need to do the reverse process: find the number N whose log is, say, 2.670. This is called "taking the **antilogarithm**." To do so, we separate our number 2.670 into two parts, making the separation at the decimal point:

$$\begin{aligned}\log N = 2.670 &= 2 + 0.670\\ &= \log 10^2 + 0.670.\end{aligned}$$

We now look at Table A–1 to see what number has its log equal to 0.670; none does, so we must **interpolate**: we see that $\log 4.6 = 0.663$ and $\log 4.7 = 0.672$. So the number we want is between 4.6 and 4.7, and closer to the latter by $\frac{7}{9}$. Approximately we can say that $\log 4.68 = 0.670$. Thus

$$\begin{aligned}\log N &= 2 + 0.670\\ &= \log(10^2) + \log(4.68) = \log(4.68 \times 10^2),\end{aligned}$$

so $N = 4.68 \times 10^2 = 468$.

If the given logarithm is negative, say, -2.180, we proceed as follows:

$$\begin{aligned}\log N = -2.180 &= -3 + 0.820\\ &= \log 10^{-3} + \log 6.6 = \log 6.6 \times 10^{-3},\end{aligned}$$

so $N = 6.6 \times 10^{-3}$. Notice that we added to our given logarithm the next largest integer (3 in this case) so that we have an integer, plus a decimal number between 0 and 1.0 whose antilogarithm can be looked up in the Table.

APPENDIX

B Selected Isotopes

(1) Atomic Number Z	(2) Element	(3) Symbol	(4) Mass Number A	(5) Atomic Mass†	(6) % Abundance (or Radioactive Decay‡ Mode)	(7) Half-life (if radioactive)
0	(Neutron)	n	1	1.008665	β^-	10.183 min
1	Hydrogen	H	1	1.007825	99.9885%	
	[proton	p	1	1.007276]		
	Deuterium	$^2_1$H	2	2.014102	0.0115%	
	[deuteron	d or D	2	2.013553]		
	Tritium	$^3_1$H	3	3.016049	β^-	12.32 yr
	[triton	t or T	3	3.015500]		
2	Helium	He	3	3.016029	0.000137%	
			4	4.002603	99.999863%	
3	Lithium	Li	6	6.015123	7.59%	
			7	7.016003	92.41%	
4	Beryllium	Be	7	7.016929	EC, γ	53.24 days
			9	9.012183	100%	
5	Boron	B	10	10.012937	19.9%	
			11	11.009305	80.1%	
6	Carbon	C	11	11.011434	β^+, EC	20.334 min
			12	12.000000	98.93%	
			13	13.003355	1.07%	
			14	14.003242	β^-	5730 yr
7	Nitrogen	N	13	13.005739	β^+, EC	9.965 min
			14	14.003074	99.632%	
			15	15.000109	0.368%	
8	Oxygen	O	15	15.003066	β^+, EC	122.24 s
			16	15.994915	99.757%	
			18	17.999160	0.205%	
9	Fluorine	F	19	18.998403	100%	
10	Neon	Ne	20	19.992440	90.48%	
			22	21.991385	9.25%	
11	Sodium	Na	22	21.994437	β^+, EC, γ	2.6027 yr
			23	22.989769	100%	
			24	23.990963	β^-, γ	14.997 h
12	Magnesium	Mg	24	23.985042	78.99%	
13	Aluminum	Al	27	26.981539	100%	
14	Silicon	Si	28	27.976927	92.223%	
			31	30.975363	β^-, γ	157.3 min
15	Phosphorus	P	31	30.973762	100%	
			32	31.973908	β^-	14.262 days

† The masses (atomic mass units) given in column (5) are those for the neutral atom, including the Z electrons (except for the proton, deuteron, triton).

‡ Chapter 30; EC = electron capture.

(1) Atomic Number Z	(2) Element	(3) Symbol	(4) Mass Number A	(5) Atomic Mass	(6) % Abundance (or Radioactive Decay Mode)	(7) Half-life (if radioactive)
16	Sulfur	S	32	31.972071	94.99%	
			35	34.969032	β^-	87.37 days
17	Chlorine	Cl	35	34.968853	75.76%	
			37	36.965903	24.24%	
18	Argon	Ar	40	39.962383	99.6035%	
19	Potassium	K	39	38.963706	93.2581%	
			40	39.963998	0.0117%	
					β^-, EC, γ, β^+	1.248×10^9 yr
20	Calcium	Ca	40	39.962591	96.94%	
21	Scandium	Sc	45	44.955908	100%	
22	Titanium	Ti	48	47.947942	73.72%	
23	Vanadium	V	51	50.943957	99.750%	
24	Chromium	Cr	52	51.940506	83.789%	
25	Manganese	Mn	55	54.938044	100%	
26	Iron	Fe	56	55.934936	91.754%	
27	Cobalt	Co	59	58.933194	100%	
			60	59.933816	β^-, γ	5.2713 yr
28	Nickel	Ni	58	57.935342	68.077%	
			60	59.930786	26.223%	
29	Copper	Cu	63	62.929598	69.15%	
			65	64.927790	30.85%	
30	Zinc	Zn	64	63.929142	49.17%	
			66	65.926034	27.73%	
31	Gallium	Ga	69	68.925574	60.108%	
32	Germanium	Ge	72	71.922076	27.45%	
			74	73.921178	36.50%	
33	Arsenic	As	75	74.921595	100%	
34	Selenium	Se	80	79.916522	49.61%	
35	Bromine	Br	79	78.918338	50.69%	
36	Krypton	Kr	84	83.911498	56.987%	
37	Rubidium	Rb	85	84.911790	72.17%	
38	Strontium	Sr	86	85.909261	9.86%	
			88	87.905612	82.58%	
			90	89.907730	β^-	28.90 yr
39	Yttrium	Y	89	88.905840	100%	
40	Zirconium	Zr	90	89.904698	51.45%	
41	Niobium	Nb	93	92.906373	100%	
42	Molybdenum	Mo	98	97.905405	24.39%	
43	Technetium	Tc	98	97.907212	β^-, γ	4.2×10^6 yr
44	Ruthenium	Ru	102	101.904344	31.55%	
45	Rhodium	Rh	103	102.905498	100%	
46	Palladium	Pd	106	105.903480	27.33%	
47	Silver	Ag	107	106.905092	51.839%	
			109	108.904755	48.161%	
48	Cadmium	Cd	114	113.903365	28.73%	
49	Indium	In	115	114.903879	95.71%; β^-	4.41×10^{14} yr
50	Tin	Sn	120	119.902202	32.58%	
51	Antimony	Sb	121	120.903812	57.21%	

(1) Atomic Number Z	(2) Element	(3) Symbol	(4) Mass Number A	(5) Atomic Mass	(6) % Abundance (or Radioactive Decay Mode)	(7) Half-life (if radioactive)
52	Tellurium	Te	130	129.906223	34.08%; $\beta^-\beta^-$	$>3.0 \times 10^{24}$ yr
53	Iodine	I	127	126.904472	100%	
			131	130.906126	β^-, γ	8.0252 days
54	Xenon	Xe	132	131.904155	26.9086%	
			136	135.907214	8.8573%; $\beta^-\beta^-$	$>2.4 \times 10^{21}$ yr
55	Cesium	Cs	133	132.905452	100%	
56	Barium	Ba	137	136.905827	11.232%	
			138	137.905247	71.698%	
57	Lanthanum	La	139	138.906356	99.9119%	
58	Cerium	Ce	140	139.905443	88.450%	
59	Praseodymium	Pr	141	140.907658	100%	
60	Neodymium	Nd	142	141.907729	27.152%	
61	Promethium	Pm	145	144.912756	EC, α	17.7 yr
62	Samarium	Sm	152	151.919740	26.75%	
63	Europium	Eu	153	152.921238	52.19%	
64	Gadolinium	Gd	158	157.924112	24.84%	
65	Terbium	Tb	159	158.925355	100%	
66	Dysprosium	Dy	164	163.929182	28.260%	
67	Holmium	Ho	165	164.930329	100%	
68	Erbium	Er	166	165.930300	33.503%	
69	Thulium	Tm	169	168.934218	100%	
70	Ytterbium	Yb	174	173.938866	31.026%	
71	Lutetium	Lu	175	174.940775	97.401%	
72	Hafnium	Hf	180	179.946557	35.08%	
73	Tantalum	Ta	181	180.947996	99.98799%	
74	Tungsten (wolfram)	W	184	183.950931	30.64%; α	$>8.9 \times 10^{21}$ yr
75	Rhenium	Re	187	186.955750	62.60%; β^-	4.33×10^{10} yr
76	Osmium	Os	191	190.960926	β^-, γ	15.4 days
			192	191.961477	40.78%	
77	Iridium	Ir	191	190.960589	37.3%	
			193	192.962922	62.7%	
78	Platinum	Pt	195	194.964792	33.78%	
79	Gold	Au	197	196.966569	100%	
80	Mercury	Hg	199	198.968281	16.87%	
			202	201.970643	29.86%	
81	Thallium	Tl	205	204.974428	70.48%	
82	Lead	Pb	206	205.974466	24.1%	
			207	206.975897	22.1%	
			208	207.976652	52.4%	
			210	209.984189	β^-, γ, α	22.20 yr
			211	210.988737	β^-, γ	36.1 min
			212	211.991898	β^-, γ	10.64 h
			214	213.999806	β^-, γ	26.8 min
83	Bismuth	Bi	209	208.980399	100%	
			211	210.987270	α, γ, β^-	2.14 min
84	Polonium	Po	210	209.982874	α, γ, EC	138.376 days
			214	213.995202	α, γ	164.3 μs
85	Astatine	At	218	218.008695	α, β^-	1.5 s

(1) Atomic Number Z	(2) Element	(3) Symbol	(4) Mass Number A	(5) Atomic Mass	(6) % Abundance (or Radioactive Decay Mode)	(7) Half-life (if radioactive)
86	Radon	Rn	222	222.017578	α, γ	3.8235 days
87	Francium	Fr	223	223.019736	β^-, γ, α	22.00 min
88	Radium	Ra	226	226.025410	α, γ	1600 yr
89	Actinium	Ac	227	227.027752	β^-, γ, α	21.772 yr
90	Thorium	Th	228	228.028741	α, γ	1.9116 yr
			232	232.038056	100%; α, γ	1.40×10^{10} yr
91	Protactinium	Pa	231	231.035884	α, γ	3.276×10^4 yr
92	Uranium	U	232	232.037156	α, γ	68.9 yr
			233	233.039636	α, γ	1.592×10^5 yr
			235	235.043930	0.7204%; α, γ	7.04×10^8 yr
			236	236.045568	α, γ	2.342×10^7 yr
			238	238.050788	99.2742%; α, γ	4.468×10^9 yr
			239	239.054294	β^-, γ	23.45 min
93	Neptunium	Np	237	237.048174	α, γ	2.144×10^6 yr
			239	239.052939	β^-, γ	2.356 days
94	Plutonium	Pu	239	239.052164	α, γ	24,110 yr
			244	244.064205	α	8.00×10^7 yr
95	Americium	Am	243	243.061381	α, γ	7370 yr
96	Curium	Cm	247	247.070354	α, γ	1.56×10^7 yr
97	Berkelium	Bk	247	247.070307	α, γ	1380 yr
98	Californium	Cf	251	251.079589	α, γ	898 yr
99	Einsteinium	Es	252	252.082980	α, EC, γ	471.7 days
100	Fermium	Fm	257	257.095106	α, γ	100.5 days
101	Mendelevium	Md	258	258.098431	α, γ	51.5 days
102	Nobelium	No	259	259.101030	α, EC	58 min
103	Lawrencium	Lr	262	262.109610	α, EC, fission	$\approx$4 h
104	Rutherfordium	Rf	263	263.112500	fission	10 min
105	Dubnium	Db	268	268.125670	fission	32 h
106	Seaborgium	Sg	271	271.133930	α, fission	2.4 min
107	Bohrium	Bh	274	274.143550	α, fission	0.9 min
108	Hassium	Hs	270	270.134290	α	22 s
109	Meitnerium	Mt	278	278.156310	α, fission	8 s
110	Darmstadtium	Ds	281	281.164510	α, fission	20 s
111	Roentgenium	Rg	281	281.166360	α, fission	26 s
112	Copernicium	Cn	285	285.177120	α	30 s
113†			286	286.18210	α, fission	20 s
114	Flerovium	Fl	289	289.190420	α	2.7 s
115†			289	289.193630	α, fission	0.22 s
116	Livermorium	Lv	293	293.204490	α	53 ms
117†			294	294.210460	α	0.08 s
118†			294	294.213920	α, fission	0.9 ms

† Preliminary evidence (unconfirmed) has been reported for elements 113, 115, 117, and 118.

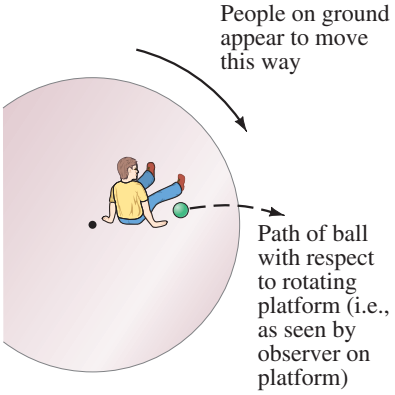
APPENDIX C

Rotating Frames of Reference; Inertial Forces; Coriolis Effect

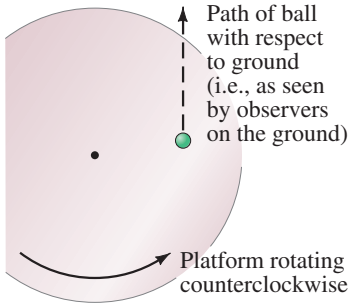
Inertial and Noninertial Reference Frames

In Chapters 5 and 8 we examined the motion of objects, including circular and rotational motion, from the outside, as observers fixed on the Earth. Sometimes it is convenient to place ourselves (in theory, if not physically) into a reference frame that is rotating. Let us examine the motion of objects from the point of view, or frame of reference, of persons seated on a rotating platform such as a merry-go-round. It looks to them as if the rest of the world is going around *them*. But let us focus on what they observe when they place a tennis ball on the floor of the rotating platform, which we assume is frictionless. If they put the ball down gently, without giving it any push, they will observe that it accelerates from rest and moves outward as shown in Fig. C–1a. According to Newton's first law, an object initially at rest should stay at rest if no net force acts on it. But, according to the observers on the rotating platform, the ball starts moving even though there is no net force acting on it. To observers on the ground, this is all very clear: the ball has an initial velocity when it is released (because the platform is moving), and it simply continues moving in a straight-line path as shown in Fig. C–1b, in accordance with Newton's first law.

But what shall we do about the frame of reference of the observers on the rotating platform? Newton's first law, the law of inertia, does not hold in this rotating frame of reference since the ball starts moving with no net force on it. For this reason, such a frame is called a **noninertial reference frame**. An **inertial reference frame** (as discussed in Chapter 4) is one in which the law of inertia—Newton's first law—does hold, and so do Newton's second and third laws. In a noninertial reference frame, such as our rotating platform, Newton's second law also does not hold. For instance in the situation described above, there is no net force on the ball; yet, with respect to the rotating platform, the ball accelerates.



(a) Rotating reference frame



(b) Earth reference frame

FIGURE C–1 Path of a ball released on a rotating merry-go-round (a) in the reference frame of the merry-go-round, and (b) in a reference frame fixed on the ground.

Fictitious (Inertial) Forces

Because Newton's laws do not hold when observations are made with respect to a rotating frame of reference, calculation of motion can be complicated. However, we can still apply Newton's laws in such a reference frame if we make use of a trick. The ball on the rotating platform of Fig. C–1a flies outward when released (even though no force is actually acting on it). So the trick we use is to write down the equation $\Sigma F = ma$ as if a force equal to mv^2/r (or $m\omega^2 r$) were acting radially outward on the object in addition to any other forces that may be acting. This extra force, which might be designated as "centrifugal force" since it *seems* to act outward, is called a **fictitious force** or **pseudoforce**. It is a pseudoforce ("pseudo" means "false") because there is no object that exerts this force. Furthermore, when viewed from an inertial reference frame, the effect doesn't exist at all. We have made up this pseudoforce so that we can make calculations in a noninertial frame using Newton's second law, $\Sigma F = ma$. Thus the observer in the noninertial frame of Fig. C–1a uses Newton's second law for the ball's outward motion by assuming that a force equal to mv^2/r acts on it. Such pseudoforces are also called **inertial forces** since they arise only because the reference frame is not an inertial one.

In Section 5–3 we discussed the forces on a person in a car going around a curve (Fig. 5–11) from the point of view of an inertial frame. The car, on the other hand, is not an inertial frame. Passengers in such a car could interpret this being pressed outward as the effect of a "centrifugal" force. But they need to recognize that it is a pseudoforce because there is no identifiable object exerting it. It is an effect of being in a noninertial frame of reference.

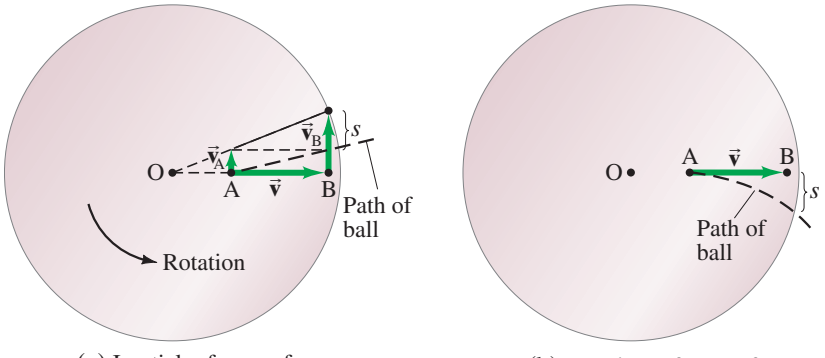
The Earth itself is rotating on its axis. Thus, strictly speaking, Newton's laws are not valid on the Earth. However, the effect of the Earth's rotation is usually so small that it can be ignored, although it does influence the movement of large air masses and ocean currents. Because of the Earth's rotation, the material of the Earth is concentrated slightly more at the equator. The Earth is thus not a perfect sphere but is slightly fatter at the equator than at the poles.

Coriolis Effect

In a reference frame that rotates at a constant angular speed ω (relative to an inertial frame), there exists another pseudoforce known as the *Coriolis force*. It appears to act on an object in a rotating reference frame only if the object is moving relative to that rotating reference frame, and it acts to deflect the object sideways. It, too, is an effect of the rotating reference frame being noninertial and hence is referred to as an *inertial force*. It also affects the weather.

To see how the Coriolis force arises, consider two people, A and B, at rest on a platform rotating with angular speed ω, as shown in Fig. C–2a. They are situated at distances r_A and r_B from the axis of rotation (at O). The woman at A throws a ball with a horizontal velocity $\vec{\mathbf{v}}$ (in her reference frame) radially outward toward the man at B on the outer edge of the platform. In Fig. C–2a, we view the situation from an inertial reference frame. The ball initially has not only the velocity $\vec{\mathbf{v}}$ radially outward, but also a tangential velocity $\vec{\mathbf{v}}_A$ due to the rotation of the platform. Now Eq. 8–4 tells us that $v_A = r_A\omega$, where r_A is the woman's radial distance from the axis of rotation at O. If the man at B had this same velocity v_A, the ball would reach him perfectly. But his speed is $v_B = r_B\omega$, which is greater than v_A because $r_B > r_A$. Thus, when the ball reaches the outer edge of the platform, it passes a point that the man at B has already gone by because his speed in that direction is greater than the ball's. So the ball passes behind him.

Figure C–2b shows the situation as seen from the rotating platform as frame of reference. Both A and B are at rest, and the ball is thrown with velocity $\vec{\mathbf{v}}$ toward B, but the ball deflects to the right as shown and passes behind B as previously described. This is not a centrifugal-force effect, because that would act radially outward. Instead, this effect acts sideways, perpendicular to $\vec{\mathbf{v}}$, and is called a **Coriolis acceleration**; it is said to be due to the Coriolis force, which is a fictitious inertial force. Its explanation as seen from an inertial system was given above: it is an effect of being in a rotating system, for which a point farther from the rotation axis has a higher linear speed. On the other hand, when viewed from the rotating system, the motion can be described using Newton's second law, $\Sigma\vec{\mathbf{F}} = m\vec{\mathbf{a}}$, if we add a "pseudoforce" term corresponding to this Coriolis effect.

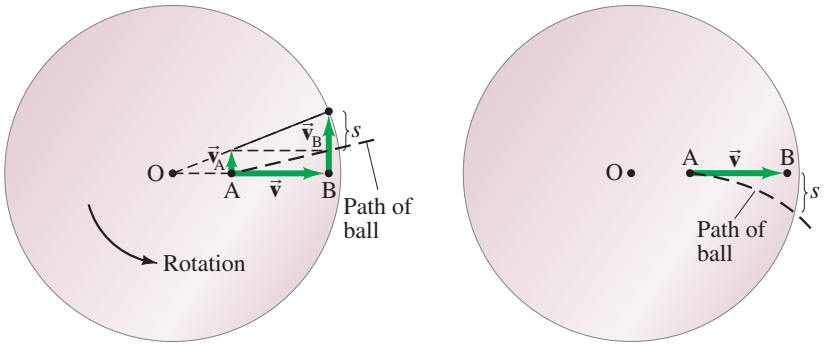


(a) Inertial reference frame

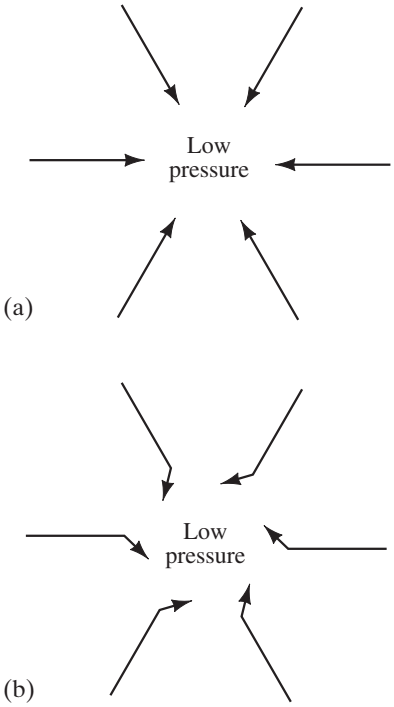
(b) Rotating reference frame

FIGURE C–2 The origin of the Coriolis effect. Looking down on a rotating platform, (a) as seen from a nonrotating inertial reference frame, and (b) as seen from the rotating platform as frame of reference.

FIGURE C–2 (Repeated.)
The origin of the Coriolis effect. Looking down on a rotating platform, (a) as seen from a nonrotating inertial reference frame, and (b) as seen from the rotating platform as frame of reference.



(a) Inertial reference frame (b) Rotating reference frame



(a)

(b)



(c)

FIGURE C–3 (a) Winds (moving air masses) would flow directly toward a low-pressure area if the Earth did not rotate; (b) and (c): because of the Earth's rotation, the winds are deflected to the right in the Northern Hemisphere (as in Fig. C–2) as if a fictitious (Coriolis) force were acting.

Let us determine the magnitude of the Coriolis acceleration for the simple case described above. (We assume v is large and distances short, so we can ignore gravity.) We do the calculation from the inertial reference frame (Fig. C–2a). The ball moves radially outward a distance $r_B - r_A$ at speed v in a short time t given by

$$r_B - r_A = vt.$$

During this time, the ball moves to the side a distance s_A given by

$$s_A = v_A t.$$

The man at B, in this time t, moves a distance

$$s_B = v_B t.$$

The ball therefore passes behind him a distance s (Fig. C–2a) given by

$$s = s_B - s_A = (v_B - v_A)t.$$

We saw earlier that $v_A = r_A\omega$ and $v_B = r_B\omega$, so

$$s = (r_B - r_A)\omega t.$$

We substitute $r_B - r_A = vt$ (see above) and get

$$s = \omega v t^2. \qquad \textbf{(C–1)}$$

This same s equals the sideways displacement as seen from the noninertial rotating system (Fig. C–2b).

Equation C–1 corresponds to motion at constant acceleration, because as we saw in Chapter 2 (Eq. 2–11b), $y = \frac{1}{2}at^2$ for a constant acceleration (with zero initial velocity in the y direction). Thus, if we write Eq. C–1 in the form $s = \frac{1}{2}a_{Cor}t^2$, we see that the Coriolis acceleration a_{Cor} is

$$a_{Cor} = 2\omega v. \qquad \textbf{(C–2)}$$

This relation is valid for any velocity in the plane of rotation perpendicular to the axis of rotation (in Fig. C–2, the axis through point O perpendicular to the page).

Because the Earth rotates, the Coriolis effect has some interesting manifestations on the Earth. It affects the movement of air masses and thus has an influence on weather. In the absence of the Coriolis effect, air would rush directly into a region of low pressure, as shown in Fig. C–3a. But because of the Coriolis effect, the winds are deflected to the right in the Northern Hemisphere (Fig. C–3b), since the Earth rotates from west to east. So there tends to be a counterclockwise wind pattern around a low-pressure area. The reverse is true in the Southern Hemisphere. Thus cyclones rotate counterclockwise in the Northern Hemisphere and clockwise in the Southern Hemisphere. The same effect explains the easterly trade winds near the equator: any winds heading south toward the equator will be deflected toward the west (that is, as if coming from the east).

The Coriolis effect also acts on a falling object. An object released from the top of a high tower will not hit the ground directly below the release point, but will be deflected slightly to the east. Viewed from an inertial frame, this happens because the top of the tower revolves with the Earth at a slightly higher speed than the bottom of the tower.

APPENDIX D

Molar Specific Heats for Gases, and the Equipartition of Energy

Molar Specific Heats for Gases

The values of the specific heats for gases depend on how the thermodynamic process is carried out. Two important processes are those in which either the volume or the pressure is kept constant, and Table D–1 shows how different they can be.

The difference in specific heats for gases is nicely explained in terms of the first law of thermodynamics and kinetic theory. For gases we usually use **molar specific heats**, C_V and C_P, which are defined as the heat required to raise 1 mol of a gas by 1 K (or 1 C°) at constant volume and at constant pressure, respectively. In analogy to Eq. 14–2, the heat Q needed to raise the temperature of n moles of gas by ΔT is

$$Q = nC_V \Delta T \qquad \text{[volume constant]} \quad \textbf{(D–1a)}$$

$$Q = nC_P \Delta T. \qquad \text{[pressure constant]} \quad \textbf{(D–1b)}$$

We can see from the definition of molar specific heat (compare Eqs. 14–2 and D–1) that

$$C_V = Mc_V$$

$$C_P = Mc_P,$$

where M is the molecular mass of the gas ($M = m/n$ in grams/mol).[†] The values for molar specific heats are included in Table D–1. These values are nearly the same for different gases that have the same number of atoms per molecule.

Now we use kinetic theory and imagine that an ideal gas is slowly heated via two different processes—first at constant volume, and then at constant pressure. In both processes, we let the temperature increase by the same amount, ΔT.

[†]For example, $M = 2$ g/mol for He, and $M = 32$ g/mol for O_2.

TABLE D–1 Specific Heats of Gases at 15°C

Gas	Specific Heats (kcal/kg · K)		Molar Specific Heats (cal/mol · K)		$C_P - C_V$ (cal/mol · K)
	c_V	c_P	C_V	C_P	
Monatomic					
He	0.75	1.15	2.98	4.97	1.99
Ne	0.148	0.246	2.98	4.97	1.99
Diatomic					
N_2	0.177	0.248	4.96	6.95	1.99
O_2	0.155	0.218	5.03	7.03	2.00
Triatomic					
CO_2	0.153	0.199	6.80	8.83	2.03
H_2O (100°C)	0.350	0.482	6.20	8.20	2.00
Polyatomic					
C_2H_6	0.343	0.412	10.30	12.35	2.05

In the constant-volume process, no work is done since $\Delta V = 0$. Thus, according to the first law of thermodynamics ($Q = \Delta U + W$, Section 15–1), the heat added (which we denote by Q_V) all goes into increasing the internal energy of the gas:

$$Q_V = \Delta U.$$

In the constant-pressure process, work *is* done. Hence the heat added, Q_P, must not only increase the internal energy but also is used to do work $W = P\,\Delta V$. Thus, to increase the temperature by the same ΔT, more heat must be added in the process at constant pressure than in the process at constant volume. For the process at constant pressure, the first law of thermodynamics gives

$$Q_P = \Delta U + P\,\Delta V.$$

Since ΔU is the same in the two processes (we chose ΔT to be the same), we can combine the two above equations:

$$Q_P - Q_V = P\,\Delta V.$$

From the ideal gas law, $V = nRT/P$, so for a process at constant pressure $\Delta V = nR\,\Delta T/P$. Putting this into the last equation and using Eqs. D–1, we find

$$nC_P\,\Delta T - nC_V\,\Delta T = P\left(\frac{nR\,\Delta T}{P}\right)$$

or, after cancellations,

$$C_P - C_V = R. \qquad \textbf{(D–2)}$$

Since the gas constant $R = 8.314\,\text{J/mol}\cdot\text{K} = 1.99\,\text{cal/mol}\cdot\text{K}$, our prediction is that C_P will be larger than C_V by about $1.99\,\text{cal/mol}\cdot\text{K}$. Indeed, this is very close to what is obtained experimentally, as shown in the last column in Table D–1.

Now we calculate the molar specific heat of a monatomic gas using kinetic theory. For a process carried out at constant volume, no work is done, so the first law of thermodynamics tells us that

$$\Delta U = Q_V.$$

For an ideal monatomic gas, the internal energy, U, is the total kinetic energy of all the molecules,

$$U = N\left(\tfrac{1}{2}m\overline{v^2}\right) = \tfrac{3}{2}nRT$$

as we saw in Section 14–2. Then, using Eq. D–1a, we write $\Delta U = Q_V$ as

$$\Delta U = \tfrac{3}{2}nR\,\Delta T = nC_V\,\Delta T \qquad \textbf{(D–3)}$$

or

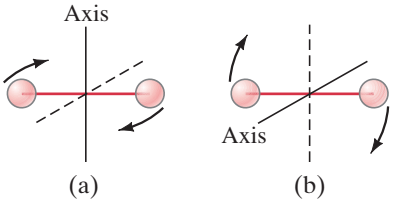
$$C_V = \tfrac{3}{2}R. \qquad \textbf{(D–4)}$$

Since $R = 8.314\,\text{J/mol}\cdot\text{K} = 1.99\,\text{cal/mol}\cdot\text{K}$, kinetic theory predicts that $C_V = 2.98\,\text{cal/mol}\cdot\text{K}$ for an ideal monatomic gas. This is very close to the experimental values for monatomic gases such as helium and neon (Table D–1). From Eq. D–2, C_P is predicted to be $R + C_V = (1.99 + 2.98)\,\text{cal/mol}\cdot\text{K} = 4.97\,\text{cal/mol}\cdot\text{K}$, also in agreement with experiment (Table D–1).

Equipartition of Energy

The measured molar specific heats for more complex gases (Table D–1), such as diatomic (two atoms) and triatomic (three atoms) gases, increase with the increased number of atoms per molecule. We can explain this by assuming that the internal energy includes not only translational kinetic energy but other forms of energy as well. For example, in a diatomic gas (Fig. D–1), the two atoms can rotate about two different axes (but rotation about a third axis passing through the two atoms would give rise to very little energy since the moment of inertia is so small). The molecules can have rotational as well as translational kinetic energy.

FIGURE D–1 A diatomic molecule can rotate about two different axes.



It is useful to introduce the idea of **degrees of freedom**, by which we mean the number of independent ways molecules can possess energy. For example, a monatomic gas has three degrees of freedom, because an atom can have velocity along the x, y, and z axes. These are considered to be three independent motions because a change in any one of the components would not affect the others. A diatomic molecule has the same three degrees of freedom associated with translational kinetic energy plus two more degrees of freedom associated with rotational kinetic energy (Fig. D–1), for a total of five degrees of freedom.

Table D–1 indicates that the C_V for diatomic gases is about $\frac{5}{3}$ times as great as for a monatomic gas—that is, in the same ratio as their degrees of freedom. This result led nineteenth-century physicists to the **principle of equipartition of energy**. This principle states that energy is shared equally among the active degrees of freedom, and each active degree of freedom of a molecule has on average an energy equal to $\frac{1}{2}kT$. Thus, the average energy for a molecule of a monatomic gas would be $\frac{3}{2}kT$ (which we already knew) and of a diatomic gas $\frac{5}{2}kT$. Hence the internal energy of a diatomic gas would be $U = N(\frac{5}{2}kT) = \frac{5}{2}nRT$, where n is the number of moles. Using the same argument we did for monatomic gases, we see that for diatomic gases the molar specific heat at constant volume would be $\frac{5}{2}R = 4.97\ \text{cal/mol}\cdot\text{K}$, close to measured values (Table D–1). More complex molecules have even more degrees of freedom and thus greater molar specific heats.

However, measurements showed that for diatomic gases at very low temperatures, C_V has a value of only $\frac{3}{2}R$, as if it had only three degrees of freedom. And at very high temperatures, C_V was about $\frac{7}{2}R$, as if there were seven degrees of freedom. The explanation is that at low temperatures, nearly all molecules have only translational kinetic energy; that is, no energy goes into rotational energy and only three degrees of freedom are "active." At very high temperatures, all five degrees of freedom are active plus two additional ones. We interpret the two new degrees of freedom as being associated with the two atoms vibrating, as if they were connected by a spring (Fig. D–2). One degree of freedom comes from the kinetic energy of the vibrational motion, and the second comes from the potential energy of vibrational motion $(\frac{1}{2}kx^2)$. At room temperature, these two degrees of freedom are apparently not active (Fig. D–3). Why fewer degrees of freedom are "active" at lower temperatures was eventually explained by Einstein using quantum theory.

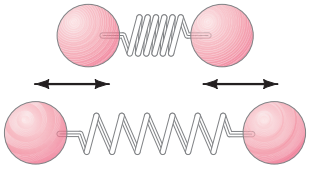


FIGURE D–2 A diatomic molecule can vibrate, as if the two atoms were connected by a spring. They aren't, but rather they exert forces on each other that are electrical in nature—of a form that resembles a spring force.

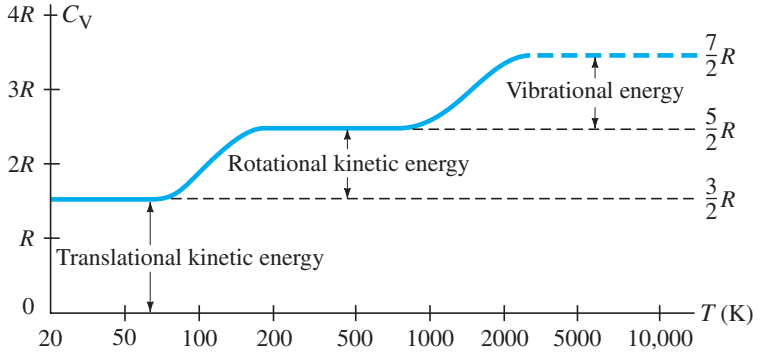
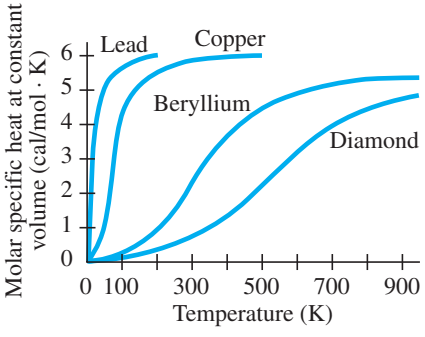


FIGURE D–3 Molar specific heat C_V as a function of temperature for hydrogen molecules (H_2). As the temperature is increased, some of the translational kinetic energy can be transferred in collisions into rotational kinetic energy and, at still higher temperature, into vibrational kinetic and potential energy. [Note: H_2 dissociates into two atoms at about 3200 K, so the last part of the curve is shown dashed.]

Solids

The principle of equipartition of energy can be applied to solids as well. The molar specific heat of any solid at high temperature is close to $3R$ $(6.0\ \text{cal/mol}\cdot\text{K})$, Fig. D–4. This is called the *Dulong and Petit value* after the scientists who first measured it in 1819. (Note that Table 14–1 gave the specific heats per kilogram, not per mole.) At high temperatures, each atom apparently has six degrees of freedom, although some are not active at low temperatures. Each atom in a crystalline solid can vibrate about its equilibrium position as if it were connected by springs to each of its neighbors (Fig. D–5). Thus it can have three degrees of freedom for kinetic energy and three more associated with potential energy of vibration in each of the x, y, and z directions, which is in accord with measured values.

FIGURE D–4 Molar specific heats of solids as a function of temperature.



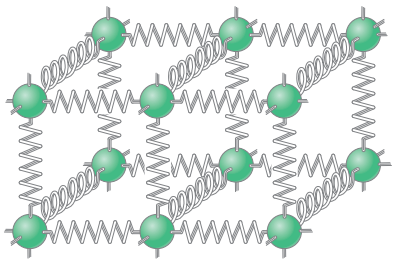


FIGURE D–5 The atoms in a crystalline solid can vibrate about their equilibrium positions as if they were connected to their neighbors by springs. (The forces between atoms are actually electrical in nature.)

APPENDIX E

Galilean and Lorentz Transformations

We examine in detail the mathematics of relating quantities in one inertial reference frame to the equivalent quantities in another. In particular, we will see how positions and velocities *transform* (that is, change) from one frame of reference to the other.

We begin with the classical, or Galilean, viewpoint. Consider two inertial reference frames S and S′ which are each characterized by a set of coordinate axes, Fig. E–1. The axes x and y (z is not shown) refer to S, and x' and y' refer to S′. The x' and x axes overlap one another, and we assume that frame S′ moves to the right (in the x direction) at speed v with respect to S. For simplicity let us assume the origins 0 and 0′ of the two reference frames are superimposed at time $t = 0$.

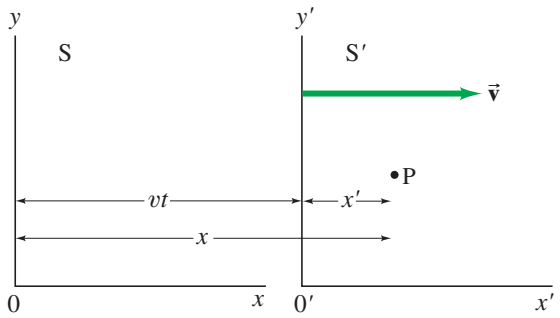


FIGURE E–1 Inertial reference frame S′ moves to the right at speed v with respect to inertial frame S.

Now consider an event that occurs at some point P (Fig. E–1) represented by the coordinates x', y', z' in reference frame S′ at the time t'. What will be the coordinates of P in S? Since S and S′ overlap precisely initially, after a time t', S′ will have moved a distance vt'. Therefore, at time t', $x = x' + vt'$. The y and z coordinates, on the other hand, are not altered by motion along the x axis; thus $y = y'$ and $z = z'$. Finally, since time is assumed to be absolute in Galilean–Newtonian physics, clocks in the two frames will agree with each other; so $t = t'$. We summarize these in the following **Galilean transformation equations**:

$$\begin{aligned} x &= x' + vt' \\ y &= y' \\ z &= z' \\ t &= t'. \end{aligned} \qquad \text{[Galilean]} \quad \textbf{(E–1)}$$

These equations give the coordinates of an event in the S frame when those in the S′ frame are known. If those in the S frame are known, then the S′ coordinates are obtained from

$$x' = x - vt, \qquad y' = y, \qquad z' = z, \qquad t' = t. \qquad \text{[Galilean]}$$

These four equations are the "inverse" transformation, and are obtained from Eqs. E–1 by exchanging primed and unprimed quantities and replacing v by $-v$. This makes sense because, as seen from the S′ frame, S moves to the left (negative x direction) with speed v.

Now suppose that the point P in Fig. E–1 represents an object that is moving. Let the components of its velocity vector in S′ be u'_x, u'_y, and u'_z. (We use u to distinguish it from the relative velocity of the two frames, v.) Now $u'_x = \Delta x'/\Delta t'$, $u'_y = \Delta y'/\Delta t'$, and $u'_z = \Delta z'/\Delta t'$, where all quantities are as measured in the S′ frame. For example, if at time t'_1 the particle is at x'_1 and a short time later, t'_2, it is at x'_2, then

$$u'_x = \frac{x'_2 - x'_1}{t'_2 - t'_1} = \frac{\Delta x'}{\Delta t'}.$$

The velocity of P as seen from S will have components u_x, u_y, and u_z. We can show how these are related to the velocity components in S′ by using Eqs. E–1. For example,

$$\begin{aligned} u_x &= \frac{\Delta x}{\Delta t} = \frac{x_2 - x_1}{t_2 - t_1} = \frac{(x'_2 + vt'_2) - (x'_1 + vt'_1)}{t'_2 - t'_1} \\ &= \frac{(x'_2 - x'_1) + v(t'_2 - t'_1)}{t'_2 - t'_1} \\ &= \frac{\Delta x'}{\Delta t'} + v = u'_x + v. \end{aligned}$$

For the other components, $u'_y = u_y$ and $u'_z = u_z$, so we have

$$\begin{aligned} u_x &= u'_x + v \\ u_y &= u'_y \\ u_z &= u'_z. \end{aligned} \qquad \text{[Galilean]} \quad \textbf{(E–2)}$$

These are known as the **Galilean velocity transformation equations**. We see that the y and z components of velocity are unchanged, but the x components differ by v. This is just what we have used before when dealing with relative velocity (Section 3–8). For example, if S′ is a train and S the Earth, and the train moves with speed v with respect to Earth, a person walking toward the front of the train with speed u'_x will have a speed with respect to the Earth of $u_x = u'_x + v$.

The Galilean transformations, Eqs. E–1 and E–2, are accurate only when the velocities involved are not relativistic (Chapter 26)—that is, much less than the speed of light, c. We can see, for example, that the first of Eqs. E–2 will not work for the speed of light, c, which is the same in all inertial reference frames (a basic postulate in the theory of relativity). That is, light traveling in S′ with speed $u'_x = c$ will have speed $c + v$ in S, according to Eq. E–2, whereas the theory of relativity insists it must be c in S. Clearly, then, a new set of transformation equations is needed to deal with relativistic velocities.

We will derive the required equations, again looking at Fig. E–1. We assume the transformation is linear and for x is of the form

$$x = \gamma(x' + vt'). \qquad \textbf{(i)}$$

That is, we modify the first of Eqs. E–1 by multiplying by a factor γ which is yet to be determined.[†] We assume the y and z equations are unchanged

$$y = y', \qquad z = z'$$

because there is no length contraction in these directions. We will not assume a form for t, but will derive it. The inverse equations must have the same form with v replaced by $-v$. (The principle of relativity demands it, since S′ moving to the right with respect to S is equivalent to S moving to the left with respect to S′.) Therefore

$$x' = \gamma(x - vt). \qquad \textbf{(ii)}$$

Suppose a light pulse leaves the common origin of S and S′ at time $t = t' = 0$.

[†] We are NOT assuming γ is $1/\sqrt{1 - v^2/c^2}$, as in Chapter 26. Our minds are open. Let's see.

Then after a time t it will have traveled along the x axis a distance $x = ct$ (in S), or $x' = ct'$ (in S′). Therefore, from Eqs. (i) and (ii) above,

$$ct = x = \gamma(ct' + vt') = \gamma(c + v)t', \quad \textbf{(iii)}$$

$$ct' = x' = \gamma(ct - vt) = \gamma(c - v)t. \quad \textbf{(iv)}$$

From Eq. (iv), $t' = \gamma(c - v)(t/c)$, and we substitute this into Eq. (iii) and find $ct = \gamma(c + v)\gamma(c - v)(t/c) = \gamma^2(c^2 - v^2)t/c$. We cancel out the t on each side and solve for γ to find

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}.$$

We have found that γ is, in fact, the value for γ we used in Chapter 26, Eq. 26–2.

Now that we have found γ, we need only find the relation between t and t'. To do so, we combine $x' = \gamma(x - vt)$ with $x = \gamma(x' + vt')$:

$$x' = \gamma(x - vt) = \gamma[\gamma(x' + vt') - vt].$$

We solve for t, doing some algebra, and find $t = \gamma(t' + vx'/c^2)$. In summary,

LORENTZ TRANSFORMATIONS

$$\begin{aligned} x &= \frac{1}{\sqrt{1 - v^2/c^2}}(x' + vt') \\ y &= y' \\ z &= z' \\ t &= \frac{1}{\sqrt{1 - v^2/c^2}}\left(t' + \frac{vx'}{c^2}\right). \end{aligned} \quad \textbf{(E–3)}$$

These are called the **Lorentz transformation equations**. They were first proposed, in a slightly different form, by Lorentz in 1904 to explain the null result of the Michelson–Morley experiment and to make Maxwell's equations take the same form in all inertial reference frames. A year later, Einstein derived them independently based on his theory of relativity. Notice that not only is the x equation modified as compared to the Galilean transformation, but so is the t equation. Indeed, we see directly in this last equation how the space and time coordinates mix.

The relativistically correct velocity equations are obtained using Eqs. E–3 (we let $\gamma = 1/\sqrt{1 - v^2/c^2}$) and $u_x = \Delta x/\Delta t$, $u'_x = \Delta x'/\Delta t'$:

$$u_x = \frac{\Delta x}{\Delta t} = \frac{\gamma(\Delta x' + v\,\Delta t')}{\gamma(\Delta t' + v\,\Delta x'/c^2)} = \frac{(\Delta x'/\Delta t') + v}{1 + (v/c^2)(\Delta x'/\Delta t')}$$

$$= \frac{u'_x + v}{1 + vu'_x/c^2}.$$

The others are obtained in the same way, and we collect them here:

RELATIVISTIC VELOCITY TRANSFORMATIONS

$$\begin{aligned} u_x &= \frac{u'_x + v}{1 + vu'_x/c^2} \\ u_y &= \frac{u'_y\sqrt{1 - v^2/c^2}}{1 + vu'_x/c^2} \\ u_z &= \frac{u'_z\sqrt{1 - v^2/c^2}}{1 + vu'_x/c^2}. \end{aligned} \quad \textbf{(E–4)}$$

The first of these equations is Eq. 26–11, which we used in Section 26–10 where we discussed how velocities do not add in our commonsense (Galilean) way, because of the denominator $(1 + vu'_x/c^2)$. We can now also see that the y and z components of velocity are also altered and that they depend on the x' component of velocity.

EXAMPLE E–1 **Length contraction.** Derive the length contraction formula, Eq. 26–3, from the Lorentz transformation equations.

APPROACH We consider measurements in two reference frames, S and S′, that move with speed v relative to each other, as in Fig. E–1.

SOLUTION Let an object of length ℓ_0 be at rest on the x axis in S. The coordinates of its two end points are x_1 and x_2, so that $x_2 - x_1 = \ell_0$. At any instant in S′, the end points will be at x'_1 and x'_2 as given by the Lorentz transformation equations. The length measured in S′ is $\ell = x'_2 - x'_1$. An observer in S′ measures this length by measuring x'_2 and x'_1 at the same time (in the S′ frame), so $t'_2 = t'_1$. Then, from the first of Eqs. E–3,

$$\ell_0 = x_2 - x_1 = \frac{1}{\sqrt{1 - v^2/c^2}}(x'_2 + vt'_2 - x'_1 - vt'_1).$$

Since $t'_2 = t'_1$, we have

$$\ell_0 = \frac{1}{\sqrt{1 - v^2/c^2}}(x'_2 - x'_1) = \frac{\ell}{\sqrt{1 - v^2/c^2}},$$

or

$$\ell = \ell_0\sqrt{1 - v^2/c^2},$$

which is Eq. 26–3a: the length contraction formula.

EXAMPLE E–2 **Time dilation.** Derive the time dilation formula, Eq. 26–1, from the Lorentz transformation equations.

APPROACH Again we compare measurements in two reference frames, S and S′, that move with speed v relative to each other, Fig. E–1.

SOLUTION The time Δt_0 between two events that occur at the same place $(x'_2 = x'_1)$ in S′ is measured to be $\Delta t_0 = t'_2 - t'_1$. Since $x'_2 = x'_1$, then from the last of Eqs. E–3, the time Δt between the events as measured in S is

$$\begin{aligned}\Delta t = t_2 - t_1 &= \frac{1}{\sqrt{1 - v^2/c^2}}\left(t'_2 + \frac{vx'_2}{c^2} - t'_1 - \frac{vx'_1}{c^2}\right) \\ &= \frac{1}{\sqrt{1 - v^2/c^2}}(t'_2 - t'_1) \\ &= \frac{\Delta t_0}{\sqrt{1 - v^2/c^2}},\end{aligned}$$

which is the time dilation formula, Eq. 26–1. Notice that we chose S′ to be the frame in which the two events occur at the same place, so that $x'_1 = x'_2$, and then the terms containing x'_1 and x'_2 cancel out.

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Answers to Odd-Numbered Problems

Chapter 1

1. (*a*) 3;
(*b*) 4;
(*c*) 3;
(*d*) 1;
(*e*) 2;
(*f*) 4;
(*g*) 2.

3. (*a*) 86,900;
(*b*) 9100;
(*c*) 0.88;
(*d*) 476;
(*e*) 0.0000362.

5. 4.6%.

7. 1.00×10^5 s.

9. 1%.

11. $(3.0 \pm 0.2) \times 10^9$ cm^2.

13. (*a*) 1 megavolt;
(*b*) 2 micrometers;
(*c*) 6 kilodays;
(*d*) 18 hectobucks;
(*e*) 700 nanoseconds.

15. (*a*) 1.5×10^{11} m;
(*b*) 1.5×10^8 km.

17. (*a*) $\dfrac{0.621 \text{ mi/h}}{1 \text{ km/h}}$;
(*b*) $\dfrac{3.28 \text{ ft/s}}{1 \text{ m/s}}$;
(*c*) $\dfrac{0.278 \text{ m/s}}{1 \text{ km/h}}$.

19. 7.3%.

21. (*a*) 3.16×10^7 s;
(*b*) 3.16×10^{16} ns;
(*c*) 3.17×10^{-8} yr.

23. 4×10^{14} kg.

25. 6×10^5 books.

27. 5×10^4 L.

29. 1×10^{11} gal/yr.

31. Second method.

33. $\left[\dfrac{M}{L^3}\right]$.

35. 10^{-35} m.

37. 3.3×10^5 chips/cylinder.

39. 46,000 years.

41. 400 jelly beans.

43. 75 minutes.

45. 5×10^5 metric tons, 1×10^8 gal.

47. 3×10^{-5}%.

49. 90 m.

51. 1.15 mi.

53. 1.18×10^9 atoms/m^2.

65. 4×10^{51} kg.

Chapter 2

1. 53 m.

3. 0.57 cm/s, no, we need the distance traveled.

5. 0.14 h.

7. (*a*) 350 km;
(*b*) 78 km/h.

9. (*a*) 3.68 m/s;
(*b*) 0.

11. 38 s.

13. 1.6 min.

15. 6.00 m/s.

17. 6.1 m/s^2.

19. 6.0 m/s^2, 0.61*g*'s.

21. (*a*) 21.2 m/s;
(*b*) 2.00 m/s^2.

23. 1 m/s^2, 110 m.

25. 260 m/s^2.

27. 112 m.

29. (*a*) 430 m;
(*b*) 42 s;
(*c*) 21 m, 19 m.

31. She will not be able to stop in time, 30.7 m past the intersection.

33. 460 m.

35. 0.70 m/s^2.

37. 61.8 m.

39. 17 m/s, 14 m.

41. (*a*) 4.8 m/s;
(*b*) 0.99 s.

43. (*a*) 17.9 m/s;
(*b*) 0.620 s, 4.28 s;
(*c*) There are two times at which the object reaches that height.

45. (*a*) 7.0×10^1 m/s, upward;
(*b*) 22 s;
(*c*) 1030 m;
(*d*) 29 s;
(*e*) −142 m/s;
(*f*) 44 s.

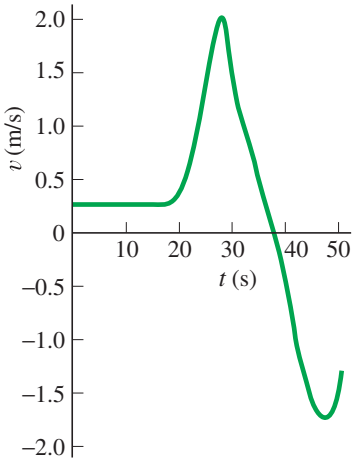
47. (*a*) 14 m/s;
(*b*) Fifth floor.

49. 1.6 m.

51. (*a*) 48 s;
(*b*) 90 s to 108 s;
(*c*) 0 s to 42 s, 65 s to 83 s, 90 s to 108 s;
(*d*) 65 s to 83 s.

53. (*a*) 0.3 m/s;
(*b*) 1.2 m/s;
(*c*) 0.30 m/s;
(*d*) 1.4 m/s;
(*e*) −0.95 m/s.

55.



57. 1.2 m.

59. 3.9×10^{-2} m/s^2.

61. 5.7 h, 850 km/h.

63. −20 m/s^2.

65. (*a*) 5.80 s;
(*b*) 41.4 m/s;
(*c*) 99.5 m.

67. She should try to stop the car.

69. 1.5 poles.

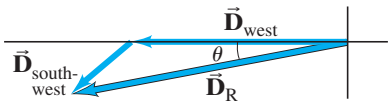
71. 30%.

73. 23.7 s, 840 km/h.

75. (*a*) 4.3×10^6 bits;
(*b*) 67%.

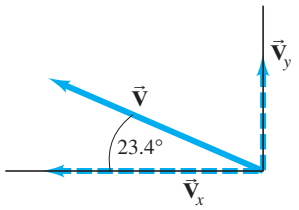
Chapter 3

1. 302 km, 13° south of west.



3. 11.70 units, −33.1°.

5. (*a*)



(*b*) −22.8 units, 9.85 units;
(*c*) 24.8 units, 23.4° above the −*x* axis.

7. (*a*) 1.3 units, positive *x* direction;
(*b*) 12.3 units, positive *x* direction;
(*c*) 12.3 units, negative *x* direction.

9. (*a*) *x* component 24.0, *y* component 11.6;
(*b*) 26.7 units, 25.8°.

11. 64.6, 53.1°.

13. (*a*) 62.6, −31.0°;
(*b*) 77.5, 71.9°;
(*c*) 77.5, 251.9°.

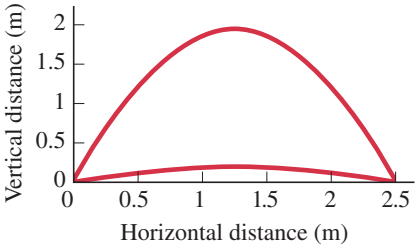
15. (−2845 m, 3589 m, 2450 m), 5190 m.

17. 3.7 m.

19. 6 times farther.

21. 14.5 m.

23. 18°, 72°.



25. 2.0 m/s.

27. (*a*) 30.8 m;
(*b*) 5.02 s;
(*c*) 136 m;
(*d*) 28.9 m/s.

29. 59.3 m.

31. 8.37 m/s down, 97.4 m/s.

33. No, 0.81 m too low; 4.6 m to 34.7 m.

35. (*a*) 4.0×10^1 m/s;
(*b*) 24 m/s.

37. 23.1 s.

39. (*a*) 10.4 m/s, 17° above the horizontal;
(*b*) 10.4 m/s, 17° below the horizontal.

41. 5.31° west of south.

43. 1.0 m/s.

45. 65 km/h, 58° west of north; 65 km/h, 32° south of east.

47. Horizontal: 3.9 m/s^2 leftward; vertical: 1.9 m/s^2 downward.

49. 6.6 m/s.

51. (*a*) 8.9 m/s;
(*b*) She will land exactly on the opposite bank, neither long nor short.

53. $v_T/\tan\theta$.

55. 1.9 m/s, 2.7 s.

57. 26.3 m/s, 18.8 m, in the "good" region, 0.714 s.

59. (*a*) 2.51 m/s, 61.4°;
(*b*) 3.60 m downstream, 6.60 m across the river.

61. (*a*) 13 m;
(*b*) 31° below the horizontal.

63. (*a*) $v_{x0} = 13$ m/s, $v_{y0} = 12$ m/s;
(*b*) 33 m.

Chapter 4

1. 77 N.

3. 1450 N.

5. -1.3×10^6 N, 39%, 1.3×10^6 N.

7. −3100 N.

9. 780 N, backward.

11. (*a*) 196 N, 196 N;
(*b*) 98.0 N, 294 N.

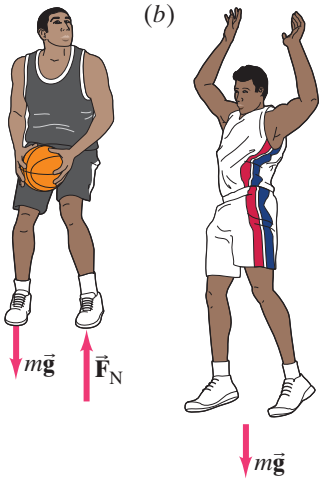
13. Descend with downward $a \geq 2.2$ m/s^2.

15. −2800 m/s^2, 280*g*'s, 1.9×10^5 N.

17. (*a*) −7.35 m/s^2;
(*b*) 1290 N.

19. (*a*) 7.4 m/s, downward;
(*b*) 2100 N, upward.

21. (*a*) (*b*)



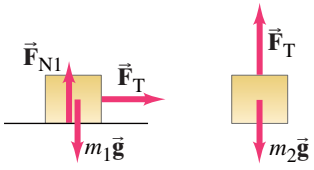
23. 1410 N.

25. (*a*) 31 N (lower cord), 63 N (upper cord);
(*b*) 35 N (lower cord), 71 N (upper cord).

27. $F_{T1}/F_{T2} = 2$.

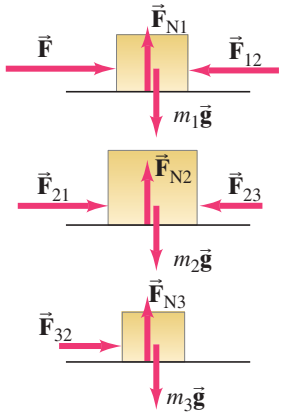
29. (*a*) 1.0×10^1 m/s^2;
(*b*) 3.3 m/s.

31. (*a*)



(*b*) $g\dfrac{m_B}{m_A + m_B}, g\dfrac{m_A m_B}{m_A + m_B}$.

33. (*a*)



(*b*) $\dfrac{F}{m_A + m_B + m_C}$;

(*c*) $\dfrac{Fm_A}{m_A + m_B + m_C}$,
$\dfrac{Fm_B}{m_A + m_B + m_C}$,
$\dfrac{Fm_C}{m_A + m_B + m_C}$;

(*d*)
$F_{BC} = F_{CB} = \dfrac{Fm_C}{m_A + m_B + m_C}$,

$F_{BA} = F_{AB} = F\dfrac{m_B + m_C}{m_A + m_B + m_C}$;

(*e*) $a = 3.20$ m/s^2,
$F_{A\,net} = F_{B\,net} = F_{C\,net} = 32.0$ N,
$F_{AB} = F_{BA} = 64.0$ N,
$F_{BC} = F_{CB} = 32.0$ N.

35. 65 N, 0.

37. 0.20.

39. −7.4 m/s^2.

41. 4.2 m.

43. 1.4.

45. (*a*) 5.0 kg;
(*b*) 1.0×10^1 kg.

47. (*a*) 1.7 m/s^2;
(*b*) 430 N;
(*c*) 1.7 m/s^2, 220 N.

49. 4.8 s.

51. 4.0×10^2 m.

53. (*a*) 2.8 m;
(*b*) 2.5 s.
55. (*a*) 0.86 m;
(*b*) 1.33 s.
57. $\mu_k = 0.21$.
59. 0.51.
61. (*a*) 1.6 m/s^2;
(*b*) 0.53.
63. 0.86 m/s^2.
65. 140 N.
67. 73 m/s.
69. 4.2°.
71. (*a*) 8.76×10^4 N, upward;
(*b*) 1.14×10^4 N;
(*c*) 1.14×10^4 N, downward.
73. 380 N (between second and last climbers), 760 N (between first and second climbers).
75. 0.36.
77. Yes, 14 m/s.
79. (*a*) 11 kg;
(*b*) 0.88 m/s^2.
81. (*b*) Yes.
83. (*a*) 16 m/s;
(*b*) 13 m/s.

Chapter 5

1. (*a*) 1.01 m/s^2;
(*b*) 22.7 N.
3. 12 m/s.
5. 13 m/s.
7. 34 m/s.
9. 24 m/s, yes.
11. 8.5 rpm.
13. 1700 rev/day.
15. 0.210.
17. $F_A = 4\pi^2 f^2(m_A r_A + m_B r_B)$,
$F_B = 4\pi^2 f^2 m_B r_B$.
19. 30 m.
21. 0.32.
23. $a_{tan} = 4.1$ m/s^2, $a_{rad} = 13$ m/s^2; 1.4.
25. (*a*) 0.930 m/s;
(*b*) 2.83 m/s.
27. (*a*) 24.0 kg on both;
(*b*) $w_{Earth} = 235$ N,
$w_{Planet} = 288$ N.
29. 3.94 kg, 0.06 kg.
31. 1.62 m/s^2.
33. 4.79×10^{20} N, $\theta = 24.5°$.
35. 2/9.
37. 9.6×10^{17} N; 2.7×10^{-5}.
39. 2.02×10^7 m.
41. 7460 m/s.
43. 2.4 m/s^2 upward.
45. 7.05×10^3 s.
47. (*a*) 568 N;
(*b*) 568 N;
(*c*) 699 N;
(*d*) 440 N;
(*e*) 0.
49. (*a*) 59 N, away from the Moon;
(*b*) 76 N, toward the Moon.
51. 9.6 s.
53. 160 yr.
55. 84.5 min.
57. 2×10^8 yr.
59. 0.38.
61. 5.97×10^{-3} m/s^2, 3.56×10^{22} N; the Sun.
63. 28.3 m/s, 0.410 rev/s.
65. 0.18; no; the wall pushes against the riders, so by Newton's third law, they push against the wall.
67. 9.2 m/s.
69. (*a*) In circular motion, they accelerate toward each other without moving toward each other.
(*b*) 9.6×10^{29} kg.
71. Yes; $\sqrt{rg}$, where r is the radius of the vertical circle.
73. $T_{inner} = 2.0 \times 10^4$ s,
$T_{outer} = 7.1 \times 10^4$ s.
75. 25.0 m/s.
77. (*a*) 7.6×10^6 m;
(*b*) 3.8×10^4 N;
(*c*) 1.2×10^6 m.
79. 1.21×10^6 m.

Chapter 6

1. 2.06×10^4 J.
3. 2300 J.
5. 1.0×10^6 J.
7. 1960 J.
9. 390 J.
11. 2 m.
13. (*a*) 2800 J;
(*b*) 2100 J.
15. 484 m/s.
17. -5.51×10^{-19} J.
19. The lighter one, $\sqrt{2}$; both the same.
21. 43 m/s.
23. (*a*) 3010 N;
(*b*) 7480 J;
(*c*) 5.42×10^4 J;
(*d*) -4.67×10^4 J;
(*e*) 7.51 m/s.
25. 1.01 m.
27. (*a*) 9.06×10^5 J;
(*b*) 9.06×10^5 J;
(*c*) yes.
29. 45.4 m/s.
31. 4.89 m/s.
33. 1.4×10^5 N/m.
35. (*a*) 7.47 m/s;
(*b*) 3.01 m.
37. 52 m.
39. (*a*) 9.19×10^4 J;
(*b*) 433 N.
41. 332 J.
43. (*a*) 15.3 m/s;
(*b*) 1.03 N, upward.
45. (*a*) 20%;
(*b*) 6.3 m/s, 5.6 m/s;
(*c*) Thermal energy.
47. (*a*) 2.8 m;
(*b*) 1.5 m;
(*c*) 1.5 m.
49. 22.0 s.
51. 510 N.
53. 480 W.
55. 5.4×10^6 J.
57. 480 W.
59. 2×10^6 W.
61. 610 W.
63. 14.9 m/s.
65. (*a*) $\sqrt{Fx/m}$;
(*b*) $\sqrt{3Fx/4m}$.
67. (*a*) 0.014 J;
(*b*) 0.039 J.
69. (*a*) -9.0×10^4 J;
(*b*) 8.2×10^4 N;
(*c*) -2.3×10^5 J.
71. 340 W.
73. (*a*) 1.0×10^4 J;
(*b*) 16 m/s.
75. (*a*) 42 m/s;
(*b*) 3.2×10^5 W.
77. (*a*) 8.9×10^5 J;
(*b*) 54 W, 0.072 hp;
(*c*) 360 W, 0.48 hp.
79. (*a*) 0.39 m;
(*b*) $\mu_s < 0.53$;
(*c*) 1.4 m/s.
81. 1.8 N.

Chapter 7

1. 0.24 kg·m/s.
3. 10.2 m/s.

5. 5.9×10^7 N, opposite the gas velocity.

7. -0.898 m/s.

9. 2500 m/s.

11. 0.99 m/s.

13. 4.9×10^6 N.

15. 2230 N, toward the pitcher.

17. (*a*) 9.0×10^1 kg·m/s;
(*b*) 1.1×10^4 N.

19. (*a*) -0.16 m/s;
(*b*) 521 N;
(*c*) astronaut: 391 J; capsule: 26 J.

21. (*a*) 290 kg·m/s eastward;
(*b*) 290 kg·m/s westward;
(*c*) 290 kg·m/s eastward;
(*d*) 340 N eastward.

23. (*a*) 5 N·s;
(*b*) 80 m/s.

25. 0.440-kg ball: 1.27 m/s, east;
0.220-kg ball: 5.07 m/s, east.

27. Tennis ball: 2.50 m/s; other ball: 5.00 m/s; both in direction of tennis ball's initial motion.

29. 2.64 *m*.

31. (*a*) 1.00;
(*b*) 0.890;
(*c*) 0.286;
(*d*) 0.0192.

33. (*a*) $\dfrac{\Delta \text{KE}}{\text{KE}_i} = \dfrac{-M}{m + M}$;
(*b*) -0.95.

35. $\text{KE}_{\text{heavy}} = 2200$ J,
$\text{KE}_{\text{light}} = 3300$ J.

37. 0.13 kg m/s, upward.

39. (*a*) 12.1 m/s;
(*b*) 56.4 J before, 13.7 J after.

41. (*a*) $p_x : m_A v_A = m_A v'_A \cos\theta'_A + m_B v'_B \cos\theta'_B$;
$p_y : 0 = m_A v'_A \sin\theta'_A - m_B v'_B \sin\theta'_B$;
(*b*) 1.23 m/s, 46.9°.

43. Along the *x* axis, $v'_A = 3.7$ m/s, $v'_B = 2.0$ m/s.

45. 6.5×10^{-11} m.

47. 2.62 m.

49. $(1.2\ell,\ 0.9\ell)$ relative to back left corner.

51. 11 kg.

53. Relative to the shoulder: $x_{\text{CM}} = 21.7$ cm, $y_{\text{CM}} = 7.6$ cm.

55. 8.1% of their height; yes.

57. (*a*) 5.8 m;
(*b*) 4.0 m;
(*c*) 4.2 m.

59. 0.45 m toward initial position of 85-kg person.

61. 2.0×10^1 m.

63. 8.

65. (*a*) $v'_{\text{A}} = 3.65$ m/s, $v'_{\text{B}} = 4.45$ m/s;
(*b*) $\Delta p_{\text{A}} = -370$ kg·m/s, $\Delta p_{\text{B}} = 370$ kg·m/s.

67. 110 km/h $\approx$ 70 mi/h.

69. 340 m/s.

71. (*a*) 2.0 m/s;
(*b*) $v'_A = -0.37$ m/s, $v'_B = 1.7$ m/s;
(*c*) $h_{\text{A}} = 6.9 \times 10^{-3}$ m, $h_{\text{B}} = 1.4 \times 10^{-1}$ m.

73. (*a*) 1.5×10^{21} J;
(*b*) 38,000.

75. $v_m = 3D\sqrt{\dfrac{k}{12m}}$, $v_{3m} = D\sqrt{\dfrac{k}{12m}}$.

77. (*a*) No;
(*b*) $m_{\text{B}}/m_{\text{A}}$;
(*c*) $m_{\text{B}}/m_{\text{A}}$;
(*d*) stays at rest.

Chapter 8

1. (*a*) 0.785 rad, $\pi/4$ rad;
(*b*) 1.05 rad, $\pi/3$ rad;
(*c*) 1.57 rad, $\pi/2$ rad;
(*d*) 6.28 rad, 2π rad;
(*e*) 7.77 rad, $89\pi/36$ rad.

3. 5.3×10^3 m.

5. (*a*) 750 rad/s;
(*b*) 23 m/s;
(*c*) 4.5×10^7 bit/s.

7. (*a*) 230 rad/s;
(*b*) 4.0×10^1 m/s, 9.3×10^3 m/s^2.

9. (*a*) 0.105 rad/s;
(*b*) 1.75×10^{-3} rad/s;
(*c*) 1.45×10^{-4} rad/s;
(*d*) 0.

11. (*a*) 464 m/s;
(*b*) 185 m/s;
(*c*) 345 m/s.

13. 3.3×10^4 rpm.

15. (*a*) 1.5×10^{-4} rad/s^2;
(*b*) $a_{\text{rad}} = 1.2 \times 10^{-2}$ m/s^2, $a_{\text{tan}} = 6.2 \times 10^{-4}$ m/s^2.

17. (*a*) -96 rad/s^2;
(*b*) 98 rev.

19. (*a*) 46 rev/min^2;
(*b*) 46 rpm.

21. 33 m.

23. (*a*) 0.53 rad/s^2;
(*b*) 13 s.

25. 1.2 m·N, clockwise.

27. $mg(\ell_2 - \ell_1)$, clockwise.

29. (*a*) 14 m·N;
(*b*) -13 m·N.

31. 0.12 kg·m^2.

33. 1.2×10^{-10} m.

35. (*a*) 7.8 m·N;
(*b*) 310 N.

37. 22 m·N.

39. (*a*) 7.0 kg·m^2;
(*b*) 0.70 kg·m^2;
(*c*) *y* axis.

41. 23 kg.

43. 31 N.

45. (*a*) $a_{\text{A}} = 0.69$ m/s^2, upward; $a_{\text{B}} = 0.69$ m/s^2, downward;
(*b*) 2%.

47. 1.36×10^4 J.

49. 48.8 J.

51. $\sqrt{\frac{10}{7} g(R_0 - r_0)}$.

53. 2.0 m/s.

55. 5.12 kg·m^2/s.

57. (*a*) Since angular momentum is conserved, an increase in rotational inertia must be accompanied by a decrease in angular velocity;
(*b*) 1.5.

59. 0.38 rev/s.

61. (*a*) 1.0×10^1 kg·m^2/s;
(*b*) -2.5 m·N.

63. (*a*) 0.43 rad/s;
(*b*) 0.80 rad/s.

65. 2.0 rev/s.

67. -0.31 rad/s.

69. 0.80 m·N.

71. (*a*) 3.5 m;
(*b*) 4.7 s.

73. 2.1 m·N, from the arm muscles.

75. (*a*) 3.1×10^{-47} kg·m^2;
(*b*) 1.9×10^{-47} kg·m^2.

77. $\dfrac{Mg\sqrt{2Rh - h^2}}{R - h}$.

79. (*a*) 4.84 J;
(*b*) $F_{4\,\text{kg}} = 26.3$ N, $F_{3\,\text{kg}} = 19.8$ N.

81. (*a*) $3g/2\ell$;
(*b*) $\frac{3}{2} g$.

83. $\ell/2$; $\ell/2$.

85. 0.965.

Chapter 9

1. 528 N, 120° clockwise from $\vec{\mathbf{F}}_{\text{A}}$.

3. (*a*) 2.3 m from vertical support;
(*b*) 4200 kg.

5. (*a*) $F_A = 1.5 \times 10^3$ N, down;
$F_B = 2.0 \times 10^3$ N, up;
(*b*) $F_A = 1.8 \times 10^3$ N, down;
$F_B = 2.6 \times 10^3$ N, up.

7. 1200 N.

9. $F_{closer} = 2900$ N, down;
$F_{farther} = 1300$ N, down.

11. (*a*) 2.3 m from the adult;
(*b*) 2.5 m from the adult.

13. $F_{left} = 260$ N, $F_{right} = 190$ N.

15. 20 N to 50 N.

17. 0.64 m to right of fulcrum rock.

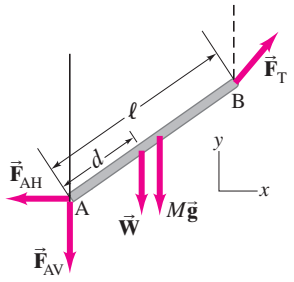
19. (*a*) 410 N;
(*b*) horizontal: 410 N;
vertical: 328 N.

21. $F_A = 1.7 \times 10^4$ N,
$F_B = 7.7 \times 10^3$ N.

23. 6.0×10^1 N; the angle is small.

25. (*a*) 230 N;
(*b*) 1.0×10^2 N.

27. (*a*)



(*b*) $F_{A_{vertical}} = 9$ N,
$F_{A_{horizontal}} = 51$ N;
(*c*) 2.4 m.

29. $F_{top_{horizontal}} = 55.2$ N, right;
$F_{top_{vertical}} = 63.7$ N, up;
$F_{bottom_{horizontal}} = 55.2$ N, left;
$F_{bottom_{vertical}} = 63.7$ N, up;

31. 970 N.

33. 1600 N.

35. Yes, 10.9 m.

37. 3.50×10^{-2} m.

39. 3.0×10^{-5} m.

41. 997.5 cm^3.

43. 9.6×10^6 N/m^2.

45. $(-2 \times 10^{-2})\%$.

47. (*a*) 393 N;
(*b*) thicker.

49. $A_A = 1.6 \times 10^{-3}$ m^2,
$A_B = 9.1 \times 10^{-3}$ m^2.

51. 2.7 cm.

53. 2.4×10^6 N.

55. (*a*) $\mu_s < \ell/2h$;
(*b*) $\mu_s > \ell/2h$.

57. (*a*) 0.78 N;
(*b*) 0.98 N.

59. 3.5×10^{-4} m.

61. A: 230 N; B: 110 N.

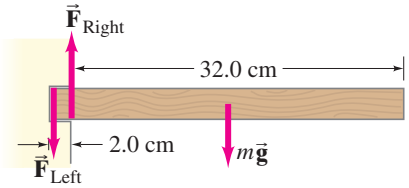
63. 570 N.

65. (*a*) $(4.5 \times 10^{-6})\%$;
(*b*) 9.0×10^{-18} m.

67. 2500 N, no.

69. 2.6×10^{-4} m^2.

71. (*a*)



(*b*) $F_{Right} = 550$ N, $F_{left} = 490$ N,
$mg = 65$ N;
(*c*) 11 m·N.

Chapter 10

1. 3×10^{11} kg.

3. 710 kg.

5. 0.8547.

7. (*a*) 5501 kg/m^3.
(*b*) 5497 kg/m^3, −0.07%.

9. (*a*) 6.1×10^6 N/m^2;
(*b*) 1.7×10^5 N/m^2.

11. (*a*) 4.5×10^5 N;
(*b*) 4.5×10^5 N.

13. 1.2 m.

15. 1900 kg.

17. (*a*) 7.0×10^5 N/m^2;
(*b*) 72 m.

19. 1.60×10^4 m.

21. 4.0×10^7 N/m^2.

23. 0.57.

25. (*a*) 1.5×10^5 N;
(*b*) 1.8×10^5 N.

27. Iron or steel.

29. 9.9×10^{-3} m^3.

31. 10.5%.

33. 32 bottles.

35. (*a*) 6.68×10^{-2} m^3;
(*b*) 1.07;
(*c*) 12%.

37. 4.4 m/s.

39. 9.6 m/s.

41. 4.12×10^{-3} m^3/s.

43. 1.6×10^5 N/m^2.

45. 9.0×10^{-3} m^3/s.

47. 3.2×10^6 N.

51. 1100 N.

53. 8.2×10^3 Pa.

55. 0.094 m.

57. (*a*) Laminar;
(*b*) 2940, turbulent.

59. 0.89 Pa/cm.

61. 2.4×10^{-2} N/m.

63. (*a*) $\gamma = F/4\pi r$;
(*b*) 1.7×10^{-2} N/m.

65. 1.5 mm.

67. (*a*) 7.6×10^{-4} N;
(*b*) 1.3 N.

69. 0.047 atm.

71. 0.32 N.

73. 4.3 m.

75. 100 N to 150 N.

77. 5.3 km.

79. 1 W.

81. 130 N.

83. 9.7×10^3 N.

85. 3.5×10^{-3} Pa·s.

87. 0.27 kg.

Chapter 11

1. 0.84 m.

3. 560 N/m.

5. (*a*) 650 N/m;
(*b*) 2.1 cm, 2.6 Hz.

7. 0.85 kg.

9. (*a*) $A_A = 2.5$ m, $A_B = 3.5$ m;
(*b*) $f_A = 0.25$ Hz, $f_B = 0.50$ Hz;
(*c*) $T_A = 4.0$ s, $T_B = 2.0$ s.

11. ±70.7% of the amplitude.

13. 0.233 s.

15. (*a*) 2.1 m/s;
(*b*) 1.5 m/s;
(*c*) 0.54 J;
(*d*) $x = (0.15\text{ m})\cos(4.4\pi t)$.

17. $\sqrt{3}:1$.

19. (*a*) 5.5×10^{-2} m;
(*b*) 0.59 m/s.

21. 312.6 m/s.

23. $\frac{1}{2\pi}\sqrt{\frac{2k}{m}}$.

25. 3.0 s.

27. (*a*) 1.8 s;
(*b*) 0.56 Hz.

29. Shorten it by 0.5 mm.

31. (*a*) $17\pi/360$;
(*b*) $\pi/15$;
(*c*) $\pi/15$.

33. 1.22 m.

35. AM: 190 m to 550 m; FM: 2.8 m to 3.4 m.
37. 0.21 s.
39. 1700 m.
41. (*a*) 0.11;
(*b*) 0.33.
43. 0.41.
45. 440 Hz, 880 Hz, 1320 Hz, 1760 Hz.
47. 60 Hz, fundamental or first harmonic; 120 Hz, first overtone or second harmonic; 180 Hz, second overtone or third harmonic.
49. 70 Hz.
51. 3.
53. 6.3 km/s.
55. 460 Hz; $f < 460$ Hz.
57. 0.11 m.
59. 0.16 m.
61. (*a*) 0.25 m;
(*b*) 0.748 Hz;
(*c*) 1.34 s;
(*d*) 0.45 J;
(*e*) KE: 0.29 J; PE: 0.16 J.
63. 3.2×10^{-2} m.
65. 8.40×10^{2} N/m.
67. 20 cm/s.
69. 0.69 m.
71. 1.59 cm.
73. 13 min.

Chapter 12

1. 430 m.
3. (*a*) 1.7 cm to 17 m;
(*b*) 1.9×10^{-5} m.
5. (*a*) 0.994 s;
(*b*) 4.52 s.
7. 33 m.
9. 62 dB.
11. 82 dB.
13. 82-dB player: 1.6×10^{8};
98-dB player: 6.3×10^{9}.
15. (*a*) 790 W;
(*b*) 440 m.
17. (*a*) 12;
(*b*) 11 dB.
19. 130 dB.
21. (*a*) 220-W: 122 dB; 45-W: 115 dB;
(*b*) no.
23. 80 Hz, 15,000 Hz.
25. 10 octaves.
27. 87 N.
29. (*a*) 360 Hz;
(*b*) 540 Hz.
31. 260 Hz.
33. (*a*) 17 cm;
(*b*) 1.02 m;
(*c*) $f = 440$ Hz, $\lambda = 0.78$ m.
35. 1.9%.
37. (*a*) 0.585 m;
(*b*) 858 Hz.
39. (*a*) 253 overtones;
(*b*) 253 overtones.
41. 4.2 cm, 8.2 cm, 11.9 cm, 15.5 cm, 18.8 cm, 22.0 cm.
43. $I_2/I_1 = 0.64$; $I_3/I_1 = 0.20$; $\beta_{2-1} = -2$ dB; $\beta_{3-1} = -7$ dB.
45. 28.5 kHz.
47. 347 Hz.
49. (*a*) 218.8 Hz or 221.2 Hz;
(*b*) 1.1% increase for 218.8 Hz, 1.1% decrease for 221.2 Hz.
51. (*a*) 0.429 m;
(*b*) Maxima are 0.429 m to the left and right of the midpoint, minimum is at the midpoint.
53. 4.27×10^{4} Hz.
55. 3.11×10^{4} Hz.
57. (*a*) Approaching: 1750 Hz; receding: 1440 Hz;
(*b*) approaching: 1880 Hz; receding: 1340 Hz;
(*c*) approaching: 1640 Hz; receding: 1530 Hz.
59. 11 km/h.
61. (*a*) 2.1°;
(*b*) 9.8°.
63. 16 km past the observer.
65. 23 dB.
67. 2.70×10^{2} W.
69. 0.013 W.
71. 1.07.
73. 1130 Hz.
75. 26 m/s.
77. (*a*) 0;
(*b*) 24 Hz;
(*c*) 0.
79. 11.5 m.
81. 50 Hz, 6th overtone.
83. 10^{6}.

Chapter 13

1. $N_{\text{gold}} = 0.548 N_{\text{silver}}$.
3. (*a*) 20°C;
(*b*) 3500°F.
5. 102.0°F.
7. −40°C = −40°F.
9. 0.08 m.
11. 2.2×10^{-6} m; $\frac{1}{60}$ of the change for steel.
13. −70°C.
15. (*a*) 5.0×10^{-5}/C°;
(*b*) copper.
17. 61.3 mL.
19. 60 minutes lost.
21. (*a*) 4300 K, 15×10^{6} K;
(*b*) 6.4%, (1.8×10^{-3})%.
23. 1000°C.
25. (*a*) 22.8 m^3;
(*b*) 2.16×10^{5} Pa.
27. 1.69×10^{8} Pa.
29. 7.4%.
31. 1.4 × original volume.
33. 181°C.
35. 2.69×10^{25} molecules/m^3.
37. (*a*) 7×10^{22} moles;
(*b*) 4×10^{46} molecules.
39. Yes; the molecules take up less than 0.1% of the volume of the gas.
41. 6×10^{3} m/s.
43. 43.9°C.
47. $\dfrac{(v_{\text{rms}})_{^{235}UF_6}}{(v_{\text{rms}})_{^{238}UF_6}} = 1.004$.
49. Vapor.
51. (*a*) Vapor;
(*b*) solid.
53. 3200 Pa.
55. 18°C.
57. 0.91 kg.
59. 33%.
61. 0.99 s, $v_{\text{diffuse}}/v_{\text{rms}} = 8.0 \times 10^{-8}$.
63. (*a*) Low;
(*b*) (2.8×10^{-2})%.
65. 18%.
67. (*a*) 1500 kg;
(*b*) 200 kg enters.
69. (*a*) Lower;
(*b*) 0.36%.
71. 910 min.
73. (*a*) 0.66×10^{3} kg/m^3;
(*b*) −3.0%.
75. 260 m/s, 3.7×10^{-22} atm.
77. (*a*) 1.61×10^{5} K;
(*b*) 2.01×10^{4} K.
79. 2.7 balls/s.

Chapter 14

1. 10.7°C.
3. 0.04 candy bars.
7. 250 kg/h.

9. 6.0×10^6 J.

11. (*a*) 3.3×10^5 J;
(*b*) 5600 s.

13. 4.0×10^2 s.

15. 42.6°C.

17. 2.3×10^3 J/kg·C°.

19. 43 C°.

21. 473 kcal.

23. 7.1×10^6 J.

25. 0.18 kg.

27. (*a*) 5.2×10^5 J;
(*b*) 1.5×10^5 J.

29. 11.2 kJ/kg.

31. 5.2 g.

33. (*a*) 71 W;
(*b*) 25 W.

35. 1.0×10^{-3} m.

37. 5.0 C°.

39. 130°C.

41. (*a*) 3.2×10^{26} W;
(*b*) 1.1×10^3 W/m^2.

43. A mixture of $\frac{1}{3}$ steam and $\frac{2}{3}$ liquid water at 100°C.

45. 2 C°.

47. 6.6×10^3 kcal.

49. 0.68C°.

51. (*a*) 8.0×10^2 W;
(*b*) 5.3 g.

53. 110°C.

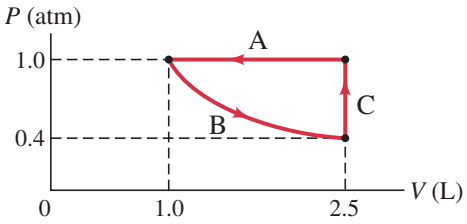
55. (*a*) 4.9×10^{16} J;
(*b*) $Q_{\text{Sun}} \approx 1.3 \times 10^4 Q_{\text{interior}}$.

57. (*a*) 3.6×10^7 J;
(*b*) 63 min.

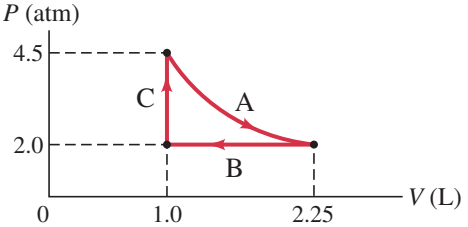
Chapter 15

1. (*a*) 0;
(*b*) 4.30×10^3 J.

3.



5.



7. (*a*) 0;
(*b*) 2630 J;
(*c*) rise.

9. −78 K.

13. (*a*) −196 J;
(*b*) 28 J;
(*c*) 168 J;
(*d*) 140 J;
(*e*) 98 J.

15. 170 W.

17. (*a*) 1.4×10^7 J;
(*b*) 3500 Cal.

19. 25.8%.

21. 8.8%.

23. 10°C decrease in the low-temperature reservoir.

25. 1.7×10^{13} J/h.

27. 1800 W.

29. 0.15.

31. 6.5.

33. (*a*) 1.0×10^3 J;
(*b*) 1.0×10^3 J;
(*c*) 230 J at 0°C, 390 J at −15°C.

35. $1,300; 11 years; $11,000.

37. 1300 J/K.

39. 4×10^4 J/K.

41. $\Delta S = \text{KE}/T_0$.

43. (*a*) $e_{\text{actual}}/e_{\text{ideal}} = 0.67$;
(*b*) 0.42 J/K;
(*c*) 0.

45. (*a*) 5/16;
(*b*) 1/64.

47. 70 m^2, yes.

49. 1.5×10^7 W.

51. −22°C.

53. (*a*) 5.0×10^4 J;
(*b*) 32 min.

55. $0.43/h.

57. (*a*) 44°C;
(*b*) 4.3×10^{-2} J/K.

59. 60 K.

61. (*a*) 13 km^3/day, possibly;
(*b*) 73 km^2.

63. 2.0×10^2 J.

65. (*a*) 110 kg/s;
(*b*) 1.1×10^8 gal/hr.

67. 160 W.

Chapter 16

1. 2.7×10^{-3} N.

3. 2.2×10^4 N.

5. (1.9×10^{-13})%.

7. 3.76 cm.

9. -4.6×10^8 C, 0.

11. $F_{\text{left}} = 120$ N, to the left;
$F_{\text{center}} = 560$ N, to the right;
$F_{\text{right}} = 450$ N, to the left.

13. 6.51×10^7 N, outward, along the line from the center of the square through the charge.

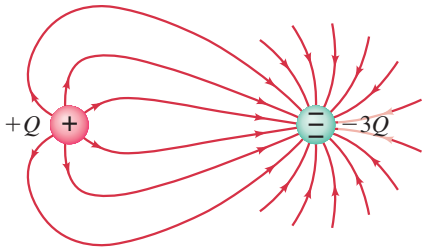
15. 4.6×10^{-6} C.

17. $0.40Q$, 0.37 ℓ from $-Q$ towards $-3Q$.

19. 6.30×10^6 N/C, upward.

21. 1.33×10^{14} m/s^2, opposite to the field.

23.



25. 1.37×10^{-10} C.

27. 1.15×10^6 N/C, towards the $-38.6\ \mu$C charge.

29. 3.7×10^7 N/C, 330°.

31. $E_A = 3.0 \times 10^6$ N/C, at 90°;
$E_B = 7.8 \times 10^7$ N/C, at 56°; yes.

33. (*a*) 5×10^{-10} N;
(*b*) 7×10^{-10} N;
(*c*) 6×10^{-5} N.

35. (*a*) -1.1×10^5 N·m^2/C;
(*b*) 0.

37. 8.3×10^{-10} C.

39. (*a*) $k\dfrac{Q}{r^2}$;
(*b*) 0;
(*c*) $k\dfrac{Q}{r^2}$;
(*d*) The shell causes the field to be 0 in the shell material. The charge polarizes the shell.

41. 4.0×10^9 C.

43. 6.8×10^5 C, negative.

45. 2.4×10^{-10} N.

47. (*a*) 1;
(*b*) $\left(\dfrac{4\ell kQ^2}{mg}\right)^{1/3}$.

49. 5×10^{-9} C.

51. 8.0×10^{-9} C.

53. (*a*) 4×10^{10} particles;
(*b*) 4×10^{-5} kg.

55. 9.90×10^6 N/C, downward.

57. -1.2×10^9 C.

59. $\dfrac{2kQa}{(x^2 + a^2)^{3/2}}$, down.

Chapter 17

1. 5.0×10^{-4} J.
3. -1.0 V.
5. 3.2×10^{4} V/m.
7. 2.4×10^{-2} m.
9. 2×10^{-5} m.
11. (*a*) 1.7×10^{7} m/s;
(*b*) 1.3×10^{7} m/s.
13. 1.63×10^{7} m/s.
15. (*a*) 4.8 keV;
(*b*) $v_e/v_p = 42.8$.
17. 1.80×10^{5} V.
19. 0.51 J.
21. 9.15×10^{6} m/s.
23. (*a*) 18 cm from $-$ charge, on opposite side from $+$ charge;
(*b*) 1.6 cm from $-$ charge, toward $+$ charge, and 8.0 cm from $-$ charge, away from $+$ charge.
25. (*a*) 1.6×10^{4} V;
(*b*) 9.9×10^{4} V/m, 64°.
27. 4.2×10^{6} V.
29. (*a*) 27 V;
(*b*) 2.2×10^{-18} J, or 14 eV;
(*c*) -2.2×10^{-18} J, or -14 eV;
(*d*) 2.2×10^{-18} J, or 14 eV.
31. (*a*) 6.6×10^{-3} V;
(*b*) 4.6×10^{-3} V;
(*c*) -4.6×10^{-3} V.
33. 19 V.
35. 7.2×10^{7} m^2.
37. 0.24 m^2.
39. 9×10^{-16} m, no.
41. $V_{2.50\,\mu F} = V_{6.80\,\mu F} = 611$ V,
$Q_{2.50\,\mu F} = 1.53 \times 10^{-3}$ C,
$Q_{6.80\,\mu F} = 4.16 \times 10^{-3}$ C.
43. 1.7×10^{-10} F.
45. 6.7×10^{-7} C.
47. 5.9×10^{-4} J.
49. 1.8×10^{3} J.
51. (*a*) $\frac{1}{2}$;
(*b*) $-\dfrac{Q^2 d}{4\varepsilon_0 A}$.
53. (*a*) 2.66×10^{-4} J;
(*b*) 1.13×10^{-4} J;
(*c*) -1.53×10^{-4} J.
55. 85.
57. (*a*) $3\frac{2}{3}$ V;
(*b*) 0110.
59. 4.
61. 3.7×10^{5} V/m.
63. (*a*) Multiplied by 2;
(*b*) multiplied by 2.
65. Alpha particle, 2.
67. Left: $-6.85kQ/\ell$, top: $-3.46kQ/\ell$, right: $-5.15kQ/\ell$.
69. (*a*) 17 cm from $-$ charge, on opposite side from $+$ charge;
(*b*) 1.1 cm from $-$ charge, toward $+$ charge, and 8.1 cm from $-$ charge, away from $+$ charge.
71. 1.03×10^{6} m/s.
73. (*a*) 8.8 nF;
(*b*) 14 μC.
75. (*a*) -2.9×10^{5} V;
(*b*) 7.3×10^{6} V/m, toward the $-$ charge
77. (*a*) 3.6×10^{3} m/s.
(*b*) 2.8×10^{3} m/s.
79. 1.7×10^{6} V.
81. 83 days.
83. (*a*) 3.3×10^{5};
(*b*) 3.5 s.
85. (*a*) 2.1×10^{-7} C, 1.2×10^{4} V/m, 5.9×10^{-9} F, 3.6×10^{-6} J;
(*b*) 6.6×10^{-7} C, 1.2×10^{4} V/m, 1.9×10^{-8} F, 1.2×10^{-5} J.

Chapter 18

1. 1.00×10^{19} electrons/s.
3. 6.2×10^{-11} A.
5. 1200 V.
7. (*a*) 28 A;
(*b*) 8.4×10^{4} C.
9. (*a*) 8.9 Ω;
(*b*) 1.2×10^{4} C.
11. (*a*) 4.8 A;
(*b*) 6.6 A.
13. 5.1×10^{-2} Ω.
15. Yes, for length 4.0 mm.
17. 2.0 V.
19. (*a*) 3.8×10^{-4} Ω;
(*b*) 1.5×10^{-3} Ω;
(*c*) 6.0×10^{-3} Ω.
21. 18C°.
23. 2400°C.
25. $R_{carbon} = 1.42$ kΩ,
$R_{Nichrome} = 1.78$ kΩ.
27. 17 Ω.
29. (*a*) 160 Ω, 0.68 A;
(*b*) 48 Ω, 2.3 A.
31. 130 A.
33. (*a*) 2.0×10^{1} Ω;
(*b*) 430 J.
35. 0.046 kWh; 6.6 cents per month.
37. 2.8×10^{6} J.
39. 24 bulbs.
41. 1.5 m; power increases 36× and could start a fire.
43. (*a*) 7.2 A;
(*b*) 1.7 Ω.
45. 0.41 A, 0.58 A.
47. 330 V.
49. (*a*) 30 V, 4 A;
(*b*) 3 V, 0.4 A.
51. 6.0×10^{-10} m/s.
53. 2.2 A/m^2, north.
55. 32 m/s (possible delay between nerve stimulation and generation of action potential).
57. 9.8 h.
59. 0.15 S.
61. (*a*) $40.70;
(*b*) 1400 kg.
63. (*a*) 19.5% decrease;
(*b*) actual decrease less than 19.5%.
65. (*a*) 190 Ω;
(*b*) 16 Ω.
67. (*a*) 1500 W;
(*b*) 12 A.
69. 2 : 1.
71. (*a*) 7.4 hp;
(*b*) 220 km.
73. (*a*) 1200 W;
(*b*) 1.0×10^{2} W.
75. 2000°C.
77. (*a*) 4×10^{6} J;
(*b*) 2×10^{4} m.
79. (*a*) 12 W;
(*b*) 4.6 W.
81. 1.34×10^{-4} Ω.

Chapter 19

1. (*a*) 5.92 V;
(*b*) 5.99 V.
3. 0.034 Ω; 0.093 Ω.
5. (*a*) 330 Ω;
(*b*) 8.9 Ω.
7. 2.
9. Connect 18 resistors in series; then measure voltage across 7 consecutive series resistors.
11. 0.3 Ω.
13. 560 Ω, 0.020.

15. 32 Ω.

17. 140 Ω.

19. $\frac{13}{8}R$.

21. 4.8 kΩ.

23. 55 V.

25. 0.35 A.

27. 0.

29. (*a*) 34 V;
(*b*) 85-V battery: 82 V;
45-V battery: 43 V.

31. $I_1 = 0.68$ A, left; $I_2 = 0.33$ A, left.

33. (*a*) $\mathscr{E}/R$;
(*b*) R.

35. 0.56 A.

37. 3 parallel sets, each with 100 cells in series.

39. 3.71×10^{-6} F.

41. 2.0×10^{-9} F, yes.

43. 1.90×10^{-8} F in parallel,
1.7×10^{-9} F in series.

45. 2:1.

47. In parallel, 750 pF.

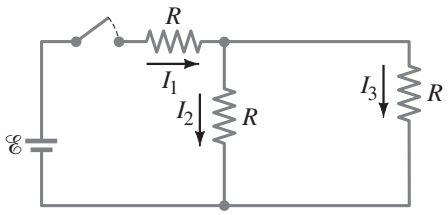
49. 29.3 μF, 5.7 μF.

51. (*a*) $\frac{3}{5}C$;
(*b*) $Q_1 = Q_2 = \frac{1}{5}CV$, $Q_3 = \frac{2}{5}CV$,
$Q_4 = \frac{3}{5}CV$; $V_1 = V_2 = \frac{1}{5}V$,
$V_3 = \frac{2}{5}V$, $V_4 = \frac{3}{5}V$.

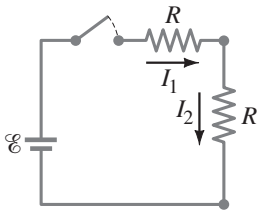
53. 1.0×10^{6} Ω.

55. 7.4×10^{-3} s.

57. (*a*) $I_1 = \dfrac{2\mathscr{E}}{3R}$, $I_2 = I_3 = \dfrac{\mathscr{E}}{3R}$



(*b*) $I_1 = I_2 = \dfrac{\mathscr{E}}{2R}$, $I_3 = 0$;



(*c*) $\frac{1}{2}\mathscr{E}$.

59. (*a*) 2.9×10^{-5} A;
(*b*) 8.8×10^{6} Ω.

61. Add 710 Ω in series with ammeter, 29 Ω/V.

63. 9.60×10^{-4} A, 4.8 V;
current: +20%, voltage: −20%.

65. 9.8 V.

67. Put 9.0 kΩ in series with the body.

69. $\frac{1}{4}C$, $\frac{2}{5}C$, $\frac{3}{5}C$, $\frac{3}{4}C$, C, $\frac{4}{3}C$, $\frac{5}{3}C$, $\frac{5}{2}C$, $4C$.

71. (*a*) 0.13 A;
(*b*) 0.13 A;
(*c*) 0.38 mA.

73. (*a*) R_2R_3/R_1;
(*b*) 130 Ω.

75. 7 resistors, 460 Ω each, in series;
7 resistors, 22.4 kΩ each, in parallel.

77. (*a*) 6.0 Ω;
(*b*) 2.2 V.

79. 11 V.

81. 100 Ω.

83. (*a*) 0.686 s;
(*b*) increase.

85. $Q_{12\,\mu F} = 1.0 \times 10^{-4}$ C,
$Q_{48\,\mu F} = 4.1 \times 10^{-4}$ C.

87. $q_1 = 15.5\ \mu$C, $q_2 = 8.5\ \mu$C,
$V_1 = V_2 = 7.1$ V.

89. 0.40.

91. 5.15×10^{-12} C.

Chapter 20

1. (*a*) 5.8 N/m;
(*b*) 3.3 N/m.

3. 1.3 N.

5. 27°.

7. (*a*) South pole;
(*b*) 3.86 A;
(*c*) 8.50×10^{-2} N.

9. 5.6×10^{-14} N, north.

11. 0.24 T.

13. (*a*) To the right;
(*b*) downward;
(*c*) into the page.

15. (*a*) 6.0×10^{5} m/s;
(*b*) 3.6×10^{-2} m;
(*c*) 3.8×10^{-7} s.

17. 0.59 m.

19. $r_{\text{proton}}/r_{\text{electron}} = 42.8$.

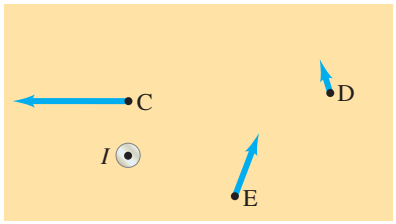
21. 1.97×10^{-6} m.

23. (*a*) Sign determines polarity but not magnitude of Hall emf.
(*b*) 0.56 m/s.

25. 2.9×10^{-4} T, about 5.8 times larger.

27. 7.8×10^{-2} N, toward other wire.

29.

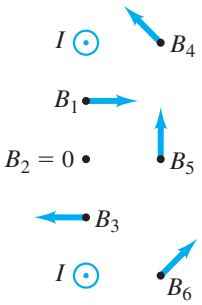


31. 5.1×10^{-6} N, toward wire.

33. 3.8×10^{-5} T, 17° below horizontal.

35. (*a*) $(2.0 \times 10^{-5}\ \text{T/A})(I - 25\ \text{A})$;
(*b*) $(2.0 \times 10^{-5}\ \text{T/A})(I + 25\ \text{A})$.

37.



39. 1.4×10^{-6} T, about 3% of Earth's field.

41. 9.5×10^{-5} T, 82°.

43. 1.19 A.

45. 0.12 N, south.

47. (*c*) No; inversely as distance from center of toroid: $B \propto 1/R$.

49. 1.18 T.

51. 69.7 μA.

53. 1.87×10^{6} V/m; perpendicular to velocity and magnetic field, and in opposite direction to magnetic force on protons.

57. 2.5 m.

59. 2.2×10^{-5} T·m/A.

61. $0.2\,\ell_{\text{wire}}$ N (assuming ℓ_{wire} in m).

63. qBr.

65. 2.0×10^{3} turns.

67. (*a*) 4.8×10^{-2} m;
(*b*) 5.5×10^{-7} s.

69. (*a*) 2.5×10^{-3} T;
(*b*) up;
(*c*) 7.0×10^{7} Hz.

71. 1×10^{9} A.

73. 3.2×10^{-4} m·N.

75. 2.1×10^{-6} *g*'s.

77. 1.3×10^{8} m/s; yes.

79. 0.03 N.

81. (*a*) M: 5.8×10^{-4} N/m, upward;
N: 3.4×10^{-4} N/m, at 300°;
P: 3.4×10^{-4} N/m, at 240°;
(*b*) 1.75×10^{-4} T, at −14°.

83. 2.9×10^{-6} T, $B_{\text{wire}} \approx 0.06 B_{\text{Earth}}$.

Chapter 21

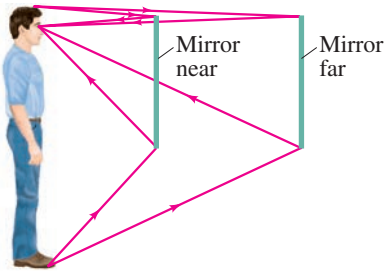
1. 560 V.
3. Counterclockwise.
5. 0.20 V.
7. (*a*) 1.0×10^{-2} Wb;
(*b*) 48°;
(*c*) 6.7×10^{-3} Wb.
9. (*a*) 1.73×10^{-2} V;
(*b*) 0.114 V/m, downward.
11. (*a*) 0;
(*b*) clockwise;
(*c*) counterclockwise;
(*d*) clockwise.
13. 0.65 mV, east or west.
15. (*a*) Magnetic force on current in moving bar, $B^2\ell^2v/R$;
(*b*) $B^2\ell^2v^2/R$.
17. (*a*) 0.15 A;
(*b*) 1.5×10^{-3} W.
19. 29 V.
21. (*a*) 99.0 A;
(*b*) 1.4×10^{-2} m^2.
23. 9.0×10^{1} V.
25. 13 A.
27. Step down, 0.33, 3.0.
29. 5.0×10^{1}, 4.8 V.
31. 450 V, 56 A.
33. 6×10^{9} m.
35. 55 MW.
37. 6.9 V.
39. 0.10 H.
41. (*a*) 1.5×10^{-2} H;
(*b*) 75 turns.
43. $\mu_0 \dfrac{N_1 N_2 A}{\ell}$.
45. 5.63×10^{-5} J, 0.537 s.
47. (*a*) 1.8 ms;
(*b*) 17 Ω.
49. V/L.
51. (*a*) 428 Ω;
(*b*) 2.57×10^{-2} Ω.
53. 10.9 Hz.
55. 5.22×10^{-2} H.
57. 2000 Ω.
59. (*a*) 3600 Ω;
(*b*) 3500 Ω.
61. 8.74 kΩ, −5.17°, 8.30×10^{-2} A.
63. (*a*) 1.77×10^{-2} A;
(*b*) −12.6°;
(*c*) $V_R = 117$ V, $V_C = 26.1$ V.
65. 3.6×10^{5} Hz.
67. (*a*) 1.3×10^{-7} F;
(*b*) 37 A.
69. (*a*) 0.032 H;
(*b*) 0.032 A;
(*c*) 16 μJ.
71. 6.01×10^{-3} J.
73. Coil radius = 1.5 cm, 10,000 turns.
75. (*a*) 0.77 A;
(*b*) 8.2.
79. Put a 98-mH inductor in series with it.
81. 82 V.
83. 93 mH.
85. 2.

Chapter 22

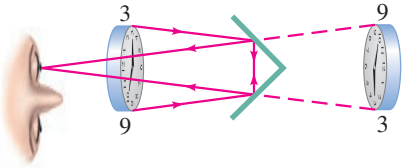
1. 1.1×10^{5} V/m/s.
3. 1.7×10^{15} V/m/s.
5. 2.4×10^{-13} T.
7. 90.0 kHz, 2.33 V/m, along the horizontal north–south line.
9. 1.25 s.
11. 4.20×10^{-7} m, violet visible light.
13. 2.00×10^{10} Hz.
15. (*a*) 1.319×10^{-2} m;
(*b*) 2.5×10^{18} Hz.
17. 4.0×10^{16} m.
19. 9600 wavelengths; 3.54×10^{-15} s.
21. 1.6×10^{6} revolutions/s.
23. 3.02×10^{7} s.
25. 5.7 W/m^2, 46 V/m.
27. 4.51×10^{-6} J.
29. 3.80×10^{26} W.
31. 7.3×10^{-7} N/m^2; 7.3×10^{-11} N away from bulb.
33. 400 m^2.
35. 0.16 m.
37. 290 pF.
39. (*a*) 1.3×10^{-6} H;
(*b*) 9.9×10^{-11} F.
41. 6.25×10^{-4} V/m;
1.04×10^{-9} W/m^2.
43. 10 ns.
45. 0.01 s.
47. (*a*) 0.07 V/m;
(*b*) 7 km.
49. (*a*) 1.2×10^{-10} J;
(*b*) 8.7×10^{-6} V/m, 2.9×10^{-14} T.
51. 61 km.
53. (*a*) 2.8×10^{-3} J/s;
(*b*) 1.0 V/m;
(*c*) 1.0 V;
(*d*) 2.0×10^{-2} V.
55. 6×10^{10} W.
57. 35 kW.

Chapter 23

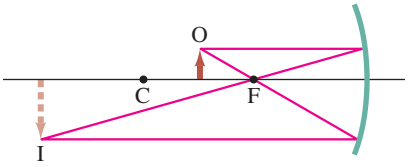
1.

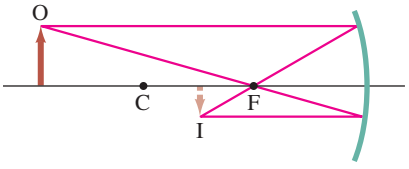


3. 11°.
5.



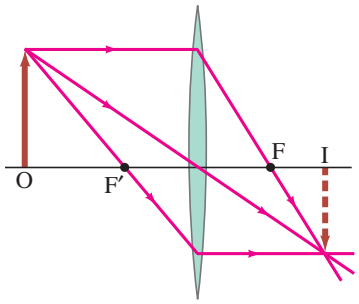
7. 37.6 cm.
9. (*a*) 12 cm;
(*b*) $d_i = 18$ cm;
(*c*) inverted.
11. Concave, 5.3 cm.
13. Convex, −38.0 cm.
15. $d_o = 2f$.
17. Convex, −16 m.
19. (*a*) Radius of curvature;
(*b*) real;
(*c*) inverted;
(*d*) −1.
21. (*a*)





23. 1.71*r*, −2.41.
25. (*a*) 2.21×10^{8} m/s;
(*b*) 1.99×10^{8} m/s;
(*c*) 1.97×10^{8} m/s.
27. 36°.
29. 38.6°.

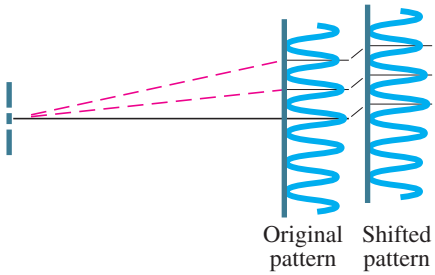
31. (*a*) 26.6°;
(*b*) 31.2°;
(*c*) 31.2°.
33. 61.0°, crown glass.
35. At least 93.5 cm away.
37. (*a*) 1.4;
(*b*) no;
(*c*) 1.9.
39. (*a*) ~500 mm;



(*b*) 478 mm.
41. (*a*) 3.08 D, converging;
(*b*) −0.148 m, diverging.
43. (*a*) 106 mm;
(*b*) 109 mm;
(*c*) 117 mm;
(*d*) 513 mm.
45. (*a*) 37 cm behind lens;
(*b*) +2.3×.
47. (*a*) 0.04 m away from lens;
(*b*) 0.007 m toward lens.
49. (*a*) −30.0 mm;
(*b*) −70.0 mm.
51. (*a*) $d_i = 151$ mm, $h_i = -0.278$ cm, real, inverted, reduced;
(*b*) $d_i = -122$ mm, $h_i = 0.226$ cm, virtual, upright, reduced.
53. 470 cm.
55. 7.2 cm beyond the second lens.
57. −10.0 cm.
59. −29.7 cm.
61. 9.0 cm.
63. $d_i = -170$ cm; +1.8×.
65. 1.00 m, 3.0 m, 4.0 m.
67. 54.6°.
69. 20.0 cm.
71. 3 m.
73. 9 cm, 12 cm.
75. $n \geq 1.60$.
77. (*a*) −0.33 mm;
(*b*) −0.47 mm;
(*c*) −0.98 mm.
79. Left: converging; right: diverging.
81. 50.0 cm.
83. −19.0 cm.

Chapter 24

1. 5.4×10^{-7} m.
3. 6.3×10^{-7} m, 4.8×10^{14} Hz.
5. 17° and 64°.
7. 1.5×10^{-4} m.
9. 3.1 cm.
11.



13. 2.2×10^{-3} m.
15. 14 mm.
17. 570 nm.
19. 0.23°.
21. 1.8°.
23. 1.0×10^{-6} m.
25. 7.38×10^{-2} m.
27. (*a*) λ;
(*b*) 400 nm.
29. 3.4×10^{-6} m.
31. 4.3°.
33. All of first order, most of second order, and part of third order.
35. 1400 slits/cm.
37. 1.3 m.
39. 462 nm, 676 nm.
41. 2.32 μm, 48.7°.
43. 630 nm.
45. (*a*) 180 nm;
(*b*) 360 nm, 540 nm.
47. (*b*) 290 nm.
49. 9.8 μm.
51. 325 nm.
53. 2.00×10^{-4} m.
55. 8.72×10^{-5} m.
57. 0.048.
59. (*a*) 35°;
(*b*) 63°.
61. 45°.
63. $(4.0 \times 10^1)°, (5.0 \times 10^1)°$.
65. $0.068 I_0$.
67. (*a*) 0.211;
(*b*) 2nd or 3rd;
(*c*) 2nd and 3rd.
69. (*a*) Constructive;
(*b*) destructive.
71. 1.35, no.
73. 544 nm.
75. (*a*) 82 nm;
(*b*) 130 nm.
77. 4.8×10^4 m.
79. 3.19×10^{-5} m.
81. 580 nm.
83. 0.6 m.
85. 658 nm; 782 slits/cm.
87. 400 nm, 600 nm.
89. (*a*) 0;
(*b*) 0.11;
(*c*) 0.

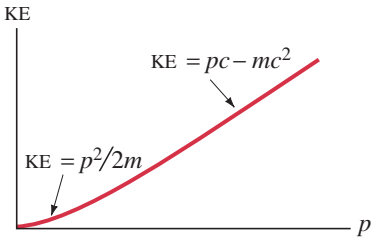
Chapter 25

1. *f*/8.
3. 3.00 mm to 46 mm.
5. 110 mm; 220 mm.
7. 5.1 m to infinity.
9. 16 mm.
11. *f*/2.5.
13. 1.1 D.
15. (*a*) −1.2 D;
(*b*) 35 cm.
17. −8.3 D; −7.1 D.
19. (*a*) 2.0 cm;
(*b*) 1.9 cm.
21. 18.4 cm, 1.00 m.
23. 1.6×.
25. 6.2 cm from lens, 4.0×.
27. (*a*) 3.6×;
(*b*) 14 mm;
(*c*) 6.9 cm.
29. 4 cm toward contract.
31. −29×; 85 cm.
33. 25 cm.
35. −22×.
37. −110×.
39. Objective: 1.09 m; eyepiece: 9.09 mm.
41. 7.0×.
43. 1.5 cm.
45. (*a*) $(8.40 \times 10^2)\times$;
(*b*) f_e: 1.8 cm, f_o: 0.299 cm;
(*c*) $d_o = 0.304$ cm.
47. (*a*) 0.85 cm;
(*b*) 210×.
49. (*a*) Converging;
(*b*) 281 cm.
51. $(1.54 \times 10^{-5})°$.
53. 1.7×10^{11} m.

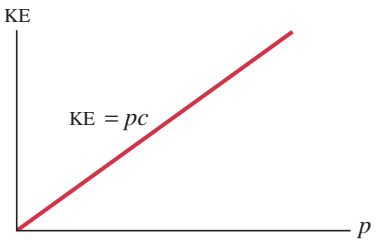
55. 1.0×10^4 m.
57. 8.5 cm, 6.2×10^{-6} rad (distance of 2.4 km).
59. (*a*) 53.8°;
(*b*) 0.19 nm.
61. (*a*) 1×;
(*b*) 1× (at back of body) to 2.7× (at front of body).
63. f_e: 3.3 cm, f_o: 25 cm.
65. 1.6 m.
67. 44 m.
69. −0.75 D (upper part), +2.0 D (lower part).
71. (*a*) −2.8×;
(*b*) +5.5-D lens.
73. +3.9 D.
75. 4.8×10^3 C°/min.
77. −15×.
79. 2.69×10^{-10} m.
81. 2 m.
83. −8.4 cm.

Chapter 26

1. 83.9 m.
3. 35 ly.
5. 0.70*c*.
7. (*a*) 0.141*c*;
(*b*) 0.140*c*.
9. (*a*) 6.92 m, 1.35 m;
(*b*) 13.9 s;
(*c*) 0.720*c*;
(*d*) 13.9 s.
11. (*a*) 3.6 yr;
(*b*) 7.0 yr.
13. $(6.97 \times 10^{-8})\%$.
15. 4.6×10^{-19} kg·m/s.
17. 0.41*c*.
19. (*a*) 220%;
(*b*) 210%.
21. 4×10^{-28} kg.
23. 939 MeV/c^2.
25. 1.6 GeV/*c*.
27. (*a*) 4.5 GeV;
(*b*) 5.4 GeV/*c*.
29. 0.866*c*, 0.886 MeV/*c*.
31. 9.0×10^{13} J; 9.2×10^9 kg.
33. 1.30 MeV/c^2.
35. 140 MeV/c^2.
37. 41.6 MeV, 283 MeV/*c*; energy: −6.3%, momentum: −4.2%.
39. 260 MeV.
43. 0.69*c*.
45. (*a*) 0.959*c*;
(*b*) 0.36*c*.
47. 0.962*c*.
49. 0.3*c*.
51. (*a*) 0.66*c*;
(*b*) 6.5 yr.
53. 0.91*c*.
55. 2.6×10^{-5} g.
57. 0.822*c*.
59. 28.32 MeV.
61. (*a*) 6.59 GeV/*c*;
(*b*) 0;
(*c*) 93.4 GeV/*c*.
63. (*a*)



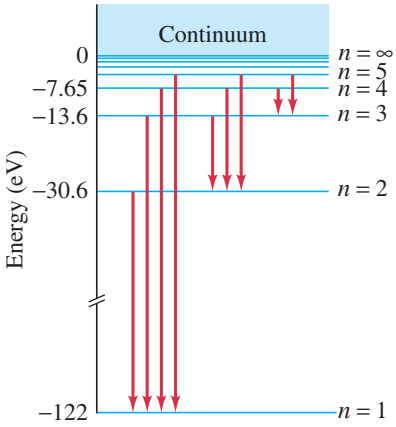
(*b*)



65. 1.8×10^{22} J.
69. (*a*) 0.986*c*;
(*b*) $(1 - 5 \times 10^{-7})c$.
71. Yes, in barn's reference frame, if his speed is ≥0.639*c*; no, in boy's reference frame.
73. 8.0×10^{-8} s.

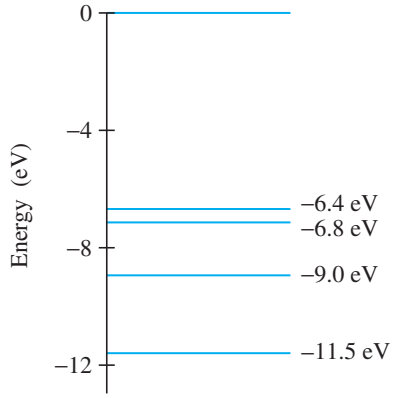
Chapter 27

1. 6.2×10^4 C/kg.
3. 5 electrons.
5. (*a*) 10.6 μm, far infrared;
(*b*) 940 nm, near infrared;
(*c*) 0.7 mm, microwave.
7. 5.4×10^{-20} J, 0.34 eV.
9. 9.35×10^{-6} m.
11. 2.7×10^{-19} J to 5.0×10^{-19} J, 1.7 eV to 3.1 eV.
13. 1.14×10^{-27} kg·m/s.
15. 2×10^{13} Hz, 1×10^{-5} m.
17. 3 photons.
19. Copper and iron.
21. 0.62 eV.
23. 3.46 eV.
25. 3.32 eV.
29. (*a*) 2.43×10^{-12} m;
(*b*) 1.32×10^{-15} m.
31. 2.62 MeV.
33. 212 MeV; 5.86×10^{-15} m.
35. 1.592 MeV, 7.81×10^{-13} m.
37. 4.7×10^{-12} m.
39. 1840.
41. (*a*) 4×10^{-10} m;
(*b*) 1×10^{-10} m;
(*c*) 3.9×10^{-11} m.
43. 6.4×10^{-12} m; yes; much less than 5 cm; no.
45. 2.30×10^{-11} m.
47. 1.51 eV.
49. 122 eV.
51. 91.2 nm.
53. Yes; from $n = 1$ to $n = 3$.
55.



57. 13,700; -7.19×10^{-8} eV.
61. 1.1 mm.
63. 1.2×10^{-10} m.
65. 3.6×10^{21} photons/m²/s.
67. 8.3×10^{-9} N.
69. 0.34 MeV for both.
71. 4.0×10^{-14} m.
73. 4.40×10^{-40}; yes.
75. 0.32 V.
77. 4.0×10^{-7} W/m²; 1.7×10^{-2} V/m.
79. 8.2×10^{-10} m.
81. $n = 4$ to $n = 3$: 1875 nm;
$n = 4$ to $n = 2$: 486.2 nm;
$n = 4$ to $n = 1$: 97.23 nm;
$n = 3$ to $n = 2$: 656.3 nm;
$n = 3$ to $n = 1$: 102.6 nm;
$n = 2$ to $n = 1$: 121.5 nm.

83. (*a*)



(*b*) Ground state;
$n = 4$ to $n = 3$: 0.4 eV;
$n = 4$ to $n = 2$: 2.6 eV;
$n = 4$ to $n = 1$: 5.1 eV;
$n = 3$ to $n = 2$: 2.2 eV;
$n = 3$ to $n = 1$: 4.7 eV;
$n = 2$ to $n = 1$: 2.5 eV.

85. (*a*) 0.39 V;
(*b*) 3.7×10^5 m/s;
(*c*) 2.0 nm.

87. 3.3×10^{52} photons/yr.

89. 7.88×10^4 K.

Chapter 28

1. 4.5×10^{-7} m.

3. 5.3×10^{-11} m.

5. 7×10^{-8} eV.

7. $3.00 \times 10^{-10}\ \text{eV}/c^2$.

9. Electron: 1.5×10^{-3} m;
baseball: 9.7×10^{-33} m;
$\Delta x_{\text{electron}}/\Delta x_{\text{baseball}} = 1.5 \times 10^{29}$.

11. 7700 m/s; 1.7×10^{-4} eV.

13. 5.53×10^{-10} m.

15. $\ell = 0,\ 1,\ 2,\ 3,\ 4,\ 5$.

17. 14 electrons.

19. (*a*) $(1, 0, 0, -\frac{1}{2})$, $(1, 0, 0, +\frac{1}{2})$,
$(2, 0, 0, -\frac{1}{2})$, $(2, 0, 0, +\frac{1}{2})$,
$(2, 1, -1, -\frac{1}{2})$, $(2, 1, -1, +\frac{1}{2})$;
(*b*) $(1, 0, 0, -\frac{1}{2})$, $(1, 0, 0, +\frac{1}{2})$,
$(2, 0, 0, -\frac{1}{2})$, $(2, 0, 0, +\frac{1}{2})$,
$(2, 1, -1, -\frac{1}{2})$, $(2, 1, -1, +\frac{1}{2})$,
$(2, 1, 0, -\frac{1}{2})$, $(2, 1, 0, +\frac{1}{2})$,
$(2, 1, 1, -\frac{1}{2})$, $(2, 1, 1, +\frac{1}{2})$,
$(3, 0, 0, -\frac{1}{2})$, $(3, 0, 0, +\frac{1}{2})$,
$(3, 1, -1, -\frac{1}{2})$.

21. $12\hbar$, or 3.65×10^{-34} J·s.

23. $n \geq 4$; $3 \leq \ell \leq n - 1$;
$m_s = -\frac{1}{2}, +\frac{1}{2}$.

25. (*a*) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6 4s^2$;
(*b*) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^4$;
(*c*) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 5s^2$.

27. (*a*) 5;
(*b*) −0.544 eV;
(*c*) $\sqrt{6}\,\hbar,\ 2$;
(*d*) $-2,\ -1,\ 0,\ 1,\ 2$.

31. $n = 3,\ \ell = 2$.

33. 36 kV.

35. 5.0×10^4 eV, 6×10^3 eV.

37. 6.12×10^{-11} m; partial shielding by $n = 2$ shell.

39. 0.017 J, 5.5×10^{16} photons.

41. 634 nm.

43. (*a*) 10^{-7} eV;
(*b*) 10^{-8};
(*c*) 122 nm, 8×10^{-7} nm.

45. 5.3×10^{-27} kg·m/s,
1.3×10^{-7} m.

47. 5.75×10^{-13} m, 115 keV.

49. 22 electrons.

53. $L_{\text{Bohr}} = 2\hbar$; $L_{\text{qm}} = 0$ or $\sqrt{2}\,\hbar$.

55. 3.7×10^{-37} m.

57. $\Delta p_{\text{electron}}/\Delta p_{\text{proton}} = 0.0234$.

59. 2.0×10^{-35} m/s; yes; 10^{34} s.

61. Copper.

Chapter 29

1. 5.1 eV.

3. HN: 110 pm, CN: 150 pm,
NO: 133 pm.

5. 4.6 eV.

9. 1.10×10^{-10} m.

11. 5.22×10^{-4} m.

13. (*a*) 680 N/m;
(*b*) 2.1×10^{-6} m.

15. 0.315 nm.

17. 2.0 eV.

19. $\lambda \leq 1.7\ \mu$m.

21. 1.2×10^6 electrons.

23. 3×10^6.

25. 1.7 eV.

27. 14 V.

29. 7.3 mA.

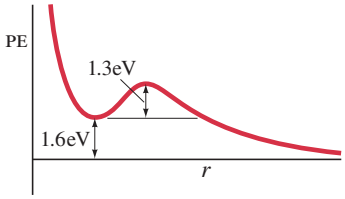
31. $I_B + I_C = I_E$.

33. 4400 Ω.

35. 0.24 mA.

37. 13 eV.

39.



41. 1.94×10^{-46} kg·m^2.

43. 5.50 eV.

45. 1.1 eV.

47. Green: 2.37 eV; blue: 2.67 eV.

Chapter 30

1. 0.149 u.

3. 0.855%.

5. 3727 MeV/c^2.

7. (*a*) 180 m;
(*b*) 3.5×10^4.

9. 30 MeV.

11. 6.0×10^{26} nucleons; no;
all nucleons have about the
same mass.

13. 550 MeV.

15. 7.699 MeV/nucleon.

17. 12.42 MeV.

19. $^{23}_{11}$Na: 8.113 MeV/nucleon;
$^{24}_{11}$Na: 8.063 MeV/nucleon.

21. (*b*) Yes, binding energy is positive.

25. 0.782 MeV.

27. (*a*) β^- emitter;
(*b*) $^{24}_{11}\text{Na} \rightarrow {}^{24}_{12}\text{Mg} + \beta^- + \bar{\nu}$,
5.515 MeV.

29. 2.822 MeV.

31. α: 6.114 MeV; β^-: 0.259 MeV.

33. 28.6 eV.

37. (*a*) $1.5 \times 10^{-10}\ \text{yr}^{-1}$;
(*b*) 6.0 h.

39. 0.1.

41. $\frac{1}{32}$.

43. 8.5×10^{19} nuclei.

45. (*a*) 4.2×10^{16} nuclei;
(*b*) 1.3×10^{15} nuclei;
(*c*) 2.8×10^{13} decay/s;
(*d*) 26 min.

47. 7.1×10^{10} decays/s.

49. 3.6 min.

51. $^{232}_{90}$Th: $^{228}_{88}$Ra, $^{228}_{89}$Ac, $^{228}_{90}$Th, $^{224}_{88}$Ra, $^{220}_{86}$Rn; $^{235}_{92}$U: $^{231}_{90}$Th, $^{231}_{91}$Pa, $^{227}_{89}$Ac, $^{227}_{90}$Th, $^{223}_{88}$Ra.

53. $\dfrac{\text{decay rate}_{218}}{\text{decay rate}_{214}} = 8.6 \times 10^{-7}$.

55. (*a*) 2.0×10^2 d;
(*b*) 2.7×10^{-14} g.

57. $^{211}_{82}$Pb.

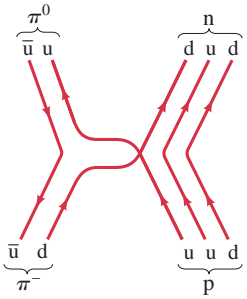
59. (*a*) 7×10^{55};
(*b*) 1×10^{29} kg;
(*c*) 3×10^{11} m/s^2.
61. (*a*) 60,000 yr;
(*b*) 8 million people.
63. $^{40}_{19}$K: 0.16 mg; $^{39}_{19}$K: 1.2 g.
65. 2.71×10^{-11} g.
67. (*a*) 4.6×10^{15} decays/s;
(*b*) 1.2×10^{4} decays/s.
69. (*a*) 0.002603 u, 2.425 MeV/c^2;
(*b*) 0, 0;
(*c*) −0.090739 u, −84.52 MeV/c^2;
(*d*) 0.043930 u, 40.92 MeV/c^2;
(*e*) $\Delta \geq 0$ for $0 \leq Z \leq 8$ and $Z \geq 85$;
$\Delta < 0$ for $9 \leq Z \leq 84$;
$\Delta \geq 0$ for $0 \leq A \leq 15$ and $A \geq 218$;
$\Delta < 0$ for $16 \leq A < 218$.
71. 0.2 decays/s.
73. (*a*) 180 m;
(*b*) 13 km.
75. (*a*) There would be no atoms—just neutrons;
(*b*) 0.083%.
77. (*a*) 1.63%;
(*b*) 0.663%.
79. Max: 4.6 MeV; min: 0; for max, 0; for min, 4.6 MeV.

Chapter 31

1. $^{28}_{13}$Al; β^-; $^{28}_{14}$Si.
3. Yes, $Q > 0$.
5. 18.000937 u.
7. (*a*) Yes;
(*b*) 20.4 MeV.
9. 4.730 MeV.
11. n + $^{14}_{7}$N → $^{14}_{6}$C + p, 0.626 MeV.
13. (*a*) He picks up a neutron from C;
(*b*) $^{11}_{6}$C;
(*c*) 1.856 MeV; exothermic.
15. 5.702 MeV released.
17. 126.5 MeV.
19. 1/1100.
21. (*a*) 5 neutrons;
(*b*) 171.1 MeV.
23. 1100 kg.
25. 260 MeV; about 30% > fission energy released.
27. 3000 eV.
31. (*a*) a: 6.03×10^{23} MeV/g;
b: 4.89×10^{23} MeV/g;
c: 2.11×10^{24} MeV/g;
(*b*) 5.13×10^{23} MeV/g;
a: 0.851; b: 1.05; c: 0.243.
33. 0.39 g.
35. No.
37. (*a*) 4.9;
(*b*) 1.5×10^{9} K.
39. 144 rads.
41. (*a*) 1.0 rad, or 0.010 Gy;
(*b*) 1.0×10^{10} protons.
45. 5.61×10^{-10} kg.
47. 9×10^{16} e$^-$.
49. (*a*) 2×10^{-7} Sv/yr,
2×10^{-5} rem/yr,
$(2 \times 10^{-4}) \times$ allowed dose;
(*b*) 2×10^{-6} Sv/yr,
2×10^{-4} rem/yr,
$(2 \times 10^{-3}) \times$ allowed dose.
51. 7.041 m; radio wave.
53. (*a*) $^{12}_{6}$C;
(*b*) 5.702 MeV.
55. 1.0043 : 1.
57. 6.6×10^{-2} rem/yr.
59. 2×10^{7} times the heat of combustion of coal.
61. (*a*) $^{226}_{88}$Ra → $^{4}_{2}$He + $^{222}_{86}$Rn;
(*b*) 4.871 MeV;
(*c*) 190.6 MeV/c for both;
(*d*) 8.78×10^{-2} MeV.
63. 8×10^{12} J.
65. (*a*) 3700 decays/s;
(*b*) 4.0×10^{-4} Sv/yr;
11% of background.
67. 7.274 MeV.
69. 0.65 J, 1.6 J.
71. Most to least dangerous: C > B > A.

Chapter 32

1. 5.59 GeV.
3. 1.3×10^{7} Hz.
5. 2.0 T.
7. Alpha particles;
$\lambda_\alpha \approx d_{\text{nucleon}}$, $\lambda_p \approx 2d_{\text{nucleon}}$.
9. 0.9 MeV/rev.
13. 33.9 MeV.
15. 1879.2 MeV.
17. 67.5 MeV.
19. 1.32×10^{-15} m.
21. 1.3×10^{-12} m.
23. 69.3 MeV.
25. 9×10^{-3} MeV.
27. 7.5×10^{-21} s.
29. (*a*) 1300 eV;
(*b*) 150 MeV.
31. 8×10^{-27} s.
33. (*a*) udd;
(*b*) $\bar{u}\bar{d}\bar{d}$;
(*c*) uds;
(*d*) $\bar{u}\bar{d}\bar{s}$.
35. $c\bar{u}$.
37.



39. 16 GeV; 7.8×10^{-17} m.
41. (*a*) 1.022 MeV;
(*b*) 1876.6 MeV.
43. 1.2×10^{10} m, 4.0×10^{1} s.
45. (*a*) Possible, strong interaction;
(*b*) possible, strong interaction;
(*c*) possible, strong interaction;
(*d*) not possible; charge is not conserved;
(*e*) possible, weak interaction.
47. 10^{-18} m corresponds to 200 GeV.
49. 64 fundamental fermions.
51. (*b*) 10^{29} K.
53. 798.7 MeV.
55. 9.3×10^{10} eV;
1.3×10^{-17} m, 0.99995c.
57. (*a*) π^0;
(*b*) $\bar{\nu}_\mu$.
61. 10^{-25} s.

Chapter 33

1. 3.1 ly.
3. Less than; by a factor of 2.
5. 2×10^{-3} kg/m^3.
7. 4.2×10^{-2} rad, or 2.4°;
about 4.5× Moon's width.
9. 1.83×10^{9} kg/m^3;
3.33×10^{5} times larger.
11. (*a*) 0.018″;
(*b*) $(5.0 \times 10^{-6})°$.
13. 280 yr.
15. $D_1/D_2 = 0.13$.
17. 3×10^{14} m.
19. 540°.
21. 3.1×10^{-16} m.
23. 2.1×10^{8} ly.
25. (*a*) 0.3 nm;
(*b*) 3.2 nm.
27. 0.058c.

29. 9×10^9 ly.

33. 3×10^{-13} m, gamma ray.

35. (*a*) 3×10^7 ly;
(*b*) 3 ly;
(*c*) 3×10^{-2} ly;
(*d*) 0.3 m.

37. (*a*) 2.1×10^{-26} kg/m^3;
(*b*) 1.0×10^{-25} kg/m^3.

39. 6×10^5.

41. 5 ly.

43. 3×10^{28} N.

45. $d_{480}/d_{660} = 1.4$.

47. (*a*) 13.93 MeV;
(*b*) 4.71 MeV;
(*c*) 5.46×10^{10} K.

49. 10^{53} kg.

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Note: The abbreviation *defn* means the page cited gives the definition of the term; *fn* means the reference is in a footnote; *pr* means it is found in a Problem or Question; *ff* means "also the following pages."

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CO–4 NASA/John F. Kennedy Space Center **4–1** Daly & Newton/Stone Allstock/Getty Images **4–4** Bettmann/Corbis **4–5** Gerard Vandystadt/Agence Vandystadt/Photo Researchers, Inc. **4–7** David Jones/Photo Researchers, Inc. **4–10** NASA/John F. Kennedy Space Center **4–33** AFP PHOTO/DIMITAR DILKOFF/Getty Images/Newscom **4–37** Lars Ternblad/Amana Japan **4–39** Prof. Melissa Vigil **4–41** Prof. Nicholas Murgo **4–42** Prof. Martin Hackworth **4–44** Kathleen Schiaparelli **4–46** Brian Bahr/Allsport Photography/Getty Images **4–71** Tyler Stableford/The Image Bank/Getty Images

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CO–9 Fraser Hall/Getty Images **9–1** AP Photo **9–21** D. Giancoli **9–23a/b** Mary Teresa Giancoli **9–26** Tarek El Sombati/iStockphoto **9–28a** D. Giancoli **9–28b** Christopher Talbot Frank/Ambient Images, Inc./Alamy Images **9–30** D. Giancoli **9–32** Richard Carafelli/National Gallery of Art **9–33** D. Giancoli **9–34** Italian Government Tourist Board **9–48a** James Lemass/Photolibrary

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CO–11 left Ford Motor Company **CO–11 right** Jonathan Nourok/PhotoEdit **11–4** Ford Motor Company **11–6** Paul Springett A/Alamy **11–11** Berenice Abbott/Photo Researchers, Inc. **11–13** D. Giancoli **11–17** Taylor Devices **11–19** Martin Bough/ Fundamental Photographs **11–20a** AP Wide World Photos **11–20b** Paul X. Scott **11–28** Art Wolfe/Getty Images **11–21a/b/c/d, 11–38** D. Giancoli **11–43** Jacynthroode/iStockphoto **11–45a/b** Richard Megna/Fundamental Photographs **11–49** Andre Gallant/Getty Images **11–59** Richard Megna/Fundamental Photographs

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CO–13 left Niall Edwards/Alamy **CO–13 right** Richard Price/Getty **13–3** Ingolfson **13–4b** Franca Principe/Istituto e Museo di Storia della Scienza, Florence **13–6** Leonard Lessin/Photolibrary **13–12** Pearson Education/Eric Schrader **13–16** D-BASE/Getty Images **13–17** Michael Newman/PhotoEdit **13–25** Paul Silverman/Fundamental Photographs **13–26a** Norbert Rosing/National Geographic Stock **13–26b** American Association of Physics Teachers/Mrinalini Modak **13–26c** Prof. Gary Wysin **13–27a/b/c** Mary Teresa Giancoli **13–30** Kennan Harvey/Getty Images **13–31** Reed Kaestner/Corbis

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CO–25, 25–5, 25–6a/b Mary Teresa Giancoli **25–08** Leonard Lessin/Photolibrary **25–19a/b** Museo Galileo - Istituto e Museo di Storia della Scienza **25–21** Yerkes Observatory, University of Chicago **25–22c** Sandy Huffaker/Stringer/Getty Images **25–22d** Inter-University Centre for Astronomy and Astrophysics/Laurie Hatch **25–24b** Leica Microsystems **25–29a/b** Reproduced by permission from M. Cagnet, M. Francon, and J. Thrier, The Atlas of Optical Phenomena. Berlin: Springer-Verlag, 1962. **25–32** Space Telescope Science Institute **25–33** David Parker/Photo Researchers, Inc. **25–37** The Burndy Library Collection/Huntington Library **25–40** Rosalind Franklin/Photo Researchers, Inc. **25–44a** Martin M. Rotker **25–44b** Scott Camazine/Alamy **25–47** Ron Chapple/Ron Chapple Photography **25–48** NOAA Space Environment Center

CO–26 "The City Blocks Became Still Shorter" from page 4 of the book "Mr Tompkins in Paperback" by George Gamow. Reprinted with the permission of Cambridge University Press. **26–1** Bettmann/Corbis **26–7** D. Giancoli **26–12** "Unbelievably Shortened" from page 3 of "Mr Tompkins in Paperback" by George Gamow. Reprinted with the permission of Cambridge University Press.

CO–27 left Paul Gunning/Photo Researchers, Inc. **CO–27 right** Professors Pietro M. Motta & Silvia Correr/Photo Researchers, Inc. **27–11** Samuel Goudsmit/AIP Emilio Segrè Visual Archives, Goudsmit Collection **27–12** Education Development Center, Inc. **27–16a** Lee D. Simon/Science Source/Photo Researchers, Inc. **27–16b** Oliver Meckes/Max-Planck-Institut-Tubingen/Photo Researchers, Inc. **27–16c** Science Photo Library/Alamy **27–22b** Richard Megna/Fundamental Photographs **27–23a/b/c** Wabash Instrument Corp./Fundamental Photographs

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CO–29 Intel **29–35** Jeff J. Daly/Fundamental Photographs **29–36** D. Giancoli **29–39** General Motors

CO–30 D. Giancoli **30–3** French Government Tourist Office **30–7** Enrico Fermi Stamp Design/2001 United States Postal Service. All Rights Reserved. Used with Permission from the US Postal Service and Rachel Fermi. **30–16** Fermilab Visual Media Services

CO–31 Peter Beck/Corbis **31–6** Archival Photofiles, [apf2-00502], Special Collections Research Center, University of Chicago Library **31–9** Tokyo Electric Power Company/Jana Press/ZUMAPRESS.com/Newscom **31–10** LeRoy N. Sanchez/Los Alamos National Laboratory **31–11** Corbis/Bettmann **31–15a** UPPA/Photoshot/Newscom **31–15b** National Ignition Facility/Lawrence Livermore National Laboratory **31–19a** Robert Turgeon, Cornell University **31–19b** Jack Van't Hof/Brookhaven National Laboratory **31–20b** Needell M.D./Custom Medical Stock Photography **31–22a/b** National Institute of Health **31–25b** Slaven/Custom Medical Stock Photo **31–27** Scott Camazine/Photo Researchers, Inc.

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CO–33 NASA, ESA, R. Ellis (Caltech), and the UDF 2012 Team **33–1a** NASA **33–1b** Allan Morton/Dennis Milon/Science Photo Library/Photo Researchers, Inc. **33–2c** NASA **33–3** U.S. Naval Observatory/NASA **33–4** National Optical Astronomy Observatories **33–5a** Reginald J. Dofour, Rice University **33–5b** U.S. Naval Observatory/NASA **33–5c** National Optical Astronomy Observatories **33–9a/b** Australian Astronomical Observatory/David Main Images **33–15a** NASA **33–23** Alcatel-Lucent **33–25** AP Photo/Fredrik Persson **33–26** WMAP Science Team/NASA **33–32** Bertil Ericson/SCANPIX/AP/Corbis

Fundamental Constants

Quantity	Symbol	Approximate Value	Current Best Value[†]
Speed of light in vacuum	c	3.00×10^8 m/s	2.99792458×10^8 m/s
Gravitational constant	G	6.67×10^{-11} N·m^2/kg^2	$6.67384(80) \times 10^{-11}$ N·m^2/kg^2
Avogadro's number	N_A	6.02×10^{23} mol^{-1}	$6.02214129(27) \times 10^{23}$ mol^{-1}
Gas constant	R	8.314 J/mol·K = 1.99 cal/mol·K = 0.0821 L·atm/mol·K	8.3144621(75) J/mol·K
Boltzmann's constant	k	1.38×10^{-23} J/K	$1.3806488(13) \times 10^{-23}$ J/K
Charge on electron	e	1.60×10^{-19} C	$1.602176565(35) \times 10^{-19}$ C
Stefan-Boltzmann constant	σ	5.67×10^{-8} W/m^2·K^4	$5.670373(21) \times 10^{-8}$ W/m^2·K^4
Permittivity of free space	$\epsilon_0 = (1/c^2\mu_0)$	8.85×10^{-12} C^2/N·m^2	$8.854187817\ldots \times 10^{-12}$ C^2/N·m^2
Permeability of free space	μ_0	$4\pi \times 10^{-7}$ T·m/A	$1.2566370614\ldots \times 10^{-6}$ T·m/A
Planck's constant	h	6.63×10^{-34} J·s	$6.62606957(29) \times 10^{-34}$ J·s
Electron rest mass	m_e	9.11×10^{-31} kg = 0.000549 u = 0.511 MeV/c^2	$9.10938291(40) \times 10^{-31}$ kg = 5.48579909(25) $\times 10^{-4}$ u
Proton rest mass	m_p	1.6726×10^{-27} kg = 1.00728 u = 938.27 MeV/c^2	$1.672621777(74) \times 10^{-27}$ kg = 1.007276467(45) u
Neutron rest mass	m_n	1.6749×10^{-27} kg = 1.008665 u = 939.57 MeV/c^2	$1.674927351(74) \times 10^{-27}$ kg = 1.008664916(45) u
Atomic mass unit (1 u)		1.6605×10^{-27} kg = 931.49 MeV/c^2	$1.660538921(73) \times 10^{-27}$ kg = 931.494061(21) MeV/c^2

[†] Numbers in parentheses indicate one-standard-deviation experimental uncertainties in final digits (2010).
Values without parentheses are exact (i.e., defined quantities).

Other Useful Data

Joule equivalent (1 cal)		4.186 J
Absolute zero (0 K)		−273.15°C
Acceleration due to gravity at Earth's surface (avg.)		9.80 m/s^2 (= g)
Speed of sound in air (20°C)		343 m/s
Density of air (dry)		1.29 kg/m^3
Earth:	Mass	5.98×10^{24} kg
	Radius (mean)	6.38×10^3 km
Moon:	Mass	7.35×10^{22} kg
	Radius (mean)	1.74×10^3 km
Sun:	Mass	1.99×10^{30} kg
	Radius (mean)	6.96×10^5 km
Earth–Sun distance (mean)		149.60×10^6 km
Earth–Moon distance (mean)		384×10^3 km

The Greek Alphabet

Alpha	A	α	Nu	N	ν
Beta	B	β	Xi	Ξ	ξ
Gamma	Γ	γ	Omicron	O	o
Delta	Δ	δ	Pi	Π	π
Epsilon	E	ϵ, ε	Rho	P	ρ
Zeta	Z	ζ	Sigma	Σ	σ
Eta	H	η	Tau	T	τ
Theta	Θ	θ	Upsilon	Υ	υ
Iota	I	ι	Phi	Φ	ϕ, φ
Kappa	K	κ	Chi	X	χ
Lambda	Λ	λ	Psi	Ψ	ψ
Mu	M	μ	Omega	Ω	ω

Values of Some Numbers

$\pi = 3.1415927$	$\sqrt{2} = 1.4142136$	$\ln 2 = 0.6931472$	$\log_{10} e = 0.4342945$
$e = 2.7182818$	$\sqrt{3} = 1.7320508$	$\ln 10 = 2.3025851$	1 rad = 57.2957795°

Mathematical Signs and Symbols

$\propto$	is proportional to	$\leq$	is less than or equal to
$=$	is equal to	$\geq$	is greater than or equal to
$\approx$	is approximately equal to	$\sum$	sum of
$\neq$	is not equal to	$\bar{x}$	average value of x
$>$	is greater than	Δx	change in x
$\gg$	is much greater than	$\Delta x \to 0$	Δx approaches zero
$<$	is less than	$n!$	$n(n-1)(n-2)\ldots(1)$
$\ll$	is much less than		

Properties of Water

Density (4°C)	1.000×10^3 kg/m^3
Heat of fusion (0°C)	334 kJ/kg (79.8 kcal/kg)
Heat of vaporization (100°C)	2260 kJ/kg (539.9 kcal/kg)
Specific heat (15°C)	4186 J/kg·C° (1.00 kcal/kg·C°)
Index of refraction	1.33

Unit Conversions (Equivalents)

Length

1 in. = 2.54 cm (defined)
1 cm = 0.3937 in.
1 ft = 30.48 cm
1 m = 39.37 in. = 3.281 ft
1 mi = 5280 ft = 1.609 km
1 km = 0.6214 mi
1 nautical mile (U.S.) = 1.151 mi = 6076 ft = 1.852 km
1 fermi = 1 femtometer (fm) = 10^{-15} m
1 angstrom (Å) = 10^{-10} m = 0.1 nm
1 light-year (ly) = 9.461×10^{15} m
1 parsec = 3.26 ly = 3.09×10^{16} m

Volume

1 liter (L) = 1000 mL = 1000 cm^3 = 1.0×10^{-3} m^3 = 1.057 qt (U.S.) = 61.02 $in.^3$
1 gal (U.S.) = 4 qt (U.S.) = 231 $in.^3$ = 3.785 L = 0.8327 gal (British)
1 quart (U.S.) = 2 pints (U.S.) = 946 mL
1 pint (British) = 1.20 pints (U.S.) = 568 mL
1 m^3 = 35.31 ft^3

Speed

1 mi/h = 1.4667 ft/s = 1.6093 km/h = 0.4470 m/s
1 km/h = 0.2778 m/s = 0.6214 mi/h
1 ft/s = 0.3048 m/s = 0.6818 mi/h = 1.0973 km/h
1 m/s = 3.281 ft/s = 3.600 km/h = 2.237 mi/h
1 knot = 1.151 mi/h = 0.5144 m/s

Angle

1 radian (rad) = 57.30° = 57°18′
1° = 0.01745 rad
1 rev/min (rpm) = 0.1047 rad/s

Time

1 day = 8.640×10^4 s
1 year = 365.242 days = 3.156×10^7 s

Mass

1 atomic mass unit (u) = 1.6605×10^{-27} kg
1 kg = 0.06852 slug
[1 kg has a weight of 2.20 lb where $g = 9.80\ m/s^2$.]

Force

1 lb = 4.44822 N
1 N = 10^5 dyne = 0.2248 lb

Energy and Work

1 J = 10^7 ergs = 0.7376 ft·lb
1 ft·lb = 1.356 J = 1.29×10^{-3} Btu = 3.24×10^{-4} kcal
1 kcal = 4.19×10^3 J = 3.97 Btu
1 eV = 1.6022×10^{-19} J
1 kWh = 3.600×10^6 J = 860 kcal
1 Btu = 1.056×10^3 J

Power

1 W = 1 J/s = 0.7376 ft·lb/s = 3.41 Btu/h
1 hp = 550 ft·lb/s = 746 W

Pressure

1 atm = 1.01325 bar = 1.01325×10^5 N/m^2
= 14.7 $lb/in.^2$ = 760 torr
1 $lb/in.^2$ = 6.895×10^3 N/m^2
1 Pa = 1 N/m^2 = 1.450×10^{-4} $lb/in.^2$

SI Derived Units and Their Abbreviations

Quantity	Unit	Abbreviation	In Terms of Base Units†
Force	newton	N	$kg \cdot m/s^2$
Energy and work	joule	J	$kg \cdot m^2/s^2$
Power	watt	W	$kg \cdot m^2/s^3$
Pressure	pascal	Pa	$kg/(m \cdot s^2)$
Frequency	hertz	Hz	s^{-1}
Electric charge	coulomb	C	$A \cdot s$
Electric potential	volt	V	$kg \cdot m^2/(A \cdot s^3)$
Electric resistance	ohm	Ω	$kg \cdot m^2/(A^2 \cdot s^3)$
Capacitance	farad	F	$A^2 \cdot s^4/(kg \cdot m^2)$
Magnetic field	tesla	T	$kg/(A \cdot s^2)$
Magnetic flux	weber	Wb	$kg \cdot m^2/(A \cdot s^2)$
Inductance	henry	H	$kg \cdot m^2/(A^2 \cdot s^2)$

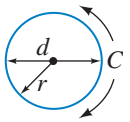
† kg = kilogram (mass), m = meter (length), s = second (time), A = ampere (electric current).

Metric (SI) Multipliers

Prefix	Abbreviation	Value
yotta	Y	10^{24}
zeta	Z	10^{21}
exa	E	10^{18}
peta	P	10^{15}
tera	T	10^{12}
giga	G	10^9
mega	M	10^6
kilo	k	10^3
hecto	h	10^2
deka	da	10^1
deci	d	10^{-1}
centi	c	10^{-2}
milli	m	10^{-3}
micro	μ	10^{-6}
nano	n	10^{-9}
pico	p	10^{-12}
femto	f	10^{-15}
atto	a	10^{-18}
zepto	z	10^{-21}
yocto	y	10^{-24}

Useful Geometry Formulas — Areas, Volumes

Circumference of circle $C = \pi d = 2\pi r$

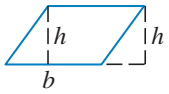


Area of circle $A = \pi r^2 = \dfrac{\pi d^2}{4}$

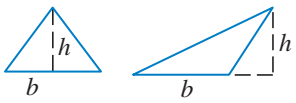
Area of rectangle $A = \ell w$



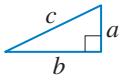
Area of parallelogram $A = bh$



Area of triangle $A = \frac{1}{2}hb$



Right triangle (Pythagoras) $c^2 = a^2 + b^2$



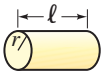
Sphere: surface area $A = 4\pi r^2$
volume $V = \frac{4}{3}\pi r^3$



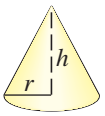
Rectangular solid: volume $V = \ell wh$



Cylinder (right): surface area $A = 2\pi r\ell + 2\pi r^2$
volume $V = \pi r^2\ell$



Right circular cone: surface area $A = \pi r^2 + \pi r\sqrt{r^2 + h^2}$
volume $V = \frac{1}{3}\pi r^2 h$



Exponents [See Appendix A–2 for details]

$(a^n)(a^m) = a^{n+m}$ [Example: $(a^3)(a^2) = a^5$]

$(a^n)(b^n) = (ab)^n$ [Example: $(a^3)(b^3) = (ab)^3$]

$(a^n)^m = a^{nm}$ [Example: $(a^3)^2 = a^6$; Example: $(a^{\frac{1}{4}})^4 = a$]

$a^{-1} = \dfrac{1}{a}$ $\quad a^{-n} = \dfrac{1}{a^n}$ $\quad a^0 = 1$

$a^{\frac{1}{2}} = \sqrt{a}$ $\quad a^{\frac{1}{4}} = \sqrt{\sqrt{a}}$

$(a^n)(a^{-m}) = \dfrac{a^n}{a^m} = a^{n-m}$ [Ex.: $(a^5)(a^{-2}) = a^3$]

$\dfrac{a^n}{b^n} = \left(\dfrac{a}{b}\right)^n$

Quadratic Formula [Appendix A–4]

Equation with unknown x, in the form

$ax^2 + bx + c = 0,$

has solutions

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

Logarithms [Appendix A–8; Table p. A-11]

If $y = 10^x$, then $x = \log_{10} y = \log y$.

If $y = e^x$, then $x = \log_e y = \ln y$.

$\log(ab) = \log a + \log b$

$\log\left(\dfrac{a}{b}\right) = \log a - \log b$

$\log a^n = n \log a$

Binomial Expansion [Appendix A–5]

$$(1 + x)^n = 1 + nx + \frac{n(n-1)}{2\cdot 1}x^2 + \frac{n(n-1)(n-2)}{3\cdot 2\cdot 1}x^3 + \cdots \qquad [\text{for } x^2 < 1]$$

$\approx 1 + nx$ if $x \ll 1$

[Example: $(1 + 0.01)^3 \approx 1.03$]

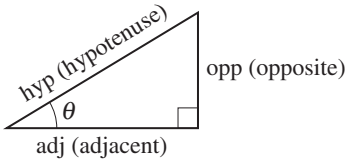
[Example: $\dfrac{1}{\sqrt{0.99}} = \dfrac{1}{\sqrt{1 - 0.01}} = (1 - 0.01)^{-\frac{1}{2}} \approx 1 - (-\frac{1}{2})(0.01) \approx 1.005$]

Fractions

$\dfrac{a}{b} = \dfrac{c}{d}$ is the same as $ad = bc$

$$\frac{\left(\frac{a}{b}\right)}{\left(\frac{c}{d}\right)} = \frac{ad}{bc}$$

Trigonometric Formulas [Appendix A–7]



$\sin\theta = \dfrac{\text{opp}}{\text{hyp}}$

$\cos\theta = \dfrac{\text{adj}}{\text{hyp}}$

$\tan\theta = \dfrac{\text{opp}}{\text{adj}}$

$\text{adj}^2 + \text{opp}^2 = \text{hyp}^2$ (Pythagorean theorem)

$\tan\theta = \dfrac{\sin\theta}{\cos\theta}$

$\sin^2\theta + \cos^2\theta = 1$

$\sin 2\theta = 2\sin\theta\cos\theta$

$\cos 2\theta = (\cos^2\theta - \sin^2\theta) = (1 - 2\sin^2\theta) = (2\cos^2\theta - 1)$

$\sin(180° - \theta) = \sin\theta$ $\quad \cos(180° - \theta) = -\cos\theta$

$\sin(90° - \theta) = \cos\theta$

$\cos(90° - \theta) = \sin\theta$

$\sin\frac{1}{2}\theta = \sqrt{(1 - \cos\theta)/2}$ $\quad \cos\frac{1}{2}\theta = \sqrt{(1 + \cos\theta)/2}$

$\sin\theta \approx \theta$ [for small $\theta \lesssim 0.2$ rad]

$\cos\theta \approx 1 - \dfrac{\theta^2}{2}$ [for small $\theta \lesssim 0.2$ rad]

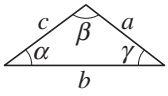
$\sin(A \pm B) = \sin A\cos B \pm \cos A\sin B$

$\cos(A \pm B) = \cos A\cos B \mp \sin A\sin B$

For any triangle:

$c^2 = a^2 + b^2 - 2ab\cos\gamma$ (law of cosines)

$\dfrac{\sin\alpha}{a} = \dfrac{\sin\beta}{b} = \dfrac{\sin\gamma}{c}$ (law of sines)



Periodic Table of the Elements[§]

[Masses averaged over natural abundances]

Group I	Group II	Transition Elements										Group III	Group IV	Group V	Group VI	Group VII	Group VIII
H 1 1.00794 $1s^1$	averaged over abundances of $^1_1$H, $^2_1$H																**He** 2 4.002602 $1s^2$
Li 3 6.941 $2s^1$	**Be** 4 9.012182 $2s^2$	Symbol: **Cl** 17 Atomic Mass[§] (averaged): 35.453 $3p^5$ — Electron Configuration (outer shells only)										**B** 5 10.811 $2p^1$	**C** 6 12.0107 $2p^2$	**N** 7 14.0067 $2p^3$	**O** 8 15.9994 $2p^4$	**F** 9 18.9984032 $2p^5$	**Ne** 10 20.1797 $2p^6$
Na 11 22.98976928 $3s^1$	**Mg** 12 24.3050 $3s^2$											**Al** 13 26.9815386 $3p^1$	**Si** 14 28.0855 $3p^2$	**P** 15 30.973762 $3p^3$	**S** 16 32.065 $3p^4$	**Cl** 17 35.453 $3p^5$	**Ar** 18 39.948 $3p^6$
K 19 39.0983 $4s^1$	**Ca** 20 40.078 $4s^2$	**Sc** 21 44.955912 $3d^14s^2$	**Ti** 22 47.867 $3d^24s^2$	**V** 23 50.9415 $3d^34s^2$	**Cr** 24 51.9961 $3d^54s^1$	**Mn** 25 54.938045 $3d^54s^2$	**Fe** 26 55.845 $3d^64s^2$	**Co** 27 58.933195 $3d^74s^2$	**Ni** 28 58.6934 $3d^84s^2$	**Cu** 29 63.546 $3d^{10}4s^1$	**Zn** 30 65.409 $3d^{10}4s^2$	**Ga** 31 69.723 $4p^1$	**Ge** 32 72.64 $4p^2$	**As** 33 74.92160 $4p^3$	**Se** 34 78.96 $4p^4$	**Br** 35 79.904 $4p^5$	**Kr** 36 83.798 $4p^6$
Rb 37 85.4678 $5s^1$	**Sr** 38 87.62 $5s^2$	**Y** 39 88.90585 $4d^15s^2$	**Zr** 40 91.224 $4d^25s^2$	**Nb** 41 92.90638 $4d^45s^1$	**Mo** 42 95.94 $4d^55s^1$	**Tc** 43 (98) $4d^55s^2$	**Ru** 44 101.07 $4d^75s^1$	**Rh** 45 102.90550 $4d^85s^1$	**Pd** 46 106.42 $4d^{10}5s^0$	**Ag** 47 107.8682 $4d^{10}5s^1$	**Cd** 48 112.411 $4d^{10}5s^2$	**In** 49 114.818 $5p^1$	**Sn** 50 118.710 $5p^2$	**Sb** 51 121.760 $5p^3$	**Te** 52 127.60 $5p^4$	**I** 53 126.90447 $5p^5$	**Xe** 54 131.293 $5p^6$
Cs 55 132.9054519 $6s^1$	**Ba** 56 137.327 $6s^2$	57–71[†]	**Hf** 72 178.49 $5d^26s^2$	**Ta** 73 180.94788 $5d^36s^2$	**W** 74 183.84 $5d^46s^2$	**Re** 75 186.207 $5d^56s^2$	**Os** 76 190.23 $5d^66s^2$	**Ir** 77 192.217 $5d^76s^2$	**Pt** 78 195.084 $5d^96s^1$	**Au** 79 196.966569 $5d^{10}6s^1$	**Hg** 80 200.59 $5d^{10}6s^2$	**Tl** 81 204.3833 $6p^1$	**Pb** 82 207.2 $6p^2$	**Bi** 83 208.98040 $6p^3$	**Po** 84 (209) $6p^4$	**At** 85 (210) $6p^5$	**Rn** 86 (222) $6p^6$
Fr 87 (223) $7s^1$	**Ra** 88 (226) $7s^2$	89–103[‡]	**Rf** 104 (263) $6d^27s^2$	**Db** 105 (268) $6d^37s^2$	**Sg** 106 (271) $6d^47s^2$	**Bh** 107 (270) $6d^57s^2$	**Hs** 108 (277) $6d^67s^2$	**Mt** 109 (278) $6d^77s^2$	**Ds** 110 (281) $6d^97s^1$	**Rg** 111 (281) $6d^{10}7s^1$	**Cn** 112 (285) $6d^{10}7s^2$	113[¶] (286)	**Fl** 114 (289) $6d^{10}7s^27p^2$	115[¶] (289)	**Lv** 116 (293) $6d^{10}7s^27p^4$	117[¶] (294)	118[¶] (294)

[¶] Preliminary evidence (unconfirmed) has been reported.

[†]**Lanthanide Series**	**La** 57 138.90547 $5d^16s^2$	**Ce** 58 140.116 $4f^15d^16s^2$	**Pr** 59 140.90765 $4f^35d^06s^2$	**Nd** 60 144.242 $4f^45d^06s^2$	**Pm** 61 (145) $4f^55d^06s^2$	**Sm** 62 150.36 $4f^65d^06s^2$	**Eu** 63 151.964 $4f^75d^06s^2$	**Gd** 64 157.25 $4f^75d^16s^2$	**Tb** 65 158.92535 $4f^95d^06s^2$	**Dy** 66 162.500 $4f^{10}5d^06s^2$	**Ho** 67 164.93032 $4f^{11}5d^06s^2$	**Er** 68 167.259 $4f^{12}5d^06s^2$	**Tm** 69 168.93421 $4f^{13}5d^06s^2$	**Yb** 70 173.04 $4f^{14}5d^06s^2$	**Lu** 71 174.967 $4f^{14}5d^16s^2$
[‡]**Actinide Series**	**Ac** 89 (227) $6d^17s^2$	**Th** 90 232.03806 $6d^27s^2$	**Pa** 91 231.03588 $5f^26d^17s^2$	**U** 92 238.0289 $5f^36d^17s^2$	**Np** 93 (237) $5f^46d^17s^2$	**Pu** 94 (244) $5f^66d^07s^2$	**Am** 95 (243) $5f^76d^07s^2$	**Cm** 96 (247) $5f^76d^17s^2$	**Bk** 97 (247) $5f^96d^07s^2$	**Cf** 98 (251) $5f^{10}6d^07s^2$	**Es** 99 (252) $5f^{11}6d^07s^2$	**Fm** 100 (257) $5f^{12}6d^07s^2$	**Md** 101 (258) $5f^{13}6d^07s^2$	**No** 102 (259) $5f^{14}6d^07s^2$	**Lr** 103 (262) $5f^{14}6d^17s^2$

[§] Atomic mass values averaged over isotopes in percentages they occur on Earth's surface; individual isotopes are found in Appendix B. For many unstable elements, mass of the longest-lived known isotope is given in parentheses. 2010 revisions.